INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

Healthcare

What the industry is, how it runs, and how to model it — for architects who have never worked in it

A working guide for an enterprise architect walking into a hospital group or a provincial health department. Why downtime here is clinical, and why the planning assumption is weeks rather than hours. The record that is a clinical instrument, a legal instrument and a billing instrument at once — and the audit finding that three of nine paid claims were conceded because nobody could find it. How to read a South African healthcare number when the official ones do not reconcile. Then the thing that makes this industry unique in the series: it has the deepest international data standards of any sector, and in South Africa almost nothing that matters is specified nationally — so the work is making, locally and explicitly, the decisions nobody has made nationally.

South African lens. Sourced throughout. October 2026.

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CHAPTER 1

The industry in one chapter

Three instincts from banking, insurance or retail will mislead you here: that an outage is a commercial event, that a system of record serves one purpose, and that the people who pay and the people who treat can be joined on a key.

What you are walking into

South Africa runs two health systems of roughly comparable financial scale, serving populations that differ by a factor of about six.

Medical schemes covered **9,13 million beneficiaries, 14,95% of the population**, in the Council for Medical Schemes' 2024/25 reporting, and paid **R239 bn** in benefits on that basis [2]. The same regulator's industry report for calendar 2024 gives benefits paid as **R259,3 bn** [3]. National Treasury treats **53,4 million people** as the public sector's responsibility, and the consolidated public health budget — national, provincial and the national entities — was reported at **R310 bn for 2026/27** [4][5]. Statistics South Africa's household survey puts scheme coverage at **14,1%**, not 14,95% [6].

Hold all four of those figures loosely. Every one is official, none is wrong, and no two of them answer the same question. Chapter 2 is about why, and it is the reason to read this book in order.

In law there is one health system, not two. Section 2 of the **National Health Act 61 of 2003** establishes "a national health system" spanning public *and* private providers [1]. That is not a drafting flourish. Private establishments are bound by the Act's records duties (ss13–17), its emergency treatment duty (s5), its consent provisions (s7) and its complaints regime (s18). The Office of Health Standards Compliance inspects private hospitals. The Health Ombud takes complaints about them. An architect arriving from banking or insurance expects the private sector to sit outside the public regulator's reach, as it largely does in those industries. Here it does not.

Three spheres of government hold health functions. The national Department of Health makes policy, sets standards and monitors; it does not, as a rule, deliver services, and most of its vote is transfers — final appropriation **R62,225 bn** in 2024/25 of which **R58,390 bn** was transfer payments [7]. Provinces hold the budget, employ the staff and run the facilities. Staff are provincial public servants paid through **PERSAL**, the national public service payroll, which is why public workforce data counts posts and persons rather than hours at a bedside [7]. Municipalities keep environmental health, local emergency services and disaster management.

So the architect commissioned "by the National Department of Health" is usually commissioned by a policy and standards body with a small operating budget and no service-delivery estate, and the architect commissioned by a province is working for the operator. Those are different engagements with different stakeholders, and confusing them wastes a quarter.

The categories of public hospital are set by **regulation**, not by policy: Government Notice R185 in Gazette 35101, current to 2 March 2012, which defines small, medium and large district hospitals, regional, tertiary, central and specialised hospitals with bed ranges [8]. The ranges **overlap** — district and regional both span 200 to 300 beds, regional and tertiary both span 400 to 800 — so a bed count does not determine a facility's category. The service package does. A central hospital must be attached to a medical school as its main teaching platform (same regulation), which makes academic health complexes a distinct governance animal in which a university and a provincial department share staff, space and records.

One further fact about the public estate belongs here, because the figure below states it. The national Department of Health has built its own primary care electronic medical record, **e-impilo**, and a tender issued on 3 March 2026 procures its rollout across fixed primary health care facilities in **eight provinces** over eighteen months. The tender's own total is **about 3 083 facilities** while its provincial figures sum to **3 213**; both are in the source document and both should be cited [37]. The Western Cape is absent, because it runs its own provincial estate. South Africa is therefore deliberately running two architectures at once, federated through the national identity service. Chapter 15 takes that properly.

Two systems, one statute, no shared mechanism

Each figure carries its own as-at date. Do not add the two populations: people use both.

PUBLIC	PRIVATE, SCHEME-FUNDED
People Treasury treats as its responsibility **53,4 million** 2026 Budget Review	Medical scheme beneficiaries **9,13 million** 14,95%, CMS 2024/25
Consolidated health budget, 2026/27 **R310 bn** reported Feb 2026	Benefits paid **R239 bn** or R259,3 bn, other basis
Hospital beds **100 656 approved / 85 126 usable** July 2022; a different series gives 88 785 for 2023	Hospital beds **Not statable nationally** last national figure 20 466, in 2011
Pathology **NHLS, one national entity** bills provinces; 140 million-plus tests a year	Pathology **Ampath, Lancet, PathCare** separate organisations from the hospital
Primary care record **e-impilo in eight provinces** a provincial estate in the Western Cape	Clinical record **Per group, per practice** specialists are not employees and keep their own

What joins them, nationally

A patient identifier: none. A facility register with a stable key: none. A tariff: none since 2010.

A procedure classification in routine use: none. A laboratory results repository: none.

Both systems are capacity-constrained and financially stressed. The connectors do not exist.

Figure 1.1 Two systems, one statute, and no national mechanism joining them. Each figure carries its own as-at date, and the two population figures must never be added, because a substantial number of scheme members use public primary care. The band at the foot is the list that makes South African health architecture different from its European and American reference models: the connectors a reference architecture assumes are national do not exist here.

Referral is a legal act, not a routing instruction. Section 44 of the National Health Act lets a user attend **any** public establishment and obliges an establishment that cannot provide necessary care to transfer the user onward [1]. Section 10 requires a discharge report with prescribed content — verbal is permitted for outpatients, **written is mandatory for inpatients** [1]. Read those two together and the position is stark: in the public system the referral letter and the discharge report are the only guaranteed clinical data interchange, and they are **documents carried by patients, not messages carried by interfaces**.

The private sector's defining structural fact is different and equally consequential: **the hospital does not employ the doctors.** Specialists hold admitting rights at one or more hospitals and bill funders or patients separately from the hospital. Netcare reported net **117 new specialists** granted admission rights in the year to 30 September 2025 [9]; Life Healthcare planned about **140** for its 2026 financial year [10]. Those are statements about relationships, not about payroll. Four consequences follow, and they are routinely underestimated:

- The hospital's clinical record and the specialist's practice record are different records, in different systems, owned by different legal entities.
- A single inpatient episode generates **multiple independent claims** — hospital, surgeon, anaesthetist, pathology, radiology, sometimes physiotherapy and a prosthesis supplier — each with its own practice number, its own tariff basis and its own submission route.
- Orders placed by a non-employee practitioner into a hospital system raise authentication, delegation and liability questions that an employed-clinician model does not.
- The hospital cannot mandate the specialist's clinical system, so interoperability at that seam is a commercial negotiation per specialty rather than an information technology decision.

Diagnostics are a separate organisation in both sectors. In the public sector pathology comes from the **National Health Laboratory Service**, a national public entity that bills provinces for tests and processes "more than 140 million tests a year" [11]. In the private sector it is Ampath, Lancet, PathCare or another group. Either way, order-to-result is an inter-organisational integration with its own identifiers, its own result formats and its own turnaround commitments, and there is no national repository of laboratory results. A provider that wants a complete result history for one patient must integrate with each laboratory and each imaging provider individually, and must resolve patient identity across every one of those boundaries.

Three listed groups dominate private acute capacity. On the Competition Commission's Health Market Inquiry measurement of 2016 shares, Netcare held **31% of beds** (**33% of admissions**), Mediclinic **26,8%** (**28,6%**) and Life **25,3%** (**28,5%**) — together about **83% of beds** — with about 60% of local facility markets highly concentrated [12]. Those shares are nearly a decade old and they are the most recent official measurement available. Volume growth in the groups is now essentially zero: Netcare's paid patient days grew **0,7%** in the year to 30 September 2025 and 0,7% in the half to 31 March 2026, and Life's went slightly negative [9][10]. Earnings growth is coming from tariff, cost control, case mix and capital structure. **The mandate in a**

private-sector engagement will therefore be efficiency, acuity mix and retention of scheme volumes, not growth, and an architect who arrives with a customer-acquisition reference model has misread the brief.

Finally, the patients. Modelled South African adult prevalence puts hypertension at **33,9%** (15,315 million people), HIV at **17,2%** (7,762 million), diabetes at **10,2%** (4,608 million), with the commonest pairs hypertension with arthritis, hypertension with diabetes, and **hypertension with HIV at about 2,17 million adults** [13]. The brief that publishes those figures flags its own caveat: several underlying prevalence sources are old, so these are modelled estimates rather than current measurements (same source).

The architectural reading of that paragraph is the one sentence to carry out of it. A person on antiretroviral therapy who also has hypertension and type 2 diabetes is **one human being who, in most South African health information architectures, exists as three disease-programme records in three vertical systems with three review cycles and three reporting chains.** That is the core data problem of the country's health system and it is not solved anywhere at national scale.

> **Two organisations appear throughout this book, and both are invented. Driehoek Health Group** is a fictional mid-size South African private hospital group — nine acute hospitals, two day clinics, about 1 750 registered beds, a shared pathology contract and an in-house pharmacy function. **Rietvlei District Health Service** is a fictional public health district — a district hospital, three community health centres, 22 fixed clinics and a mobile service. They carry the private-sector and public-sector teaching respectively. Real organisations are named normally, and every figure attributed to one carries its source and its date.

The first thing another industry does not prepare you for: downtime here is clinical

In banking an outage costs money and reputation. In healthcare an outage delays a diagnosis, delays an antibiotic, loses a blood test result, sends an ambulance somewhere else and can contribute to a death. That last clause is not rhetoric and it is not a thought experiment. It is a finding.

In June 2024 the Qilin group attacked **Synnovis**, a pathology provider to several National Health Service trusts in London. Thousands of appointments and procedures were cancelled. One patient died unexpectedly during the disruption, and an NHS patient safety investigation found several contributing factors, **including a long wait for a blood test result caused by the attack on pathology services**. A South East London Integrated Care Board figure recorded **170 patients harmed**, mostly classed as low harm [14].

That single citation does more work than any argument about confidentiality. It establishes that availability of the diagnostic loop is a patient safety control, which means it is a clinical governance obligation rather than an information technology service level.

The duration evidence is the second surprise. **Ascension**, a large and well-resourced United States health system, detected unusual activity on 8 May 2024 and confirmed ransomware affecting its hospitals on 13 May. Hospitals were still on downtime procedures on 24 May. Electronic health record access took roughly **six weeks** to restore. During it, staff used pen and paper; some hospitals diverted ambulances; pharmacies closed and many hospitals could not fill prescriptions; nurses reported heavier patient loads and concerns about medication errors; and **laboratory results that normally took 30 to 60 minutes sometimes took several hours** [15].

South Africa has its own case, and every architect working here should be able to describe it. The **National Health Laboratory Service** was attacked in June 2024. Laboratories fell back on manual processes to collate and communicate results, creating a backlog [16]. The NHLS initially projected full operation by 15 July 2024; some critical services returned in a phased manner through August; **operations were announced as fully restored on 21 August 2024, after a rebuild of systems and infrastructure** [16][17]. That is approximately two months, in the organisation that provides pathology for most of the country's public health sector, during an mpox outbreak.

Then, twenty months later, it happened again for a different reason. On **Monday 16 March 2026 at about 21:00** NHLS information systems failed; public health facilities nationwide could not access laboratory results; and **TrakCare**, named as the system underpinning national laboratory operations, was back up in the early hours of **Wednesday 18 March**. The NHLS attributed the disruption to "erratic power supply challenges" around its Sandringham head office, called it an infrastructure issue and said it was not a cybersecurity incident. The chief executive of the South African Medical Research Council linked it to a power outage at the head office where the servers holding patient results are housed and **said the NHLS appeared to have no uninterruptible power supply or other backup system**. Specimen registration was mainly affected, new investigations had to be deferred, emergency services were reportedly hardest hit, and staff relied on printouts from laboratories [18].

Three conclusions follow, and they are the ones to carry into every design conversation in this industry. Chapter 19 develops all three with the full set of sourced incidents.

First, the planning assumption is weeks, not hours. Every incident in the record with a sourced restoration timeline ran to weeks or months. The widely used contingency control set asks for enough paper forms on each unit to care for patients for at least **eight hours** [19]. Read against a six-week reality that is a replenishment unit, not a reserve, and an architecture needs a *supply* plan for extended paper operation including printing capacity that does not depend on the failed estate.

Second, the read path is the one that must survive. In a clinical outage the write path can be substituted by paper at a cost in effort. The read path cannot be substituted at all: a clinician without the record does not know the allergies, the current medications, the last potassium result or the resuscitation decision. Paper can capture what happens next; it cannot reconstruct what came before. The published control set asks for a **read-only backup record, updated at least hourly, tested weekly, and able to print** [19]. It is the highest-value single resilience investment in a clinical estate and it is cheap relative to the primary, because a read-only copy needs no transactional integrity and no high availability on write — provided it is **independently reachable**, with separate authentication, separate network path and separate hosting. A read-only copy inside the same failure domain as the primary is decoration.

Third, the dominant failure mode is supplier concentration, and the architecture must name the single points of failure it does not own. Synnovis was a supplier. The NHLS is a supplier. The clinical impact in both cases was distributed across many providers who had no control over the compromised entity and could not have mitigated it alone. In South Africa the list of shared services whose failure stops clinical work is short and

nameable: the **NHLS** for public-sector pathology, the **claims switch network** for private-sector revenue, **HPRS** for national identity, and **SyNCH** for chronic medicines dispensing. A provider architecture that does not declare those dependencies has not described the system it is actually running.

> Downtime in healthcare is a clinical risk, with a documented patient death and 170 recorded harms behind the claim. Derive recovery objectives from how long it takes a patient to come to harm, not from financial loss per hour. Then design the degraded state deliberately, because it is a state the organisation will occupy for weeks.

The second thing: one record, three instruments

In most industries the system of record for a transaction serves one purpose and acquires others by accident. In healthcare the patient record is, simultaneously and by design, three different instruments governed by three different bodies of rule.

It is a clinical instrument. It is how the next clinician to see this patient knows what the last one found, decided and prescribed. The professional guidance sets out the required content with unusual specificity: identifying particulars; history including **allergies and idiosyncrasies**; the **time, date and place of each consultation**; assessment, proposed management, medication and dosage; referrals; reactions to treatment including adverse effects; test and imaging results; sick-leave periods and reasons; and written proof of informed consent where applicable [20]. The 2022 revision adds that **critical information such as allergies must be prominently recorded and easily accessible** [21]. That is a user-interface requirement stated by a professional council.

It is a legal instrument. The same guidance says **nothing may be removed from a record**; errors are struck through so the original stays legible, with the correction dated, signed in full and the **reason recorded**; later additions are dated and signed [20]. The 2022 revision carries those rules into electronic records: entries must be **attributable**, and where a system will not identify the practitioner it must be signed by electronic means; **late entries and addenda must be marked and their reasons given**; alterations must be **fully attributable to the person who made them**, with the **reason for the amendment recorded** [21].

Read that set together and the professional rule describes an **append-only record with a full, attributed, reasoned audit trail of every change**. It is a demanding specification, and most clinical systems in use do not meet it. The requirement that is almost never implemented is the last one: the *reason* for an amendment, as a captured field. An architect who writes nothing else into a target state should write that.

Two further legal facts raise the stakes. First, **unauthorised access to a health record is a criminal offence** under National Health Act s17(2), punishable by a fine or up to one year's imprisonment, independently of any privacy statute [1]. That is the strongest available argument for access auditing in this industry, and it is not a privacy argument — it is a criminal-liability argument, which lands differently in a boardroom. Second, the access rules are age-sensitive in ways most consent models cannot express: patients aged **12 and older** may request a copy, abstract or direct access; for patients **under 16** a parent or guardian may apply but access requires **the patient's** written authorisation; and information about termination of pregnancy may be disclosed only to the patient, **whatever her age** [20]. A system whose consent model is "a parent sees a child's record until 18" is non-compliant in three separate ways.

It is a billing instrument. This is the part an architect from another industry will not predict, and it is where the South African specifics bite. Diagnostic coding in private claims is required not by a health statute but by **Regulation 5(f) of the Medical Schemes Act 131 of 1998** — a funder regulation — which requires every claim to carry the relevant diagnostic code and provides that claims without it may not be considered for payment [22]. ICD-10 was adopted as the national diagnostic coding standard on **1 July 2005**, but the operationally decisive date is **1 July 2014**, when phase 4.1 introduced age and gender validation with full claim rejection on clinical data [23].

Notice what that instrument does and does not do. The obligation is **conditional on payment, not penal**: an uncoded claim is unpayable, not unlawful. There is no national clinical coding regulation; the Department's own status document recorded one as "in development" as at 2015 [22]. So the only enforced coding discipline in the country is the one a funder enforces by refusing to pay, which means **the code that reaches a national data set is the code that got the claim paid**. Every downstream analysis of South African private-sector morbidity inherits that incentive. Chapter 12 works through what an architect must therefore write down locally.

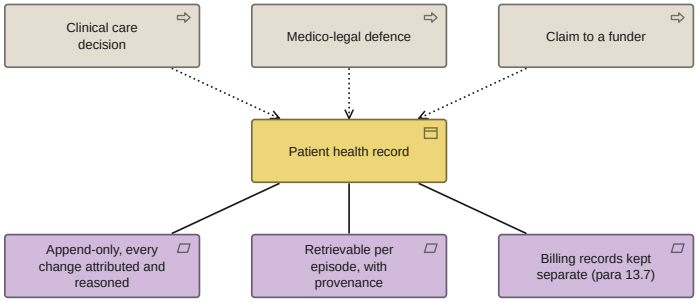


Figure 1.2 One object, three instruments, three sets of rules. The same record is read by clinical care, by medico-legal defence and by the claim to a funder, and each reader imposes requirements the others do not. The three requirements in the bottom row come from different instruments: the append-only and attribution rules and the separation of billing from care records from HPCSA Booklet 9, and retrievability from the Auditor-General's findings on conceded claims. Modelling the record for only one of the three readers is the commonest structural error in this industry.

The three instruments pull in different directions, and the professional guidance says so explicitly. **Booklet 9 paragraph 13.7 requires billing records to be kept separate from patient care records** [20]. That is an architectural segregation requirement, stated by the professional regulator, which almost no integrated hospital information system honours. For an architect it is unusually useful: it is published authority for a design principle that is otherwise

hard to sell, namely that the revenue cycle and the clinical record are different stores with different access populations, and that the clinical record should not be reachable from a billing role.

One more sentence from the practitioner guidance deserves to be read aloud in any meeting about a cloud platform or an outsourced scanning bureau: **outsourcing record-keeping does not transfer accountability.** The practitioner remains responsible for the record's integrity and safety. An operator agreement allocates work, not liability. Chapter 13 takes the record as an object end to end, and chapter 18 takes the regulatory frame around it.

The third thing: the money and the care are governed separately, and share no identifier

This is the structural fact that most often produces a wrong architecture, because it has no analogue in banking, insurance or retail. **Two parallel regulatory stacks govern the same episode of care, and they are joined by nothing.**

The money is governed by the **Medical Schemes Act 131 of 1998** and its regulations, administered by the **Council for Medical Schemes**. The care is governed by the **National Health Act 61 of 2003**, with quality enforced by the **Office of Health Standards Compliance** against gazetted norms and standards, professional conduct by the **Health Professions Council of South Africa** and the **South African Nursing Council**, and medicines and devices by **SAHPRA**. Privacy sits across both under POPIA and the Information Regulator, with a dedicated health-information regulation in force since **6 March 2026** that binds funders, administrators, managed-care organisations, employers and pension funds — and **not** providers [24].

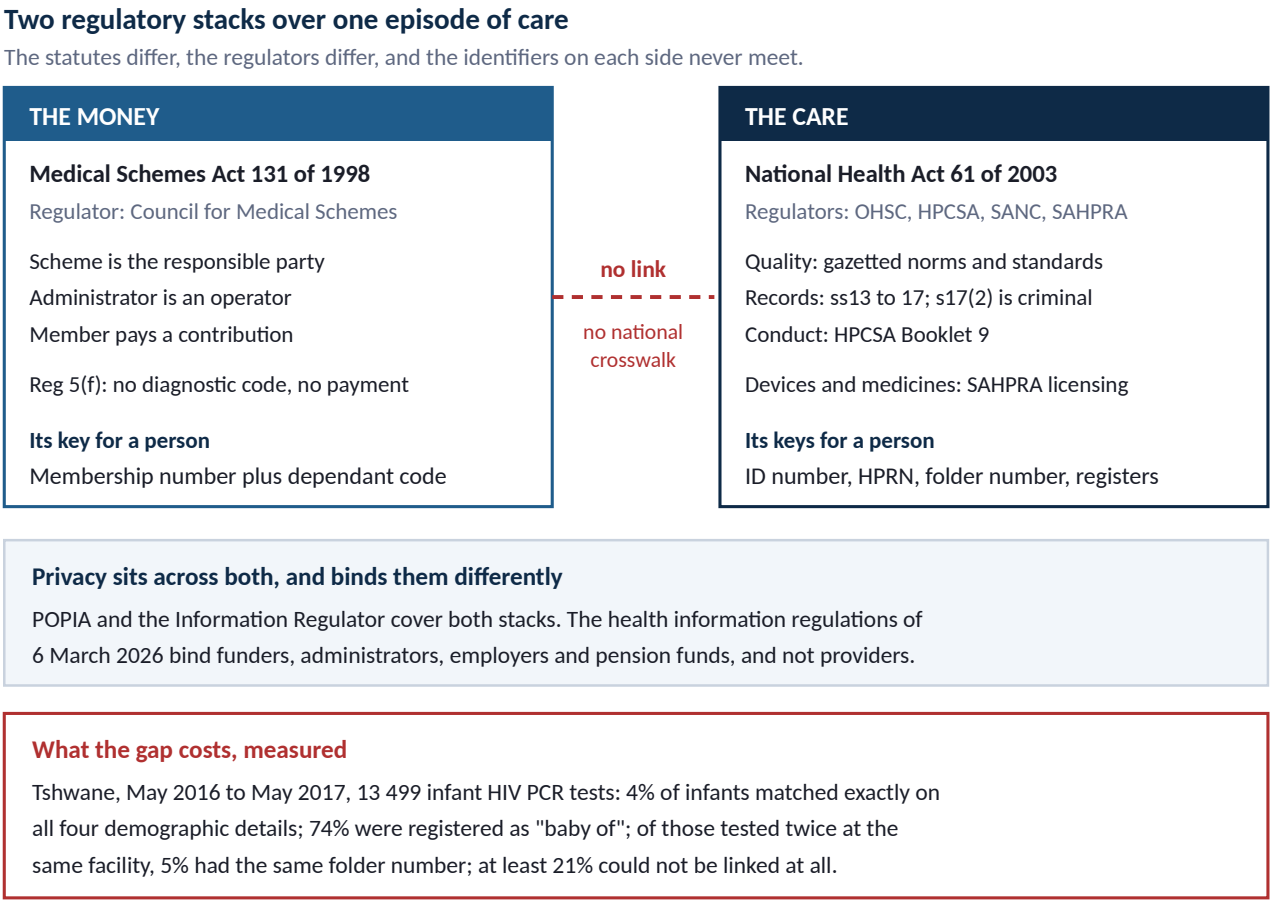


Figure 1.3 The two stacks and the gap between them. The money and the care are governed by different statutes, enforced by different regulators, and keyed on different identifiers. The dashed line is the whole problem: there is no national crosswalk between a membership number and a folder number, and the measured consequence at the foot of the figure is published South African evidence rather than an analogy from elsewhere.

A medical scheme is not an insurer, and this is a legal distinction rather than a semantic one. Schemes are not-for-profit, community-rated within an option, open-enrolment for open schemes, without health-status underwriting, with mandatory prescribed minimum benefit cover, statutory solvency requirements and rules approved by the Council. A member pays a **contribution**. A scheme is also a different legal entity from its administrator and from its managed care organisation, and under POPIA the **scheme** is the responsible party while the administrator is an operator. An architect who arrives with an insurance reference model and maps the administrator as the data controller has the accountability inverted before the first workshop. Chapter 6 takes the money end to end.

Now the part that cannot be designed around. **There is no identifier shared between the two stacks, and no national crosswalk.** The same human being holds:

1. a **national identity number**, frequently absent, wrong or belonging to somebody else;
2. a **ten-digit health patient registration number** issued by the public sector's Health Patient Registration System, if the facility has HPRS;
3. a **facility folder number**, which is scoped to one facility;
4. **programme register numbers** — the antiretroviral register, the tuberculosis register, the antenatal card, the Road to Health booklet;

and, if they are on a scheme, a **membership number plus a dependant code** that lives entirely in the funder's world and reaches the provider only on a claim.

HPRS has reached **3 266 establishments** — 3 150 primary health care facilities, 113 hospitals and **three general practitioner practices** — as at 31 March 2025, with an automated biometric identification system at **82** primary care facilities [7]. The private sector is not covered at all.

The published South African evidence on what happens when you try to link without an identifier is better than the generalities usually offered, and an architect should quote it rather than import matching statistics from elsewhere. Haeri Mazanderani and colleagues studied infant identification in the prevention-of-mother-to-child-transmission programme in Tshwane between May 2016 and May 2017, across **13 499 birth HIV PCR tests**. No identifier is issued to an infant at birth; infants are registered by first name, surname, sex, date of birth and a facility folder number. **Only 4% of infants had exact matches on all four demographic details** across tests. First name matched in **5%**, because **74% of birth tests were registered as "baby of" or a similar prefix** rather than a name. Of infants tested twice, 18% were tested at the same facility both times — and **only 5% had the same folder number on both occasions**. At least **21% could not be linked at all** [25].

That is a published, quantified, South African rebuttal to "we will link on demographics", and chapter 11 is built on it.

The consequence for the architecture is immediate and it is not a data quality problem. **A provider cannot read a patient's history off the record**, because the record is facility-scoped. A funder cannot connect its claim history to a clinical outcome, because it holds a membership number and the provider holds a folder number. A province cannot compute a readmission rate that means what the word implies, because a readmission to a different facility is invisible; absent a cross-facility identifier, a provider-level readmission rate measures patient loyalty to one facility. And no national reporting layer can deduplicate, because the aggregate systems that carry national reporting have no patient in them at all.

Any design that assumes a single patient key exists is a design that will be rebuilt.

The paradox at the centre of the work

Here is the claim this volume is organised around, and it is counter-intuitive enough to be worth stating twice.

Healthcare has the deepest and most mature set of international data standards of any industry in this series — and in South Africa almost nothing that matters is specified nationally.

The international stack is genuinely rich: HL7 version 2 and FHIR for messaging and application programming interfaces, SNOMED CT for clinical meaning, LOINC for observations, DICOM for imaging, the IHE profiles for cross-enterprise document sharing and identity queries, openEHR for model-driven clinical content. Chapters 12 and 16 take them properly.

Now the national position, item by item, each sourced in the chapters that follow:

- **No national procedure classification in routine use.** Private-sector procedure coding is a licensed derivative of an American code set, whose own maintainer positions it for data rather than for billing and expressly does not recommend it as a national standard. The public sector has the Uniform Patient Fee Schedule, which is a fee instrument.
- **No national tariff since 28 July 2010**, when the High Court set aside the regulations behind the reference price list and the Director-General was interdicted from publishing one [26]. Price in a South African private-sector data model is therefore `(funder, provider, service code, effective date)` and nothing less. A single `price` column on a service master is a defect.
- **No national medicines dictionary** with virtual-product or dose-form structure for prescribing, which blocks closed-loop administration and forces every ePrescribing project to build its own.
- **No case-mix grouping in routine use**, and the industry committee's own grouping subcommittee is dormant.
- **No present-on-admission flag** in either the public morbidity coding standards or the private claim message, so hospital-acquired condition rates are not computable and comorbidity-adjusted outcome models silently adjust for complications.
- **No national patient-identity merge policy**, which is the subject of chapter 11.
- **No national retention instrument.** Every retention period in circulation traces to a professional guideline, and that guideline disagrees with itself across versions: the 2016 text says at least six years from dormancy, while the September 2022 revision says **ideally indefinite, particularly if electronic** [20][21]. "The law says six years" is wrong twice over.

And the sharpest illustration of all. The gazetted **Health Normative Standards Framework**, published by the Director-General under section 74(1) of the National Health Act in Gazette notice of 21 October 2022, is built on the IHE profiles with HL7 version 2.3.1, 2.5 and 2.6, version 3, CDA Release 2, FHIR Release 4 and Release 3, and DICOM. It **names neither SNOMED CT nor LOINC**, and ICD-10 appears only in its acronym list [27]. Three and a half years later, on or about **14 May 2026**, South Africa joined **SNOMED International** [28]. The National Release Centre is not yet operating: there is no national extension, no reference sets and no published maps, and the announcement names no centre.

> The status of the SNOMED CT National Release Centre was unresolved at the research date for this volume and is the kind of fact that moves quietly. **Check it before relying on anything in this book about national terminology governance**, starting with SNOMED International's South Africa member page and the Department's own standards publications.

So any clinical platform procured from the middle of 2026 onwards must be able to add a terminology binding it does not currently have, to a code system its vendor may not support, against a national extension that does not yet exist. That is an awkward position, and it has one genuinely useful consequence: the framework contemplates **conformance testing at a laboratory approved by the national Department, with an attestation issued** [27]. Whether such a laboratory is operating could not be established. If it is, it is the most useful procurement lever in the country.

The conclusion is the spine of this volume. **The architect's work in South African healthcare is making, locally and explicitly, the decisions that nobody has made nationally — and writing them down in a form that survives the architect leaving.** Chapter 12 is where that becomes a method.

What is not the reason for the work

National Health Insurance will be put to you as the reason for the programme. Treat it as a contingency that validates the work rather than as the reason for it, and the architecture will age better.

The **National Health Insurance Act 20 of 2023** is on the statute book and **no section of it is in force.** Not "some sections": none. By letter of 20 February 2026 the State Attorney confirmed the President would not commence any section before the Constitutional Court ruled on the public participation challenges; the Presidency confirmed the agreement on 24 February 2026, and it was made an order of court the same day [29]. In October 2026 the Department told Parliament's Select Committee on Social Services that there would be no NHI-specific implementation before judgment [30]. There is no Fund, no accreditation regime, no contracting units established under the Act, no central payment and no change to medical schemes.

The scale of the current digital commitment is worth stating next to any vendor's estimate of a national claims and registration platform. In the October 2026 adjustment the direct NHI grant for 2026/27 carried **R6,5 million** for "Digital Health System" [30].

A second fact belongs here because it is the one most often written in the present tense by mistake. **Certificates of need are dead.** In *Solidarity Trade Union and Others v Minister of Health and Others* (CCT 240/24), ZACC 19 of 2026, handed down **18 May 2026**, a unanimous Constitutional Court declared sections 36 to 40 of the National Health Act invalid and severed them. The scheme had never commenced and no regulations were ever made under it [31]. The state had described it as a central pillar of NHI implementation; the court did not rule on the NHI Act. Write about certificates of need in the past tense, and note that Parliament may legislate again within the constraints the judgment sets.

> The Constitutional Court's judgment in the public participation challenges to the NHI Act — CCT 251/25 and CCT 269/25, heard 5 to 7 May 2026 — was **still reserved as at 31 October 2026** [29]. It may have been delivered by the time you read this. **Re-verify the status before any statement about NHI reaches a client deliverable.**

Now the useful part. Every requirement an NHI architecture would impose is **already mandated today**, by instruments in force today, with a named enforcement record:

Requirement	Already required by	In force
Identity resolution and breach notification	POPIA s22; National Health Act s17 access control	Yes
A facility and provider register	OHSC certification; provincial licensing; practice numbering	Yes
Coded encounter capture	Medical Schemes Act Regulation 5(f)	Yes
Effective-dated benefit reference data	Prescribed minimum benefit regulations; UPFS provincial notices	Yes
Retrievable records	HPCSA Booklet 9; the Auditor-General's findings on claims	Yes

And the enforcement record is no longer theoretical. Six POPIA administrative fines were reported at the Information Regulator's briefing of **31 August 2026**, including **Lancet Laboratories, R100 000, paid** — the health sector's first. On **26 and 27 October 2026** the Regulator fined **First National Bank R1 million** and a technical and vocational college R500 000. The dominant findings across the record are **section 22 notification failure** and **section 15(1) incompatible further processing**, and the Regulator has named **health among its priority sectors** for its own compliance assessments in 2026 and 2027 [32]. Chapter 7 takes NHI properly and chapter 18 takes the regulatory frame.

Where the business case actually is

An architect arriving in a South African provincial health department will be handed strategy documents about universal coverage, digital transformation and patient experience. The business case is somewhere else entirely, and it is arithmetic.

Provincial health departments' recorded medico-legal claims balance fell from a peak around **R110 billion in 2019/20** to roughly **R50 billion at 31 March 2026** — published by the Auditor-General as **R50,2 bn** and by National Treasury as **R50,6 bn**, two measurement bases from two institutions on the same date [33][34]. The balance is falling. The cash is not: **R1,57 bn was paid to claimants in 2025/26**, and Treasury's figure is that provinces spend an average of **R1,5 billion a year** settling claims [33][4]. About **80% of claim quantum is future medical care**, often costed at private hospital rates [35], and awards in serious birth-injury cases "often exceed R20 million" [36].

Then the finding that writes the business case. The Auditor-General audited **27 facilities across nine provinces** and found [33]:

Finding	Facilities
Reported missing or lost records	15 of 27
Did not properly track records after issue	17 of 27
Inadequate storage conditions	25 of 27
Too few records-management staff	18 of 27

And then this. **In three of nine paid claims reviewed, departments conceded because they could not locate the medical record** [33].

A third of the paid claims reviewed were conceded for want of a record. Set that against a payout stream averaging R1,5 billion a year, and against serious birth-injury awards that often exceed R20 million. A records-retrieval capability that prevented **one** R20 million concession would fund more digital health than the national programme has budgeted for the year, by a factor of three.

That is not a rhetorical flourish; it is R20 million against the R6,5 million digital health line in the 2026/27 direct grant. The comparison is crude — one is capital avoided in a province, the other is a national grant line — and it is still the most arresting number in this book.

Note the Auditor-General's own word. The recommendation is to "fill critical records and claims vacancies, train staff, **secure retrievable records**, and act on non-compliance" [33]. The operative word is **retrievable**. The requirement is not storage. It is the ability to produce the complete record of a specific episode of care, for a specific patient, from up to twenty-one years ago, under litigation, with provenance. That is a records-management specification, and it is a different system from an electronic medical record. Chapter 13 opens with this arithmetic and specifies what satisfies it.

One honest caveat, because this book would rather state a limit than smooth one over. The audit periods differ: the records audit is 2021/22, the clinic audit 2022/23 and the claims balance is at 31 March 2026 [33]. They should not be read as one snapshot, and the three-of-nine finding is a review of nine claims, not a population estimate. The direction is unambiguous and the magnitude is not precisely known.

What the rest of this book does

Part I finishes the industry: how to read its numbers (chapter 2), the two systems in detail (chapter 3) and the vocabulary (chapter 4), which is where the terms this chapter has used loosely — encounter, episode, spell, record, scheme, administrator — are defined properly.

Part II is how a provider runs: who does what, how the money moves, NHI in full, the operating model and the clinical cadence, and five process models.

Part III is data and systems: the conceptual data model, identity as a chapter of its own, coding and the specifications that do not exist, the clinical record as an object, and the application and interoperability landscape.

Part IV is the constraints: technology providers and how to buy, regulation and the records duty, and safety, downtime and supplier concentration.

Part V is modelling it, end to end, and what to do in a first ninety days.

The three claims in this chapter are the ones to carry forward. Downtime is clinical. The record is three instruments at once. The money and the care are governed separately and share no identifier. Every architectural argument in the rest of the book descends from one of those three.

THE CHAPTER IN TEN LINES

- South Africa runs two health systems of comparable financial scale for populations differing by about six times: 9,13 million scheme beneficiaries against 53,4 million people Treasury treats as the public sector's responsibility.
- In law there is one national health system: the National Health Act's records, consent, emergency and complaints duties bind private establishments too, and provinces rather than the national department hold the budget and run the facilities.
- In the private sector the hospital does not employ the doctors, so one inpatient episode generates several independent claims and several clinical records in different legal entities.
- Diagnostics are always a separate organisation, so order-to-result is always an inter-organisational integration with no national results repository behind it.
- Downtime in healthcare is a clinical risk. A ransomware attack on a pathology supplier was found to be a contributing factor in a patient death, with 170 recorded harms.
- Plan for weeks of degraded operation, not hours, and treat an independently reachable read-only record as the highest-value resilience investment.
- Name the shared services whose failure stops clinical work and that the organisation does not own: the NHLS, the claims switch network, HPRS and SyNCH.
- The patient record is a clinical, a legal and a billing instrument at once, and the professional guideline requires billing records to be kept separate from care records.
- Unauthorised access to a health record is a criminal offence under National Health Act s17(2), which is a stronger argument for access auditing than any privacy clause.
- The money is governed by the Medical Schemes Act and the Council for Medical Schemes; the care by the National Health Act, the OHSC, the HPCSA, the SANC and SAHPRA. No identifier joins them.
- Healthcare has the deepest international standards of any industry in this series, and in South Africa almost nothing that matters is specified nationally. Making those decisions locally is the work.
- No section of the NHI Act is in force, and its 2026/27 digital health grant line is R6,5 million. Everything an NHI architecture would need is already mandated today.
- Three of nine paid medico-legal claims reviewed by the Auditor-General were conceded because the department could not find the record. That is the business case.

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CHAPTER 2

How to read a South African healthcare number

Two official figures for the same quantity on the same date are not an error to be resolved; they are two measurement bases, and knowing which is which is the whole of the skill.

Why a chapter about numbers

An architect's first deliverable in a health engagement is usually a baseline. Somebody wants to know how many beds there are, how many admissions, what the occupancy is, what the sector spends, how many people are covered. The request sounds like a data-gathering task. It is not. In South African healthcare it is a measurement-governance task, because the sector's own official figures do not reconcile, the gaps are large, several of them are recent, and nobody owns the reconciliation.

That is not a complaint. It is the subject of this chapter.

Three examples to fix the scale of the problem before the method arrives. Provincial medico-legal liability at 31 March 2026 is published by the Auditor-General as **R50,2 billion** and by National Treasury as **R50,6 billion** [1][2]. Public hospital beds have been officially reported as **100 656 approved**, **85 126 usable** and **88 785**, depending on the series [3][4]. Medical scheme coverage is **14,1%** of the population or **14,95%**, both official, both current, both correct for their own purpose [5][6].

None of those pairs is a mistake. Each is two measurement bases, from two institutions, often on the same date. An architect who averages them has destroyed information. An architect who picks one without recording which has built a reporting layer that nobody downstream can audit. An architect who asks the right eight questions before accepting any of them has done the actual job.

By the end of this chapter you should be unable to quote a South African healthcare figure without asking for its denominator and its date, and faintly uncomfortable when somebody else does.

The interrogation

Eight questions. They take about four minutes to ask and they catch almost everything.

Eight questions, and the defect each one catches

Four minutes of interrogation before a figure enters a baseline document.

ASK the question	IT CATCHES the failure it is there to find
1 What is the denominator?	Approved, usable, staffed or open tonight
2 As at when, for what period?	Extraction date standing in for measurement date
3 Which institution, measuring what?	Audited balance against fiscal contingent liability
4 The thing, or a register of the thing?	Nurse registrations read as a workforce
5 Measured, or derived?	An unwritten formula doing the arguing
6 Aggregate, or patient-level?	Questions that are unanswerable, not merely hard
7 Is there a second official value?	A silent choice between two correct figures
8 Who benefits from this value?	Theatre minutes as revenue; incidents as reporting

If question 7 has an answer, carry both values and say they disagree. Do not average and do not pick.

Figure 2.1 The eight questions, and what each is there to find. The order matters: the denominator first, because that is where almost every contested figure in this sector is contested, and the beneficiary last, because it changes how a figure should be read rather than whether it can be used. Question seven usually has an answer in South African healthcare, and the correct response to it is to carry both values.

1. What is the denominator? Not the numerator. Almost every contested figure in this sector is contested at the denominator: available beds, eligible admissions, expected reports, the registered population, the insured population. The numerator is usually a count somebody can produce.

2. As at when, and for what period? A contingent liability is at a date. A benefit payment is for a period. A coverage percentage is for a survey year. The most common defect in a health dashboard is a figure whose as-at date is the date of extraction rather than the date of the measurement.

3. Which institution published it, and what does that institution measure? The Auditor-General measures a recorded balance under audit. Treasury measures a contingent liability for fiscal purposes. A regulator measures regulatory returns. A household survey measures what people say. These are different instruments and they will not agree.

4. Is this a count of the thing, or a count of a register of the thing? Nurse numbers count registrations. Facility licences count authorisations. Bed licences count permissions. None of those is the physical population they are taken to describe.

5. Is it measured, or derived? A derived figure is a formula applied to entities, and the formula is where the argument is. Ask for the formula. If nobody can produce it in writing, the figure is an opinion with a decimal point.

6. Is the source aggregate or patient-level? If the source is an aggregate reporting system, whole classes of question are categorically unanswerable, not merely difficult.

7. Does a second official value exist? In this sector the answer is usually yes. Find it, carry both, and say they disagree. Do not average and do not choose.

8. Who benefits from the number being what it is? Theatre minutes are a billing input under one of the statutory funders. Reported patient safety incidents reflect reporting behaviour. A contingent liability is a balance-sheet item in a department under fiscal pressure. None of that makes a figure dishonest; all of it makes a figure directional.

The remaining sections are worked cases. Each is a real South African disagreement, each is sourced, and each teaches a different failure.

Case one: the liability that halved while the payments stayed flat

This is the best teaching case in the sector because it contains four separate measurement problems in one subject.

The problem of the series. Here are published totals for provincial medico-legal exposure, with their bases:

Figure	Date	Basis
About R110 bn	2019/20 peak	Contingent liabilities
R125 bn, 15 148 claims	31 March, year ambiguous in the source	**All** claims against government
R75,3 bn	March 2023	Recorded claims balance
R91,09 bn, 9 567 active claims	31 May 2025	National active claims
R62,5 bn falling to R57,6 bn	2023/24 to 2024/25	Contingent liabilities
R50,2 bn	31 March 2026	Recorded claims balance
R50,6 bn	2025/26	Contingent liabilities

Sources, in the order of the rows: [2]; [7]; [1]; [8]; [9]; [1]; [2].

Those are not seven points on one line. "All claims against government" is a different population from medico-legal contingent liability, which is different again from active claims — and "active" is itself a classification, because Gauteng alone was reported with 1 397 dormant claims alongside 707 active [8]. Plotting them as a trend produces a chart that is wrong in a way no reader can see.

The defensible statement, and it is worth copying verbatim into a baseline document: provincial health departments' recorded medico-legal claims balance fell from a peak around R110 billion in 2019/20 to roughly R50 billion at 31 March 2026, on figures published by the Auditor-General and National Treasury that differ by about R400 million on the same date [1][2].

The problem of the ranking. Provincial exposure at 31 March 2026, as the Auditor-General reported it: Eastern Cape **R18,6 bn**, Limpopo **R9,3 bn**, Gauteng **R9,1 bn** — those three being 74% of the total — then Mpumalanga R6,1 bn, KwaZulu-Natal **R2,8 bn**, Free State R2,1 bn, North West R1,6 bn, Northern Cape R0,5 bn and Western Cape **R0,1 bn** [1].

Now the 2024 ordering, as it was given to Parliament: KwaZulu-Natal about **R29 bn** and highest; Gauteng about R24 bn; Eastern Cape about R20 bn; Limpopo fourth with no figure given [9].

KwaZulu-Natal goes from R29 billion and first place to R2,8 billion and fifth. Eastern Cape goes from third to first. **That is a complete reversal of the provincial ranking in two years, and neither source explains it.** A change of that magnitude is a recognition or measurement change, not a change in underlying litigation. Any reporting layer built on provincial medico-legal liability must carry the measurement basis as an attribute, and any architect asked to build a provincial comparison should say in writing that the ranking is not stable across published sources.

The problem of the same province. A report of 20 October 2026 gives Gauteng contingent medico-legal liabilities of about **R21,2 bn** and Free State claims of about R5,7 bn [10]. Neither reconciles with the Auditor-General's **R9,1 bn** for Gauteng and R2,1 bn for the Free State at 31 March 2026 [1], and neither reconciles with the **about R24 bn** given to Parliament in 2024 [9]. Three Gauteng figures, from three publications, two of them within six months of each other, and no source explains the difference. Be careful of the secondary summaries of this subject too: at least one of them restates the 2024 Gauteng figure as about R21 bn rather than about R24 bn, which is close enough to the October 2026 figure to look like agreement and is not. The

disagreement is reported here rather than resolved, because it cannot be resolved from the public record, and because an architect will meet exactly this situation and needs to see it handled honestly.



Figure 2.2 Four measurement problems in one subject. Provincial medico-legal exposure is the best teaching case in the sector because the published figures are incommensurable in four different ways at once: the series mixes three populations, two institutions give two values for the same date, the provincial ranking reverses without explanation between publications, and the balance falls while the cash does not. Every figure here is published and none is disputed.

The problem of the balance against the cash. The recorded balance is falling. The money leaving the system is not. **R1,57 bn was paid to claimants in 2025/26**, and R4,6 bn over three years across the nine provinces, with Eastern Cape R474 m, Gauteng R456 m and North West R230 m in 2025/26 [1]. Treasury's figure is that provinces spend an average of **R1,5 billion a year** settling claims [11]. Gauteng's budget-versus-payout record is the sharpest illustration: budgeted R236 m against R392 m paid in 2021; R227 m against R369 m in 2022; R300 m against R500 m in 2023; R179 m against about R700 m in 2024 [12].

The province budgets roughly a third of what it pays, every year, and the shortfall comes out of medical supplies, maintenance and operating costs [12]. A falling contingent liability with flat cash payments and chronic under-budgeting is a measurement story at least as much as a management story, and the only honest baseline presents the balance and the cash as two series with two meanings.

One last detail, because it explains the quantum. **About 80% of claim value on average is future medical care, often costed at private hospital rates** [9]. The claim is largely the cost of a lifetime of private care for an injured child, which is why awards in serious birth-injury cases "often exceed R20 million" [13]. That is a composition fact, and it means the liability is driven by private-sector price inflation as much as by clinical events.

Case two: how many beds

Bed counts are where an architect gets into trouble fastest, because the question sounds trivially answerable.

Public. A parliamentary reply of 14 October 2022 gives **100 656 approved** and **85 126 usable** public hospital beds, usable as at July 2022 — a 15,4% gap between the two [3]. A separate series gives public hospital beds as 85 063 for 2020, **55 913** for 2021, 88 556 for 2022 and **88 785** for 2023 [4]. The 2021 value is almost certainly a definitional or collection artefact rather than a 34% collapse and recovery; the publisher's own notes flag multiple labelling errors in that annexure. **Do not use it.** That instruction is itself the lesson: a series with one obviously impossible value has an unknown number of quietly impossible values.

Private. The last national figure located is **20 466 registered private beds in 2011**, split Gauteng 9 412, Western Cape 3 252, KwaZulu-Natal 2 972 and other provinces 4 830, with the industry association putting private hospitals at about 26% of total hospital beds [14]. Six years later Gauteng alone

reported **17 052 private beds** against 18 833 public beds across its five districts [15] — already far above the 2011 national figure for that province. Either the 2011 figure counted something narrower, or private capacity in one province grew by 80% in six years. The sources do not say which.

The defensible position is a refusal. Beds cannot be stated nationally with confidence for either sector. Coverage and spend can. An architect asked for a national bed figure should supply the published values with their bases and dates, and say that they do not reconcile.

And there is a structural reason behind the mess. In the private sector a bed number is a **licensing quantity**: a provincial licence authorises a specific bed count at a specific site. Gauteng reported in 2017 that it had approved **48 facilities not yet built, representing 2 796 beds and 306 theatres**, of which only five were under construction [15]. Licensed beds, built beds, commissioned beds and beds open tonight are four different populations, and the group's own asset register is usually a fifth.

That matters beyond the baseline, because the bed count is the denominator of occupancy. Occupancy is patient days divided by available beds times days, and "available beds" in South Africa is one of approved, usable, commissioned, staffed or open tonight. Published occupancy figures in this sector are therefore not comparable between organisations unless the denominator convention is stated, and it almost never is. Private acute occupancy was 65,0% at Netcare for the year to 30 September 2025 [16] and like-for-like 67,5% at Life Healthcare for the six months to March 2026 [17]; public inpatient bed utilisation for 2023 was 66,9% [4]. Those three numbers look comparable. They are not.

Case three: how many people are covered

Four official figures, four instruments, four purposes.

- Statistics South Africa's Income and Expenditure Survey 2022/23: medical aid coverage **14,1% of the population** [5].
- The Council for Medical Schemes, 2024/25 reporting: **9,13 million beneficiaries, 14,95% of the population** [6].
- The Competition Commission's Health Market Inquiry, for its study period: **8,8 million people covered by private insurance out of a population of 53,7 million** [18].
- National Treasury, 2026 Budget Review: provinces are responsible for healthcare for **53,4 million people without private medical insurance** [11].

Scheme beneficiaries are not the same population as people with any private cover, which includes gap cover products, health insurance products, employer clinics and cash-paying users of private general practice. The Statistics South Africa figure is self-reported household data; the Council figure is a regulatory return. Both are correct for their purpose.

The practical rule is worth stating as a rule. **Use the Council's figure for scheme-funded volumes and Treasury's uninsured figure for public-sector demand, and never add them, because the sum double-counts people who use both.** That is not a theoretical caution: the same household survey shows 71,6% of people relying on the public sector for usual care, including 65,7% using public clinics, while only 1,4% use private clinics and 1,8% private hospitals — against 14,1% holding medical aid [19]. A substantial number of scheme members use public primary care.

And a national average is useless for capacity planning, because coverage varies by more than a factor of three across provinces: Western Cape **23,8%**, Gauteng 19,6%, Northern Cape 14,3%, Free State 14,0%, Eastern Cape 10,2%, North West 9,6%, KwaZulu-Natal 9,1%, Mpumalanga 8,4%, Limpopo **7,3%**. It also splits sharply by settlement type — urban 19,1%, farm settlements 12,1%, traditional settlements 3,9% [19]. Any private-sector demand model built on the national percentage is wrong in both directions at provincial level.

Case four: how much money

Scheme benefits paid in a year are published as **R259,3 bn** for calendar 2024, up 8,52% [20], and as **R239 bn** in the Council's 2024/25 annual reporting, alongside contributions excluding medical savings of R226,94 bn and a further R24,30 bn into personal medical savings accounts [6]. Different bases, different periods, both from the same regulator. A model that uses one of those figures in one place and the other in another will not balance, and the discrepancy will be found three months later by somebody reconciling a dashboard.

Household out-of-pocket spending is published as **R31,5 bn**, averaging R1 481 per household, with the wealthiest quintile accounting for R22,5 bn of it [19] — and as **R43,3 bn** for 2023 by the Council [6]. Two instruments, two definitions, two numbers.

Then the clearest single illustration in the sector, because both parties are arms of the same state. Provinces collectively owed the National Health Laboratory Service **R10,7 bn** on one Treasury figure, **R10,6 bn as at June** on another, and **about R11 bn as at March 2026** per the portfolio committee [2] [21][22]. The provincial breakdowns are worse than the totals. One account gives KwaZulu-Natal about **R3,94 bn** and Gauteng about R3,3 bn at end-May 2026 [22]; the other gives Gauteng **R3,5 bn**, KwaZulu-Natal **R1,2 bn** and Eastern Cape R1 bn as at June 2026 [21]. The KwaZulu-Natal line differs by about R2,7 billion and neither source explains it; the laboratory service itself acknowledged billing disputes and invoice-processing issues [21].

An intercompany reconciliation difference of roughly R2,7 billion between two arms of the same state is what happens when two organisations bill each other without a shared service catalogue and a shared identifier. It is also the single most useful fact an architect can put in front of a provincial finance director, because it is their own number.

Case five: the list that is not a list

Prescribed minimum benefits are the most consequential rule in private funding, and a reader would reasonably expect the list of them to be settled. It is not.

The Council for Medical Schemes' own page describes "a limited set of **271** medical conditions" in the diagnosis and treatment pairs and **26** chronic conditions on the chronic disease list [23]. A scheme's 2026 member guide gives **271** pairs and **27** chronic conditions [24]. A third source, reporting the funders' representative body, gives 271 conditions and **26** chronic conditions [25].

So: state the pair count as **271**, and say explicitly that published sources give the chronic disease list count as both 26 and 27, and that the discrepancy was not resolvable from public sources. This is marked unverified in the research behind this volume; what would settle it is the current gazetted

annexure to the Medical Schemes Act regulations, read directly rather than through a summary.

One further detail, because it is diagnostic of how sector figures spread. The commonly repeated "270 diagnosis and treatment pairs" appears in **neither** primary source. It is a number with no publisher, in wide circulation. When a figure in this sector has no traceable source, that is usually because it never had one.

Case six: the same number printed twice

Internal inconsistency inside a single document is commoner here than an architect expects, and it is the failure that does most damage, because a single source feels authoritative.

The national annual report. The Department of Health's 2024/25 annual report records complaints logged as **26 811**, up from 24 724, with 25 643 resolved. It separately records patient safety incidents reported as 27 865, of which **26 811** were closed [26]. Two different indicators, in two different programmes, producing the identical value. Either it is coincidence or it is a transcription error. It is not resolvable from the published report, and the two figures must not appear in the same table without the collision being flagged.

The national nursing workforce report. It states that most nurse specialists are over 55 while its own table shows about 24%. Its nurse specialist total in the age table is **17 941**, while the payroll figures in the same document give filled posts as **15 318** and total posts as **18 108**. One age table labelled "professional nurses" totals 102 669, which matches nothing else in the document [27]. This is a national workforce planning document whose internal figures do not reconcile, and anyone building workforce analytics on it has a reconciliation work package before they have a model.

A listed hospital group. Mediclinic's own operating divisions page describes the southern African division as 50 hospitals, 5 subacute hospitals, 6 mental health facilities, 14 day case clinics and about 8 856 inpatient beds, while a summary panel on the same page gives the group as 72 hospitals, 21 day case clinics, 459 theatres and 11 646 inpatient beds — figures that do not fully reconcile with the division-level list beside them [28].

A digital programme's own benefit figure. Netcare attributes cumulative gross savings and cost avoidance of **R587 m since the 2022 financial year** to its digital programme [16]; a report of the same programme gives **R706 m since 2022** [17]. The two are at different dates and must not be used interchangeably — which matters, because this is the kind of figure an architect is asked to reuse in a business case.

Derived numbers: where the formula is the argument

Everything so far has been a count. Derived measures are worse, because the disagreement moves from the data into the arithmetic, where it is invisible.

The governing claim, and the most important sentence in this chapter: **a healthcare measure without a versioned, published definition is not a measure, it is an opinion with a decimal point.**

Theatre utilisation is the best-documented case and should be the one an architect uses to win the argument, because it has multiple published definitions, a quantified spread between them, and a peer-reviewed critique of the national benchmark. A regulator's published guideline requires in-anaesthetic, in-theatre, procedure start, procedure finish and out-of-theatre timestamps, plus scheduled session start and end, and defines raw utilisation, adjusted utilisation with a nominal 15-minute changeover allowance, starting on time, average changeover, finishing on time and several cancellation measures. Its own worked example shows the trap: a 240-minute list with two cases and one 20-minute changeover tops out at about **92%** utilisation, while a team doing eight cases with 10-minute changeovers tops out near **71%**. The high-throughput team looks worse [29].

Then the critique. Reviewing a national benchmarking service's calculations, Zhang and Pandit show that **10 hours scheduled with 8 hours worked gives 80% raw utilisation, and with 2 hours of intercase downtime the touch-time utilisation falls to 60%** — the same list, two defensible methods, twenty percentage points apart. More damaging, their figure drawn from one specialty in one trust over fifteen weeks shows **no simple mathematical relationship between the four calculation methods**, which means an organisation cannot convert between them or infer one from another. Their other findings generalise: pooling unused time implies that **five minutes wasted on each of a hundred lists is 500 minutes of usable surgical capacity**, which it is not; and dividing unused capacity by an average case duration produces a count of extra cases that is arithmetically correct and operationally fictional [30].

The same structure appears everywhere else.

- **Average length of stay.** A hospital with substantial internal transfer reports roughly half the mean length of stay on episodes that it reports on spells. Day cases in or out moves it again. Mean against median moves it again, because the distribution is strongly right-skewed. Those terms are defined properly in chapter 4; the point here is that three unstated conventions can halve a published figure.
- **Bed occupancy.** A midnight census counts patients in a bed at midnight and misses day cases, emergency observation, theatre recovery and the patient admitted at 06:00 and discharged at 20:00. Computed from admission and discharge timestamps they all contribute. The two methods diverge most in exactly the units management is trying to manage.
- **Readmission within 30 days.** "Unplanned" requires a planned-readmission algorithm: a long, versioned, published rule set built from procedure and diagnosis codes [31]. **South Africa has no published national planned-readmission algorithm**, so every South African readmission rate either includes planned readmissions or uses a locally invented rule, and is in neither case comparable. And absent a cross-facility identifier, a provider-level readmission rate measures patient loyalty to one facility.
- **Mortality.** The decisive evidence against casual standardised mortality comparison is a study of four English hospitals with published ratios from 88 to 143, which found that the two variables doing most of the adjustment work are the two most contaminated by local practice. Mean coding depth was **2,12 codes per record at the lowest-ratio hospital against 1,42 to 1,62 elsewhere**, and under-coding lowers expected deaths and so inflates the ratio [32]. A change in coding depth looks exactly like a change in mortality.
- **Present on admission.** Neither the public morbidity coding standards nor the private claim message mandates a present-on-admission flag. So hospital-acquired condition rates are not computable from routine South African data, and a comorbidity-adjusted outcome model here is silently adjusting for complications — which biases toward making poor care look like sick patients.

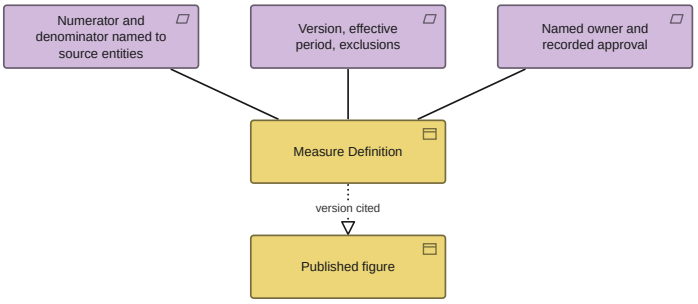


Figure 2.3 The object that fixes derived measures. A Measure Definition is a first-class, versioned business object, not a footnote in a report specification. It names the numerator and denominator against source entities and attributes, carries an effective period and exclusions, and has an owner who approved it. A published figure then cites a definition version, which makes a change of definition a new version rather than a silent restatement. The repository an architect is handed in this industry has nowhere to put it.

The architectural answer is one object. A **Measure Definition** carries an identifier, a version, an effective period, numerator and denominator specifications naming the source entities and attributes, inclusion and exclusion criteria, the handling of missing values, an owner and an approval. A published figure then cites a measure definition version, and a change of definition becomes a new version rather than a silent restatement. The supplied repository you will be handed has nowhere to put it; chapters 10 and 12 return to that, and the entity is introduced properly in chapter 13.

There is a published South African acknowledgement of the whole problem, from a provincial dashboard team. "Facilities calculate bed occupancy differently. For example, some include the admission and discharge days and others do not. Facility managers may therefore distrust dashboard figures that differ from their own" [33]. That is the cleanest statement available of why the measure definition has to be an object: the dashboard and the facility are both right under their own rules, and nobody can tell, because the rules are not attached to the numbers.

Aggregates cannot answer questions nobody anticipated

The country's routine public-sector reporting runs on **DHIS2**, which is an aggregate reporting system. Its unit of data is a facility, a period, a data element and a value. **There is no patient in it.**

The principal published assessment is the single best illustration in this chapter of why a completeness figure needs its denominator. Assessing data from April 2017 to March 2020 across 265 public health facilities and 77 data elements, Thabethe and Mathews report **health facility reporting completeness of 99,5%** — 9 489 of 9 540 expected reports submitted — and **data element reporting of 49,7%**. Missing values numbered 121 199, an average missing rate of 16,50%; unaccounted values numbered 248 153; missing plus unaccounted averaged **50,3%**. The overall data completeness score was **74,6%, rated poor**, against internal consistency of 95,1%. **Timeliness was not assessed at all** [34].

The gap between 99,5% and 49,7% is the whole finding. A regime that measures *reporting* completeness will report 99,5% and declare success while half the values inside those reports are absent. Quoting either number alone misrepresents the system. And the "unaccounted" category — larger than the missing-value count — means the assignment of data elements to facilities is itself unmanaged reference data, so about a quarter of the expected-value space is in dispute. A denominator you cannot define is a completeness measure you cannot compute.

Three categorical consequences follow from aggregation, and they are worth stating as refusals rather than as caveats. You cannot disaggregate an aggregate: any analytical question not anticipated when the indicator set was designed is unanswerable. You cannot link: no cohort, no longitudinal follow-up, no readmission measure, no risk adjustment. You cannot audit back to source, because an aggregate value is a human-entered count and the only check is the paper register at the facility.

One practical consequence for anyone building a dashboard on it. Aggregate returns are entered monthly with a submission window, and late data arrive in later revisions — so **a figure for a recent month is systematically incomplete and rises for several months**. Any dashboard plotting the latest month against prior months shows a decline that is an artefact. Plot a vintage, at a fixed lag, and show the lag.

Registers are not workforces, and "compliant" has four meanings

Two last traps, both of the "count of a register" kind from question four.

Nurse-to-population ratios describe a register. The South African Nursing Council states explicitly that its numbers count people registered with it, not people employed in nursing, including employed nurses, job seekers and registered people not working in nursing, across both sectors [35]. A peer-reviewed analysis adds that the registers cannot separate practising nurses from retirees or those working overseas, computes a ratio of **47 nurses per 10 000 population against a World Health Organization benchmark of 25 per 10 000** — so South Africa is *above* the benchmark on that basis — and concludes that the figures "do not accurately reflect the nursing situation in the country" [36]. The same paper's own literature review cites 52 per 10 000 from the same underlying source, an inconsistency it acknowledges.

And there is no South African safe-staffing norm to implement. The national workforce report asks the Department to fast-track staffing norms for hospitals and names no method [27]. An architect cannot build to a norm that does not exist; the deliverable is instrumentation so that the client can establish a baseline.

"Compliant" has at least four meanings and the institutions that grant them do not agree. A facility can hold Ideal Clinic status, which is a departmental self-assessment programme; an Office of Health Standards Compliance certificate, which is statutory inspection against gazetted norms; a voluntary third-party accreditation; and, if private, a provincial licence. In 2019, **33,3% of primary health care facilities achieved Ideal Clinic certification**; in the 2019/20 inspection audit **only 14,7% of inspected facilities met the minimum standard**, ranging from 46% in Gauteng to **0% in**

North West and Limpopo [37]. Over 2019/20 to 2022/23 the inspectorate inspected 2 410 facilities and certified **473, or 19,6%**, described as far below the self-assessment rate [38]. The Department has separately reported **59% of primary healthcare facilities** as having achieved Ideal Clinic status nationally [1].

Three instruments, incompatible results, the same facilities. If a client says "we are compliant", establish which instrument, which assessor and which year before writing it into a baseline.

The habit

The discipline this chapter teaches is small, cheap and almost never practised.

Attach to every figure you publish: its source, its as-at date, its period, its denominator convention, and the second published value if one exists. Record the formula for anything derived, as a versioned object with an owner. Say in writing when a figure cannot be stated — beds nationally, admissions nationally, the chronic disease list count, timeliness in routine reporting — and name what would settle it. Refuse to average two official values. Refuse to plot a series whose rows have different bases.

> Four refusals worth rehearsing before a steering committee. Do not average two official values. Do not plot figures with different measurement bases as one series. Do not add scheme beneficiaries to the uninsured population. Do not accept a derived measure without its formula. Each of those refusals will, at some point, be unpopular, and each of them is the reason the baseline survives audit.

Two closing notes on tone, because this chapter can be misread.

None of this is evidence of a dishonest sector. It is evidence of a sector with many legitimate measurement instruments and no owner for the reconciliation between them. That absence is an architectural fact, and it reappears in every chapter of this book: no national patient identifier, no national facility register, no national tariff, no national procedure classification, no national measure definitions. Chapter 12 is where the pattern is named and turned into a method.

And the discipline is portable. An architect who has learned to ask these eight questions about a South African bed count will ask them about a customer count, a claim count or a network element count, and will be better at those too. This is the most transferable chapter in the volume, which is a strange thing to say about a chapter on provincial liability balances, and it is true.

THE CHAPTER IN TEN LINES

- The sector's official figures do not reconcile, the gaps are large and recent, and nobody owns the reconciliation. That is the subject, not a complaint.
- Eight questions catch almost everything: the denominator, the as-at date, the publishing institution, thing against register, measured against derived, aggregate against patient-level, the second official value, and who benefits.
- Provincial medico-legal liability at 31 March 2026 is R50,2 bn on the Auditor-General's basis and R50,6 bn on Treasury's. Both are right.
- The provincial ranking of medico-legal exposure reverses completely between the 2024 and 2026 publications, and neither source explains it.
- The recorded liability balance is falling while cash payments stay near R1,5 bn a year, and Gauteng budgets about a third of what it pays.
- Public beds are 100 656 approved, 85 126 usable or 88 785 depending on the series. Private beds cannot be stated nationally at all.
- Scheme coverage is 14,1% or 14,95%, on a household survey and a regulatory return. Never add scheme beneficiaries to the uninsured population.
- Scheme benefits paid are R239 bn or R259,3 bn, both from the same regulator on different bases and periods.
- Prescribed minimum benefits are 271 diagnosis and treatment pairs, with the chronic disease list published as both 26 and 27 conditions.
- Single documents contradict themselves: one annual report prints 26 811 as both complaints logged and patient safety incidents closed.
- A derived measure without a versioned published definition is an opinion with a decimal point. Theatre utilisation is 80% or 60% on the same list, with no conversion between methods.
- Routine aggregate reporting shows 99,5% facility reporting completeness against 49,7% data element reporting, and timeliness was never assessed.
- Nurse ratios describe a register, not a workforce, and there is no South African safe-staffing norm to build to.

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- **31.** National Quality Forum, all-cause admissions and readmissions measure specifications (planned-readmission algorithms as long, versioned, published rule sets) https://www.qualityforum.org/Measure_Specs_Table/All-Cause_Admissions_and_Readmissions_Fall_Cycle_2020_SpecsTable.aspx
- **32.** Mohammed, Deeks, Girling, Rudge, Carmalt, Stevens and Lilford, BMJ 2009;338:b780 (four English hospitals, standardised mortality ratios from 88 to 143; coding depth 2,12 codes per record at the lowest-ratio hospital against 1,42 to 1,62 elsewhere; hospital-by-variable interactions for emergency admission and comorbidity) https://pmc.ncbi.nlm.nih.gov/articles/PMC2659855
- **33.** South African Health Review account of the Western Cape mental health admissions dashboard (facilities calculating bed occupancy differently, and managers distrusting the provincial figure as a result; the episode-boundary rule; admitting specialty used where ICD-10 capture was unreliable) https://sahr.hst.org.za/article/141580-developing-a-dashboard-for-surveillance-of-mental-health-admissions-in-the-western-cape-province-of-south-africa
- **34.** Thabethe and Mathews, South African Journal of Information Management 2024;26(1) a1903, 29 November 2024 (265 facilities, 77 data elements, April 2017 to March 2020: facility reporting completeness 99,5%, data element reporting 49,7%, missing plus unaccounted 50,3%, overall completeness 74,6% rated poor, internal consistency 95,1%, timeliness not assessed) https://scielo.org.za/pdf/sajim/v26n1/55.pdf

- **35.** South African Nursing Council, statistics frequently asked questions (the published numbers count people registered, not people employed in nursing, including job seekers and those not working in nursing, across both sectors) https://www.sanc.co.za/statistics-faq/
- **36.** Assessing the composition of nurses in South Africa against the WHO perspective, peer-reviewed analysis (registers cannot separate practising nurses from retirees or emigrants; 47 nurses per 10 000 against a WHO benchmark of 25 per 10 000; 52 per 10 000 cited in the same paper's own review; the conclusion that the figures do not accurately reflect the nursing situation) https://pmc.ncbi.nlm.nih.gov/articles/PMC12587092/
- **37.** Rapid review of the Ideal Clinic programme, Gauteng (33,3% Ideal Clinic certification in 2019, 1 145 of 3 467 facilities; 14,7% of inspected facilities meeting the minimum standard in the 2019/20 OHSC audit, from 46% in Gauteng to 0% in North West and Limpopo) https://www.heroza.org/wp-content/uploads/2025/05/Rapid-review-of-the-ideal-clinic.pdf
- **38.** District health system performance monitoring analysis, SAMJ and South African Health Review (2 410 facilities inspected over 2019/20 to 2022/23 with 473 certified, 19,6%, described as far below the Ideal Clinic self-assessment rate) https://scielo.org.za/scielo.php?script=sci_arttext&pid=S0256-95742023001300005

CHAPTER 3

The two systems

Two health systems of roughly comparable financial scale, serving populations that differ by a factor of about six, with almost nothing connecting them.

South Africa runs two health systems. One is funded from general revenue, delivered by nine provincial departments through roughly three and a half thousand establishments, and responsible — in National Treasury's own framing — for 53,4 million people [5]. The other is funded by medical scheme contributions and out-of-pocket payment, delivered by three listed hospital groups, a tail of independent hospitals and day clinics, and tens of thousands of private practices, and covers about 9,1 million beneficiaries [30]. The two are of roughly comparable financial scale. They serve populations that differ by a factor of about six. Almost nothing connects them.

An architect arriving from banking, insurance or retail will reach for the wrong model twice.

The first wrong model is that these are two markets, a public one and a private one. In South African law they are one system with two funding mechanisms. Section 2 of the **National Health Act 61 of 2003** establishes a national health system that spans **both public and private providers** [1]. The private sector is inside it. Private establishments are subject to the Act's emergency treatment duty (s5), its consent provisions (s7), its records duties (ss13–17), its quality standards (s47) and its complaints regime (s18). The Office of Health Standards Compliance inspects them. The Health Ombud takes complaints about them. An architect who treats the private sector as a lightly regulated commercial space has misread the statute.

The second wrong model is that because both systems are large and regulated, both are measured. They are not. Two of the four measures you will instinctively reach for to size the sectors — beds and admissions — cannot be stated nationally with confidence for **either** sector. Coverage and spend can, with caveats. This chapter says so where it arises, and the last section says so in one place, because the temptation to fill the gap with a plausible round number is the single most common way an architecture document in this industry becomes untrue.

The system the law describes

Three spheres of government hold health functions.

The **national Department of Health** develops policy, sets national norms and standards under s21, may prescribe by regulation under s90, and monitors provincial performance. As a rule it does not deliver services. **Provincial departments** manage the provincial health budget, deliver services, run facilities and employ staff. **Municipalities** retain environmental health — water quality, waste management, business inspection, pest control, pollution — plus local emergency medical services and disaster management; clinical services have largely moved to provincial departments [2].

Section 3 sets the national duty: within available resources, protect and promote health, set policy, ensure prescribed essential services including **at least primary health care**, and prioritise services equitably. Section 4 provides free services at state-funded facilities to pregnant and lactating women, to children under six who are not on a medical scheme, and others, including free primary care for most people outside medical scheme cover, and free termination of pregnancy services under s4(3) [1].

The intergovernmental body is the **National Health Council** (ss22–23): chaired by the Minister, with the deputy minister, the nine provincial MECs, a municipal councillor, the Director-General and deputy directors-general, the nine provincial heads of department, a local government representative, and the head of the military health service. It advises the Minister on policy, norms and standards, proposed legislation, district guidelines, integrated plans and the research strategy, deciding by consensus or majority with a quorum of half plus one [1].

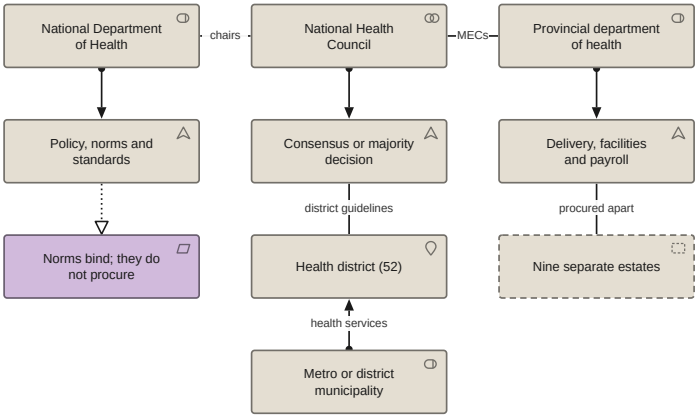


Figure 3.1 The three spheres and the intergovernmental body. The National Health Council is chaired by the Minister and includes the nine provincial MECs; it decides by consensus or majority with a quorum of half plus one. National sets norms; provinces procure, employ and deliver; municipalities supply municipal health services into the district under a service level agreement. Nothing in the structure gives any one party ownership of the system's data.

Read that composition as an architect rather than as a constitutional lawyer. The body that agrees national health policy is a committee of the people who do not have to implement it and the people who do, and it decides by consensus or majority. There is therefore **no single owner of the health system's data.** The national department can issue norms and prescribe regulations. It does not employ the clinicians, own the facilities, or control the provincial information technology estate. Nine provinces procure independently. A national system in South African health care is always a federation problem

before it is a technology problem, and the Auditor-General's 2025 finding that the National Digital Health Strategy is poorly implemented, relying on manual and error-prone processes, is in part a statement about that structure rather than about any particular programme [7].

One procedural trap in the Act itself, because it will cost you a week. The consolidated text of the National Health Act already carries amendments made by the **National Health Insurance Act 20 of 2023 that are not in force**. Section 31A, on District Health Management Offices, is an uncommenced amendment; several s25 provincial functions are marked for deletion or amendment under the uncommenced Act [1]. No section of the NHI Act is in operation, by a presidential undertaking of 20 February 2026 made an order of court on 24 February 2026 [40]. Reading a consolidated version without checking commencement will give you a system that does not exist. The Constitutional Court heard the public participation challenges on 5 to 7 May 2026 and judgment was still reserved as this chapter was written; **re-check the outcome before you rely on any statement here about NHI's legal status.** Chapter 7 carries that subject.

The public sector: who employs whom, who buys what, who holds the budget

Employment. Clinical and non-clinical staff in public facilities are **provincial public servants**, paid through **PERSAL**, the national public service payroll. Medical interns and community service officers are placed nationally and employed provincially.

That single fact has a consequence most architects miss. Workforce data for the public sector comes from a payroll system, not from a clinical rostering system. It therefore counts **posts and persons**, not shifts or hours at the bedside. The national nursing workforce report is explicit that its figures are PERSAL data as at June 2023 [4]. If a provincial client asks for a staffing dashboard, the first question is whether any system in the estate knows who was actually on the ward last night. Usually none does.

Purchasing. Goods and services are procured provincially under the Public Finance Management Act and provincial supply chain management policy, with significant national contracts for pharmaceuticals — the national department reported ten pharmaceutical contracts worth R7,423 billion in 2024/25, with R3,289 billion, or 44,31%, awarded to local manufacturers [3].

Laboratory services are the exception that shapes everything downstream. They are not procured facility by facility. They come from the **National Health Laboratory Service**, a national public entity that bills provinces for tests and processes more than 140 million tests a year [24]. A provincial hospital does not choose its pathology provider and cannot change it.

Budget holding. The province holds the budget, and the money arrives in two structurally different ways. The bulk is the **provincial equitable share**, an unconditional transfer the province may spend on any provincial function — so the health allocation is a provincial political decision, not a national one. The total provincial equitable share across all functions is R670,323 billion for 2026/27. On top sit **conditional grants** ring-fenced by national government: District Health Programmes R29,549 billion, National Tertiary Services R16,700 billion, Health Facility Revitalisation R7,717 billion, Human Resources and Training R5,909 billion, and National Health Insurance (direct) R475 million, plus an indirect NHI grant of R2,560 billion [5]. The consolidated health budget — national, provincial and entities — was reported at **R310 billion for 2026/27**, from a revised R295 billion in 2025/26 [6].

The architectural consequence is that a provincial health department's financial systems must satisfy **two different accountability chains**: Provincial Treasury for the equitable share and the national department for grant conditions and grant-specific reporting, with different reporting calendars and different data definitions for the same spend. Reconciling them is a permanent work package in every provincial reporting layer, and it is not a defect to be fixed.

The national department's own vote is small relative to its apparent importance, because most of it is transfers: a final appropriation of R62,225 billion in 2024/25 with actual expenditure of R61,882 billion, of which **transfer payments were R58,390 billion**, fully spent [3]. Programme actuals for the same year were Administration R640,2 million; National Health Insurance R1 168,7 million; Communicable and Non-Communicable Diseases R25 405,6 million; Primary Health Care R3 295,5 million; Hospital Systems R23 860,6 million; Health System Governance and Human Resources R7 511,1 million [3]. The two large lines are overwhelmingly conditional-grant transfers to provinces, not national spend.

> **WHO IS COMMISSIONING YOU**
>
> An architect commissioned "by the National Department of Health" is usually being commissioned by a policy and standards body with a small operational budget and no service-delivery estate. An architect commissioned by a province is being commissioned by the employer, the facility owner and the budget holder. The same words in a terms of reference mean different things depending on which it is, and the difference determines whether your target architecture can be implemented at all.

One more fiscal fact that will appear in your integration design whether you want it there or not. Accruals in provincial health peaked at almost R25 billion in 2024/25 and were reduced to R20,5 billion in 2025/26 with additional equitable share allocations [8]; the Auditor-General put accumulated unpaid bills above R21 billion and reported that **nine of the ten health departments audited had liabilities exceeding assets** [7]. Unpaid invoices are not an accounting curiosity in this sector. They are why a supplier integration goes quiet, why a laboratory threatens to suspend a service, and why a provincial finance system's open-item ledger is the most politically sensitive data in the department.

The district health system

Chapter 5 of the National Health Act creates the district health system. The MEC establishes a **district health council** for each health district, with municipal councillors, an MEC nominee and up to five others, chaired by the metropolitan or district municipal councillor (s31). A **district health manager** prepares a **district health plan** for the council and the MEC (s33(1)), and the MEC must ensure each district develops a human resources plan (s33(2)). Metropolitan and district municipalities must ensure appropriate **municipal health services** are provided, assigned through **service level agreements** that must set out the services, the resources, the performance standards and the termination conditions (s32) [1].

The geography: nine provinces, 278 municipalities — eight metropolitan, 44 district and 226 local — and 4 277 wards, with typical ward populations of 5 000 to 10 000 and metropolitan wards exceeding 50 000 [2]. Health districts align broadly to district and metropolitan municipalities, giving **52 health**

districts. That figure is corroborated indirectly rather than stated in a register: HPV testing was introduced as primary cervical screening in 43 of 52 health districts by March 2025 [3].

Reporting. Districts set targets in the annual district health plan and report against the **National Indicator Data Set**, which carries about 250 indicators and several hundred data elements, grouped mainly by disease programme and reviewed every second year with the provinces. The indicators are unevenly distributed: HIV has over 70, child health 19, non-communicable diseases 6. Much of the collection remains manual. Since 2005 the **District Health Barometer** has published the results to rank and compare districts [9].

For an architect the National Indicator Data Set is the de facto national reporting contract for public-sector facilities, and it is **disease-programme shaped, not patient shaped**. Any data warehouse built for a provincial department will spend most of its effort reconciling programme aggregates to each other, because the same encounter is counted differently by the HIV programme, the TB programme, the maternal health programme and the facility's own headcount return. That is not untidiness. It is the consequence of a national information backbone that holds aggregates, and chapter 15 shows where the consequence lands.

The facility tiers, and who refers to whom

The categories of public hospital are set by **regulation, not policy**: Government Notice R185 in Gazette 35101, made under sections 35 and 90 of the National Health Act after consulting the National Health Council, current to 2 March 2012 [10]. The service package, not the bed count, determines the category — which matters because the bed ranges overlap.

Category	Bed range	Service package, in summary
Small district hospital	50 to 150	District package 24 hours; inpatient, ambulatory, emergency; primary care through general practitioners and clinical nurse practitioners
Medium district hospital	over 150 to 300	As above; specialist services limited to paediatrics, obstetrics and gynaecology, internal medicine, general surgery, family medicine
Large district hospital	300 to 600	As above
Regional hospital	200 to 800	Four core specialties 24 hours, plus at least one of orthopaedic surgery, psychiatry, anaesthetics or diagnostic radiology; trauma and emergency; short-term ventilation
Tertiary hospital	400 to 800	Regional services plus subspecialties; intensive care under a specialist or intensivist; referrals from regional hospitals without provincial limits
Central hospital	maximum 1 200	Tertiary plus central referral services; may provide national referral services; must train providers and conduct research; must be attached to a medical school
Specialised hospital	maximum 600	Psychiatric, tuberculosis, infectious disease and rehabilitation services

Note the overlaps. District and regional both span 200 to 300. Medium and large district both include 300. Regional and tertiary both span 400 to 800. A facility master that derives category from bed count will be wrong, and there is no national register to check it against.

National referral services are listed in the regulation as super-specialised and very expensive: heart, lung, bone marrow and liver transplants, and cochlear implants [10]. The national department directly manages ten major teaching hospitals according to one secondary source — a headline-only claim that should be re-sourced before it reaches a client deliverable [11] — and in July 2025 the Minister described the Limpopo Central Academic Hospital as the country's eleventh central or academic hospital, then at 26% of construction [12].

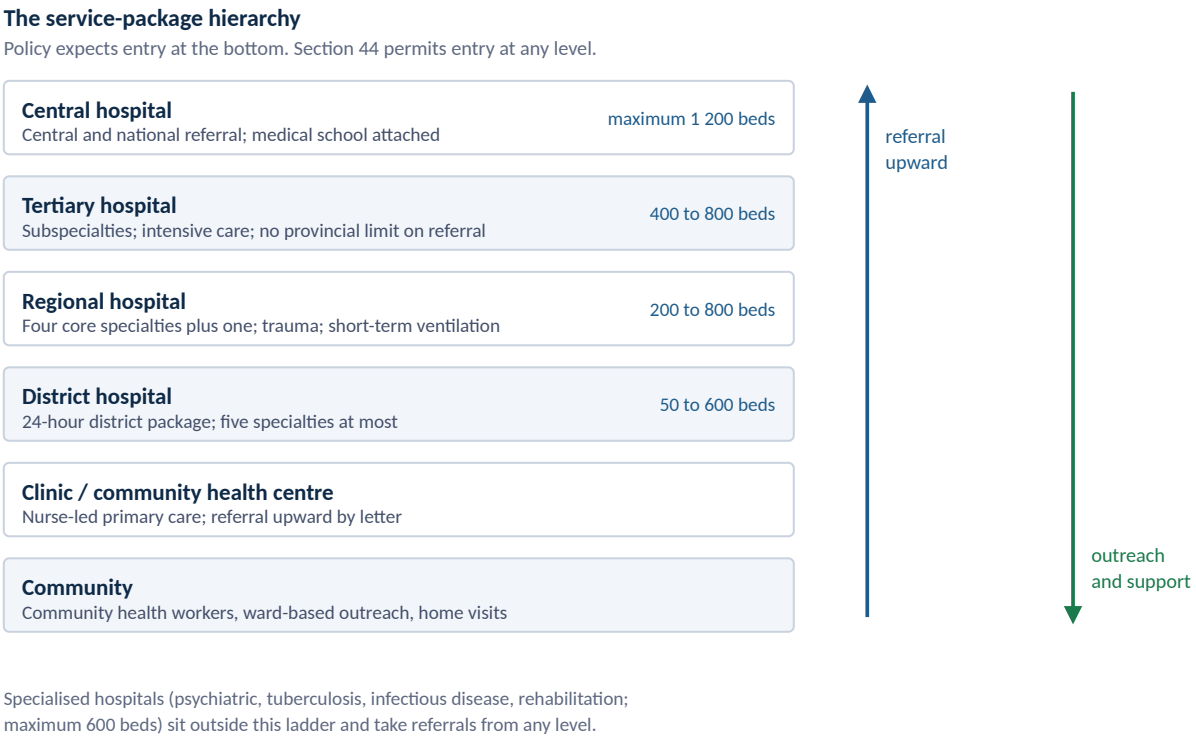


Figure 3.2 The public service-package hierarchy, with the bed ranges from the hospital categories regulation. The ranges overlap, so a bed count does not determine a category. Referral runs upward and specialist outreach runs downward, but section 44 lets a user attend any public establishment, which is why triage sits at the front door of a district hospital.

Referral in law is not what the tier diagram implies. Section 44 provides that users may attend **any** public establishment, and that an establishment which cannot provide necessary care **must transfer** the user to an appropriate facility [1]. Policy expects patients to enter at primary care and be referred upward; in practice many use hospital emergency units after hours, which is precisely why formal triage exists at the front door [2]. The tiers are a service-package hierarchy with a statutory right of direct access cutting across them.

The architectural consequence is the one to carry forward. A referral is not a routing instruction between two systems with a shared patient record. It is a **transfer of clinical responsibility between two legally distinct establishments** that may be on different patient administration systems in different districts, accompanied by a paper letter. Section 10 requires a **discharge report** with prescribed content: verbal is permitted for outpatients, written is mandatory for inpatients [1]. The discharge report and the referral letter are the only guaranteed clinical data interchange in the South African public system, and **they are documents, not messages.** Chapter 16 returns to why that distinction decides an integration strategy.

What cannot be counted, and why that is the finding

This is where an architect gets into trouble fastest, so the figures are given with their provenance and their disagreements intact.

Beds. Parliamentary reply NW3455 of 14 October 2022 gives **100 656 approved** and **85 126 usable** public hospital beds as at July 2022 — a gap of 15,4% — and lists 30 new hospitals in the pipeline across eight provinces [13]. A separate series gives public hospital beds as 85 063 (2020), 55 913 (2021), 88 556 (2022) and 88 785 (2023) [14]. The 2021 value is almost certainly a definitional or collection artefact rather than a real collapse and recovery, and the publishing institute's own notes flag labelling errors in that annexure. **Do not use it.**

The reason the numbers disagree is that "available beds" in South Africa is one of *approved*, *usable*, *commissioned*, *staffed* or *open tonight*, and most sources do not say which they mean. In the private sector there is a sixth: a bed is a **licensing** quantity, because a provincial licence authorises a specific bed count at a specific site. Gauteng reported in 2017 that it had approved 48 facilities not yet built, representing **2 796 beds and 306 theatres**, of which only five were under construction [15]. Licensed beds, built beds and open beds are three different populations, and the hospital group's own asset register is usually a fourth.

Facilities. The candidate denominators do not reconcile either. The Office of Health Standards Compliance's inspection universe of public establishments was **3 741** in 2023/24 [16]. The Ideal Clinic denominator for primary care facilities in 2019 was **3 467** [17]. A book chapter citing the Department of Health, year not stated, gives over 3 000 public clinics and over 470 public hospitals, plus over 1 500 private clinics and over 260 private hospitals, for a combined 5 809 [18]. A 2014 table gives 3 863 public clinics, 407 public hospitals, 610 private clinics and 203 private hospitals, totalling 5 083 — without an inline citation in its source, so headline only [11].

> **THE MOST RELIABLE PREDICTION IN THIS BOOK**
>
> There is no single authoritative, current, publicly available register of South African health establishments with a stable identifier. The nearest things are the Office of Health Standards Compliance's inspection universe, the Department's Ideal Clinic denominators, Health Patient Registration System installations, and provincial licensing registers for private facilities. They do not reconcile. If your project needs a facility master, **it has to be built and reconciled, and that reconciliation is a work package rather than an assumption.**

One more number that reframes the public estate, because architects consistently design for the wrong condition. Against the 2024/25 Ideal Clinic assessment covering 3 333 primary health care facilities, the department reported 52 facilities (1,5%) without functional computers, 173 (5,2%) without internet access, and 278 (8,3%) without functional fixed-line telephones [43]. A 2023 qualitative study quotes a participant saying nearly 40% of public health facilities lack a *reliable* internet connection [38]. Those two figures are not contradictory — one measures presence and the other reliability — and the gap between them is the design brief. **Design for intermittency, not for absence.** Designing for absence produces a paper fallback. Designing for intermittency produces offline capability, and offline capability without conflict resolution is where duplicate patient records come from. Chapter 11 has the documented South African example.

The private sector: groups, day clinics, independents

Three listed groups dominate acute private hospital capacity. The Competition Commission's Health Market Inquiry measured 2016 shares by beds, with admissions in brackets: **Netcare 31% (33%), Mediclinic 26,8% (28,6%), Life 25,3% (28,5%)** — together about **83% of beds**. The **National Hospital Network**, an association of independents, held 13,6% of beds but only 7,7% of admissions; independent hospitals outside it held 2,3% of beds and 2,2% of admissions. About 60% of local facility markets were highly concentrated, and national concentration exceeded "highly concentrated" thresholds on even the most conservative benchmarks [19].

A 2025 report gives the big three about **83% of available private beds and 87% of admissions** [20]. The bed figure agrees with the Health Market Inquiry; the admission figure does not, and neither source explains the difference. Cite both and say they disagree. Note also the bed-to-admission divergence inside the inquiry's own data: the National Hospital Network's share of admissions is roughly half its share of beds, which is consistent with lower occupancy, shorter stays, or both, among independents.

Current group scale, with as-at dates that differ, so do not sum across them.

- **Mediclinic Southern Africa**, as at 18 October 2024: 50 hospitals including three in Namibia, 5 subacute hospitals, 6 mental health facilities, 14 day case clinics of which four are operated by Intercare, and about 8 856 inpatient beds. A summary panel on the same page describes the group as a whole as 72 hospitals, 21 day case clinics, 459 theatres, 11 646 inpatient beds, more than 840 000 admissions and more than 37 000 employees — **figures that do not fully reconcile with the division-level list on the same page** [21].
- **Netcare**, for the year to 30 September 2025: revenue R26 344 million, up 4,5%; acute occupancy 65,0%; acute average length of stay 4,5 days; paid patient days split 51,2% surgical and 48,8% medical; a net 117 new specialists granted admission rights; four Level 1 trauma facilities [22].
- **Life Healthcare**, six months to March 2026 on continuing operations: revenue R12,4 billion, up 2,4%; like-for-like weighted average occupancy 67,5%, down from 68,8%; average tariff increase 3,3%; imaging volumes up 4,2%; renal treatments up 9,3%; the 140-bed Life Paarl Valley Hospital under construction [23].
- Other named groups, by "locations" as at January 2026 from a secondary source: Lenmed 19, Clinix Health Group 8, Melomed 6 — a count that plainly includes non-hospital sites and is not comparable to the hospital counts above [11].

Current hospital and registered bed counts for Netcare and Life Healthcare could not be sourced: the groups' investor documents were not retrievable and their results statements do not state them. If you need them, they must come from the groups directly or from the Hospital Association of South Africa, and they should not be inferred from the "locations" figures above.

What the groups' own numbers tell an architect. Volume growth is essentially zero. Netcare's paid patient days grew 0,7% in the year to September 2025 and 0,7% in the half to March 2026; Life's went slightly negative [22][23]. Both groups are generating earnings growth from tariff, cost control, case mix, complementary services and capital structure. Netcare attributes cumulative gross savings and cost avoidance of R587 million since 2022 to its digital programme, with an internal rate of return above 25% [22]; a separate report of the same programme gives R706 million since 2022 [23]. The two figures are at different dates on possibly different measures and should not be used interchangeably.

> **THE MANDATE YOU WILL ACTUALLY BE GIVEN**
>
> In a private group the mandate will be efficiency, acuity mix and retention of scheme volumes, not growth. Netcare notes consumers shifting toward lower-cost, restricted-network schemes [22]; Life describes a specialist development programme expanding to 115 specialists and sub-specialists over nine years at nearly R500 million for a stated 22% return [23]. A target architecture written around patient growth is answering a question nobody in the room is asking.

The doctors the hospital does not employ

The defining structural fact about South African private hospitals is that **the hospital does not employ the doctors.** Specialists hold **admitting rights** at one or more hospitals and bill the funder or the patient separately from the hospital. Netcare's "net 117 new specialists granted admission rights" and Life's planned "about 140 new specialists" are statements about a relationship, not about a payroll [22][23].

The architectural consequences are severe and routinely underestimated.

- The hospital's clinical record and the specialist's practice record are **different records in different systems owned by different legal entities**, and the specialist's rooms may use any of dozens of practice management packages.
- A single inpatient episode generates **several independent claims** — hospital, surgeon, anaesthetist, pathology, radiology, sometimes physiotherapy and a prosthesis supplier — each with its own practice code number, its own tariff basis and its own submission route.
- **Orders** placed by a non-employee practitioner into a hospital system raise authentication, delegation and liability questions that an employed-clinician model does not.
- The hospital cannot mandate the specialist's clinical system, so interoperability at that seam is a commercial negotiation per specialty rather than an information technology decision.

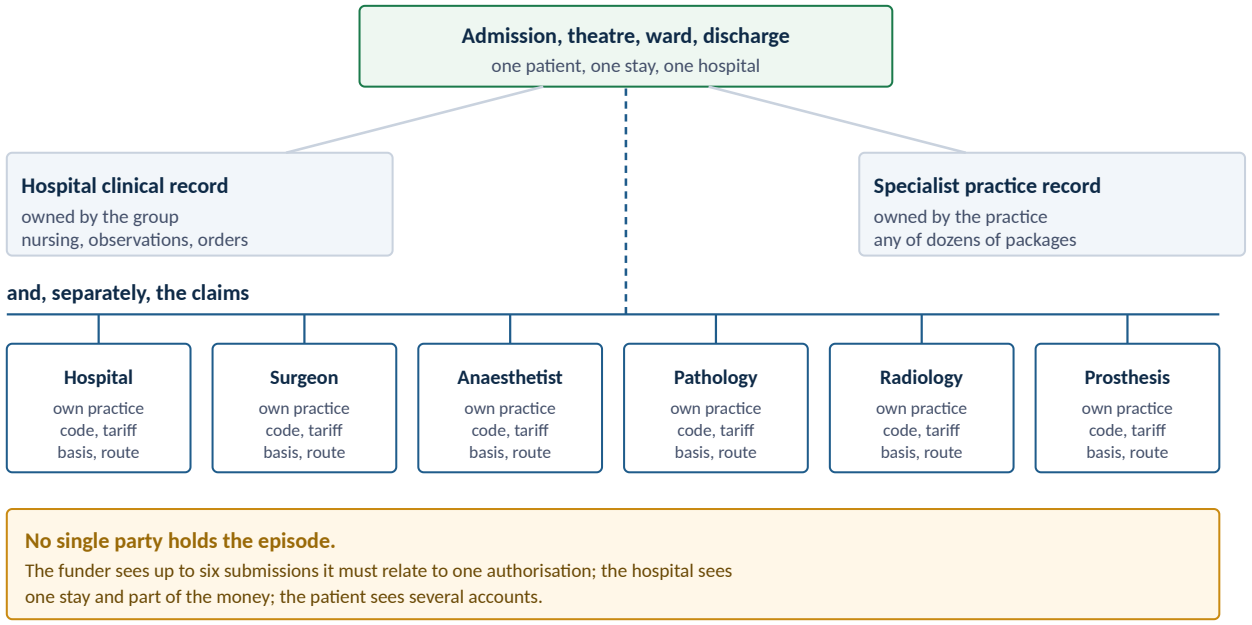


Figure 3.3 The admitting-rights model, drawn as data. One stay, two clinical records in two legal entities, and up to six independent claims each with its own practice code number, tariff basis and submission route. Every integration at this boundary is a commercial negotiation per specialty rather than an architecture decision.

The inquiry measured **1,75 private practitioners per 1 000 insured people**, with general practitioners spread fairly evenly at just under one per 1 000 insured and **specialists concentrated in provincial capitals and metropolitan areas, with some areas having none**. It found that the three large groups use inducement programmes to bind top specialists to their hospitals, with exclusionary effects on newcomers, and that they dominate designated service provider networks [19].

It also found hospital admission rates, high-care and intensive-care admissions and length of stay higher than the local burden of disease explains, and that holding other factors constant, more practitioners — except obstetricians — led to more hospital admissions, which it took as evidence of **supply-induced demand** [19]. Whether or not one accepts the inference, it is the published official finding and it shapes every policy proposal now on the table. An architect who builds a utilisation dashboard for a private group is building an instrument that the competition regulator has already said measures the wrong thing if it measures only volume.

The diagnostic layer, in both sectors

In both sectors the diagnostic provider is a **separate organisation from the ordering clinician and the facility.** Order-to-result is therefore always an inter-organisational integration, with its own identifiers, its own result formats and its own turnaround service levels. There is no national laboratory results repository. A hospital that wants a complete result history for a patient must integrate with each laboratory and each imaging provider individually, and must reconcile patient identity across each boundary without a national identifier.

Public. The National Health Laboratory Service is the near-monopoly provider of diagnostic pathology to state hospitals and clinics, processing more than 140 million tests a year and funded by **billing provinces for tests** rather than by direct appropriation [24]. That funding model is the source of a large and unreconciled intra-state debt. Provinces collectively owed the service R10,7 billion (Treasury), R10,6 billion as at June (Treasury to Parliament) or about R11 billion as at March 2026 (Portfolio Committee) [8][24][25]. The provincial breakdowns are worse: KwaZulu-Natal about R3,94 billion and Gauteng about R3,3 billion as reported to the Portfolio Committee at end-May 2026 [25], against Gauteng R3,5 billion, KwaZulu-Natal R1,2 billion and Eastern Cape R1 billion in Treasury's figures to Parliament as at June 2026 [24]. The KwaZulu-Natal line is irreconcilable between the two — R3,94 billion against R1,2 billion — and neither source explains it; the service itself acknowledged billing disputes and invoice-processing issues [24].

Pause on that. An intercompany reconciliation defect of roughly R2,7 billion between two arms of the same state is what happens when two organisations bill each other without a shared service catalogue and a shared identifier. It is the best available illustration in the country of why reference data governance is a financial control.

Governance has deteriorated on the record: an unqualified audit in 2022/23, qualified in 2023/24, a **disclaimer in 2024/25** — the worst outcome, meaning records were in too much disarray for an opinion — and in 2025/26 the service missed the statutory deadline for submitting annual financial statements. On 9 October 2026 the Minister appointed a three-person team, Eric Buch, Barry Kistnasamy and Anban Pillay, to address the governance problems. The service states the audit findings mainly concerned financial and supply chain management and did not question the scientific accuracy of test results [24].

Private. Private pathology is concentrated among a small number of groups — Ampath, Lancet and PathCare are the three usually named — and private radiology among independent practices and group-affiliated imaging businesses. Funders have publicly disputed private pathology pricing, including seeking refunds of what they called excessive COVID-19 testing charges, a claim reported secondhand and not re-sourced here [27]. **Lancet Laboratories** appears in the Information Regulator's published list of administrative fines at R100 000, paid [26] — the only named private health care business among the first six POPIA administrative fines, and therefore the single most useful precedent for a South African health data governance business case. The

grounds of the fine are not stated in the source that names it; obtain the enforcement notice before you build an argument on it. Chapter 18 carries the regulatory detail.

Where the two systems touch

Six real interfaces, each a source of architectural work.

1. **Dual practice.** Public-sector clinicians may do remunerative work outside the public service. Treasury has flagged possible savings in rural allowances, medical intern remuneration and this arrangement, and the Minister announced a review by a ministerial advisory committee expected to finish by April 2027 [8]; it was also a theme of the Health Market Inquiry's licensing and quality examination [41]. Operationally it means the same practitioner appears in a public rostering system and a private practice management system, and a public hospital's theatre list can be constrained by a consultant's private commitments.
2. **Scheme members using public facilities.** The Uniform Patient Fee Schedule classifies scheme members and their dependants as **full-paying patients**, and provides that members who were means-tested without disclosing scheme membership must be reassessed retrospectively [28]. That is a revenue-recovery process requiring eligibility verification against scheme membership — an integration almost no public hospital has.
3. **Designated service provider arrangements.** A scheme may designate providers for prescribed minimum benefit purposes. The Council for Medical Schemes states that the public sector cannot be designated unless the scheme ensures the service will be available [29].
4. **Referral upward from private general practitioners into public hospitals.** Many people without scheme cover use private general practitioners, who may refer them to district or regional government hospitals; contact between private practitioners and government hospitals is limited, though many do sessional work in district hospitals [2].
5. **Statutory funds.** Compensation for occupational injuries and the Road Accident Fund pay for care delivered in either sector, at their own tariffs, with their own code sets and claim formats. Chapter 6 has the money.
6. **The training platform.** Central hospitals must be attached to a medical school as the main teaching platform [10], and all medical training happens on the public platform. That makes the academic health complex (s51) a distinct governance animal: a university and a provincial department share staff, space and records, which means two responsible parties over one record set.

Where they do not touch

The list of what is missing is shorter and more consequential than the list of what connects.

There is **no shared patient identifier.** The Health Patient Registration System had reached 3 266 establishments by 31 March 2025 — 3 150 primary health care facilities, 113 hospitals and **three general practitioner practices** — and an automated biometric identification system had reached 82 primary health care facilities [3]. The private sector is not covered. No national crosswalk exists.

There is **no national facility register** with a stable identifier, as the previous section established. There is **no national tariff**: the Reference Price List regulations were declared invalid and set aside on 28 July 2010, and nothing has replaced them [42]. Price is now a property of a contract, and each group publishes its own schedule for self-payers [44]. There is **no national results repository**, no national procedure classification in routine use, and no shared clinical record.

The practical test is simple. Any measure that requires following a person across the boundary is not computable. A scheme member's public primary care attendances are invisible to the scheme. A public patient's private pathology history is invisible to the hospital that admits them. A readmission to a facility in the other sector is invisible to both. Chapter 11 is a chapter rather than a section because of this, and chapter 12 is built on the decisions it forces a local architect to make.

Relative size: the defensible statement

Putting the sourced figures together, here is what can be said without inventing anything — and what cannot.

Measure	What can be stated	Source and date	Status
Population covered, private	9,13 million beneficiaries, 14,95% of the population	Council for Medical Schemes, 2024/25 reporting	Two official values
Population covered, private	14,1% of the population	Statistics South Africa, Income and Expenditure Survey 2022/23	Household self-report
Population, public responsibility	53,4 million people without private medical insurance	National Treasury, 2026 Budget Review	Stated, not measured
Spend, private	R259,3 billion benefits paid, calendar 2024; or R239 billion on the regulator's annual report basis	Council for Medical Schemes	Two bases, two periods
Spend, public	About R310 billion consolidated, 2026/27	Budget 2026 reporting	Budget, not outturn
Beds	Not statable nationally for either sector	—	See below
Admissions	Not published nationally for either sector	—	See below

On **coverage**, the two official values are 14,1% from a household survey and 14,95% from a regulatory return [32][30]. Both are correct for their purpose. Scheme beneficiaries are not the same population as people with any private cover, which also includes gap cover, health insurance products, employer clinics and cash-paying private general practitioner users. **Never add the coverage figure to the uninsured figure**, because the sum double-counts: Statistics South Africa's own data show 71,6% of people relying on the public sector for usual care, including 65,7% using public clinics, against only 1,4% using private clinics and 1,8% private hospitals — while 14,1% hold medical scheme cover [33]. A substantial number of scheme members use public primary care.

Coverage also varies by a factor of more than three across provinces — Western Cape 23,8%, Gauteng 19,6%, Northern Cape 14,3%, Free State 14,0%, Eastern Cape 10,2%, North West 9,6%, KwaZulu-Natal 9,1%, Mpumalanga 8,4%, Limpopo 7,3% — and by settlement type, from 19,1% urban to 3,9% in traditional settlements [33]. A private-sector demand model built on a national percentage will be wrong in both directions at provincial level.

On **money**, schemes paid R259,3 billion in benefits in calendar 2024, up 8,52% [31]; the regulator's 2024/25 annual report gives contributions excluding medical savings of R226,94 billion with a further R24,30 billion into personal medical savings accounts, and benefits paid of R239 billion [30]. The two benefit figures are on different bases and different periods and this is one of the clearest traps in the sector. Household out-of-pocket spending was R31,5 billion, averaging R1 481 per household, with the wealthiest quintile accounting for R22,5 billion of it [33]; the regulator reports out-of-pocket payments of R43,3 billion for 2023 [30]. Again: two instruments, two definitions, two numbers.

On **workforce**, public-sector doctors and nurses by province on PERSAL as at January 2025 give national ratios, excluding about 8,95 million insured people, of **1 doctor to 2 230 people** (4,484 per 10 000) and **1 nurse to 762 people** (13,123 per 10 000) [35]. Private-sector density is the inquiry's 1,75 practitioners per 1 000 insured [19]. Older secondary figures give roughly 79% or 70% of doctors in the private sector depending on source and year; they conflict and neither is current [11]. The honest statement is that **a large majority of specialists work at least partly in the private sector, serving under a sixth of the population, and the precise share cannot be stated from publicly available current data.**

> **THE TWO NUMBERS YOU WILL BE ASKED FOR AND CANNOT SUPPLY**
>
> **Beds.** Public figures disagree between series by up to a quarter, and the last national private bed figure is 20 466 registered private beds from 2011 [34], contradicted by Gauteng's own 2017 report of 17 052 private beds in that one province [15]. **Admissions.** Not published nationally for either sector. The private groups report *paid patient days*, not admissions; the Health Market Inquiry's admission shares are 2016 and a 2025 report gives a different share [19][20]. When a sponsor asks for "beds and admissions, public against private", the correct answer is the sourced coverage and spend figures, plus a statement of what is not measured and what would settle it — a national facility and bed register with a stable identifier, and a national admissions series with a published definition.

The defensible paragraph, for reuse. South Africa runs two health systems of roughly comparable financial scale serving populations that differ by a factor of about six. Medical schemes paid R239 to R259 billion in benefits for 9,1 million beneficiaries; the consolidated public health budget is around R310 billion for 53,4 million people. The private system holds roughly a quarter of hospital beds — on figures that are old and do not reconcile — and a disproportionate share of specialists, diagnostics and equipment. Both systems are capacity-constrained. Both are financially stressed. And the mechanisms that would connect them — a national patient identifier, a national facility register, a national tariff, a national results repository — do not exist.

The structure nobody describes: two architectures, federated

There is one more structural fact, and it is the most recent and least understood.

South Africa's public primary care architecture has already been decided, and it is not what most architects assume. **e-impilo** is the National Department of Health's own bespoke electronic medical record. The 2024/25 annual report records the first phase, covering tuberculosis and HIV, as deployed across 37 pilot facilities in March 2025, tested in Dihlabeng in the Free State, with a Single Patient Viewer scoping exercise completed as a forerunner to the longitudinal patient record [3].

A 2026 tender — bid NDoH-04 of 2025/2026, issued 3 March 2026 — invites **non-governmental and non-profit service providers**, not systems integrators, to roll the department's bespoke record out across fixed primary health care facilities in **eight provinces** within 18 months. The document states 3 083 facilities in total; its own provincial figures sum to 3 213. **That discrepancy is in the source document** and should be cited rather than silently resolved. Its stated integration requirements are explicit: the record must integrate with the **Health Patient Registration System**, the **laboratory information system**, **SyNCH** and the **Master Health Facility List** [36].

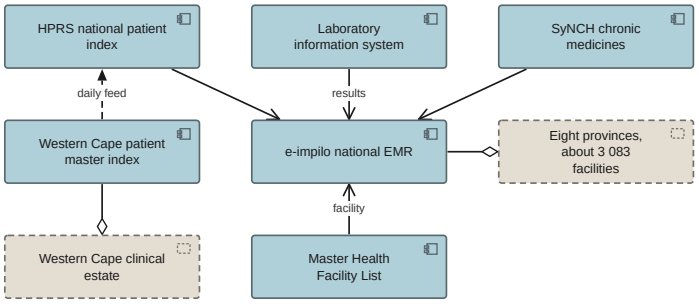


Figure 3.4 The two-architecture federated model, as stated by the Department of Health's own 2026 rollout tender. The national record must integrate with four national shared services: the patient index, the laboratory information system, the chronic medicines system and the facility list. The Western Cape is not in the rollout; it runs its own estate and feeds its provincial patient master index into the national index daily. The federation point is identity, not the record.

Three readings of that tender, in ascending order of importance.

First, the integration map is now explicit, and those four are the national shared services a South African primary care record must consume. That is the figure above, and it is stated by the department rather than inferred.

Second, the eighth province is the point. The Western Cape is absent, consistent with its own provincial estate, and it has sent daily updates from its provincial patient master index to the national index since June 2020 [37]. South Africa is therefore deliberately running a **two-architecture federated model**: a national bespoke record in eight provinces, a provincial estate in the ninth, federated at the identity layer through the national patient index. That is a legitimate and under-discussed federated pattern. It is not an anomaly to be rationalised, and an architect who proposes rationalising it has misunderstood what is being federated — identity, not the record.

Third, what the tender does not say. It declares that the record already complies with the national normative standards framework, and states that no integration or interoperability work is needed or included in the scope of the rollout. That is a claim to report, not to endorse: it names no implementation guide, no message or resource version and no conformance attestation. And **none of the system's technical architecture is public** — not its hosting, not its data residency, not its standards conformance, not who built it [36]. Say so in your assessment rather than assuming. Chapter 15 carries the systems; chapter 16 carries what governs each of those four seams.

THE CHAPTER IN TEN LINES

- In South African law the private sector is inside the national health system, not outside it: the National Health Act's duties on emergency care, consent, records, confidentiality and quality apply to private establishments.
- National sets policy and norms; provinces hold the budget, own the facilities and employ the staff; municipalities keep environmental health and local emergency services. There is no single owner of the health system's data.
- Public workforce data comes from a payroll system, so it counts posts and persons, not shifts at the bedside.
- Provincial money arrives as an unconditional equitable share plus ring-fenced conditional grants, which means two accountability chains with different calendars and different definitions for the same spend.
- Hospital categories are set by regulation and the bed ranges overlap, so the service package determines the category and a facility master cannot derive it from bed count.
- A referral is a transfer of clinical responsibility between legally distinct establishments, carried by a letter. The discharge report and the referral letter are the only guaranteed clinical data interchange in the public system, and they are documents.
- There is no authoritative current register of South African health establishments. Building and reconciling a facility master is a work package, not an assumption.
- Beds and admissions cannot be stated nationally with confidence for either sector. Coverage and spend can, each with two official values. Never add scheme coverage to the uninsured figure.
- The hospital does not employ the doctors. One inpatient episode produces several independent claims and at least two clinical records in different legal entities.
- The diagnostic provider is always a separate organisation, so order-to-result is always an inter-organisational integration, and there is no national results repository.
- Provinces owe the National Health Laboratory Service roughly R10,6 to R11 billion, and the provincial breakdowns are irreconcilable at the KwaZulu-Natal line. Two arms of one state billing each other without a shared catalogue or identifier.
- Design the public estate for intermittent connectivity rather than for its absence, and remember that offline capability without conflict resolution is where duplicate patients come from.
- The country is deliberately running two architectures federated at the identity layer: a national bespoke primary care record in eight provinces, the Western Cape's own estate in the ninth, joined by a daily feed into the national patient index.

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CHAPTER 4

The vocabulary

Five words for the boundary of one hospital stay, two unrelated meanings of claim, and four entities wearing the name of one.

Healthcare is the industry in which the entity definitions themselves are contested. Two systems in the same hospital will each report a correct average length of stay and disagree by half. A funder and a provider will both count admissions and get different answers, and neither will be wrong.

An architect's usual defence — agree a glossary in the first week — fails here, for a specific reason. These words are not vague. They are precise and incompatible. *Spell* has a published definition. *Episode of care* has a published definition. They cut the same patient's care at different places, both correctly, for different purposes. A glossary that picks one and declares it the standard has silently destroyed the other, and somebody's statutory return depends on it.

This chapter therefore states each definition once, precisely, in a form you can quote into a deliverable — and where two legitimate definitions exist it says so, and says what the choice costs. Later chapters use these definitions and do not restate them.

Three ways the words go wrong

Conflation under a clinical-sounding name. "Medication" looks like one thing because clinicians say "the patient's medication". It is at least four objects with four identifiers, four lifecycles, four owners and four legal footings. A model that collapses them cannot answer "was the drug given?", which is the only question that matters at a mortality and morbidity meeting.

Borrowing a billing entity to carry a clinical meaning. The classic is using the claim line as the record of what was done. Claim lines exist to be reimbursed; they are shaped by what is payable, not by what happened. In South Africa this is institutionalised, because Regulation 5(f) of the Medical Schemes Act requires a diagnostic code on the claim, and the national ICD-10 task team's own circular complains that providers were defaulting whole disciplines to a single code off cheat sheets assembled from scheme call-centre information [8]. A code that exists to clear an edit is not clinical data, and an analytics layer built on it measures reimbursement behaviour and calls it morbidity.

Treating a boundary as a fact. Encounter, episode, visit, spell and admission are all boundary constructs. Each is a *decision* about where to cut a continuous stream of care, made differently by the patient administration system, the clinical record, the funder and the statistician. Every length-of-stay, occupancy, readmission and mortality figure in the organisation depends on which cut was used, and almost none of them is published alongside the figure.

A fourth, less often named: **the entity that exists only as a row.** Consent and medical device commonly appear in a repository as one attributeless row. An entity with no attributes is a placeholder for an argument nobody has had yet.

The boundary words

These are the definitions the rest of the volume rests on. Take them exactly.

- **Encounter.** The atomic interaction between a patient and the health system in which care is delivered or assessed, with a start, an end, a setting, a responsible practitioner and a reason. Encounters nest and overlap, because real care does.
- **Visit.** An administrative construct: one attendance at one place on one day. It is the unit of registration — what the patient administration system opens and closes — and in outpatient settings it frequently coincides with an encounter, which is why it is mistaken for one.
- **Ward stay** (or **bed occupancy period**). The location-time fact: this patient, in this bed, in this ward, from this time to that time. It is the only construct from which occupancy can be computed.
- **Admission.** An *event*, not a duration: the event that opens an inpatient spell, with a method, a source and a timestamp. "Number of admissions" and "number of spells" are the same count only if nobody was ever admitted twice.
- **Spell.** The continuous stay: one patient's unbroken period as an inpatient of one provider. The NHS Data Model and Dictionary defines a hospital provider spell as the total continuous stay of a patient using a hospital bed on premises controlled by a health care provider, starting when a consultant, nurse or midwife assumes responsibility after a decision to admit — which can be *before* formal admission completes and before the patient reaches a ward — and ending at the provider discharge [5].
- **Consultant episode.** A subdivision of the spell by clinical responsibility: the period a patient spends in the continuous care of one consultant. A patient cannot have concurrent consultant episodes; a transfer within the spell ends one episode and starts another; and where several consultants share care, one takes overriding responsibility and only one episode is recorded [6].
- **Episode of care.** The association of a patient with a provider organisation over a period, for a problem or a programme, spanning many encounters. The FHIR resource of that name exists precisely because the encounter construct could not carry it: a chronic disease programme, an antenatal course of care, an oncology treatment plan [4].
- **Separation.** The event that ends a stay: a discharge, a death or a transfer out, and the denominator of average length of stay.

> **TWO LEGITIMATE DEFINITIONS, AND THE CHOICE IS YOURS**
>
> The spell and consultant episode definitions above are **English**, from the NHS data dictionary. South Africa has no equivalent published national definition of either. They are worth quoting because they are the only precise, citable statements of the constructs available — but an architect who adopts them must say so in the document, or a reader will assume a South African authority that does not exist. Defining the boundary locally is the legitimate alternative, and more work.

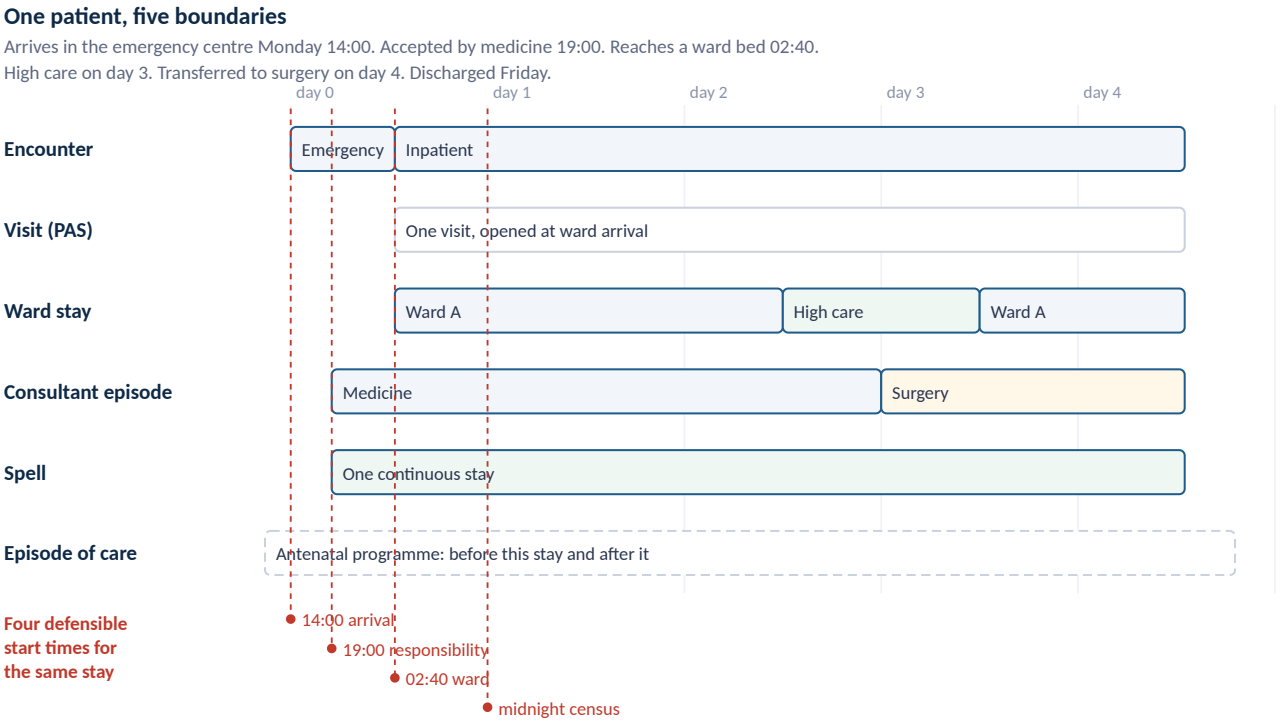


Figure 4.1 One patient's stay, cut five ways. Every bar is correct. The encounter is the timestamped contact; the visit is what the patient administration system opened; ward stays are the location-time facts; consultant episodes subdivide by clinical responsibility; the spell is the continuous stay; and the episode of care extends beyond both ends of it. The four red markers are the four defensible start times for the same inpatient stay, and the choice between them moves this one patient's length of stay by more than twelve hours.

The conflicts, concretely

The boundary in question	What moves	What it costs
Emergency centre to ward	Four defensible start times: arrival, the moment responsibility was assumed, ward arrival, or the following midnight census	Twelve hours and forty minutes on one documented patient; across a busy emergency centre, the difference between a hospital that looks efficient and one that does not
Internal transfer between specialties	Count episodes and an eleven-day stay becomes two admissions of four and seven days; count spells and it is one of eleven	Mean length of stay on episodes is roughly half that on spells wherever internal transfer is common, and both numbers are right
Ward transfer within one specialty	A ward stay closes and opens; the consultant episode and the spell do not	An occupancy report built on ward stays and a length-of-stay report built on spells never reconcile, and the reason is invisible
Leave of absence	Whether home leave interrupts the episode is a rule, not a fact (the English rule exempts 28 days or less)	Decisive in psychiatry and rehabilitation, and South African provincial systems vary
Day case and same-day discharge	Length of stay is zero or one day depending on whether the calculation subtracts dates or adds one	A hospital with 25% day-case activity shifts its mean length of stay by a quarter of a day by changing nothing but the query
Midnight census against continuous timestamps	Occupancy is counted at a fixed hour; length of stay is computed from timestamps	A patient in a bed from 08:00 to 21:00 contributes no occupied bed-day. Theatre recovery, day wards and observation units are invisible in occupancy and visible in cost
The funder's unit	A scheme has no spells or episodes. It has claims grouped by **authorisation**, and an event that extends a stay produces a second authorisation	The funder's count of admissions is a count of authorisations, which matches neither spells nor episodes
Episode of care across providers	A pregnancy is one episode of care and typically four to six providers, each opening its own spell	No South African provider has an episode construct spanning them, so perinatal data is reassembled from mortality audit rather than read off the record

The damage is predictable. Two dashboards both labelled "average length of stay", differing by 30 to 50%, each defended by its owner, neither wrong. A year-on-year improvement attributable entirely to a system upgrade that changed the admission timestamp source. Benchmarking against an external figure whose definition is unstated. A clinical audit that cannot find its patients, because its unit of analysis is the episode of care and the record's unit is the visit.

The remedy is not to pick one. Hold **Encounter** as the timestamped contact and **Ward Stay** as the location-time fact. *Derive* Spell by a documented rule from contiguous occupancy and *derive* Episode of Care by a documented rule from clinical linkage. Compute every measure from the lowest-level timestamped facts rather than from a pre-cut aggregate. The derived boundaries then become views rather than data, and a definitional change is a re-computation rather than a migration. The corollary is the most useful advice in this chapter: **bed-days are additive and boundary-insensitive; averages are not.** Model bed requirement from bed-days, never from separations multiplied by an average.

That remedy needs one more word, and it is the word nobody asks for.

- **Measure Definition.** The versioned, effective-dated, owned rule by which a derived figure was computed: its numerator, denominator, exclusions, boundary rule and the period over which it applies. It is an entity, not an audit field.

The evidence that it must be an entity rather than a convention is published. On theatre utilisation — the best-documented measure in the field, because theatre time is expensive enough that people have argued about it in journals — there is **no simple mathematical relationship between the four calculation methods**, with raw utilisation of 80% and touch-time utilisation of 60% on the same list [23]. There is no conversion, and the same holds for occupancy, length of stay, readmission and mortality. The five theatre timestamps that make the computations possible are routinely collapsed to two [42]. Chapter 2 teaches reading these numbers; this entity makes them traceable.

The money words

Nothing here is more likely to be got wrong by an architect arriving from financial services, because the words look familiar and are not.

Word	What it is	What it is not
Medical scheme	A not-for-profit mutual entity registered under the Medical Schemes Act 131 of 1998, supervised by the Council for Medical Schemes, owned by its members, which holds the money, carries the risk and owns the member relationship in law	An insurer. It may not underwrite on health status, must community-rate within an option, may not refuse membership on health grounds, must cover prescribed minimum benefits in full, and holds statutory reserves
Insurer	A licensed insurance undertaking regulated under insurance law by the Financial Sector Conduct Authority and the Prudential Authority, selling products such as gap cover and hospital cash plans	A scheme. The Council for Medical Schemes has no power over these products
Contribution	What a member pays a medical scheme: a function of benefit option, dependant composition and in some schemes an income band	A risk-rated price. There is no underwriting engine in the actuarial sense and no individual risk score driving the amount
Premium	What the insured pays an insurer for an insurance product	A word to use about a medical scheme. If you write it, name the insurance product it belongs to
Administrator	A company accredited by the Council to perform membership, contribution collection, claims adjudication, payment and reporting on a scheme's behalf; an **operator** under POPIA	The owner of the data. The scheme is the responsible party, so an administrator change is a data migration with a regulatory deadline
Managed care organisation	A separately accredited entity performing managed health care: protocols, pre-authorisation, case management, provider networks, formularies, utilisation review	The scheme, or the administrator, even when one corporate group owns all three
Member and dependant	The funder's subject: scheme, then benefit option, then member, then dependant, with dependant code 00 conventionally the principal member	The patient. It is a separate entity with a lifecycle the provider does not own and must re-verify at every encounter
Benefit option	The priced benefit design within a scheme, whose rules and contribution table require the Council's approval before they change	A product in the retail sense. Network restriction is a pricing input, so the provider network master is rating-relevant data
Tariff	Contract data: the combination of funder, provider, service code and effective date	An attribute of a service. A single price column on a service master is a defect

Two of those rows carry more weight than the rest.

Scheme against insurer is a legal distinction, not a semantic one, and getting it wrong signals an architect who has read insurance material and assumed it transfers. At 31 March 2025 there were 71 registered medical schemes — 16 open and 55 restricted — 33 accredited administrators and 43 accredited managed care organisations [14], and managed care arrangements covered **99,19% of beneficiaries** [16], so pre-authorisation is near-universal in the private sector and has no public-sector equivalent. The data consequence: the **scheme** is the POPIA responsible party and the administrator is an operator, so the claims data belongs to a not-for-profit entity that typically employs nobody beyond the principal officer's office [14].

Tariff is a data-architecture fact rather than a commercial one. There has been no national tariff since the Reference Price List regulations were declared invalid and set aside on 28 July 2010 [39]. What exists is scheme rates, bilateral contracts, group schedules for self-payers, and three statutory tariffs. So the tariff in a South African model is the tuple of payer, provider, service code and effective date. Chapter 6 has the money.

The funder's own model is visible in the claim message, which carries on one message a membership number, the principal member's surname and initials, a benefit option number, a scheme number, a patient account number, the patient's name, a dependant number, the relationship to the principal member, an identity number, a date of birth and a sex marker [11]. Read that as four levels the provider does not own, cannot validate, and must nevertheless store accurately or the claim is rejected. A model using the membership number as a patient identifier loses the patient at every policy change, and merges siblings who share a membership number if the dependant code is dropped.

Claim, and the other claim

The word *claim* is central to this industry in **two unrelated senses**, and the research is unambiguous that they must never share a bare noun.

- **Invoice** (or **account**). The provider's assertion of amounts due for services rendered, directed at whoever is liable. One stay can produce several invoices; one invoice can span several encounters.
- **Claim**, in the funding sense. A submission to a third-party funder requesting payment of part of an invoice under a benefit arrangement. It is not the invoice. In the private sector it travels as an EDIFACT message through a **switch**, an intermediary that validates and routes claims between providers and funders [11][12].
- **Claim line.** The granular request: a code, a quantity, an amount and modifiers. The South African line-item qualifier can carry tariff codes, modifiers, surcharges, non-chargeable codes, laboratory and fixed-fee codes, several clinical and product code systems, a medicine and consumable group, and dental tooth numbers [11]. One line is a small polymorphic object, and a single code column with a type discriminator carries more than ten meanings.
- **Authorisation.** The funder's prior approval of an admission, an expected length of stay and often a procedure. It is the funder's unit of counting.

- **Remittance** (or **payment advice**). The funder's statement of what it has paid, declined or adjusted, line by line, with reason codes. It is the entity organisations most often fail to model, and the South African standard is the reason: the claim message specification **covers the claim side only**, defining no remittance message and no link from a claim response to a payment [11]. So remittance arrives in whatever format each funder chooses, which is why denial-rate analysis is hard here.
- **Claim**, in the medico-legal sense. A demand for damages against a provider or a provincial department arising from alleged negligent care: an unrelated object with an unrelated lifecycle, counterparty and evidential standard, and the one the records duty answers to. Chapter 13 is built on it.

> NEVER USE THE BARE WORD
>
> "Claims data", "claims volume" and "claims management" are ambiguous here, and will be read in both senses by different people in the same workshop. Write **funder claim** and **medico-legal claim**. The extra word costs nothing; the ambiguity costs a programme scoped against the wrong counterparty.

Medication: four entities, not one

The reason to insist on four is that a medication safety incident is almost always a mismatch between two of them, and a model that holds one cannot see a mismatch.

- **Medication order** (prescription). A prescriber's instruction that a patient should receive a medicine, with a dose, a route, a frequency, a duration and an indication. The dose expression is a structured thing masquerading as text, and it is where dosing errors live. The transition that breaks models is **stopped**: a medicine stopped on day three of a seven-day course has an active order and a stop event, and a model holding only a start date and a duration reports seven days of exposure.
- **Dispense.** A pharmacy event in which a quantity of a specific product is issued against an order, to a patient or to a ward. The key difference from the order is that a dispense is against a **product**, not an ingredient: the order says a drug and a strength, the dispense says a specific manufacturer's pack with a lot number and an expiry date. Those two attributes are what every product recall depends on, and they are the ones most often dropped on the way into a warehouse because they look like logistics data.
- **Administration.** The event in which a dose was actually given, or deliberately not given. It is the entity most often omitted entirely and the only one that answers the clinical question. Its most valuable attribute is **the reason not given** — refused, absent, nil by mouth, unavailable, held on clinical instruction — which makes stock-outs visible and which a paper chart records as a blank box.
- **Medication reconciliation.** A point-in-time assertion of what the patient is actually taking, compiled by a practitioner from the patient's account, the previous record, the dispensing history and whatever containers the patient brought in. It is **not** a view over the orders. It is an independent clinical assertion with an author, a date, a set of sources consulted and a set of **discrepancies** — and the discrepancies are the clinically valuable output.

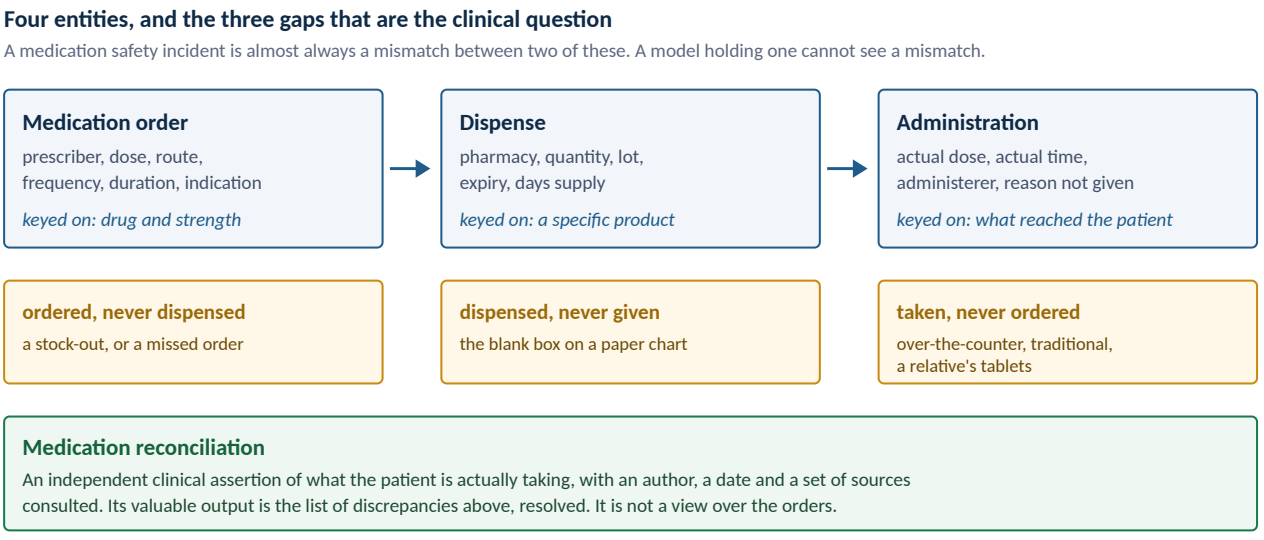


Figure 4.2 The four medication entities, keyed differently and owned differently. The order is keyed on a drug and a strength; the dispense on a specific product with a lot number; the administration on what actually reached the patient. The three amber gaps are what a single medication table with a status column cannot represent, and they are the only clinically interesting cases.

Without an electronic medicines administration record the administration entity has no owner and no data, and the organisation cannot answer "was the antibiotic given within an hour of the sepsis alert?" Where public-sector pharmacy runs a stock system rather than patient-level dispensing there is no dispense record at all, and the only evidence a medicine was issued is a stock movement: the national Stock Visibility System is supply chain, not a medication record [33].

The people words

Word	What it is	What it is not
Practitioner	A person authorised to deliver health care in a defined scope	The registration. A practitioner may hold more than one, and a registration identifies the authorisation rather than the person
Registration	The authorisation itself, with its own identity and lifecycle: register, category, number, first registration date, status, conditions and endorsements, continuing education compliance, scope of practice	A static attribute. Status is live: a lapsed practitioner may not practise, their orders should not be actionable and their claims should not be payable
Practitioner role	The practitioner at one organisation, in one specialty, at one location, for a period, with a set of privileges	The practitioner. One practitioner has many roles, and one clinical event can carry up to six roles that may be the same person
Practice code number	The identifier that routes the money. It names a **billing entity**, which may be an individual, a group practice, a hospital or a laboratory	A clinician identifier. Conflating the two produces reports in which a group practice performed four thousand procedures and no individual performed any
Health care provider	The National Health Act's term for a **person** who provides health care	A place or an organisation. "Provider" unqualified covers a person, a practice and an organisation, and should never be used where one of the three is meant
Health establishment	The Act's term for the **place**: the regulated unit that holds a licence or certificate, owes the records duty and may be inspected	A ward, a service or a legal entity
User	The Act's word for the patient: the subject of its consent, records, confidentiality and complaints provisions	A system user. Say "patient" or "system user" and never the Act's word for the other one
Organisation	The legal entity: holds licences, is party to contracts, holds the practice code, is the responsible party under POPIA, and is what an accreditation applies to	A site, a hospital or a department

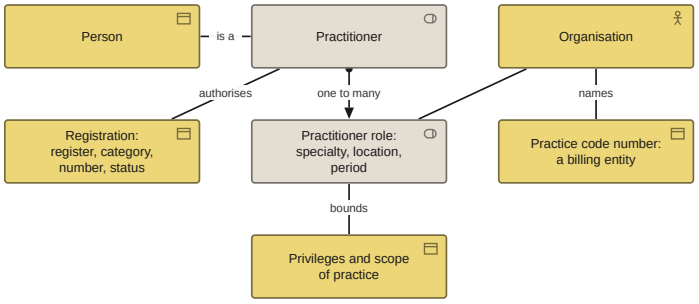


Figure 4.3 Practitioner, registration and practitioner role are three entities, and the practice code number is a fourth that names a billing entity rather than a clinician. One clinical event can carry up to six roles — specialist, anaesthetist, assistant, referring, treating, admitting — some of which may be the same person. A single doctor table with a name, a specialty and a code can represent none of this.

The registration identifiers are real and usable. The statutory councils publish verification registers — the South African Nursing Council's online register being the clearest example [18] — and the registration number is the best available practitioner identifier in the country. Scope of practice is data, not policy, because it is what a prescribing or ordering authorisation check must consult, and the nursing council's own regulation framework defines what a given category may do [19].

Two honest limits. The governing document on practice code numbering, discipline codes and place-of-service codes could not be retrieved for this volume, so the account above is descriptive rather than sourced to its own standard; obtain the Council for Medical Schemes standardisation manual before building to it. And **there is no single plausible owner of Practitioner**: human resources owns the employment relationship, credentialling owns registration verification, the clinical directorate owns privileges, billing owns practice codes. Four owners, who disagree, which is why a practitioner who has left is still an active prescriber in one system and an active biller in another. The deliverable is the governance decision, not the answer.

The place words

Four things, routinely collapsed into one, and the collapse is why nobody can count beds.

- **Care setting** (or **service**). The *kind* of care delivered: emergency, acute inpatient, day case, outpatient clinic, operating theatre, intensive care, high care, maternity, psychiatric inpatient, rehabilitation, community, home-based, virtual. It is a classification of service, **not a place**. Including "virtual" in a concept defined as a place is exactly where the collapse hurts: a teleconsultation has a setting and either no physical location or two.
- **Location.** The physical place, hierarchically from site to building to department to room to bed, with addresses, capacities, physical characteristics — negative pressure, piped oxygen, radiation shielding — and an operational status.
- **Ward.** An *organisational unit* with a nursing establishment, a bed complement and a specialty, which occupies a location but is not one. Wards are reconfigured without moving: a surgical ward becomes a medical ward overnight in a winter surge.
- **Bed.** The unit of capacity, and not a physical object. A bed in the occupancy sense is a **funded, staffed capacity unit**, which is why a hospital can have more physical beds than beds.

That last definition resolves an argument you will certainly have. There are at least four bed counts — **physical**, **commissioned**, **staffed or available**, and **occupied** — and a fifth in the private sector, the **licensed** count a provincial licence authorises for a site. Which one a report means is the commonest

cause of occupancy figures nobody can reproduce. Commissioning, closure, reconfiguration and decommissioning are all *dated* events, so bed complement is a time series and not a number.

A South African wrinkle on the **visit**: the Uniform Patient Fee Schedule assesses a subsidised outpatient attending several clinics on one day **once, at the most expensive clinic**, while full-paying and higher-band patients are assessed per visit per clinic [43]. One attendance is one charge or three depending on a means-test band, so a visit model assuming one encounter per attendance will mis-bill.

The movement words

- **Referral.** A request from one practitioner or service to another to assume or share care, carrying the referrer, the intended recipient — which may be a service rather than a named person — the reason, a clinical summary, an urgency and supporting documents. **Its auditable states are `sent` and `acknowledged`** : a referral sent and never acknowledged is a patient lost to follow-up, a safety incident with a data cause. In South Africa it is the entity on which the whole public referral pathway rests, carried by a letter rather than a message [1].
- **Appointment.** A booked future encounter. It is *not* an encounter: it may become one, be cancelled, or be not attended. The relationship is zero-or-one to zero-or-one and both zeros matter — the appointment with no encounter is the did-not-attend, the encounter with no appointment is the walk-in. Deleting cancelled appointments destroys the denominator for non-attendance, utilisation and cancellation analysis. No published South African did-not-attend rate could be found for this volume, partly because much public outpatient attendance is not appointment-based.
- **Triage.** The act of assigning a clinical priority at the point of unscheduled presentation. In South Africa the instrument is the **South African Triage Scale**: five categories derived from a composite physiological score, with separate adult and child discriminator lists that escalate a patient regardless of the score [27]. Three properties make it architecturally important. It is a **deterministic, documented, scored algorithm with override rules**, so it can be implemented faithfully or not at all. **Re-triage is mandatory and continuous**, so a triage record is a *series*, not an event, and the clock for time-to-treat runs from the most recent category rather than the first. And its published performance thresholds — 95%, 80%, 75% and 70% for the four treated categories — are the only sourced, published, numeric service-level targets for any South African clinical process found in this research [28]. Two cautions: the practice guideline dates from March 2010 with a review due in February 2012, so **obtain current versions from the Emergency Medicine Society of South Africa before implementing it**; and the manual is internally inconsistent about the scoring range per vital sign [27]. Triage is also not chargeable under the public fee schedule [43], so the first clinical record of an emergency visit has no billing event behind it.
- **Pathway.** The *instantiation* for one patient of a standardised route through care: where the patient is, which steps are done, which are overdue, and which were deviated from. **Deviation with a documented reason is the clinically and legally important fact**, not an exception to suppress.
- **Protocol.** An organisation-adopted, evidence-based specification of how a defined clinical situation should be managed. Not patient-specific. It is identified by an identifier **plus a version** — version is part of the identity, not an audit field — because "was this patient managed according to protocol?" is meaningless without knowing which version was in force on the date of care. The national source objects are citable and dated: the Primary Healthcare Standard Treatment Guidelines and Essential Medicines List, eighth edition, 2024 [29], and the National Integrated Maternal and Perinatal Care Guideline signed on 4 October 2024 [30]. A provider's protocol register should carry the national version it derives from *and* the **local modification** made to it, because a hospital that has modified a national guideline has taken on the clinical risk of the modification.
- **Care plan.** A patient-specific, forward-looking statement of intended care: goals with targets and target dates, planned activities with statuses and scheduled times, responsible parties, the conditions it addresses, and a review date. A patient has one care plan assembled from several pathways — a diabetic with hypertension and a pregnancy is on three. The planned activity and the actual event are **different objects**, linked, which is the only way to see what was planned and not done.

The coding words

This is the distinction that governs the most downstream work, and the one most often got wrong.

- **Terminology.** A code system whose purpose is to let a clinician record what was observed or done, as precisely as they can express it, with a defined and computable meaning: poly-hierarchical, formally defined, built for *capture* and *inference*. SNOMED CT is the canonical example.
- **Classification.** A code system whose purpose is to sort every case into exactly one mutually exclusive, exhaustive category so that cases can be counted. Mono-hierarchical by design, with residual categories to achieve exhaustiveness, organised around statistical reporting needs. ICD is the canonical example.
- **Code list** (or **identifier scheme**). A scheme that names a specific thing unambiguously: a medicine product, a billing provider, an observation. LOINC names the *question* an observation answers, distinguishing terms by several axes so that two terms differing on any one of them are different terms [22].
- **Payable-unit nomenclature.** A code set that names a payable unit of work, which is not the same object as a clinical act. One procedure generates several payable lines — surgeon, assistant, anaesthetist, theatre time, consumables, prosthesis — and one line can cover several procedures.
- **Post-coordination.** The assembly of a meaning from several terminology concepts using a defined grammar, so that a meaning with no single pre-existing code can still be expressed. Two different expressions can then mean the same thing and only a logic classifier can tell; querying requires subsumption testing rather than a list of values; and somebody must decide what expressions are allowed. A field sized for one code cannot hold a post-coordinated expression.
- **Coded against recorded.** The recorded text is the clinician's *assertion*. The code is a classification of it, made later, by someone else, for a purpose that is usually not clinical. A model storing only the code has thrown away the assertion; a model storing only the text cannot aggregate. Hold both, linked, with the coder, the coding date, the classification version and the coding purpose.
- **Present on admission.** The flag stating whether a condition existed at admission or arose during the stay. It is the only thing that distinguishes a hospital-acquired complication from a reason for admission. **Neither the South African public morbidity coding standards nor the private claim message mandates it** [7][11], so hospital-acquired condition rates are not computable from routine South African data. That is a finding rather than a gap in the research, and chapter 12 says what to do about it.

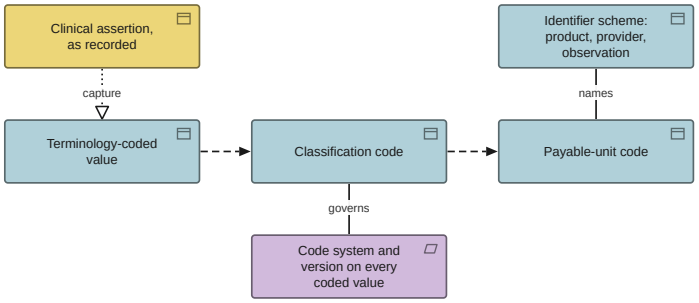


Figure 4.4 Capture in a terminology, aggregate through a classification by governed mapping, bill through a payable-unit nomenclature, and name specific things with identifier schemes. The mapping from terminology to classification is one-way, many-to-one and lossy by design; the reverse direction invents precision it never had. Every coded value carries its code system and version, because a code's meaning is a function of the release in force on the date of care.

The correct architecture follows from the definitions: **capture in a terminology, aggregate through a classification by governed mapping, bill through a payable-unit nomenclature, and keep all three with their mappings under version control** [32]. The mapping from terminology to classification is one-way, many-to-one, lossy by design and carries rule conditions rather than single targets; the reverse direction loses nothing because there was nothing to lose — it *invents* precision.

The classic error is using a classification as a terminology, and its symptom is a clinical system whose problem list is an ICD-10 picker. The South African evidence on what that costs is unusually good. In 223 analysed discharge records from internal medicine wards at a Western Cape central hospital, where clinicians coded their own diagnoses, **complete code sets appeared in 45,3%** and **accurate primary codes in 74,0%** — and each additional comorbidity *reduced* the odds of complete coding [24]. The sicker the patient, the less completely the record was coded, so any analysis of outcomes by comorbidity burden built on that data is measuring documentation effort. The follow-up study is sharper still: a training and supervision package moved complete code sets from 27,1% to 68,1% at the intervention site against 22,5% to 25,5% at the control, while primary-code accuracy moved from about 71% to 79,1% and was **not statistically significant** [25]. Completeness is an effort problem; accuracy is a competence and classification problem.

One further South African definition belongs here. The **main condition** is the condition diagnosed at the end of the episode of care that is primarily responsible for the patient's need for treatment or investigation, with a tie-break on clinical severity, then resource consumption, then order of recording; **other conditions** are additional conditions affecting care [7]. Note what that does to the model: *primary* is not a property of a diagnosis, it is a property of a diagnosis **within an episode**, assigned at the end of that episode by a rule. The same condition is primary in one admission and secondary in the next.

TWO THINGS TO RE-CHECK BEFORE YOU RELY ON THEM

The terminology position has moved and is not finished. South Africa joined SNOMED International on or about 14 May 2026 [20][21], which changes the licensing economics, because a member licence covers use within the member country. But **the National Release Centre is not yet operating**: no national extension, no national reference sets, no published national map, representatives listed as to be advised [20]. Nothing is mandated. Be ready rather than early — store the code system and version on every coded value, keep the clinical assertion separate from the classification, and build the terminology service as a service rather than as tables in an application database. **Re-check the status of the National Release Centre before designing to it.**

The coding instruments may have moved too. The national morbidity coding standards retrieved for this volume are version 6 of April 2014, which states that it is revised annually; no later version was found, while the industry body's own addendum is at version 7 of October 2024 [7][13]. The Master Industry Table referenced is the January 2014 edition on a stated biennial cycle, and no later edition was located [9][10]. Confirm both with the Department of Health before quoting either as current.

Four more words the later chapters need

- **Person against Patient. Person** is the human being. **Patient** is a *role* a person occupies in relation to a provider organisation. The distinction is not pedantry: it is what lets a model represent a person with two folder numbers and one body, a person who is both patient and staff member, a neonate whose first identifiers derive from the mother's, and the unidentified trauma case who must be registered, treated, imaged and billed before anyone knows who they are. "Deceased" is a state of the **Person**, not of the Patient, which is why a model that puts it on Patient has a patient who dies at one facility and is alive at the next. A deceased person is also no longer a data subject under POPIA, which protects living natural persons, while their record still exists and is still legally regulated — an unresolved question in the South African literature rather than a settled rule [34]. Chapter 11 owns identity; this is the definition it starts from.
- **Specimen.** The sampled material, identified by a specimen or container identifier, with a type, a collection date-time, a collector, a collection site, a condition on receipt and a chain of custody. It is an entity rather than an attribute because one order may require several specimens and one specimen may serve several orders. It is also what makes mislabelled samples visible — and a mislabelled sample is an identity failure, which is why specimen labelling and patient matching are the same problem seen from two ends.
- **Diagnostic order against diagnostic result.** The **order** is a clinician's request, identified by an order identifier at the placer *and* an accession number assigned by the laboratory on receipt — two identifiers owned by two organisations, and the mapping between them is where results get lost. The **result** is one or more observations plus an interpretation plus a report, and the report and the observations have different lifecycles: observations are corrected individually, reports are reissued as versions. Collapsing the two makes the unordered result — a reflex test, an add-on, an external referral — impossible to represent, and makes turnaround time unmeasurable because there is only one timestamp. And an observation carries at least four times, of which the one most often missing is *when it became visible to a clinician*: during the 2024 national laboratory outage results were delivered by telephone, message, email, fax, courier and paper, every route producing an observation with a correct effective time and an unrecorded available time [26].
- **Consent.** At least three entities, not one. **Consent to treatment** is a clinical-legal act about an intervention, and withdrawal stops the intervention. **Consent to disclosure** is a data-protection act about information, recipients and purposes, and withdrawal stops *future* disclosure but cannot unsend what was sent. **Consent to research** is a third object with its own ethics approval reference, scope and withdrawal semantics. Two South African constraints must be modelled rather than assumed: section 14 of the National Health Act permits third-party disclosure only with the user's consent **in writing** [1], so a verbal consent does not satisfy it; and under section 129 of the Children's Act a child over 12 with sufficient maturity and capacity may consent to their own treatment [17], so consent capacity is a function of age *and* an assessed maturity, and the assessment is itself a clinical act that must be recorded. Chapters 13 and 18 carry the rest.

THE FOUR WORDS FOR THE RECORD, IN ONE PLACE

Patient administration system: the system of record for the administrative identity of the patient and for the encounter — arrival, ward, responsible team, discharge. It owns the institutional patient index, the bed state, the waiting list and the clinic diary, and no clinical content. **Electronic medical record** or **electronic patient record**: the clinical record held by *one* provider organisation. **Electronic health record**, used strictly: a longitudinal, cross-provider record of a person assembled from many organisations — a different architectural animal, with no single owner, a consent model and an identity-resolution problem an organisational record does not have. **The clinical record as a legal construct**: what the provider must retain, can be compelled to produce, and must be able to reconstruct as it stood at a point in time — not identical to any one system's contents and not identical to a screen [2][35]. Accepting a requirements document that says "we need an EHR" without forcing one of these four is how scope is lost in week one. Chapter 13 owns the legal record.

A closing discipline. Where two legitimate definitions exist — spell against episode, or the national aggregate reporting assessment's two completeness figures, 99,5% for facility reporting against 49,7% for data element reporting [37] — state both and name the choice rather than averaging or picking quietly. The alternative is a repository in which every number is defensible and no two agree.

THE CHAPTER IN TEN LINES

- These words are not vague; they are precise and incompatible. A glossary that picks one definition and calls it the standard has destroyed another one that somebody's statutory return depends on.
- Encounter is the timestamped contact. Visit is one attendance at one place on one day, the unit of registration. Ward stay is the location-time fact. Admission is an event, not a duration.
- Spell is the continuous stay; consultant episode subdivides it by clinical responsibility; episode of care spans many encounters for one problem or programme. The spell and episode definitions in circulation are English, and South Africa has no national equivalent.
- Hold encounter and ward stay as facts and derive spell and episode by a documented, versioned rule. Bed-days are additive and boundary-insensitive; averages are not.
- Measure Definition is an entity: the versioned rule by which a derived figure was computed. There is no conversion between theatre utilisation methods, and the same is true of occupancy, length of stay, readmission and mortality.
- A medical scheme is not an insurer. A member pays a contribution, never a premium. The scheme is the responsible party under POPIA and the administrator is an operator.
- Tariff is contract data: funder, provider, service code, effective date. A single price column on a service master is a defect.
- Claim has two unrelated senses, a funder claim and a medico-legal claim. Never use the bare word. Invoice, claim, claim line, authorisation and remittance are five entities, and the South African claim standard defines no remittance message.
- Medication is four entities: order, dispense, administration and reconciliation. A safety incident is a mismatch between two of them, and only the administration record answers whether the drug was given.
- Practitioner, registration and practitioner role are three entities, and the practice code number names a billing entity rather than a clinician. Practitioner has four candidate owners and no single plausible one.
- Care setting is a kind of care, not a place. Ward is an organisational unit, location is the place, and a bed is a funded staffed capacity unit, which is why there are at least four bed counts.
- Referral is auditable through its sent and acknowledged states. Triage is a series, not an event. Protocol is identified by identifier plus version; pathway is its instantiation for one patient; care plan is assembled from several pathways.
- Terminology captures, classification counts, a code list names a thing, and a payable-unit nomenclature names a unit of work. Capture in the first, aggregate through the second by governed mapping, bill through the fourth, and version all of them.
- Present on admission is not available in South African routine data, so hospital-acquired condition rates are not computable from it.

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CHAPTER 5

Who does what

An organogram will not tell you who may do what inside a health establishment. Registration, a licence and an accreditation will, and none of the three follows the reporting lines.

An architect arriving in a new industry asks for the organogram. In healthcare the organogram is close to useless as a guide to authority, because what a person may do is not set by their place in a reporting line. It is set by their registration with a statutory council, the scope that registration carries, and — for the establishment they work in — a licence and a certificate issued by bodies that appear nowhere on the chart.

This chapter is the cast list. Confusing two of these roles produces a predictable class of defect: an access-control model that cannot express the rule it approximates, an entity mastered in the function that bought the first system, a data-ownership line drawn to someone with no authority over the data. The last section is the hardest version of that problem — the entities for which there is no single plausible owner, where asserting one teaches a falsehood.

Two invented organisations carry the worked examples through this volume. **Driehoek Health Group** is a mid-size South African private hospital group: nine acute hospitals, two day clinics, about 1 750 registered beds, a shared pathology contract and an in-house pharmacy function. **Rietvlei District Health Service** is one public health district: a district hospital, three community health centres, 22 fixed clinics and a mobile service. Both are fictional; every other organisation named in this book is real.

Clinicians are defined by registration, not by job title

A clinician in South Africa is a person on a register. Medical practitioners, dentists, psychologists, physiotherapists, radiographers, emergency care practitioners and the rest are registered with the **Health Professions Council of South Africa** under the **Health Professions Act 56 of 1974**, which operates through twelve professional boards [1]. Nurses and midwives are registered with the **South African Nursing Council** under the **Nursing Act 33 of 2005**; pharmacists, pharmacy support personnel and pharmacy premises with the **South African Pharmacy Council** [1]. Registration is a precondition of lawful practice and, in the public service, of employment.

Registration is not a flag on a person record. It is an object with its own identity and lifecycle: a council, a register, a category, a number, a first-registration date, a status — registered, suspended, erased, lapsed, restored — conditions and endorsements, fee currency, continuing professional development compliance, and a scope. The status is live and consequential: a practitioner whose registration has lapsed may not practise, their orders should not be actionable and their claims should not be payable, so a system that caches registration status without an expiry and a refresh has built a compliance defect. Both councils publish an online register for verifying an individual; SANC's is the eRegister [2].

Scope of profession and scope of practice are different things

Two concepts, routinely conflated, and the distinction matters for authorisation design [3]:

- **Scope of profession** is the broad field a profession covers, defined under **section 33 of the Health Professions Act** and applying to every profession registrable with the HPCSA.
- **Scope of practice** is the boundary of an **individual** practitioner's work, defined under **ethical rule 21** of the HPCSA's ethical rules, with additional scope-of-practice regulations for some professions under their own board's rules.

Rule 21 provides that a practitioner may perform, except in an emergency, only a professional act "for which he or she is adequately educated, trained and sufficiently experienced", performed "under proper conditions and in appropriate surroundings". New interventions require further board-approved training, and short courses count only towards skills in a field in which the practitioner is already registered. In an emergency involving a threat to life or limb, where no appropriately trained professional is available, the practitioner **must** intervene to the best of their ability. And a practitioner **cannot be compelled by an employer** to perform acts outside their scope [3].

Read that against a role-based access-control matrix and the mismatch is plain. Scope of practice is an attribute of a named individual, graded by training and experience, and it varies between two people holding the same job title. A clinical system that grants prescribing or order-entry rights by job title is approximating a legal rule it cannot express. The workable compromise is three layers: **role** for the coarse grant, **credential** for the registration check, and **privilege** for the individual permissions a credentialling committee has granted at this establishment, which is what those committees already do by hand.

The emergency clause has its own consequence. Rule 21 requires intervention beyond the usual boundary in a life-threatening emergency, and **section 5 of the National Health Act 61 of 2003** separately forbids any provider, worker or establishment from refusing emergency medical treatment, with no payer precondition [4]. Emergency override is legally mandated, not an exception to be designed out, so the pattern is break-glass with audit. A system that hard blocks will be bypassed with a shared password, and the audit trail will then be worthless.

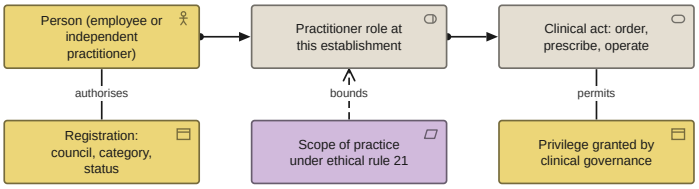


Figure 5.1 Authority over a clinical act is assembled from three separate objects. A job title carries none of them, which is why role-based access control alone cannot express scope of practice.

The mid-level cadre is where job-title authorisation breaks completely

Clinical associates are a South African mid-level medical cadre, registered with the HPCSA and trained to work in district hospitals under a medical practitioner's supervision [5]. Their permitted acts overlap a doctor's in some areas and not others, and the overlap is defined by regulation rather than by organisational convenience. For Rietvlei's district hospital that is not an edge case but a staffing model, and any authorisation design that cannot express "may do A and B but not C, under supervision by a person of category D" will be worked around within a month of go-live.

One clinician, at least three identifiers

The council registration number is the best practitioner identifier in the country, and a practitioner **may hold more than one**. In the private sector the identifier that routes money is different again: the **practice code number** issued for billing, which providers call the BHF number. It identifies a *billing entity* — an individual, a group practice, a hospital or a laboratory — and conflating it with the practitioner is the standard way to produce a report in which a group practice performed four thousand procedures and no individual performed any. Add the public-sector employee number from **PERSAL** and one clinician carries three identifiers from three unrelated authorities for three unrelated purposes. Chapter 11 takes identity further.

The councils keep registers, and a register is not a workforce

The councils' aggregate numbers are quoted constantly in South African health debate and mean less than they appear to.

For the HPCSA, current registration statistics by board and category could not be obtained for this book. The one secondary figure that exists — **72 207 registered professionals on the Medical and Dental Board**, year not stated [6] — should not be used: it appears to be a board-level total, not a count of doctors. The HPCSA's own annual report would close the gap.

For SANC the problem is interpretation, and the Council says so itself. SANC states plainly that its published figures count **people registered with SANC, not people employed in nursing** — including job seekers and registered people not working in nursing, across both sectors [7]. A peer-reviewed analysis adds that the registers cannot separate practising nurses from retirees or emigrants [8].

The consequence is worth stating bluntly, because the opposite is asserted in public almost weekly. On SANC register data the South African nurse-to-population ratio is **47 per 10 000 against a WHO benchmark of 25 per 10 000** — and the same paper's own literature review cites 52 per 10 000 from the same underlying source, an inconsistency the authors acknowledge. Their conclusion is the useful one: the ratio is above the WHO standard, but "these figures do not accurately reflect the nursing situation in the country" [8]. **A nurse-to-population ratio describes a register, not a workforce, and cannot be used for capacity planning.**

> Use the councils' registers for what they are good for: verifying that a named individual is currently registered, in which category, with what conditions. Never as a workforce measure.

What *is* useful from SANC for architecture is the scope framework: **Regulation R.195** defines what a given nursing category may do [9].

Nursing management, and the shape of the workforce

Nursing is the workforce. The public-sector categories run from auxiliary nurses through enrolled nurses to professional nurses, with primary health care and other nurse specialists above them and a separate line for nurse educators. The national nursing workforce report gives the post position from **PERSAL, June 2023** [10]; the vacancy rates below are calculated from the report's own filled and vacant figures.

Category	Filled	Vacant	Total posts	Vacancy rate
Auxiliary nurses	34 028	2 351	36 379	6,5%
Enrolled nurses	29 281	2 717	31 998	8,5%
Professional nurses (general)	48 096	4 315	52 411	8,2%
Primary health care nurse specialists	10 191	1 900	12 091	15,7%
Other nurse specialists	15 318	2 790	18 108	15,4%
Community service nurses	2 425	767	3 192	24,0%
Nurse educators	969	323	1 292	25,0%
Managers: nursing practice	2 364	391	2 755	14,2%
Managers: nursing education	226	71	297	23,9%

Two rows matter more than the rest. **The highest vacancy rates in the table are nurse educators at 25,0% and managers of nursing education at 23,9%.** The posts that produce the next generation of nurses are the emptiest in the system — a statement about the next decade rather than a staffing inconvenience.

The two management rows are the nursing management layer an architect will meet. At an establishment that layer is a nursing service manager or chief nursing officer, with operational managers running wards and units. They own the roster, the duty register, the local escalation and — in practice — the ward's bed complement, which is why they reappear at the end of this chapter.

That report is a teaching example in its own right. Its nurse-specialist total appears as 17 941 in one table, 15 318 as PERSAL filled and 18 108 as total posts, and one age table labelled "professional nurses" totals 102 669, matching nothing else in the document [10]. **This is the national nursing workforce planning document, and its internal figures do not reconcile.**

And there is no norm to implement. The same report recommends that the Department "fast-track development of staffing norms for hospitals" and **names no method** [10]. There is no South African safe-staffing standard, and an architect cannot implement one that does not exist. The deliverable is instrumentation: actual staffing against actual activity, at shift granularity, so the client can establish and defend its own baseline.

A last constraint, documented rather than inferred. A Western Cape study of patient safety incident reporting found that austerity measures had left **quality assurance manager posts unfilled**, pushing the load onto other managers [11]. The posts that operate the quality system are the posts that get cut, and an architecture whose operating model depends on a quality assurance manager depends on a post that may not be filled.

In a private hospital, the doctors are not employees

This is the structural fact about South African private healthcare that most misleads an architect arriving from another industry. **The hospital does not employ the specialists.** They hold **admitting rights** at one or more hospitals and bill the funder or the patient separately from the hospital. Netcare reported **net 117 new specialists granted admission rights** in the year to 30 September 2025 [12]; Life Healthcare planned **about 140 new specialists** for its 2026 financial year [13]. Those are statements about a relationship, not a payroll. The consequences are severe and routinely underestimated:

- The hospital's clinical record and the specialist's practice record are **different records, in different systems, owned by different legal entities**, and the rooms may run any of dozens of practice management packages.
- A single inpatient episode generates **multiple independent claims to the funder** — hospital, surgeon, anaesthetist, pathology, radiology, sometimes physiotherapy and a prosthesis supplier — each with its own practice code number, tariff basis and submission route. Chapter 6 follows that money.
- **Orders** placed into a hospital system by someone who is not the hospital's employee raise authentication, delegation and liability questions an employed-clinician model does not.
- The hospital **cannot mandate** the specialist's clinical system, so interoperability at that seam is a commercial negotiation per specialty rather than an IT decision.

So Driehoek can specify its own record, identifiers and integration standards for everything it owns, and must negotiate for everything a specialist holds. A target architecture that assumes otherwise will be rejected in the first clinician workshop.

The clinical directorate, and the one entity with an unambiguous owner

Clinical governance in a hospital of any size runs through a clinical directorate or its local equivalent: clinical heads of department, a clinical governance committee, morbidity and mortality review, protocol adoption and — critically — privileging. The clinical governance function decides which procedures a named practitioner may perform at this establishment, on that practitioner's training and demonstrated competence. It is doing by committee what no access-control matrix can express, and its decisions are what a credential-plus-privilege model has to carry.

The clinical directorate also owns the **clinical protocol** — the one entity in a healthcare data model whose owner is genuinely unambiguous, and which is nevertheless almost always unmastered, living as PDF files on an intranet with the version in the filename and the approval in an email. A protocol's identity includes its **version**, not as an audit field but as part of the key: "was this patient managed according to protocol?" is meaningless without knowing which version was in force on the date of care, and "why did outcomes change in April?" is frequently answered by "the protocol changed in April".

The clinical coder, and why there are not enough of them

Clinical coding is a profession in countries that treat morbidity data as an asset. South Africa treats it as a billing function privately and a clinician's afterthought publicly, and the data quality follows.

A clinical coder reads the record, identifies the conditions and procedures documented, applies the classification's sequencing and tie-break rules, **queries the clinician where the record is ambiguous**, and then assigns and sequences the codes. The query is what separates coding from data entry, and it requires standing to ask a consultant a question. In the private sector coding happens at or near billing — practice staff, a hospital case manager, or a dedicated coder in the larger groups. In the public sector it is done by clinicians, usually junior ones, at discharge.

The strongest South African evidence is a pair of Western Cape studies. The first, in a central hospital's internal medicine wards, found clinicians coding their own inpatient diagnoses rather than using dedicated coders as the private sector does: of the 33 clinicians involved, **54,6% were interns, 30,3% registrars and 15,2% medical officers**. Complete code sets appeared in **45,3%** of records, accurate primary codes in **74,0%**, and **75,4% of 875 diagnostic clinical concepts** were coded at all [14]. Two findings deserve lifting out:

- **The odds of complete coding fell with each additional comorbidity (odds ratio 0,6).** The sicker and more complex the patient, the worse the coding. Comorbidity-adjusted analysis on this data is therefore systematically biased: the patients whose comorbidities most need adjusting for are the ones whose comorbidities are least likely to be recorded.
- **Paying patients had lower odds of accurate primary coding (adjusted odds ratio 0,4; p = 0,03).** In a public hospital, the patients for whom a claim would be generated were *less* accurately coded. The authors do not explain the mechanism, and the finding stands as a caution against assuming a payment incentive improves coding accuracy.

The second study tested an intervention — orientation, on-site coding training, senior review of juniors' discharge summaries, an online course and **an on-site case manager**. Completeness moved from **27,1% to 68,1%** at the intervention site, a difference-in-differences of 38,0 percentage points, adjusted odds ratio 6,6 (95% CI 3,5–16,2). Accuracy of the primary code moved from about 71% to **79,1%**, a difference-in-differences of 6,6 points, adjusted odds ratio 1,9 (95% CI 0,97–3,6), **not statistically significant** [15].

That pair gives a rule worth carrying: **completeness responds to process and supervision; accuracy responds only to expertise.** Mandatory fields, senior review and a case manager fix completeness. Nothing but a trained coder fixes accuracy. So an organisation that needs accurate coded data must employ coders, and one that will not employ coders should not build analytics that depend on coding accuracy.

A clinical coding qualification is registered on the South African qualifications framework [16] — the precondition for a labour market. No statute requires anyone to employ a qualified coder, which is why there is not much of a market. Three consequences:

1. **Coding latency is a data-availability constraint.** A case-mix report cannot be produced for a month that has not been coded, so an organisation with a coding backlog has an analytics backlog whether it knows it or not.
2. **Coding is a single point of failure for every downstream measure.** Case mix, risk adjustment, complication rates, comorbidity indices and prescribed minimum benefit identification all key off coded diagnoses.
3. **The clinician-as-coder arrangement puts classification rules inside the clinical workflow**, competing with clinical work for attention. A design defect, not a training defect.

Chapter 12 covers the instruments; what belongs here is the person.

The people who look after the record

Records management is a distinct function with its own staff, and the public-sector evidence on it is the most quotable material in this book. The Auditor-General audited 27 facilities across nine provinces and found **15 reporting missing or lost records, 17 not tracking records after issue, 25 with inadequate storage and 18 with too few records-management staff**. In **three of nine paid claims reviewed, departments conceded because they could not locate the medical record** [17].

Note where the statutory duty sits. **Section 13 of the National Health Act** requires the person in charge of a health establishment to ensure a prescribed health record is created and maintained for each user, and **section 17(1)** requires that person to put controls in place against unauthorised access [4]. The duty is on the person in charge of the establishment, not on the clinician. In provincial hospitals, HPCSA guidance records that records fall under the **clinical manager's** care and control, with access subject to the Promotion of Access to Information Act and to conditions approved by the superintendent [18].

The records function is therefore a management accountability, and asking **who holds it, by name** is a good first-week question. Chapter 13 treats the record as an object. What matters here is that the person accountable for the record's existence and the person accountable for its lawful processing are usually two different people, and the next section is about the second.

The information officer, who is the chief executive unless someone says otherwise

POPIA's accountability condition requires a responsible party to have an **Information Officer**, and the Act makes that the head of the organisation unless the function is delegated. **Deputy information officers** must be designated, and both the Information Officer and the deputies must be **registered with the Information Regulator** [19].

This is not a paper formality. The Regulator's first enforcement notice of 2026, against Central Johannesburg TVET College, found breaches of POPIA's further-processing, security-safeguards and security-compromise-notification provisions **and** a failure to register the Information Officer and to designate and register deputies. The remedial steps it ordered, each with a deadline, are the most useful artefact available for scoping a remediation plan [19]:

Required action	Deadline
Register the Information Officer, with proof	31 days
Designate and register deputy information officers, with proof	31 days
Notify affected data subjects of the security compromise, with proof	31 days
Send a written apology by email and every other channel used, with proof	31 days
Submit a POPIA compliance framework, or develop and submit one	31 days, or 120 if developing
Take action against the employee whose conduct caused the unlawful sharing	60 days
Run POPIA awareness and training for all employees, with proof	90 days

The framework must comprise a privacy policy, a **retention policy and schedule**, an incident response policy, and an information privacy and security policy [19]. A client that cannot produce those artefacts inside a month has no defence.

The Information Officer belongs in a chapter about who does what because of the gap the role creates. It is a named natural person carrying personal statutory duties over the lawfulness of processing, and in most South African health organisations that person is not a clinician and does not sit inside the clinical governance structure. So the two accountabilities for the same object — the record as a clinical and legal instrument, and the record as personal information lawfully processed — sit in different places and meet nowhere on the organogram. Drawing that gap, with the names in it, is a legitimate first-fortnight deliverable.

Who runs the establishment

The public sector: the accounting officer is the audience

Clinical and non-clinical staff in public facilities are **provincial** public servants paid through PERSAL; medical interns and community service officers are placed nationally and employed provincially [10]. The province holds the budget. The national Department of Health makes policy, sets norms and standards and monitors — it does not, as a rule, deliver services or own facilities [4]. Inside that structure the roles to be able to name are:

- The **MEC** for health, politically accountable, and the **head of department**, normally the **accounting officer** under the Public Finance Management Act — personally accountable for the department's expenditure and the person to whom an Auditor-General finding is addressed. This is the audience for a records or resilience business case, and the reason the medico-legal arithmetic in chapter 13 is the most persuasive argument available.
- The **hospital chief executive officer**, with a **clinical manager**, a **nursing service manager** and a finance and support function beneath. **Hospital boards** are appointed by the Minister for central hospitals and by the MEC for others, with composition set out in section 41(7) and three-year terms; section 41 also empowers the Minister or MEC to determine the services an establishment provides, its **admission and referral procedures** and its **fee schedules** [4].
- The **district health manager**, who prepares the district health plan for the **district health council** and the MEC, and the clinic and community health centre committees established under section 42. Municipalities retain environmental health functions and provide municipal health services under **service level agreements** setting out services, resources, performance standards and termination conditions [4].
- The **National Health Council**, the intergovernmental body chaired by the Minister with the nine MECs, the Director-General, provincial heads of department, a local government representative and the head of the military health service [4].

That structure has one consequence to internalise before costing anything: **there is no single owner of the health system's data.** The national department issues norms, standards and guidelines and prescribes by regulation, but it does not employ the clinicians, own the facilities or control the provincial IT estate. Nine provinces procure independently, so a national system is always a federation problem.

And the counterparty is frequently unstable. For 2025/26, of nine provincial health departments **five received qualified audit opinions, three were unqualified with findings, and only the Western Cape achieved a clean audit** [20]. The Auditor-General recorded KwaZulu-Natal operating with an **acting head of department from February 2025 alongside acting chief information officer and chief audit executive posts**, North West's chief financial officer post vacant for at least three years, and **most departments achieving below 60% of their targets** [21]. That is not a reason to refuse the work. It is a reason to design for governance that does not depend on a stable counterparty: decisions documented, standards written down, no critical dependency on one person's tenure.

The private sector: two keys to the door

A private group has a group executive, a hospital manager per site, a nursing service manager, a finance and revenue-cycle function, and — separately, and not on the payroll — the independent practitioners. What is less obvious is who authorises the establishment to exist. Private facility licensing is **provincial**, and a licence is **site-specific and bed-count-specific**: KwaZulu-Natal's licences run for **12 months from issue and not beyond 31 December of that year**, relicensing inspections are planned annually, and floor plans for renal units, chemotherapy units and oncology bunkers must be approved **before construction begins** [22]. Chapter 18 deals with that regime and its shaky legal basis.

One detail on the same page is worth noticing: a facility may open only after receiving the approval letter "and other necessary approvals, such as from the Board of Healthcare Funders" [22] — because a private facility needs a BHF practice code number to claim. The licence to operate is a **two-key arrangement**, and the person who can authorise one more bed at a Driehoek hospital is a provincial official, not anyone in the group.

The money side: scheme, administrator and managed care organisation are three entities

Clients conflate these three constantly. **The scheme** is the fund: it holds the money, carries the risk, owns the member relationship in law, and is the **responsible party** for the claims data. It is governed by a board of trustees of which at least half are member-elected, plus a principal officer, and

registered with the Council for Medical Schemes — typically with no staff beyond the principal officer's office. **The administrator** is a commercial company accredited by the Council to run administration on the scheme's behalf, and is an **operator** processing under the scheme's authority. **The managed care organisation** is separately accredited for managed health care: protocols, pre-authorisation, case and disease management, provider networks, formularies and utilisation review. At 31 March 2025 there were **71 registered medical schemes — 16 open and 55 restricted — 33 accredited administrators, 43 accredited managed care organisations, 2 223 broker organisations and 7 773 individual brokers** [23].

One corporate group frequently owns the administrator and the managed care organisation and may sponsor the open scheme, which is lawful and is exactly why the data-ownership lines must be written down. **The scheme's data is the scheme's, not the administrator's**, and an administrator change is a data migration with a regulatory deadline. There is a fourth party in the chain — the **switch**, the electronic clearing intermediary a claim travels through. It is an operator too, it sees diagnosis codes, and it owns the message format. Chapter 6 follows the claim through it.

The regulators, and who inspects what

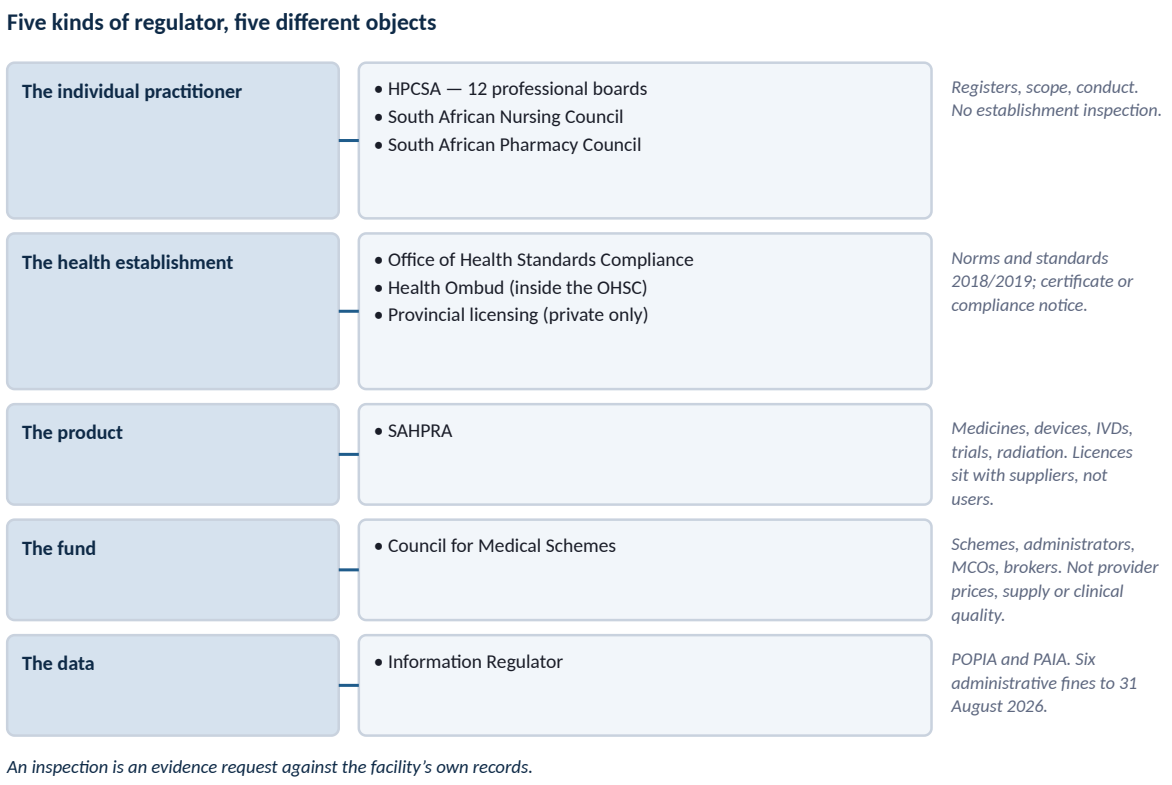


Figure 5.2 Each body regulates a different object. Assuming that one of them covers another is the commonest error an architect makes about South African health regulation.

Five kinds of body regulate five different things, and the common error is to assume any one covers the others. **The professional councils regulate individuals** — they register practitioners, define scope and discipline conduct, and they do not inspect establishments.

The Office of Health Standards Compliance regulates establishments, public and private. It inspects against the **norms and standards regulations promulgated in 2018 and effective in 2019 — seven chapters and 22 regulations** [24], across five domains: user rights; clinical governance and clinical care, including **health records** and responses to adverse events; clinical support services, including medicines, diagnostics, blood and **medical equipment**; facilities and infrastructure; and governance and human resources [25]. Each inspection ends in a certificate or in a compliance notice stating the nature and severity of the failure with timeframes for correction, followed by a re-inspection. An inspection is an evidence request against the facility's own records, and a compliance notice is a dated remediation workflow almost no facility has a system for.

Two things about the OHSC matter. The first is throughput. In 2023/24 it inspected **734 public health establishments — 19,62% of a universe of 3 741 — and 60 private establishments: Netcare 17, the National Hospital Network 26, Mediclinic 7, Life Healthcare 8 and Clinix 2**, issuing **618 certificates of compliance** and **45 written warnings**, with **132 funded posts** [26]. The February 2026 Budget redirected **R24 million** to the OHSC to fill posts and increase inspections [27] — a signal that regulatory inspection is a growing consumer of a facility's own operational data. The second is that the regime is **incomplete rather than merely new**: as recently as 2022 there were no finalised inspection tools for private acute hospitals, GP practices or emergency medical services [25]. **Design to the domains, which are stable, not to a tool version.**

The Health Ombud sits inside the OHSC structure and takes serious complaints about poor-quality care in public **and** private facilities. Its throughput is the binding constraint again: **34 of 63 user complaints, 54%, were resolved through assessment within 30 working days against a 70% target**, a shortfall the OHSC attributed to having only four assessors [26].

The Council for Medical Schemes regulates the funding side. It registers schemes and approves their **rules**, including benefit option design and contribution tables, so a scheme cannot change a benefit or a price without the Council approving the rule amendment. It accredits administrators, managed care organisations and brokers; supervises solvency; handles members' complaints against schemes; and maintains the prescribed minimum benefit definitions [23]. What it does **not** regulate is the structural hole at the centre of the private system: **provider prices, provider supply, clinical quality in providers, and health insurance products that are not medical schemes** [23].

SAHPRA regulates products, not providers — medicines, biologicals and vaccines, **medical devices and in-vitro diagnostics**, clinical trials, radiation control, and pharmacovigilance including recalls and field safety corrective actions [28]. Two consequences for a hospital. **Device establishment licences are held by suppliers, not by users**, and a valid licence is a prerequisite for bidding on national and provincial tenders — so supplier licence status is a procurement control belonging on the supplier master with an expiry date. And device **registration is not yet operating**, so there is no national device registry and a hospital's device master is its own, while recalls still arrive from the regulator and must be matched against the hospital's device and lot holdings, which is impossible without lot-level traceability [28].

The Information Regulator regulates personal information and access to information, and its enforcement record changed most recently: six administrative fines by its 31 August 2026 media briefing, including **Lancet Laboratories, R100 000, paid** — the health sector's first — and on 26 and 27 October 2026 fines of R1 million against First National Bank and R500 000 against a technical and vocational education college [29]. It has named **insurance and health among its priority sectors** for its own compliance assessments [30].

Beyond those five: **provincial health departments** licence private facilities; the **Competition Commission and Tribunal** regulate the private market's structure; the **Forum of Statutory Health Professional Councils** under section 50 of the National Health Act coordinates the councils [4]; and **COHSASA** offers voluntary third-party accreditation, with a published list of **49 accredited facilities across ten countries, 29 of them in South Africa** [31] — against roughly 3 700 public establishments plus several hundred private ones.

> "We are compliant" has at least four meanings in South African healthcare, granted by four bodies that do not agree. Ideal Clinic status is a departmental self-assessment; an OHSC certificate is a statutory inspection outcome; COHSASA accreditation is voluntary and third-party; a provincial licence applies to private facilities only. They diverge sharply: Ideal Clinic certification reached **33,3% of primary health care facilities in 2019**, while in the 2019/20 OHSC audit **only 14,7% of inspected facilities met the minimum standard** [32]. Establish which instrument, which assessor and which year before accepting the sentence.

Where a complaint goes

Section 18 of the National Health Act gives anyone the right to lay a complaint about the manner in which they were treated, requires the MEC and municipal councils to establish complaint procedures, requires those procedures to be **displayed**, and requires referral of misdirected complaints [4]. There are seven routes, and a complaint may enter at any of them.

Seven routes a complaint can take, and the clocks on them

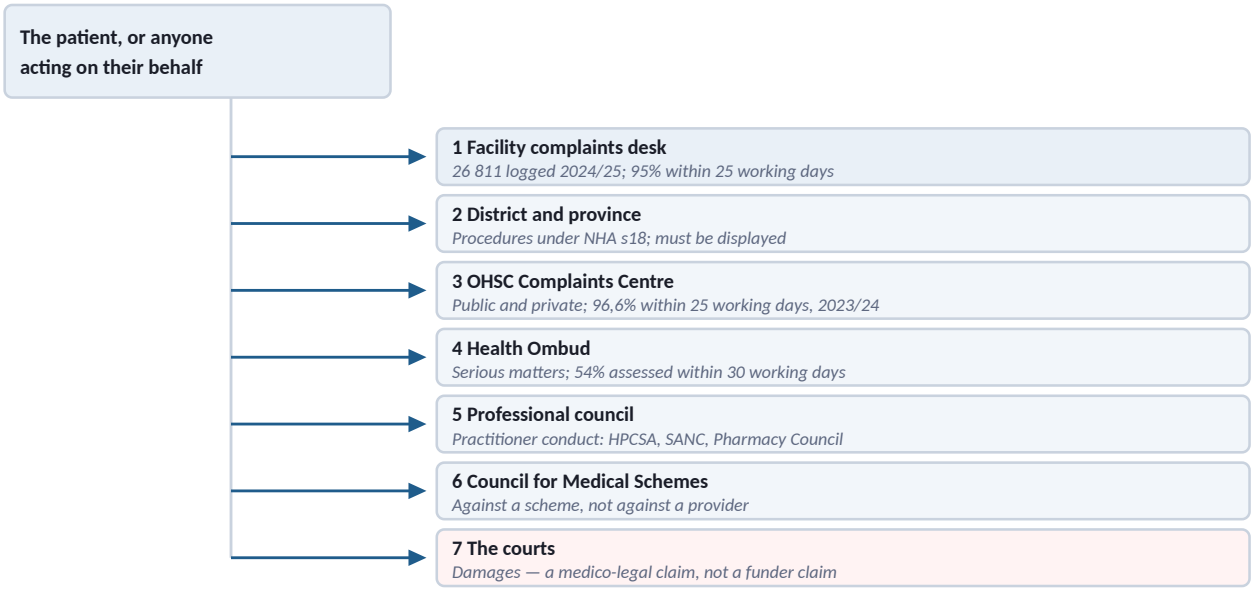


Figure 5.3 A complaint may enter at any of seven points and must be routed on if it arrives in the wrong place. Route 7 is a *medico-legal* claim and shares nothing with a claim to a funder.

The service standards become requirements: **26 811 complaints were logged nationally in 2024/25, 25 643 (96%) resolved and 24 425 (95%) within 25 working days** [33]. So a register with a working-day clock; routing to and from the OHSC and the Ombud; a publication capability; and **linkage, but not merger, with the patient safety incident register**, because the two are different records with different owners, legal consequences and access rules.

Note also the two unrelated senses of "claim" running through this chapter and the next. A **claim to a funder** requests payment under a benefit arrangement; a **medico-legal claim** is a civil action against a provider. They share no data, no owner and no lifecycle, and a requirement written with the bare word will be built wrong.

For several things, there is no single owner — and saying there is teaches a falsehood

The standard architecture deliverable asks for one owner per entity. For several of the most important entities in healthcare there is no single plausible owner, and asserting one is an error rather than a simplification. It belongs here because it is a fact about the organisation rather than the data model, and

chapters 10 and 20 build on it.

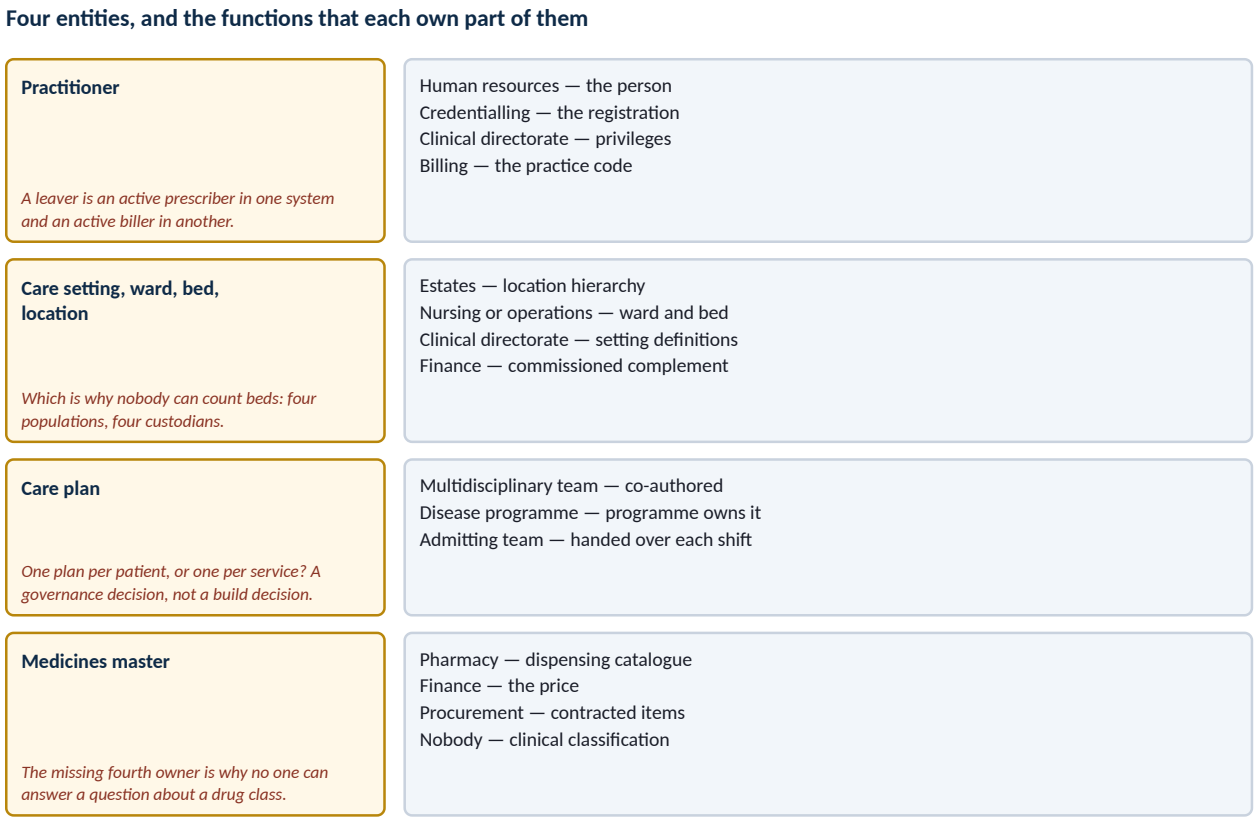


Figure 5.4 For these four entities the ownership column cannot honestly hold one name. Asserting one teaches a falsehood and will be contradicted by the first reader who does the work.

Practitioner has four candidates. Human resources owns the employment relationship and genuinely is the master of the *person*; credentialling owns registration verification; the clinical directorate owns privileges; billing owns practice code numbers. And the four disagree in a way visible in any hospital: **a practitioner who has left is still an active prescriber in one system and an active biller in another.** The error behind the supplied repository's defect of mastering medicines management in human resources is not that HR is irrelevant — it is the assumption that mastering the person means mastering the practitioner.

Care Setting, Ward, Bed and Location have four. Estates owns Location — the physical hierarchy of site, building, floor, department, room and bed. Nursing or operations owns Ward and Bed in practice, because they open and close capacity. The clinical directorate owns the definitions of Care Setting, such as what counts as high care rather than intensive care. Finance owns the commissioned complement. Which is exactly why nobody can count beds: physical, commissioned, staffed and occupied beds are four populations with four custodians, and the bed complement is a dated time series rather than a number.

Care Plan is genuinely contested. In a multidisciplinary setting it is co-authored; in a disease programme the programme owns it; in a hospital the admitting team owns it and hands it over at every shift. The real question — **one care plan per patient, or one per service** — bears directly on whether a patient's care can be seen whole, and should be asked explicitly rather than discovered in build.

The medicines master has three partial owners and no clinical-classification owner at all. Pharmacy owns the dispensing catalogue and finance the price; procurement arguably owns the contracted items, though that third owner is an inference rather than a sourced finding — and nobody owns the therapeutic classification that would let the organisation answer a clinical question about a drug class.

So what does an architect deliver here? Not an answer. The deliverable is the governance decision, in four parts: name the candidate owners; state what each demonstrably owns today, with the system that proves it; state what breaks if nobody decides; and get a decision minuted with a date and a named person. One page, and worth more than the rest of the ownership column put together.

> An architecture document that asserts a single owner where four exist will be contradicted by the first person who reads it who does the work. That credibility cost is not recoverable inside a quarter, and a quarter is all the time you have.

THE CHAPTER IN TEN LINES

- Authority comes from registration with a council, a provincial licence and a funder accreditation, not from a position on the chart.
- Scope of profession is set by statute for a whole profession; scope of practice is set by ethical rule 21 for one named individual.
- Build authorisation as role plus credential plus privilege, with break-glass and audit, because emergency override is legally required.
- Registration status is a live attribute with an expiry; a cached copy without a refresh is a compliance defect.
- A nurse-to-population ratio describes a register, not a workforce. SANC says so itself.
- The two highest nursing vacancy rates in the country are nurse educators at 25,0% and nursing education managers at 23,9%, from PERSAL June 2023.
- There is no South African safe-staffing norm, so the deliverable is instrumentation rather than a standard.
- A private hospital does not employ its specialists. One episode produces several independent claims to the funder from several legal entities.
- Completeness of clinical coding responds to process and supervision; accuracy responds only to a trained coder.
- Section 13 of the National Health Act puts the records duty on the person in charge of the establishment, not on the clinician.
- The Information Officer is the head of the organisation unless delegated, must be registered with the Regulator, and sits outside clinical governance.
- The professional councils regulate individuals, the OHSC establishments, SAHPRA products, the CMS funds and the Information Regulator data. None covers another.
- For Practitioner, Ward, Bed, Location, Care Plan and the medicines master there is no single owner. Deliver the governance decision, not a guess.

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CHAPTER 6

How the money moves

Two flows with nothing in common, and one fact that governs both: price is not an attribute of a service.

Two money flows run through South African healthcare and they have almost nothing in common. The private flow begins with a member's monthly contribution to a medical scheme and ends, several intermediaries later, as a partial payment against an invoice the provider raised. The public flow begins with a national revenue allocation to a province and ends as salaries, pharmaceuticals and an electricity bill, with a trickle of fee revenue coming back the other way. Different statutes, regulators, code sets and calendars; no shared identifier and no shared price.

One statement covers both and is the single most useful thing in this chapter: **price is not an attribute of a service.** In the private sector there is no national tariff and has not been one since 28 July 2010. In the public sector the fee schedule is provincial, means-tested and issued nine times over in nine gazettes. A `price` column on a service master is not a simplification; it is a defect, and it will be found during reconciliation rather than during design.

This chapter traces both flows and states what each requires of an architecture. It leaves three things to their owners: chapter 7 has National Health Insurance, chapter 12 has coding and the national specifications that do not exist, and chapter 18 has what the regulators can do to you.

A medical scheme is not an insurer, and the difference is architectural

A South African medical scheme is a distinct species of entity created and governed by the **Medical Schemes Act 131 of 1998** and supervised by the **Council for Medical Schemes** [1]. It is not an insurer, and the distinction is legal rather than semantic: schemes are not-for-profit mutual entities owned by their members, may not underwrite on health status in the ordinary insurance sense, must set **community-rated** contributions within a benefit option, may not refuse membership on health grounds if open, and must fund **prescribed minimum benefits** in full on every option irrespective of that option's limits.

An architect who has worked on an insurance book should notice what is missing. There is **no underwriting engine** and no individual risk score driving price: contribution is a function of option, dependant count and dependant type, and in some schemes an income band — never of the member's health. The permitted substitutes are late-joiner penalties and waiting periods, both rule-bound and time-limited. In place of capital adequacy there is a statutory solvency ratio, accumulated funds as a percentage of gross annual contributions: **40,87% at the end of 2024** [2], and **43,45%** on the Council's 2024/25 annual-report basis dated to 2023, having fallen for three consecutive years [1][3]. Two figures on two bases; do not use them interchangeably.

Vocabulary follows from the legal form, and getting it wrong signals an architect who has read insurance material and assumed it transfers. **A member pays a contribution.** The word *premium* belongs only to an actual insurance product — gap cover, a hospital cash plan, a primary-care insurance product — regulated under insurance law by the Financial Sector Conduct Authority and the Prudential Authority rather than by the Council, and reaching the market through the demarcation arrangements discussed below [1].

Chapter 5 separated the three entities in the private flow. The money consequence is that the **scheme** is the fund and the **responsible party** for the claims data, the **administrator** that adjudicates and pays is an **operator** acting under the scheme's authority, and the **managed care organisation** is separately accredited. At 31 March 2025 there were **71 registered schemes, 33 accredited administrators and 43 accredited managed care organisations** [1]; schemes paid **R6,18 bn for accredited managed healthcare services in 2024** [2], and managed care arrangements covered **99,19% of beneficiaries** [3].

Two scale figures, with their traps. At the end of 2024 the Council reported **9 168 534 beneficiaries**, up from 9 127 453 in 2023 — and Table 5 of the same publication gives **9 168 533** for the same total [2]. A one-beneficiary discrepancy inside a single regulatory publication is the cleanest available illustration of why reconciliation tolerances exist. And the book is ageing: average beneficiary age rose to **34,2 years in 2024** from 32,5 in 2016, and the pensioner ratio to **9,8%** from 7,9% [2]. A book that ages while its membership stagnates is one whose claims intensity rises faster than its contribution base.

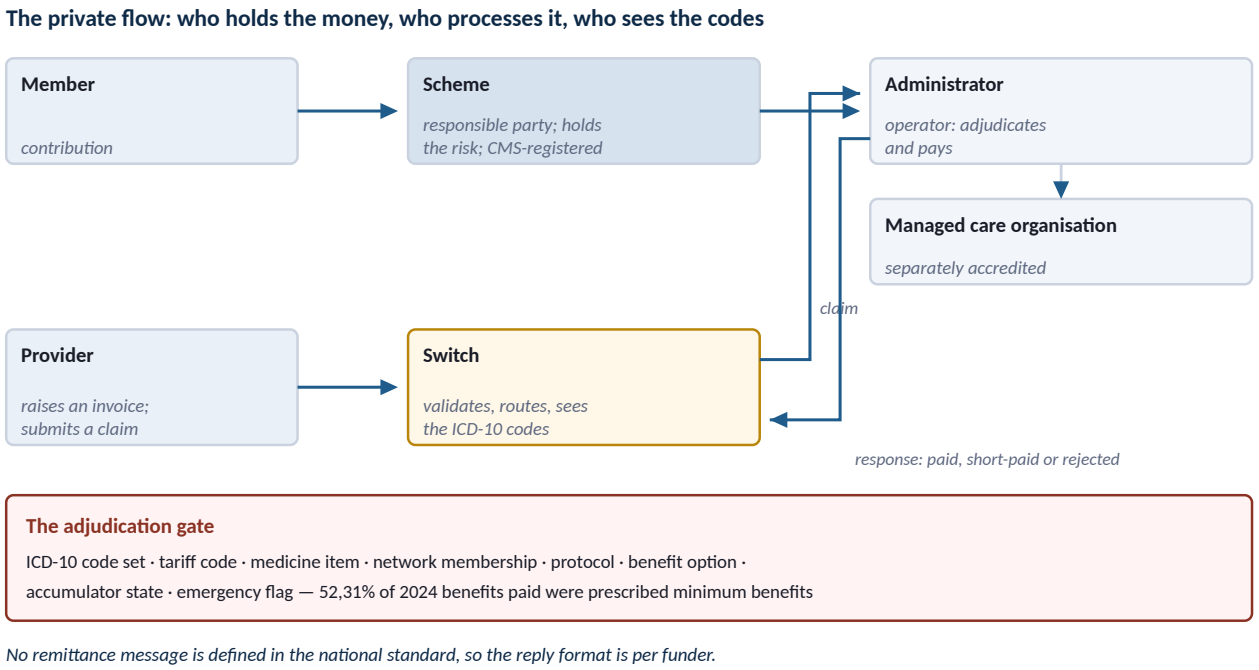


Figure 6.1 Four parties and one gate. The scheme carries the risk and owns the data, the administrator adjudicates as an operator, and the switch sees the diagnosis codes on the way through.

One structural feature changes a data model. **Efficiency discount options** — a lower contribution in exchange for a restricted provider network — grew from 50 to 71 between 2017 and 2023, with enrolment rising from 20% to 37% [3]. Network membership is therefore a **pricing input**, which makes the provider network master a rating-relevant asset rather than a reference list. On the provider side, benefit option cannot be modelled as an attribute of the member: it is an effective-dated contract with its own network, formulary, accumulators and prescribed-minimum-benefit treatment, and a model that gets this wrong will mis-state revenue.

Prescribed minimum benefits: the rule that governs more than half the money

Prescribed minimum benefits are created by **regulations under the Medical Schemes Act 131 of 1998** [4]. Every registered scheme, on every option, must fund them in full regardless of annual limits or available savings. There are three components: **any emergency medical condition**, a defined list of **diagnosis-treatment pairs**, and a defined list of conditions on the **Chronic Disease List**.

And here the official numbers disagree. The Council's own page gives **271** diagnosis-treatment pairs and **26** chronic disease list conditions [4]; a scheme's 2026 member guide gives **271 and 27** [5]; a report of the Board of Healthcare Funders gives 271 and 26 [6]. The frequently repeated "270 pairs" appears in none of them. **State the pair count as 271, and state that the chronic disease list count is published as both 26 and 27 and could not be resolved from public sources.** A benefit-rules engine built against a list whose length two authoritative sources disagree about needs that list versioned and dated, not hard-coded.

Operationally, entitlement is harder than a list lookup, and the scheme guide is explicit about why [5]:

- The condition must be a listed prescribed-minimum-benefit condition **and** the treatment must match the defined clinical treatment for it.
- **ICD-10 codes are "an industry guide to which conditions may qualify". A code alone does not confirm cover.** The diagnosis must still meet the provision description — the guide's own example carries a life-threatening requirement — and the treatment must meet the criteria.
- Approved treatment is defined by an approved medicine list, specified consultations, procedures and investigations, and national and evidence-based protocols. Treatment outside that basket of care does not automatically qualify.
- **Designated service provider** use gives full cover with no co-payment. Where a member uses a non-designated provider **voluntarily**, the scheme pays up to **80% of its own rate** and the member pays the difference; involuntary use, no available designated provider, or an emergency attract full cover.
- There is an appeal route: member and doctor, then the scheme's dispute process, then the principal officer, then the Council.

> A prescribed-minimum-benefit determination is a rules evaluation over an eight-part input: ICD-10 code set, procedure or tariff code set, medicine item, provider network membership, clinical protocol, benefit option, accumulator state and emergency flag. It cannot be derived from the diagnosis code alone, nor from the benefit option alone. It is also the largest single source of dispute in the industry.

The Council's complaint data confirms that reading: short payment of claims for prescribed minimum benefits has been the steady top theme for five years, and disputes over **level of care** and funding liability for high-cost new treatments are rising [1].

The scale is what makes this the centre of the private system. Prescribed-minimum-benefit expenditure was **R135,66 bn in 2024**, up 9,09% from R124,36 bn in 2023, which is **52,31% of total benefits paid** [2]; per-beneficiary cost was about **R1 242**, up 9% year on year and highest among those over 85 [6]. **More than half of all private healthcare benefit spend in South Africa is governed by a rule set that must be paid in full, at a price nobody regulates.**

The rule set is also overdue for revision. The Board of Healthcare Funders states that the review began in **2017**, that the regulations have not changed since 2003 despite a legal requirement for biennial review, and that the Minister declined an earlier review in 2023 for lacking a primary healthcare element [6]. That is a claim by an interested party, and the biennial-review obligation could not be verified against the regulation for this book. What follows for design is the same either way. **Effective-dating of the prescribed minimum benefit rule set is the single most predictable change in the sector**, and a design that compiles the pair list, the chronic disease list or the chronic algorithms into application logic rather than into versioned, dated reference data will be rewritten when the review concludes.

The tariff vacuum

This is the material an architect most needs and least often gets, because the honest answer — *there is no national tariff* — sounds like a research failure. The history, dated, because the sequence explains why nothing has replaced it:

Date	What happened
To 1994	Fee-for-service tariffs negotiated by the funders' association and **gazetted annually**; abolished by regulatory amendment in 1994 [7]
2004	The Competition Commission declared the centralised reference schedules of the funder, hospital and medical associations a **restricted practice**, because they were set collusively [7]
2006	The **National Health Reference Price List** was set by the Council with the Department as an interim cost-based schedule, and the HPCSA adopted it as its **Ethical Tariff** [8]
23 July 2007	**Government Notice R681** published the regulations governing determination and publication of a reference price list, under sections 90(1)(u) and (v) of the National Health Act [9]
24 November 2008	The HPCSA Council **scrapped the Ethical Tariffs**, with effect from the Department's publication of its own reference price list [8]
24 December 2008	The 2009 Reference Price List was published [9]
28 July 2010	Ebersohn AJ in the North Gauteng High Court **declared GN R681 invalid**, set the reference price list aside, and interdicted the Director-General from publishing one [9]

The common account of that judgment is wrong about what was set aside. **The court did not rule on prices. It declared the enabling regulations invalid** — for want of admissible evidence of the consultation with the National Health Council that section 90(1) requires; for conflating the information-gathering power in section 90(1)(u) with the price-determination power in section 90(1)(v); for unlawful sub-delegation, the Minister having left methodology and guidelines to the Director-General; and for vagueness and irrationality, "stakeholder" being undefined and the representation rules effectively excluding the applicants [9]. The price list fell because its regulations fell.

The step usually omitted is the one that matters. The HPCSA had already scrapped its own Ethical Tariffs in November 2008, conditional on the Department publishing a reference price list. When the list was set aside in 2010 the profession was left with **no fee benchmark at all** — so the relevant Board and the HPCSA's own ombudsman had nothing against which to assess an overcharging complaint. The Council therefore built **2012 Guideline Tariffs** by taking the **2006** reference price list and applying an inflator of **46,66%**, under **section 53(3)(d) of the Health Professions Act 1974**, which lets the Board publish a fee as a benchmark for overcharging complaints; **section 53(1)** separately requires a practitioner to disclose their intended fee before treatment if asked or if it exceeds what is usually charged [8].

So what does a practice bill against in 2026? In descending order of practical importance: each scheme's own published rate; a bilaterally negotiated rate where the provider has a contract with the funder, standard for hospital groups; the hospital group's own private tariff schedule for self-payers, published and dated — one group's took effect on 1 January 2026 [10]; the HPCSA guideline tariff, a disciplinary benchmark rather than a price; and the statutory tariffs. **Balance billing** is the gap between the scheme's rate and the provider's charge, and the patient pays it: the Council's 2010 discussion document records specialists balance-billing at times up to **300% of the reference price list** [7].

One appendicectomy, one theatre, one patient — five price regimes

Medical scheme	Scheme rate or a bilaterally negotiated rate. No national tariff since 28 July 2010.	*Claim via a switch, EDIFACT MEDCLM*
Self-payer	The hospital group's own published private tariff schedule.	*Invoice to the patient*
Public facility	Uniform Patient Fee Schedule, means-tested into H0 to H3, issued per province.	*Facility revenue, 12-hour or 30-day blocks*
COIDA	Gazetted annually; per diem plus carve-outs; theatre billed per minute.	*Invoice supported by hospital records*
Road Accident Fund	Externally funded full-paying in a public facility. Tariff basis not sourced.	*Not established*

Each regime has its own code set, billing rules, documentation and settlement cycle — and which one applies is sometimes not known at the point of care.

Figure 6.2 Five simultaneous price regimes on the same clinical act. A single *price* attribute on a service master cannot represent any of them.

For the data model this resolves to one sentence. **A tariff code is not a price.** It is a key into a price that depends on payer, provider, contract, effective date and sometimes practitioner seniority — the Council's 2010 document lists seniority-based differentials among the items a negotiation would cover [7]. Tariff is **contract data, per funder, per agreement, per period**, and that is a data-architecture fact before it is a commercial one. A claim line carries a tariff or procedure code, an ICD-10 code set, a practice code number, a place and date of service, a quantity or time, and modifiers; **the price is resolved by the payer, not asserted by the provider** — which is why short payment rather than rejection is the dominant dispute [1].

The replacement mechanism, which is still a draft

The **Draft Interim Block Exemption for Tariffs Determination in the Healthcare Sector** was published on **14 February 2025**, comments closed 31 March 2025, and it was **not in force** as at this book's compile date [11]. It would exempt three categories of conduct for both prescribed and non-prescribed benefits: collective determination of tariffs; collective determination of standardised **diagnosis, procedure, medical device and treatment codes**; and collective determination of quality measures, formularies and treatment protocols. Two bodies would be created — a Tariffs Governing Body providing oversight and a Multilateral Negotiating Forum setting maximum tariffs — for three years [11]. The design details matter more than the principle: the **Director-General of Health appoints Forum members**, the Governing Body sits **inside the National Department of Health**, agreed tariffs would be **binding on the parties**, **bilateral deals outside the Forum could only reduce tariffs**, and **general practitioners and specialists are covered while hospitals are excluded** by reference to private hospitals' market power [12][13]. A separate track runs alongside: the Board of Healthcare Funders applied in 2021 for an exemption allowing schemes to negotiate collectively with willing providers, the Commission rejected it, and an appeal to the Competition Tribunal is pending — an account given by that body's own managing director, so cite the procedural facts rather than the argument [14].

> Why an architect should care about a draft. If the second category proceeds, South Africa will have a **nationally governed clinical and device code set with a named custodian and a change process** for the first time. Every private-sector claims, billing, formulary and protocol system in the country is currently built on code sets maintained bilaterally or by vendors. A national code authority would be the largest master-data event in South African private healthcare, and the systems that will have to absorb it are the ones being specified now.

Payment mechanisms, and why "capitation or fee-for-service" is the wrong question

Fee-for-service dominates, and the official position is that it should not. The Council notes that managed care covers 99,19% of beneficiaries "but the fee-for-service model still rewards volume over value" [3]. The Health Market Inquiry recommended **ending fee-for-service**, with transition mechanics: bilateral negotiation only where contracts carry a value component and risk transfer, facility-funder contracts moving to alternative risk-based contracting within three years, fee-for-service networks open to **any willing provider**, and **closed networks only for value-based contracts on three-year terms** [15][13].

Five mechanisms are in use, in rough order of prevalence: **fee-for-service** per tariff code against a scheme or negotiated rate, dominant for specialists, pathology, radiology and most hospital line items; **per-diem or case fee** arrangements; **global fees** for defined episodes, common in maternity and some orthopaedic and cardiac programmes; **capitation**, a fixed amount per enrolled member per month for a defined scope; and **alternative reimbursement models**, the industry's term for risk sharing, bundles, outcome-linked payment and shared savings.

Framing the first and the fourth as a choice understates the problem. A fee-for-service claims system is a transaction counter: an event, a code, a price, an eligibility check. A capitated arrangement needs five capabilities continuously, and South Africa does not operate any of them at scale.

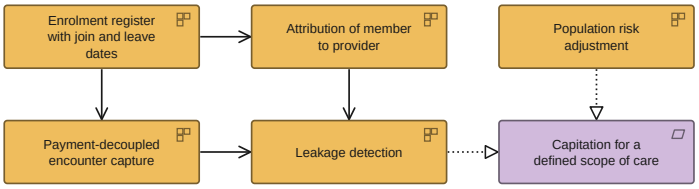


Figure 6.3 The five capabilities a capitated arrangement needs continuously. South Africa operates none of them at scale, which is why the choice between payment models is not the real problem.

The last two are where capitation architectures fail, and the failure is behavioural. In fee-for-service a missing claim costs the provider money and somebody chases it. In capitation a missing encounter record costs nobody anything today, so it will not be captured unless the system compels it. **Any capitation design that does not solve the "why would anyone type this in" problem produces a population you are paying for and cannot see.** The Department reports a draft primary health care contract pending finalisation of the benefits package and the capitation model [16]; chapter 7 deals with what that means.

The claim as a message

A South African private healthcare claim does not travel directly from provider to funder. It goes through a **switch** — an electronic clearing intermediary, a switching house in the local idiom.

The names commonly given are **MediKredit, MediSwitch, Interpharm and Zieto**, between them reaching roughly 151 medical schemes through a single practitioner-facing account, with **Healthbridge** the other long-standing name [17]. That comes from an undated trade source and the current market structure could not be established for this book — a gap to close with the client's own switch contracts. What a switch does is better documented: the practice or pharmacy system submits the claim, the switch submits it to the scheme on the member's behalf, the scheme responds **back through the switch**, and the switch also supplies electronic remittance advice, membership status checks, claim tracking, and reversals and cancellations [17]. In retail pharmacy real-time adjudication at the point of dispensing is the norm: the pharmacist knows before the patient leaves whether the claim is paid, short-paid or rejected, and why.

The published national standard for the message itself is **UN/EDIFACT MEDCLM** — message type MEDCLM, version 0, release 912, controlling agency ZA, revision 13.4, November 2016, maintained by the Private Healthcare Information Standards Committee [18]. Not XML. Not FHIR. Its structure runs: a heading section with the message header, a beginning-of-message segment carrying a batch number, a documentary-requirements segment carrying **adjustments, reversals, resubmissions and interim accounts**, and the admission, discharge and accident dates; a party and reference group for the supplier, the member and the doctors; a detail section with service date and references; line-item groups, one per tariff item with modifiers, payment terms and VAT, plus a group for medicines and consumables; and a closing section with control totals [18].

UN/EDIFACT MEDCLM v0 release 912, agency ZA, revision 13.4, November 2016

What the message carries

Batch number; adjustments, reversals, resubmissions, interim accounts
Admission, discharge and accident dates
Member, plan, scheme, administrator and patient account numbers; dependant number
Six separate doctor roles: specialist, anaesthetist, assistant, referring, treating, admitting
ICD-10 for diagnosis, admission, discharge
SNOMED II and III for pathology
Tariff, CPT, ICPC, dental, optometry and NAPPI codes; control totals

What it does not carry

No remittance or payment-advice message
No link from a claim response to a payment
No present-on-admission flag
Internal inconsistency: one section allows one claim per message, another allows multiple items per patient

What is actually on the wire

The switch's own XML, validated against its published XSDs, certified in its test environment, batched or real-time by funder

The published standard and the operative contract are not the same artefact. Design to both.

Figure 6.4 The absence on the right is the consequential half. With no remittance message defined, denial-rate analysis has to be rebuilt per funder.

Three things about that message are worth an architect's attention.

It carries a great deal of identity: membership, plan, scheme, administrator and patient account numbers, patient name, dependant number and relationship to the principal member, identity number, date of birth and sex; supplier number, group practice number, council registration number, and **separate identifiers for the specialist, anaesthetist, assistant, referring, treating and admitting doctors** [18]. Those last six are not six practitioners on one claim but up to six **roles**, some of which may be the same person.

It carries clinical data, which makes the switch an operator processing special personal information. ICD-10 codes for diagnosis, admission and discharge; complaint codes; **SNOMED II and III codes for pathology**; and line-item qualifiers spanning tariff codes, modifiers, surcharges, laboratory and fixed-fee codes, ICD-10, ICPC, CPT, dental and optometry codes, NAPPI codes with a fallback where none exists, European Article Numbers, generic

and mixture codes, and dental tooth numbers [18]. The SNOMED fields are a fossil: the South African claim standard has carried one for decades, referencing a long superseded version of that terminology, and South Africa joined SNOMED International only in May 2026. Any data-flow map that stops at "sent to the medical aid" is incomplete.

It does not carry a remittance. The specification covers the claim side only — amounts claimed, amounts due by the scheme, patient-paid amounts to be reimbursed, deposits, discounts, levies, VAT and control totals — and **defines no remittance or payment-advice message and no link from a claim response to a payment** [18]. The remittance therefore arrives in whatever format each funder chooses. **That single absence is why denial-rate analysis is hard in South Africa**, and it is why an architect told "the claim standard is sorted" should immediately ask what the *response* standard is. The specification also carries an unresolved inconsistency — one section says a message may contain only one claim while another allows multiple items per patient — so expect scheme-specific interpretations [18].

> One correction to make before designing anything. The published national standard is EDIFACT, but the **operative contract is the switch's own interface**. The largest switch publishes **XSDs that software vendors use to validate their XML requests and responses**, supplies an integration pack, an accreditation step after integration, and a vendor test environment available continuously that mirrors live funder rules and pricing with dummy beneficiary data; and it supports **either batched or real-time switching depending on each funder's preference** [19]. The standards committee's own remit likewise covers standardised claim forms, paper layouts **and XML schemas** alongside the EDIFACT message [20]. So: EDIFACT is the published standard, XML against the switch's schemas is frequently the wire format, the switch certifies the practice system, and batched against real-time is a funder-by-funder variable rather than a platform setting. Model the switch as an external standard with an external owner, versioned on somebody else's schedule.

That continuously available test environment is an asset worth naming: it gives integrators the repeatable, safe conformance test most healthcare interfaces never have, so ask every vendor whether theirs exists and whether it mirrors production rules. The standards committee is reachable too, with subcommittees for clinical coding, message standards and pharmaceutical coding, a **dormant** diagnosis-related-group subcommittee, and the national Department attending [20]. **Attending those meetings is a better route to the operative rules than reading the published standards.**

ICD-10 as a payment gate

The instrument that requires a diagnosis code on a South African claim is not a health statute and is not penal. It is **Regulation 5(f) made under the Medical Schemes Act 131 of 1998** — a *funder* regulation — requiring every claim to carry the relevant diagnostic code for the service rendered and providing that claims without it **may not be considered for payment** [21]. The national morbidity coding standards cite a second instrument for the same requirement: the regulations to the Medical Schemes Act published in *Government Gazette* No. 20556 in October 1999 [22]. Either way the obligation is **conditional on payment**.

That distinction explains the entire coding-for-billing pathology. **A provider who submits a claim without an ICD-10 code has not committed an offence. They have submitted an unpayable claim.** There is no national clinical coding regulation; the Department's own status document recorded one as "in development" as at 2015 and that position could not be updated for this book [23]. So a funder regulation about claims is doing the work of a national clinical coding mandate.

The dates are commonly given wrong. The coding standards record ICD-10 as **adopted by the Department in 1996 and implemented in July 2005** [22]; other departmental material gives adoption as **1 July 2005** [21]. Both are in print, describing different events. The operationally decisive date is neither: it is **1 July 2014**, when Phase 4.1 introduced **clinical validation, beginning with age and gender checks against the current Master Industry Table, with full claim rejection on clinical data** [23][24]. That is the day an ICD-10 code stopped being a form field and became a machine gate.



Figure 6.5 Five machine gates and two benefit gates. The first five produce a rejection; the last two produce a short payment, which is the outcome almost nobody measures.

Three mechanics of that gate have architectural consequences.

Capacity and position differ between hospitals and everyone else. Practitioner claims allow one primary plus up to nine secondary codes **per line item**; hospital claims allow up to 30 codes **at header level** [24], and referring-provider diagnoses sit at a summary level, unlinked to individual procedure lines [22]. A model giving both the same cardinality will either truncate hospital claims or over-provision practitioner ones, and a warehouse that flattens hospital header codes onto line items has invented linkages the standard says do not exist. Position is also semantic: the dagger-and-asterisk relationship between two diagnosis rows is encoded **only in their order** on an electronic claim, so a pipeline that sorts codes or stores them as a set rather than a sequence has destroyed information — and the technical guide requires switches to preserve code integrity and order [24].

The Master Industry Table is the most important South African coding artefact, and most systems hold a stale private copy of it. It is the industry-standard listing of every ICD-10 code used in the country, compiled by the Ministerial ICD-10 Task Team and published by the Department, carrying per code the three-character and fully specific codes with descriptions, validity flags for clinical use, primary position, asterisk, dagger and sequelae, **edit indicators** in the form of an age range and a gender field, and an add, modify or discontinue status with **South African start and end dates** [24]. Three consequences. It is **effective-dated reference data** that most implementations treat as current state, so a 2019 claim must be validated against the 2019 table and a warehouse joining historical claims to today's table will silently drop and mis-describe codes. The flags are a ready-made constraint set belonging in the data layer rather than in application logic. And it has a stated change cadence: **biennial update on 1 July**, three months allowed for operational change and six months for system change [22]. Chapter 12 goes further; what belongs here is that it is a payment control.

Some rules protect the provider, so a funder-side data-quality rule can itself be non-compliant. The technical guide states that unspecified, other-specified, sign-and-symptom and default codes **are valid and cannot be rejected by schemes**, and that schemes must identify prescribed-minimum-benefit codes **whether primary or secondary** [24] — the second a direct mitigation of primary-position gaming, which tells you the behaviour was observed. A separate provision requires codes to be submitted **without descriptions, to protect privacy** [22], so the funder receives a classification and not a narrative.

One more thing the gate does not do: **there is no national procedure code set to gate against.** Private-sector procedure coding uses a South African adaptation of an American classification whose own maintainer positions it "for Data and not for Billing" and declines to recommend it as a national standard, while the public sector bills on the Uniform Patient Fee Schedule [25][20]. The money consequence is immediate: a provider cannot price a procedure against a national schedule, cannot group an episode into a case-mix class, and cannot compare its case mix to anyone else's. Chapter 12 carries that argument in full.

Rejection, short payment, and what each tells you

Claims fail in two structurally different ways, and the two need different instrumentation.

Machine rejections never reach adjudication: a missing or invalid ICD-10 code; **age or gender validation failure** against the Master Industry Table, from 1 July 2014 [23][24]; a code in an invalid position — an external cause, asterisk, sequela or causative-organism code as primary; formatting violations, including descriptions left in, wrong separators, or codes on only the first line of a multi-line claim [22][24]; membership not active at the date of service; and submission outside the four-month window [4].

Benefit decisions are adjudicated and then short-paid. The dominant reasons are the **wrong benefit pool**; a **level-of-care dispute**, where the scheme accepts the condition qualifies but disputes the setting or intensity of care; **non-designated-provider use**, at up to 80% of the scheme rate; a **non-formulary** medicine or treatment outside the basket; a **tariff differential** where the provider charged above the scheme rate and the balance falls to the patient; rule-based exclusions; and membership terminations [1][5].

The wrong-benefit-pool case deserves its worked example, because it is the clearest demonstration that a clinical code is a financial instrument. A commercial coding guide gives it: one ICD-10 code for Graves' disease draws on day-to-day benefits or the medical savings account and will not pay if those are exhausted, while the code for thyroid storm is paid in full from risk as a prescribed minimum benefit [26]. **Same patient, same consultation, a different fifth character, a different payer and a different patient liability.** That is a commercial reference rather than a primary instrument and should be checked before it goes into a client document.

What cannot be stated is the distribution. **No national breakdown of claim rejection reasons by volume could be sourced for this book**; the Council reports complaint *themes*, not claim-line rejection statistics [1]. What would close the gap is a client's own switch response-code data, which exists, is granular, and is almost never analysed.

> The dominant failure mode in South African private healthcare claims is not rejection. It is **short payment** — a disagreement about rules applied to data, not a transmission failure. An architecture that measures success as "claim accepted" measures the wrong thing. The four measures worth building are **first-pass full payment rate**, **short-pay rate by reason code**, **patient liability generated per episode** and **days to full settlement**. Expect none of them to exist in the client's current reporting.

Two timing rules set the envelope for everything built on claims data: accounts must be submitted **within four months** and schemes **should pay within 30 days** of receiving a claim [4]. A four-month submission window means claims data for a period is not complete for a third of a year, which matters for every management report, every actuarial reserve and every "how are we doing this month" dashboard. Say so before the dashboard is specified, not after the first month-end.

Three dependencies in the private flow that the provider does not own

The medicine identifier. The national product code used for private claiming is **NAPPI**, and MediKredit — a claims switch — states that it **owns and maintains** it [19]. So every medicines service, interaction check, formulary and medicines analytic in the private sector sits on a licensed dependency held by a commercial party whose primary business is adjudicating claims. Establish the licence terms for use, redistribution and derived works before designing a medicines service, and do not assume that because the code is in every system it is freely usable.

The medicine price. The **single exit price** is the gazetted maximum private-sector medicine price. The Department published a **5,25% adjustment on 17 November 2024** [16], and a further **2,88% increase took effect on 1 October 2026**, which the Health Funders Association estimates adds **R1 bn in unbudgeted medicine costs for schemes in 2027** [6]. A single exit price adjustment is an annually gazetted, item-level, effective-dated price event changing thousands of items on one date. Any system holding medicine prices must be built for that, not for a maintenance screen.

The exemption under the insurance products. The Council, with other regulators, approved a two-year extension of the **Exemption Renewal Framework** for insurers conducting medical scheme business, **until 31 March 2027**, pending a policy determination on low-cost benefit options that has not been made [1]. Read that carefully: a class of health product is lawfully sold **under a time-limited exemption that expires inside a normal delivery window**. Any architecture for a primary-care insurance product, an employer clinic scheme or a hospital cash plan sits on it — a dated, named, sourced risk that belongs in the risk register.

The statutory funders, which are routinely forgotten

COIDA. The **Compensation for Occupational Injuries and Diseases Act 130 of 1993** funds treatment of injured or diseased employees, and its tariffs are gazetted annually. For 2026, *Government Gazette* 54356 of **20 March 2026**, Notice 7259, issued by the Minister of Employment and Labour under section 97, prescribing the fees-for-medical-aid scale under section 76, **effective 1 April 2026**, with tariffs up **6%** and fees excluding value-added tax, though certain hospital per-diem tariffs are treated as VAT-inclusive [27].

The structure is instructive because it is the opposite of fee-for-service: **per diem plus carve-outs**. Fees are grouped by facility type — acute, day, acute and sub-acute rehabilitation, psychiatric, and frail care, palliative or hospice — each with its own practice code, and within each there are ward accommodation rates, special care unit rates, emergency and theatre unit charges, and rehabilitation and home health care rates. Prostheses, fixators and other artificial items are billed separately on bases such as the lowest available manufacturer's price, as are discharge medication and serious burns. Billing rules turn ward fees on whether admission or discharge falls before or after **12:00**, generally include pharmaceuticals supplied in the accommodation, **do not permit running accounts**, and require invoices to be supported by hospital records [27].

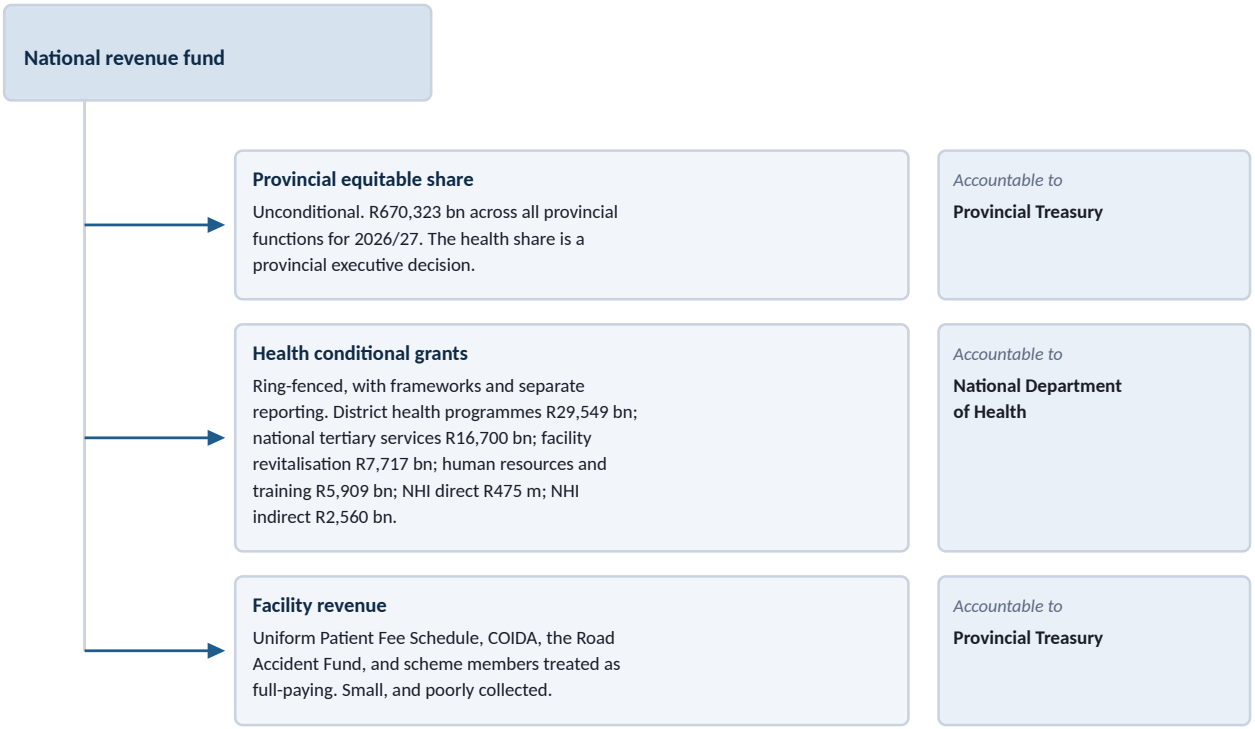
> One line in that structure changes a system design. **Minor and major theatre charges are billed per minute.** Theatre time in a COIDA case is therefore not an efficiency metric, it is a **revenue field**, and it must be captured to the minute from a defensible source. An architect who models theatre time only as a utilisation input has missed a billing input, and any theatre management system in a South African private hospital must produce auditable start and stop times rather than estimates.

There is a second, subtler link between COIDA and the clinical record. Every injury or poisoning code in the ICD-10 S and T chapters must carry an **external cause code**, coded to full specificity, whose fifth character records the activity — and activity code "2" means work for income, which **links directly to COIDA eligibility** [22]. The payer determination for an injury is therefore **partly encoded in the clinical coding**, so an architecture that treats payer identification as a registration-desk question and coding as a billing afterthought will get both wrong on exactly the cases where the money is largest.

The Road Accident Fund compensates for road accident injury and pays for associated medical care, and the Uniform Patient Fee Schedule classifies its patients as **externally funded full-paying** in public facilities [28]. Beyond that, the Fund's medical tariff basis and claim volumes could not be sourced for this book — an honest gap, closed by its own published tariff instrument.

The public flow: three sources of money with three sets of rules

The public flow: three sources, two accountability chains, 2026/27



Two chains, two calendars and two sets of data definitions for the same spend. Both views are legitimate, so the design has to hold both rather than reconcile one away.

Figure 6.6 The equitable share is unconditional and the grants are not, which is why a provincial health department reports the same spend twice, on two calendars, to two masters.

The provincial equitable share is an **unconditional** transfer from national revenue to each province, allocated by formula, which the province may spend on any provincial function. The health share of a province's budget is therefore a **provincial executive decision**. The total across all functions is **R649,339 bn (2025/26 estimate), R670,323 bn (2026/27), R698,626 bn (2027/28) and R720,409 bn (2028/29)**, averaging 3,5% annual growth [29]. The formula contains a health component whose **weighting is not stated** in the budget chapter retrieved for this book — a gap worth noting, because every provincial allocation argument turns on it.

Health conditional grants are ring-fenced, with frameworks, business plans, conditions and separate reporting to the national Department. For 2026/27, with three-year totals, in rand million [29]:

Grant	2025/26 adjusted	2026/27	2027/28	2028/29	Three-year total
District health programmes	28 929	29 549	30 790	31 787	92 126
National tertiary services	15 995	16 700	17 410	17 980	52 090
Health facility revitalisation	7 569	7 717	8 177	8 389	24 283
Human resources and training	5 650	5 909	6 174	6 385	18 468
National health insurance (direct)	467	475	497	513	1 485

On top sits an **indirect** national health insurance transfer of **R2,560 bn for 2026/27**, where "indirect" means national government spends on the provinces' behalf rather than transferring cash [29]. Two naming traps live in that table: the budget chapter lists **no separate HIV and TB grant** — HIV and TB sit inside **district health programmes** — and the "HIV and AIDS (life skills education)" grant in the wider tables is an **education** grant, not a health one [29].

Facility revenue is the third source: fees under the Uniform Patient Fee Schedule, statutory funders, and scheme members treated as full-paying. It is a small share of a provincial budget and collection is poor — the fee schedule itself provides for retrospective reclassification of scheme members who did not disclose membership [28], which tells you the leakage is known and material.

For context: the consolidated health budget — national plus provincial plus entities — was reported as **R310 bn for 2026/27**, from a revised R295 bn in 2025/26, growing 4,2% a year against a 3,3% average inflation estimate, with **R21,3 bn of R32,3 bn in additional provincial allocations** over the medium term going to compensation, employing doctors and shortfalls in goods and services [30][29].

The architectural consequence shows up in every reporting layer a provincial department builds. **Its financial systems must satisfy two accountability chains** — Provincial Treasury for the equitable share, the national Department for grant conditions and grant-specific reporting — with **different reporting calendars and different data definitions for the same spend**. That is not a data-quality problem to clean up. It is two legitimate views of one transaction, and the design has to hold both.

The Uniform Patient Fee Schedule, which is a workflow before it is a tariff

The fee schedule is the public sector's tariff and means-test instrument, and it is issued **provincially**. The Western Cape's 2026 instrument is Provincial Notice 114/2026, published 31 March 2026 and in operation from 1 April 2026 [28]. Gauteng's 2025 instrument was gazetted in February 2025, approved by the Provincial Legislature and effective 1 April 2025, with a **4,4% increase across all patient categories** and a **30% increase in standby tariffs for emergency medical services at major events** [31][32]. Limpopo issues its own departmental revenue circular [33].

Patients fall in two groups [28]. **Subsidised patients** are means-tested into **H0, H1, H2 and H3**. **Full-paying patients** pay the full tariff and split into **externally funded** — a medical scheme, the Road Accident Fund, workers' compensation, another state department, a foreign government, a research trial or an unregistered scheme — and **self-funded**. And some subsidised patients are **reclassified** as full-paying: scheme members and their dependants, patients treated by their own private practitioner, and patients covered by workers' compensation, a road accident or another state department.

The means test, on annual gross income, is three bands [28]: **H1** for an individual below **R70 000** or a household below **R100 000**; **H2** for an individual from R70 000 to under **R250 000** or a household from R100 000 to under **R350 000**; and **H3** at or above those upper figures. Married patients are assessed on combined income. **H0** patients are fully subsidised and receive all services free except listed exclusions, covering recipients of specified social grants and the formally unemployed supported by the Unemployment Insurance Fund, on proof — while **social grant recipients presenting only a grant card and an identity document are treated as H0 for three visits and then means-tested**. **Triage is not chargeable.** Some services are always charged at full rates whoever the patient is: medical reports, copies of X-rays, forms and certificates, cosmetic surgery, paternity testing, foreign travel immunisations, work evaluations, autopsies and mortuary fees [28].

The billing rules are where a system design is won or lost [28]. Inpatient tariffs depend on ward type and length of stay, in **12-hour periods** for full-paying and H2 and H3 patients, while **H1 patients are charged per 30-day period**. Admission or discharge before or after **12:00** determines a half or a full day, and a same-day admission and discharge before 23:00 draws the day-patient tariff. The facility fee includes consumables and ward stock; theatre, radiology, laboratory, physiotherapy, high-cost drugs and discharge medication are billed separately. On the outpatient side there is a consultation tariff per consultation and an emergency tariff for emergency or trauma visits; **H1 patients attending several clinics on the same day are assessed once, at the most expensive clinic, while H2, H3 and full-paying patients are assessed at each clinic**; outpatients later admitted are charged **both** fees; and **laboratory services are charged to full-paying patients regardless of where the service is given**.

Five reasons that is an architecture problem rather than a billing problem:

1. **The means test is a workflow**, performed at the front desk with documentary evidence, and it determines the price of everything that follows. It must be re-performed on a change of circumstance and, for grant-card-only patients, after three visits — a stateful, date-aware, document-backed classification sitting in front of every clinical encounter.
2. **The classification is retrospectively mutable.** A patient reclassified as a scheme member must be re-billed retrospectively, so a system treating classification as immutable at admission will understate revenue and fail audit.
3. **Charging granularity is 12-hour blocks for some patients and 30-day blocks for others**, with half-day rules keyed to a noon boundary. This is not a per-diem.
4. **Nine provinces, nine instruments, nine effective dates and nine increase percentages.** A national product cannot compile the fee schedule into code.
5. **Triage is free**, so the first clinical act at the emergency door generates no charge and does generate a record. The record cannot be driven by the billing event — the public-sector mirror of the private-sector lesson that a claim is not an activity record.

There is a reputational edge to this: public anger over a fee levied for opening a patient file at an Eastern Cape hospital made national coverage [34]. In the public sector the fee schedule is a politically live document, and so is any system that enforces it.

What unpaid invoices do to an architecture

Public-sector arrears are not an accounting curiosity. They are an operational constraint that appears in integration designs.

Accruals in provincial health peaked at almost **R25 bn in 2024/25** and were reduced to **R20,5 bn in 2025/26** only by adding to the equitable share [30]. The Auditor-General put accumulated unpaid bills above **R21 bn**, reported that **nine of the ten health departments audited had liabilities exceeding assets**, and found departments funding medico-legal claim payments by diverting money from medical supplies, maintenance and operating costs because few made specific budget allocations for claims [35]. Treasury extended the Eastern Cape a **R1 bn overdraft** against about R22 bn in liabilities, and Gauteng carries **R5 bn to R8 bn of accruals annually** [36][35].

Then there is the largest intercompany balance in the sector. Provinces collectively owed the **National Health Laboratory Service R10,7 bn** on one Treasury figure, **R10,6 bn as at June** on another, or **about R11 bn as at March 2026** on a third — three figures for the same debt [30][37]. The breakdowns are worse: one account to the health portfolio committee at end-May 2026 gives **KwaZulu-Natal about R3,94 bn and Gauteng about R3,3 bn**, while Treasury's figures to Parliament as at June 2026 give **Gauteng R3,5 bn, KwaZulu-Natal R1,2 bn and the Eastern Cape R1 bn** [38][37]. The two are irreconcilable at the KwaZulu-Natal line, the discrepancy is unexplained in both sources, and the laboratory service has acknowledged billing disputes and invoice-processing issues [37].

> An intercompany reconciliation difference of roughly R2,7 bn between two arms of the same state is what happens when two organisations bill each other without a shared service catalogue and a shared patient identifier. It is also the best available answer to a client who asks why a facility master and a service catalogue are a work package rather than an assumption.

Three consequences follow. **A province that cannot pay cannot be assumed able to procure**, so a roadmap whose second year depends on a capital allocation should carry that as a named risk. **A maintenance budget that exists on paper and is diverted in practice** is why asset and maintenance data in a provincial hospital does not describe reality. And **the licence to spend is not the licence to transact**: irregular expenditure fell from R10 bn to R6 bn annually while the cumulative total kept rising, and **90% of the sector had material non-compliance with legislation** [36]. A procurement workflow there operates where compliance failure is normal, so automating procurement without enforcing the compliance gates automates the finding.

What the money flow requires of the architecture

Eight requirements, each traceable to something sourced above.

1. **Model tariff as contract data**: payer, provider, agreement, service code, effective period. No `price` column on a service master.
2. **Effective-date every code set and benefit rule set**, including the Master Industry Table with its per-code South African start and end dates, and validate historical claims against the table that was current when they were raised.
3. **Keep the clinical assertion and the code as separate, linked objects**, each with its own author, date and purpose; never let a billing code update a clinical one.
4. **Model the remittance**, because the national claim standard does not. Without it nobody can compute a denial rate, analyse denial reasons or reconcile cash to activity.
5. **Model the claim as a versioned object**, with adjustments, reversals, resubmissions and interim accounts as first-class states, because the message carries all four and keeping only the latest state destroys the analysis that would fix the root cause.
6. **Capture theatre start and stop times to the minute, from a defensible source**, because under COIDA they are a revenue field.
7. **Treat patient classification as a dated, evidenced, retrospectively mutable process** — in the public sector because the means test is, in the private sector because benefit option and network membership are.
8. **Name the dependencies the organisation does not own**: the switch's interface specification, the medicine product identifier held by a claims switch, the gazetted medicine price event, the Master Industry Table, and for insurance products an exemption expiring on 31 March 2027.

THE CHAPTER IN TEN LINES

- A medical scheme is not an insurer. Members pay contributions, there is no underwriting engine, and prescribed minimum benefits must be paid in full on every option.
- The scheme is the responsible party for the claims data; the administrator that adjudicates and pays is an operator.
- Prescribed minimum benefits were 52,31% of benefits paid in 2024, at R135,66 bn; state 271 diagnosis-treatment pairs, and note the chronic disease list count as both 26 and 27.
- There has been no national tariff since 28 July 2010. The court struck down the enabling regulations, not the prices.
- Tariff is contract data: payer, provider, agreement, service code, effective period. There is no defensible price column.
- Capitation needs five capabilities South Africa does not operate at scale, so the choice between payment models is not the real problem.
- The claim is a UN/EDIFACT MEDCLM message that defines no remittance, which is why denial-rate analysis has to be rebuilt per funder.
- The operative contract is the switch's own XML schema, not the published EDIFACT standard, and batched against real-time is a per-funder variable.
- ICD-10 is required by a funder regulation and the obligation is conditional on payment: an uncoded claim is unpayable, not unlawful. The decisive date is 1 July 2014.
- Short payment, not rejection, is the dominant failure mode, and almost nobody measures it.
- Under COIDA theatre time is billed per minute, so it is a revenue field and not only a utilisation metric.
- The public flow is an unconditional equitable share plus ring-fenced grants plus thin facility revenue, reported twice to two masters.
- The Uniform Patient Fee Schedule is a means-test workflow with retrospectively mutable classification, nine provincial instruments and free triage.

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CHAPTER 7

NHI, and why it is not the reason for the work

A statute with no section in force, a digital health budget line of R6,5 million, and a requirement set that is already law.

The one-sentence version

The National Health Insurance Act 20 of 2023 is law. No section of it is in force. Its own digital health budget line for 2026/27 is **R6,5 million**. Two of the mechanisms the state itself described as central are either struck down or still undefined. And every single requirement an NHI architecture would impose on a provider is **already mandated today**, by instruments in force today, with a named enforcement record behind each one.

That is the argument of this chapter, and it has a practical consequence. An architect who justifies a records programme, an identity programme or a coded-encounter programme by reference to NHI has anchored the business case to a court judgment nobody can date. An architect who justifies the same programme by reference to POPIA section 22, National Health Act section 17, Office of Health Standards Compliance certification, Regulation 5(f) of the Medical Schemes Act and the Auditor-General's findings on missing records has anchored it to things that are true this morning.

Build the second case. NHI then becomes a contingency that validates the work rather than the reason for it, and the work survives whichever way the judgment goes.

> Everything in this chapter is stated as at **31 October 2026** and the most important fact in it is unresolved. The Constitutional Court heard the two public-participation challenges on 5 to 7 May 2026 and reserved judgment [7]. As at 31 October 2026 it had not been delivered [8]. **Re-check the Constitutional Court's judgment in CCT 251/25 and CCT 269/25 before you use any page of this chapter in a client deliverable.** The published list of that year's Constitutional Court judgments is the quickest check [8]; the court's own case listing is the authoritative one [6]. If judgment has been delivered, the whole of the next three sections needs re-reading, and the argument in the last two sections does not change at all.

Law, regulation, policy, annual report, advocacy

The single most common factual error in architecture documents about South African health care is reading a consolidated statute without its commencement notes. The second most common is quoting a departmental annual report as though it were an Act. Before any of the substance, fix the vocabulary.

Six distinctions, and they produce opposite build decisions:

- **Law** — an Act of Parliament. Binding when, and only when, the relevant sections have commenced.
- **Regulation** — a gazetted instrument made under an Act. Binding, and usually where the operative detail lives.
- **Policy** — government's stated intention. Not binding on third parties. Frequently overtaken by events without being withdrawn.
- **Guideline** — professional or departmental guidance. Persuasive, and often enforceable through a disciplinary route rather than a court.
- **Departmental report** — an annual report or a parliamentary presentation. Evidence of what the department says it has done. Not an instrument.
- **Advocacy** — an assertion by an interested party. Useful for understanding the argument. Worthless as a fact.

What is actually in force, and what is something else

Eight provenance tiers. An NHI statement is only as strong as its tier.



Figure 7.1 The provenance ladder applied to NHI. Each tier carries a different build decision, and conflating them is the commonest error in architecture documents about South African health care.

Apply that ladder to NHI and the picture resolves quickly.

The Act is law. The Bill was passed by Parliament in December 2023 [31] and assented to by the President on 15 May 2024 [5].

Nothing of it has commenced. This is stronger than most commentary implies. It is not that some sections are in force and others are not. By letter dated 20 February 2026 the Office of the State Attorney in Pretoria confirmed that the President "will not commence any sections of the Act" before the Constitutional Court rules on the public-participation challenges; the Minister had said the day before that he would not ask the President to promulgate any sections while the cases were pending [1]. The Presidency confirmed on 24 February 2026 that the President, after consulting the Minister, had agreed to delay the proclamation of any sections until judgment [2]. That day Judge Brenda Neukircher made the agreement an order of court, the challengers agreeing to stay their cases and not to bring new challenges without the court's leave in exceptional circumstances [3]. One of the applicants characterised the outcome as the state conceding to suspension [37] — that characterisation is advocacy; the order is a fact.

It was restated in October 2026: the National Department of Health told Parliament's Select Committee on Social Services that the committee "will not see the implementation of NHI-specific interventions until the Constitutional Court has issued a judgment", and that full rollout has effectively been on hold since February 2026 [4].

So, plainly: **as at 31 October 2026 the National Health Insurance Act is on the statute book and no section of it is in force.** There is no NHI Fund, no accreditation regime, no contracting units established under the Act, no central payment, and no change to medical schemes.

And there is a trap inside the trap. The consolidated National Health Act carries *uncommenced* amendments made by the NHI Act — an amendment to the "health agency" definition by section 58(1) of the NHI Act, to take effect by proclamation [9], plus amendments or deletions to many section 25 provincial functions and the insertion of a section 31A creating District Health Management Offices [12]. A consolidated statute read without commencement notes will describe a health system that does not exist. If a strategy document in the client's repository describes District Health Management Offices as part of the organisational structure, somebody read the consolidated text.

The litigation, and what the one decided case actually decided



Figure 7.2 The NHI litigation to 31 October 2026. The matter decided (red) concerned the National Health Act, not the NHI Act; the two matters reserved concern public participation.

Two matters are before the Constitutional Court on public participation: **CCT 251/25, *Board of Healthcare Funders NPC v Speaker of the National Assembly and Others***, brought by the body representing medical schemes and administrators; and **CCT 269/25, *Premier of the Western Cape Government v Chairperson of the National Council of Provinces and Others***. Both were heard 5 to 7 May 2026 and judgment was reserved [7][6]. One commentator expected judgment around November 2026 [31] — that is a prediction, not a court listing, and should be presented as such.

The arguments matter because they shape what a remedy could require. The Board of Healthcare Funders argues that there were no changes of significance between the 2018 and 2019 Bills, that insufficient information on the benefit package, operation and costing was put before the public, that Parliament approached the matter with a closed mind, and that the Public Participation Framework was not complied with [6]. The respondents' record is substantial: hearings in all nine provinces, 33 in the National Assembly and 60 in the National Council of Provinces, about **338 891 written submissions**, **961 oral submissions** heard in the National Assembly, twelve clause-by-clause meetings and **35 changes** to the Bill [6]. On costing, the Department argued that costing a 25-year project was futile; the Board says costing could readily have been sourced; the respondents say costing belongs in budget and Money Bill processes rather than the legislative phase [6]. The Western Cape seeks the entire Act set aside and sent back to Parliament for re-enactment; the respondents say that if a defect exists the remedy should be remittal to the National Council of Provinces only [32].

That difference is the one an architect should watch. A remittal to one House is a delay. An order setting the whole Act aside for re-enactment restarts the policy argument.

The case that was decided did not concern the NHI Act at all, and it is the more instructive one. In *Solidarity Trade Union and Others v Minister of Health and Others*, reported as (2026) ZACC 19 (CCT 240/24), heard 9 September 2025 and decided 18 May 2026, a unanimous nine-judge court, Savage J writing, confirmed the High Court's declaration of invalidity and **severed sections 36 to 40 of the National Health Act 61 of 2003** — the certificate-of-need scheme — with costs against the Minister and Director-General [9][8]. The applicants were Solidarity, the Alliance of South African Independent Practitioners Association, the South African Private Practitioners Forum, several doctors, and the Hospital Association of South Africa [11].

The scheme was struck down as irrational and as an unjustifiable limitation of section 22 of the Constitution, the right freely to choose a trade, occupation or profession [9][10]. Severance caused no administrative disruption because the provisions had never operated; no regulations were ever made under them. Parliament may legislate again, within constitutional limits [10].

> **Three of the court's four grounds are reusable against any replacement instrument, and an architect should know them, because they predict what a future licensing or accreditation regime can and cannot do.** First, leaving the scheme's scope to regulations that were never made was an impermissible delegation. Second, the thirteen statutory factors the decision-maker had to weigh omitted providers' own rights and interests. Third, the state offered no objective evidence that existing licensing and norms-and-standards mechanisms could not achieve the same quality aims [9][10]. The third is the one with teeth for a data architect: it means a future instrument has to be justified against the evidence the *existing* instruments already produce. Which makes the evidence the existing instruments consume — inspection records, compliance notices, incident registers, maintenance schedules — more valuable, not less.

What it means is disputed, and both positions are advocacy. The Minister and Director-General had described certificate of need as "a central pillar in the implementation of the National Health Insurance Act". Afterwards the Department's spokesperson said the judgment does not affect the NHI Act, that the sections were passed 23 years ago and never brought into effect, and that no part of the NHI Act had been declared unconstitutional [11]. Solidarity and others say a central pillar has been removed. The verifiable facts are narrower and sufficient: the sections are gone; the state had called them central to NHI; the court did not rule on the NHI Act.

There is one more consequence worth stating because it changes a constraint most architects assume is fixed. With certificates of need severed, **private health establishment supply in South Africa is now regulated only by Regulation R158 of 1980**, made under the Health Act 63 of 1977, section 44 of which was wholly repealed in 2012, with provincial authority delegated by Proclamation R152 of 1994 — and two provinces cite different amendment years for R158. A bed count or a theatre count in a private group's licence is governed by a forty-six-year-old regulation under a repealed section.

The wider volume of litigation is itself a planning fact. The Minister told Parliament the department had set aside **R74 million for 2026/27** for NHI legal challenges, across fifteen cases averaging R2,5 to R3 million each, engaging six senior counsel, ten junior counsel and attorneys [14]. Another report cites fourteen court challenges and departmental fears of fifteen to twenty years of hearings, judgments and appeals [15]. The Department reported four High Court applications lodged immediately after assent [5]. The challenged grounds, as listed in October 2026, are the public participation process, the rationality of signing the Act into law, various sections relating to private medical schemes, sections relating to specialist services, and overall constitutionality [4].

What can be sourced to the Act, and what cannot

Here is a correction that matters more than it looks. The Fund's architecture is routinely described in five parts: user registration, accreditation, contracting units for primary health care, a formulary, and payment mechanisms. **Only one of those five can be sourced to the Act.**

The best retrievable source on the Act's provisions in this research was a Constitutional Court filing in CCT 251/25 [6]. It describes accreditation, at section 39. It **does not describe user registration, contracting units for primary health care, the formulary, or payment mechanisms** — four of the five. What it does describe, with section numbers:

Provision	What the filing says it does	Confidence
s57(1)(a)–(b)	Implementation in two phases, coverage phased in gradually, subject to financial resource availability	From a court filing [6]
s25(5)	A Benefits Advisory Committee determines and reviews benefits and service types at each level of care, and sets treatment guidelines	From a court filing [6]
Chapter 10	Financial matters and the sources of funding	From a court filing [6]
s39	Accreditation of service providers; the 2018 Bill's link to certification under National Health Act s36 was removed in 2019	From a court filing [6]
s33	The role of medical schemes	From a court filing [6]
s35	Ambulance and emergency medical services definitions	From a court filing [6]

Two cautions attach to that whole table. First, the Act's own text could not be retrieved in the research behind this book, and the filing discusses the Act and earlier Bill versions interchangeably, so **every section number above must be verified against the gazetted Act before it reaches a client deliverable**. That is an unverified item, and what would settle it is the gazetted Act itself. Second, note what the section 39 history tells you: the Bill's drafters had once linked NHI accreditation to the certificate-of-need regime and then de-linked it in 2019, five years before the regime was struck down.

Contracting units for primary health care are real, and they are sourced to the Department's annual report rather than to the statute. The Department records a draft primary health care package "being phased in at Contracting Units for Primary Health Care (CUP) sites across the country", a draft accreditation framework awaiting Technical Committee and National Health Council review, a draft primary health care contract pending finalisation of **the benefits package and the capitation model**, a user manual in preparation, and Governance Regulations for the Board and two advisory committees translated into three languages and published for comment [5]. Attribute each of those to the annual report, not to the Act. They are evidence of departmental activity, at a stated level of completion, and that is exactly what makes them useful: a draft is a statement about what is not yet decided.

Section 33 and medical schemes. The provision is widely described as confining medical schemes to complementary cover once NHI is "fully implemented" [31][11]. Both of those are secondary accounts. The trigger condition — what "fully implemented" means and who determines it — is the entire commercial question for the scheme industry, and it should be read off the section rather than off a paraphrase of it.

The money, which settles the argument faster than the law does

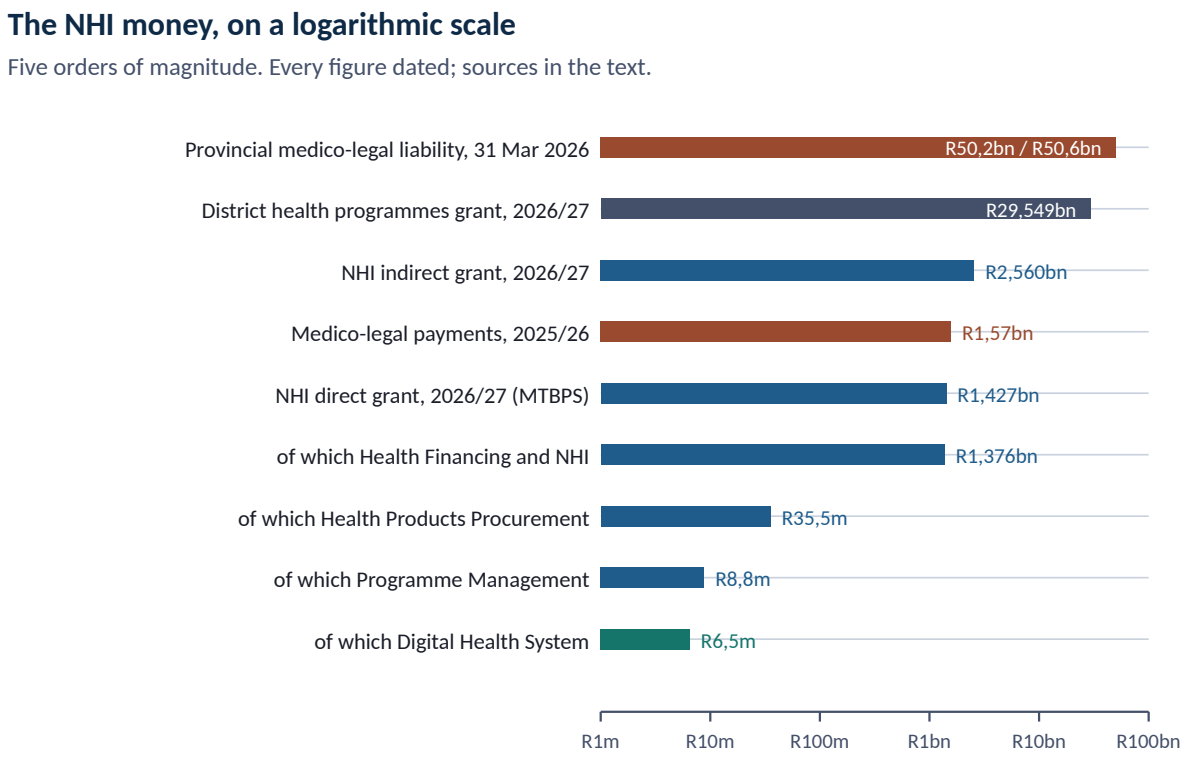


Figure 7.3 The NHI allocations for 2026/27 after the October 2026 Medium Term Budget Policy Statement, on a logarithmic scale, with the medico-legal liability and the annual claim payments for scale.

The 2026 Budget allocates health conditional grants for 2026/27 as follows: District Health Programmes R29,549 billion, National Tertiary Services R16,700 billion, Health Facility Revitalisation R7,717 billion, Human Resources and Training R5,909 billion, and National Health Insurance (direct) **R475 million**, plus an indirect NHI transfer of **R2,560 billion** [13]. "Indirect" means national government spends on provinces' behalf rather than transferring cash.

The October 2026 Medium Term Budget Policy Statement adjusted the 2026/27 direct allocation to **R1,427 billion**, a net reduction of R7,3 million, split: Health Financing and NHI R1,376 billion; Health Products Procurement R35,5 million; Programme Management R8,8 million; and **Digital Health System R6,5 million** [4]. On the indirect side the Health Facility Revitalisation grant was R275 million higher, at just under R8 billion [4].

Two honest notes. The direct-grant figures as reported are hard to reconcile with the Budget Review's own table, and that source's own account of direct funding is internally inconsistent — it gives R1,43 billion over three years in one place and "around R500 million over the medium term" in another, without explanation [4]. That inconsistency is unresolved in this research and what would settle it is the Division of Revenue Act schedules. Cite both figures or neither.

What is not in dispute is the digital health line. **R6,5 million.** Put that beside the only other sourced NHI technology number: a Council for Scientific and Industrial Research contract for an NHI information system, with **38% of R84 million used** as at 2024/25 [5]. And beside the Department's NHI programme spend for 2024/25: a final appropriation of R1 355,5 million against actual spend of R1 168,7 million [5].

Now set R6,5 million against the arithmetic this book keeps returning to. In an Auditor-General audit of 27 facilities across nine provinces, fifteen reported missing or lost records, seventeen did not track records after issue, twenty-five had inadequate storage, and **in three of nine paid claims reviewed, departments conceded because they could not locate the medical record** [19]. Provincial medico-legal contingent liability at 31 March 2026 is published as R50,2 billion by the Auditor-General [19] and R50,6 billion by National Treasury [20] — two measurement bases, same day, both official. Payments were R1,57 billion in 2025/26 [20].

A records-retrieval capability that prevented one conceded claim would fund more digital health than the national programme has budgeted for the year. That is not rhetoric; it is division.

There is no official total cost figure for NHI and the book should say so. The Department argued that costing a 25-year project was futile; the Board of Healthcare Funders disputed that; the court heard argument on it, and one judge is reported to have called the scheme "frightfully expensive" from the bench [31]. A remark from the bench, reported secondhand, is not a finding.

What is being built anyway, and under which authority

The Minister's position, stated repeatedly, is that the pause on the Act is not a pause on preparation: "When we paused, we did not say we would stop our preparation" [15]. On 9 October 2026 he told the portfolio committee that hospital infrastructure, equipment and human resource capacity should not be put on hold, that "whether NHI is going to be there or not, the health system needs to be strengthened", and that the department "would fail public healthcare users if it did not introduce electronic health records" [16].

The named workstreams continuing are infrastructure for three academic hospitals serving historically black medical schools, in eThekwini, Tshwane and Mthatha, with Limpopo "already done" — the Limpopo Central Academic Hospital was at 26% of construction in July 2025 and is described as the country's eleventh central or academic hospital [17]; seventeen major hospital projects in design or development [15]; the CSIR patient-tracing and health information system [5][15]; private healthcare price regulation [15]; and the identity, pharmacy and programme systems covered elsewhere in this book — the Health Patient Registration System, biometric identification, the TB and HIV electronic medical record at 37 pilot facilities as at March 2025, and centralised chronic medicine dispensing [5].

> **Opposition framing to be aware of, and not to repeat.** The MK Party questioned an NHI allocation citing a claim that 79% of health facilities are non-compliant; the Democratic Alliance described the core problem as managerial incompetence and corruption and warned of five to ten years of litigation [17]. The 79% figure is a party claim reported in a parliamentary summary. The comparable sourced figures are the Office of Health Standards Compliance's certification rate of **19,6%** over 2019/20 to 2022/23 [25] and the finding that **14,7%** of inspected facilities met the minimum standard in 2019/20. Do not reproduce 79% as a fact.

The important point about this list is what it establishes about authority. None of it is being done under the NHI Act, because the NHI Act is not in force. Every item is being done under the ordinary powers of a national department and the ordinary conditions of a conditional grant. Which is the chapter's argument arriving from the other direction: the work does not need the Act.

If it proceeds: the requirement set, and what exists

Suppose judgment goes the state's way and sections commence. What would a provider actually have to build or change? This can be answered honestly from the sourced provisions plus the Department's own draft artefacts, and the honest answer is that the information requirements are substantially harder than the financing requirements.

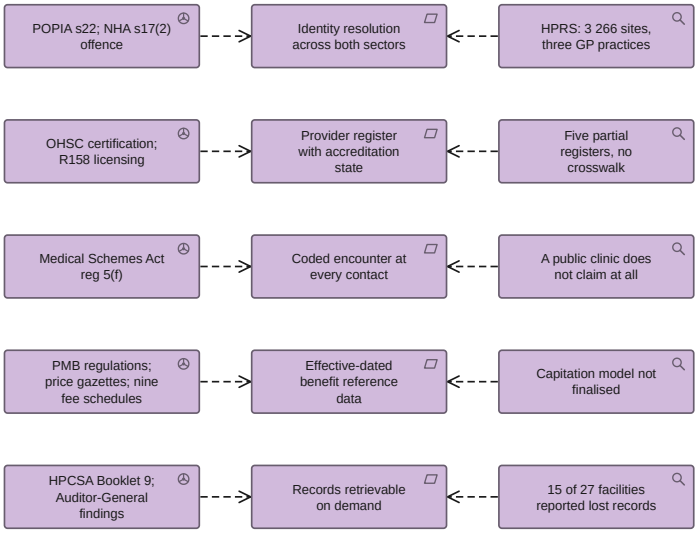


Figure 7.4 Five of the requirements an NHI architecture would impose. Left: the instrument that already requires it today. Middle: the capability. Right: what exists now.

1. A resolvable national person register. Driven by the single-purchaser design. The substrate exists only partially: the Health Patient Registration System had reached **3 266 establishments by 31 March 2025 — 3 150 primary health care facilities, 113 hospitals and three GP practices** — with an Automated Biometric Identification System at **82 primary care facilities** [5]. The private sector is not covered at all, and **no national crosswalk exists**. The work is not "install a registry". It is identity resolution across nine provincial estates, roughly 3 700 public establishments, the scheme and administrator population, the claims switches and tens of thousands of private practices. That is chapter 11's subject, and the point to carry here is that **the identifier problem is a precondition of NHI, not a deliverable of it.**

2. A provider and facility register carrying accreditation state. Driven by section 39 [6]. What exists is five partial registers with different keys: practice code numbers administered for the private sector, PERSAL for public employees, professional council registers for individuals, provincial licensing registers for private facilities, and Office of Health Standards Compliance certificates for both sectors. None of these is a provider register. One of them — the certificate — would be the gating attribute. An architect should expect to build both the crosswalk and the accreditation state machine: applied, inspected, certified, notice issued, re-inspection due, suspended.

3. Contracting units for primary health care as entities with contracts, catchments and money. A contracting unit is a new kind of thing in South African health data: a contracting entity spanning a district hospital and its referring clinics, with an enrolled population, a capitation rate and a service scope. Nothing in a current provincial system models it, because facility hierarchies model geography and ownership, not contracting [5].

4. A capitation engine, with everything capitation needs. Driven by the Department's own statement that the primary care contract is pending finalisation of the benefits package and the capitation model [5]. "Capitation versus fee-for-service" is the wrong framing. Capitation needs five capabilities South Africa does not operate at scale in either sector: enrolment with join and leave dates; attribution rules; risk adjustment; **encounter**

capture decoupled from payment; and leakage detection. Fee-for-service dominates today despite managed care arrangements covering **99,19% of beneficiaries** [33]. Naming the five missing capabilities is more useful than describing two payment models as a choice.

5. A national benefits authority with versioned, effective-dated reference data. Driven by section 25(5): a Benefits Advisory Committee determines and reviews benefits at each level of care — primary care facilities and district, regional and tertiary hospitals — with the basket not fixed in the legislation and determined incrementally, depending on resources [6]. Read that as a data requirement and it is severe: every claim, authorisation and decision-support rule must be evaluated against the version in force **on the date of service, at the level of care where the service was given**.

6. Central payment with provider-side reconciliation. The existing national pipeline is the private switch network; the public sector has nothing comparable, because a public clinic today does not submit claims at all. Making a public primary care facility into a claiming entity — producing a coded, priced, attributable encounter record for every contact — is the largest single change NHI would impose on the public sector, and it is a behaviour change long before it is a software change.

7. Certification as a hard gate, at scale. In 2023/24 the Office of Health Standards Compliance inspected **734 public establishments, 19,62% of 3 741**, and 60 private establishments; issued 618 certificates of compliance; and ran 79 re-inspections [18]. At that rate, certifying the public estate once takes roughly five years, before private facilities and GP practices are reached. If certification gates accreditation and accreditation gates payment, the inspection pipeline becomes the rate limiter on the entire reform, and inspection throughput becomes an architecture problem — scheduling, evidence capture, re-inspection tracking, certificate validity.

8. Clinical coding capacity in the public sector. The private sector has coded to ICD-10 since 1 July 2005 because claims depend on it [22]. A public clinic has no such incentive. The Department's own outstanding list records a **national coding regulation "in development"** and a national coding qualification being updated [21], and that status statement is itself more than a decade old.

9. The missing national specifications. Mapped against a reference component set, a single-payer purchaser would need a beneficiary registry carrying entitlement, a provider registry, a facility registry, a product and service catalogue with prices, a claims and payment platform, a clinical data substrate sufficient to adjudicate clinically, a terminology service, fraud and abuse analytics, and a consent and access-control regime. The current estate satisfies perhaps two of those nine — a national patient index and an aggregate reporting system. The four genuinely missing dependencies are **a governed national procedure code set with prices** [28], an **operating terminology release centre**, a patient-level clinical record at national coverage, and a provider accreditation registry. Each is a multi-year programme with no public delivery date.

> **The second item carrying a re-check instruction.** South Africa joined SNOMED International on or about 14 May 2026 [29]. As at the research date the **National Release Centre is not yet operating** — no national extension, no reference sets, no maps, representatives listed as to be advised. That status must be re-checked before any terminology commitment is made on the strength of membership. Meanwhile the gazetted interoperability framework names neither SNOMED CT nor LOINC [30], which is the contradiction chapter 16 carries.

And the data requirements themselves are not specified. The Act mentions health records only a few times; section 6(c) lets users access records held by the Fund to exercise or protect their rights; section 34(3) requires health workers and facilities to follow National Health Act rules on access to and protection of records; and the Fund's information platform must support the portability and continuity of services [27]. The published assessment of those provisions is that they "offer very little detail as to how electronic health records should be created, retained, stored and protected" and that the Act lacks detail on audit trails, encryption and disaster recovery [27]. The only published statements of intent are the 2017 White Paper's national health information repository drawing a subset of data from a national electronic health record [26], and the Department's own note that the diagnostic, treatment and billing modules an electronic health record would need under NHI are not yet developed [26].

So: do not design to a published NHI data dictionary, because there is not one. Design to the two published national specifications that do have force — the identity system and the interoperability framework — and expect a defined subset extraction rather than a full feed.

The argument: it is all already required

Here is the table that should replace the NHI slide in every architecture deck.

What an NHI architecture would need	What already requires it today	Enforcement record
Identity resolution and access control	POPIA s22 breach notification; National Health Act s17(2), under which unauthorised access to a health record is a **criminal offence** carrying a fine or up to one year	Six administrative fines reported at the Regulator's 31 August 2026 briefing, including **Lancet Laboratories, R100 000, paid** — the health sector's first [23]
A facility and provider register	OHSC certification; provincial licensing; practice code numbering	734 public and 60 private establishments inspected, 618 certificates, 45 written warnings, 2023/24 [18]
Coded encounter capture	**Regulation 5(f) of the Medical Schemes Act 131 of 1998** — a claim without the diagnostic code may not be considered for payment	Age and gender validation from 1 July 2014, with **full claim rejection on clinical data** [21]
Effective-dated benefit reference data	Prescribed minimum benefit regulations; single exit price gazettes; nine provincial fee schedules	A gazetted price adjustment is a dated, item-level event affecting thousands of lines on one day [34][35]
Retrievable records	HPCSA Booklet 9, a professional guideline [24]; National Health Act s13, which makes record-keeping subject to the National Archives Act and the access-to-information regime	**Three of nine paid claims conceded for want of a record**, across 27 facilities in nine provinces [19]

Four things about that table are worth saying out loud.

First, the coding mandate is not a health statute and it is not penal. It is a funder regulation, and the obligation is conditional on payment: an uncoded claim is unpayable, not unlawful [22][21]. That single distinction explains the whole coding-for-billing pathology this book describes in chapter 12, and it also explains why the public sector has no coding capacity — nobody has ever made it a condition of anything a public clinic wants.

Second, the access-control argument is a criminal-law argument, not a privacy argument. Unauthorised access to a health record is an offence under the National Health Act independent of POPIA. That is the strongest sentence available to an architect proposing access auditing, and it costs nothing to say.

Third, the enforcement record is now real and it is recent. POPIA enforcement against private bodies has begun. The dominant findings across the reported matters are **section 22 notification failure** and **section 15(1) incompatible further processing**; later in October 2026 a bank was fined R1 million and a college R500 000 [23]. An earlier matter against a government department — a R5 million fine for, in substance, letting antivirus, intrusion-detection and monitoring licences lapse — is the most directly usable precedent in the country for an argument about unglamorous security hygiene [23].

Fourth, and this is the one that changes behaviour: the POPIA health regulations in force since 6 March 2026 do not bind health care providers. The Regulations Relating to the Processing of Data Subjects' Health Information by Certain Responsible Parties, published in *Government Gazette* 54268 on 6 March 2026 and effective the same day with no grace period, bind insurers, medical schemes, scheme administrators, managed healthcare organisations, administrative bodies, pension funds, employers and institutions acting for them [36]. A hospital is not on that list. A starter pack that told hospital architects the new health regulations applied to them would be wrong, and the reason is structural: the regulation-making power is confined to two subsections of POPIA section 32, which is chapter 18's subject.

What to do with this chapter

A decision rule, and then a posture.

The rule. For any proposal justified by NHI, ask one question: *which instrument in force today requires this?* If the answer names an Act, a regulation or a professional guideline with an enforcement record, the proposal stands on its own and NHI is a bonus. If the answer is "NHI will require it", the proposal is a bet on a reserved judgment, and should be presented that way to the person signing it.

The posture. Build what is valuable on its own terms and would also be necessary under NHI. Every item in the requirement set above — identity resolution, a facility and provider register, coded encounter capture, effective-dated benefit reference data, and records that survive an inspection and a lawsuit — is justified today by POPIA, the National Health Act, the standards-compliance regime, the medico-legal exposure and the Medical Schemes Act regulations, **without any reference to NHI at all**.

The certificate-of-need judgment is the cautionary tale, and it is worth keeping in mind for a decade. The state described the scheme as a central pillar of NHI. It was never commenced, no regulations were ever made under it, and on 18 May 2026 it was severed [9][10]. Any architecture anchored to a named mechanism rather than to a named requirement loses its anchor when the mechanism goes. Requirements are more durable than mechanisms, and in South African health care they have been more durable than most architects expect.

Driehoek Health Group, the fictional private hospital group this book uses for its private-sector examples, has a slide in its 2026 strategy pack titled "NHI readiness". The slide lists five capabilities. All five are good. None of them needs the slide's title, and the title is the only part of it that will be wrong in a year.

THE CHAPTER IN TEN LINES

- The NHI Act 20 of 2023 is law and no section of it is in force, by presidential undertaking of 20 February 2026 made an order of court on 24 February 2026.
- The Constitutional Court heard the two public-participation challenges on 5 to 7 May 2026 and judgment was still reserved at 31 October 2026. Re-check it before you rely on this chapter.
- Certificates of need are dead: National Health Act sections 36 to 40 were declared invalid and severed on 18 May 2026, unanimously. They had never operated.
- Three of the court's four grounds are reusable, including the absence of evidence that existing licensing and standards mechanisms could not do the job.
- Of the five fund-architecture elements commonly listed, only accreditation can be sourced to the Act. Contracting units are sourced to a departmental annual report as live sites.
- Attribute every NHI statement to its tier: law, regulation, policy, departmental report or advocacy. A consolidated statute read without commencement notes describes a system that does not exist.
- The 2026/27 direct NHI grant's Digital Health System line is R6,5 million. The only other sourced technology figure is 38% of an R84 million CSIR contract.
- Provincial medico-legal liability at 31 March 2026 is R50,2 billion on one official basis and R50,6 billion on another. Payments were R1,57 billion in 2025/26.
- If NHI proceeds, the information requirements are harder than the financing requirements, and the four genuinely missing dependencies have no public delivery dates.
- Identity resolution is a precondition of NHI, not a deliverable of it: 3 266 public establishments, three GP practices, 82 biometric sites, no private coverage, no national crosswalk.
- Every requirement an NHI architecture would impose is already mandated by an instrument in force, with a named enforcement record, including the health sector's first POPIA fine.
- The POPIA health regulations in force since 6 March 2026 bind funders and employers, not providers. Do not tell a hospital architect otherwise.
- For any NHI-justified proposal, ask which instrument in force today requires it. If none does, it is a bet on a reserved judgment.

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CHAPTER 8

The operating model and the clinical cadence

A provider runs continuously. Everything that observes it runs in periods. That mismatch is where the defects live.

A provider has no closed period

Everything an architect is used to designing around assumes a period: a trading day, a month-end, a financial year with a close, a freeze and a quiet fortnight in which to deploy.

A hospital has none of that. It runs continuously, and the things that observe it run in periods. That mismatch is the source of a repeating family of defects: metrics that move when nothing changed, reports that disagree with the ward, release calendars that collide with statutory deadlines, and reconciliation differences nobody can close because the two sides were never counting the same interval.

This chapter is about time — not the processes, which chapter 9 takes as the five journeys, and not where the money comes from, which is chapter 6. It asks four questions of every part of a provider's operation: what is its cadence, what is its clock, what is its planning horizon, and what has to be decided before it.

Three kinds of clock run at once, and they are worth naming because people confuse them.

- **The clinical clock.** Minutes to days, continuous, driven by physiology and the shift pattern. It never stops and has no reporting period.
- **The administrative clock.** The reporting month, the inspection year, the audit cycle, the budget year. Discrete, with a close and a cut-off.
- **The statutory clock.** Fixed intervals attached to named obligations, in hours, days, working days or months, each under its own instrument with its own consequence for breach.

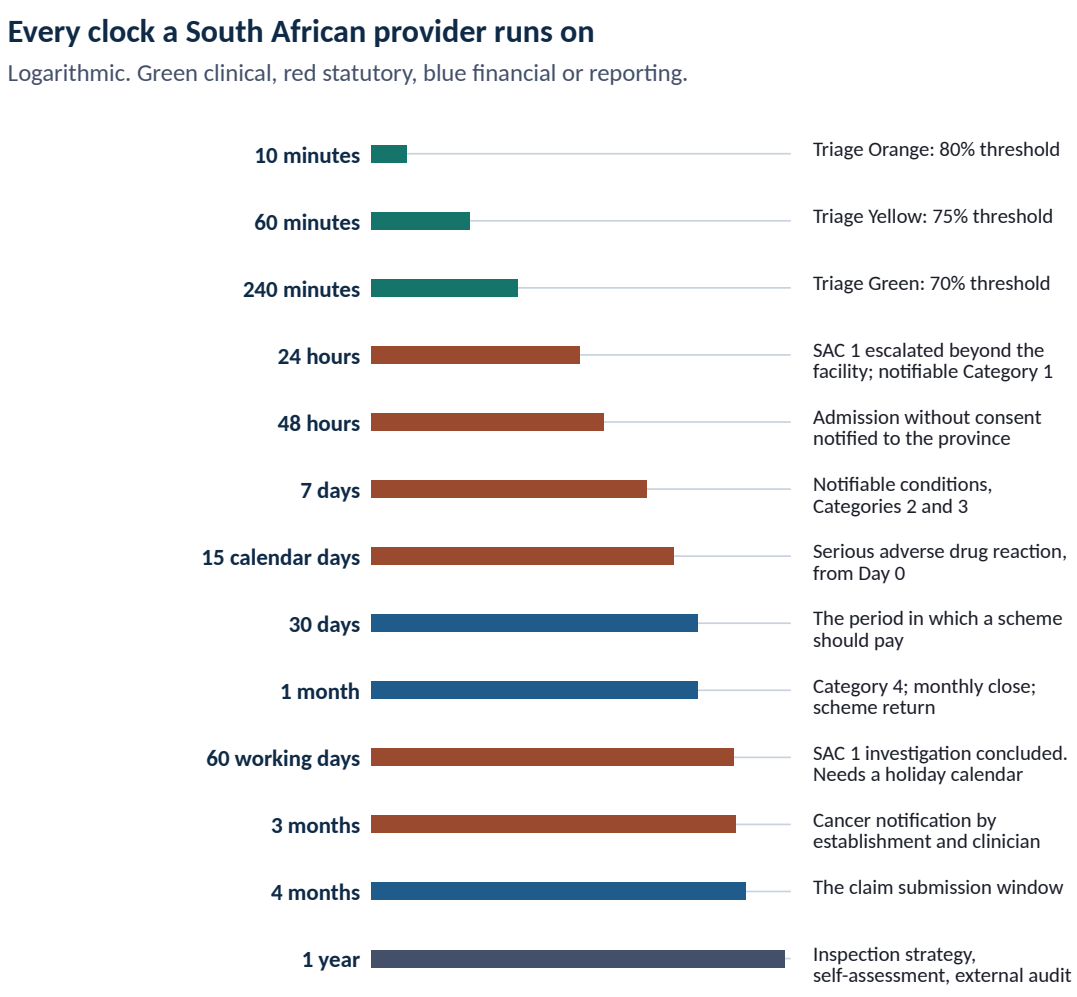


Figure 8.1 Thirteen clocks a South African provider runs on simultaneously, on a logarithmic scale. Each is attached to a named obligation with its own consequence for breach.

Rietvlei District Health Service — the fictional public district this book uses for its public-sector examples — runs a district hospital, three community health centres, 22 fixed clinics and a mobile service. At any moment it is inside a clinical shift, a reporting month, a 60-working-day investigation clock,

an inspection year and a fiscal year ending on a different date from its private neighbour's. Nobody holds those calendars in one place. That absence is the artefact this chapter asks you to produce.

The clinical day and the shift

The unit of continuity in a hospital is the shift, and the riskiest moment is the handover between two of them. Three consequences follow for information design.

First, access control in a hospital is really assignment control. A doctor should be able to read the records of patients on their ward this week. That is a roster statement, not a role statement. Without a roster feed, role-based access degrades to "all clinicians can see everything", compensated after the fact by access auditing. That is defensible — in an emergency the cost of a blocked read exceeds the cost of an unnecessary one — but it must be a decision, with the audit capability actually built.

Second, routing follows the roster too. Results, referrals, tasks and alerts have to reach whoever is covering, not whoever ordered. Published guidance on test-result reporting makes the point concretely: responsibility should default to the supervising clinician on a change of service, and a surrogate needs inbox access during an absence [43]. An unacknowledged abnormal result in the inbox of a doctor on leave is the commonest silent failure in a clinical estate.

Third, the South African workforce data does not describe shifts. The public-sector system of record is a payroll. It describes posts and persons, not rosters, and shift-level data where it exists is local [41]. The Department's own analysis of priority interoperability use cases ranks aligning the aggregate reporting system, financial management and the payroll among the most important national use cases [44] — so the seam matters and is not yet closed.

Two South African specifics make rostering harder than it looks. **Commuted overtime decouples the cost of a doctor's time from the hours rostered**: it is a lump-sum arrangement rather than payment for hours worked. Treasury has said better management of it "could free money to hire more doctors", and has flagged savings in rural allowances, intern remuneration and remunerative work outside the public service, with a ministerial review of the last due by **April 2027** [21]. Any cost-per-shift model built on salary divided by rostered hours will be wrong, in the direction that makes night cover look cheap. And **remunerative work outside the public service puts one practitioner in two calendars**, maintained by two organisations with no shared scheduling system.

> Two honest gaps. **No sourced figure for agency or locum use in South African hospitals could be found.** And **there is no South African safe-staffing norm to implement**: the national nursing workforce report asks the Department to fast-track staffing norms for hospitals and names no method [41]. An architect cannot build to a norm that does not exist; the deliverable is instrumentation, actual staffing against actual activity, so the client can establish its own baseline. What would settle both gaps is a published staffing norm and a sector survey of agency use.

The ward round

A ward round is not a meeting. It is a synchronisation event: a fixed hour each day at which the record must be complete and current for every patient on the ward, and the plan for the next twenty-four hours is set.

Treat it as a deadline and the requirements fall out. By the time the round reaches a bed, four things must be present: the overnight observations, every result returned since the last round, the medication administration record and the current plan. Miss one and the round produces a repeat order instead of a decision — the main mechanism by which duplicate pathology requests appear in a hospital's data.

The binding dependency is results, and the acknowledgement step in the order-to-result loop has the least instrumentation of anything in a provider's estate. An unactioned abnormal result is a delayed diagnosis, and delayed diagnosis is a named driver of medico-legal claims against South African providers [46].

In the public sector the laboratory is always a different organisation. The national laboratory service processes more than 140 million tests a year [48], bills provinces, and was owed **R10,6 billion, R10,7 billion or about R11 billion** on three official accounts of the same debt [21][47]. So the turnaround time the ward round depends on is set inside an organisation the hospital does not control and may not be paying.

> **No South African laboratory turnaround-time standard could be sourced for either sector.** The book declines to import a figure from elsewhere; what would settle it is a published service-level schedule from the national laboratory service or a private pathology group. Meanwhile, instrument the loop and measure your own: order to collection, collection to receipt, receipt to verification, verification to acknowledgement. The fourth interval is the one nobody has, and the only one with a clinical outcome attached.

Design result arrival around the round, not the other way around. A result landing twenty minutes after the consultant has left the ward is, operationally, a result that arrives tomorrow.

Theatre lists and scheduling horizons

Theatre is where the scheduling horizon is longest and the number of organisations highest.

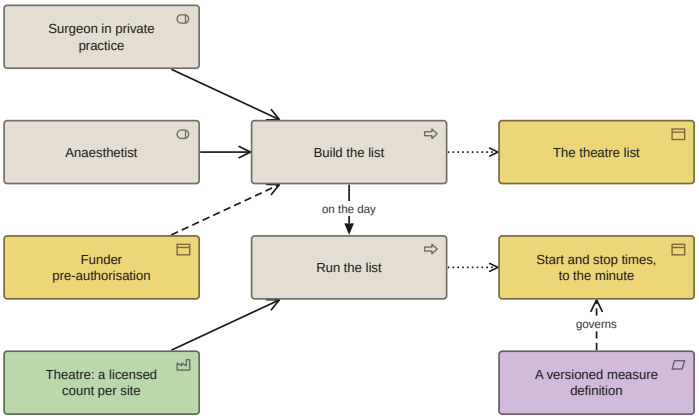


Figure 8.2 The four calendars behind one private theatre list, and the two records the list produces. Theatre time is a revenue field, billed per minute, not operational telemetry.

In the private sector the surgeon is not an employee. A theatre list is the intersection of four calendars in four organisations: the hospital's capacity, the surgeon's private availability, the anaesthetist's availability and — where the practitioner also holds a public post — their public-sector commitments. No single system holds all four, and the booking clerk is the integration layer.

The funder's decision precedes the list. Managed care arrangements cover **99,19% of beneficiaries** [29], so pre-authorisation is effectively universal in the private sector. The schedule is constrained by a decision made in another organisation, on that organisation's clock.

Capacity is a licensed count, not an engineering one. Theatre numbers in a private facility are authorised by a provincial licence for a specific site. Gauteng reported 48 approved-but-unbuilt facilities representing 2 796 beds and **306 theatres**, five under construction carrying 420 beds and nine theatres [25]; KwaZulu-Natal requires provincial approval of floor plans for theatres, dialysis units, chemotherapy units and oncology bunkers **before construction begins** [26]. You cannot add a room to relieve a list without a licence variation, and the variation has its own calendar.

And theatre time is money, to the minute. Under the injury-compensation tariff, minor and major theatre charges are **billed per minute**, running accounts are not permitted and long stays are billed monthly [24]. Recorded start and stop times are therefore revenue fields with an evidentiary standard, not operational telemetry. An architect who models theatre time only as a utilisation metric has missed a billing input, and a theatre system that produces estimated times has produced an unbillable record.

The timestamps, and why collapsing them destroys the measure

The best-documented specification of the timestamps a theatre record needs is Australian — a Queensland Health guideline of January 2017, labelled as such because there is no South African equivalent [7]. It requires in-anaesthetic, in-theatre, procedure start, procedure finish, out-of-theatre, the scheduled session start and end, and ready-for-discharge and discharge from recovery. **Collapsing those to "start" and "end" makes every measure below uncomputable**, and most theatre systems in the field are configured that way.

From them the guideline derives raw utilisation; a comparative elective utilisation adding a **nominal 15-minute changeover allowance per changeover**; starting on time against the scheduled session start; average changeover time against a 15-minute benchmark; finishing on time, with overruns defined as the last case leaving 30 or more minutes after the scheduled end and underruns as 45 or more minutes before it; day-of-surgery cancellations, split into preventable and other; and **unplanned session closures, defined as sessions closed within 48 hours' notice** [7]. That last definition is the one to steal: it converts "lists keep getting cancelled" into a dated, countable event with an owner.

The measure does not have one answer, and that is the teaching

Raw utilisation is a trap, and the guideline shows why with its own arithmetic: a 240-minute list with two cases and one 20-minute changeover tops out near **92%**, while a team doing eight cases with 10-minute changeovers tops out near **71%**. The high-throughput team looks worse [7].

A peer-reviewed critique of the way one national health service computed the measure finds four defects that all generalise [8]. Two are worth carrying. Pooled aggregation implies small gaps can be combined into usable blocks, and they cannot: **five minutes wasted on each of a hundred lists is not five hundred minutes of spare surgical capacity**. And the definitional spread is large — ten hours scheduled and eight worked gives **80% raw utilisation**, and with two hours of intercase downtime **touch-time utilisation falls to 60%** on the same list. Across one specialty in one organisation over fifteen weeks the authors found **no simple mathematical relationship between the four calculation methods** [8], so a hospital cannot convert between them or infer one from another.

The conclusion is not a formula. It is that the measure definition must be an object in the architecture, versioned and owned, and that any utilisation figure quoted without its definition and its vintage is unusable. Chapter 12 carries that as a principle; theatre is where it is proved.

> **No South African theatre utilisation, start-time, turnaround-time or cancellation-rate benchmark could be sourced in the research behind this book.** Do not import one. What can be stated is what the metric is, which timestamps it needs, and what distorts it. And for context on what "theatre improvement" means in parts of the public sector: the Auditor-General reported to Parliament that **one Free State hospital has lacked an operating theatre for 28 years** [20].

The outpatient clinic template

The clinic template is the quietest source of operational pain in a provider estate. A real booking consumes several resources at once — a clinician, a room, sometimes equipment — and the same appointment may need to appear in three systems: the administration diary, the departmental system and the patient-facing channel. Slot templates change weekly. Two failure modes are so common they should be assumed present.

Resource definition drift. The clinic template in the administration system and the session structure in the departmental system are maintained by different people, diverge within weeks, and present as "double-booked lists" rather than an integration fault. Because the symptom looks operational it is never escalated to the integration team, and it persists for years.

Asymmetric event propagation. Where the two are integrated by messaging, the scheduling event family carries new bookings, reschedules, cancellations, no-shows and modifications [38]. Cancellations and no-shows are not propagated as reliably as bookings, so the downstream system accumulates phantom appointments and every capacity figure drawn from it is overstated.

The South African twist: much of the public sector has no appointments

Non-attendance is the measure every client asks for and few can compute. The denominator alone has four defensible definitions — all appointments ever booked for the period, appointments live at the start of the day, appointments live at the appointment time, or available slots — and they answer different questions. Systems that delete cancelled appointments destroy the denominator; systems that overwrite an appointment on rebooking destroy the history.

Then the South African fact. A clinic that tells patients to arrive at 07:00 and sees them in arrival order **has no appointments, and therefore no non-attendance rate**. Much public outpatient attendance is not appointment-based at all, so an architect asked to report non-attendance across a district will find the measure undefined for most contacts. There is a countervailing asset: reminder channels exist at national scale — one national maternal messaging service had more than 2,5 million mothers registered as at 31 March 2019 [36] — so reminder-event data exists where appointment data does not.

No published South African non-attendance rate could be found. Say so, and offer the actionable split instead: patient cancellation with notice, which recovers the slot; without notice, which does not; provider cancellation, which is the organisation's own failure; and non-attendance without contact. Only the last is the colloquial "did not attend", and each needs a different response.

The fee schedule changes what counts as a visit

One more trap: a billing rule that reaches into the clinical data. Under the public fee schedule a fully subsidised **H1 patient attending several clinics on the same day is assessed once, at the most expensive clinic**, while H2, H3 and full-paying patients are assessed for each visit at each clinic, and an outpatient later admitted is charged **both** fees [3]. A visit model that assumes one encounter per attendance will mis-bill, and a clinic-activity report built on charges will not match the clinic's own count of people seen.

The on-call roster and the after-hours shape

Policy expects patients to start at primary care and be referred upward. In practice many use hospital emergency units after hours, which is why triage exists at the front door of a district hospital [42]. The triage instrument is the most implementable clinical decision rule in the country, and its cadence properties are why it belongs here rather than in chapter 9.

Re-triage is mandatory and continuous. The duty sister must continually reassess waiting patients and re-triage them if their condition changes [5]. So **a triage record is a series, not an event**, and the time-to-treat clock runs from the most recent category, not the first. A data model with one triage row per attendance has lost the measure.

The targets are published, and they are the only ones of their kind. The practice guideline sets target times to treat of immediate, 10 minutes, 60 minutes and 240 minutes for the Red, Orange, Yellow and Green categories, with performance thresholds of **95%, 80%, 75% and 70%** [4]. Those are the only sourced, published, numeric clinical service-level targets for any South African clinical process found in the research behind this book. Start an emergency department dashboard there, because every other service level in this sector has to be invented locally. The guideline also specifies the audit: a task team of one doctor and one professional nurse auditing **25 to 50 randomly selected patients** regularly, and benchmarking by category distribution, mortality, intensive-care admission and waiting time [4].

> **Both triage documents are long past their own review dates.** The practice guideline was published in March 2010 with review due February 2012 [4]; the training manual retrieved is dated 2012 [5]. They also contain at least two internal inconsistencies: the per-parameter scoring range differs between the manual's glossary and its charts, and one category's target time is rendered differently in different places. **Obtain the current versions from the professional society before you implement the arithmetic**, and build the target times and thresholds as configuration rather than as code.

Two more cadence facts about the front door. **Triage is not chargeable** under the public fee schedule [3], so the first clinical record of a visit has no billing event behind it: the capture must be justified clinically or it will not happen — the opposite of the private-sector instinct that the claim drags the record along behind it. And **the week is not flat**: the Department told a parliamentary committee that weekday occupancy in private hospitals was **22% above weekend occupancy** [37]. That figure is from 2011 and is dated, but the shape is stable and visible in any provider's own data. A capacity model built on a weekly average understates the weekday peak, and the weekend, the only period that looks like a maintenance window, is when the emergency unit's share of the work is highest.

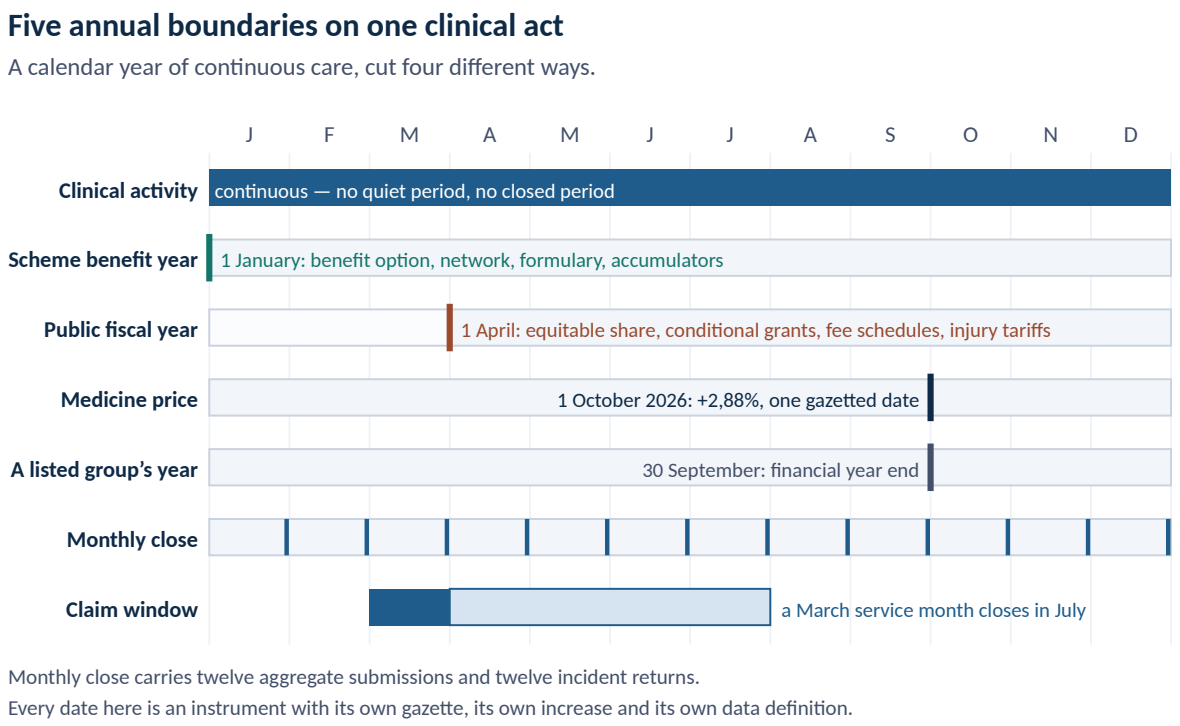


Figure 8.3 Five annual boundaries cutting one year of continuous clinical activity. Each is a separate instrument with its own gazette, its own increase and its own data definition.

The private claim cycle has a four-month tail. Accounts must be submitted within **four months** and schemes should pay within **30 days** of receiving a claim [6]. So claims data for a service month is not complete for a third of a year. Every management report, every reserve and every "how are we doing this month" dashboard inherits that, and a release that changes claim content cannot be measured for four months.

The dominant failure is short payment, not rejection, and short payment is a disagreement about rules applied to data rather than a transmission failure. The useful measures are first-pass full payment rate, short-pay rate by reason code, patient liability per episode and days to full settlement. Expect none to exist in the client's reporting — and there is a structural reason the last is hard. The South African medical scheme claim is an EDIFACT message, version 0, release 912, agency ZA, revision 13.4, November 2016, and the specification **covers the claim side only: it defines no remittance or payment-advice message and no link from a claim response to a payment** [35]. Days-to-settlement cannot be computed from the standard. It has to be reconstructed per funder, which is why settlement-time and denial analysis is harder here than architects arriving from other markets expect.

The public billing cycle has a different shape entirely. The inpatient tariff is calculated in **12-hour periods** for full-paying and H2/H3 patients and in **30-day periods** for H1 patients, with admission or discharge before or after **12:00** determining a half or a full day [3]. So "a day" in the public sector is a billing construct with a 12:00 boundary, and any length-of-stay figure derived from billing is measuring something else. And **patient classification is retrospectively mutable**: a patient reclassified as a scheme member must be re-billed retrospectively, and the schedule expressly provides for reclassifying members who did not disclose membership [3]. A billing system that treats classification as immutable at admission will understate revenue and fail audit.

The reporting month and the submission clock

The aggregate reporting system is monthly. A facility counts events from its registers and enters monthly totals per data element; there is a submission window, then the period closes, and late data is captured in subsequent revisions.

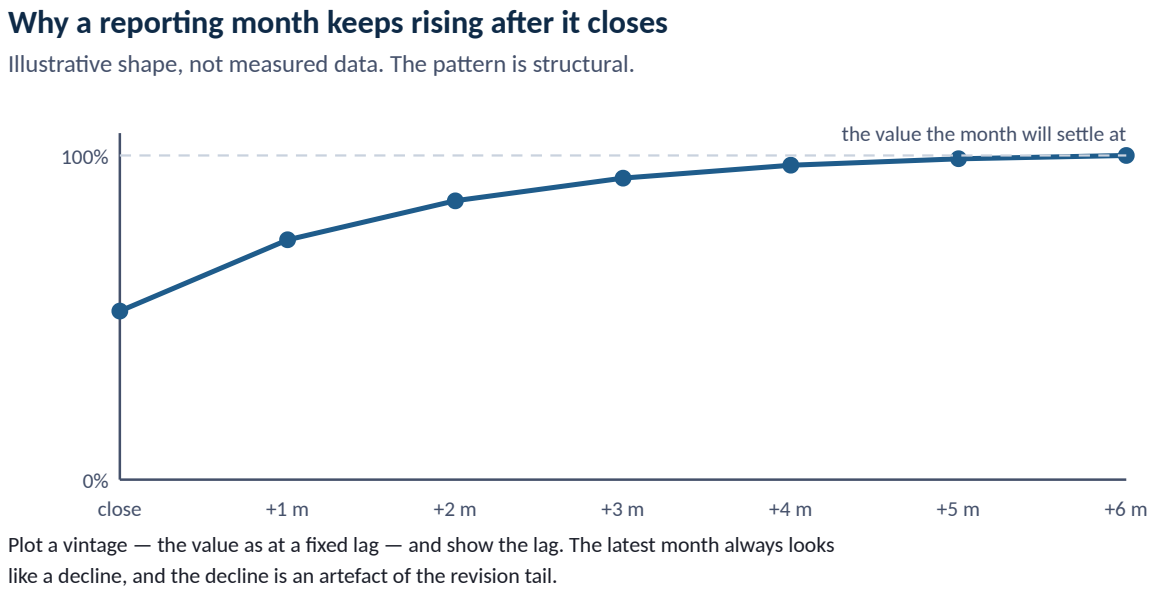


Figure 8.4 The revision tail on a monthly aggregate return. Illustrative shape, not measured data; the pattern is structural and makes the latest month always look like a decline.

That revision tail is the most under-managed fact in South African health reporting. **A figure for a recent month is systematically incomplete and rises for months.** Any dashboard plotting the latest month against prior months shows a decline that is an artefact. The remedy is standard and rarely implemented: plot a **vintage**, the value as at a fixed lag, and show the lag.

The quality picture underneath is worse than the compliance picture suggests. The principal published assessment of the national aggregate system, covering 265 public facilities and 77 data elements over April 2017 to March 2020, found **facility reporting completeness of 99,5%** and **data element reporting of 49,7%** — essentially every facility submitted essentially every expected report, and about half the values inside those reports were absent. Missing plus unaccounted values averaged **50,3%**; the overall completeness score was **74,6%, rated poor**; internal consistency was **95,1%**; and **timeliness was not assessed at all** [9].

> Quoting either completeness figure alone misrepresents the system. The gap between 99,5% and 49,7% *is* the finding, and it is the failure mode of every compliance-measured pipeline: **the thing that is measured is submission, and submission is not content.** The second finding is nearly as important — the unaccounted category was larger than the missing category, because it is unclear which data elements a given facility was expected to report. That makes the data-element-to-facility assignment unmanaged master data, and about a quarter of the expected-value space is in dispute [9].

The outward-flow register, and its clocks

A provider is a node in at least a dozen statutory and programmatic reporting flows. Each has its own instrument, recipient, clock, form and identifiers, and almost none is modelled as part of the provider's architecture, because each was built as a departmental obligation rather than an interface. The result is a clinician completing a form that duplicates data already in three systems. The first half of the register, with the clock that governs each:

Flow	Recipient	Trigger	Clock
Notifiable condition, Category 1	Focal person, then the national institute	Diagnosis or clinical suspicion	Immediate by phone, written within **24 hours** [10]
Notifiable condition, Category 2	As above	Clinical or laboratory diagnosis	**7 days** [10]
Notifiable condition, Category 3 (laboratories)	As above	Laboratory diagnosis	**7 days** [10]
Notifiable condition, Category 4 (laboratories)	As above	Laboratory diagnosis	**1 month** [10]
Outbreak or unusual cluster	Focal person	Detection	Immediate, written within **24 hours** [10]
Medical scheme surveillance return	National department	Claim received	**Monthly** [10]
Cancer notification, establishment	National cancer registry	Confirmed diagnosis	**3 months** [13]
Cancer notification, clinician, under 15	As above	Diagnosis	**3 months** [13]
Cancer notification, laboratory	As above	—	**Quarterly** [13]
Death certification	Home Affairs, then the statistics agency	Death	At death, before burial [15]
Patient safety incident, SAC 1	District or province	The incident	**24 hours**, investigation concluded in **60 working days** [1]

And the second half:

Flow	Recipient	Trigger	Clock
Patient safety incident statistics	Province	Monthly close	**Monthly**, except where a shared web system is used [1]
Provincial safety statistics	National department	Quarterly close	**Quarterly** [1]
Maternal death	Confidential enquiry committee; also a notifiable condition	Death	Per the notification form; **7 days** as a notifiable condition [11][16]
Perinatal and child death	The respective audit programmes	Death	Per programme [16]
Serious adverse drug reaction	Certificate holder, then the regulator	Reaction report	Holder: **15 calendar days** from the day the four minimum data elements arrive [14]
Foreign safety issue	The regulator	Action by a listed authority	**5 calendar days** [14]
Disease programme indicators	Aggregate reporting system	Monthly register close	**Monthly** [36]
National indicator data set	Aggregate reporting system	Monthly facility close	**Monthly** [36]
Medical scheme claim	Switch, then scheme	Service rendered	Four-month submission window, 30-day payment period [6]
Security compromise notification	The Information Regulator and the data subjects	Discovery	"As soon as reasonably possible" — there is **no 72-hour deadline**
Access-to-information annual report	The Information Regulator	Annual	Per the Regulator's call

Twenty-two flows. Filling that table in for a real organisation, with the instrument and the current owner against each row, is the single highest-value workshop in a South African healthcare engagement, and it usually takes a day. Two details in it repay attention.

The finest-grained published clock in South African health reporting is a day-of-week clock. Districts send weekly notifiable-condition line lists upward by **Wednesday 12:00**, provinces by **Thursday 12:00**; **zero-case weeks must still be reported**; and national reports issue **every Monday, one week behind** [12]. That is what a genuinely enforced weekly cadence looks like, and it runs on a three-part carbonised form with a web and mobile application alongside [12].

The 60-working-day clock needs a public-holiday calendar. Severity Assessment Code 1 incidents must be reported outside the facility within 24 hours and the investigation concluded within 60 **working** days [1], which makes a holiday calendar a shared service rather than a utility function in one application. The national figures suggest the clock is being met for closed incidents — 27 865 incidents reported in 2024/25, 26 811 closed, 26 552 of those within 60 working days [2] — while the Western Cape evidence shows where the slippage is: **20,4% of SAC 1 incidents were reported two or more days after the event**, one in five breaching the 24-hour rule [23].

The audit and accreditation cycle

An inspection is an evidence request against the facility's own records, on somebody else's calendar. Treating it so turns a compliance headache into a design requirement.

The statutory inspection cycle. The standards regulator publishes an **annual inspection strategy**, and selected establishments are **notified in advance**. Each inspection ends in compliance or non-compliance. Compliant establishments receive a certificate; non-compliant ones receive a **compliance notice stating the nature and severity of the non-compliance, with timeframes for correction**; after the timeframes lapse a **re-inspection** is scheduled; then certification or enforcement follows [18].

Read that as a workflow specification. A compliance notice is a dated remediation workflow with named owners, deadlines, evidence of closure and a re-inspection trigger. Almost no facility has a system for it, and the piece always missing is the evidence-of-closure link back to the operational record that proves the fix.

The throughput is the binding constraint. In 2023/24 the regulator inspected **734 public health establishments, 19,62% of 3 741**, and **60 private establishments**; identified non-compliance leading to **79 re-inspections**; issued **618 certificates of compliance**, 563 of them (91,0%) within 15 days of the final report against a 100% target; and issued **45 written warnings**, enforcement initiated within 10 days in every case [17] — all with **132 funded posts, 126 filled** [17]. The 2026 Budget redirected **R24 million** to the regulator to fill posts and increase inspections [3], which means more notices, more re-inspections and more demand for the evidence an inspection consumes.

The self-assessment cycle runs in parallel on its own clock. The departmental primary-care programme tracks progress through **annual reviews** awarding Silver, Gold or Platinum status, with facilities meeting none classed as not achieved [19]. A clinic is therefore inside an annual self-assessment cycle and an unpredictable statutory inspection cycle at once, measured against different instruments.

> **Three quality instruments run at once and they disagree.** The 2011 national core standards were used for inspections until 2019; the **norms and standards regulations** — promulgated 2018, effective 2019, seven chapters and 22 regulations — are what the statutory regulator now inspects against; and the departmental self-assessment programme runs in parallel. In 2019 the self-assessment produced **33,3% certification (1 145 of 3 467 facilities)** while the statutory audit for 2019/20 found **14,7% of inspected facilities met the minimum standard**, from 46% in one province to **0%** in two others [19]. Three instruments, incompatible results, the same facilities. If a client says "we are compliant", establish which instrument, which assessor and which year.
>
> And the regime is incomplete, not merely new: as recently as 2022 there were **no finalised inspection tools for private acute hospitals, GP practices or emergency medical services** [18]. **Design to the domains, which are stable, not to a tool version.**

The external audit cycle is annual, and its results are the governance context an architect inherits. For 2025/26, of nine provincial health departments, five received qualified audits, three unqualified with findings and only one a clean audit [21]. Most achieved below 60% of their targets; thirty audited infrastructure projects had average delays of **29 months**; and the Auditor-General cited leadership instability, senior vacancies and acting appointments as weakening oversight [20][22]. An architect may be engaging an organisation whose information-technology leadership post is filled in an acting capacity. Design for governance that does not depend on a stable counterparty.

One last cadence fact, and the most important here. There is a documented account of managers **borrowing equipment to present at assessments** [19], and of 23 of 27 audited clinics having no maintenance schedule and 24 not having determined required stock levels [22]. **An assessment date is therefore a data-integrity event.** Any measure built on an asset register, an equipment-availability indicator or a stock-on-hand figure must assume it is being managed around the assessment calendar, and must capture location-attributed, time-stamped data to mean anything.

The mortality and morbidity review

Morbidity and mortality review is a departmental meeting cycle: clinicians reviewing deaths and complications, usually monthly, educational and confidential to the professional group. Incident reporting is a different thing — managerial, statutory in its bases, escalating, auditable.

They produce different records and must not be merged. A merged system destroys the candour of the review meeting and creates a discoverable record of clinical opinion never intended as a finding of fact. It is one of the few places in a health architecture where the right answer is a deliberate separation with no join, and an architect should say so when asked to consolidate "all the quality systems".

No national instrument governing morbidity and mortality review could be sourced, and the one provincial adverse-event guideline identified could not be retrieved. Provincial guidelines may add categories, forms, timeframes and routing, **so a national product needs provincial configuration** — build the categories, the clocks and the routing as reference data from the start.

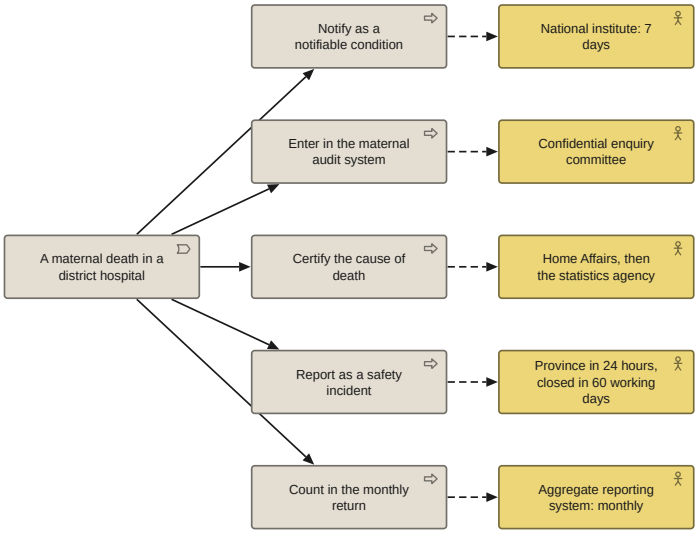


Figure 8.5 What one death in a district hospital owes outward: five recipients, five clocks, five forms, and two national systems that disagreed by 85 deaths in four provinces in 2023.

What one death owes outward

A single death in a district hospital can generate obligations to five recipients on five clocks: notification as a notifiable medical condition, where maternal death now appears in Category 2 with a seven-day clock [11]; entry into the national maternal audit system for confidential enquiry [16]; certification of the cause of death on the statutory notice, completed by hand before burial [15]; an incident report if the death meets the severity threshold, within 24 hours and investigated within 60 working days [1]; and a count in the monthly aggregate return [36].

And the two national systems disagree. The 2023 maternal mortality report entered **929** deaths during pregnancy, childbirth and the puerperium, **902** after excluding 27 coincidental deaths, with **839 of the 929** (**90,3%**) in health facilities and an uncorrected institutional ratio of **102,1 per 100 000 live births** computed on 883 244 live births in public facilities taken from the aggregate reporting system. After correction the ratio is **111,7** and the count **987** [16].

Here is the finding. The aggregate reporting system recorded **more** deaths than the purpose-built clinical audit system in four provinces — Eastern Cape +13, Gauteng +42, Limpopo +24, KwaZulu-Natal +6 — which the report attributes to under-reporting to the confidential enquiry and calls "of great concern", since that system was designed to be the more accurate one. The aggregate system's institutional ratio (**105,9**) exceeded the audit system's uncorrected ratio (**102,1**), and the report notes possible over-reporting to the aggregate system also needs checking [16].

Two national systems, the same event, the same facilities, the same year, disagreeing by 85 deaths in four provinces. **The remedy is not a third system.** It is that a death is one identified event in the record and both reports are derived from it with a documented rule, which requires the death to be a first-class event rather than a form. One more detail from the same report, the purest cadence defect in this research: the database closure date is given as **June 2024** in the executive summary and **June 2023** in the methods section [16]. A reporting cut-off that is not a recorded data attribute is one nobody can reconcile.

The budget year, and the five annual boundaries

Five annual boundaries cut the same clinical act, each a separate instrument with its own gazette, increase percentage and data definition.

1. **1 January — the scheme benefit year.** Benefit option, network, formulary, accumulator and prescribed-benefit treatment all change together, and a private group's own tariff schedule takes effect on the same date [33]. Benefit option configuration is the fastest-moving reference data in the private sector, and it changes the revenue of every provider downstream.
2. **1 April — the public fiscal year.** The equitable share and the conditional grants start, and with them two accountability chains: provincial treasury for the unconditional money and the national department for grant conditions, **with different reporting calendars and different data definitions for the same spend**. The public fee schedule is provincial, so nine instruments, nine effective dates and nine increase percentages — one province's 2026 schedule was published on 31 March 2026 and came into operation on 1 April 2026 [3]; another's 2025 schedule was gazetted in February, open for comment to 13 March, and effective 1 April with a **4,4%** increase across all patient categories and a **30%** increase in standby tariffs for emergency services at major events [32]. The injury-compensation tariff also starts on 1 April, gazetted on 20 March 2026 with tariffs up **6%** [24].
3. **A gazetted date — the medicine price.** The single exit price is adjusted by gazette, not by year boundary: **5,25% on 17 November 2024** and **2,88% from 1 October 2026** [2][34]. A bulk, dated, item-level price event affecting thousands of lines on one day; one funder body estimates the 2026 adjustment adds about **R1 billion** in unbudgeted medicine costs for schemes in 2027 [34].
4. **30 September — one listed private group's financial year end** [27], against another whose interim period ends in March [28]. Two groups treating the same population close their books six months apart.
5. **The monthly close**, twelve times a year, carrying the aggregate return and the incident statistics.

A single appendicectomy in April therefore sits inside a benefit year that started in January, a fiscal year and a tariff that started that month, a medicine price set the previous October and a group financial year ending in September. Nobody designed that. It is what five instruments made by five bodies with no shared calendar produce.

Where the cadences disagree, and what that does to a release calendar

There is no quiet period, so every candidate window is somebody's deadline, and the collisions are predictable:

- **The first week of April** carries the fiscal year start, the public fee schedule's effective date and the injury tariff's effective date at once — a reference-data change and a year-end in the same week.
- **The first week of January** carries the scheme benefit year. Every benefit option, network and formulary change in the private sector lands together, and provider revenue depends on getting it right on the first claim.
- **Month-end** carries the aggregate close, the incident statistics, the scheme surveillance return and the claim batch.
- **A medicine price gazette** lands on a date chosen by somebody else and moves thousands of reference-data rows in one day.
- **The audit and inspection periods** consume the same operational evidence a release would disturb, and an inspection is notified in advance but not scheduled by you.
- **The weekend**, the obvious-looking maintenance window, is when the emergency unit's share of the hospital's work is highest.

Six rules follow, and they are cheap.

1. **Publish a cadence calendar as an architecture artefact**, with the instrument behind every date and the owner of every clock. It is the outward-flow workshop's deliverable, and the first thing a new architect will find does not exist.
2. **Separate reference-data effective dating from software release.** If tariffs, benefit options, fee schedules and medicine prices are effective-dated data, a 1 April change lands without a deployment. If they are configuration baked into a release, every statutory date becomes a release date. That one decision removes most of the collisions above.
3. **Name the deadline, not the month.** "No release in April" is lazy and will be overridden. "No release in the week of the fee schedule's effective date, because the re-billing of reclassified patients runs then" survives a steering committee.
4. **Treat the public-holiday calendar as a shared service.** Working-day clocks are statutory, and a holiday calendar embedded in three applications will disagree with itself.
5. **Measure a change against a vintage, not the latest month**, or the revision tail will be attributed to your release in one direction or the other.
6. **Freeze the measure definition before the release, not after**, or nobody will be able to say whether the movement came from the definition or the system — and the client will assume the system.

Decision order is not teaching order

A last point about sequence — the order decisions are taken in, not which products to buy.

This book teaches in a sensible reading order: the industry, how a provider runs, data and systems, constraints, modelling. That is the right order to learn in. **It is not the order to decide in.** The evidence says the decision order is **identity, terminology and integration strategy first, application strategy after**. Each of the first three has a long lead time, an organisation-wide blast radius and no cheap reversal. An application choice made before them constrains all three, and the constraint surfaces in the second year.

The evidence says one more thing about sequencing, and for someone with a small budget it is the most useful sentence in this chapter. **In a low-maturity estate, event-driven notification is the highest-return first capability — higher than a shared record.** The pattern is narrow: when something clinically significant happens to a person, tell the people who need to know. It needs an event source, which is the admission-and-discharge feed that already exists; a subscription registry recording who wants to know about whom; a notification service; and a delivery channel. No shared record, no document repository, no terminology harmonisation. Two of the relevant profiles are already named in the gazetted national interoperability framework [39].

Its failure modes are cadence failures, which is why it belongs here. **Subscription list currency**: a stale list notifies the wrong practice, which is a privacy breach rather than a nuisance. **Notification fatigue**: an unfiltered feed is ignored within weeks. And **no acknowledgement path**, so nobody knows whether the notification was acted on — the same unclosed loop as the result nobody signed off, arriving a second time.

Build the notification, build the acknowledgement with it, instrument both from day one. Then go back and decide identity.

THE CHAPTER IN TEN LINES

- A provider runs continuously and everything that observes it runs in periods. Three clocks run at once: clinical, administrative and statutory.
- Access control in a hospital is assignment control, so it depends on a roster feed. Without one it degrades to everyone seeing everything, compensated by audit.
- Commuted overtime decouples the cost of a doctor's time from the hours rostered, so a cost-per-shift model built on salary divided by hours will be wrong.
- A ward round is a synchronisation event: a daily deadline at which observations, results, administration records and the plan must all be present.
- A private theatre list is the intersection of four calendars in four organisations, and theatre capacity is a licensed count that cannot be flexed without a licence variation.
- Theatre time is billed per minute under the injury-compensation tariff, so start and stop times are revenue fields with an evidentiary standard.
- Raw theatre utilisation of 80% and touch-time utilisation of 60% describe the same list, and there is no conversion between the four published methods.
- Much South African public outpatient attendance is not appointment-based, so non-attendance is undefined for most contacts. No published South African rate exists.
- A triage record is a series, not an event, and the time-to-treat clock runs from the most recent category. The 95/80/75/70 thresholds are the only published clinical service levels in the country.
- The private claim has a four-month submission window and a 30-day payment period, so a service month is not complete for a third of a year.
- Aggregate reporting has a revision tail: a recent month is systematically incomplete and rises for months. Plot a vintage at a fixed lag and show the lag.
- Facility reporting completeness is 99,5% and data element reporting is 49,7%. The gap is the finding, and timeliness was never assessed.
- Twenty-two outward statutory flows, each with its own clock, from 24 hours to annual. The finest-grained published clock is a day-of-week clock.
- Five annual boundaries cut the same clinical act: 1 January, 1 April, a gazetted medicine price date, two different private financial year ends, and twelve monthly closes.
- Decision order is not teaching order: identity, terminology and integration strategy are decided before application strategy, and event-driven notification is the highest-return first capability.

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CHAPTER 9

The process model — five journeys

Five named value streams, twenty numbered stages, and not one stage with a name, a performer or a system behind it.

Five named streams, numbered stages, no names

The supplied repository names five value streams — referral to treatment, order to result, prescribe to administer, encounter to payment, incident to improvement — with reasonable scope lines: referral through triage, appointment, treatment and discharge; diagnostic order through collection, analysis and reported result; and so on. Beneath each sits a set of processes called *stage 1*, *stage 2*, *stage 3* and *stage 4*. Three of the five declare a stage count the layer below does not contain — five declared against four present in two cases, six against four in a third. No stage has a name, a performer, or a system.

That is the normal condition of a repository populated to satisfy a template, and the first deliverable in a process workstream is therefore not a new model. It is names. A stage called *stage 3* cannot be argued with; a stage called **accept, redirect or decline** can, and the argument is the architecture.

This chapter names the stages of the five streams the repository carries, then does the same for three it omits that every provider runs. Each stage answers four questions: **who performs it**, by role rather than job title; **which system the step touches**, named not described, since chapter 15 describes the estate; **where the handoff fails**, because almost every documented failure here is a handoff, not a step; and **what the stage requires of the data**.

Value streams and their stages are modelled in ArchiMate, as `Value Stream` and `Process` elements with roles assigned to them. The level-2 models here are BPMN, for one reason: ArchiMate has no swimlanes, and in every stream below the lane boundary *is* the finding. A level-2 model has a short half-life and must be maintained by whoever owns the process. Four are drawn here; a provider does not need thirty.

Two invented organisations carry the examples, both fictional: **Driehoek Health Group**, a mid-size private hospital group of nine acute hospitals and two day clinics with about 1,750 registered beds, a shared pathology contract and an in-house pharmacy function; and **Rietvlei District Health Service**, one public health district — a district hospital, three community health centres, 22 fixed clinics and a mobile service.

> **NOT CLINICAL GUIDANCE**
>
> This chapter describes clinical processes as an architect must implement them. Nothing in it is a dose, a protocol or a decision rule for use on a patient. The South African Triage Scale appears later as a worked example of a deterministic algorithm with override rules, described as a mechanism. Its published practice guideline is dated March 2010 with review due February 2012; the current instrument must be obtained from its publisher before any system is built to it.

Referral to treatment

The repository's four stages collapse seven, three of which have no system behind them at all in much of the public sector.

Stage	Who performs it	System the step touches	What the data must carry
Recognise and decide	Referring clinician; in Rietvlei a nurse practitioner	Paper folder, or the primary care record	Reason, level of care, decision date
Compose	Referring clinician	The same record; in practice a handwritten letter	Summary, question asked, urgency, referrer registration
Transmit	The patient, in most cases	None	Nothing — which is the finding
Accept, redirect or decline	Gatekeeper at the receiving service	Receiving patient administration or waiting list	Referral identifier, received timestamp, acknowledgement
Book	Bookings clerk	Clinic diary	Slot, service, specialty, priority, link to the referral
Attend and treat	Treating clinician	Receiving clinical record	The encounter, linked to its referral
Back-refer and close	Treating clinician	Receiving record; letter to the referrer	Outcome, plan, responsibility returned

Three provisions shape this stream, none of them a standard. A user may attend **any** public establishment, under section 44(1) of the National Health Act; an establishment that cannot provide necessary care **must transfer** the user to one that can, under section 44(2); and the hospital categories regulation gives tertiary hospitals regional referrals **without provincial limits** and central hospitals patients from more than one province [1][2]. So the destination may be wrong by statutory design, and a tertiary or central referral may cross a provincial boundary — an integration between two separately procured estates with no shared patient identifier. Policy expects patients to start at primary care and move upward; in practice many present at emergency units after hours, which is why triage at a district hospital's front door spends much of its time on primary-care presentations [24]. The private stream differs in kind: the patient chooses, may self-refer, and is steered by scheme design rather than a letter [19].

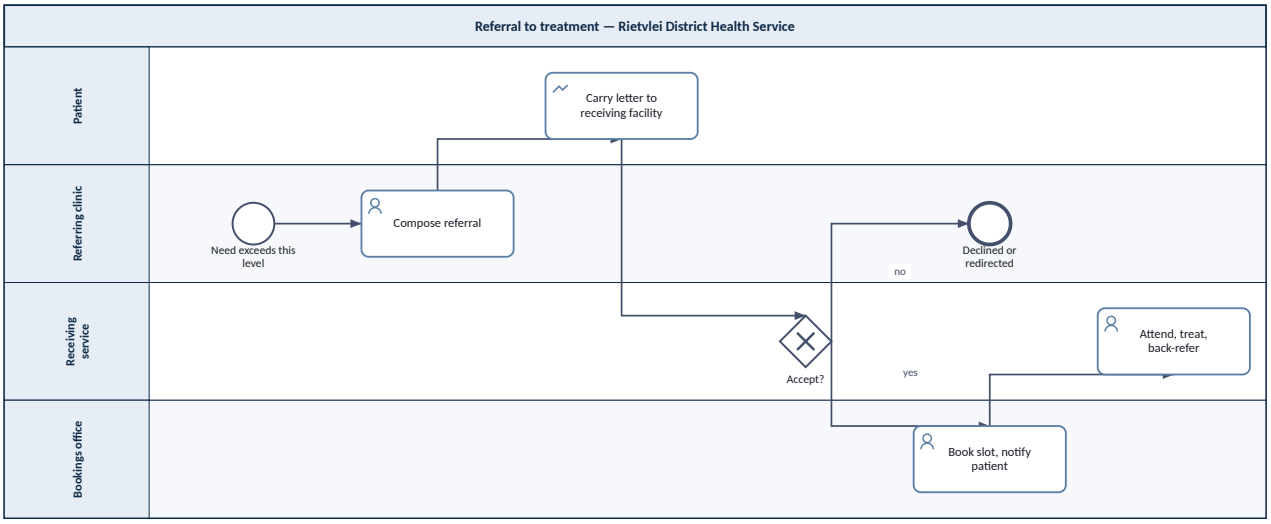


Figure 9.1 Referral to treatment, level 2. The transmit stage has no system and no lane of its own: the patient carries the letter. The decline branch returns to the referrer, and the acknowledgement that would make the referral auditable is the state most estates do not record.

Where it fails

A referral is a document, not a message. There is no national referral standard in routine use. Clinical responsibility transfers on a letter, often handwritten, usually accompanying the patient rather than preceding them — so the receiving service learns of the referral when the patient arrives holding it.

Sent is not acknowledged. The two states that make a referral auditable are `sent` and `acknowledged`, and a referral sent and never acknowledged is a patient lost to follow-up. Most estates record only the first.

There is no clock. No South African referral-to-treatment time standard could be sourced anywhere in the research behind this volume, so there is no national breach reporting and no waiting-time dataset to benchmark against. Waiting times appear only as a sub-domain of the health standards regulator's user-rights inspection domain, and referral systems as an action area in the core standards framework [33][37]. An architect used to a published clock will have to define one locally and say so in writing.

What the stream needs from the data is small and absent. **Referral** must be a tracked request with states and timestamps, not a document with an attachment: model it as a letter and the interval between two states cannot be computed, so the stream has no measure. **Appointment** must be separate from **Encounter**, because the appointment that produced no encounter is the non-attendance numerator and the encounter with no appointment is the walk-in — and cancelled appointments must never be deleted.

Order to result

Stage	Who performs it	System the step touches	What the data must carry
Request	Ordering clinician	Order communications; paper requisition in much of the public estate	Orderable item, clinical question, priority
Collect and label	Nurse, phlebotomist or clinician	Label printer at the patient	Identity verified at the patient; type, collection time
Transport	Facility or district logistics	Usually nothing	Chain of custody; time and temperature where needed
Accession and check	Laboratory reception	The laboratory information system — in the public estate, the NHLS's	Accession number, cross-referenced to the order identifier
Analyse and authorise	Scientist or pathologist	Analyser middleware into the laboratory system	Value, unit, reference range, authoriser
Report and route	The laboratory system	Interface to the placer, or a web viewer	Result status, named recipient, amended-result signal
Acknowledge and act	The clinician responsible, or whoever covers	Nothing, in most deployments	An acknowledgement event, with an actor and a timestamp

The repository declares five stages here and carries four processes; the two missing are transport and acknowledgement.

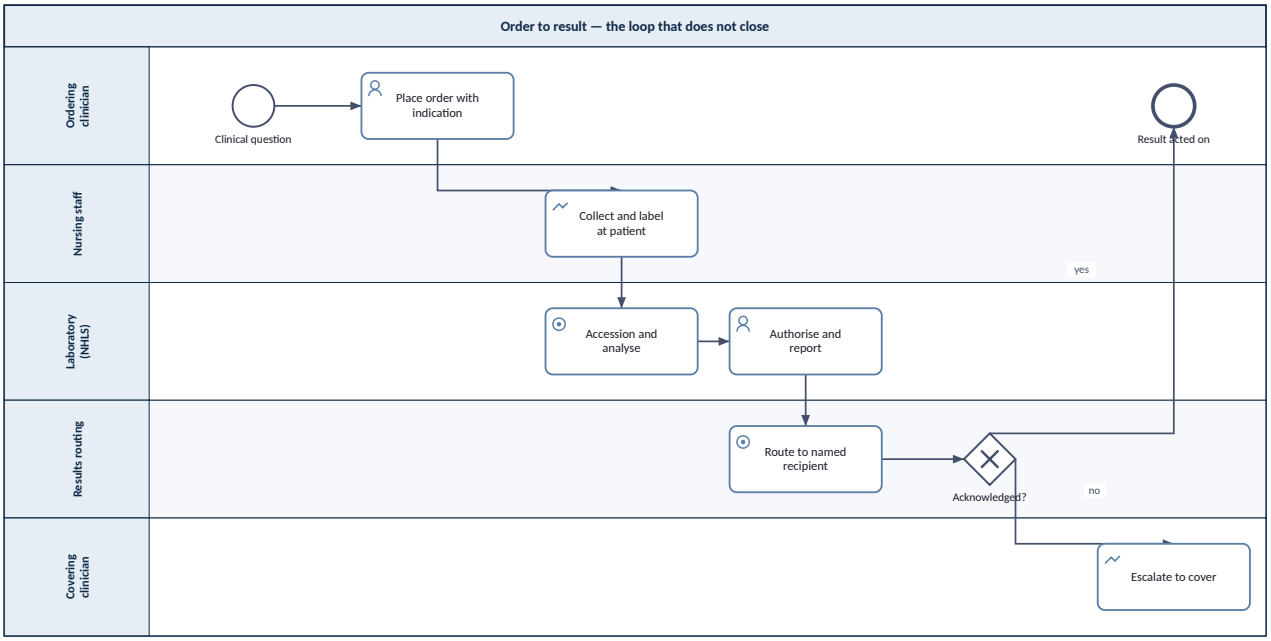


Figure 9.2 Order to result. Every stage up to the report is instrumented; the acknowledge-and-act stage usually touches no system at all, which is why the escalation lane exists on paper and rarely in a deployment.

The laboratory is always a different organisation. In the public sector it is the National Health Laboratory Service, a national entity that bills provinces and processes more than 140 million tests a year [21]; privately it is Ampath, Lancet, PathCare or another group, each with its own result formats. Rietvlei's pathology comes from an entity whose availability Rietvlei does not control, and which lost its laboratory system and central data warehouse to ransomware in June 2024 [21]. This stream therefore has a stage performed by an organisation the repository does not contain.

Where it fails

The failure is not transmission. It is acknowledgement and follow-up, and the control set is well articulated: track every order through collected, completed, reported and acknowledged; identify both the ordering clinician and whoever is responsible for follow-up; monitor time to acknowledgement; escalate critical results automatically when nobody responds; and document delegation of follow-up, *accepted* by the delegate [17]. Two are routinely missed. Referred-out tests must be tracked from the point of ordering, which where a national laboratory performs most referred testing is not an edge case. And trainee responsibility defaults to the supervising clinician on a change of service, so result routing depends on the roster — a dependency almost never drawn on an architecture diagram [17].

- **Amended and cancelled.** If a correction arrives as a new observation rather than a replacement related to the original, the record holds two contradictory truths and the clinician sees whichever sorts first. And where the placer cancels after the laboratory has started, a result returns against a cancelled order and is dropped by inbound rules.
- **Catalogue drift.** The laboratory adds a test, the orderable list catches up weeks later, and meanwhile clinicians order the nearest thing.
- **Unmatched results.** A result arrives with an identifier the placer cannot resolve and lands in an exception queue — the most dangerous object in a hospital integration estate, and one that belongs in the architecture as a named, staffed, monitored function.

What it requires of the data

Order and result are two entities. Collapsing them makes the unordered result impossible to represent — the reflex test, the add-on, the external referral — and makes turnaround time unmeasurable, because there is one timestamp where there need to be four. **Specimen is an entity**, not an attribute, because one order may need several specimens and one specimen may serve several orders, and specimen is what makes rejected and mislabelled samples visible — a mislabelled specimen being an identity failure, which is why labelling and patient matching are one problem from two ends. Result status needs the state **appended**, which microbiology requires and most models lack, and the unit must be stored with the value. And **no South African laboratory turnaround-time standard could be sourced** for either sector, so the target is a local decision that must be written down.

Prescribe to administer

This is the one stream where the repository's declared count — six — matches the real shape, though four processes sit beneath it.

Stage	Who performs it	System the step touches	What the data must carry
Prescribe	Prescriber	Electronic prescribing, or a paper chart	Product, dose, route, frequency, duration, indication
Verify	Pharmacist	Pharmacy system	The review event, the reviewer, any intervention
Dispense or supply	Pharmacist or assistant	Pharmacy dispensing and stock; the chronic medicine programme	Product, quantity, batch, expiry, days supply, pharmacist
Administer, or record not given	Nurse or clinician	Medicines administration record, where one exists	Actual dose, route, time; practitioner; the reason not given
Monitor	Nurse, clinician, pharmacist	Clinical record; the national pharmacovigilance route	Reactions to treatment, including adverse effects
Reconcile	Pharmacy or admitting clinician	Clinical record	The reconciliation event, sources consulted, discrepancies

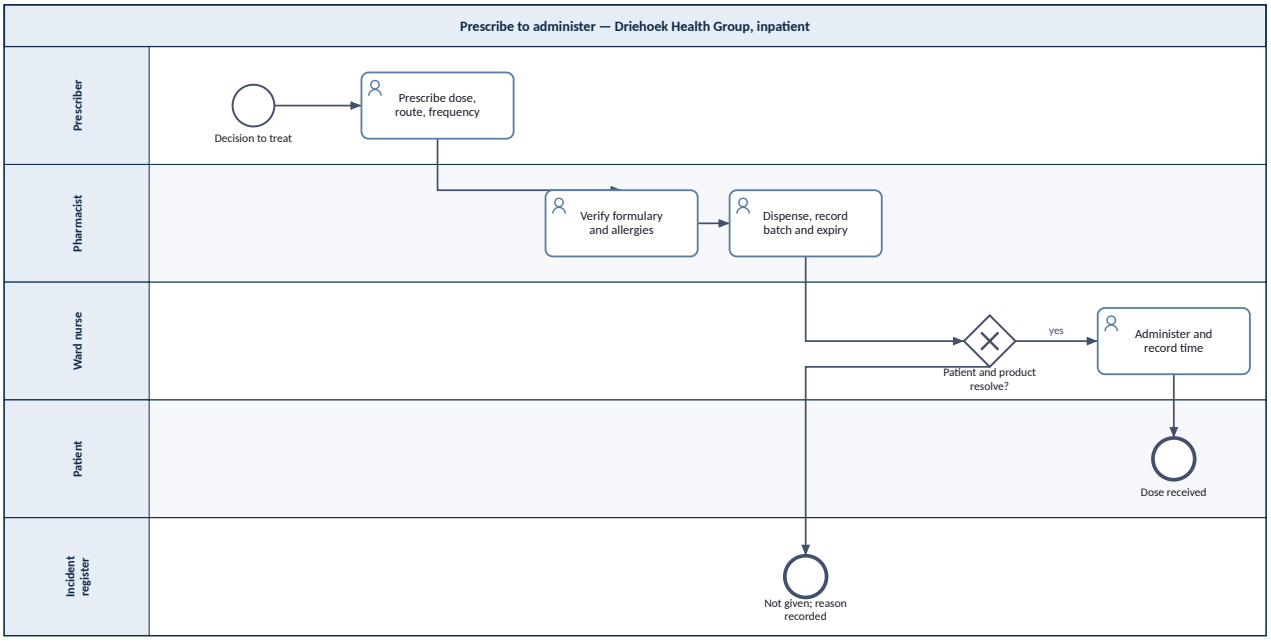


Figure 9.3 Prescribe to administer. The gateway is the closed loop: if the scan does not resolve the patient and the product against the active prescription, the dose is not given and the reason is the record. The not-given path feeds the incident register, which is where the fifth stream begins.

Two South African constraints that cap the design

Schedules 5 to 8 must be handwritten. Under the professional rule carried in the Health Professions Council's record-keeping guidance, prescriptions for Schedules 1 to 4 may be typed or pre-printed, those for Schedules 5 to 8 must be handwritten, and both require the practitioner's personal original signature [5]. An electronic prescribing target state without an explicit paper path for scheduled substances is wrong, and whether any instrument displaces that requirement could not be established, so the paper path must be designed.

There is no national medicines dictionary with virtual-product and dose-form structure for prescribing. NAPPI is a product identifier for claims and pricing; the anatomical-therapeutic-chemical classification is a consumption classification. So every electronic prescribing implementation in the country builds its own dictionary, and the absence blocks the product-resolution step on which closed-loop administration depends [29].

The chronic-medicine path is a distribution network rather than a pharmacy system, at logistics scale: 2,912 pick-up points, 8,978,126 parcels delivered and 3,576 facilities enrolled across eight provinces in 2024/25, with 1,185,875 patients still collecting from facilities [14]. Its unit of work is a parcel to a pick-up point against an enrolment and a prescription validity period, on its own national system, which began on 5 September 2016 and reported 1,537,123 beneficiaries since launch [22][23] — treat that as "dispensing" and the model misses three entities. And the gazetted single exit price is a bulk, item-level price change on one day — 5,25% on 17 November 2024, 2,88% from 1 October 2026 [14][35] — so every price-dependent calculation downstream must be effective-dated.

CLOSED-LOOP ADMINISTRATION IS A CONTROL WITH A COST, NOT A BENEFIT

Barcode verification of patient, product and practitioner against the active prescription at the bedside is usually put to a hospital board as a safety improvement with an evidence base. In the settings that describe most hospitals outside North America — where medicines are not supplied as pre-packaged single doses — that evidence base is thin and mixed. The only systematic review restricted to those settings found **four** before-and-after or controlled cohort studies, **no randomised trials**, and error definitions too heterogeneous to pool. Total errors fell from 8,6% of opportunities to 4,4% in one study; wrong-dose errors fell from 29 to 5 and fast intravenous bolus errors from 31 to 5, while **wrong-route errors rose from 2 to 6**. A controlled study found reductions in wrong intravenous rate of 5,7 per 100 administrations, wrong timing 3,4, wrong strength 1,1, wrong dosage 0,4, and wrong formulation and extra dose 0,2 each — alongside an **increase of 4,3 per 100 in wrong solvent volume** for injectables. A fourth found **timeliness worse**: 81% of ordered doses given on the barcode ward against 89% on the comparison ward, with mean delay 67,5 minutes against 60,3. Three of the four studies were graded low certainty and one moderate [16].

The reading is not "do not do it". It is that closed-loop administration **moves error and costs time**, so the design must budget for the time cost at the bedside and instrument the error categories that get worse. An architecture claiming it as an unqualified safety benefit will be contradicted by the hospital's own incident data inside a year.

What it requires of the data

Four entities, not one. Medication Order, Dispense, Administration and Medication Reconciliation have separate state machines, and a medication safety incident is almost always a mismatch between two of them, which a single table with a status column cannot see. The order names an ingredient and a dose; the dispense names a manufacturer's pack with a batch number and an expiry date; the administration says what was actually given, when, by whom, and **why it was not given** if it was not. That last attribute is the one a paper chart records as a blank box, and reason not given — refused, absent, nil by mouth, medicine unavailable, held on clinical instruction — is what makes stock-outs visible in clinical data.

Medication Reconciliation is not a view over active orders but an independent clinical assertion with an author, a date, sources consulted and discrepancies found, because the patient is taking things nobody prescribed and not taking things that were. Deriving current medications from active orders will be wrong for most chronic patients, and wrong in the dangerous direction. Two state transitions break models: **stopped**, because a medicine stopped on day 3 of a 7-day course has an active order and a stop event, so a model holding only a start date and a duration reports it as given for seven days; and **on hold**, which must not look like a stop.

Encounter to payment

Stage	Who performs it	System the step touches	What the data must carry
Establish liability	Admissions clerk	Private: membership enquiry via a claims switch. Public: the means test	Member identity, option, effective dates; or fee classification
Authorise	Managed care organisation	Pre-authorisation through the switch	Authorisation number, authorised stay and procedure
Deliver and document	Treating clinicians	Clinical record	The clinical facts, with the content the rule requires
Code	The diagnosing clinician	Clinical record, or a coding workbench	ICD-10 to maximum specificity, in valid positions
Assemble and price	Revenue cycle	Billing system, against the contract in force on the service date	Tariff code, quantity, modifiers, product codes, version
Submit	Revenue cycle	The switch	A claim distinct from the invoice, with its own version
Adjudicate	Funder	Funder adjudication, response through the switch	Response and reason codes, amounts paid and declined
Reconcile and resubmit	Revenue cycle	Billing system	Remittance matched line by line; resubmission as a new version

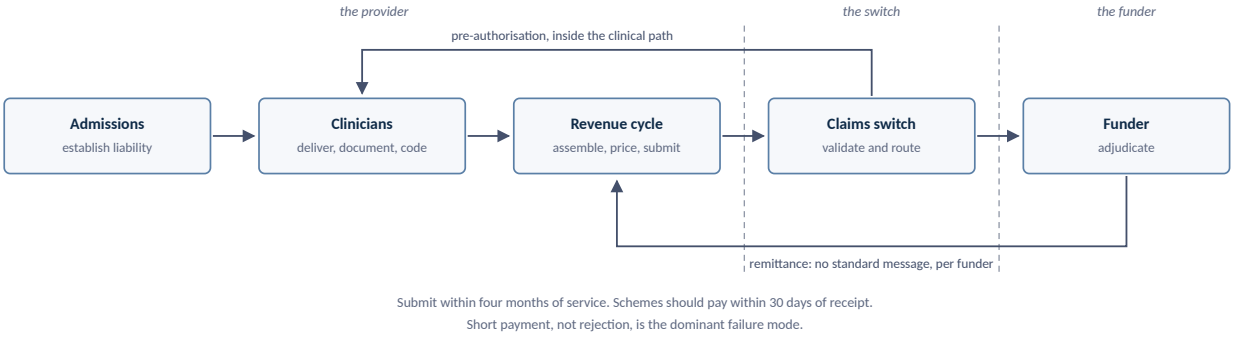


Figure 9.4 Encounter to payment, across four organisations. Two loops matter more than the forward chain: pre-authorisation, which is the only funder call inside the clinical path, and the remittance, for which the South African claim standard defines no message.

Coding is a clinical act with a compliance rule attached. The requirement that a claim carry a diagnostic code sits in **Regulation 5(f) of the Medical Schemes Act 131 of 1998** — a funder regulation, not a health statute — and the obligation is **conditional on payment**: an uncoded claim is unpayable, not unlawful [6][7]. There is no national clinical coding regulation; one has been in development for a decade. The operationally decisive date is 1 July 2014, when age and gender validation against the national code table began rejecting claims outright [7]. And the code must be assigned by a clinician:

the departmental circular holds that administrative and call-centre staff may not help a prescriber assign one, and that short "cheat sheets" built from call-centre codes are not compliant [6]. A dropdown of common diagnoses presented to a receptionist is therefore a compliance exposure, not a usability improvement.

The capacities differ by claim type. A hospital claim carries up to 30 ICD-10 codes at header level — one primary and 29 secondary — while a non-hospital practitioner claim carries one primary and nine secondary **per line item** [8][9], so the funder's view of the diagnosis sits at a different granularity from its view of the activity. **And there is no response standard.** The private claim travels as a UN/EDIFACT MEDCLM message — version 0, release 912, agency ZA, revision 13.4, November 2016 — and that specification **covers the claim side only**: no remittance or payment-advice message, and no defined link from a claim response to a payment [27]. Remittance arrives in whatever format each funder chooses, which is why denial-rate analysis is hard here and why Remittance is the entity organisations most often fail to model.

Short payment, not rejection, is the dominant failure mode. Machine rejections — invalid or missing code, age or gender validation failure, a code in an invalid position — are the visible failures. The expensive ones are benefit decisions: the wrong benefit pool, a dispute about the level of care for a prescribed minimum benefit, non-network use paid at a reduced rate, a non-formulary medicine, a tariff differential left with the patient [19][28][32]. Short payment is a disagreement about rules applied to data, not a transmission failure, so an architecture measuring success as "claim accepted" measures the wrong thing. What matters is first-pass full payment rate, short-pay rate by reason code, patient liability per episode, and days to full settlement. And two statutory windows set the envelope: accounts must be submitted within **four months**, schemes should pay within **30 days** [10], so activity data for a period is incomplete for a third of a year.

Authorisation sits inside the clinical path, the only part of the revenue cycle that does. Managed care covers **99,19% of beneficiaries** [20], so if the transaction is slow or unavailable the patient waits — which makes a commercial integration a clinical availability requirement.

The unit of charge is not the unit of care. Chapter 6 carries the money; a process model must absorb that under the public patient fee schedule inpatient billing runs in **12-hour periods** for full-paying and partially subsidised patients and **30-day periods** for the fully subsidised, with the **12:00 boundary** deciding half a day or a whole one [11]; and that under the compensation fund tariff minor and major theatre are **billed per minute** [12]. Recorded theatre times are therefore revenue fields with an evidentiary standard, and an architect who models theatre time only as a utilisation metric has missed a billing input.

Invoice, Claim and Remittance are three entities, one encounter routinely produces several invoices — hospital, surgeon, anaesthetist, pathology — and because reversals and resubmissions travel in the claim message itself, a claim is a **versioned** object.

Incident to improvement

Stage	Who performs it	System the step touches	What the data must carry
Detect and report	Any staff member who witnessed it	Incident form and register; a provincial web system where one exists	The minimum field set: patient, date and time, ward, section
Take statements	Staff, patients and others present	The same register	Third-party personal information, with its own access treatment
Classify	Section manager, then the facility committee	Register	A Severity Assessment Code, which drives routing and clocks
Escalate	Facility management	Report to sub-district, district and province	A code-1 report within 24 hours, mandatory
Investigate	Investigating manager	Register plus investigation workflow	Chronology, interviews, root causes, incident type, outcomes
Decide and act	Facility patient safety committee	Register and an action tracker	Actions to prevent recurrence, with an owner and a date
Verify closure	Facility and provincial committees	Register	Closure date, and evidence the action was implemented
Aggregate	Facility, province, national	Monthly and quarterly returns	Counts by type, severity, outcome, with a denominator

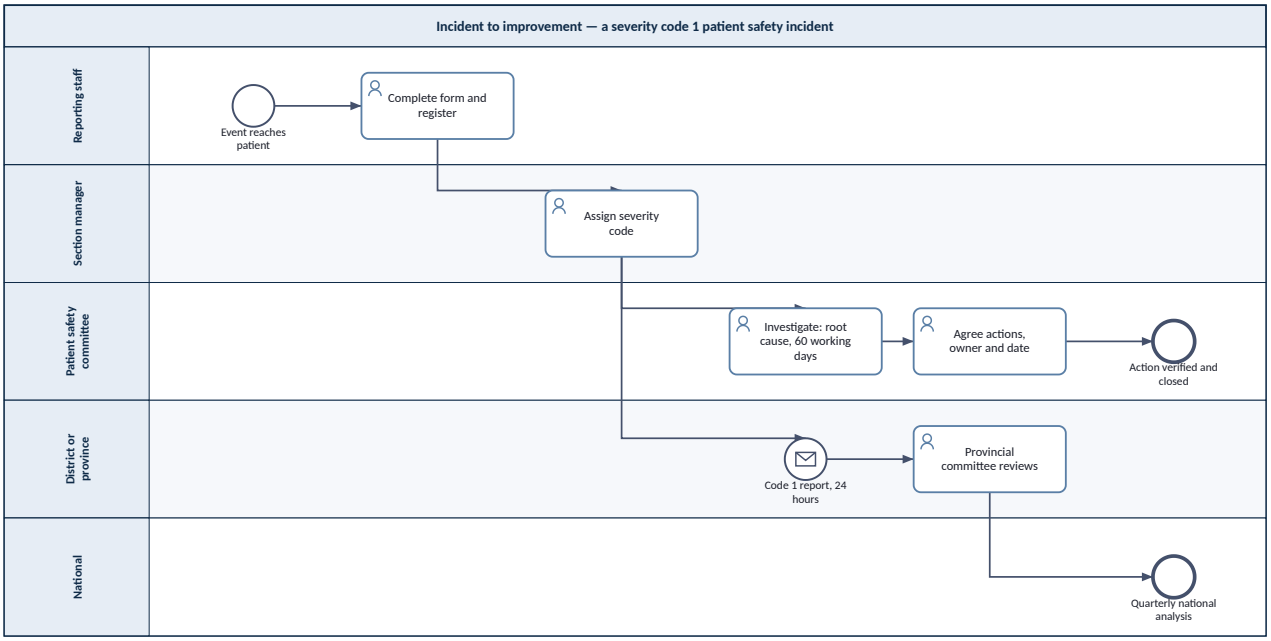


Figure 9.5 Incident to improvement for a severity code 1 incident. The severity code is assigned once and then drives two parallel paths with different clocks — a 24-hour report out of the facility, and a 60-working-day investigation inside it.

The vocabulary does not match the international one

The controlling instrument is the national guideline for patient safety incident reporting and learning, version 2, 2022 [13] — a guideline rather than a regulation, with statutory legal bases, which is why it has effect, and an architect importing a vocabulary from elsewhere will mis-map every term in it. The unit is a **patient safety incident** — an unplanned or unintended event or circumstance that could have resulted or did result in harm to a patient while in a facility's care, explicitly **not** harm due to the underlying condition or natural disease progression; an **adverse event** is the harmful subset, not a synonym. **"Sentinel event" does not appear in the guideline at all**; the nearest concept is a **Severity Assessment Code 1** incident. And the guideline does **not** define "concise" and "comprehensive" investigation types. That exclusion for the underlying condition is the hard part of the model, because it means a clinical judgement decides whether an event is reportable at all.

A code-1 incident must be reported to management and to the district or provincial office **within 24 hours**; its report must be finalised and its investigation concluded **within 60 working days**. Codes 2, 3 and 4 go to executive management inside the facility and the guideline sets **no timeframe** [13]. Working days is not a detail: the clock needs a public-holiday calendar as a first-class reference object.

In 2024/25, **27,865** patient safety incidents were reported nationally; **26,811 (96%) were closed**, and **26,552 (99%) of those within 60 working days** [14]. Two cautions. Reported incidents fell from **29,552** in 2023/24 and no source explains why; given documented under-reporting, a fall cannot be read as improvement. And 26,811 appears in the same annual report as **both** incidents closed and complaints logged — a coincidence or a transcription error that could not be resolved [14].

Where it fails

The best evidence is a 2025 study of Western Cape district health services for calendar 2023: **577** incidents from **16** facilities, counts ranging from **0 to 148** with a median of 19; **district hospitals submitted about 91% of reports and primary care about 9%**; one metro primary care facility reported **none at all** in a full year. About **69%** of incidents reached the patient and **33,5%** caused harm. **Code 1 was 17,9%**, and **76,7% of those were linked to patient behaviour, mainly absconding**. Code-1 incidents were reported quickly, median under one day, but **20,4% two or more days after the event**. Median time to close was **4 days** (interquartile range 1 to 12,8; range 1 to 77), against national data for 2020/21 putting code 1 at 23%, code 2 at 27% and code 3 at 49% [15].

Three-quarters of the most serious reported incidents in that district were absconding patients — what a reporting culture produces when it captures what is easy to classify and misses what implies clinical fault. **A patient safety incident dataset is not epidemiology.** The dominant risk is non-reporting, so a facility reporting zero is a facility with a broken process; a dashboard must treat a low rate as a warning and must carry a denominator — admissions, headcount or bed days — or a raw count comparison across facilities of different size will be used to punish the honest. One named barrier is squarely architectural: **implementing recommendations and following up action plans were weak points** [15].

What it requires of the data

The guideline's minimum information model is a mandatory field set and should be implemented as one rather than as a form design: patient name, file number, gender and age; date and time; ward, department and section; classified contributing factors; incident type from an eleven-category taxonomy; outcomes; actions; reporter [13]. Three consequences follow. **Action tracking to closure is where the value is**, because investigation happens and follow-through does not. **A just-culture process and a disciplinary referral are contradictory access requirements over one record** — as are statements from patients and relatives, which put third-party personal information in a safety record. And where a shared web-based system is used, the guideline removes the monthly submission requirement because the province reads the system directly, which is on its own the business case for a provincial incident system [13].

And one separation to hold: morbidity and mortality review is clinician-led and confidential to the professional group, while incident reporting is managerial, escalating and auditable. **They must not be merged**, because a merged system destroys the candour of the review meeting and creates a discoverable record of clinical opinion never intended as a finding of fact.

Three streams the repository does not have

Each crosses a boundary the repository cannot see: the first begins before the patient has an identity, the second has a derived boundary rather than a recorded one, the third two organisations and no shared identifier.

Emergency arrival and triage

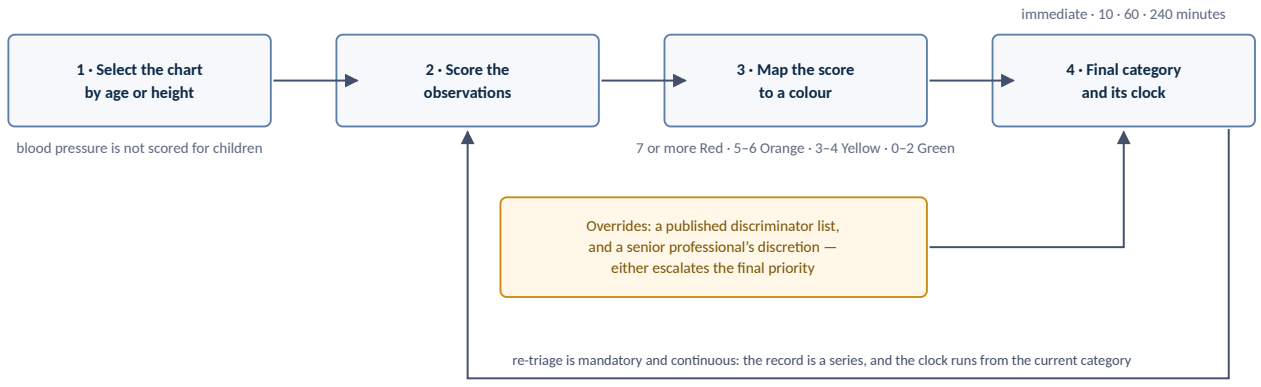
Stage	Who performs it	System the step touches	What the data must carry
Arrive	Patient, or an ambulance crew	Pre-hospital record, created before registration	A pre-hospital identifier, and the step that links it to the encounter
Register, or open an unknown record	Registration clerk	Patient administration and the identity index	A provisional identity with provenance, and an audited merge path
Triage	A trained provider; a dedicated triage nurse at all times	Triage paperwork and a coloured sticker; a triage module where one exists	Observations, score, chart used, discriminator checks, category, timestamp
Re-triage	The duty sister, continually	The same record	A series of categories, each timestamped — not one value
Stream and treat	Emergency centre clinicians	Tracking board and the emergency clinical record	Tracking state, location, responsible clinician
Dispose	Emergency centre clinician	Admission, discharge, transfer or certification	The disposition, and the handover to the inpatient record

Providers, health workers and establishments **may not refuse emergency medical treatment** (section 5 of the National Health Act [1]); **any emergency medical condition is a prescribed minimum benefit**, exempt from the designated-provider requirement [19]; and the public fee schedule provides an emergency tariff — but **triage is not chargeable** [11]. The first clinical record of the visit therefore has no billing event behind it, so data capture at triage must be justified clinically or it will not happen.

The South African Triage Scale as a mechanism

The instrument is the South African Triage Scale, published as a practice guideline by the emergency medicine society and supported by a training manual [3][4]. It is the most implementable clinical decision rule in the country, and its arithmetic is simple enough to run at the triage desk on any device.

Its shape, as a mechanism. **A composite score** is computed from scored physiological observations, mobility and trauma among them; the chart is selected by **age or height** — younger child, older child, adult — and the parameters differ, blood pressure being scored for adults and **not used for children**. **The score maps to a colour by fixed thresholds**: 7 or more is Red, 5 or 6 is Orange, 3 or 4 is Yellow, 0 to 2 is Green [4]. **A published list of discriminators overrides the score**, escalating a patient regardless of the arithmetic. **A senior health professional's discretion can override the final priority**, restricted to experienced professional nurses and emergency doctors [3]. **Re-triage is mandatory and continuous**, the duty sister reassessing waiting patients as their condition changes [3]. And **some categories are assigned by rule rather than by score**: neonates, under one month or under 50 cm, are triaged Red [4].



Mechanism only. Not a triage instruction. Confirm the current instrument with its publisher before building to it.

Figure 9.6 The South African Triage Scale as a mechanism: a scored composite, fixed score-to-colour thresholds, two override routes, and a mandatory re-triage loop that makes the triage record a series rather than a value. Described as an implementation target, not as clinical guidance.

The practice guideline sets a target time to treat and a performance threshold for each colour — the only sourced, published, numeric service-level targets for any South African clinical process found anywhere in the research behind this volume [3].

Category	Target time to treat	Performance threshold
Red — emergency	Immediate	95%
Orange — very urgent	10 minutes	80%
Yellow — urgent	60 minutes	75%
Green — non-urgent	240 minutes	70%

Blue denotes a dead patient referred for certification. The training manual's extracted text renders a time against Blue and the practice guideline does not; the two could not be reconciled, so no time should be quoted [3][4].

Three consequences follow. **The clock runs from the current category, not the first**: re-triage being mandatory, a triage record is a *series*, so a model storing one category per attendance cannot compute the target it is measured against. **Benchmarking is specified in the instrument** — audits of **25 to 50 randomly selected patients**, and cross-facility benchmarking on **category distribution, mortality by category, intensive care admission by category, and waiting times by category** [3]: a ready-made emergency department data mart specification, published, free, and almost never implemented. And **the instrument itself must be confirmed before it is built**: the practice guideline is dated **March 2010 with review due February 2012**, the manual retrieved is dated 2012, and the manual's glossary says each vital sign scores 0 to 2 while its charts allow up to 3 — **an internal inconsistency in the published instrument**, and the reason the charts rather than the glossary must be the implementation reference [4]. Obtain the current version from the publisher and store its version identifier against every triage record, because when the instrument changes the historical scores stop being comparable.

Where it fails

Identity. The emergency department opens unknown-patient records routinely — trauma, unconscious patients, children without a guardian — which makes it the main source of the duplicates an identity index must resolve, and the place where a false merge is most likely to be catastrophic, because the patient cannot correct the record. Merge must be a first-class audited operation rather than data repair, and a product that cannot unmerge must not be allowed to merge.

The pre-hospital boundary. The ambulance record is created *before* the patient is registered, so the pre-hospital identifier can never be the institutional identifier and the design needs an explicit linkage step. Crews may take patients to the most appropriate facility rather than the nearest, and ambulance provider organisations numbered 834 in 2024 [18][24] — the pre-hospital layer is not one counterparty. Then **the admission handover**, where the emergency record becomes an inpatient record and information is re-entered rather than carried. Ambulance response is a named action area in the core standards framework [37], but **no national response-time standard could be sourced** — state the gap rather than fill it.

Admission to discharge

Stage	Who performs it	System the step touches	What the data must carry
Decide to admit	Emergency or clinic clinician	Clinical record; bed state	The decision, its timestamp, the accepting service
Authorise (private)	Managed care organisation	Pre-authorisation through the switch	Authorised stay and procedure, conditions
Admit	Admissions and admitting clinician	Patient administration; consent record	Consent under section 7, the surrogate's authority, admission method
Occupy	Nursing operations	Bed state and ward stay records	Ward stay and bed occupancy period, each with a start and end
Treat	Ward, theatre, diagnostics, pharmacy	The streams above	The clinical facts, and the links between them
Plan discharge	Multidisciplinary team	Clinical record	Expected date, destination, what is blocking it
Discharge	Discharging clinician and pharmacy	Clinical record; discharge medication; the account	A **written** discharge report, discharge medication, follow-up
Transition	Receiving service or primary care provider	A letter, carried by the patient	The receiving party's acknowledgement — usually absent

Admission requires consent under section 7 of the National Health Act, with the surrogate order where the user lacks capacity. If a user is admitted **without** consent, the establishment must notify the head of the provincial department **within 48 hours**, unless the user consents within 24 hours [1] — a statutory notification with a clock, triggered by a state in the admission record.

Where the stay begins is a decision, not a fact



Figure 9.7 One patient, four defensible start times for the inpatient stay. Which event opens the stage is a documented decision, not a fact, and the decision has to be recorded with the rule version that produced it.

A patient arrives at 14:00, is triaged and treated in the emergency centre, is accepted by the medical service at 19:00, and reaches a ward bed at 02:40. There are four defensible start times, and the length-of-stay difference across them on this one patient is twelve hours and forty minutes. The British data dictionary's rule starts the stay where clinical responsibility was assumed — 19:00 — and includes the waiting time in the first consultant's episode [25]; most patient administration systems start it at ward arrival, because that is when a bed is allocated. The internal transfer does the same: a patient moving from surgery to medicine on day 4 of an 11-day stay is one continuous stay and two episodes of clinical responsibility, so counting episodes gives two admissions of four and seven days while counting stays gives one of eleven.

Chapter 8 owns the cadence and the clocks; chapter 10 owns the entities. The process-model consequence belongs here: **do not pick one boundary and bury it.** Hold the encounter as the timestamped contact and the ward stay as the location-time fact, derive the continuous stay and the episode of care by documented rules, and record the rule version behind each derived boundary — so a definitional change is a re-computation rather than a migration. The episode of care is separate — FHIR carries a distinct `EpisodeOfCare` resource because the encounter could not hold the concept [36] — and a pregnancy is one episode across four to six providers, which no South African provider's model spans. This is also why **Measure Definition** has to be an entity, and the repository has no place for it. On one specialty in one hospital over fifteen weeks, four methods of computing theatre utilisation had no simple mathematical relationship to each other, raw utilisation reading 80% and touch-time 60% **on the same list** [26] — and the same plurality applies to length of stay, occupancy, readmission and mortality. The deliverable is not the right formula; it is a versioned definition attached to every figure.

The discharge summary is a statutory artefact

Section 10 of the National Health Act requires a discharge report with prescribed content: for an outpatient it may be verbal, **for an inpatient it must be written** [1]. The professional record-keeping guidance sets what a record must contain, from identifying particulars and allergies to referrals and sick-leave periods, and the discharge summary is where most of it is assembled for the receiving clinician [5].

Discharge is also the hardest break in the medicines stream: the model changes from administration instructions to supply instructions, the dictionary from hospital stock to community products, and the recipient from a nurse to a patient and a community pharmacist. Discharge medicines reconciliation is the most-studied failure point in the chain and the one most often descoped, because it crosses an organisational boundary. Discharge medication is billed separately from the facility fee under both the public fee schedule and the compensation fund tariff [11][12] — a useful tell, because a separately priced thing is a distinct object.

And length of stay is set as much by **discharge destination availability** as by clinical readiness, so a model treating discharge as an event inside the acute hospital rather than a transition requiring capacity elsewhere will produce a figure that blames the ward for a system constraint. The compensation fund tariff maps that receiving estate, pricing rehabilitation, psychiatric, frail, palliative and hospice care as separate per-diem categories [12].

Transfer between facilities

Stage	Who performs it	System the step touches	What the data must carry
Decide and justify	Treating clinician	Clinical record	The clinical reason and the statutory basis
Find and secure a bed	Referring clinician, or a bed bureau	Telephone, in most of the country	Nothing — which is why this stage has no measure
Obtain acceptance	Receiving clinician	Receiving facility's record	An acceptance, with actor and timestamp
Compose the transfer summary	Referring clinician	Clinical record	Identifiers, active medicines, results outstanding, what was done
Move	Ambulance crew	Pre-hospital record	The transfer period and the clinical state during it
Hand over and acknowledge	Receiving clinician	Receiving record	The acknowledgement — the state almost nobody records
Reconcile identity	Receiving registration and clinician	Receiving identity index	The link between two facility identifiers for one person

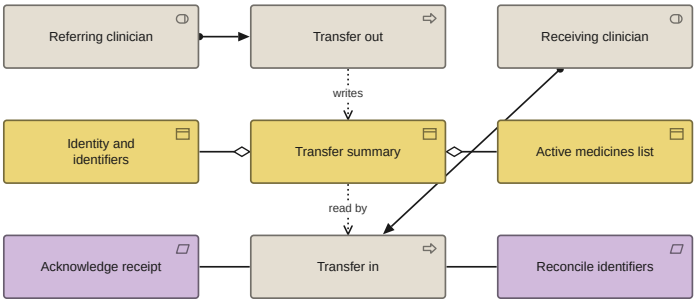


Figure 9.8 Transfer between facilities. The transfer summary is the only object that crosses, so it has to carry the sending facility's identifiers as data — with their namespace and provenance — because the receiving facility will mint its own.

Section 44(2) makes the transfer a duty [1], and the mechanism is a phone call and a letter. Four things break, and all are identifier problems wearing different clothes.

There is no shared identifier across estates. The national identity index is deliberately demographic-only — identity and the existence of an encounter, not clinical content [34] — and covered 3,266 public establishments and three general practitioner practices at 31 March 2025, with biometric identification in 82 primary health care facilities [14]. The private sector is not covered. So a transfer from Rietvlei to a provincial hospital is a linkage problem, from Rietvlei to a Driehoek hospital a linkage problem with no index at all, and across a provincial boundary a linkage problem between two separately procured estates. The transfer summary must therefore carry **the sending facility's identifiers as data**, with their namespace and provenance, because the receiving facility will mint its own.

A transfer ends one stay and begins another, which breaks two measures at once. Length of stay is split across two providers and neither reports the whole; and absent a cross-facility identifier a readmission to a *different* hospital is invisible, so a readmission rate measures the patient's loyalty to one facility rather than the quality of the discharge.

Results outstanding at the moment of transfer have nowhere to land, because the order was placed by the sending facility against its own encounter and the result returns there. If the acknowledgement loop is weak inside one facility, across a transfer it does not exist. And **in the private sector the transfer is also a commercial event**: a transfer to another facility, including one in the same group, produces a new authorisation and usually a new claim, so the funder's count of admissions is a count of authorisations, matching neither stays nor episodes.

Underneath all of it sits the records problem. A transfer is a request to another organisation for a record the sender may not be able to retrieve — in an Auditor-General audit of 27 facilities across nine provinces, 15 reported missing or lost records and 17 did not track records after issue [31] — so the **transfer summary must be composable from the system rather than assembled from a folder**.

What to take to the client

The five streams the repository names are real and the names are right. What is missing is what makes them computable, and most of it is cheap. Name the stages. Timestamp the state transitions, because a stream with one timestamp has no measure and most of the measures a sponsor wants are intervals between two states. Name the acknowledgement in every stream that crosses an organisational boundary, since each has a documented failure at exactly that point: referral sent and never acknowledged, result reported and never acted on, transfer handed over and never accepted. Give the exception queue an owner, and write down the measure definition and version it.

One sequencing recommendation follows, and the evidence supports it more strongly than the instinct to build the record first: in a low-maturity estate, **event-driven notification is the highest-return first capability**. It needs only the admission, discharge and transfer feed that already exists plus a subscription list, and its named profiles sit in the gazetted national framework, so it is procurable [30]. Design for two failure modes: a stale subscription list sends clinical information to the wrong practice, which is a breach rather than a nuisance, and without an acknowledgement path it reproduces the open loop it was built to close.

THE CHAPTER IN TEN LINES

- The repository names five value streams and numbers their stages; naming the stages is the first deliverable.
- Each stage needs a performer by role, the system it touches, the handoff that fails, and what it asks of the data.
- Level-2 models go to BPMN because ArchiMate has no swimlanes, and the lane boundary is the finding in every stream.
- A referral is a document, not a message: sent is recorded, acknowledged is not, and there is no national referral clock.
- Order and result are two entities, specimen is a third, and the acknowledge-and-act stage usually touches no system.
- Schedules 5 to 8 must be handwritten, and there is no national medicines dictionary to resolve a product against.
- Closed-loop barcode administration moves error and costs time; the evidence in non-unit-dose settings is thin and mixed.
- Coding is a clinical act with a funder regulation attached: an uncoded claim is unpayable, not unlawful.
- The claim standard defines no remittance message, and short payment rather than rejection is the dominant failure.
- A severity code 1 incident carries a 24-hour report and a 60-working-day investigation, in working days.
- A low patient safety incident count is a warning, and any rate needs a denominator.
- The South African Triage Scale is a deterministic algorithm with override rules and the only published clinical targets.
- Where an inpatient stay begins is a documented decision; record the rule version that produced every derived boundary.

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CHAPTER 10

The conceptual data model

Fourteen declared entities, sixteen missing ones, and six for which naming a single owner teaches a falsehood.

The conceptual data model is the Phase C deliverable with the longest life. Applications are replaced, integration is rebuilt, the organisation restructures — and the entity model, if it was drawn honestly, is still being argued with five years later. In healthcare it is also the deliverable most often handed over as a list of nouns with a diagram attached, and a list of nouns is exactly what this industry defeats.

Chapter 4 did the definitions. It settled what an encounter is, what a ward stay is, why admission is an event and not a stay, why there are four medication entities, and why *claim* must always be qualified. Those definitions are not restated here. This chapter does the other four columns: **what identifies each entity, what it relates to and with what cardinality, what states it moves through, and who owns it** — and then the three things the fourth column cannot be answered with.

The supplied repository you will be handed declares **fourteen data entities**: Patient, Encounter, Clinical Observation, Diagnosis, Procedure, Medication Order, Diagnostic Order/Result, Care Plan, Clinical Protocol, Practitioner, Care Setting, Claim, Consent, Medical Device. That is a reasonable first cut and a bad final one. Several of the fourteen are two or four entities wearing one name. **Sixteen load-bearing entities are missing**, and the last section of this chapter names all sixteen and says what each one holds up. Two of them — Disclosure event and Measure Definition — are introduced properly in chapter 13, and are referred forward here rather than defined.

Three rules govern the order of work, and they are worth stating before the model rather than discovering during it.

Identity before structure. Every relationship in the model resolves to an identifier, and in South Africa the patient identifier is not a solved problem. Chapter 11 is a whole chapter on it for that reason. A conceptual model drawn before the identity question is asked will have one box called Patient and a key nobody can supply.

Derived before stored. Several of the most-used entities in healthcare reporting — the continuous stay, the episode of care, every average — are *computed* from lower-level facts by a rule. Model them as views with a named rule version, not as tables. The alternative is a migration every time somebody changes a definition.

Ownership is a question, not a column. For six of the entities below there is no single plausible owner. Chapter 5 set out why, organisation by organisation. This chapter says what that does to the model, which is more than it looks.

The spine, and the four entities it hangs on

Almost every argument in a healthcare data model reduces to four entities and the relationships between them. Get these right and the rest follows; get them wrong and no amount of detail recovers it.

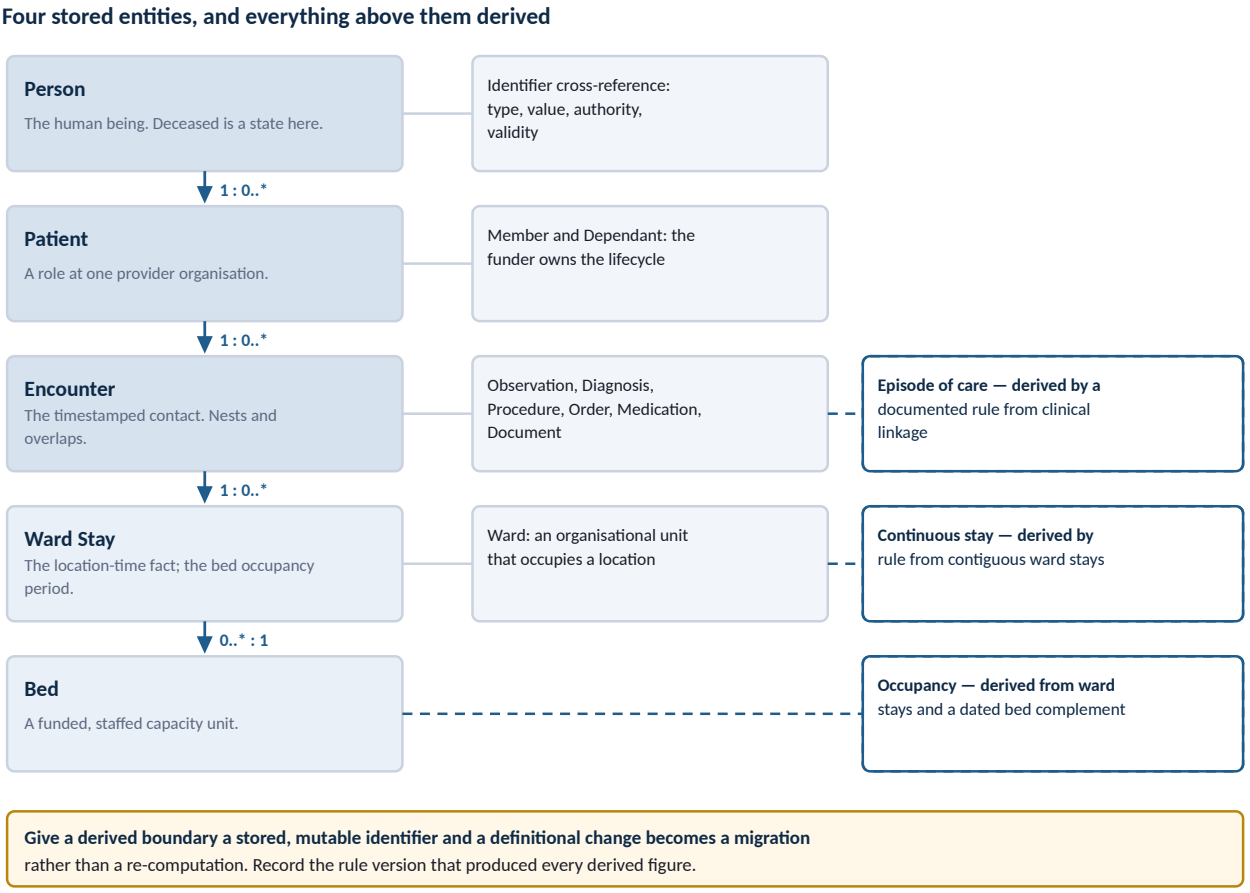


Figure 10.1 The spine. Person, Patient, Encounter and Ward Stay are stored facts; the continuous stay, the episode of care and occupancy are computed from them by a named rule. The cardinalities on the left are the ones an architect from another industry gets wrong.

Person is the human being. **Patient** is a role a person occupies in relation to a provider organisation. **Encounter** is the timestamped contact. **Ward Stay** — equivalently the bed occupancy period — is the location-time fact. Everything else in the clinical model attaches to one of those three lower boxes, and every boundary construct above them is derived.

The cardinalities on that figure are the point of it.

Person to Patient is one to zero-or-more. A person who has never presented is still a person: a next of kin, a guardian, a research subject, a staff member, a donor. A person who has presented at three facilities in one group has three patient roles if the folder numbers are facility-scoped, which in South Africa they usually are. Making Patient the root entity and hanging demographics off it produces a person with three dates of birth and no way to tell that two of them are wrong.

Deceased is a state of the Person, not of the Patient. A model that puts it on Patient has a patient who dies at one facility and is alive at the next, which is not a hypothetical: the fact of death arrives from at least three independent directions — the clinical record, the death notification process, and the funder — and they do not arrive together. There is an unresolved legal question sitting underneath it. POPIA protects an identifiable, *living*, natural person, so a deceased person is no longer a data subject, while their record still exists and is still regulated by the National Health Act and by professional guidance. Prinsen flags exactly this as unsettled in South African law: it is unclear whether records of deceased patients must be destroyed, de-identified, or treated as outside POPIA entirely [1]. Hold the state on Person, record the source and date of the death fact, and expect the retention rule for a deceased person's record to be decided locally. Chapter 18 has the retention instruments.

Patient to Encounter is one to zero-or-more, and Encounter to Encounter is many to many. Encounters nest and overlap because real care does: a patient on a ward has an inpatient encounter and, inside it, a theatre encounter, a radiology encounter and three consultant reviews. The specification allows them to nest and overlap deliberately, which is also why a separate episode-of-care resource had to exist at all [3]. A model that forbids it will represent the theatre visit as a separate attendance, and the theatre list and the ward census will disagree by exactly the number of operations performed.

Encounter to Ward Stay is one to zero-or-more; Ward Stay to Bed is many to one. An outpatient encounter has no ward stay. An inpatient encounter has as many ward stays as the patient had wards, each with its own start and end. This is the entity the supplied repository does not have, and its absence is why occupancy cannot be computed. A `ward` column on the admission record gives one ward per stay, so transfers vanish; no bed, so occupancy is uncomputable; no dates, so the bed complement is treated as a constant; and no distinction between the ward's specialty and the treating specialty, which differ in every hospital that has ever had a bed shortage.

Member and Dependant is a separate entity from Patient, with a lifecycle the provider does not own. The funder claim message makes the funder's model explicit: scheme, then benefit option, then member, then dependant, with the patient occupying the dependant position and dependant code 00 conventionally the principal member [4]. The link from Patient to Member is an *assertion* that must be re-verified at every encounter — which is what a membership enquiry and a benefit check actually are. A model that uses the membership number as a patient identifier loses the patient at every policy change, and merges siblings who share a membership number if the dependant code is dropped.

THE TWO RULES THAT DO MOST OF THE WORK

Hold **Encounter** as the timestamped contact and **Ward Stay** as the location-time fact. Derive everything above them. And keep **Person** as the root of identity, with Patient as a role — because the alternative is a model that cannot represent the trauma case with no name, the staff member who is also a patient, the neonate whose first identifiers come from the mother, or the person with two folder numbers and one body.

What identifies each entity

The honest answer for most clinical entities is a surrogate key, and the interesting part is the natural key that *nearly* works and the reason it fails. That reason is usually the modelling defect waiting to happen.

Entity	What identifies it	Where the natural key fails
Person	A provider-minted enterprise identifier, cross-referenced to national and documentary identifiers	No universal national identifier exists at the point of care; chapter 11 is about this
Patient	A provider-assigned patient identifier — in South African practice the folder number	The folder number is facility-scoped, so it is almost never a candidate key for Person
Encounter	A provider-assigned encounter identifier, unique within the provider	Not unique across providers, and the funder has no encounter at all
Ward Stay	Encounter identifier plus sequence, or its own surrogate	A patient can occupy two beds in one day; sequence must be dated, not ordinal only
Clinical Observation	Surrogate. Natural key is patient, code, effective time and the performing context	Drop the performing context and the ward glucometer result deduplicates into the laboratory result
Diagnosis	Surrogate	The same condition is asserted, revised, refuted and re-asserted; there is no natural key
Procedure	Surrogate	Patient, code, time and performer fails for bilateral procedures, repeats in one session, and team operating

Entity	What identifies it	Where the natural key fails
Medication Order	A prescription or order identifier	In private practice the prescription is also a legal document carrying the prescriber's registration number
Dispense	A dispense identifier, against a *product*	The order names a drug and a strength; the dispense names a pack with a lot or batch and an expiry
Administration	An administration identifier	An infusion with rate changes is one administration with a duration or a series; the choice changes every exposure figure
Diagnostic Order	An order identifier at the placer **and** an accession number assigned by the laboratory	Two identifiers owned by two organisations; the mapping between them is where results get lost
Specimen	A specimen or container identifier	One order may need several specimens and one specimen may serve several orders
Practitioner	The statutory council registration number, per register	It identifies the *registration*, not the person; one person may hold several
Device Instance	A serial number plus a provider asset number	The two are rarely the same value and both must be held
Claim, to a funder	A claim number, versioned	A funder claim is reversed, resubmitted and appealed; the latest state is not the object

Three of those rows carry consequences beyond themselves.

The identity number must not be the primary key of Person. It breaks the moment a patient presents without one, presents with somebody else's, or has a corrected one issued — and all three happen daily. It belongs in an identifier cross-reference table with a type, an assigning authority and a validity period.

The council registration number is the best practitioner identifier available in the country, and it is an identifier of an authorisation rather than of a human being. Both councils publish an online register for verification [20]. A provider that does not verify registration numbers against the register at credentialling and periodically afterwards has declined to use the best identity data in the health system. Scope of practice is data, not policy, because it is what an ordering or prescribing authorisation check must consult, and the nursing council's own framework defines what a given category may do [21].

The practice code number identifies a billing entity, not a clinician. It may name an individual, a group practice, a hospital or a laboratory. Conflating the two produces reports in which a group practice performed four thousand procedures and no individual performed any. One honest limit: the governing document on practice code numbering, discipline codes and place-of-service codes could not be retrieved for this volume, so this account is descriptive rather than sourced to its own standard. Obtain the Council for Medical Schemes standardisation manual before building to it.

The cardinalities that are not what an architect expects

Four relationships in this model fan out where an architect from another industry would expect one to one, and each fan-out is where a real failure lives.

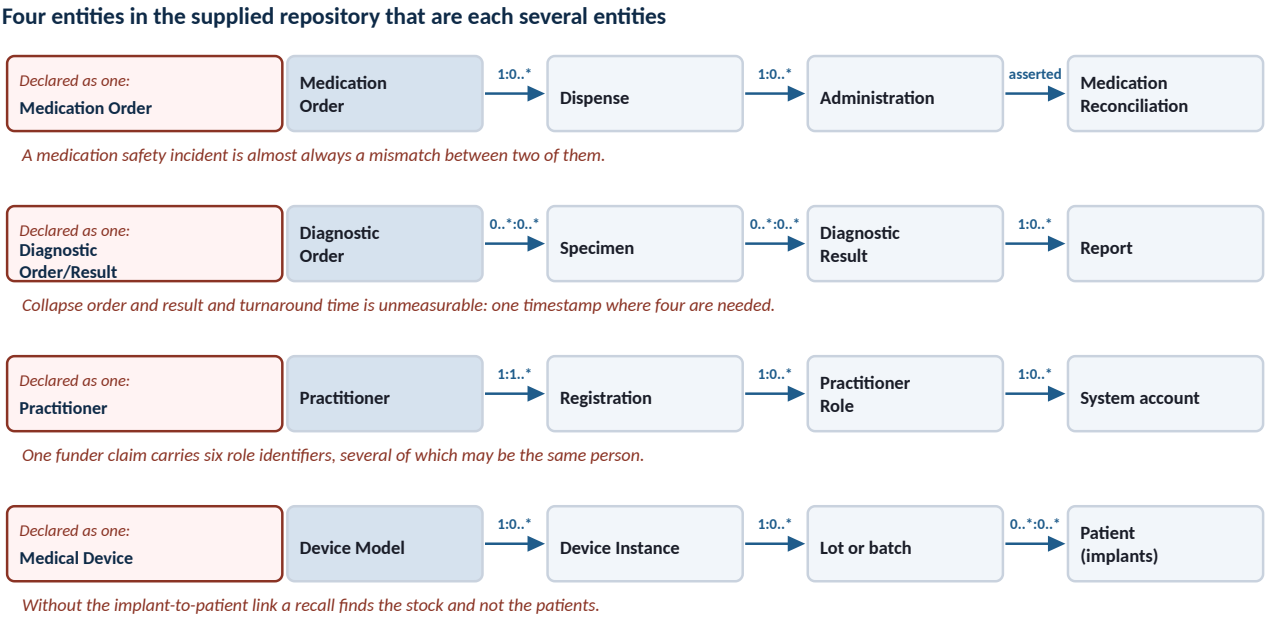


Figure 10.2 The four fan-outs. Each left-hand box is one declared entity in the supplied repository; each row is what it has to become before the question underneath it can be answered. The italic line is what the collapse costs.

Medication is four entities, and the relationships between them are the model. Order to Dispense is one to zero-or-more: a partial dispense is a completed dispense for a smaller quantity, so a model that assumes dispense quantity equals order quantity over-counts exposure. Order to Administration is one to zero-or-more, and the zero is the clinically important case — a dose not given, with a reason. Medication Reconciliation is **not** a view over the orders; it is an independent clinical assertion with an author, a date, a set of sources consulted and a set of discrepancies, and the discrepancies are the valuable output. Derive "current medications" from active orders and you will be wrong for most chronic patients, and wrong in the dangerous direction, because the patient is taking things nobody here prescribed and not taking things that were.

The reason to hold all four is diagnostic: a medication safety incident is almost always a mismatch between two of them. A model holding one cannot see a mismatch. And where public-sector pharmacy runs a stock system rather than patient-level dispensing, there is no dispense record at all and the only evidence a medicine was issued is a stock movement — the national stock visibility system is supply chain, not a medication record [13].

Diagnostic Order, Specimen and Result are three entities with two many-to-many relationships. Order to Specimen is many to many. Result to Order is many to one and sometimes many to many, because reflex testing produces a confirmatory result nobody ordered. Collapsing order and result — which the supplied repository does, under the name Diagnostic Order/Result — makes the unordered result impossible to represent and makes turnaround time unmeasurable, because there is one timestamp where there need to be four.

Those four timestamps are worth naming, because they are the single most common omission in a clinical observation model. **Effective time** is when the specimen was taken or the measurement made. **Issued time** is when the result was produced. **Recorded time** is when it entered the system. **Available time** is when it became visible to a clinician. Clinical safety depends on the first, audit on the third, and turnaround-time measurement on the gap between the first and the fourth. A model with one `date` column on observations can measure none of them. The South African illustration is exact: during the 2024 national laboratory outage, results were delivered by telephone, message, email, fax, courier and paper, and every one of those routes produces an observation with a correct effective time and an unrecorded available time [6]. Chapter 19 owns the incident; this is its data-model residue.

Practitioner is three entities in a chain. Practitioner to Registration is one to one-or-more; Registration carries the live status. Practitioner to Practitioner Role is one to zero-or-more, where the role is the practitioner at one organisation, in one specialty, at one location, for a period, with a set of privileges. Then the funder claim shows what that means operationally: one message carries a supplier number, a group practice number, a council registration number, and separate identifiers for the specialist, anaesthetist, assistant, referring, treating and admitting doctors [4]. Those are not six practitioners. They are up to six *roles* on one event, several of which may be the same person. A single `doctor` table with a name, a specialty and a code cannot represent any of it, and cannot support the authorisation check that scope of practice requires.

Medical Device is three levels, not one. Device Model is a product type with a manufacturer, a classification, a regulatory status and an intended use. Device Instance is a specific physical unit with a serial number, a location, a maintenance history and a calibration state. Lot or batch is the third level and is the unit of recall. The relationship that matters most is the one almost always absent: **Device Instance to Patient, for implants**, which creates a permanent association and is the basis of every device recall. Hold devices only in the asset register and a recall cannot be executed — the organisation can find the stock and not the patients.

Three smaller cardinalities, each with a cost attached.

- **Appointment to Encounter is zero-or-one to zero-or-one, and both zeros matter.** The appointment with no encounter is the non-attendance; the encounter with no appointment is the walk-in. Deleting cancelled appointments destroys the denominator for non-attendance, utilisation and cancellation analysis in one stroke. No published South African non-attendance rate could be found for this volume, partly because much public outpatient attendance is not appointment-based, so an organisation that wants the figure will have to be the one that computes it.
- **Observation to Observation is many to many**, for panel membership and for derived values. A derived observation's provenance includes the demographic attributes used in the derivation — an estimated glomerular filtration rate depends on age and sex — which is why changing a patient's

recorded sex silently changes historic values in some systems.

- **Invoice to Encounter is many to many.** In South African private practice one stay routinely produces a hospital account, a surgeon's account, an anaesthetist's account and a pathology account, from four legal entities. One invoice can also span several encounters. Model it one to one and the revenue model fits a single-employer hospital that does not exist here.

The funder claim line is a polymorphic object

One row in the money model deserves its own warning, because it is where a tidy design meets a standard that is not tidy. The South African private funder claim line can carry tariff codes, modifiers, surcharges, non-chargeable codes, laboratory and fixed-fee codes, several clinical and product code systems, a medicine and consumable group with its own quantities and amounts and a fallback code where no product code exists, and dental tooth numbers [4]. One line is therefore a small polymorphic object, and a data model giving it a single `code` column with a type discriminator has more than ten possible meanings on one column.

The capacities differ by type of funder claim as well: a hospital funder claim carries up to thirty diagnostic codes at header level while a non-hospital practitioner funder claim carries one primary and nine secondary *per line item* [4][5], so the funder's view of the diagnosis sits at a different granularity from its view of the activity. And the funder claim is a **versioned** object, because the standard explicitly carries adjustments, reversals, resubmissions and interim accounts [4]. A model that overwrites it loses the record of what was claimed first, which is the only thing a rejection analysis can be built on.

Care Plan, Pathway and Protocol — three objects, and the argument between them

These three are modelled as one entity more often than any other group in the catalogue, and the collapse costs the organisation its only view of what was planned and not done.

Clinical Protocol is the standard: an organisation-adopted, evidence-based specification of how a defined clinical situation should be managed. It is not patient-specific, and it is identified by an identifier **plus a version**, because "was this patient managed according to protocol?" is meaningless without knowing which version was in force on the date of care — and "why did outcomes change in April?" is frequently answered by "the protocol changed in April". The national source objects are citable and dated: the Primary Healthcare Standard Treatment Guidelines and Essential Medicines List, eighth edition, 2024 [15], and the National Integrated Maternal and Perinatal Care Guideline signed on 4 October 2024 [16]. A provider's protocol register should carry the national version it derives from *and* the local modification made to it, because a hospital that has modified a national guideline has taken on the clinical risk of the modification, and the modification must be traceable.

This is also the one entity in the catalogue whose owner is unambiguous — clinical governance — and which is nevertheless almost always unmastered, living as documents on an intranet with the version in the filename and the approval in an email. The second-order defect is worse: decision support rules implemented without a traceable link to the protocol they implement, so the protocol is updated and the rule is not, and the system now enforces a superseded standard. That defect is invisible until it appears in a medico-legal report.

Pathway is the instantiation for one patient: where they are on the route, which steps are done, which are overdue, and which were deviated from. The pathway instance carries the protocol **version** it instantiates, the entry date, the current step, the expected and actual step dates, and the deviations — and **deviation with a documented reason is the clinically and legally important fact**, not an exception to suppress.

Care Plan is the patient-specific forward-looking statement: goals with targets and target dates, planned activities with statuses and scheduled times, responsible parties, the conditions addressed, and a review date. A patient has one care plan assembled from several pathways; a patient with diabetes, hypertension and a pregnancy is on three.

Two modelling rules follow. **Review date is a first-class attribute**, not a diary entry, because an unreviewed care plan past its review date is a measurable governance failure and the measure needs the date in the model. And **the planned activity and the actual event are different objects, linked** — the planned activity relates to the Procedure, Medication Order, Diagnostic Order, Appointment or Encounter that fulfilled it. Model them as one row with a `completed` flag and the organisation has destroyed its ability to see what was planned and not done, which is the clinically important case. The other common error is modelling the care plan as a document: a document cannot be queried for overdue activities, cannot drive a task list and cannot be reported on.

The lifecycles, and the states that models leave out

A lifecycle in a healthcare model is not decoration. Several entities have a state whose absence makes a legal obligation unmeetable, and the pattern is consistent enough to be checkable in an afternoon.

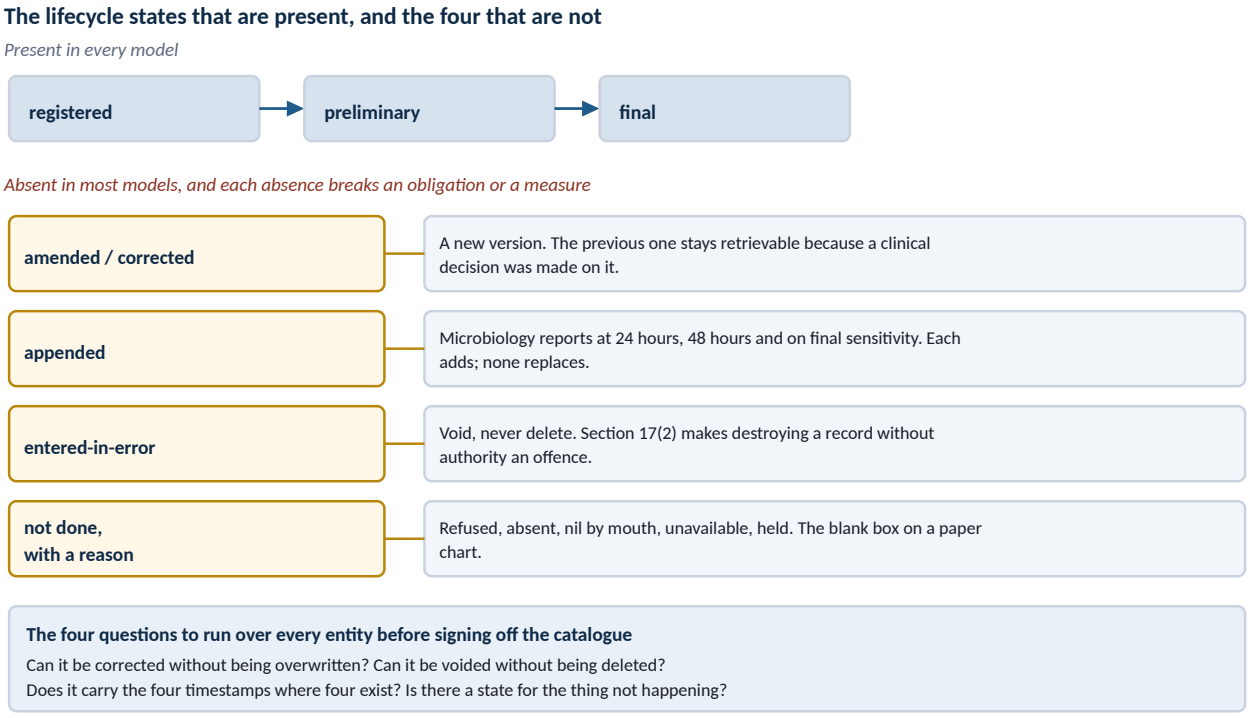


Figure 10.3 Lifecycles. The three states every model has, the four it usually lacks, and what each absence costs. Two of the four are legal requirements rather than design preferences.

Entity	States everyone models	The state whose absence breaks something
Encounter	planned, arrived, in progress, finished	**cancelled** (the non-attendance numerator) and **entered-in-error** (opened on the wrong patient)
Clinical Observation	registered, preliminary, final	**amended** and **corrected** as new versions, and **entered-in-error**
Diagnostic Result	registered, preliminary, final	**appended** — microbiology reports at 24 hours, 48 hours and on final sensitivity, each adding to the last
Diagnosis	provisional, confirmed, resolved	**refuted** — a refutation is clinical information, not a deletion
Medication Order	draft, active, completed	**stopped** and **on hold** — stopped on day three of a seven-day course is not a seven-day exposure
Administration	completed	**not done, with a reason** — refused, absent, nil by mouth, unavailable, held on clinical instruction
Consent	active	**withdrawn**, with a date and an actor, and **expired** where it was time-bounded
Clinical Protocol	draft, active	**superseded** — a retired protocol is not deleted, because care delivered under it must stay interpretable
Bed and Ward	open, closed	commissioning, reconfiguration and decommissioning are **dated events**, so bed complement is a time series

Two of those rows are legal requirements rather than preferences, and they are the reason this section exists.

A correction is a new version, not an update. Section 17(2) of the National Health Act makes it an offence to falsify a health record, or to create, change or destroy one without authority, punishable by a fine, imprisonment of up to a year, or both [9]. The professional rule is more specific. HPCSA Booklet 9, in the September 2016 edition retrieved for this volume, states at paragraph 8 that nothing may be removed from a record, that an error is corrected by a line through the entry leaving the original legible, with the correction dated and signed in full and the reason recorded, and that later additions are dated and signed [2]. The September 2022 revision goes further for electronic records: entries must be attributable, electronic alterations must be fully attributable to the person who made them, late entries and addenda must be marked with their reasons, and the reason for any amendment must be recorded [2].

Read those forward and the professional rule describes an **append-only record with a full, attributed, reasoned audit trail of every change**. An observation table with `UPDATE` semantics is non-compliant on its face. The requirement that the *reason* for an amendment be recorded is the one almost never implemented, and it is the cheapest of the set to add.

> **WHICH EDITION OF BOOKLET 9 IS IN FORCE COULD NOT BE ESTABLISHED**
>
> The retrievable published version is September 2016, and a version marked "Revised: September 2022" is also in circulation [2]. The 2022 text is materially stronger on electronic records, and the two differ on retention in a way chapter 18 deals with. Design to the stricter reading, and obtain the current booklet from the Council before quoting a clause number to a client.

A corrected result is not an updated result. A potassium value corrected after the fact is a new version, and the previous version must remain retrievable, because a clinical decision was made on it. That is also why the imaging report and the observations behind it have different lifecycles: observations are corrected individually, reports are reissued as versions. A model holding only the report cannot aggregate; a model holding only the observations cannot show a clinician what the radiologist actually said.

Two further lifecycle facts, both South African, both easy to miss.

An **asserted** diagnosis and a **coded** diagnosis have different lifecycles and different authors. The recorded text is the clinician's assertion; the code is a classification of it, made later, by someone else, for a purpose that is usually not clinical. Hold both, linked, with the coder, the coding date, the classification version and the coding purpose. The South African evidence on the gap between them is unusually good and chapter 4 carries it; the modelling consequence is that `diagnosis_code` on the encounter row is not a shortcut, it is a different entity with the wrong name.

And **primary is a property of a diagnosis within an episode, not of the diagnosis.** The national morbidity coding standards define the main condition as the one diagnosed at the end of the episode of care that is primarily responsible for the patient's need for treatment or investigation, with a tie-break on clinical severity, then resource consumption, then order of recording [5]. The same condition is primary in one admission and secondary in the next. So `is_primary` as a boolean is correct only if the row is scoped to the episode and the rule version that assigned it is recorded — which is the next section.

Derived, not stored

Four of the most-quoted constructs in a provider's reporting are not data. They are computations.

The continuous stay is derived from contiguous ward stays by a documented rule. **The episode of care** is derived from clinical linkage by a documented rule. **Occupancy** is derived from ward stays and a dated bed complement. **Every average** is derived from a numerator, a denominator, a set of exclusions and a boundary rule.

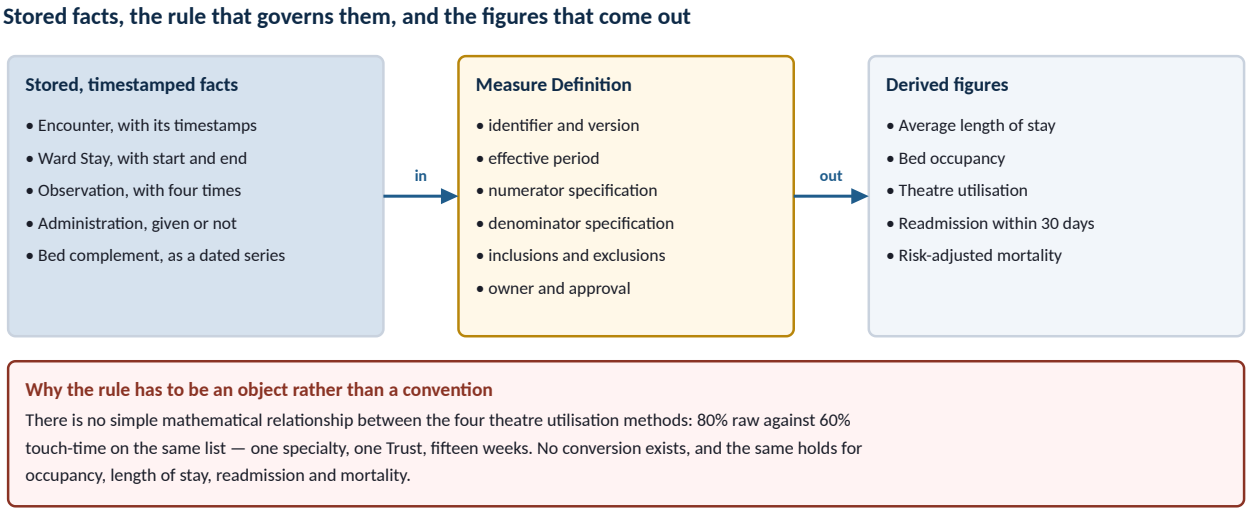


Figure 10.4 The derived layer. Compute every figure from the lowest-level timestamped facts through a versioned Measure Definition, so that a change of definition is a re-computation and a published figure can cite the rule that produced it. Chapter 13 introduces the entity.

Giving a derived boundary a stored, mutable identifier is the trap, because it converts a definitional change into a migration. Hold the lowest-level timestamped facts, compute the boundaries as views, and make the rule a first-class object so that any figure can be traced to the rule that produced it. The derived boundaries then become views rather than data, and a definitional change is a re-computation.

The published evidence that the rule must be an object rather than a convention sits on theatre utilisation, because theatre time is expensive enough that people have argued about it in journals. Zhang and Pandit show **no simple mathematical relationship between the four calculation methods**, with raw utilisation of 80% and touch-time utilisation of 60% on the same list [10]. Carry the denominator with that finding: it is **one specialty, in one Trust, over fifteen weeks**, and chapter 2 teaches readers not to accept a figure without one. There is no conversion between the methods, and the same holds for occupancy, length of stay, readmission and mortality.

The object that solves it is the **Measure Definition**: the versioned, effective-dated, owned rule by which a derived figure was computed, with its numerator, denominator, exclusions, boundary rule and effective period. Chapter 4 named it; **chapter 13 introduces it properly**. The point to register here is a repository fact: the supplied model has nowhere to put it. Fourteen entities, none of which is a rule, so every derived figure in the organisation is defined in a query somebody wrote and nobody versioned.

The Western Cape Provincial Health Data Centre is the South African proof that the derived-layer approach works at scale. It builds a master list of encounters spanning household visits to admissions, then *infers* episodes of health conditions by weighting evidence from several sources with a certainty score, then builds cohorts as virtual continuously-updated views [14]. The reason it infers rather than demands is worth taking seriously: diagnosis coding in the source systems is limited, so condition presence has to be triangulated. **Infer the clinical layer rather than requiring the source systems to code better** is a legitimate strategy and not a compromise. Chapter 22 reads the data centre as a worked reference design, including its published costs.

The ownership question, and why it is the deliverable

The standard entity catalogue has an owner column and expects one name in it. Chapter 5 set out, function by function, why that expectation fails here. This section says what it does to the model.

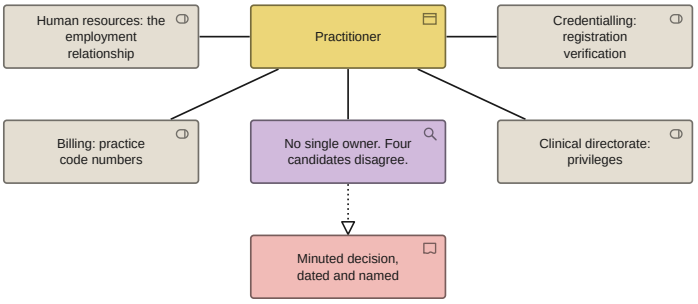


Figure 10.5 The ownership question as a deliverable rather than a column. The same shape applies to Care Setting, Ward, Bed, Location, Care Plan and the medicines master. What the architect produces is the decision, with a date and a name on it.

For six entities, naming a single owner teaches a falsehood. Practitioner has four candidates: human resources for the employment relationship, credentialling for registration verification, the clinical directorate for privileges, billing for practice code numbers. Care Setting, Ward, Bed and Location have four between them: estates for Location, nursing or operations for Ward and Bed, the clinical directorate for the definitions of setting, finance for the commissioned complement. Care Plan is genuinely contested between services, and the question underneath it — one care plan per patient or one per service — bears on whether a patient's care can be seen whole.

The medicines master needs a careful statement, because the loose version of it is wrong. The research names **two** sourced owners and an absence: **pharmacy owns the dispensing catalogue, finance owns the price, and nobody owns the clinical classification** [17]. A third owner — procurement, through the public sector's master procurement catalogue — is a defensible inference and not the research's statement [19]. Say two and an absence, or say the third is an inference, and do not assert three.

The absence is the expensive part. South Africa has a commercial pack-level product identifier in NAPPI, maintained by a funder claims switch [17]; a regulated single-exit-price register, which makes price an effective-dated attribute of a product set by regulation rather than negotiated [18]; a public-sector procurement catalogue on its own item numbering that does not reconcile to NAPPI without a crosswalk [19]; a national formulary in the essential medicines list, which is a list of what may be used and a different object again [15]; and **no national ingredient-level medicines terminology**. The consequence is specific: a South African provider cannot answer "which patients are on an angiotensin-converting enzyme inhibitor?" from its medicines master without building its own classification, because the only identifier it reliably holds is a pack code. Chapter 12 carries what to do about that.

So what does the model carry instead of an owner? **Four fields and a date.**

1. **The candidate owners**, named as functions rather than people.
2. **What each demonstrably owns today**, with the system that proves it — not what the organisation chart implies.
3. **What breaks if nobody decides**, stated concretely. For Practitioner it is that a leaver remains an active prescriber in one system and an active biller in another. For Bed it is that nobody can count beds. For the medicines master it is that no clinical question about a drug class can be answered.
4. **The decision**, minuted, with a date and a named person.

That is one page per contested entity, and it is worth more than the rest of the ownership column together. An architecture document that asserts a single owner where four exist will be contradicted by the first reader who does the work, and that credibility cost is not recoverable inside a quarter.

> **THE DE FACTO MASTER IS A SEPARATE PROBLEM**
>
> Where ownership is undecided, something becomes the master anyway, and it is usually whatever was integrated first. The supplied repository's defects are exactly this pattern pushed to caricature — Patient mastered in the prescribing system, Clinical Observation in the billing system, a funder claim mastered in the clinical record. The last of those is a precise inversion: revenue cycle owns the funder claim unambiguously, and Booklet 9 paragraph 13.7 requires billing records to be kept **separate** from patient care records [2]. Chapter 21 names the documented pattern behind each defect. The audit question for any estate is short: for each entity, which system's value wins when two disagree, and who decided that?

One more owner worth naming, because architects routinely leave it out. **Document and Clinical Note is owned by the records function** — the registry or records office in a South African public facility — and the published evidence is that the function is widely failing its statutory obligations. Kolawole reports that numerous South African public health facilities fail to meet their record-keeping obligations as required by law, that the non-compliance undermines medico-legal claims, and calls on the courts to compel compliance [11]. The architectural reading is that the records function is a first-class stakeholder in the data-architecture phase, not a consumer of whatever the clinical systems happen to produce.

The sixteen entities the repository omits

This is the list that makes the chapter useful on a Monday morning. Each entry says what the entity is load-bearing for — what question becomes unanswerable, or what obligation unmeetable, without it.

One precision before the list, because the count is quoted loosely. There are sixteen *gaps*, and several of them hold more than one entity: Dispense and Administration are one gap and two entities, Device Model, Device Instance and Lot or Batch are one gap and three, and the three consents are one gap and three. Counted as entities rather than as gaps the number is nearer twenty-six. Use sixteen when citing the research and twenty-six when sizing the work.

Entity	What it holds up
Person, distinct from Patient	Identity across facilities, the staff member who is also a patient, the unidentified trauma case, the neonate, the deceased
Member and Dependant	The funder's subject, with a lifecycle the provider does not own and must re-verify at every encounter
Ward Stay / Bed Occupancy Period	Occupancy, in any form. Without it the number cannot be computed from the record at all
Episode of Care	Longitudinal care for a condition or programme: antenatal, chronic disease, oncology — none of which fits in an encounter
Specimen	The laboratory pathway, rejected and haemolysed samples, and mislabelling, which is an identity failure seen from the other end
Dispense, and Administration	What product, from which lot or batch, with what expiry and in what quantity; and then whether the dose was given, and the reason if not
Medication Reconciliation	The gap between what was prescribed and what the patient is taking, expressed as discrepancies

Entity	What it holds up
Remittance	The denial rate, denial-reason analysis, and reconciliation of cash to activity. The South African funder claim standard defines no remittance message at all [4]
Registration, and Practitioner Role	The live authorisation status that makes an order actionable and a funder claim payable; and who did what, where, under which privileges
Device Model, Device Instance, Lot or Batch	Recall execution, calibration state, software version, and the implant-to-patient link
Appointment, Slot and Referral	Capacity utilisation, non-attendance, and the referral states `sent` and `acknowledged` on which lost follow-up turns
Document with structured metadata	Retention rules, access requests, and finding the discharge summary for an audit
Consent to treatment, to disclosure, to research	Three different legal bases, scopes and withdrawal effects. One flag can carry none of them
Protocol Version as part of protocol identity	The question "was this patient managed according to protocol?", which is meaningless without the version in force on the date of care

Two more, and they are the two nobody asks for.

Disclosure event — the record that specified information was given to a named recipient on a date for a purpose. Without it the organisation cannot tell a patient who received their data and cannot act on a withdrawal selectively, so POPIA's openness and data-subject-participation conditions cannot be satisfied in practice [1]. **Chapter 13 introduces it** and chapter 18 carries the obligations behind it. Register it in the model now, because retrofitting a disclosure log over years of existing disclosures is not possible.

Measure Definition — the versioned rule behind every derived figure, as above. **Chapter 13 introduces it.**

> THE TEST FOR AN ENTITY CATALOGUE
>
> Run four questions over every entity before you sign the catalogue off. Can it be *corrected* without being overwritten? Can it be *voided* without being deleted? Does it carry the **four timestamps** where four exist? And is there a state for the thing *not happening* — the dose not given, the appointment not attended, the referral not acknowledged, the consent withdrawn? Most repositories fail at least two.

The two entities that exist only as a row

Consent and Medical Device commonly appear in a repository as one attributeless row each. That is not a random omission; it is the commonest real-world state of the two entities carrying the most legal risk. **An entity with no attributes is a placeholder for an argument nobody has had yet**, and naming the argument is the useful contribution.

Consent is at least three entities and arguably four. Chapter 4 defines them and chapters 13 and 18 carry the law. The modelling rules are what belong here, and there are six.

1. **Three entities minimum**, not one with a type code, because the attributes genuinely differ: an intervention reference on the first, a recipient-and-purpose set on the second, a protocol and ethics approval reference on the third.
2. **Versioned and effective-dated**, because an act is governed by the version in force at the time of the act. The model must answer "what consent was in force on 14 March 2024?", which rules out a current-state flag.
3. **Withdrawal is a state with a date and an actor**, and it must be queryable, because the organisation's defence is the record that it stopped.
4. **Scope is structured, not free text** — a set of information classes, recipients, purposes and a period. Free text makes enforcement impossible; triples make it enforceable by a policy decision point.
5. **The capacity assessment is part of the consent record.** Under section 129 of the Children's Act a child over 12 with sufficient maturity and the mental capacity to understand the benefits, risks and implications may consent to their own treatment [12]. Capacity is therefore a function of age *and* an assessed maturity, and the assessment is itself a clinical act that must be recorded. A `patient_consented` boolean cannot represent any of it.
6. **Consent is not the access-control mechanism.** The lawful basis for clinical processing in a provider is most likely the statutory authorisation for treatment and care, not consent; consent governs disclosure outside the care relationship and research [1][9]. A system that gates clinical access on a consent flag has mis-modelled the law and will block emergency care. A system with no consent object at all cannot govern disclosure or research. Both failures are common, and they are opposite.

And the fourth family, which is a different modelling problem: **advance directives, refusals and proxies** — an advance refusal of a specific treatment, a durable power of attorney for health decisions, an organ donation registration, a resuscitation instruction. Each has a different authority and a different

asserter, and each must be findable in under a minute at three in the morning. That is an availability requirement, and it belongs in the model as one.

Medical Device needs its regulatory attributes or there was no point recording it. The South African regulatory frame sets them. Devices are regulated under the Medicines and Related Substances Act through the general regulations for medical devices and in-vitro diagnostics, published in *Government Gazette* 40480 on 9 December 2016, and SAHPRA's classification guideline — version 5, dated 27 February 2025 — sets four risk classes from A to D for non-IVD devices with a parallel scale for IVDs, classified by rule, with the higher class applying where more than one rule is engaged and accessories classified separately [7].

The architecturally decisive fact is that **product registration is not yet operating**. SAHPRA's own communication of 26 September 2025 states that it has not begun product registration of devices and IVDs and that the product-registration regulation has not been called up; what is in force is the establishment licence regime for manufacturers, distributors and wholesalers [8]. So a South African provider's device master cannot be populated from a national product register, because there is not one. The provider's own asset register plus the supplier's declarations are the only available source of truth, and the model should record which of the two each attribute came from.

The attributes that must be there and usually are not: risk class; intended use, which the guideline says is taken from the instructions for use, labelling, advertising and technical documentation — making it a *claimed* attribute with a provenance rather than a fact [7]; calibration due date and last calibration; maintenance schedule and history; software version for anything containing software; patch level; decontamination status for reusable instruments; and the patient link for implants. Chapter 20 gives the modelling pattern that keeps the regulatory constraint attached.

> **THE ASSET REGISTER IS AN ASSERTION, NOT A FACT**
>
> Treat equipment data as an assertion with a source until it has been counted. The published South African evidence includes managers borrowing equipment to present at facility assessments [22], and an audit in which 23 of 27 clinics had no maintenance schedule and 24 had not determined required stock levels [23]. Design for location-attributed, timestamped data, and record the provenance of every attribute.

One more absence, and it is structural

There is a layer of South African health data with no patient entity in it at all. The national aggregate reporting system holds counts by facility, data element and period. It is downstream of the record, it is not an electronic health record, and it cannot deduplicate a patient because there is no patient in it to deduplicate.

That absence explains a landscape rather than a system. The HIV programme has its own patient register and register number; the tuberculosis programme has its own; the antenatal card has its own; the Road to Health booklet has its own. Each exists because the facility identifier did not serve the programme's longitudinal need, and each now constitutes another identifier type that has to be held and matched. **Reframing the vertical registers as the consequence of one architectural absence changes the recommendation** from rationalisation by decree to supplying the missing capability — a patient-level longitudinal record with a resolvable identity. Chapter 11 is about the identity half of that, and chapter 15 about the systems half.

What to take to the client

The conceptual model is a short document and a long argument. Nine things make it useful rather than decorative.

1. **Person and Patient as separate entities**, with Patient as a role and an identifier cross-reference under Person. Everything in chapter 11 depends on it.
2. **Encounter and Ward Stay as stored facts; the continuous stay and the episode of care as derived views** with a named rule version.
3. **Four timestamps on every observation**, and a named owner for the available time, which is the one nobody collects.
4. **The four medication entities**, with the administration entity's "not given, with a reason" state present even where the organisation has no electronic administration record — so that the gap is visible as a gap.
5. **Order, Specimen and Result as three entities**, with the placer's order identifier and the laboratory's accession number both held and mapped.
6. **Practitioner, Registration and Practitioner Role as three entities**, with registration status carrying an expiry and a refresh.
7. **Every entity's correction path**: version, void, reason, author, timestamp. Section 17(2) and Booklet 9 paragraph 8 make it a legal requirement, not a design preference [2][9].
8. **The sixteen missing entities added**, with Disclosure event and Measure Definition registered even if they will not be populated this year.
9. **For each contested entity, the four-field ownership record** — candidates, what each owns today, what breaks without a decision, and the minuted decision with a date and a name.

And one thing to refuse. Do not accept "we will fix the data model in the next release". A conceptual model is cheap to argue about and expensive to retrofit, and the entities that get retrofitted last — Disclosure event, Measure Definition, Administration, Specimen — are precisely the ones that cannot be reconstructed from history. Everything else in Part III rests on this chapter: identity next, then coding, then the record itself.

THE CHAPTER IN TEN LINES

- Chapter 4 settled the definitions; this chapter does identifiers, relationships, cardinalities, lifecycles and owners.
- The supplied repository declares fourteen entities. Several are two or four entities under one name, and sixteen are missing.
- Person is the root of identity and Patient is a role; deceased is a state of the Person, never of the Patient.
- Hold Encounter as the timestamped contact and Ward Stay as the location-time fact, and derive everything above them.
- The identity number must never be the primary key of Person. It belongs in an identifier table with a type and an authority.
- Medication is four entities, order and result are three, Practitioner is three, and Medical Device is three.
- A correction is a new version, not an update. An observation table with update semantics is non-compliant on its face.
- The four states models omit are amended, appended, entered-in-error, and the thing not happening with a reason.
- There is no simple relationship between the four theatre utilisation methods: 80% raw against 60% touch-time on one list.
- For six entities there is no single owner. The deliverable is four fields and a minuted decision, not a guess.
- The medicines master has two sourced owners and an absence: pharmacy, finance, and nobody for clinical classification.
- Consent is three entities with structured scope; Medical Device needs its regulatory attributes or there was no point recording it.
- Disclosure event and Measure Definition are the two entities nobody asks for. Register them now; they cannot be retrofitted.

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CHAPTER 11

Identity

Four per cent of infants matched exactly on four demographic details, and the national linking algorithm found 4,7% of the true matches.

This is a chapter rather than a section for one reason. South Africa has a published, quantified, peer-reviewed measurement of national record-linkage failure, on the country's highest-priority clinical programme, and it is better evidence than anything an architect could bring from elsewhere. Most material on patient matching argues from first principles or imports American statistics. South Africa does not need either.

Haeri Mazanderani and colleagues studied infant identification in the prevention-of-mother-to-child- transmission programme in Tshwane between May 2016 and May 2017, across **13,499 birth HIV polymerase chain reaction tests** [1]. No identifier is issued to an infant at birth; infants are registered by first name, surname, sex, date of birth and a facility folder number. The measured results:

- **Only 4% of infants had exact matches on all four demographic details** across tests.
- **First name matched in 5%** — because **74% of birth tests were registered under a "baby of" or similar prefix** instead of a name.
- Surname matched in **61%**. Sex and date of birth each matched in **91%**.
- Of infants tested twice, 18% were tested at the same facility both times — and **only 5% had the same folder number on both occasions**.
- Linkage succeeded by folder number or exact surname-and-date-of-birth for **51%**, needed looser criteria for a further **28%**, and **at least 21% could not be linked at all**.

Then the finding that settles the argument. The same study measured the National Health Laboratory Service's own linking algorithm against barcoded Road to Health booklet identifiers as a gold standard, and found **sensitivity of 4,7%**, specificity 99,1%, positive predictive value 40,5%, negative predictive value 88,4% and **accuracy 87,7%** [1].

Read those last two numbers together, because the pairing is the lesson. An algorithm that found 4,7% of the true matches scored 87,7% accurate. That is not a paradox; it is what accuracy does in a linkage problem. Almost every candidate pair in a population is a genuine non-match, so an algorithm that matches nothing scores well on accuracy and is useless. **Accuracy is the wrong metric for record linkage, and an MPI vendor quoting one should be asked for sensitivity, positive predictive value, and the gold standard they were measured against.**

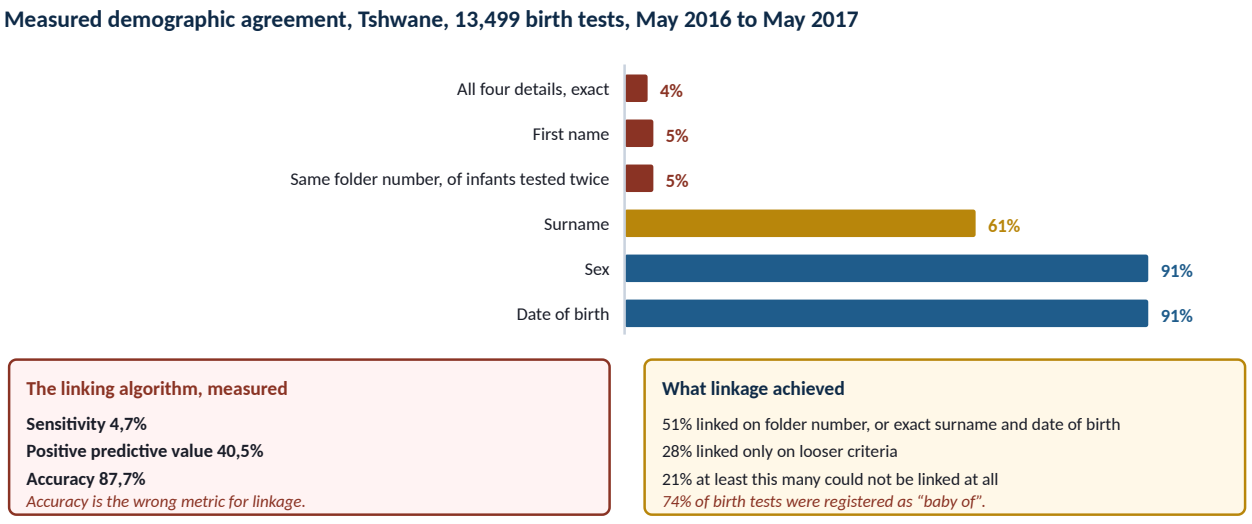


Figure 11.1 The published South African rebuttal to matching on demographics. Note the pairing at bottom left: an algorithm that found 4,7% of true matches scored 87,7% accurate, because almost every candidate pair in a population is a genuine non-match.

Chapter 1 stated the headline. This chapter is the mechanics, and it ends somewhere an architect does not expect: with the patients for whom reliable linkage is a risk rather than a benefit.

> **WHAT THE TSHWANE STUDY LICENSES YOU TO SAY**
>
> "We will link on demographics" has a published, dated, South African rebuttal with a denominator. Quote it rather than arguing. And note the denominator honestly: this is neonates in one metropolitan district over twelve months, which is the hardest population in the country to identify, because neonates have no identifier, no name on the day of the test and a mother who may be registered under three spellings. Adult matching is better. It is not good.

Four identifier layers, not two

The architect's instinct is that there is a national identifier and a local one. There are at least four layers, and they do not nest cleanly.

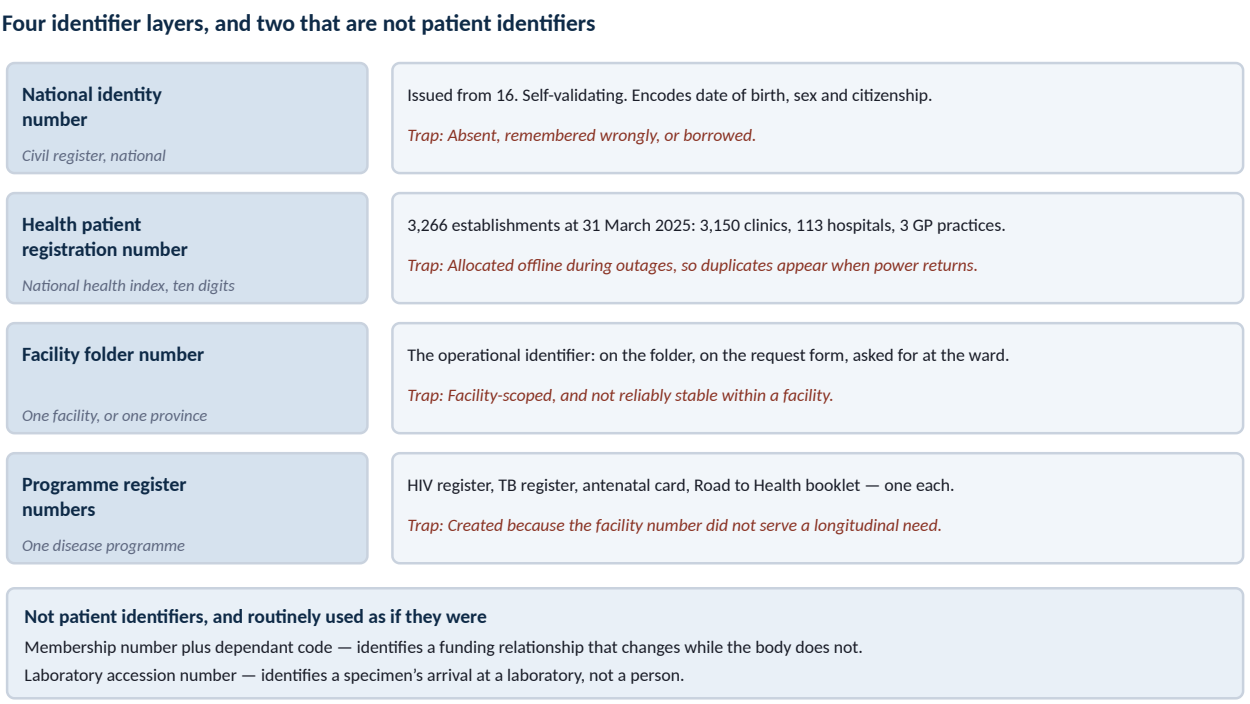


Figure 11.2 The identifier is a table, not a column: type, value, assigning authority and validity period. A folder number without a facility is meaningless and a membership number without a scheme is ambiguous.

The national identity number is the civil register's identifier, and it is the strongest attribute available when present and correct. **The health patient registration number** is a ten-digit identifier issued by the national identity index [4]. **The folder number** is the facility's operational identifier. **Programme register numbers** are the fourth layer: the HIV programme has its own patient register and register number, the tuberculosis programme its own, the antenatal card its own, the Road to Health booklet its own. Each was created because the facility identifier did not serve the programme's longitudinal need, and each is now another identifier type that must be held, matched and reconciled.

A fifth and sixth exist on the money side and must not be mistaken for patient identifiers: the medical scheme membership number plus dependant code, and the laboratory accession number. Neither identifies a person. The membership number identifies a funding relationship that can change after a divorce while the body does not; the accession number identifies a specimen's arrival at a laboratory.

The design rule that follows is simple and almost never implemented. **The identifier is not a column. It is a table** of tuples — identifier type, identifier value, assigning authority, validity period — each linked to an enterprise person identifier. Assigning authority is not optional: a folder number without a facility is meaningless, and a membership number without a scheme is ambiguous. And the identifier type enumeration in a South African provider must include, at minimum, national identity number, passport plus issuing country, asylum or refugee permit, birth certificate reference, health patient registration number, facility folder number per facility, membership number plus dependant code, programme register numbers, laboratory-derived patient keys, and a **provisional identifier** for unidentified patients.

The South African identity number, and the four traps in it

The number is thirteen digits in the pattern year, month, day, a four-digit sequence, a citizenship digit, a twelfth digit and a check digit [2]. The sequence range distinguishes women from men. The eleventh digit is citizenship status: citizen, permanent resident not born in the country, or refugee. The twelfth digit was historically a racial classifier and is now randomly assigned. The thirteenth is a Luhn check digit. Identity documents are issued from age **16**. The smart identity card replaced the green bar-coded book from **2013**; the Department of Home Affairs issued **3,42 million** smart cards in **2024** and about **4 million** in **2025**, a 17% increase, and from **March 2026** began recording Khoisan traditional names on smart cards, birth certificates and passports for the first time after IT upgrades allowed the required characters to be printed [2].

One honest limit on the paragraph above. The account of the number's structure comes from a public description of the identity card rather than from a Home Affairs specification, which could not be retrieved [2]. It is a headline-level source and it does not carry an argument here — the architectural consequences below stand on the structure being as described, which is easy to confirm against any live document, and the structure should be confirmed before a validator is written to it.

That last fact is worth more than it looks. **The character set of a national identifier system is itself a dated, changeable design decision**, and it changed in 2026. Every system that stores a name inherited a constraint from a different era, and a name that could not previously be printed could not previously be matched either.

Four properties, each with a trap attached.

It is self-validating, and the check is almost never implemented. The Luhn digit means a typographical error is detectable most of the time with no external lookup, which makes it the cheapest data-quality control available at a registration desk. The reason given for not implementing it is always that it rejects "real" numbers — which is usually a sign that the system is storing numbers that were never valid in the first place. Implement it as a warning at capture, not a hard reject, and count the warnings: the count is a free measure of registration quality.

It encodes date of birth, so the two attributes are not independent. If both are present and disagree, one is wrong. If both are present and agree, they provide much less combined discriminating power than two independent attributes would. A matching engine that weights them independently overstates its own confidence, and the overstatement is systematic rather than random.

It encodes a sex marker, with the same dependency and one extra consequence. A transgender patient whose number has not been amended will fail a consistency check that a naive validator treats as a data error. Home Affairs provides an amendment process, which means the record must hold **both** the historical and the current number, because the historical number appears on historical funder claims and historical laboratory results.

It encodes citizenship, which is special-category-adjacent information embedded in a primary identifier. A system that displays the number displays the citizenship status. A report grouped by the eleventh digit has produced a report by immigration category, probably without anyone intending it. A de-identification routine that truncates the number without considering the eleventh digit has not removed what it thinks it has removed. The historical twelfth digit is the same problem in retrospect, and legacy data from before randomisation still carries it. This is a concrete privacy finding rather than a theoretical one, and chapter 18 carries the obligations.

Who has one, and who does not

This is the question that determines the design, and the honest answer is that a material fraction of patients presenting at a South African facility do not have a usable national identifier at the moment of presentation.

Children. No identity document is issued until 16. An identity number is allocated at birth registration, but the *child* carries no document and the number is obtained in practice from a parent who may not have the certificate with them. The gap between live births and births registered within a year is the size of the problem, and the civil registration profile is the reference point for it [15]. The structural fix is already in progress: online birth registration was live in **299 facilities** as at 31 March 2025 [6], which makes the identity number available at the point of birth rather than weeks later. That is the single highest-value identity intervention in the country, and it is a registration service rather than a matching algorithm.

Foreign nationals. A passport number plus issuing country, an asylum-seeker permit number, or a refugee identity number. Each has a different format, usually no check mechanism, and a different renewal cycle — and **asylum permits are renewed frequently, with a new number each time**, so a patient on one may present with a different identifier at every visit. There is no validator and no authoritative lookup available to a facility. Two consequences for the model: the identifier field must accept multiple types, each with its own validation rule and its own reliability weight in matching; and the permit number is a **poor** matching attribute precisely because it changes. The national chronic medicine dispensing system's design accommodates a national identity number, a passport **or** an asylum-seeker number [16], which is the right pattern — several identifier types with provenance, not one mandatory field.

Undocumented patients. Patients with no document of any kind, who must be registered because care cannot be withheld, and who are registered on demographics alone. They are also the patients most likely to present under variant names, to be unable to confirm a date of birth, and to use several facilities.

Patients who present without their document. Numerically the largest group. The number is recorded from memory or not at all, and a remembered thirteen-digit number fails the check digit more often than a transcribed one.

Patients who present with someone else's. Both benignly — a relative's scheme membership — and fraudulently. This produces the most dangerous record state in the whole system: **correct identity number, wrong body.** No amount of identifier hygiene detects it; only contradiction detection inside the record does, which is a point the threshold section returns to.

The folder number, which is not going away

Every South African facility maintains its own patient numbering, and the folder number is the operational identifier: it is on the physical folder, it is what the clerk writes, it is what the ward asks for, and it is what the laboratory request form carries. It has three properties that matter.

It is facility-scoped. The same patient at two facilities has two folder numbers that do not reconcile. This is the structural reason a provider cannot read a patient's history off the record. And the Tshwane finding shows folder numbers are not even reliably stable: of the infants tested twice, 18% were tested at the same facility both times and only 5% had the same folder number on both occasions [1]. Note the denominator carefully, because the stronger version of this statistic circulates: the 5% is of all infants tested twice, not of the 18% tested twice at one facility.

It is a physical-file locator as well as an identifier. Folder numbers in many facilities encode the filing terminal digits, which is why they have internal structure and why re-numbering is resisted. The number is how the file is found on the shelf.

It will not go away. Any target architecture that assumes the folder number can be retired has misunderstood what it is for. Treat it as one identifier type among several, scoped to an organisation, with a validity period, in the identifier table — never as the primary key, and never assumed unique outside its scope.

There is a documented South African improvement on it, and it is cheap. The Western Cape issues a **shared nine-digit folder number whose last digit is a modulus-ten checksum**, printed on barcoded labels, and the provincial data centre reports that this materially helps linkage [10]. Three design features in one sentence: a *shared* namespace across facilities rather than per-facility numbering; a *check digit* so transcription errors are caught at capture rather than discovered at merge; and *barcode* presentation so the number is machine-read at specimen collection and medication administration rather than retyped. A provider group with facility-scoped numbering and no check digit has three improvements available to it that require no new product.

The national index, and what it does and does not resolve

The **Health Patient Registration System** is the national answer. It is owned by the National Department of Health, was developed under contract with the CSIR, performs identity verification and recording of clinic visits using the national identity number as its unique identifier with passports and other legal identification supported, acts as the national index, and holds **demographics only** [3][14]. Patients receive a ten-digit health patient registration

number [4]. The national digital health strategy describes it as an initial prerequisite for a patient electronic health record, and records that the diagnostic, treatment and billing modules such a record would need had not been developed [3].

That restraint is an architectural decision worth naming. A national index holding identity and the *existence* of an encounter, but no clinical content, is far easier to govern, far easier to secure and far more likely to be adopted than a national clinical repository. It is the right shape.

Now the coverage, with the breakdown rather than the conjunction. As at **31 March 2025** the system was installed at **3,266 health establishments: 3,150 fixed primary health care facilities, 113 hospitals and three general practitioner practices**, with an automated biometric identification system in **82 primary health care facilities** [6]. A 2023 study reported at least 44 million people registered [14]. The private sector is not covered at all, and **no national crosswalk exists** between the public index and private-sector identifiers.

Three things follow, and the third is the teaching example.

Biometrics are a pilot, not a capability. Eighty-two facilities out of 3,266 is about two and a half per cent. Any design that assumes biometric identity resolution in South African public primary care is designing for something that does not exist at scale [6].

Integration is partial. The index links to the national laboratory service and to the HIV programme register — the latter to a *local* copy on the same local area network — and is **not** integrated with the maternal messaging service, the stock visibility system or community health worker systems [4]. There is no direct link described to the national aggregate reporting system. So the identifier that is supposed to resolve the patient does not reach several of the systems holding the patient's data. Provincial progress is uneven in ways that matter to a design: the Eastern Cape reported 98% facility implementation alongside poor rural connectivity affecting service delivery, and the Western Cape has sent daily updates from its own provincial index to the national system since June 2020 [17]. That Western Cape arrangement — a province running its own index and *feeding* the national one — is the federated pattern working, and it is the worked example to use rather than a hypothetical.

And a nationally-issued identifier here cannot be globally unique by construction. On the Free State's own evidence to the National Council of Provinces on **25 June 2025**: rollout began in 2014; as at 26 May 2025, 253 clinics and hospitals — 100% coverage — used the system, with **3,860,000 patients registered and 37,490,000 facility visits recorded**; and **duplicate entries exist because the system works offline during power outages, with duplicates appearing when power returns**, remediated by periodic national database clean-ups [5]. The Western Cape reported that potential duplicates are flagged and confirmed directly with patients. The committee chairperson observed that the number of recorded patients exceeded the province's resident population and asked about duplicates and out-of-province patients [5]. The landscape report attributes duplicate numbers partly to unstable connectivity preventing a real-time check for an existing number, and records that unique identification still fails within facilities and when patients move between them [4].

> **OFFLINE ALLOCATION AND GLOBAL UNIQUENESS ARE INCOMPATIBLE**
>
> This is the best teaching example in the country of offline-first design without a conflict-resolution design, and the finding to put in front of a design review is one sentence: **an identifier allocated by a system that must function offline is not a unique identifier; it is a locally-issued identifier with a national format.** The pattern that reconciles them exists and must be explicit — allocate a *provisional* identifier offline, in a declared provisional state, with a reconciliation process on reconnection and a merge audit trail. Offline capability is not the defect. The missing provisional state is.

Two honest gaps, both of which an architect should name rather than fill. **No published national duplicate rate for the index could be found, and no national merge or unmerge policy** [5][4]. **Whether the ten-digit number carries a check digit or any validation structure is not published**; the length comes from the landscape report [4]. What would settle both is a request to the National Department of Health for the index's data dictionary and its duplicate-management standard operating procedure. Until one exists, the provider's own architecture must supply the merge policy, and should expect it to be superseded — the legal position is that **there is at present no national instrument specifying how patient identity must be resolved, what a merge requires, or who is accountable for a mis-merge** [12].

Master patient index mechanics

An index is three things: an identifier cross-reference, a demographic golden record, and a matching engine with a governance process attached. Organisations buy the first two and neglect the third, and the third is where the risk lives.

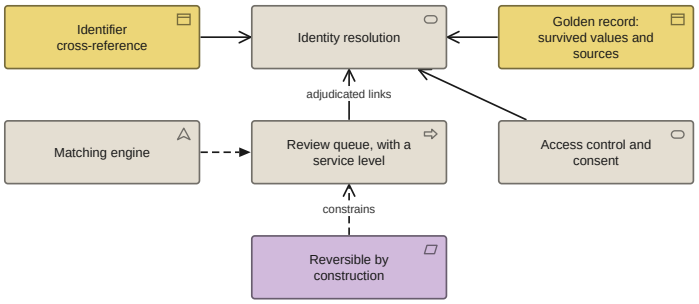


Figure 11.3 The index is three parts and a governance process. The access-control and consent service on the right is the half of the design most often left out, and without it the patients who most need linkage are the ones who will refuse it.

The cross-reference is the identifier table of the previous sections. Nothing clever, and it is the part that survives every subsequent redesign.

The golden record is the demographic set the enterprise asserts, and its one real design decision is **survivorship**: when two source systems disagree about a date of birth, which wins? The options are source precedence, recency, frequency across sources, or completeness, and each produces different errors. The only defensible arrangement is that the survivorship rules are explicit, versioned, and that the **source values are retained alongside the survived value** so a change of rule can be re-applied. Which is to say the golden record is a derived view, exactly as the continuous stay is in chapter 10.

The matching engine is where the choices are, and there is a documented South African answer that differs from the product default.

The Western Cape Provincial Health Data Centre runs a hybrid at provincial scale. Linkage is **mainly deterministic and assigns a certainty level**: matches on canonical identifiers — the civil identity number, the folder number — with corroboration rank highest, and fuzzy text comparison using edit distance supplements them. A **probabilistic score exists and is not yet in production, pending validation**. Blacklists and whitelists handle known recurring errors, including identity numbers known to be placeholders. **Links are stored in a lookup table**, chains are revised through transitive closure, and historical links are reused [10].

Five design lessons sit in that paragraph, and all five generalise.

1. **Store the certainty level with the link.** A link is not a boolean. A research extract can then accept lower-certainty links than a clinical record view.
2. **Keep a link table, not a merged record.** Merging destroys information; linking is reversible. Transitive closure over a link table means a wrong link is removed and the chains recomputed. This is the single most consequential engineering decision in index design, and products that physically merge get it wrong.
3. **Blacklists are necessary, not fastidious.** Every national identifier scheme has values entered when the real one is unknown — all zeros, all nines, a staff member's own number, a well-known placeholder. Without a blacklist these become super-nodes linking thousands of unrelated people into one chain. That is the classic catastrophic failure and it is entirely preventable.
4. **Validate before promoting a probabilistic scorer to production.** The data centre built one and has not deployed it. That restraint is correct and worth holding up.
5. **Reuse historical links.** Matching is expensive; a decision once made and reviewed should persist.

Deterministic against probabilistic, and what actually discriminates

Deterministic matching compares specified attributes character by character in a tiered rule set: identity number plus date of birth; or surname plus first name plus date of birth plus sex; or local identifier plus facility. It is explainable, auditable, fast and defensible to a clinician and a regulator. It is also brittle: one transposition defeats it, and married names, inconsistent transliteration and approximate dates of birth defeat it routinely. The American federal review of patient matching describes the character-by-character mechanism and its weighted variant, and cites a study in which simple deterministic methods performed *worse* than probabilistic or hybrid ones [8].

Probabilistic matching, in almost every commercial implementation, is a variant of one framework: for each comparison attribute, estimate the probability of agreement given a true match and given a non-match, take the log of the ratio as a weight, sum the weights, and compare the total to thresholds. The same review confirms that most products use a version of that base algorithm and that most combine deterministic and probabilistic methods [8].

Three mechanics drive the configuration decisions an architect will be asked to approve, and all three are usually set by an implementation consultant and never revisited.

The weight of an attribute depends on the value, not just the attribute. Agreement on a common surname is much weaker evidence than agreement on a rare one, because the chance of coincidental agreement is far higher. Frequency-based, value-specific weighting is the difference between an engine that works in South Africa and one that does not, because South African surname and first-name distributions are highly concentrated. **An engine configured on a European name distribution will systematically over-match** — and over-matching here means false merges, which is the error that harms patients.

Blocking sets a recall ceiling. No engine compares every record to every other; it partitions the data into blocks — same date of birth, same phonetic key on surname, same first three characters plus sex — and compares within blocks. Any true match whose records disagree on every blocking key is invisible to the engine regardless of threshold. Blocking strategy is therefore the ceiling on what the engine can ever find, and it is almost never reviewed.

The comparator must suit the languages in use. Soundex was designed for English and performs poorly on isiZulu, isiXhosa, Sesotho and Afrikaans orthography. Edit-distance and prefix-weighted comparators handle transposition and prefix agreement reasonably. The comparator is a configuration item with a measurable effect, and nobody in the governance chain usually knows which one is running.

Which attributes actually discriminate is an empirical question, and the best available dataset is American. An analysis of **398,939 confirmed duplicate pairs** from ten states and more than a hundred data-cleanup projects, published in 2016 on data collected in **2014**, measured discrepancy rates across six fields: **middle name 58,1%**; the national number field **53,4%**; **last name 23,7%**; **first name 22,6%**; **date of birth 6,2%**, down from 14,9% in the equivalent 2007 study; and **gender 6,0%**, *up* since 2007 [9]. Of the middle-name discrepancies, 62,0% had a blank in one record. Of the national-number discrepancies, 58,2% had a blank in one record and 50,2% involved a default value. Misspellings and transpositions accounted for 53,14% of first-name and 33,62% of last-name mismatches, with first-name and last-name swaps also common [9].

That is United States data from 2014 and must be labelled as such. No South African equivalent could be found. Its conclusion transfers, though, and it is counter-intuitive: the attributes that *feel* most identifying — a full middle name, the national number — were the least reliable, because they were so often blank, defaulted or inconsistent.

The surrounding American figures are worth knowing as a sanity check on vendor claims, and carry the same label — a 2013 to 2014 environmental scan of more than fifty organisations. Systems with tightly managed data-quality programmes reached match rates of 90% or higher. One large integrated provider achieved above 90% within each of its seventeen instances of the same product and roughly **50 to 60% when sharing between regions or with outside partners on the same product**. A study of 112 indexes found a mean duplication rate of 8%, with a quarter above 10%, and eleven enterprise indexes ranging from 7% to 39%. Registration entry error rates were about 7% for first names and 5% for last names. One organisation estimated about

US$60 in operational cost to fix a single duplicate record. And a survey of chief information officers found **one in five reporting at least one adverse event in the prior year arising from mismatched records** [8].

> **THE SOUTH AFRICAN READING OF ALL OF THAT**
>
> The identity number is the strongest single attribute **when present and correct**, and its check digit makes correctness partly testable — but presence is the problem, and a defaulted or borrowed number is worse than a blank. Date of birth is reliable and low-entropy, and partly redundant with the identity number. Surname is moderately reliable and low-entropy in the South African distribution. First name fails worst for neonates: 5% exact agreement in the Tshwane cohort [1]. **Mobile telephone number is underused and is in practice one of the better discriminators here**, with the caveat that it is shared within households and changes often, so it needs history rather than a single column. **Address is weak**, because informal settlement addressing is not standardised and the same dwelling is described differently at every visit. **Mother's name and mother's identity number are strong discriminators for paediatric records**, and are the main reason the Road to Health booklet was tried as an identifier [1].

Thresholds, and the asymmetry of error

A probabilistic engine produces a score, and the configuration sets two thresholds: an **auto-link threshold** above which records are linked without human involvement, and a **review threshold** below it, above which records queue for human adjudication. Below the review threshold, records are treated as different people. The band between the two is not a failure of the algorithm. It is the algorithm working correctly by declining to decide.

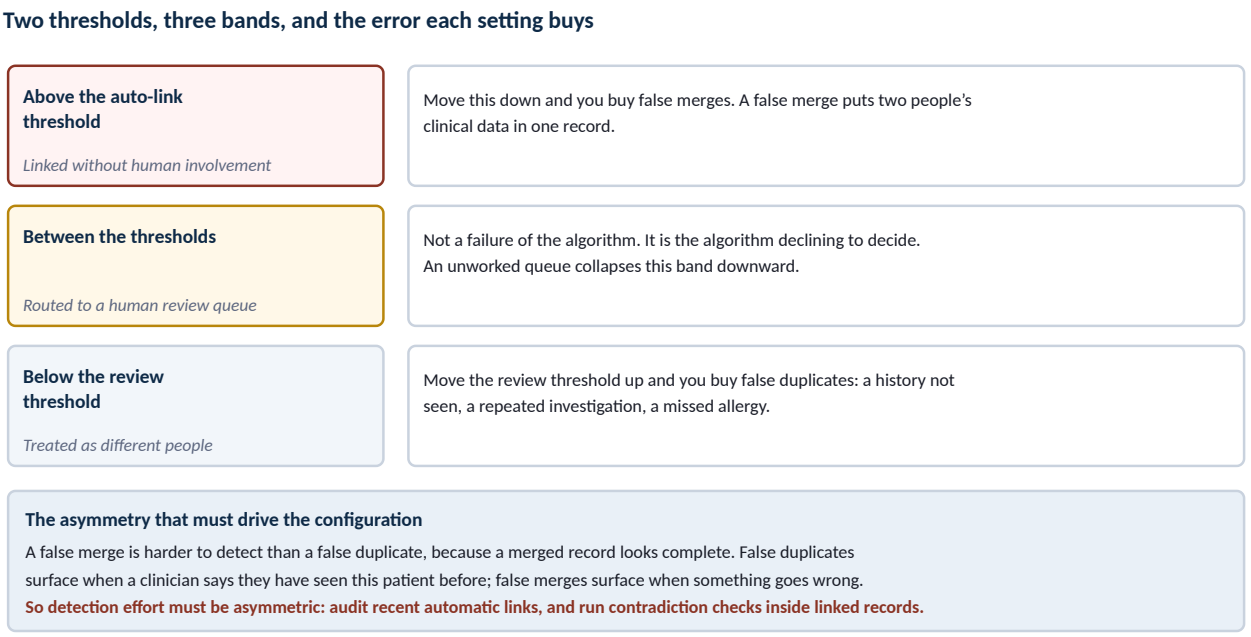


Figure 11.4 The threshold structure deployments actually use, and the error each movement buys. The thresholds are a clinical safety decision; the governance record should show that a named clinical risk owner approved them.

The asymmetry that must drive the configuration is the thing that separates this from commercial master data management, where it is inverted.

A false merge puts two people's clinical data in one record. The consequences are clinical: a wrong blood group, a wrong allergy, a wrong medication list, a diagnosis attributed to the wrong person, a result acted on for the wrong patient. The federal review states the position plainly — wrongly merging two patients' records is generally considered a larger patient-safety risk than failing to link them, because the wrong care may be given — and notes that health systems therefore set matching to **create duplicates for review rather than automatically overlay records**, which keeps manual review necessary [8].

A false duplicate splits one person's data across two records. The consequences are also clinical and differently shaped: a history that is not seen, a repeated investigation, a missed allergy because it is in the other record, a chronic patient who presents as new. And manual review carries its own risk of missing a result sitting in a record nobody opened [8].

Neither error is acceptable and they trade against each other. Five consequences for design.

1. **The thresholds are a clinical safety decision, not a technical one**, and the governance record should show that a named clinical risk owner approved them. In practice they are set during configuration and never revisited, which is the defect to look for in week one.
2. **The review queue must be resourced, measured and carry a service-level target.** An unworked queue converts the whole probabilistic design into "treat as different people", which is the false-duplicate-maximising setting. **Queue depth and queue age are the two operational measures that tell you whether the index is working**, and they cost nothing to publish.
3. **A false merge is harder to detect than a false duplicate**, because a merged record looks complete. False duplicates surface when a clinician says "I know this patient has been here before"; false merges surface when something goes wrong. Detection effort must therefore be deliberately asymmetric: periodic audit sampling of recent automatic links, plus **automated contradiction detection** inside linked records — two different blood

groups, two different sexes, two live births on the same day, an implausible age for a diagnosis. That contradiction check is also the only thing that catches the borrowed-identity-number case, where the identifier is right and the body is wrong.

4. **High-risk data should gate the link.** A link that would combine two records with different ABO blood groups, or combine a record carrying a recorded fatal allergy, should not link automatically regardless of score.
5. **Thresholds should differ by context.** A link decided at a registration desk with the patient standing there is cheap to verify — ask them. A link decided in a nightly batch run is not. The Western Cape's reported practice of flagging potential duplicates and confirming them with the patient is the right instinct [5]: **the cheapest verification is the patient.**

The review queue is an architectural element and needs specifying as one. It wants a staffed team with clinical and health-records competence; a service level, because an unresolved identity is a clinical risk; tooling that presents the candidate pair *with the evidence that produced the score*; an unambiguous outcome of link, do not link, or escalate; an audit record of who decided and on what basis; and feedback into threshold tuning. It also needs an unmerge path — and that is the test of whether a design is serious. **If the product cannot unmerge, the architecture must not merge.**

Merge and unmerge as operations

A merge is not an update. It is a structural transformation of the record with legal and clinical consequences, and it must be designed as a first-class operation.

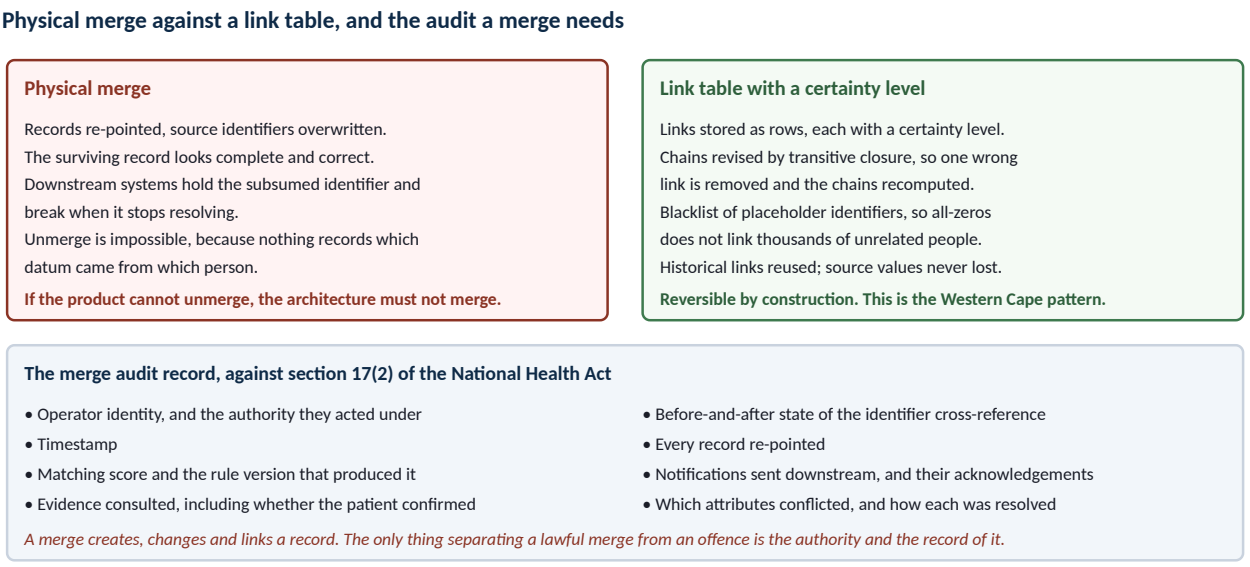


Figure 11.5 Merge as a first-class operation. The rule that governs the whole design is that source identifiers and source demographic values are never overwritten, only superseded, because that is what makes unmerge possible at all.

What a merge must do. Choose a surviving enterprise person identifier and **retain the subsumed one as a permanently resolvable alias**, because external systems hold the old identifier and a merge that invalidates it breaks every downstream reference. Re-point all clinical, administrative and financial records to the surviving identifier **without altering their own content or timestamps**. Resolve demographic conflicts by the survivorship rule, retaining the losing values with their sources. Preserve the provenance of every datum — which source system, which original patient identifier, which encounter. Record the merge as an event. Propagate it downstream, which requires those systems to support a merge notification, and most do not, so merges in a heterogeneous estate produce a period of divergence that must be monitored rather than assumed away. And handle the financial consequences: two accounts, possibly two schemes, possibly funder claims already submitted under the subsumed identifier.

What an unmerge must do is all of that backwards, which is strictly harder. It must separate co-mingled records, which requires that each record still carries its original source patient identifier. **If the merge overwrote source identifiers, unmerge is impossible.** That is the single most important design rule in this chapter: a merge must be **reversible by construction**, which means source identifiers and source demographic values are never overwritten, only superseded. It must deal with data created *after* the merge, which was recorded against the merged person and may pertain to either underlying person — there is no algorithmic answer, it requires clinical adjudication, and the model must support a record being *in dispute* during it. And it must deal with decisions already made on the merged record: results viewed, medicines prescribed, a care plan written. **Unmerging does not un-prescribe.**

The audit trail a merge needs, and why it is not optional. Section 17(2) of the National Health Act makes it an offence to create, change or destroy a health record without authority, to fail to create or change one when properly required, and to link a user's identifying details to their condition, treatment or history without authority — on conviction, a fine, imprisonment of up to one year, or both [11]. **A merge does all three of those things, legitimately.** The only thing separating a lawful merge from an offence is the authority and the record of it.

So the merge audit record carries: the operator's identity and the authority under which they acted; the timestamp; the matching score and the rule version that produced it; the evidence consulted, including whether the patient confirmed; the full before-and-after state of the identifier cross-reference; every record re-pointed; and the notifications sent. The paper analogue is the right mental model — Booklet 9's rule that nothing may be removed, that an error is struck through leaving the original legible, and that the correction is dated, signed in full and its reason recorded [13].

And who may merge is a named role, with training, not "anyone in the records office". The role that may unmerge must be **distinct** from the role that may merge, because unmerge is the recovery path for an abuse of merge.

One note on practitioner identity, because the same mechanics apply with three differences. The identifiers are better: council registration numbers come from a single national authority per profession and are verifiable against a public register [18]. The multiplicity is worse: a council number per registration, a practice code number per billing entity, an employee number per employer, a domain account per system, a prescriber identifier per prescribing system, and one funder claim message carrying a supplier number, a group practice number, a council registration number and six role-specific doctor numbers [19]. And the consequence of error is different in kind: a mis-identified practitioner does not receive the wrong treatment, they receive the wrong **rights** — to prescribe, to order, to view, to sign, to bill. Practitioner identity therefore belongs architecturally with identity and access management as much as with the clinical record.

The patients who will refuse linkage

Everything above treats resolution as the goal. For some patients it is the risk.

Rampilo, Phalane and Phaswana-Mafuya, publishing on 30 September 2025 from stakeholder mapping, a thirteen-document review and twenty in-depth interviews conducted in Gauteng between August 2024 and January 2025, report two findings an architect must hold together [7].

The first is an ordinary interoperability problem. Implementing partners use organisation-specific unique identifier code formats for key populations that do not align with facility formats, so clients referred from those programmes into public facilities cannot be tracked and are recorded as general population. The authors also note that identifier codes are assigned within individual facilities and that computers are not networked, which limits cross-facility tracking regardless of identifier design [7].

The second is not an interoperability problem at all. Their interviewees raised that **identifying key populations on a government system would expose them to law enforcement**, and said key populations are **unlikely to permit health workers to identify them on government systems at all** [7].

Identity resolution is therefore not an unqualified good. For some patients, being reliably linked across the health system is a risk they will decline, and a design that maximises linkage without an access-control story attached is a design that will not be used by the people it was built for. The clinical consequence of non-use is worse than the analytical consequence of non-linkage: a patient who avoids registration avoids care.

That turns a technical chapter into a design constraint with three parts.

An index design without an access-control and consent counterpart is half a design. The same project that specifies the matching engine must specify who can see that two records are linked, what a clinician sees when they open a linked record, and what is logged. Unauthorised access to a health record is a criminal offence in South Africa, independent of privacy law [11], which is the strongest available argument for access auditing and is not a privacy argument at all.

Disclosure must be recorded as an event, not inferred. Chapter 13 introduces the Disclosure event entity and chapter 18 carries the obligations. Its relevance here is specific: a patient who asks who has seen their linked record is asking a question the index cannot answer on its own.

And some identifiers should be designed not to resolve. A programme serving a population at risk of criminalisation may legitimately hold a programme identifier that is deliberately not linkable to the civil register, with linkage occurring only on explicit, recorded, withdrawable consent. That is a worse analytical architecture and a better clinical one, and the trade is a governance decision rather than a technical default. Say which you chose, and why, in the document.

What to take to the client

1. **Quote the Tshwane study.** Four per cent exact agreement on four demographic details; 4,7% sensitivity on the national linking algorithm; 87,7% accuracy on the same algorithm [1]. It ends the "we will match on demographics" conversation in one slide, with a South African source and a date.
2. **Build the identifier table**, not an identifier column: type, value, assigning authority, validity period, with a provisional type for unidentified patients.
3. **Implement the check digit as a warning at capture, and count the warnings.** Free measure, free control.
4. **Link, do not merge.** Link table, certainty level stored with the link, transitive closure, blacklist of placeholder identifiers, historical links reused [10].
5. **Get the thresholds approved by a named clinical risk owner**, and publish queue depth and queue age monthly.
6. **Add contradiction detection** inside linked records, because it is the only control that catches a false merge and the only one that catches a borrowed identity number.
7. **Prove unmerge before go-live**, on real linked data, and never overwrite a source identifier or a source demographic value.
8. **Specify the merge audit record** against section 17(2), and make the merge and unmerge roles different people [11].
9. **Design the access-control and consent counterpart in the same project**, and write down which patients the design has chosen not to resolve, and why [7].

THE CHAPTER IN TEN LINES

- South Africa has a published, quantified measurement of linkage failure. Quote it rather than importing statistics.
- In 13,499 Tshwane birth tests, only 4% of infants matched exactly on all four demographic details.
- The national linking algorithm scored 4,7% sensitivity and 87,7% accuracy. Accuracy is the wrong metric for linkage.
- There are four identifier layers: national identity number, health patient registration number, folder number, programme registers.
- The identity number encodes date of birth, sex and citizenship, so it is not independent of them and it leaks immigration status.
- The check digit is the cheapest data-quality control at a registration desk, and it is almost never implemented.
- The index reached 3,266 establishments at 31 March 2025: 3,150 clinics, 113 hospitals and three general practitioner practices.
- Offline allocation and global uniqueness are incompatible unless the offline identifier is explicitly provisional.
- An engine configured on a European name distribution will over-match, because South African name distributions are concentrated.
- Link, do not merge. Store the certainty level with the link, blacklist placeholder identifiers, and recompute chains.
- If the merge overwrote source identifiers, unmerge is impossible. Reversibility is a construction rule, not a feature.
- A merge creates, changes and links a record, which section 17(2) makes an offence without authority. The audit trail is the authority.
- For some patients reliable linkage is a risk they will refuse. An index without an access-control counterpart is half a design.

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CHAPTER 12

Coding, classification, and the specifications that do not exist

Nine decisions nobody has made nationally, and the register in which an architect records the answers.

Healthcare has the deepest and most mature set of international data standards of any industry in this series: a formally defined clinical terminology with description logic under it, a statistical classification maintained by the World Health Organization and used by every member state, a royalty-free naming scheme for clinical and laboratory observations, a classification of functioning, a cancer registry classification, a consumption classification for medicines, and a set of integration profiles that compose them into specified transactions. No other industry in this series has anything comparable.

And in South Africa, almost nothing that matters is specified nationally.

No national procedure classification in routine use. No national tariff since 28 July 2010. No national medicines dictionary with dose-form structure for prescribing. No case-mix grouping in force. No present-on-admission flag, so hospital-acquired conditions are not computable from routine data. No national planned-readmission algorithm. No national observation code subset. No national patient-identity merge policy. And a gazetted interoperability framework that names neither of the two code systems a modern clinical record depends on, three and a half years before the country joined the organisation that publishes one of them.

That is not a complaint and not a research failure. It is the shape of the work. The architect arriving in a South African hospital group or provincial department is not there to adopt a national specification, because for most of what matters there is not one. They are there to **make, locally and explicitly, the decisions nobody has made nationally — and to write them down in a form that survives the architect leaving.**

This chapter is organised around those decisions. For each one it says the same three things: what exists, what does not, and what the architect must therefore write down. The third is the deliverable. A decision taken in a workshop and not recorded as a specification with an owner and an effective date has not been taken; it has been deferred to whoever next opens the system.

Two fixed points, and a register

Before the absences, the two things that *are* specified, because a design that ignores them is wrong for a different reason.

The first is **diagnosis coding**: a national classification position, a national rule set, a national reference table and a national override. Chapter 6 covered the payment side — the funder regulation that requires a code, the 1 July 2014 validation gate, the code capacities. This chapter covers the reference data, which is where the architecture is.

The second is **identity**: a published national demographics-only index, which chapter 11 owns, and which supplies an identifier without supplying a policy for what to do when two of them turn out to be one person.

Everything else on the list is an absence.

What is specified nationally, and what the architect must specify instead

South Africa, at 31 October 2026. Each row is a decision an architecture cannot avoid taking.

The decision	What exists nationally	Status	Who decides instead
Diagnosis classification	Classification, standards, Master Industry Table	specified	National task team
Patient identity	National demographics-only index	specified	Department of Health
Procedure coding	A licensed private adaptation, "for Data not for Billing"	not national	The provider
Tariff	Nothing since 28 July 2010	absent	Each contract
Case mix grouping	Nothing; subcommittee dormant	absent	The provider
Present on admission	Neither instrument mandates it	absent	The provider
Planned readmission	No published algorithm	absent	The provider
Medicines terminology	A pack code and a consumption classification	partial	The provider
Observation naming	Nothing national; code system is royalty-free	absent	The provider
Clinical terminology	Membership 14 May 2026; no release centre	in motion	The provider, for now
Identity merge policy	No published national policy	absent	The provider

Two rows specified. Nine to be written down locally, each with an owner, an effective date and a version.

Every locally written specification has an expiry: it stands in for a national one that does not exist yet.

Figure 12.1 The nine decisions, and the two things that are specified. Two rows carry a national instrument; nine do not, and each of those nine is a specification the provider must write, own, date and version. The status column is the chapter: absent is not the same as not yet mature, and a locally written specification standing in for a national one has an expiry rather than a conclusion.

The output of working through the list is a single artefact, worth naming before the detail: a **register of local specification decisions**. One row per decision — what it is, what national instrument exists (often none), what was decided locally, who owns it, the date it takes effect, the version, and what

would cause it to be revisited. It is the most valuable thing an architect leaves behind in this industry, because every one of these decisions will eventually be superseded by a national specification and the organisation needs to know what it chose and why.

Diagnosis: the Master Industry Table, and why effective dating is architecture

The **ICD-10 Master Industry Table** is the most important South African coding artefact and it is routinely left out of accounts of South African health data: the industry listing of every ICD-10 code used in the country, compiled by the Ministerial ICD-10 Task Team, published by the National Department of Health as a spreadsheet and a comma-delimited file, and imported by providers and funders alike [3]. Three properties make it more than a code list.

It carries validity flags that are a ready-made constraint set. Per code, indicators state whether the code may be used clinically at all, whether it may occupy the primary position, and whether it is of the asterisk, dagger or sequelae type; alongside them sit an age range and a sex field drawn from the classification's own volumes [2][3]. An organisation holding the table properly can validate its own coding without writing a line of clinical logic. One holding a code list stripped of the flags cannot, and will discover this when a funder claim is rejected for a code its own system accepted.

It is a shared public-private artefact. Provider and funder validate against the same table, which is unusual internationally and a genuine strength: the argument about whether a code is valid has a single authority behind it. It also makes the table a sector-wide dependency, so its publication cadence is everyone's cadence.

It is effective-dated reference data, and most implementations treat it as current state. Each code carries South African start and end dates, a status of added, modified or discontinued, and a revision history [3]. The table updates biennially on 1 July to track classification version updates, with ad hoc updates permitted, three months allowed for operational change and six months for system change from publication [2].

That last property is the architecture requirement, and the one most often missed.

> **EFFECTIVE-DATED CODE VALIDITY IS AN ARCHITECTURE REQUIREMENT, NOT A REFERENCE-DATA CHORE**
>
> A funder claim or a record from 2019 is validated against the table in force in 2019, not A warehouse joining historical coded data to the current code table will silently drop codes discontinued since, and silently mis-describe codes whose description or validity changed. The fix is ordinary — a temporal dimension on the code table and point-in-time joins — and it is routinely not done for clinical code sets, because people think of them as static.
>
> The rule generalises beyond this one table. **Every coded value in a healthcare record carries, as part of its identity, the code system and the version of that system in force on the date of care.** A model that stores a code without its system and version has recorded a string, not a meaning.

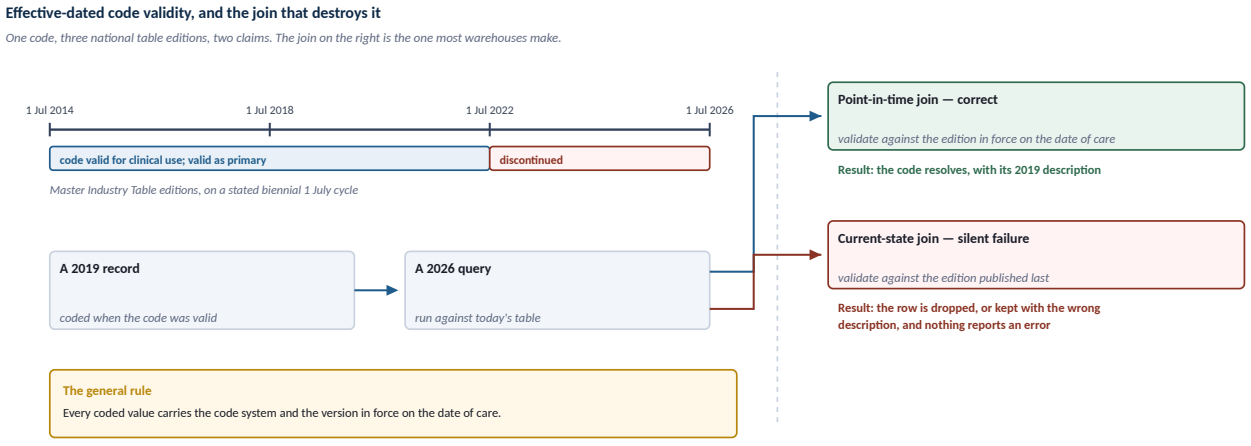


Figure 12.2 Effective-dated code validity, and the join that destroys it. The national code table carries South African start and end dates per code on a stated biennial 1 July cycle, so a 2019 record must be validated against the 2019 edition. A current-state join drops codes discontinued since and mis-describes codes whose description changed, and neither failure reports an error.

A second, quieter requirement sits in the same document set. **Serialisation rules are part of the standard**: the national technical guide specifies where the decimal point goes, which characters are permitted in which position, how placeholder characters are used, how codes are separated on an electronic claim and how on a paper one, and that the primary code comes first [2]. A system that normalises codes internally and re-serialises on output will produce codes that fail a funder's edit even though the clinical content behind them is right. Store the canonical form; derive the wire format; test the derivation against the guide.

And then the thing that makes this harder than it sounds. The operative rule set for coding a diagnosis in South Africa is not one document. It is the World Health Organization's classification, plus the national morbidity coding standards, plus the national technical user guide, plus the current Master Industry Table with its effective dates, plus the private sector's own addendum maintained by a separate industry body [1][2][3][4]. **Five documents, three maintainers, and no single machine-readable source that combines them.** The national standards also carry an override worth putting to any vendor claiming international conformance: in the South African environment, options the classification marks as optional are treated as **mandatory** [1]. South African ICD-10 is a national variant with stricter rules than the base classification, and a system validating against base ICD-10 will accept codes the South African standard rejects.

TWO EDITIONS TO CONFIRM BEFORE QUOTING EITHER AS CURRENT

The national morbidity coding standards retrieved for this volume are **version 6 of April 2014**, and the document itself states that it is revised annually; no later version was found [1]. The Master Industry Table referenced in the technical guide is the edition published **1 January 2014** for implementation from 1 July 2014, on a stated biennial cycle, and no later edition was located [2][3]. Meanwhile the private sector's addendum is at **version 7 of October 2024** [4], which suggests the national documents may not have kept pace. **Both are unverified for this volume.** Confirm the current editions with the National Department of Health before building a validation service against either, and record the edition you built against in the register.

The other reason to hold the clinical assertion and the code as separate linked objects — chapter 4's distinction between **coded** and **recorded** — is that the coding act has a *purpose*, and the purpose changes the code. Chapter 4 carried the South African evidence on what that costs: complete code sets in **45,3%** of 223 analysed discharge records where clinicians coded their own diagnoses, accurate primary codes in **74,0%**, and each additional comorbidity *reducing* the odds of complete coding [5]; and a training, supervision and case-manager package moving completeness from **27,1% to 68,1%** while primary-code accuracy moved from about 71% to **79,1%** and was not statistically significant [6]. The reading for an architect is one sentence: completeness responds to process and supervision, accuracy responds only to expertise, and an organisation that will not employ coders should not build analytics that depend on coding accuracy.

So the first entry in the register is already written: **record the coding purpose on every coding act** — reimbursement, morbidity statistics, clinical registry, research — and never let a billing-purpose code update a clinical-purpose code.

Procedure coding: the absence that blocks the most

Diagnosis coding in South Africa is specified. Procedure coding is not, and the consequences run further than most architects expect, because case mix, activity-based costing, clinical registries and any national purchasing arrangement sit on top of it.

What exists in the private sector. Complete CPT for South Africa, a local adaptation of the American Medical Association's procedural terminology structure. The copyright position matters before the content does: **CCSA copyright is held by the South African Medical Association and CPT copyright by the American Medical Association**, reprinting the American content in a product or publication requires a licence from its owner, and training material using the local adaptation is subject to the South African association's review terms [7]. Licensed classification content cannot be loaded into a system, shipped in a data extract or handed to a cloud analytics provider on the assumption that it is public reference data.

Then two statements from the maintainers that should be read twice, because they are routinely attributed to the same body and are not.

The first is the **code set maintainer's own positioning**: the standards apply to coding "for Data and not for Billing" [7]. The second is the position of the **industry standards body** that publishes them: it does **not** recommend the local adaptation as a national procedural coding standard [8]. Two bodies, one of them a copyright holder, and between them a code set that is officially for data, unofficially for money, used inside funder-provider contracts for reimbursement, and declined as the national standard by the body that maintains its standards document. Some practitioners bill on relative value units instead, and specialty societies publish their own coding guidance inside that landscape [7][9].

What exists in the public sector. The Uniform Patient Fee Schedule, which chapter 6 covers as a workflow before it is a tariff. It is a fee schedule and not a procedure classification, and it does not support clinical or epidemiological analysis of what was done.

What does not exist. A national procedure classification in routine use. Research has examined the World Health Organization's International Classification of Health Interventions as a candidate, including a published study of the public health value of coding surgery in South Africa with it [10]; it is free, maintainer-backed and natively post-coordinated in its axis structure, which makes it the obvious candidate. Nothing found for this volume indicates adoption.

What the architect must write down. Three things.

1. A **procedure-coding abstraction layer**, because the same clinical act must be expressed in at least three vocabularies — the clinical record, the funder claim and the public-sector fee schedule — and the mappings between them are neither published nor stable. Build it as a mapping service with governed, versioned mappings and a named owner, not as translation logic inside each interface; per-interface translation means the same act is mapped differently on three seams and nobody can say which is right.
2. The **licence position** on every procedure code set held, with permitted uses, permitted recipients and expiry. This is the one code set where an extract to a research partner may be a licensing breach rather than a privacy question.
3. A statement that the organisation's procedure data **is not comparable** to anyone else's, because there is no national classification to be comparable against. That belongs in the metadata of every report built on procedure counts.

Tariff: contract data, and chapter 6 has the history

Chapter 6 has the sequence from 1994 to the 2010 judgment that set the enabling regulations aside [11]. What belongs here is the data consequence, stated once.

A tariff is not an attribute of a service. It is the tuple of funder, provider, service code and effective date, resolved against a contract. A single price column on a service master is a defect, a single national price list is a thing that does not exist, and a price asserted by the provider is not the price that will be paid. The register entry is the **effective-dated contract rate structure**: which funder, which agreement, which codes, which period, which version, and which rule applies where an agreement is silent.

Case mix: a dormant subcommittee and a stack of preconditions

Case mix classifies episodes into clinically and resource-similar groups so that activity can be compared across providers with different patient populations. Every activity-based purchasing arrangement needs it, and it is the clearest instance in healthcare of a measure that is **more misleading**

when computed on poor data than not computed at all, because the adjustment creates an appearance of comparability.

What exists. Nothing. The industry standards body records that its Diagnosis Related Group subcommittee, which ran a task team with the National Department of Health from March 2008, has been **declared dormant** [8]. The body of work that would have produced a national activity-based classification is not active.

Why the absence compounds. Case mix sits on top of every other absence in this chapter. A grouper needs complete and correctly sequenced diagnosis coding, procedure coding, a resolved episode boundary and the demographic attributes. South Africa has no national procedure classification, no national tariff to attach weights to, morphology coding the national standards describe as encouraged but not mandatory [1], no present-on-admission flag, and public-sector diagnosis coding measured at 45,3% completeness in the best-studied setting [5]. A grouper fed with that data produces groups, and the groups mean what the coding means.

What the architect must write down. The decisions that move a case-mix figure, before anyone computes one: **which grouper and which version**, because groupers are revised and the revisions move weights; **whether the weights are local or imported**, because an imported weight set encodes another country's cost structure and a case-mix-adjusted cost comparison using it compares South African activity to foreign relative costs; **whether complications are counted from all secondary diagnoses or only those present on admission** — which South Africa cannot do, so the answer is all of them and that must be stated; **how outliers are trimmed**; and **whether the episode is the spell or the episode of care**, because a transfer splits the diagnosis set between two groups and therefore changes the group, not only the length of stay.

And one sequencing rule: do not build case-mix reporting until diagnosis coding completeness and depth are measured and acceptable, and then publish the coding-depth measure alongside every case-mix figure permanently.

Present on admission: the flag that is not there

Chapter 4 defined it. **Present on admission** is the flag stating whether a condition existed at admission or arose during the stay, and the only thing that distinguishes a hospital-acquired complication from a reason for admission.

What exists. Neither the national public morbidity coding standards nor the private claim message mandates it [1][12]. That is the whole position. It is a finding, not a gap in the research.

What follows. Two consequences, both measurement consequences that look like clinical ones.

First, **hospital-acquired condition rates are not computable from routine South African data.** Not difficult — not computable. An organisation that reports a hospital-acquired infection rate derived from coded diagnoses is reporting something else, and it should say which something.

Second, and less often noticed, **South African comorbidity-adjusted outcome models are silently adjusting for complications.** A model that adjusts hospital mortality for comorbidity without being able to separate pre-existing conditions from acquired ones has adjusted away part of the thing it is trying to measure, and the direction of the bias is known: it makes poor care look like sick patients.

The international literature is a warning rather than a reassurance. The flag's accuracy has been questioned wherever money attached to it — a Swiss tertiary-centre study found reliability varying substantially by diagnosis group [13], and the American programme that used it commissioned a formal accuracy review of its own coding [14]. The flag is hard to collect well even where it is mandatory, which is an argument for designing its collection carefully rather than for leaving it out.

What the architect must write down. If the organisation wants hospital-acquired condition rates it must specify the flag itself, because nobody else will: the **values**, including an explicit "not determinable" rather than a default of absent, since a default of absent manufactures hospital-acquired conditions; **who sets it and when**, because a flag set by a coder from a discharge summary is an inference about an inference while one set by the admitting clinician at the admitting assessment is a clinical assertion, and the two are different data that should not share a field; the **audit**, because a flag nobody checks drifts toward whichever value is easier; and the **statement in every report** that the flag is locally defined and not nationally comparable.

Planned readmission: an algorithm somebody has to write

The 30-day readmission rate is the measure most often requested and least often specified. Every word in "unplanned readmission within 30 days of discharge" is a decision, and chapter 2 taught the discipline of asking which one.

What exists elsewhere. In jurisdictions that use the measure for payment, a **planned-readmission algorithm** is a long, versioned, published artefact: a rule set built from procedure and diagnosis codes that classifies a readmission as planned, with its own change history. The specification families collected for the United States programmes are the clearest published example [15].

What exists here. No published South African planned-readmission algorithm was found for this volume, and no published South African readmission rate. So any South African readmission figure either includes planned readmissions, which inflates it, or applies a locally invented rule, which makes it incomparable. There is no third option. And underneath that sits a second problem whose cause chapter 11 owns: the measure needs patient-level linkage across admissions, so where the folder number is facility-scoped and no cross-facility identifier is in general use, **a provider-level readmission rate measures patient loyalty to one facility rather than discharge quality** — and the undercount is largest exactly where patient mobility is highest.

What the architect must write down. The measure, as an object: the **index-admission eligibility rules** and every exclusion — death in the index admission, transfer out, admissions where readmission is part of the plan; the **planned-readmission rule set**, as a versioned artefact with an owner rather than a `WHERE` clause; the **30-day arithmetic**, from discharge date or discharge date-time, inclusive of day 30 or not; the **multiple-readmission rule**, whether a patient readmitted three times is one event or three; and the **scope** — this provider, this group, this district, this country — with the undercount that follows from it.

One published South African rule is worth borrowing because it is explicit. The Western Cape provincial data centre's mental-health admissions work treats admissions beginning **within a day of a prior discharge** as referrals and joins them into the prior stay for length-of-stay purposes [16]. That is a defensible rule. It is a *rule*. And the figure it produces differs from the figure a different rule produces, which is the entire point.

Medicines: a product code, a consumption classification, and no dictionary

Medicines have more identifier layers than any other clinical entity, because the same substance is a therapeutic class, an ingredient, a dose form, a product, a pack and a lot. Different business processes key off different layers, and South Africa supplies some layers and not others.

What exists.

- **NAPPI**, the National Pharmaceutical Product Index code, is the de facto South African product identifier for medicines and surgical consumables and is what the funder claim carries [17][12]. Two properties matter. It is **commercially maintained, by a claims switch**, so a provider's medicines master depends on a licensed feed from a party in the payment chain. And it identifies a **pack**, not an ingredient or a dose, so it serves dispensing and claiming and does not serve prescribing, decision support, allergy checking or exposure analysis.
- **ATC**, the World Health Organization's Anatomical Therapeutic Chemical classification, sorts active substances by the organ or system they act on and by therapeutic, pharmacological and chemical properties, in five levels, with codes assigned by *main therapeutic use* [18]. It is for measuring and comparing consumption — stewardship, prescribing variation, pharmacovigilance signal detection, procurement analysis. **You cannot prescribe from it.** It names neither a product, nor a strength, nor a form, nor a dispensable item, and because each product gets one code by main use it does not reflect secondary uses [18].
- **The single exit price register**, which makes medicines the one part of the South African cost picture with an authoritative national reference: price as an effective-dated attribute of a product set by regulation rather than negotiated [19].
- **The Master Procurement Catalogue** for the public sector, keyed on its own item numbering, which does not reconcile to the commercial product code without a crosswalk [20].
- **The Essential Medicines List and Standard Treatment Guidelines**, eighth edition 2024 for primary care [21] — a *formulary* plus protocol, which is a different object from a product catalogue and a different object again from a terminology.
- A **registration register** under the medicines legislation, where registration status is a regulatory attribute of a product with its own effective dates.

What does not exist. A national medicines dictionary with virtual-product and dose-form structure for prescribing — no governed hierarchy linking ingredient, dose form, strength, virtual product, actual product and pack with the relationships between them. The British dictionary of medicines and devices is the reference pattern, and two things about it are the lesson rather than the content: it refreshes **weekly**, so a medicines dictionary is a high-velocity data product and not a reference table, and it is formally integrated with a clinical terminology, so a prescription is expressible in the same vocabulary as the rest of the record [22].



Figure 12.3 The five medicines levels and what South Africa supplies at each. Level 5 is specified and commercially maintained; level 4 has a registration register and a regulated price; level 2 has a consumption classification nobody mandates; levels 1 and 3 have nothing, and level 3 is the prescribable one. That is why a provider cannot answer which patients are on a given drug class from its medicines master without building its own classification.

The consequence is the question to ask in the first fortnight: **a South African provider cannot answer "which patients are on an ACE inhibitor?" from its medicines master without building its own classification**, because the only identifier it reliably holds is a pack code. The same absence blocks closed-loop administration, because without a dictionary relating the dispensable pack to the prescribable virtual product, the barcode on the item cannot be verified against the prescription except by local mapping. Chapter 9 carried the evidence on what closed-loop administration is and is not worth; this is why the South African version is harder.

What the architect must write down. A five-level medicines model, and who maintains each level:

1. **Substance or ingredient** — the join point for allergy checking, interaction checking and exposure analysis.
2. **Therapeutic class or classes** — held separately, because a substance has more than one, and the consumption classification is the obvious free candidate and is not mandated here [18].
3. **Virtual therapeutic product** — ingredient plus form plus strength. This is the prescribable level and the one South Africa does not supply.
4. **Actual product** — a manufacturer's product.

5. **Pack** — where the commercial product code attaches [17].

Then map the formulary to the virtual level, the product code to the pack level, the procurement catalogue to whichever level its item numbering supports, and expect to maintain all of it, because nobody else will. And record the ownership honestly: the evidence for this volume shows **pharmacy owning the dispensing catalogue, finance owning the price, and nobody owning the clinical classification**. A third owner by way of public-sector procurement is a defensible inference and not what the evidence states. Two owners and an absence is the accurate statement, and the absence is the deliverable.

Observations: the cheapest win nobody has taken

Classifications sort conditions. Terminologies name clinical concepts. **LOINC names the questions that observations answer** — what was measured, not the value and not the interpretation. A term is defined by a fixed set of axes covering the component, the property measured, the time aspect, the specimen, the scale and, where it changes the meaning, the method; two terms differing on any axis are different terms [23].

Why it matters architecturally. A laboratory's local test codes are local. Two hospitals' "glucose" are not the same test, and the same hospital's ward glucometer glucose and laboratory plasma glucose are genuinely different observations. A common identifier for the *question* is the precondition for aggregating results across laboratories, which is the precondition for multi-site analytics, for a research dataset and for any prediction model trained on more than one site's data. **A clinical data warehouse without observation mapping is a warehouse in which no cross-site comparison is valid**, and that is the commonest fatal defect in one.

What exists. The code system itself, maintained by the Regenstrief Institute, updated twice a year in February and August, royalty-free [23][24]. Royalty-free is a material architectural fact: no licensing negotiation, which makes it the lowest-friction semantic improvement available to a provider.

What does not exist. No national South African mandate, subset or mapping was found for this volume. The national laboratory service has internal test coding; private pathology groups have their own. The Western Cape provincial data centre harmonises its data points to standard code sets and uses this one for laboratory tests "where possible" [16] — both the best South African practice found and an honest statement of its limits. The gazetted framework does not mention it at all [25].

What the architect must write down. The mapping project, scoped honestly:

- It is a **local project with no national specification**, and it is the single highest-value piece of unglamorous data work available to a provider that wants multi-site analytics. The effort is real: a hospital laboratory catalogue is typically a few thousand terms, and the work requires a laboratory scientist rather than a data engineer.
- The **three classic errors** belong in the specification as prohibitions: mapping on test name alone, which collapses specimen and method distinctions; mapping the local panel rather than its constituent analytes; and mapping once and never re-mapping when the laboratory changes assay.
- What the code system does **not** give you: two laboratories reporting the same code with different assays may produce systematically different values, units must be handled and converted separately with the conversion recorded, and reference ranges are properties of the producing laboratory rather than of the code. An observation is typically *two* coded values — the question and, for ordinal and nominal results, the coded answer — and many models provide one slot.

Clinical terminology: a membership, no release centre, and a framework that names neither

This is the live contradiction in South African health data, and it has a date.

What exists, as of May 2026. South Africa joined SNOMED International, announced on **14 May 2026**, following a 2024 government announcement that the country would introduce new health terminological systems and classifications, including this one, for its national health information system [26] [27]. Membership is attributed to the National Department of Health and is stated to extend access across all health sector entities, public and private [27].

The licensing consequence is immediate and large. A member licence covers use within the member country, so South African providers and vendors can deploy the terminology without individual affiliate licensing [28]. That removes the largest practical barrier to clinical terminology adoption in the country, which was a cost conversation rather than a technical one.

What does not exist. A **National Release Centre is not yet operating.** The Department is described as establishing one, with a departmental clinical coding advisory technical working group coordinating implementation [27]. Until it operates there is **no South African extension, no national reference sets, no national subsets and no national maps**, and the announcement names no release centre and no exact effective membership date [26][27]. Nothing is mandated: no requirement to record in the terminology, no national subset to conform to, no conformance requirement.

And the gazetted framework names neither this terminology nor the observation code system. The Health Normative Standards Framework published in the *Gazette* of 21 October 2022 is built on integration profiles with several messaging, document and imaging standards, and mentions neither; the national classification appears in it only in an acronym list [25]. Chapter 16 owns the framework, its conformance regime and what that regime is worth. What belongs here is the terminology consequence, and it is sharp.

THE CONTRADICTION, STATED AS A PROCUREMENT PROBLEM

A platform procured in South Africa from mid-2026 onward must be able to add a terminology binding it does not have, to a code system the vendor may not support, against a national extension that does not exist, under a gazetted framework that does not mention the code system at all. All four clauses are true at the same time.

The consequence is contractual. The requirement to state is not "supports SNOMED CT", which is a claim no national specification currently tests. It is: the product stores a code system identifier and a version on every coded value; terminology content is **configuration rather than embedded in application logic**; reference sets can be re-pointed without a release; and the vendor will accept a national extension and national maps when they appear, at a stated commercial basis. That last clause is the one to negotiate now, because it will be expensive later.

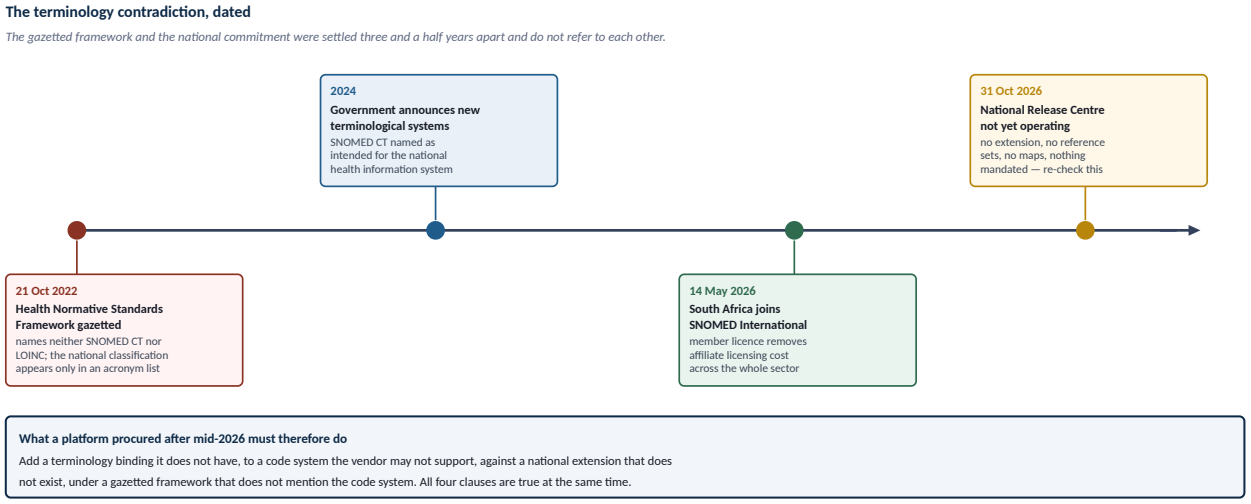


Figure 12.4 The terminology contradiction, dated. The gazetted framework of 21 October 2022 names neither SNOMED CT nor LOINC; the country joined SNOMED International on 14 May 2026; and at 31 October 2026 the National Release Centre is not yet operating, so there is no extension, no reference set and no map. The status of the Release Centre must be re-checked before an architecture is designed around it.

Three mechanisms an architect should understand before designing to any of this, because they are properties of a terminology rather than of a code list.

Content changes monthly, including by inactivation. Release notes record hierarchy remodelling, concept edits, map additions and retirements, and the inactivation of large numbers of indicators [29]. Concept identifiers are permanent and concepts are inactivated rather than reused, with a historical association to a replacement — better than a classification, where codes are reused and redefined between revisions. But a repository must handle inactivation, and the only safe design stores the identifier **and the release version in which it was recorded**, never silently re-pointing to a replacement. It also means a cohort definition written against a subtype hierarchy may return a different cohort after a release, so **cohort definitions must be versioned against a release**.

Post-coordination is expressive and creates three new problems. Two different expressions can mean the same thing, and determining that needs a logic classifier rather than a string comparison; querying requires subsumption testing and an expression constraint language rather than a list of values; and somebody must decide which expressions are allowed, because the grammar permits nonsense as readily as precision. A field sized for one code cannot hold a post-coordinated expression. The practical control is the **reference set** — a curated list of permitted concepts for a given field, for a given purpose, at a given time — and a reference set needs an identifier, a version, an effective period, a purpose, a binding to a specific data element on a specific form, an owner, a review cadence, and a recorded expansion, which is the explicit member list a historical record must be interpreted against. An organisation with four hundred clinical forms has several hundred reference sets and typically versions none of them, which is why a question about what a field's options were in 2021 cannot be answered.

The map to the classification is one-way and rule-bearing. Maps are published inside terminology releases and versioned against both ends [29][30]. A single terminology concept may map to different classification codes depending on age, sex, other conditions present or whether the condition was the reason for admission, so the map carries rule conditions rather than single targets, and an implementation that takes the first target has thrown the rules away. The direction matters too: terminology to classification is supported, and the reverse is not meaningfully possible, because classification categories are coarser and include residual groupings that correspond to no single clinical concept.

"WE WILL CODE IN SNOMED CT AND MAP TO ICD-10 FOR CLAIMS" IS TWO-THIRDS TRUE

The mapping gives you candidate codes. It cannot give you the **main condition**, because chapter 4's definition of that — the condition diagnosed at the end of the episode of care primarily responsible for the patient's need for treatment or investigation, with a tie-break on clinical severity, then resource consumption, then order of recording [1] — is an episode-level judgement with a resource criterion in it. No map makes that judgement. So automated classification coding from terminology-coded clinical data **cannot produce a compliant funder claim without human judgement about sequencing**, and an architecture that assumes otherwise has built a coding queue it did not plan for.

And the two clocks do not align: the national reference table updates biennially on 1 July [2] while the international terminology releases far more often [29], so a national map, when there is one, will have windows in which neither end matches the other.

What the architect must write down. Four decisions, and then one instruction.

1. **The terminology service is infrastructure**, not a set of tables in an application database. A subsumption query sits inside a clinical workflow, so it has availability and latency requirements. Design it as a service in the application layer with its own service levels.
2. **Code system and version on every coded value** — the same requirement the effective-dated reference table imposes, arrived at from the other end.
3. **Reference sets as versioned, owned configuration**, with the expansion recorded per version.
4. **Be ready rather than early.** Nothing is mandated, so an organisation adopting the terminology in 2026 or 2027 is an early adopter with no national specification to conform to, and every reference set it builds will have to be reconciled when national content appears. Design for re-pointing.

And the instruction: **re-check the status of the National Release Centre before designing to it.** Whether a release centre has been designated, whether a namespace has been allocated, whether any national subset, extension or map has been published, and the exact effective date of membership are all open as at 31 October 2026 [26][27]. An enquiry to SNOMED International and to the National Department of Health would settle them, and the answers change decision four.

The next classification, and why not to wait for it

ICD-11 was adopted by the World Health Assembly in 2019 and member-state reporting was expected from 1 January 2022 [31]. Its architecture differs in ways that matter to a data architect: a foundation component from which linearisations are derived, support for post-coordination from a stem code plus extension codes where its predecessor is pre-coordinated only, and distribution through an interface rather than primarily as a file. That makes it more terminology-like than its predecessor, which blurs the capture-against-aggregate distinction chapter 4 set out.

What exists in South Africa. ICD-10, and nothing dated beyond it. The country's collaborating centre for the classification family states that ICD-10 is the current national standard for morbidity and mortality coding and that ICD-11 is in its work plan, without stating what has been done about it [32]. **There is no South African ICD-11 migration in any dated, published sense.** A reform argument published outside government proposes phased adoption over five to ten years in parallel with the current classification, with mapping governance, tooling and validation [33] — a proposal, not a plan. International transition is slow and uneven, and published analyses discuss national pathways rather than completed transitions [34].

What the architect must write down. Do not build for the next classification now, and do not hard-code the shape of the current one. A diagnosis model should allow more than one code per clinical assertion, because post-coordination requires it; allow a code to be a composite of a stem plus extensions; and hold the pre-coordinated code and the clinical assertion separately, so that a re-coding exercise is a new coding act rather than a migration.

National Health Insurance data requirements: there is not a specification

Chapter 7 covered the legal position. The data position is narrower and should be stated plainly, because architects are routinely asked to design to it.

What exists. Very little. The Act mentions health records only a few times: users of funded services may access records held by the Fund to exercise or protect their rights; health workers and facilities must follow the National Health Act's rules on access to and protection of records; and the Fund's information platform must support the portability and continuity of services [35][36]. The analysis closest to the question concludes that these provisions offer very little detail on how electronic records should be created, retained, stored and protected, and that the Act lacks detail on audit trails, encryption and disaster recovery [36].

Outside the Act, the best available indication of intent is the National Digital Health Strategy, which records an integrated national repository and data warehouse that will **extract a subset of data** from a national electronic record and supply data and analytics to the Fund; the national patient index as a prerequisite for the patient record; the diagnostic, treatment and billing modules needed for such a record as **not yet developed**; and the gazetted interoperability framework as a minimum requirement with conformance testing required of all health information systems [37].

What does not exist. A published national data dictionary, schema or extract specification an architect could design to. As at 31 October 2026 there is not one.

What the architect must write down. Three positions, defensible precisely because they are modest. **Design to the two published national specifications that have any force** — the national patient index for identity and the gazetted framework for interoperability [37][25], which chapters 11 and 16 own. **Expect a subset extraction, not a full feed**, because the stated intent is an extract with a defined schema, which is a familiar and manageable pattern and a very different design from publishing the whole record. And **store the code system and version on every coded value**: the May 2026 terminology membership is the most concrete published signal of intended direction [26], and an architecture that records system and version is positioned for whatever the specification turns out to be.

Patient-identity merge policy: a national identifier without a national policy

Chapter 11 owns identity matching — the attributes that discriminate, deterministic against probabilistic rules, thresholds, the asymmetry of error, merge and unmerge as operations. What belongs in a chapter about missing specifications is narrower and easier to miss: **South Africa has a national patient identifier and no national policy for what to do when two of them are the same person.**

What exists. A national demographics-only index, installed at 3,266 health establishments as at 31 March 2025 — 3,150 fixed primary health care facilities, 113 hospitals and three general practitioner practices [38]. It uses the national identity number as its unique identifier and issues its own number alongside.

What does not exist. No published national duplicate rate and no published national merge or unmerge policy was found for this volume. And the index cannot guarantee uniqueness by construction: on a province's own evidence to a parliamentary committee on 25 June 2025, it **allocates offline during power outages and produces duplicates when power returns**, remediated by periodic central clean-ups [39]. A nationally issued identifier allocated offline is not globally unique, and a periodic clean-up is a remediation rather than a policy.

What the architect must write down. Not the matching algorithm — chapter 11 has that — but the *policy* it serves, because the policy is the thing no national instrument supplies: the **merge decision authority**, meaning who may approve a merge at what confidence and who may approve an unmerge,

which is a different and rarer authority; the **merge record** as data, holding both source identifiers, the decision, the decider, the evidence, the timestamp and the reversal path, because a merge that cannot be described after the fact cannot be undone; the **downstream contract**, meaning which systems are told about a merge, by what mechanism, and what each must do about documents, orders, results and funder claims already issued under the superseded identifier; and the **reconciliation cadence** against the national index, with a stated position on what happens when a national clean-up merges two records the organisation deliberately kept apart.

Measure Definition, and why the repository has no place for it

One more absence, and it is an absence inside the organisation rather than inside the country.

Chapter 4 defined **Measure Definition** as the versioned, effective-dated, owned rule by which a derived figure was computed: its numerator, denominator, exclusions, boundary rule and the period over which it applies. Chapter 13 introduces it as an entity. What belongs here is the coding consequence, because almost every measure in this chapter resolves to a coded value.

Case mix depends on a grouper version and a weight set. Readmission depends on a planned-readmission rule set. Hospital-acquired condition rates depend on a locally defined flag. Comorbidity indices depend on which secondary diagnoses were recorded and by whom. Every one of those is a versioned artefact, and in a typical repository none is modelled at all — the supplied repository used in chapter 20 has no place to put one. So the figures are computed, published, argued about, and nobody can reproduce last year's. The published evidence that this must be an entity rather than a convention comes from theatre utilisation: on one specialty in one trust over fifteen weeks, there is **no simple mathematical relationship between the four calculation methods**, with raw utilisation of 80% and touch-time utilisation of 60% on the same list [40]. Carry that denominator, and note that the finding generalises to occupancy, length of stay, readmission and mortality — every one of which appears in this chapter as a coding-dependent figure.

The register, which is the deliverable

Nine decisions, one artefact. The table is its shape; the content is the engagement.

Decision	What exists nationally	What the architect writes down
Diagnosis coding	Classification, national standards, Master Industry Table with validity flags and effective dates, private-sector addendum — five documents, three maintainers	The edition built against; point-in-time validation; coding purpose on every coding act
Procedure coding	A licensed private-sector adaptation positioned for data, not recommended as national; a public fee schedule	The mapping service, its governed versioned mappings, the licence position, and a non-comparability statement
Tariff	Nothing since 28 July 2010	Effective-dated contract rates by funder, agreement, code and period
Case mix	Nothing; the industry subcommittee is dormant	Grouper and version, weight provenance, complication rule, outlier rule, episode definition
Present on admission	Nothing in either sector's instrument	The flag, its values, who sets it and when, its audit, and the report caveat
Planned readmission	Nothing	Index eligibility, exclusions, the planned rule set as a versioned artefact, the 30-day arithmetic, the scope
Medicines terminology	A commercial pack code, a consumption classification, a price register, a procurement catalogue, a formulary	The five-level model, the owner of each level, and the named absence of a clinical-classification owner
Observation naming	Nothing national; the code system itself is royalty-free	The mapping project, its resourcing, the three prohibited shortcuts, and the two-coded-value model
Clinical terminology	Membership from 14 May 2026; no release centre, no extension, no reference sets, no maps; not named in the gazetted framework	Terminology as a service, system and version on every value, reference sets as owned configuration, and the vendor clause

Two notes on how to use it.

Every row has an expiry. Each decision is a local specification standing in for a national one that does not exist yet. When a national specification arrives, the row becomes a migration with a known starting point — which is the whole reason to write the register rather than carry the decisions in people's heads.

The rows are not independent. Case mix needs diagnosis and procedure coding. Readmission needs identity. Hospital-acquired conditions need the flag and the coding. Medicines decision support needs the ingredient level and the terminology. Sequence the register by dependency rather than by urgency, and chapter 8's statement of decision order applies: identity and terminology are decided before the application strategy, not after it.

The gap between what international healthcare standards can express and what South Africa specifies nationally is the largest single difference between this industry here and this industry in a reference architecture written elsewhere. An architect who treats that gap as a maturity deficit will spend a quarter writing a gap analysis. An architect who treats it as a **specification duty** will spend the same quarter writing the nine specifications, and will leave behind something the next architect can read.

THE CHAPTER IN TEN LINES

- Healthcare has the deepest international data standards of any industry in this series, and in South Africa almost nothing that matters is specified nationally.
- The architect's work is making, locally and explicitly, the decisions nobody has made nationally, and writing them down with an owner, an effective date and a version.
- Two things are specified: diagnosis classification and national patient identity. Nine are not.
- The ICD-10 Master Industry Table is the most important South African coding artefact: a shared public-private table with validity flags, age and sex edits, and South African start and end dates per code.
- Effective-dated code validity is an architecture requirement. A 2019 record is validated against the 2019 edition, and a current-state join fails silently.
- Every coded value carries the code system and the version in force on the date of care.
- There is no national procedure classification. The private adaptation is licensed, positioned by its maintainer for data and not for billing, and not recommended as a national standard by the standards body.
- There is no case-mix grouping in force and the industry subcommittee is dormant. Case mix sits on top of every other absence in this chapter.
- Present on admission is mandated by neither sector's instrument, so hospital-acquired condition rates are not computable and comorbidity-adjusted models silently adjust for complications.
- There is no national medicines dictionary with virtual-product and dose-form structure. The prescribable level is the one nothing supplies.
- The observation code system is royalty-free and has no national subset, which makes mapping the laboratory catalogue the highest-return unglamorous data work available.
- South Africa joined SNOMED International on 14 May 2026, the National Release Centre is not yet operating, and the gazetted framework names neither that terminology nor the observation code system. Re-check the Release Centre before designing to it.
- There is no published national ICD-11 transition plan, no published NHI data dictionary, and no published national patient-identity merge policy.

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CHAPTER 13

The clinical record

One object with three jobs, one of which has a measured price attached.

In three of nine paid medico-legal claims the Auditor-General reviewed, the department conceded because it could not locate the medical record [1].

That sentence is the business case for everything in this chapter, and it is worth taking slowly. Not "the record was incomplete". Not "the clinical care was indefensible". The department had a medico-legal claim against it, could not produce the record of the episode of care, and therefore paid. The clinical question was never reached.

The arithmetic, and why it comes first

Chapter 1 stated the arithmetic. This chapter is built on it, so the figures are repeated once, with their sources and dates, and then used.

The Auditor-General audited **27 facilities across nine provinces** and found **15** reporting missing or lost records, **17** not properly tracking records after they were issued, **25** with inadequate storage conditions and **18** with too few records-management staff [1]. Those findings are from the 2021/22 records audit; a separate clinic audit covers 2022/23 and the liability balance is at March 2026, so they are not one snapshot and should not be presented as one [1].

Provincial health departments' recorded medico-legal claims balance was **R50,2bn at 31 March 2026** on the Auditor-General's figures and **R50,6bn for 2025/26** on National Treasury's — two official numbers about R400 million apart on substantially the same thing, down from a peak around **R110bn in 2019/20** [1][2]. Chapter 2 taught why both are quoted rather than averaged.

What actually leaves the fiscus is smaller and more useful. **R1,57bn was paid to claimants in 2025/26** and **R4,6bn over three years across the nine provinces**, with Eastern Cape at R474m, Gauteng at R456m and North West at R230m in 2025/26 [1]. Treasury puts the long-run average at **about R1,5bn a year** [2]. Gauteng's own pattern is the sharpest illustration: in 2024 it budgeted **R179m** and paid out **about R700m**, and the shortfall comes out of medical supplies, maintenance and operating costs [3].

The quantum is dominated by one thing. **About 80% of medico-legal claim value, on average, is future medical care, costed at private rates**, and awards in serious birth-injury cases "often exceed R20 million" [4][5]. The R20 million figure comes to this volume through a secondary report of a practitioner's remark rather than from a judgment, so treat it as an order of magnitude rather than a measured average [5].

Now set that against the money available for the thing that would prevent it. The direct National Health Insurance grant for 2026/27 carries a line for **"Digital Health System" of R6,5 million** [6].

> **TWO READINGS OF THE SAME ALLOCATION, AND THEY DISAGREE**
>
> The R6,5 million line is presented inside a 2026/27 direct allocation of **R1,427bn** [6]. The 2026 Budget Review gives the direct National Health Insurance grant for 2026/27 as **R475m**, with **R1,485bn** as the three-year medium-term total [7]. R1,427bn is therefore most plausibly a three-year total presented as a single year — which would make R6,5m a medium-term figure too. Both are published. **Cite both and say they disagree**, which is what chapter 2's discipline requires.
>
> The argument is unaffected on either reading. Against a R50bn liability and a R1,5bn annual payout, the national digital health line is a rounding error, and a single prevented R20 million concession would fund more records capability than the national programme has budgeted.

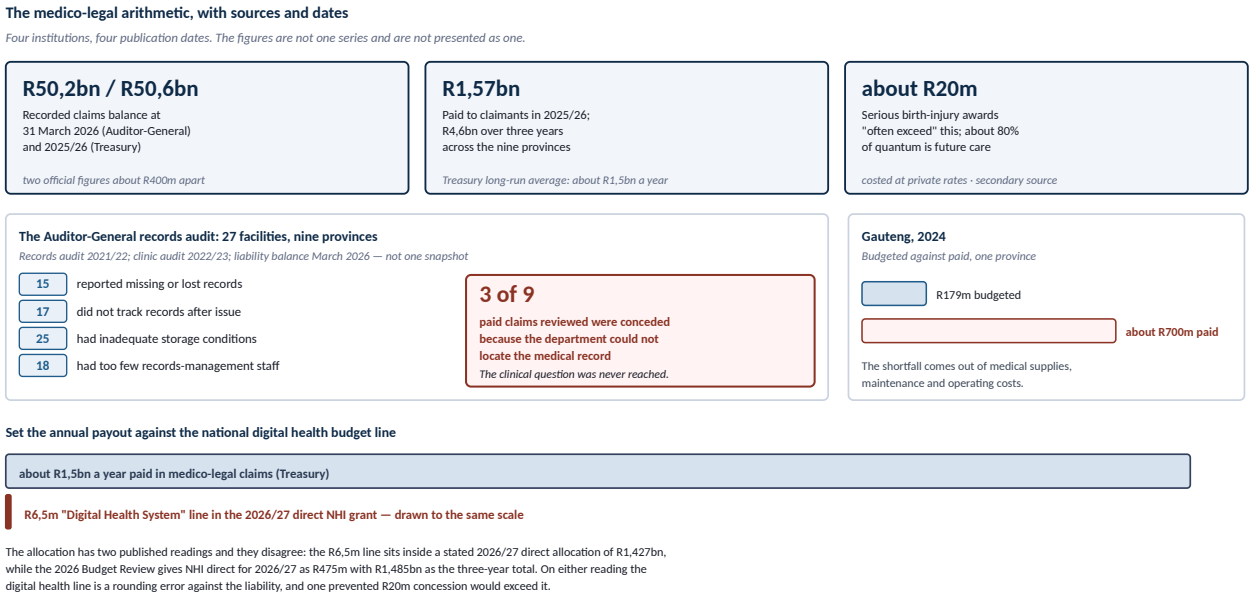


Figure 13.1 The medico-legal arithmetic, with sources and dates. The liability figures are two official numbers about R400 million apart on the same date and are not presented as one series. The audit panel is the 2021/22 records audit; the liability balance is at March 2026. The red sliver at the foot is the national digital health budget line drawn to the same scale as the annual payout, and the allocation has two published readings that disagree.

The operative word in the Auditor-General's own recommendation is **retrievable**: fill critical records and claims vacancies, train staff, **secure retrievable records**, and act on non-compliance [1]. That is not a storage requirement. It is the ability to produce the complete record of a specific episode of care, for a specific patient, from up to twenty-one years ago, under litigation, with provenance. That is a records-management capability, and it is a different system from an electronic medical record.

One more reason the arithmetic drives the design rather than the other way round. The exposure period is **indeterminate**. The general prescription period runs from when the claimant knew or ought reasonably to have known of the cause of action, which in a delayed-diagnosis case can be many years after the consultation; a minor has until three years after majority, which is where the commonly quoted age of 21 comes from; and no prescription period runs while incapacity continues [8]. A South African Law Reform Commission review found that **75% of the cases it examined took more than five years to finalise** — a figure available to this volume only through a secondary account and therefore unverified [9]. A matter that takes five years to finalise may have been issued years after the event, against a record approaching the end of a six-year retention period. **A six-year retention period does not guarantee the record will exist when the medico-legal claim arrives.**

One object, three instruments

Chapter 1 landed the framing: the record is simultaneously a clinical instrument, a legal instrument and a billing instrument. This chapter makes it operative, because each of the three imposes requirements the other two do not, and a design that optimises for one breaks the others.

As a clinical instrument the record must be legible, current, complete enough to hand over, and fast. Its users are clinicians under time pressure, and every control that slows entry degrades the clinical function. Critical information — allergies above all — must be prominently recorded and easily accessible [10].

As a legal instrument the record must be attributable, immutable, reconstructable as it stood at a point in time, and retrievable years later. Its users are a court, a council's disciplinary committee, a regulator and the organisation's own defence. Nothing may be removed; corrections are made visibly; late entries are marked; and the reason for every amendment is recorded [10][11].

As a billing instrument the record must support a funder claim: the coded diagnosis, the procedure, the authorisation, the practitioner, the dates. Its users are a revenue cycle function and a funder's adjudication engine. And here the professional rule does something almost nobody designs for: it requires that **billing records be kept separate from patient care records** [10].

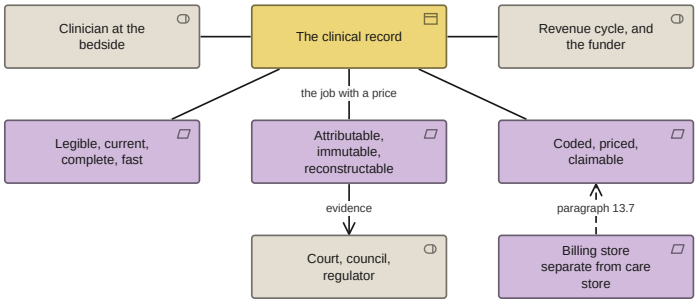


Figure 13.2 One record, three jobs, three sets of requirements, three sets of users. The clinical job wants speed; the legal job wants attribution and immutability; the billing job wants codes and prices. The professional rule that billing records be kept separate from patient care records is the explicit authority for a separation almost no integrated system honours, and it is the one requirement in this figure that comes with a regulator behind it.

The three sets of requirements conflict in predictable places, and the conflicts are the design work:

- **Speed against attribution.** Shared logins, ward passwords and a nurse entering a doctor's order under her own credential all make the clinical instrument faster and the legal instrument worthless.
- **Completeness against minimisation.** The legal instrument wants everything retained; the data protection duty wants nothing retained longer than its purpose requires. Section 9 of this chapter is about where that resolves.
- **Billing convenience against clinical truth.** A coded diagnosis that cleared a funder edit is not a clinical assertion, and a system that lets the billing code overwrite the clinical one has corrupted the clinical instrument to serve the billing one. Chapter 12 carries the coding side.

What must be in it

The South African answer is not a statute. The National Health Act requires the person in charge of a health establishment to ensure a health record is created and maintained **for every user**, containing "the prescribed information" — and no regulation prescribing that information could be located for this volume [12]. That absence is itself worth knowing: a reader sent looking for the records regulations will not find them, and the item is unverified rather than settled.

What fills the gap is professional guidance, and it reads like a specification. The Health Professions Council's record-keeping guidance defines a health record broadly and then lists both the components of a record and the minimum entries it must contain [10]. The 2022 revision in circulation defines the patient health record as the **longitudinal collection** of an individual's personal and health information, recorded by or at the direction of a practitioner, **regardless of form or medium** [11].

The components read as a document taxonomy: contemporaneous notes, notes from previous practitioners and discharge summaries, referral letters, laboratory, histology, cytology and imaging records, audiovisual records, clinical research and trial data, other forms including insurance, disability and injury-on-duty forms, and death certificates and autopsy reports [10]. The minimum entries read as a data requirements specification: identifying particulars; bio-psychosocial history **including allergies and idiosyncrasies**; the **time, date and place of every consultation**; assessment; proposed management; medication and dosage; specialist referrals; reactions to treatment including adverse effects; test and imaging results; **periods of sick leave and the reasons for them**; and **written proof of informed consent where applicable** [10].

Three of those have consequences architects miss.

Consent is part of the record, not a separate filing system. "Written proof of informed consent where applicable" is a required record element. A consent management product that holds consent outside the record has satisfied neither the record requirement nor the retention requirement.

Place of consultation is a required entry. Which means a teleconsultation has to record a place, and chapter 4's definition of care setting as a kind of care rather than a place is what makes that expressible.

Sick leave and its reasons are a required entry, which puts an occupational-health-adjacent data class with its own access population inside the general clinical record.

And the obvious structural point, since the record is substantially documents: hold **Document** as an entity with metadata — type, title, subject, encounter, authors, authenticator, three distinct dates for the clinical event, the authoring and the signing, status, version, confidentiality level, format and custodian. Storing documents as files with a patient identifier and a filename leaves no type, no encounter link, no author, no version and no retention class, and therefore no ability to apply a retention rule, to answer a subject access request or to find the discharge summary for the audit.

Who may write in it, and under whose name

The record is a set of **attributed assertions**. Attribution is the property that makes it evidence, and it is the property most often lost in implementation.

The professional rule is explicit in the revision now in circulation. Electronic entries must be **attributable**; where an electronic record will not identify the practitioner, it must be **signed by electronic means**; and electronic alterations must be **fully attributable to the person who made them** [11]. The underlying definition permits a record to be made by a practitioner **personally or under their direction** [10], which is what allows a scribe, a transcriptionist or a student to produce an entry — and means the model needs both an author and a responsible practitioner, which are not always the same person.

Four consequences for design.

One credential per human, with no exceptions that are not themselves recorded. A shared ward credential is not an authentication failure, it is an evidence failure: every entry made under it is unattributable and therefore worth very little when the matter is litigated.

Signature and co-signature are separate events. A document signed by an intern and countersigned by a consultant has two authentications, with two signatories and two timestamps, and the model must hold both. A single `signed_by` column collapses the supervision relationship that the legal instrument depends on.

Verbal and telephone orders need a named mechanism, because they are made by one person and entered by another. The entry records who gave the order, who took it, when, and when it was confirmed. Without that, the prescriber in the record is the person at the keyboard.

And some things cannot be electronic at all. The professional record-keeping guidance carries a rule that prescriptions for the more tightly scheduled substances must be handwritten with a personal original signature [10]. Two honest limits on that statement: the rule itself was not retrievable for this volume and is available only second-hand inside the Council's record-keeping guidance, and whether the Electronic Communications and Transactions Act displaces it for medicines is **open** [10]. Both are unverified. The design consequence holds under either reading and should be written down: **an electronic prescribing target state must carry a paper path**, with the paper document becoming part of the record, which means a scanning, indexing and retrieval step rather than an exception.

How it is corrected, and never overwritten

This is the clearest instance in the whole volume of a professional rule describing a technical architecture, and of most systems failing it.

The paper rules are precise. Records are kept in **non-erasable ink** and **erasure fluid is not permitted**; **nothing may be removed** from a record; an error is **struck through so the original stays legible**, with the correction **dated, signed in full, and the reason recorded**; and later additions are dated and signed in full [10]. The revision in circulation adds, for electronic records, that **late entries and addenda must be marked and their reasons given** and that **the reason for any amendment must be recorded** [11].

Read the older media provisions forward and the intent is unmistakable. The 2016 text permitted storage on **write-once media that allowed additions but not overwriting**, with encryption and password protection, a read-only working copy held locally, and a backup kept at a **physically separate site** so that the two could be compared if tampering were suspected [10]. The technology is dated. The requirements are not: immutability, encryption at rest, a geographically separate copy, and a tamper-comparison capability.

> **THE PROFESSIONAL RULE DESCRIBES AN APPEND-ONLY RECORD, AND `UPDATE` ON A CLINICAL NOTE IS NON-CONFORMANT**
>
> Taken together — nothing removed, errors struck through so the original stays legible, alterations fully attributable, late entries marked, and the reason for every amendment recorded — the professional rule specifies **an append-only record with a full, attributed, reasoned audit trail of every change**.
>
> Most clinical systems in use do not meet it. The provision almost never implemented is the last one: **the reason for the amendment**. A system that versions a note and captures who changed it, but not why, has built three-quarters of the control and failed the part a court will ask about.
>
> And the stakes are professional rather than merely technical. Altering notes without indicating that they were amended may amount to unprofessional conduct, misrepresentation or fraud, or obstruction of justice [4].

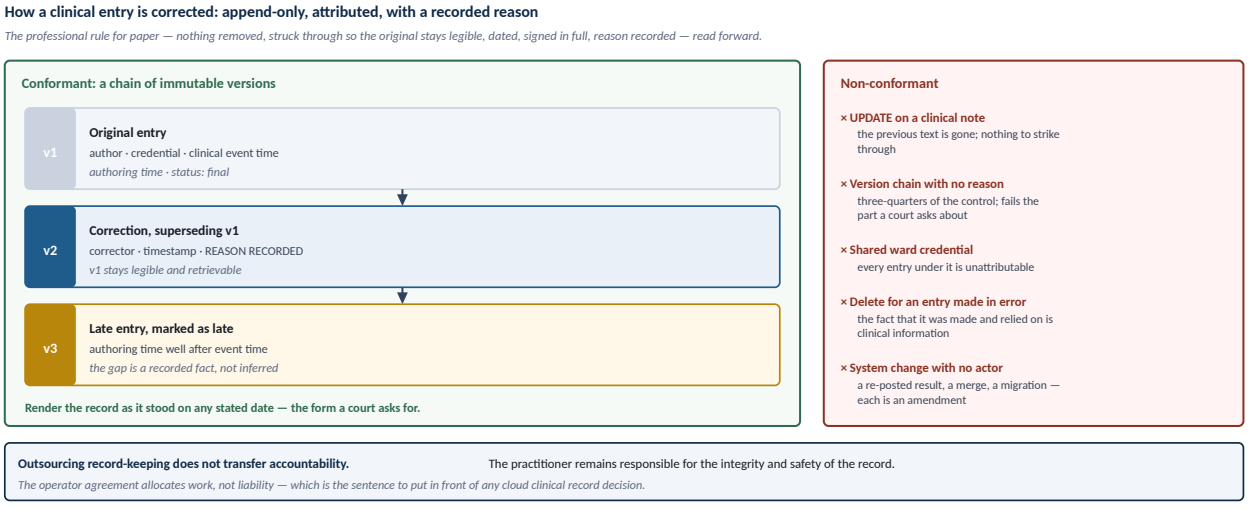


Figure 13.3 How a clinical entry is corrected. Every statement is an immutable version; a correction supersedes rather than replaces and carries its corrector, timestamp and recorded reason; a late entry is marked and the gap between event time and authoring time is a recorded fact. The right-hand column is the set of implementations that fail the rule, and the one at the top is the commonest: an in-place update leaves nothing to strike through.

The model that satisfies this is not exotic. Every clinical statement is an immutable version. A correction creates a new version that supersedes the previous one, carrying the corrector, the timestamp, the reason, and a link to what it supersedes. A **late entry** is a version whose authoring time is materially after its clinical event time, and the gap is a recorded fact rather than an inferred one. An entry made in error is marked as such rather than deleted, because the fact that it was made and relied upon is itself clinical information. And the whole chain is reconstructable: the system can render the record **as it stood at a stated point in time**, which is the form a court asks for and the form almost no clinical system produces without a project.

Two further requirements that fall out of the same rule.

Outsourcing record-keeping does not transfer accountability — the practitioner remains responsible for the integrity and safety of the record [4]. That sentence belongs in front of any client contemplating a cloud clinical record or an outsourced scanning bureau. The operator agreement allocates work, not liability.

An amendment made by a system is still an amendment. An interface that re-posts a corrected laboratory result, a merge that re-points documents from one identifier to another, a migration that reformats a note — each is a change to a clinical record, each needs an actor and a reason, and the actor may legitimately be a named system account. What is not legitimate is a change with no actor.

The criminal offence, and why it is not a privacy argument

Section 17 of the National Health Act requires the person in charge of a health establishment to set up control measures preventing unauthorised access to records and to the storage facilities or systems holding them. Section 17(2) then makes a list of things an **offence**, carrying a fine, imprisonment of up to one year, or both [12].

Read the list as a data architecture requirements specification, because that is what it is. It is an offence to fail to perform the duty to set up control measures; to falsify a record; to create, change or destroy a record without authority; to fail to create or change a record when properly required; to provide false information intending it to be included in a record; to copy part of a record without authority; to link a user's identifying details to their condition, treatment or history without authority; to gain unauthorised access to a record or to the system holding it, **including intercepting transmitted information**; to connect a record-keeping system to other systems or terminals without authority; and to modify or impair the operating system or the programme used to keep records [12].

Four of those are unusual enough to be worth drawing out.

Failing to create or change a record when properly required is an offence. Under-documentation in South Africa is not merely poor practice; it is capable of being an offence. This is the statutory backing for the argument, made in the South African literature, that courts should compel the state and public facilities to meet their record-keeping obligations — an argument raised precisely because the study's author found numerous public health facilities failing those obligations, and found the failure undermining medical negligence claims [13].

Linking identifying details to condition, treatment or history without authority is an offence. That is a *linkage* offence. An architect building a patient index, a clinical data warehouse or a research linkage must be able to point at the authority for the linkage. Record linkage in South African health data is not a neutral technical act.

Connecting a record-keeping system to other systems or terminals without authority is an offence. Integration requires authority, which makes an integration project's authorisation record a legal artefact. "We connected the laboratory feed to the warehouse" without a recorded authority is, on a literal reading, inside the section.

Modifying or impairing the programme used to keep records is an offence, which brings unauthorised configuration change and unmanaged patching into scope.

> **THIS IS THE STRONGEST AVAILABLE ARGUMENT FOR ACCESS AUDITING, AND IT IS NOT A PRIVACY ARGUMENT**
>
> An access-control and audit programme pitched as a privacy measure competes for attention with every other privacy measure and loses to clinical delivery. The same programme pitched as **a control against a criminal offence that the establishment's own head is on the hook for** is a different conversation with a different sponsor.
>
> Note what the offence does *not* require: harm, complaint, or a data subject. It attaches to the access, the copy, the linkage and the connection. So the control it demands is not "protect privacy"; it is **know who accessed which record, when, and under what authority, and be able to show it.** That is an audit requirement expressed in criminal law, and it is enforceable against an individual rather than only against the organisation.
>
> Chapter 18 carries the data protection regime and the regulators' powers. This section is the independent statutory duty that sits underneath it, and it binds providers directly where some of the newer data protection instruments do not.

The access rule itself is narrower and vaguer than people assume. A health worker or health care provider with access to a user's records may disclose personal information to others **where it is necessary for a legitimate purpose within the ordinary course and scope of their duties and where disclosure serves the user's interests** [12]. The Act does not define a legitimate purpose, and emergency treatment may qualify [14]. For architecture the three conditions map onto three access-control questions: is the accessor a health worker or provider, which is identity and role; is the purpose within the ordinary course of their duties, which is purpose binding and break-glass justification; and does disclosure serve the user's interests, which is a clinical judgement that cannot be automated and must therefore be logged and auditable. **An access model granting all clinicians access to all records does not satisfy that provision. An access model with break-glass, a mandatory reason and an audit does.**

Access by the patient, by a parent, by a proxy, and after death

Four rights, four legal bases, and a system that implements one has implemented none of the others.

By the patient. Two routes: an access-to-information request, and the data protection right to be told what is held, to be given a record or description of it, and to request correction or deletion [14]. The professional guidance states that patients **aged 12 and over** may request a copy, an abstract or direct access to their records [10]. Access may not be refused over an unpaid account; a reasonable fee may be charged; originals are retained by the institution; and records are released without unreasonable delay but only on written consent or where the law requires it [4].

And there is a safety valve that almost no portal design carries. The access-to-information regime allows an information officer to **consult a health care practitioner** where disclosure of a record about the requester's own physical or mental health could cause **serious harm**, with access still grantable if

counselling or other safeguards are arranged; and it requires refusal where disclosure would involve the unreasonable disclosure of a third party's personal information, with exceptions for certain minors and incapacitated persons [14]. The analysis closest to the question raises the design consequence directly: patients viewing electronic records online **without an intermediary safeguard** could be harmed, and calls it an issue requiring clarification [14].

> **FOR A PATIENT PORTAL, THIS IS THE PARAGRAPH THAT MATTERS**
>
> South African law contemplates a mediated-disclosure safety valve for health records, implemented by a human being consulting a practitioner. **A portal that publishes results to patients in real time has removed that valve.**
>
> So the design question is not "should patients see their records". It is **which record classes release immediately, which release after clinician review, and which release only with an arranged conversation** — and that is a classification attribute on the Document and Observation entities, not a feature flag. The classes that conventionally need it: a new cancer diagnosis, an HIV result, a genetic result, mental health notes, child protection notes, and anything about a third party.
>
> There is also a practical constraint the same analysis names and architects routinely ignore: socio-economic, educational and computer-literacy barriers limit who can use a portal at all [14]. A portal is an additional access route, not a replacement for the records office.

By a parent or guardian. For a patient **under 16**, a parent or guardian may apply — **but access requires the patient's written authorisation** [10]. Read that twice. The parent of a 13-year-old may request the record, and the 13-year-old must authorise the disclosure. It follows from the Children's Act consent architecture, under which a child over 12 with sufficient maturity may consent to their own treatment [15], and it means **parental access is a request requiring the child's authorisation, not a relationship-based entitlement**. A portal feature giving a parent full access to a 14-year-old's record is non-conformant.

Two absolute carve-outs sit alongside it. Information about a **termination of pregnancy** may be disclosed **only to the patient, whatever her age** [10]. And a child's HIV status is restricted by the Children's Act's own confidentiality provisions [16].

So the access decision has four dimensions that interact: a **record class** dimension, a **requester relationship** dimension, a **patient age** dimension, and a small set of absolute rules. That is why role-based access control is insufficient in healthcare, and why attribute- or policy-based access control is the right pattern: the decision depends on attributes of the record, of the subject, of the requester and of the relationship between them.

Four access rights, four legal bases, and the rules that override all of them

A system that implements one of these has implemented none of the others.

Who is asking	On what basis	What is required	Where it breaks
The patient, aged 12 or over	Access-to-information request; data-subject participation rights	Copy, abstract or direct access. No refusal over an unpaid account. A reasonable fee. Originals retained.	A portal that releases every result in real time
A parent of a patient under 16	Access-to-information regime, read with the Children's Act	The parent may apply — and access requires the PATIENT'S written authorisation.	A "parent proxy" feature giving full access to a 14-year-old
A proxy for a patient without capacity	Curatorship order, power of attorney, next-of-kin hierarchy, advance directive	A Proxy Authority: who, for whom, under what instrument, what scope, from when to when, verified by whom.	A next-of-kin name and a telephone number — a contact, not an authority
An executor, after the patient's death	Two readings. Both give the same architecture.	Documented adjudication, not a self-service route. The record stays and stays protected.	A proxy relationship that still opens the portal after death

Absolute rules that override the table above

Information about a termination of pregnancy may be disclosed only to the patient, whatever her age.
A child's HIV status is restricted by the Children's Act's own confidentiality provisions.

Four interacting dimensions: record class, requester relationship, patient age, and a small set of absolute rules.
Which is why role-based access control is insufficient here, and attribute- or policy-based access control is the right pattern.

Figure 13.4 Four access rights with four legal bases, and the absolute rules that override them. The second row is the one portal designs get wrong: a parent of a patient under 16 may apply, and access requires the patient's own written authorisation. The fourth is the one systems get wrong structurally, because granting access on a proxy relationship after death requires the system to know the patient has died.

By a proxy. A person authorised to act for a patient who lacks capacity, or who appointed them. The authority may be a curatorship order, a power of attorney, a statutory next-of-kin hierarchy or an advance directive naming a proxy — and each has a document, a scope and a period. The entity is a **Proxy Authority**: who, for whom, under what instrument, for what scope, from when to when, verified by whom. Modelling it as a next-of-kin name and a telephone number, which is what almost every patient administration system does, provides a **contact** rather than an **authority**, and the two are routinely conflated at three in the morning.

After death. The hardest case and the one with the least guidance. The competing considerations are all real: the data protection statute protects living persons, so its access rights do not obviously run to an executor; the Act's confidentiality duty is not expressed as lapsing on death; professional confidentiality survives death; an executor has duties that may require the record; a family may have a legitimate interest in a hereditary condition; a negligence claim may be brought by the estate; and a life insurer may seek the record under a policy term. The practical South African route is an access-to-information request by the executor, together with the statutory basis for disclosure where a law or a court order requires it [12][14].

Both readings should be given. On the first, the record of a deceased patient falls outside the data protection statute and access is governed by the access-to-information regime, the Act and professional duty, so the executor's request is the route and the confidentiality duty is weighed. On the second, the confidentiality duty is independent and survives death absolutely, absent a court order or a statutory requirement, so disclosure to family or insurers

requires a legal instrument. **The architectural answer is the same under both, which makes it safe to implement:** a deceased patient's record stays, stays protected, and access requests on it go through a documented adjudication rather than a self-service route. Which means the portal must *not* grant access on the strength of a proxy relationship once the patient is deceased — and therefore the system must know the patient is deceased, which is why chapter 4 puts "deceased" on Person rather than on Patient.

Disclosure event

Here is the entity nobody asks for, and the one that makes several other things possible. **It is introduced in this chapter because it is a property of the record rather than of a consent form.**

A **Disclosure event** is the record that specified information about a specified patient was given to a **named recipient**, on a **date**, for a stated **purpose**, under a stated **authority**, by a named actor. Not a log line. An entity, with relationships to the patient, the documents or data disclosed, the recipient, the consent or legal basis relied on, and the request that prompted it.

Without it, three things cannot be done.

The organisation cannot tell a patient who received their data. The openness and data-subject-participation conditions of the data protection regime assume the organisation can answer that question [14]. A professional body's own 2022 digital health statement recommends that patients be able to access, correct and restrict the use of their records and **be told who has received their data** [14] — which is the Disclosure event restated as a patient right.

A withdrawal cannot be acted on selectively. Withdrawing consent to disclosure stops *future* disclosure; it cannot unsend what was sent. But an organisation that does not know what was sent, to whom, cannot even identify which future flows to stop, and cannot tell the patient what cannot be recalled.

The organisation cannot prove it stopped. The defence to a complaint about an unlawful disclosure is the record that disclosure ceased on a date. That record is the absence of further Disclosure events after a withdrawal event, which is only evidence if the events were being recorded in the first place.

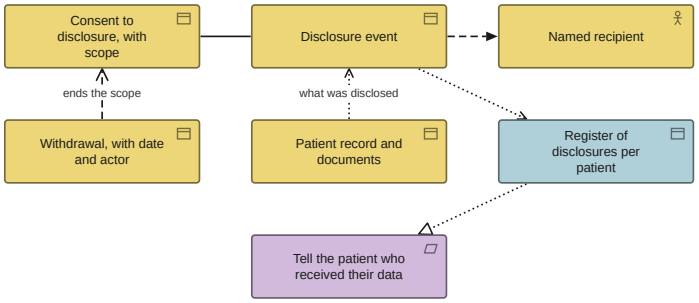


Figure 13.5 Disclosure event, introduced here as an entity rather than a log line. It records that specified information about a patient went to a named recipient, on a date, for a purpose, under an authority, by a named actor. Without it the organisation cannot tell a patient who holds their data, cannot act on a withdrawal selectively, and cannot prove that disclosure stopped.

Two design notes. First, the **scope of a disclosure consent is a set of triples** — information classes by recipients by purposes, over a period. Modelling scope as free text makes enforcement impossible; modelling it as structured triples makes it enforceable by a policy decision point, and makes a proposed disclosure checkable against the consent in force on the date of the disclosure. Second, **consent is not the access-control mechanism**. The lawful basis for clinical processing is a statutory authorisation plus the Act's access rule; consent governs disclosure outside the care relationship, and research. A system that gates clinical access on a consent flag has mis-modelled the law and will block emergency care. A system with no consent object at all cannot govern disclosure or research. Both failures are common and they are opposite. Chapter 18 carries the regime; what belongs here is that the evidence of a disclosure lives in the record.

Retention, and a guideline that disagrees with itself

There is no South African health-record retention instrument. Every period in circulation traces to a **professional guideline**, and the guideline disagrees with itself between versions.

The periods, as the September 2016 edition gives them [10]:

Category	Period	Authority the guideline cites
General adult records	At least **6 years** from the date the record became dormant	None cited
Minors, including obstetric records	Until the **21st birthday**	A minor has up to 3 years after majority to bring a civil claim
Mentally incompetent patients	The **lifetime** of the patient	None cited
Occupational health records	**20 years** after treatment	Occupational Health and Safety Act 85 of 1993
Long-latency exposure conditions	Recommended not less than **25 years**	The guideline's own recommendation; it states there is no clear-cut rule

Plus three provisions that matter more than the periods: where a statute sets a longer period it must be complied with; records in provincial hospitals and clinics may be destroyed only with the authorisation of the relevant **Deputy Director-General**; and retention decisions should weigh storage costs against the risk of a negligence defence being weakened by missing records, together with academic and research value [10].

And then the disagreement. The **September 2022 revision** keeps the same specific periods but changes the general position: general records should be retained **ideally indefinitely, particularly if electronic**, and otherwise for at least six years from dormancy [11].

> **"THE LAW SAYS SIX YEARS" IS WRONG TWICE OVER**
>
> It is wrong because the six-year period is a **professional guideline** rather than a statute or a regulation — no regulation under the National Health Act prescribing retention periods could be located for this volume, and the item is unverified rather than settled [12].
>
> And it is wrong because the same guideline's later version says **ideally indefinite**. The two editions of one instrument give different headline positions [10][11], and which edition is currently in force could not be established for this volume: a September 2016 edition and a revision marked September 2022 are both in circulation, and a 2025 practitioner briefing describes its material as current without giving a version number [4][10][11]. **Obtain the current booklet from the Council before relying on a period**, and record in the retention schedule which edition it was built against. A retention schedule built on a cited secondary source is a compliance artefact built on sand.

"Ideally indefinite, particularly if electronic" is a change in professional posture and it is architecturally consequential, because it converts retention from a schedule into an archive strategy on the reasoning that electronic storage is cheap. It does not address the data protection duty not to retain information longer than necessary for the purpose [14]. **Both cannot be satisfied by a single default.**

The resolution that makes the architecture tractable is to notice that the professional periods attach to the **patient record as the profession understands it**, and do not obviously attach to every derived copy — the warehouse extract, the analytics mart, the de-identified research dataset, the integration engine's message store, the ten-year backup, the access log. That yields the single most useful retention rule available to a healthcare data architect:

> **THE RETENTION PERIOD OF A COPY IS A FUNCTION OF THE COPY'S PURPOSE, NOT OF THE SOURCE RECORD'S CLASS**
>
> A message queue holding clinical messages for replay has a purpose measured in days. It is routinely retained for years, because nobody ever assigned it a purpose. Assign every copy a purpose, a period and a disposal method, and the retention problem shrinks to the one record class the professional rule actually governs.

Destruction authority, which is where the decision sits

The retention argument is where architects spend their time. The **destruction** authority is where the decision actually sits, and it is the part most often missed.

The Act's record-keeping duty is expressly **subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act** [12]. That is the constraint. A public body may not destroy public records without the authority of the National Archivist, against approved retention schedules — and for a provincial facility the professional guideline's Deputy Director-General authorisation sits alongside it [10]. The text of the Archives Act could not be retrieved for this volume and its precise effect on provincial hospital record disposal is therefore **unverified**; obtaining it, and the applicable disposal authority, is the first step in any public-sector retention project.

So a provincial hospital's record of an adult treated in 2010 and dormant since 2011 is simultaneously: deletable on a purpose-limitation reading, because the purpose is served; retainable on the same statute's own exception, because another law requires or authorises retention; required to be kept under the professional guideline until 2017 and arguably beyond on its balancing provision; and **not destroyable at all** without a Deputy Director-General authorisation and a disposal authority [10][12][14]. The architecture decision is not what the period is. It is what the organisation is permitted to do when the period expires, and who signs.

What to build, concretely:

- A **retention schedule as data**, not as a policy document: record class, the authority for the period, the period itself, the trigger event (last contact, dormancy, 21st birthday, date of treatment, date of death), the disposal method (destruction, deletion, de-identification), the approval required, and the evidence of disposal.
- A **retention rule relationship** from the Document and record entities to that schedule, so that the schedule is enforceable rather than aspirational.
- The thing almost nobody has: **evidence of disposal**. A regulator's own enforcement order has named a required retention policy and schedule as a remedial step, and practitioner guidance recommends a **signed disposal authority** and a **register of destroyed records**, with destruction by shredding or incineration for paper and overwriting or physical destruction for electronic media, under a confidentiality agreement with any outside contractor [17][18].

> **A REGISTER OF DESTROYED RECORDS IS WHAT LETS A FACILITY SAY "THIS RECORD NO LONGER EXISTS, AND HERE IS THE LAWFUL REASON"**
>
> Without it, **a missing record is indistinguishable from a lost record.** And in three of nine paid medico-legal claims the Auditor-General reviewed, that distinction cost the state the case [1]. The register is cheap. The alternative has a measured price.

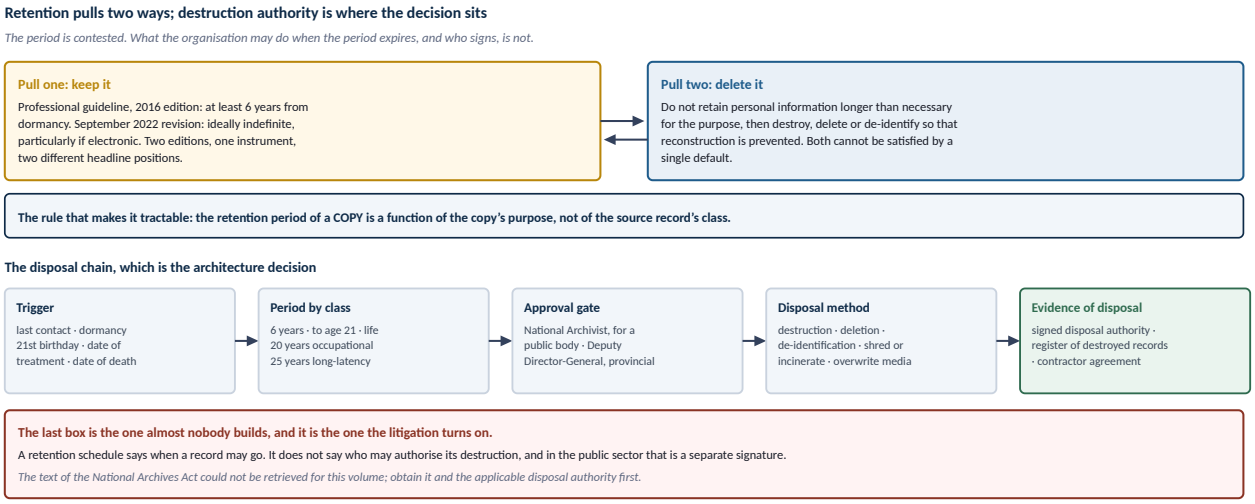


Figure 13.6 Retention pulls two ways and the disposal chain is where the decision sits. One instrument gives two headline positions in two editions; the data protection duty pulls the other way; and neither settles what a public body may actually do, because destruction requires an approval the retention schedule does not grant. The last box in the chain is the one almost nobody builds, and it is what distinguishes a lawfully destroyed record from a lost one.

The record as evidence

A litigated record has requirements a clinical record does not, and they are specifiable.

Completeness for a defined episode. Not "the patient's record" — the complete record of a specific episode of care, including the documents that arrived from elsewhere, the results, the observation charts, the consent forms, the operation note, the nursing notes and the medicines chart. A retrieval that returns the discharge summary and nothing else has not answered the request.

Point-in-time reconstruction. What the record said on the date of the alleged negligence, not what it says now. Which is the amendment model of section 5 doing its real work.

Provenance for every item. Who authored it, who authenticated it, when, under what credential, and what it superseded.

A chain of custody after issue. The Auditor-General found **17 of 27 facilities not properly tracking records after they were issued** [1]. A record issued to a ward, a clinic, an attorney or a court and not tracked back is a record the organisation cannot produce on the next request.

A retention horizon set by the longest applicable period, which for obstetric records is the patient's 21st birthday [10], and in practice longer, because the exposure period is indeterminate.

Two South African facts about the litigation environment belong with that specification. First, the state is building a case management function for these matters: the national department reports a Case Management System for medico-legal claims rolled out in **five provinces** and **not fully implemented** [19]. Second, the procedural route has changed: a Gauteng High Court directive of **22 April 2025** means civil matters in that division cannot obtain a trial date without a report from an accredited mediator, and medical negligence claims against provincial health departments are civil trials inside it [20]. Mediation brings the record forward in the timeline — it is examined earlier, by fewer people, with less time to assemble it. Retrieval latency becomes a litigation variable.

Two cases are cited in the secondary literature on quantum for children with cerebral palsy — *MEC for Health, Gauteng Provincial Government v PN obo EN* (2024) and *MEC for Health, Gauteng v DZ obo WZ* (2023) — and both citations reach this volume through a secondary source and should be read before being relied on [5].

The billing and clinical inversion

One defect deserves naming on its own, because it is the one the professional rule expressly prohibits and the one a supplied repository is most likely to contain.

The record-keeping guidance requires that **billing records be kept separate from patient care records** [10]. That is an explicit segregation requirement from the professional regulator, and almost no integrated hospital information system honours it.

The inversion that follows is a real pattern, not a hypothetical. Clinical observations mastered in the billing system, because the billing system was integrated first and had a convenient table. The funder claim mastered in the clinical record, because that is where the coder sat. Both are caricatures of documented patterns — entities mastered where the first system was integrated rather than where they belong — and chapter 20 names the pattern behind each. What belongs here is the authority: this is not a preference an architect has to argue for on general principles. **The revenue cycle and the clinical record are different stores, with different access populations, and the clinical record should not be reachable from a billing role** — and a professional guideline says so in terms.

It also has an access-audit consequence. A billing role that can reach the clinical record is a population of users with access they cannot justify under the Act's "ordinary course and scope of their duties" test [12], which brings the section 17(2) offence into view for the organisation that granted the access.

Measure Definition

The record is also the evidential basis for every number the organisation publishes about itself, and that is the second reason it needs a companion entity.

Chapter 4 defined **Measure Definition** as the versioned, effective-dated, owned rule by which a derived figure was computed: its numerator, denominator, exclusions, boundary rule and the period over which it applies. **As an entity it needs**: an identifier and a version; an owner; an effective period; the numerator, denominator and exclusion specifications; the boundary rule it relies on, which is a reference to one of chapter 4's constructs; the source entities it reads; and a relationship to every published figure computed under it.

It belongs in this chapter because it has the same property the clinical record has: **a figure, like a record, must be reproducible as it stood.** A published occupancy rate, length of stay, readmission rate or mortality figure is an assertion with an author and a date, and the defence of it is the rule that produced it. Chapter 12 showed how much of that rule is coding-dependent. The evidence that it must be an entity rather than a convention is chapter 4's: on one specialty in one trust over fifteen weeks, there is **no simple mathematical relationship** between the four theatre utilisation calculation methods, with raw utilisation of 80% against touch-time utilisation of 60% on the same list [21]. There is no conversion. Carry that denominator, and note that the finding generalises.

The repository consequence is blunt. The supplied repository used in chapter 20 has no place for a Measure Definition, which is why its figures cannot be reproduced, and no place for a Disclosure event, which is why its consent model cannot be made operable. They are the two entities experienced healthcare architects add and nobody asks for.

What to take to the client

The record is one object with three jobs, and the legal job is the one with a measured price attached. Lead with the arithmetic, not with the clinical benefit: a third of the paid medico-legal claims the Auditor-General reviewed were conceded for want of a record, provinces pay about R1,5bn a year, and the national digital health budget line is R6,5 million on either reading of the allocation [1][2][6].

Then ask five questions, in this order, and write down the answers.

1. **Can you render this patient's record as it stood on a stated date?** If not, the organisation has a clinical system and not a legal one.
2. **For every amendment in the last year, who made it and why?** The reason is the field almost nobody captures and the one a court asks about.
3. **Who accessed this record, when, and under what authority?** Not for privacy. For the offence.
4. **What is the retention schedule, which edition of the guideline is it built against, and where is the evidence of disposal?** A missing record and a lawfully destroyed record must be distinguishable.
5. **Where does a disclosure get recorded?** If the answer is an email folder, the organisation cannot tell a patient who holds their data and cannot act on a withdrawal.

None of those five is a technology question, and all five are answered by architecture.

THE CHAPTER IN TEN LINES

- In three of nine paid medico-legal claims the Auditor-General reviewed, the department conceded because it could not locate the medical record.
- The recorded claims balance was R50,2bn at 31 March 2026 on the Auditor-General's figures and R50,6bn for 2025/26 on Treasury's. R1,57bn was paid in 2025/26, about R1,5bn a year on average.
- The national direct NHI grant line for Digital Health System is R6,5 million, and the allocation it sits inside has two published readings that disagree.
- The record is one object with three jobs, and the three sets of requirements conflict in predictable places.
- The professional rule describes an append-only record with a full, attributed, reasoned audit trail. The field almost nobody captures is the reason for the amendment.
- Unauthorised access to a health record is a criminal offence under section 17(2), independent of the data protection regime, and so are failing to record when required, linking identity to condition without authority, and connecting a record system to another without authority.
- That offence is the strongest available argument for access auditing, and it is not a privacy argument.
- A parent of a patient under 16 may apply for the record, and access requires the patient's own written authorisation. Termination-of-pregnancy information goes only to the patient, at any age.
- Retention rests on a professional guideline that gives six years from dormancy in one edition and ideally indefinite in the next. Which edition is in force could not be established.
- Destruction authority is where the architecture decision sits: the record-keeping duty is subject to the National Archives Act and the access-to-information regime, and a public body needs an approval the retention schedule does not grant.
- A register of destroyed records with a signed disposal authority is what makes a lawfully destroyed record distinguishable from a lost one.
- Billing records must be kept separate from patient care records, which is explicit authority for a separation almost no integrated system honours.
- Disclosure event and Measure Definition are the two entities experienced healthcare architects add and nobody asks for.

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CHAPTER 14

The conceptual application model

What survives a change of vendor is a list of functions, the data each one owns, and the contract at every boundary between them.

A conceptual application model is not a system inventory. The inventory is the next chapter's job, and it dates. This chapter is the layer above it: the functions that must exist in any provider of this kind, what each one owns, what each one must never own, and where each one ends. That list survives a change of vendor, a merger, a provincial reorganisation and a decade. The products do not.

Healthcare needs the distinction more than most industries, for a reason that is historical rather than technical. The product boundaries in a provider estate were set by purchasing sequence. The patient administration system came first, because it replaced the admissions register and the appointment book and because it was the system funders and statisticians needed. The laboratory bought a laboratory system when analyser interfacing became possible. Radiology bought an archive when film became expensive. Pharmacy bought a dispensing system because stock is money. Almost none of those boundaries records an architectural decision. They record who had a budget in which year.

An architect who models the estate as it was sold inherits a sales history and calls it an architecture. The remedy is to model functions first, then ask which boundaries between them are real.

The three questions that make the model

For every function, three questions, and the answers are the model:

- **What does it own?** Which data is authored here, mastered here, and corrected here.
- **What must it never own?** Stated as a prohibition, because the failure mode is always accretion: a system that is asked a question often enough starts answering it authoritatively.
- **Where does it end?** The boundary at which its authority stops and another function's begins.

Chapter 4 has already applied that triplet to the four words for the record — the patient administration system, the electronic medical or patient record, the electronic health record used strictly, and the clinical record as a legal construct — and this chapter does not restate those definitions. It applies the same discipline to everything else, and then does the harder thing: it asks which of the boundaries you can see in a real estate are load-bearing and which are packaging.

What the decision order does to the application model

Chapter 8 states the sequencing fact: identity, terminology and integration strategy are decided before application strategy. This is the application consequence of each, and it is the practical reason the order matters.

Identity decided first means the model holds identity as a consumed service. Every function that touches a person references an identifier it did not mint, in a namespace it does not own, with a provenance marker saying where the identifier came from. Decide application strategy first and the chosen product's internal patient index becomes the identity architecture by default, which is how an estate ends up with four indexes and no index. What a decided identity looks like is documented in South Africa: the Western Cape issues a nine-digit folder number shared across government facilities, whose last digit is a modulus-10 checksum, printed on barcoded labels, and the provincial data centre states that this materially helps linkage [1]. Three decisions in one sentence — a shared namespace, a check digit so transcription errors are caught at capture rather than discovered at merge, and machine reading at the point of collection. Nationally the Health Patient Registration System holds demographics only [2], which is a restraint worth naming: a national index that carries identity and the existence of a visit, and no clinical content, is far easier to govern and far more likely to be adopted than a national clinical repository. Chapter 11 owns identity; the application consequence is that identity is a function with its own product category, not an attribute of the record.

Terminology decided first means every coded value in the model carries its code system and its version, and every function consults a service rather than a table. Decide application strategy first and each product's internal code tables become the terminology architecture, and the mapping burden lands in the integration layer, where it is invisible to clinical governance. South Africa makes this concrete and uncomfortable. The gazetted interoperability framework names neither SNOMED CT nor LOINC, and ICD-10 appears in it only in the acronym list [3]. The country joined SNOMED International on or about 14 May 2026 [4][5], and the National Release Centre is not yet operating — no national extension, no national reference sets, no published national map, with representatives listed as to be advised [4]. So a platform procured now must be able to add a terminology binding it does not have, to a code system the vendor may not support, against an extension that does not exist. That is a contractual requirement before it is a technical one, and it belongs in the application model as a stated capability of the terminology function rather than as a hope about a product roadmap. **Re-check the status of the National Release Centre before designing to it.** Chapter 12 carries the specifications that do not exist; chapter 16 carries what governs each seam.

Integration strategy decided first means the model has a named integration function with a stated topology. The defensible position is hub by default, point-to-point by exception with a documented reason, and a hard rule that device telemetry never enters the clinical integration engine. Decide application strategy first and the topology becomes whatever interfaces the chosen products happen to support, which is point-to-point with extra steps.

Only then does application strategy become answerable, and the honest form of the question is not whether to buy a single suite or assemble an estate of specialist products. It is which kind of integration risk the organisation prefers to carry. Between products the risk is visible, contractible and ownable: there is an interface, a specification, an owner and an inventory entry. Inside a product it is invisible, non-contractible and owned by the vendor. The best published evidence for that claim comes from a single-product estate. A prospective audit of electronic chemotherapy prescribing at St Bartholomew's Hospital found a regimen library of over 400 regimens with everything preset except the cytotoxic dose, some regimens requiring manual dose-unit

calculation, and a dose field that accepted a number without a unit — which permitted a thousand-fold methotrexate underdose despite an unambiguous protocol; carboplatin doses wrongly adjusted by the system itself in five cases; and no modelled state for a prescription issued before the blood results that would release it, because that is how outpatient clinics run and the software did not represent it [6]. Four of those five defects are seams. They are inside one product, and nobody would have called them integration defects.

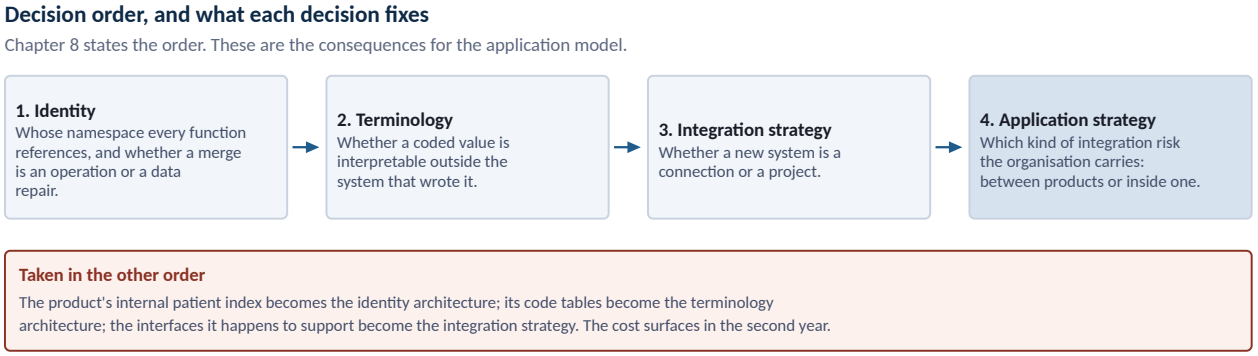


Figure 14.1 The decision order and its application consequences. Each of the first three has a long lead time, an organisation-wide blast radius and no cheap reversal, which is why an application choice taken before them settles all three by default.

WHAT EACH EARLIER DECISION FIXES

Identity fixes whose namespace every function references, and whether a merge is an operation or a data repair. Terminology fixes whether a coded value is interpretable outside the system that wrote it. Integration strategy fixes whether a new system is a connection or a project. Application strategy then chooses products inside those three constraints. Taken in the other order, the product chooses all three for you, and the choice surfaces in the second year.

The functions

What follows is the function list this book uses. Twenty-seven functions in five groups: those that sit beneath the record and are referenced by everything, those that are the record, those that supply clinical depth or acquire data from machines, those that sit around the record and consume it, and three that nobody budgets for.

Two cautions before the tables. First, a function is not a product: one product may implement several functions and one function may be spread across three products, and the vendor's names for your functions are not these names. Second, the "must never own" column is the useful one. It is what an architect reads out in a design review, and every row of it is a documented failure.

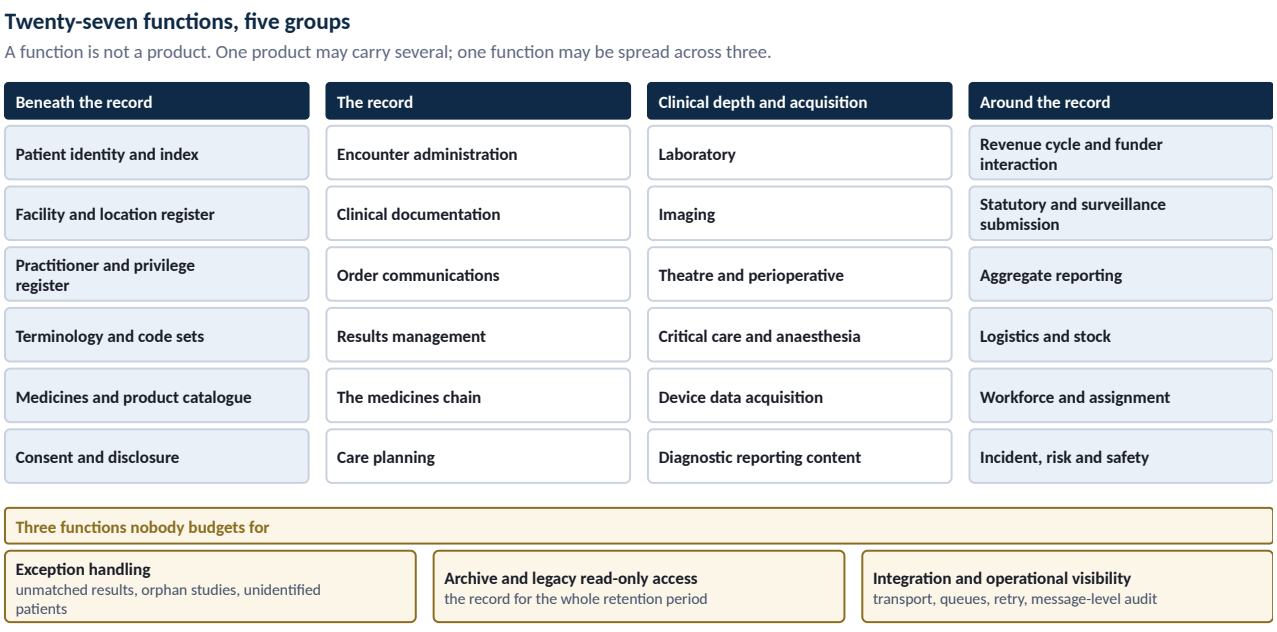


Figure 14.2 The conceptual application model: the functions a provider of this kind must have, grouped by their position relative to the record. The register functions in the left column are the ones programmes discover eighteen months in. The amber band holds the three that appear in no business case and in every incident review.

Beneath the record

Function	Must own	Must never own
Patient identity and index	The person, the identifier namespaces, the links between them with their certainty, and the merge and unmerge history	Clinical content. An index that starts holding diagnoses becomes a record with no author
Facility and location register	The health establishment, its site, building, department, room and bed hierarchy, its physical characteristics and its dated operational status	The ward. A ward is an organisational unit with a nursing establishment that occupies a location and is not one
Practitioner and privilege register	The person, each registration with its own lifecycle and status, and the practitioner role at an organisation with its privileges	The roster. Assignment is a workforce function, and conflating the two makes a locum unable to prescribe
Terminology and code-set service	Code systems, versions, value sets, maps, and the effective dates on which validity changes	The clinical assertion. A terminology service answers questions about codes; it does not hold the note
Medicines and product catalogue	Product identity, its classification, its pack and form, and the mapping to each external coding scheme	Price, which is contract data, and the patient's medication list, which is clinical
Consent and disclosure	Consent to treatment, consent to disclosure and consent to research as separate objects, and the record of every disclosure made	Access control enforcement. Consent states a permission; the enforcement point is elsewhere and must be able to read it

The register functions are the ones programmes discover eighteen months in. The OpenHIE reference architecture is useful here precisely because it names them as components rather than assuming them: a client registry, a facility registry, a health worker registry, a terminology service and a product catalogue, each separate from the repositories and from the exchange layer [7]. South African national policy reads as that component set re-expressed in policy language — a master patient index, a facility list, a terminology service, registries, repositories and analytics behind a shared integration layer [8] — which makes the reference architecture an unusually good reading aid for a South African architect.

The record

Function	Must own	Must never own
Encounter administration	Registration, the visit, the ward stay, the bed state, the clinic diary, the waiting list and the separation event	Clinical truth. A diagnosis code entered here for billing must never become the source of the problem list
Clinical documentation	The clinical assertion, with its author, their role, the time of authoring, and every correction as an addition rather than an overwrite	The code. The code is a later classification of the assertion, made by someone else, for a purpose that is usually not clinical
Order communications	The order as a long-lived stateful object, its clinical indication, its priority and the identity of whoever must follow it up	Order state after the filler has accepted it, which in practice the filler owns
Results management	The inbound result, its status, its acknowledgement event with an actor and a timestamp, and the escalation when nobody acknowledges	The result value's authorship. The authorising professional in the performing organisation owns that
The medicines chain	The order, the dispense, the administration and the reconciliation as four linked entities, including the reason a dose was not given	Stock. A dispensing record is evidence a product left a shelf, not evidence a patient took it
Care planning	The patient's goals with target dates, planned activities with statuses, the responsible parties and the review date	The protocol. A protocol is an organisational artefact with a version; a care plan is patient-specific

Clinical depth and acquisition

Function	Must own	Must never own
Laboratory	The specimen, the accession identity, the worklist, autoverification, the authorising professional, the report and the archive	The patient demographic master. It subscribes, and it must be able to represent a result for a test the clinician did not order
Imaging	The study, its identity inside every image object, the reading workflow, the report as a versioned document, and the archive that outlives the viewer	Typed identity. Nothing is typed at a modality: the worklist supplies the demographics, and the fallback is an audited reconciliation path, not free typing
Theatre and perioperative	The list, the case, the operative record, the implant log with its serial or lot number, and the cancellation reason	The definitive problem list. It is the best source of the coded procedure and still not the owner of the diagnosis
Critical care and anaesthesia	The high-frequency flowsheet, the fluid balance, scoring, ventilation and sedation records and the handover document	The medication prescription, which must come from the enterprise prescribing function or the two diverge and a patient is double-dosed at step-down
Device data acquisition	The sampling policy, the artefact annotation, the device identity behind every value and the provenance flag	The clinical interpretation. A monitor value captured while a patient is repositioned is an observation with a correct timestamp and no clinical meaning
Diagnostic reporting content	The report as an attested, versioned artefact with an author and a status	The assumption that findings arrive coded. Structured radiology reporting is real and narrowly deployed, and a data strategy that assumes coded findings is planning on something untrue

The imaging row carries the single most useful operational rule in this chapter, and it is worth stating on its own. The modality worklist exists because a radiographer once had to type a patient's name and number into a scanner, and a mis-identified study is extraordinarily hard to correct because the identifier is embedded in every image object. The standard's answer is a query service against a worklist information model, with a complementary mechanism by which the modality reports back what it actually did [9]. The architectural rule that follows is absolute: nothing is typed at a modality.

Around the record

Function	Must own	Must never own
Revenue cycle and funder interaction	The invoice, the funder claim, the authorisation, the remittance and the reconciliation	The clinical record, the problem list or the terminology. It consumes all three
Statutory and surveillance submission	The submission event, its acknowledgement where one exists, and the reconciliation of a pre-confirmation notification against the later laboratory result	The case definition, which is published nationally and versioned by its publisher
Aggregate reporting	The data element, the organisation unit, the period and the category disaggregation	A person. There is no person in an aggregate model, and a clinical question cannot be asked of it
Logistics and stock	Stock on hand by location and lot, expiry, cold chain, the controlled-substance register and the issue to a ward	The clinical medication list, and the inference that a cabinet removal is an administration
Workforce and assignment	The establishment, the roster, leave, cover and time and attendance	Professional registration status, which belongs to the practitioner register, and the clinical identity used for prescribing
Incident, risk and safety	The patient safety incident, its severity assessment, the investigation and the improvement action	An open join to clinical identity. The linkage that makes system-level analysis possible is one-way and pseudonymised

The aggregate reporting row is the one that most often causes an expensive mistake, and the reason is worth stating because it generalises beyond any one product. An aggregate model holds a value for a data element, for an organisation unit, for a period, optionally disaggregated. It cannot answer whether *this* patient attended, what *this* patient's last result was, or whether *this* patient is lost to follow-up. It cannot be de-duplicated, so coverage denominators built on aggregate attendance over-count service delivery in a mobile population. And the governance changes completely at the boundary, because aggregate data is generally not personal information and patient-level data is. The consequence for the model is a rule: do not build clinical capability on an aggregate platform, and do not build national statistics on a clinical platform. Everything about the South African vertical-programme register landscape follows from one organisation breaking that rule in the only direction available to it — because the national routine system held aggregate data, every programme that had to follow individuals built its own patient-level register [2][23].

Three functions nobody budgets for

Function	Must own	Must never own
Exception handling	The unmatched result, the orphan study, the unidentified-patient record and the undeliverable message, each with an owner and a service level	Silence. An exception queue with no staff and no target is a clinical risk with an audit trail
Archive and legacy read-only access	The record for the whole retention period, independent of the system that created it, constructible as it stood at a point in time	A licence dependency on a product that has been replaced
Integration and operational visibility	Transport, queueing, retry, per-destination persistence, and the message-level answer to "did this reach the ward system"	Meaning. Transformations may change representation; anything that changes meaning is a separate, owned, documented component

The exception queue deserves the attention it never gets. A result arrives with an identifier the placer cannot resolve; a study is acquired against an emergency identity; a funder claim rejects on a code-set version the provider has not adopted yet. Each lands in a queue. Those queues are where patients are lost, and they are not a technical artefact — they are a business function with a workload, a competence requirement and a service level, and they should appear in the architecture with a performer against them. The test of whether an organisation believes this is whether anyone can say, without looking, how many items are in the queue today and who owns the oldest one.

Where boundaries are real

Five tests. A boundary that passes one of them is load-bearing and must be engineered. A boundary that passes none of them is packaging, and consolidating across it is legitimate.

It crosses a legal entity. This is the commonest real boundary in healthcare and the one architects from other industries miss, because in most industries the systems inside a building belong to one company. In a South African public facility, pathology is performed by the National Health Laboratory Service, a separate national entity that bills provinces [10]. In a private hospital, the admitting doctors are usually not employees. A seam that crosses a legal entity carries a contract, a consent question under section 14 of the National Health Act, and an availability the organisation does not control.

It crosses a regulatory boundary. Where a device and its companion software are a single regulated product, interposing a hub may invalidate the manufacturer's intended configuration, and the duty to manage the risk of connecting devices to a network sits with the healthcare delivery organisation rather than the manufacturer [11]. That makes the device boundary real in a way no amount of protocol work changes.

The two sides must survive each other's failure. Two functions that must be independently available cannot share an instance, a database or a failure domain. This is the test that produces the read-only record, and chapter 19 owns it.

They change on different calendars. Reference data moves on an external cadence the organisation does not set: the single exit price of a medicine changes on a gazetted day [12], and the ICD-10 Master Industry Table carries effective dating and validity flags that determine whether a code is valid and payable on a given date [13]. A function whose content changes on somebody else's calendar is a real boundary, because a shared release cycle across it is impossible.

One side attests and the other mutates. The legal record is a view that must be constructible, versioned and attestable as it stood at a point in time [14]. A function that attests must be separable from one that corrects rows in place, or the attestation means nothing. Chapter 13 owns the record as a legal object.

Boundaries that are an artefact of who sold what first

Patient administration against the clinical record. The split exists because the administrative system was bought a generation earlier. The real boundary is between administrative identity and encounter on one side and clinical assertion on the other, and that boundary can sit inside one product perfectly well. What cannot move is the prohibition: the administrative system must not become the source of clinical truth, which it does the moment a code captured for a funder claim becomes the problem list [16].

The radiology information system against the archive against the viewer. Three products, and only one of the two boundaries is real. The archive must outlive the viewer, which is the whole reason a vendor-neutral archive exists: the older generation of image archives stored in proprietary formats, so changing the viewer meant a multi-year data migration, and the newer pattern stores standard objects and rewrites attributes so studies produced under one institution's conventions serve another's viewer [15]. That rewriting is a data-altering function applied to a regulated clinical artefact, so the architecture must record what was changed, by which rule, and when. The boundary between the departmental workflow system and the viewer, by contrast, is mostly packaging — and the archive's own promise should be tested rather than assumed, by asking for a named customer who has changed viewers over the incumbent archive and what proportion of studies needed remediation.

Pharmacy. One name, two systems: a clinical dispensing function and a logistics function with a regulatory overlay. The real boundary runs through the middle of the product, not around it, and an architect who models "the pharmacy system" as one component cannot explain why the organisation can see stock and cannot see administrations.

The emergency department system. Genuinely separate for a genuine reason — its workflow is time-driven and location-driven in a way no other department's is, and it is the department where identity is most often unknown on arrival. But that reason dictates what it must never own, which is definitive identity. The emergency department is the main source of the duplicates an index must resolve, and the place where a false merge is most likely to be catastrophic because the patient cannot correct the record.

The patient-facing channel. A portal is a view over the record with a small write path; a digital front door is a channel strategy that operates before any record exists. Neither is a function in its own right, and where a telehealth or engagement platform becomes a parallel estate the organisation acquires clinical encounters that do not appear in the patient's record. South African professional guidance closes that door explicitly: telehealth records must meet the same standard as face-to-face records, and electronic communications must be filed in the patient's medical record [14].

Billing holding clinical codes. The most damaging inversion in provider estates, and in South Africa it is institutionalised rather than accidental, because the obligation that produces a diagnosis code on a private claim is a funder regulation and the obligation is conditional on payment [16]. A code that exists to clear a payment edit is not clinical data, and an analytics layer built on it measures reimbursement behaviour while reporting morbidity.

And one inversion that is real, documented and South African. Where patient administration and patient accounting live inside an enterprise resource planning platform — Gauteng is recorded in a parliamentary reply as running a resource-planning-based hospital information system [17] — the master patient index is an enterprise resource planning object, the encounter is an enterprise resource planning object, and the clinical system is a module hanging off them. That is not wrong. It is an inverted dependency, and every integration an architect designs has to pass through the platform's transaction model, release cycle and change control. Model it as it is rather than as the reference architecture says it should be. Chapter 15 has the estate.

The integration engine is a function, and it exists for four reasons

Every provider above a few hundred beds has an integration engine, it is usually run by two or three people, and it is usually the least documented and most critical component in the estate. The intuitive explanation — healthcare lacks standards — is wrong, and getting it wrong leads to the wrong remedy. Healthcare has an abundance of standards. What it lacks is enforced constraint, which is what conformance profiles and implementation guides exist to supply, and which is chapter 16's subject [18].

The engine exists for four specific properties of a clinical estate:

1. **Version heterogeneity.** The same standard runs at several versions at once with no prospect of a coordinated upgrade. South Africa's own gazetted framework names three versions of one messaging standard and declines to name a single version of another [3]. The engine version-bridges because nobody can afford to synchronise the endpoints.
2. **Dialect translation.** Two conformant implementations of the same standard can be mutually unintelligible. The engine maps fields, restructures segments, translates codes and strips or injects local extensions.
3. **Fan-out to independently available endpoints.** One registration feed must reach a dozen or more systems, each of which can be down on its own and each of which must receive every message eventually. The engine queues, retries and persists per destination. This is the property an application programming interface gateway does not have and cannot fake.
4. **Message-level operational visibility.** Somebody must be able to answer "did this result reach the ward system, and if not, where is it". That is a message-level audit requirement, and the engine is the only place in the estate where it can be satisfied.

It is therefore a message broker with a transformation language, per-destination queues, persistence, retry, alerting and an operational console. It is not an enterprise service bus in the sense that phrase carries in banking, it is not an application programming interface gateway, and it is not a data platform, whatever it is marketed as.

Four things go wrong with it, in order of frequency. **It becomes the business logic layer**, as clinical rules, routing decisions, identifier assignment and data enrichment accumulate in transformation scripts where they are invisible to clinical governance, untested, and understood by one person. **There is no interface inventory** — the freely available contingency-planning guidance requires that interface inventories be reviewed every six to twelve months, which tells you how often it happens [19]. **Buffers are mishandled on restart**: the same guidance has a dedicated practice on emptying buffers before stopping an interface, sizing them to prevent loss, alerting users when an interface fails, and documenting the start, stop, test and monitor procedures, and the failure mode is a restart that silently discards a queue so that the missing results are missing without anyone knowing [19]. **And two or three people know the engine**, which the architecture cannot fix but can mitigate, by insisting that transformations sit in version control and that interfaces be specified before they are built.

The seam is an element of the model

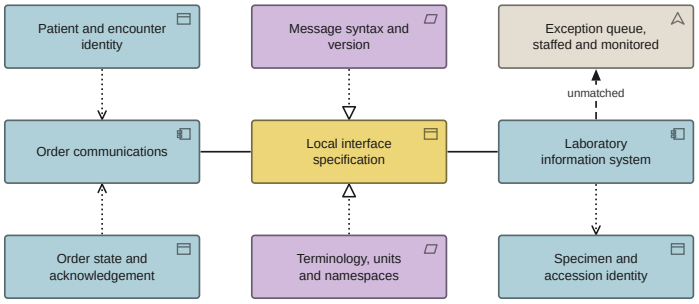


Figure 14.3 One seam, modelled. The contract is the local interface specification, not the standard: the standard supplies syntax and the bindings supply meaning, and both are realised in a document with a version and an owner. The exception queue is where the seam fails, so it is drawn as a function rather than left implicit.

Treat each seam as a first-class object with an owner, a version and an inventory entry, and record four things against it.

Whose identity crosses, and in which namespace. Including what happens to a merge: a merge that is not propagated leaves downstream systems holding a pre-merge identifier, which is the commonest integration defect in a provider estate and one that only shows up clinically.

Which system owns state. The order is a long-lived stateful object, and the common messaging standard models state transitions poorly [24], so in practice the filler rather than the placer becomes the system of record for order state. That is why "where is my test?" is answered by logging into the laboratory system rather than the record, and it is a fact to model rather than a defect to fix.

The terminology, units and identifier bindings. A numeric result without its unit and its reference range, for that analyser, for that patient's age and sex, is clinically uninterpretable, and a consolidated record that stores the number and discards the unit is the most common route to a clinically significant data error.

The actual contract, which is almost never the standard. Most seams are governed by two or three standards at different layers at once, plus a profile, plus a local interface specification — and the local specification is the contract. Chapter 16 says which layers govern which seam; the point here is that the contract is an artefact with a version and an owner, and if the organisation does not hold it, the supplier does.

The three closed loops that must be designed at seams rather than assumed are results acknowledgement, referral acknowledgement and transfer acceptance. The published control set for the first is specific: track every order through collected, completed, reported and acknowledged; identify the ordering clinician and whoever is responsible for follow-up; monitor time to acknowledgement; flag abnormal results distinctly and critical values differently again; escalate automatically when nobody responds; forward unacknowledged abnormal results to an alternate provider after a set time; and document delegation, accepted by the delegate [20]. Two of those are routinely missed, and both are architectural. Referred-out tests must be tracked from the point of ordering, which where a national laboratory performs most referred testing is the normal case, not an edge case. And trainee responsibility defaults to the supervising clinician on a change of service, which makes results routing dependent on the roster — a dependency almost never drawn on an architecture diagram [20].

> **WHERE RESULTS ARE DELIVERED BY PORTAL, THE CLOSED LOOP IS STRUCTURALLY UNAVAILABLE**
>
> In the South African public sector, laboratory results reach clinicians substantially through a web results viewer rather than an interface into a clinical system [22]. A viewer is a reasonable delivery channel and a poor safety channel, because it cannot support acknowledgement tracking, escalation of unacknowledged abnormal results, or roster-aware routing. The controls above are not weakened in that design; they are absent. The risk has to be carried somewhere else, explicitly, and named in the architecture rather than assumed away.

What survives a change of vendor

The test of a conceptual application model is what is still true after the products change. These things are:

- the function list, and the three answers against each function;
- the data each function owns, and the prohibitions;
- the seam inventory, with each seam's identity namespace, state owner, bindings and contract;
- the identity, terminology and integration decisions, written down with their reasons;
- the measure definitions behind every number the organisation reports;
- the exception and identity review queues, with their owners and service levels;
- the archive, and the record's constructibility as it stood at a point in time.

These things do not: module boundaries, screens, licence metrics, the supplier's names for your functions, and any statement beginning "the platform handles that".

Two deliverables follow, and they are small. A one-page function map with an owner against each function. And a seam inventory with, for every seam, the two systems, what crosses, the contract and its version, the owner, and what breaks. If an organisation cannot produce both by the sixth week of an engagement, that absence is the first finding, and the inventory is usually the highest-value cheap thing an architect can build. Most of the systems in a provider estate were bought by departments, at different times, from different budgets, on different terms, and nobody holds the aggregate.

One closing caution about the public-sector case, because it changes what the model is for. Where a national programme declares its own integration already complete — as the South African primary care record rollout does, stating that the record is already interoperable with existing health information systems and that no integration or interoperability work is needed or included in scope [21] — the conceptual model is not a design instrument. It is an assessment instrument. The functions still exist, the seams still exist, and the four questions above are the questions to ask about a claim rather than about a design. Chapter 15 reports that claim and what can and cannot be established behind it.

THE CHAPTER IN TEN LINES

- A conceptual application model is a function list with three answers against each function: what it owns, what it must never own, and where it ends. That list survives a change of vendor; the products do not.
- Product boundaries in a provider estate record purchasing sequence, not architecture. Model the functions first, then ask which boundaries between them are load-bearing.
- Identity, terminology and integration strategy are decided before application strategy. The consequence is that identity is a consumed service, every coded value carries its code system and version, and the integration topology is stated rather than inherited.
- The application question is not suite against assembled estate. It is which integration risk you prefer: between products, where it is visible and contractible, or inside one, where it is invisible and the vendor owns it.
- Five tests make a boundary real: it crosses a legal entity, it crosses a regulatory boundary, the two sides must survive each other's failure, they change on different calendars, or one side attests while the other mutates.
- The patient administration and clinical record split, the radiology workflow and viewer split, and the single pharmacy product covering two systems are packaging. The archive boundary, the device boundary and the laboratory boundary are real.
- A code captured to clear a payment edit is not clinical data. An analytics layer built on it measures reimbursement behaviour and reports morbidity.
- Hospitals have an integration engine because of version heterogeneity, dialect translation, fan-out to independently available endpoints, and message-level operational visibility. Not because healthcare lacks standards: it has an abundance, and no enforced constraint.
- Transformations may change representation, not meaning. Anything that changes meaning is a separate, owned, documented component.
- Three functions appear in no business case: exception handling, archive with legacy read-only access, and integration with operational visibility. Each has an owner and a service level or it has neither.
- Model each seam with four facts: whose identity crosses, which system owns state, the terminology and unit bindings, and the actual contract, which is the local interface specification rather than the standard.
- Where results are delivered by a web viewer rather than an interface, acknowledgement tracking, automatic escalation and roster-aware routing are structurally unavailable. Name the residual risk instead of assuming it away.
- Two small deliverables: a one-page function map with an owner per function, and a seam inventory. If an organisation cannot produce both by week six, that is the first finding.

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CHAPTER 15

The application landscape

The real estate, system by system, and the seam where each one meets the record.

A chapter called "who sells what" would miss most of South African healthcare. The country's most consequential health systems are not sold by anyone. The national primary care record is built by the National Department of Health and described in its own tender as bespoke [1]. The electronic record running in most Eastern Cape hospitals names no developer, vendor or contractor in its own public presentation [2]. The flagship private platform is a ten-year internal build on a licensed foundation [3]. The national aggregate reporting platform and the integration mediator in the country's best-documented provincial data centre are open source [4][5]. The national patient index is state-built under contract to a state research council [6].

So the first question in a landscape chapter is not which vendor to buy from. It is which of four routes applies to each capability, and what each route costs in capability the organisation must hold permanently.

Four ways to get a capability

Route	What the South African estate shows	What you must hold permanently	What you give up
Build	e-impilo, HMS², Netcare's CareOn, the Western Cape data centre	A product capability: analysts, developers, testers, a release process and clinical governance of your own	Nothing to a roadmap, and every defect is yours
Buy	The suites, the departmental products, the practice systems	Contract management, a specification you own rather than the supplier, and the evidence that conformance claims are true	Roadmap influence, which for a small buyer in a small market is close to none
Adopt open source	DHIS2 and OpenHIM	A standing engineering team and a relationship with the community that maintains it	A single accountable supplier, and the maintainer may be a small organisation on project funding
Consume a state service	HPRS, SyNCH, the Master Health Facility List, the national laboratory service	An integration, a dependency register entry and a degraded-mode design	All influence over the service's availability, change cadence and scope

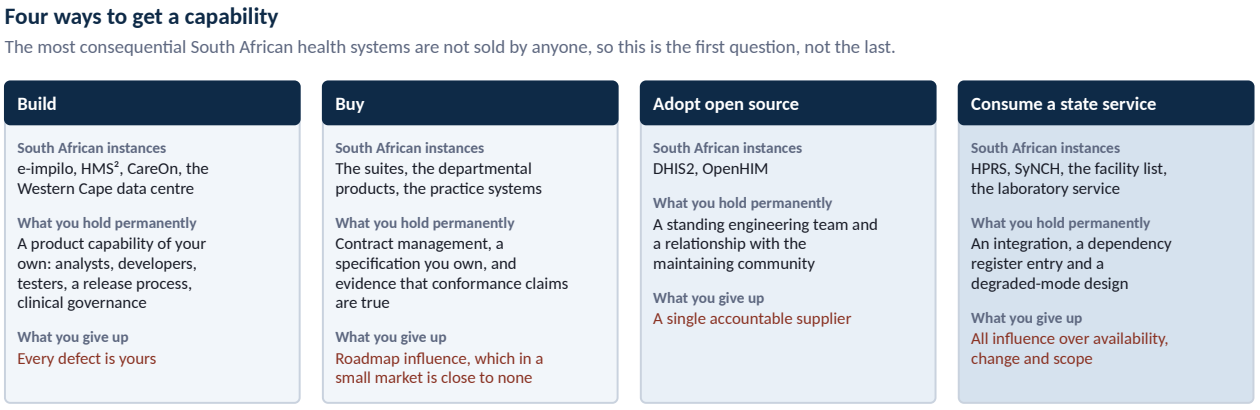


Figure 15.1 The four-way sourcing question. A South African provider uses all four routes at once, and the hard seams are where one route meets another. The fourth branch is the one architects from other industries have no instinct for: there is no contract to negotiate and the only lever is the degraded-mode design on your own side.

Three things about that table matter more than the examples in it.

The four routes are not alternatives at the estate level. A South African provider will use all four at once, and most of the hard seams in this chapter are where one route meets another: a bought hospital system consuming a state identity service, an open-source mediator carrying data from a built record into a bought warehouse.

"Consume a state service" is the branch architects from other industries have no instinct for. There is no contract to negotiate, no service level to agree and no escalation that works on a project timescale. The dependency is real, it is named in the architecture, and the only lever is the degraded-mode design on your own side.

Build is more common here than the textbooks allow, and the reason is capability rather than ideology. The most sourceable build in the country took a decade. Netcare's chief executive set a goal in 2014 of becoming largely paper-light by 2020; the record investigation began in 2017; the board approved the business case in March 2018; the pilot went live at one hospital in September 2019; the system was live at 14 hospitals by June 2022; and phase one, "digitally enabled", was completed in April 2024 [3][7][21]. Any business case promising an enterprise clinical record in three years is being compared against the only documented South African group-scale example, which took ten, with capital, executive sponsorship and a single owner.

Chapter 17 takes the buy branch as a commercial subject: who owns which product, how a licence is metered, what an interface costs and what an exit clause has to say. This chapter describes what the systems are and where they join.

The national primary care record

This is the most important recent development in South African health information, it post-dates most published commentary, and it is not widely understood outside the department.

What it is. e-impilo is the National Department of Health's own electronic medical record for primary health care: departmental, described as modular, linking to existing core registries, and letting a clinician retrieve a patient's care history by a unique identifier [9]. Phase 1, covering tuberculosis and HIV, **went formally live on 28 March 2025**, after beta testing at **36 primary health care facilities and one general practitioner practice** across the Free State, KwaZulu-Natal, Northern Cape, Mpumalanga and North West between October 2024 and February 2025, with scale-up expected from January 2026 [6]. As at November 2025 it was **in use at 55 facilities, with about 3,300 more planned over the following two years** [9]. A departmental annual report describes the same first phase as deployed across **37 pilot facilities** in March 2025, tested in Dihlabeng in the Free State, with a Single Patient Viewer scoping exercise completed as a forerunner to the longitudinal patient record [10]. The 36-plus-one and the 37 are the same programme counted differently; cite whichever you use and say which.

The name is not stable, and that is an architecture problem before it is an editorial one. The same programme appears as e-impilo in a donor publication [9], e-Mpilo in a parliamentary committee record [11], and "the Electronic Medical Record" in a parliamentary reply [6] and in two tender documents [1][12]. An academic evaluation refers to an "Impilo Platform"; that paper was not retrieved and it cannot be confirmed to concern the same system, so it is unverified and should be checked against the department before use. An architect writing a current-state catalogue for a South African public-sector client must record every variant name against one asset, or the catalogue will double-count the most important system in the country.

The rollout, and the two facility counts. Two tenders are on the record. The first, bid NDoH-04 of 2025/2026, invited non-governmental and non-profit service providers — not systems integrators — to roll the record out across fixed primary health care facilities in **eight provinces within 18 months**, excluding the Western Cape, with awards made separately per province; the compulsory briefing was 23 February 2026 and the closing date 9 March 2026 [12][13]. Its stated per-province counts are Eastern Cape 779, KwaZulu-Natal 608, Limpopo 476, Gauteng 372, North West 312, Mpumalanga 289, Free State 216 and Northern Cape 161. **The document states 3,083 facilities in total; its own provincial figures sum to 3,213** [12]. The discrepancy is in the source. Cite both. The second tender, NDoH 11/2026-27, was issued 19 June 2026 and closed 13 July 2026, covering rollout to **about 3,083 registered fixed primary health care facilities** in the same eight provinces over 18 months, with at least two days per facility for training and go-live, a post-implementation visit and a follow-up about a month later, priced per facility per province in rand, and **estimated to exceed R50 million** [1].

Three sentences in the second tender that an architect should read twice, all from the same document. The record "is already interoperable with other existing health information systems." "No integration or interoperability work is needed or included in the scope of successful bidders." And it "complies with the HNSF and is also compliant with the Draft Data Governance Framework" [1]. Those are claims to report, not to endorse. They name no system, no implementation guide, no message or resource version and no conformance attestation. The national framework does contemplate conformance testing at a departmental approved laboratory with an attestation issued [14], which makes the compliance claim testable — ask for the attestation. Whether an approved laboratory is operating could not be established in the research behind this book, and that matters, because the framework's prescriptive force rests on it. Chapter 16 takes the standards question.

And none of the system's technical architecture is public. Not its hosting, not its data residency, not its standards conformance beyond the framework-level claim, not who built it. The second tender excludes network procurement from scope and the first includes "troubleshooting of computer and network issues before training" [1][12]. Read the second as a statement about the binding constraint and the first as a statement about the condition of the estate. Say the absence plainly in any assessment rather than inferring an architecture from the function list.

Two further facts from the same documents are worth carrying because they are design constraints rather than commentary. Bidders "will have no access to any Personally Identifiable Information", which implies synthetic training data, strict environment separation and support tooling that masks content [1]. And "all data captured on the EMR digital solution belongs to the patient" [1] — an unusual formulation that is not the same as saying the data belongs to the department, the province or the facility, and that does not resolve the access, portability and research questions it raises. Do not treat it as a settled legal position.

The four mandatory integrations

The department has stated the integration map itself, which is rare and valuable. The record must integrate with the **Health Patient Registration System**, the **laboratory information system**, **SyNCH** and the **Master Health Facility List** [12]. That is identity, results, chronic medicines and location — and it is the architecture of South African public primary care, stated by its owner rather than inferred.

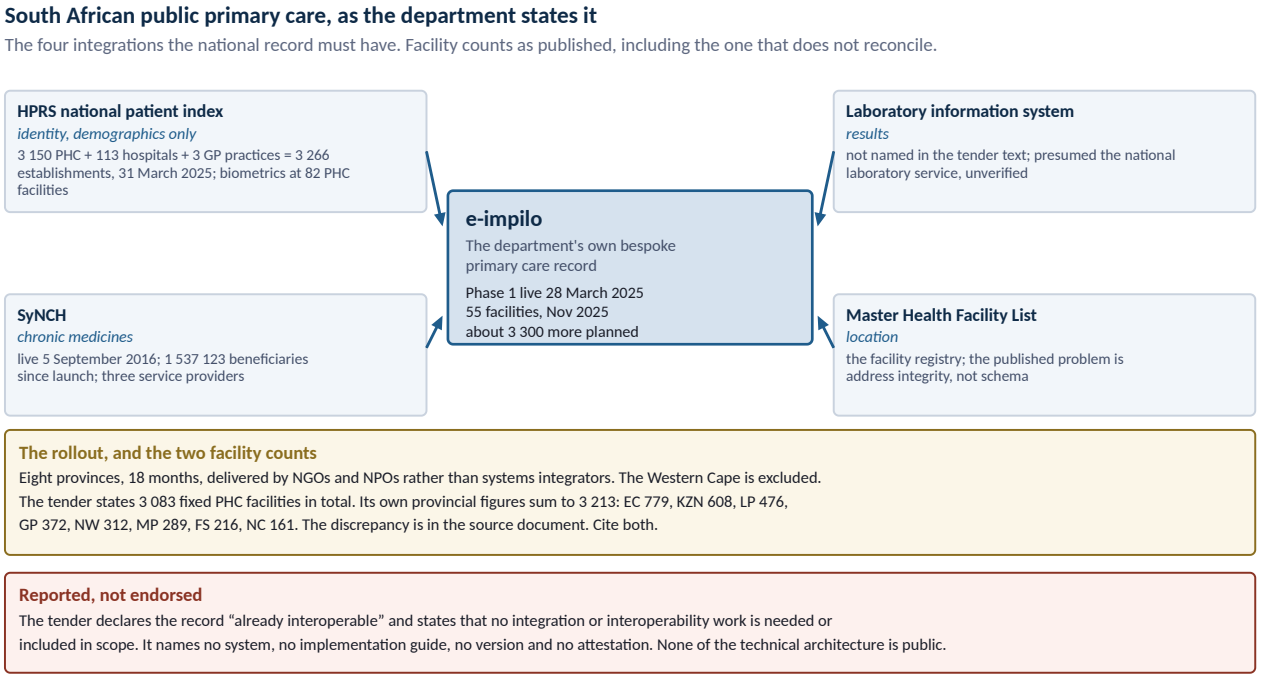


Figure 15.2 The architecture of South African public primary care, stated by its owner rather than inferred: the national bespoke record and the four national shared services it must consume. Both published facility counts appear because the source document contains both. The interoperability declaration is reported as a claim; the conformance attestation the national framework contemplates is the thing to ask for.

What sits behind each of the four:

- **HPRS** is the national patient index: departmental, developed under contract with the Council for Scientific and Industrial Research, using the South African identity number with passports and other documents as alternatives, and **holding demographics only** [4][39]. At 31 March 2025 it was installed at **3,266 health establishments — 3,150 fixed primary health care facilities, 113 hospitals and three general practitioner practices** — with the Automated Biometric Identification System in **82 primary health care facilities** [10]. A 2023 study reports at least 44 million people registered [4]; no current national registration total could be sourced. Biometrics at 82 of 3,266 is a pilot, not a capability. A committee record notes that the research council "recovered the HPRS data and hosts a secondary site" [11], which is a short sentence worth pausing on: the national index has required recovery, and every downstream system depends on it.
- **The laboratory information system** is not named in the tender text reviewed. It is presumably the National Health Laboratory Service's, and that inference is unverified; the departmental enquiry is worth making, because the identity and vintage of that interface determine what the results seam can carry.
- **SyNCH** digitises the central chronic medicine dispensing and distribution programme end to end: online registration against an identity number, passport or asylum-seeker number, selection of an approved pick-up point, electronic prescriptions, transmission to dispensing and distribution providers, scanning of issued parcels and automated reporting. It started on 5 September 2016 and reports 1,537,123 beneficiaries since launch and three service providers [15]. Its unit of work is a parcel to a pick-up point against an enrolment and a prescription validity period, which is not a dispense against a prescription, and a model that treats it as one loses three entities.
- **The Master Health Facility List** is the facility registry. The departmental account of it covers training to reduce support tickets, its use in the Free State's 2021 vaccine rollout, and the integrity of its address data [8] — which is the usual shape of a facility register problem: not the schema, the addresses.

Four tiers, not a layer

It is common to describe the South African public estate as provincial systems with an HPRS, DHIS2 and Stock Visibility System layer above them. That is a mis-description, and it obscures the distinction most worth teaching. HPRS is an identity service **beneath** the clinical systems, not a record above them: it captures demographics only [4]. The national aggregate platform is **downstream** of the clinical systems. The Stock Visibility System is an electronic stock management application reporting medicine availability from clinics and hospitals [39] — a **parallel** logistics channel that never touches the clinical record. Three systems in three architectural positions.

Four tiers describe it properly.

Tier	What it holds	South African instances
Identity	The person and the existence of a visit, with no clinical content	HPRS; provincial patient master indexes; the Eastern Cape index with its own patient reference number allocation; biometric verification against the civil register
Clinical record	The clinical assertion, patient by patient	e-impilo nationally; HMS² in the Eastern Cape and Free State; Clinicom, PHCIS and PREHMIS in the Western Cape; the resource-planning estate in Gauteng; the KwaZulu-Natal hospital system
Reporting and surveillance	Aggregates, programme registers and notifications, downstream of care	webDHIS; TIER.Net; the notifiable medical conditions system; SINJANI in KwaZulu-Natal; the Western Cape provincial health data centre
Logistics and support	Commodities, parcels, facilities, people and money	Stock Visibility System; the National Surveillance Centre; RxSolution, MEDSAS and Unisolv; SyNCH and the chronic medicines programme; the payroll system

Where the three national systems actually sit

Identity is beneath the clinical systems, reporting is downstream of them, and logistics runs in parallel and never touches the record.

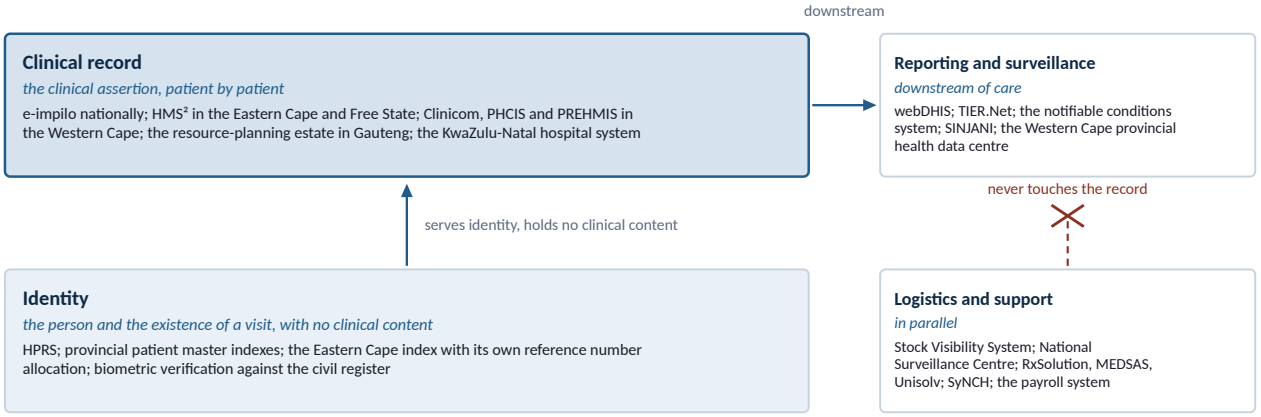


Figure 15.3 The three national systems most often described as one layer above the provincial estate sit in three different architectural positions. The patient index is beneath the clinical systems and holds demographics only; the aggregate reporting platform is downstream of them; the stock system is a parallel logistics channel. Calling them a layer obscures the distinction most worth teaching.

Two of those instances need a caution. **SINJANI**, the KwaZulu-Natal provincial reporting system, could not be sourced to any primary technical description: not its data model, not its relationship to the national reporting platform, not its supplier, not its vintage. The national strategy and the gazetted framework do not mention it at all. An architect working in that province must obtain the specification from the provincial department, and the absence of public documentation is itself a finding about the estate. And in the logistics tier, **RxSolution** is run by the National Department of Health, was rolled out incrementally at hospitals from 2014, is personal-computer based on an in-house server, and is now used across the country's facilities, alongside MEDSAS at depot level and Unisolv in some facilities; by October 2021 some 3,826 facilities had used the Stock Visibility System, RxSolution or another online system, which one estimate puts at roughly 90% of public hospitals and clinics [40].

The tier model earns its place because it predicts the failure modes. A system in the identity tier that starts holding clinical content becomes a record with no author. A system in the reporting tier asked a clinical question gives a wrong answer confidently, because there is no person in an aggregate model. A logistics system treated as a medication record reports what left a shelf as though it were what a patient took. And the proliferation of programme registers in the third tier is not untidiness: because the national routine system held aggregates, every programme that had to follow individuals built its own patient-level register, and because the programmes are vertically funded the registers do not interoperate [4][16]. The department's own material names TIER.Net, the electronic drug-resistant tuberculosis register and the notifiable conditions system among "fragmented, non-interoperable" systems [9]. The fix is not rationalisation by decree. It is to supply the missing capability — a patient-level record consuming a national identity service — after which the registers can become views or retire on their own timetable.

The provincial estate, and its vintage

The most authoritative single inventory is a parliamentary reply, which carries accountability a vendor page does not. As at 1 August 2025 [6]:

Province	Hospital systems on the record
Eastern Cape	HMS² in at least 44 hospitals, at various stages
Free State	Meditech in 3 hospitals; HMS² in 16 hospitals since March 2023
Gauteng	A resource-planning-based hospital information system
KwaZulu-Natal	An "eHealth system" in 42 public hospitals since March 2020; no product or vendor name could be established
Western Cape	Clinicom in 53 hospitals, PHCIS in 308 primary care facilities, PREHMIS in City of Cape Town primary care facilities, and a newly started Electronic Continuity of Care Record
Mpumalanga, North West, Limpopo, Northern Cape	Recorded as not currently implementing patient information or record systems at hospitals

Two corroborating figures and one that does not reconcile. A 2023 study puts the Western Cape at 54 of 55 hospitals with an electronic record and **Gauteng at 2 of 37**, with five of nine provinces having some operational record in public hospitals [4]. The Western Cape figures are consistent with the parliamentary reply. The Gauteng pair is not: 2 of 37 against "a resource-planning-based hospital information system in hospitals". The likely explanation is that the resource-planning estate is administrative rather than clinical, and that is an inference rather than a finding. Say so if you use it.

On vintage, the honest position is narrow. **Clinicom** is the Western Cape's hospital patient administration system and the core source of the provincial patient master index, built on the Caché hierarchical database, covering both hospital and primary care patients [5]. **PHCIS** was developed in-house by the province for primary care registration, assessments and visit tracking [4]. Installation dates and current version levels for Clinicom, Medicom, PHCIS and Meditech deployments could not be sourced for this book, in any province. What can be said is that the technology base and the nine-digit folder number with a modulus-10 check digit place Clinicom in an earlier generation [5], and that the provincial data centre **queries it daily** rather than consuming an interface [5]. Treat provincial hospital administration systems as long-lived systems of record with limited interface surfaces; design integration around scheduled extraction where that is what exists; and obtain version and date detail from the provincial department rather than assuming a modern application programming interface.

Two further facts set the operating envelope for anything built on this estate. Against the Ideal Clinic assessment for 2024/25, covering **3,333 primary health care facilities**, **52 (1,5%) lack functional computers, 173 (5,2%) lack internet access and 278 (8,3%) lack functional fixed-line phones**, with all public hospitals reported as having computers, internet and telephones [6]. And a participant in peer-reviewed research put the proportion of public facilities without a reliable internet connection at **nearly 40%** [4]. Those two are not contradictory — one measures presence, the other reliability — and the gap between them is the most important number in South African health architecture. Design for intermittency, not for absence. Designing for absence produces a paper fallback; designing for intermittency produces offline capability, and the conflict resolution is where the duplicates come from. The Free State told a committee that HPRS **allocates offline during power outages and produces duplicate entries when power returns**, remediated by regular central database clean-ups [11]. Chapter 11 builds on that.

HMS², the record with no vendor

HMS² is described in its own public material as an electronic medical record developed in South Africa for South African hospitals, an open platform following the national normative standards framework, part of a wider provincial ecosystem — and the only team it names is "the ICT Team" [2]. No developer, software vendor or contractor appears. Frere Hospital is the named originating site, with releases 1.0 in 2013, 1.1 in 2014/2015, 1.2 in 2016, 1.3 in 2017 and 1.4 in 2018 [2].

Its module list is unusually complete for a public-sector system and worth reading as a statement of what a provincial record actually covers: patient registration, outpatient and inpatient management, theatre, documents, orthopaedics, wheelchair management, billing, admission-discharge-transfer, clinical discharges and electronic registers; electronic requests for scripts, radiology, allied health referrals and beds; a clinical kiosk, patient document access, bed management, radiology order entry and electronic capital asset requests; and a corporate set covering performance management, human resources, employee verification and dashboards [2].

The integration inventory is the most valuable part for an architect, because it is the only published seam list for a South African provincial record: HPRS, a master patient index with patient reference number allocation, an integration component, the national laboratory service, the picture archive, the aggregate reporting system, a named integration component, the pharmacy systems, digital radiology, and medical aid and debtor systems [2]. Its stated standards position is the normative framework with patient-identifier cross-reference and demographics query profiles, a messaging engine with web services, ICD-10 for diagnosis and the Uniform Patient Fee Schedule for procedures, with role-based user management, audit trails and encryption [2].

Two cautions. The Eastern Cape told a committee in June 2025 that HMS² was in **40 hospitals (phase 1)** [11], against at least 44 in the parliamentary reply [6] — two counts within two months, and both should be cited. And the published submission carries a cost and savings table whose **units are not stated**: development 500,000, implementation 80,000, licensing and hosting 100,000, operational savings to date 1,040,000 [17]. If those are rand, a hospital record developed for R500,000 is implausible against every comparator in this book. **Do not quote the figures until the units are resolved.**

The Western Cape, briefly

The province runs its own estate and is excluded from the national rollout, which means South Africa is deliberately operating two primary care record architectures federated at the identity layer. Chapter 3 carried that structure. The landscape facts are that Clinicom provides the provincial patient master index across its hospitals, PHCIS covers provincial primary care, the City of Cape Town runs PREHMIS, a continuity-of-care record has recently started [6], and the province has sent **daily updates from its patient master index to the national index since June 2020** [18]. Above the transactional estate sits the Provincial Health Data Centre, which draws on Clinicom as its core source, the national laboratory service for pathology, pharmacy and dispensing data, bespoke programme registers, community and household data and an antenatal messaging programme, and which links more than twenty systems [4][5]. It uses an open-source integration mediator, OpenHIM, developed by a Cape Town non-profit, to load data into holding databases [5]. Chapter 22 reads the data centre as a worked reference design; here it is enough to place it in the reporting tier and to note that it holds report metadata while images and formatted reports stay on the source systems [5] — a deliberate scope decision that keeps the platform small enough to operate.

The private estate

CareOn, on iMedOne

CareOn is Netcare's electronic medical record, covering its hospitals, its mental health business and its emergency service [3]. The single most misattributed fact in South African private healthcare is what it runs on. **The core platform underneath CareOn is iMedOne, from Deutsche Telekom Clinical Solutions — not Cerner and not Epic** — with the supplier's teams in South Africa, Germany and India refining, customising and developing it, and with CareOn adding medical device integration, digital scripting, drug interaction and safety content, and resource-planning integrations built by Netcare's own teams and partners. CareOn data is held in Netcare's own data centre [7]. The product is current and actively marketed by the supplier's parent, with English and German product material dated April 2026 [19].

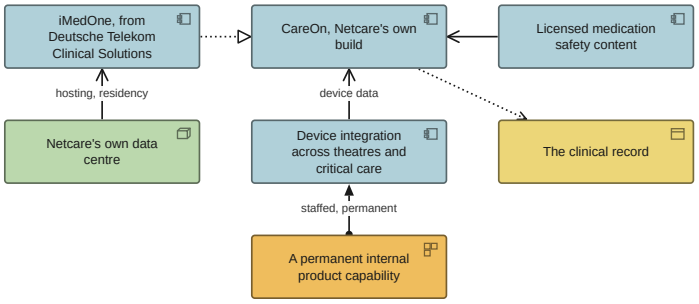


Figure 15.4 The best-documented South African private clinical platform, drawn as what it is: a ten-year internal build on a licensed foundation, with a licensed content dependency, device integration in scope from the start, and data held in the group's own facility. It is not Cerner and it is not Epic, and the organisation that bought the foundation became a development organisation.

Scale, as reported. About 34,000 users across seven South African provinces, more than 13,000 tablet devices in 45 hospitals, about 1,8 million in-hospital scripts a year, and development involving 40 workstreams across three countries over seven years [20]. Netcare's own report calls the programme "the highest risk project ever undertaken at Netcare", states that the rollout required integrating all medical equipment into the record, and names the linking of radiology and pathology requests and reports [3].

Five readings for an architect, and only the first is about software.

It is a build on a platform, not a product purchase. The organisation bought a foundation and became a development organisation, which needs a permanent product capability — and that is an organisational design consequence usually absent from the business case.

Device integration was in scope from the start, not deferred to a later phase, which is the opposite of the usual sequencing. It is also the clearest South African statement of what device integration involves at group scale, and it came from a hospital group rather than a supplier.

The medication safety content is a licensed third-party dependency, named in Netcare's own material by its pre-2022 owner's brand [7][21]. The content has a renewal date, it keys on its own medicine identifiers rather than yours, and its redistribution to affiliated practices is very likely outside the licence. Chapter 17 takes the contract; the architectural point is that a safety function with a renewal date must fail loudly rather than silently serve stale content.

Data residency was decided deliberately, in a market where most suppliers would prefer to host.

And the hardest part was not technical. Netcare ran change programmes for nurses with little prior computer experience and involved doctors in design, tracking adoption to resolve problems quickly [3]. The group's own phasing is explicit about what follows a record: phase two, data-driven clinical efficiency, started in financial year 2024 and runs two to three years, including analytics and a funder portal; phase three, person-centred care, runs three to four years from the same start and overlaps phase two [3]. The record is the beginning of the programme, not the end of it.

The rest of the group's named estate is useful because it shows what a private group's landscape looks like once the record is done: a separate oncology system, a renal system, a primary care system, an occupational health system, and a patient-facing application with phased improvements planned to financial year 2027 [3]. Five clinical platforms, one group, each with a seam to the record.

What cannot be established, and why saying so matters

Mediclinic Southern Africa's clinical platform could not be established from any retrievable source. The case material on a supplier's South African web domain describes a 182-bed facility in Dubai and a Middle East deployment programme, which is a different business from Southern Africa; conflating the two would be an error [22]. **No enterprise clinical record for Life Healthcare could be established either.** What is on the record is its chief information officer describing it as one of the first hospital groups in Africa to implement a general-purpose customer relationship management platform's health product, used for dialysis patient care [7]. That is a single-domain deployment rather than an enterprise record, and it is architecturally more interesting than it looks: chronic disease management is a relationship and scheduling problem with a clinical overlay, and the seam is the clinical one — a renal programme in a relationship platform needs laboratory results, dialysis session data and medication from systems it does not own.

Smaller groups show a different shape. Busamed consolidated several on-premises data centres into a hosted private cloud, and chose private cloud **partly because many suppliers billed in United States dollars, which made hosting costs hard to forecast against rand volatility** [7]. That is a legitimate, documented reason a South African organisation picks an architecture a European reference model would call retrograde, and it belongs in any option appraisal written in rand. Lenmed, the independent hospital network grouping, Clinix, Joint Medical Holdings and the day-clinic operators could not be established in the research behind this book.

The resource-planning estate, and the Gauteng question

A distinctively South African pattern, poorly understood outside the country. A supplier states that it had launched enterprise resource planning healthcare solutions at over 33 hospitals across South Africa by the end of 2019, public and private, naming a stack of finance, materials, plant maintenance, billing and medical aid claims; human resources, payroll and shift planning; an industry solution for healthcare covering patient management and patient accounting; and a clinical information system integrated with both. The same article gives two counts — over 33 hospitals in total and over 20 for the industry solution specifically — and names no hospital groups. **It was issued on the supplier's behalf, so all of it is the supplier's own claim**, and the widely repeated "fifteen of the top twenty hospitals" figure in it has no independent corroboration [23]. The independent corroboration that does exist is one line in a parliamentary reply: Gauteng runs a resource-planning-based hospital information system in hospitals [6].

The architectural consequence is the one chapter 14 names as a real but inverted dependency. Where patient administration and patient accounting live inside the resource-planning platform, the master patient index is a resource-planning object and the clinical system hangs off it, so every integration

passes through that platform's transaction model, release cycle and change control. It also explains why Netcare's own built record includes resource-planning integrations [7]: even where a group builds its clinical platform, the administrative and financial spine is likely to be the same product family.

The practice layer, and the switch

Most private clinical encounters in South Africa are recorded in a practice system, and this is the least documented layer in the country. There is no reliable public market-share data for South African practice management software, and that absence is itself a finding: a supplier's share claim here cannot be checked.

What can be established is the shape. A South African private practice stack is assembled rather than bought, from six parts: a practice management system holding the demographic, the appointment and the account; a clinical note capability that is frequently thinner than the billing capability, because billing is what the practice pays for; one or more switch connections for eligibility, authorisation, funder claim submission and remittance; a reference data subscription for tariffs, scheme rules, medicine and coding files; inbound result feeds from pathology and radiology, often point to point; and increasingly a bolt-on for scheduling, telehealth, patient engagement or documentation. Named examples an architect will meet, each from the supplier's own material, include Altron HealthTech's practice management, switching and engagement products [24]; GoodX, a South African developer of practice and hospital management software founded in 1985 [25]; Healthbridge, founded in 1999, with practice, billing, benefit-checking and hospital products [26]; and Eminance, from Health Focus, whose published partner list names practice funding, cash-flow management, pathology report linking, two claims switches and a reference data supplier as separate integrations [27]. **Every one of those six parts is a separate contract with a separate renewal and a separate data position.**

The architectural conclusion follows directly. **You do not integrate with "the practice system". You integrate with the switch**, because the switch is the only component the practice cannot replace cheaply.

A South African healthcare switch is a message broker and adjudication gateway between providers and funders, and it is the most load-bearing and least discussed component of the private estate. The largest describes its own function with unusual precision: real-time, end-to-end adjudication through one specification covering eligibility checks, membership and authorisation rule downloads, claims, reversals, resubmissions and re-sends; published schema definitions that software suppliers validate their requests and responses against; connections to pharmacies, hospitals, doctors, specialists and funders; **either batch or real-time switching depending on each funder's preference**; a vendor test environment available continuously that mirrors live funder rules and pricing with dummy beneficiary data; and an integration pack with a live-environment accreditation step [28].

Four consequences. **The switch owns the integration contract, not you**: it certifies the practice system against its own specification, versioned on its schedule and tested in its environment, so model it as an external standard with an external owner. **The test environment is the asset** — a continuously available environment mirroring live rules is the only thing that settles an interoperability claim, and here is a South African example of one existing. **Batch against real-time is a funder-by-funder variable**, not a platform setting, so the provider architecture must carry both concurrently and reconcile a mixture. And **the national medicine coding system used for private claiming is owned and maintained by a switch** [29], which makes every medicines service, interaction check, formulary and analytic in the private sector dependent on a licensed asset held by a commercial party whose primary business is adjudicating funder claims. Establish the licence terms for use, redistribution and derived works before designing a medicines service. Chapter 12 carries the coding consequences; chapter 17 carries the contract.

Pathology and radiology

The public sector is one organisation and one system, and that is the single most important fact in South African clinical informatics. The National Health Laboratory Service is the near-monopoly provider of diagnostic pathology to state hospitals and clinics, processing more than 140 million tests a year and funded by billing provinces rather than by direct appropriation [30]. Its laboratory system of record is InterSystems TrakCare [31], and it runs a separate data warehouse [32]. Separate the supplier as platform from the supplier as application here: the same company sells an integration and data platform family and a laboratory and hospital application, and they are different procurement conversations. The concentration is a landscape fact with consequences chapter 19 owns: one organisation's information systems mediate pathology for most of the public estate, and no individual hospital architecture can mitigate that, so the dependency has to be declared in every provincial and facility architecture.

Delivery to clinicians in the public sector is substantially by **web results viewer** rather than by interface into a clinical system [33]. That is a reasonable delivery channel and a poor safety channel, for the reasons chapter 14 gives.

The private pathology market is three groups — Lancet, Ampath and PathCare — and **no laboratory information system could be established for any of them** in the research behind this book. That is a significant gap, because between them they process the private sector's pathology and their result-delivery interfaces are the ones every private practice and private hospital consumes. It has to be obtained from the groups directly.

In radiology, the product an architect is most likely to meet in South African private practice is eRAD's radiology information system, selected by a Cape Town group described by the supplier as 32 radiologists across 11 centres, announced 28 November 2017, with integration to the group's archive and billing system through the supplier's own messaging engine [34]. Note the shape rather than the name: a local practice's core operational system supplied by a subsidiary of a foreign imaging business, which means roadmap influence near zero and makes version-support and data-extraction terms the only real protection. Chapter 17 has the ownership ledger.

CareConnect

South Africa has one private health information exchange, and its composition is more instructive than its technology. CareConnect is a not-for-profit exchange founded in 2018 by six organisations — three hospital groups and three scheme administrators — to carry authorised information between hospitals, clinicians and funders [35]. **Three providers and three funders in one governance body, in a market where provider and funder interests are structurally opposed**, is the architecturally decisive fact. It is built on InterSystems HealthShare, came online in late 2021 and was officially unveiled on 7 April 2022 [36].

The design positions that are published are good ones. Clinicians see consolidated clinical information across a patient's care; administrative staff see demographic data and not clinical data; access requires patient consent, is limited to what is medically necessary, and **consent can be revoked at any time**; and the stated expansion direction is radiology practices, pathology laboratories and clinicians' practice management systems [36]. Revocable patient-mediated consent is the right position under South African privacy law and it creates a hard technical requirement: a revocation must propagate to every participant and be enforceable on a query already in flight. Ask how.

Two honest gaps. **No health-specific interoperability standard is named** in its public material — no messaging standard, no resource specification, no integration profile — and the compliance list is information-security and privacy standards instead [36]. The underlying platform supports the health standards natively; the public material does not say which are used, so ask, and ask for the implementation guide. And the current participant count, record volume and query volume could not be established. The founding six are documented; nothing beyond that is.

For an architect the lesson is about sequence. There was no private exchange in South Africa before 2021, and the reason was not that the platform did not exist. Study the governance before the architecture.

The departmental systems, and where each meets the record

Every one of these exists because a clinical department needed depth the enterprise record did not provide, and every one creates a boundary across which patient identity, order identity, result identity and clinical responsibility must pass without loss. The failure modes are overwhelmingly identity failures and state failures at these boundaries, not transport failures.

System	What it owns	Where it meets the record, and what breaks
Laboratory	Specimen, accession, worklist, autoverification, authorisation, report	Order identity: placer and filler numbers both must survive, and a reflex or add-on test produces a result with no order in the placer. Units and reference ranges. Corrected results that overwrite rather than supersede
Radiology workflow and archive	Study identity, reading workflow, the report as a document, long-term storage	Three identifiers and three systems diverge on every unscheduled study, portable film and imported disc. Importing an outside study means rewriting identifiers inside image headers, and doing it badly puts another patient's images in this patient's record
Digital pathology	The slide image, the case, the block and the stain	It sits between the laboratory system and the archive and is usually integrated with neither properly
Cardiology	The waveform, the study, the measurement set	The electrocardiogram is simultaneously a waveform, an image, a measurement set and a report, and most records store one of the four
Ophthalmology	Device-acquired images and measurements	Devices frequently emit proprietary formats rather than standard objects, so the archive seam is the problem
Oncology	The regimen instance: protocol, cycle, day, calculated dose	Dosing is a calculation with a release gate on blood results. The calculator is regulated clinical function, not infrastructure
Theatre	The list, the case, the operative record, the implant log	Implant traceability for the life of the device, the procedure code's authoritative source, and the cancellation sequence that reporting depends on
Anaesthesia and critical care	The high-frequency record, ventilation, infusions, scoring	Device gateway: sampling policy, who validates machine-sourced values before they become the legal record, and what the record looks like when the gateway is down
Pharmacy and automation	Stock by location and lot, expiry, the controlled-substance register, dispensing	A cabinet knows what was removed, not what was given. Treating removal as administration is a common and wrong shortcut
Emergency	Attendance, triage, tracking state	The unidentified patient. Merging a temporary identity into a real one, carrying imaging, results, medicines and transfusion records, is the hardest routine operation in a clinical estate
Telehealth	The remote encounter	Identity of an unverified patient, and whether the encounter lands in the record at all
Patient flow and command centre	A read-only view of every other system's state	It is a consumer of state it does not own, so its accuracy is the accuracy of the worst feed
Rostering	Establishment, roster, leave, cover	Access and routing both depend on assignment, so without a roster feed access degrades to all clinicians seeing everything
Revenue cycle	Invoice, funder claim, authorisation, remittance	The encounter to funder claim boundary, where the clinician records what happened, the coder what is codeable, the biller what is billable and the funder pays what is payable. Four states of one event
Decision support content	Interaction, dose-range, allergy and duplicate-therapy content	Licensed content keyed on the supplier's own medicine identifiers; the mapping to yours is a maintained artefact or a silent gap
Terminology service	Code systems, value sets, maps, versions	Nothing in South Africa mandates a clinical terminology, so the binding is an organisational decision that must survive a later national one
Patient engagement	Booking, reminders, messaging, results release, record access	Identity proofing, proxy access with age transitions, and result release policy, which has no technically correct answer

One South African piece of evidence belongs under the last row because it contradicts an assumption almost every business case makes. A prospective cohort study at a Cape Town tertiary hospital clinic enrolled 150 participants to use a United Kingdom patient-held record platform and compared them with 49 United Kingdom users of the same platform. **42% of the South African participants registered, 26% logged in and 20% completed the usability questionnaire, against 96% of the United Kingdom participants registering and logging in.** The authors conclude that implementing the platform in the current South African public environment is not feasible without adaptations, and recommend allowing access without an email address, integrating with existing infrastructure and streamlining hospital digital notes into the record [37]. The requirement to work without an email address never appears on a feature matrix and determines whether the product works at all.

Six categories a landscape chapter usually omits



Figure 15.5 The six categories absent from most application-landscape work. Each one creates a seam the record depends on, and each is routinely discovered after a programme has been scoped.

Medical device integration and the device gateway. A distinct capability with its own suppliers, its own failure modes and a biomedical engineering owner rather than an information technology one. Netcare's own report states that the record rollout required integrating all medical equipment [3], which is the clearest available South African statement of the scope. Three questions follow for any design: the sampling and retention policy, because storing every second of every anaesthetic forever is a capital decision; who validates a machine-sourced value before it becomes part of the legal record; and what the record looks like when the gateway is down, because a gap in an anaesthetic record is a medico-legal exposure.

The integration engine itself. A product category with named suppliers, and in a South African context the only part of the estate the architect usually still controls. Chapter 14 says why it exists and what it must never own.

Identity and the master patient index as a product category, separate from the record. HPRS, the Western Cape provincial index, the Eastern Cape index with its own reference number allocation and the departmental master household index described as resource-specification-based for linking patient data across systems are all identity products [2][5][6][8]. The household index is potentially the most significant undocumented deployment of a modern health interchange specification in South African government, and only a chapter description of it was retrievable; treat it as unverified and pursue it with the department.

Archiving, retention and legacy read-only access. A statutory retention obligation that outlives the system which created the data is a product category and a cost line, and it is the reason the real question is never the cost of a second instance but the cost of the instance set: production, training with realistic data, functional test, integration test, disaster recovery, reporting and read-only legacy for the retention period, each with licence, hosting, support and a recurring de-identified refresh. Chapter 13 owns the retention duty; chapter 17 owns the pricing.

Consent management. Deserves its own treatment for two reasons. The private exchange's whole proposition rests on a revocable patient consent [35]. And consent is routinely modelled as a single attributeless row, which is a placeholder for an argument nobody has had: consent to treatment, consent to disclosure and consent to research are three objects with three lifecycles, and third- party disclosure under the National Health Act requires the user's consent in writing.

Ambient documentation and the clinical overlay. The fastest-moving category of 2025 and 2026 and the one with the least settled contracting pattern. Evidence of scale exists in other markets — a regional health service procuring ambient voice technology across 1,239 general practices and 15 trusts [38] — and at least one of those products appears in the navigation of a South African practice system supplier's site, which indicates distribution into the local private market on terms this research could not establish [27]. Four reasons an architect must be in the room early. It processes the consultation, which is the richest and most sensitive data the organisation holds, and it processes it before anything has been entered into the record, so lawful basis, consent, retention and processing location all have to be established for audio and transcript as well as for the record. It is frequently bought by clinicians departmentally or individually on a per-user subscription, so it arrives without an architecture review or a processing agreement and the first task is discovery rather than design. The output is a draft a clinician signs, so responsibility stays with the clinician while the error mode moves into the software — and an omission in a drafted note, a negative finding not recorded or an allergy not carried forward, is a different class of defect from a transcription error, so the governance has to be designed for omission. And the seam is write access: a product that writes a note is a record-integrated product with every identity, authorship and audit obligation that implies, while a product that produces text a clinician copies and pastes is not integrated at all and the copy-paste is a safety hazard. Establish which you are buying.

What to build in the first six weeks

The deliverable that pays for itself fastest in this landscape is not a target state. It is a catalogue, and it is cheap.

- Every system, with **every variant of its name** against one asset.
- Its tier: identity, clinical record, reporting and surveillance, or logistics and support.
- Its sourcing route: built, bought, open source, or a state service consumed.
- Its owner, its contract end date and the funding source of its **running** cost, because a grant-funded component has a scheduled failure date. Donor money is real and discontinuous: the Eastern Cape told a committee that about **R435 million of donor funding issued in January 2025 was expected**

to be withdrawn in September 2025 [11], and the accelerator supporting the national record's interoperability work is tied to a 2023 to 2025 grant cycle [9]. Running costs on two- and three-year cycles, systems with ten- to twenty-year lifetimes.

- Its interface inventory, and for each interface the contract and its version.
- Its extraction position: what you can get out, in what format, at what notice.

Most of the systems in a provider estate were bought by departments, at different times, from different budgets, on different terms, and nobody holds the aggregate. Building that aggregate is frequently the highest-value thing an architect does in the first six months. Chapter 16 takes the standards that govern each of these seams; chapter 17 takes the suppliers as businesses.

THE CHAPTER IN TEN LINES

- Ask the four-way sourcing question before the vendor question: build, buy, adopt open source, or consume a state service. The most consequential South African health systems are not sold by anyone.
- Each route leaves you holding something permanently. Build leaves a product capability; buy leaves contract management and a specification you must own; open source leaves a standing engineering team; a state service leaves an integration and a dependency you cannot influence.
- e-impilo is the department's own bespoke primary care record. Phase 1 went live 28 March 2025 after beta at 36 facilities and one general practice; it was at 55 facilities in November 2025 with about 3,300 more planned.
- The rollout tender covers eight provinces in 18 months, excludes the Western Cape, is delivered by non-governmental and non-profit organisations rather than systems integrators, and states 3,083 facilities while its own provincial figures sum to 3,213. Cite both.
- The tender declares the record already interoperable and no integration work needed or in scope. Report that as a claim and ask for the conformance attestation the national framework contemplates. None of the system's technical architecture is public.
- The four mandatory integrations are identity, results, chronic medicines and location: the patient registration system, the laboratory system, SyNCH and the Master Health Facility List.
- Four tiers, not a layer: identity beneath the clinical systems, the clinical record, reporting and surveillance downstream, and logistics and support in parallel. The patient index holds demographics only.
- The provincial estate is uneven by design of nobody: four provinces are recorded as not running hospital patient information systems at all, and vintage for the long-lived provincial systems could not be sourced in any province.
- Design for intermittency rather than absence. Against a 2024/25 assessment of 3,333 primary care facilities, 5,2% lacked internet access, while peer-reviewed research puts unreliable connectivity at nearly 40%.
- Netcare's CareOn is a ten-year internal build on iMedOne from Deutsche Telekom Clinical Solutions. It is not Cerner and it is not Epic, and this is the most misattributed fact in South African private healthcare.
- Do not integrate with the practice system. Integrate with the switch, because the switch is the only component a practice cannot replace cheaply, and the switch owns the integration specification.
- CareConnect's hard part is its governance, not its technology: three provider groups and three funders in one body. Its consent is patient-mediated and revocable, and its public material names no health interoperability standard.
- Six categories are missing from most landscape work: device integration and the gateway, the integration engine, identity as a product category, archiving and legacy read-only access, consent management, and ambient documentation.
- The cheapest high-value deliverable is a catalogue: every variant name against one asset, its tier, its sourcing route, its owner, the funding source of its running cost, its interfaces and its extraction position.

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CHAPTER 16

Interoperability, and what governs each seam

Three standards, a profile, and a spreadsheet nobody versions. Only the spreadsheet is the contract.

An architect handed the seam map from chapter 15 asks the obvious question: which standard governs this interface? It is the right instinct, and in healthcare it produces a wrong model.

Take the order-to-result seam between a hospital's clinical system and a laboratory. HL7 version 2 governs the message structure and the transport. LOINC governs what the observation is, and UCUM the unit it is expressed in. A conformance profile or implementation guide governs which fields are mandatory, which repeat, and which value sets apply. Beneath all three sits a site-specific interface specification — a spreadsheet, usually, listing segments, fields, locally defined segments, code mappings and acknowledgement behaviour — and that is the document the two engineering teams actually built to.

Four layers. Three of them are standards. The fourth is the contract.

> **THE QUESTION TO ASK INSTEAD**
>
> Not *which standard governs this seam* but **which layers govern this seam, and what is the actual contract**. The second question has an answer you can put in a procurement document, hold a supplier to, and hand to whoever inherits the estate. The first has an answer that is true and useless.

Two seams, four layers of governance — and only the bottom row is enforceable

	Order to laboratory result	Provider to funder (chapter 6)
Syntax and transport	HL7 v2.5 ORM / ORU over MLLP	UN/EDIFACT MEDCLM is the published standard
Semantics	LOINC for the test, UCUM for the unit	ICD-10 via the Master Industry Table; NAPPI for the product
Profile or implementation guide	A conformance profile fixing usage, cardinality and value sets	PHISC rules; the funder benefit rules
Local interface specification *the actual contract*	The site spreadsheet: segments, fields, Z-segments, code maps, acknowledgements	The switch's own XML, validated against its published XSDs

Nothing above the bottom row can be tested at a connectathon and then relied on at this site. The bottom row is the one nobody versions.

Figure 16.1 Two seams, four layers each. Only the bottom row can be enforced at this site, and it is the row nobody versions.

Chapter 6 worked this through on the funder-claim seam, and it is worth restating as the pattern: the published national standard for a medical scheme claim is a UN/EDIFACT message, while the operative wire format at the switch is frequently the switch's own XML, validated against the switch's published schemas, with an integration pack and a live-environment accreditation step [1]. Both statements are true, and an architect who builds to the published standard will be wrong at the first integration. **The standard tells you the shape of the problem; the local specification tells you what you are being paid to build.**

Three kinds of standard, and the one procurement forgets

Syntax and transport standards say how a message or resource is structured and moved: HL7 v2, CDA, FHIR, DICOM. **Semantic** standards say what the content means: SNOMED CT, LOINC, ICD, ATC, UCUM — mostly not interchangeable in purpose, and chapter 12 owns them. **Profiles and implementation guides** are constrained, use-case-specific selections from the first two, with named actors and named transactions. Conflating the three is the commonest error in healthcare architecture documents.

The third category makes systems interoperate, and procurement documents omit it. "We are HL7 compliant" is a claim about the first category and carries almost no information. "We conform to this named implementation guide, at this version, and here is our conformance test report from a named test facility" is a claim worth something. South Africa has institutionalised that distinction, which is a real strength almost nobody uses: the gazetted framework states that Implementation Guides — which it also calls National Interoperability Specifications — "are prescriptive in nature", are assessed for conformance, and are subject to conformance testing at a laboratory approved by the National Department of Health, with an attestation issued [2]. Hold that sentence; the end of this chapter returns to it.

The best statement of how a profile relates to a standard comes from IHE itself: profiling means limiting a standard to what a particular use case needs and removing the non-essential parts, and the same problem may be profiled against different standards over time — patient identifier cross-referencing and demographics query began on HL7 v2, were adapted to HL7 v3, and were later adapted to FHIR [3]. That explains why there appear to be four standards for one job: there is one *problem*, profiled repeatedly as the technology generation changed, with every generation still deployed somewhere.

One further distinction runs through everything below. **What a standard requires, what a product claims and what a deployment actually does are three different statements**, and only the third determines whether a result reaches a ward. Chapter 17 gives the evidence tests for the second.

HL7 v2.x, and why it is still the backbone

HL7 version 2 is a message standard first published in the late 1980s, with versions from 2.1 to 2.9 still in production. It remains the backbone of integration inside a hospital for structural rather than nostalgic reasons. It is good enough for the events that dominate hospital traffic — admission, discharge, transfer, order, result, scheduling, charge, document — which have a small, stable payload for which a pipe-delimited message is a reasonable encoding. It is cheap at the endpoint, needing no web stack, which matters for an analyser interface or a bedside device gateway. It is universally implemented, which cannot be said of FHIR even in 2026. And its transport survives a poor network better than an HTTPS interface with token refresh.

The message families an architect actually meets [4]:

Family	What it carries, and what to check
ADT	Admission, discharge, transfer, registration, update, merge, cancel. Check whether merge, unmerge, cancel-admit and cancel-discharge are implemented downstream. Usually not
ORM, OML, OMG, OMI	Orders — the older general form and the newer laboratory, general and imaging forms. Check which the pair agreed, and a cancellation that races a collection
ORU	Observation results, unsolicited. Check whether an amended result replaces or appends
SIU, DFT, MDM	Scheduling, a charge, and a document with its status. Check whether cancellations and non-attendances travel at all, and whether an addendum is a new document or a revision
RDE, RDS, RAS	Pharmacy order, dispense and administration. Check whether administration is transmitted — usually not, which is why chapter 4's fourth medication entity has no data
MFN	Master file notification, for synchronising catalogues. Almost universally unimplemented, hence catalogue drift
ACK	Acknowledgement. Accept against application: whether the sender learns the message was processed or only received

That last row is the one to argue about in week one. A sender receiving only an accept acknowledgement knows its message reached a socket, not that a readable result reached a chart. Chapter 9 named the acknowledgement state in every stream crossing an organisational boundary; this is the field where it lives or does not.

The Z-segment problem

HL7 v2 permits locally defined segments, conventionally prefixed with Z. They are not part of the standard, carry whatever fields an implementer needs, and may be placed anywhere in a message — often appended at the end so that a system parsing only standard segments keeps working [5].

The mechanism is not the problem; what it permits is. Z-segments carry three different things. **Genuinely local data** that no standard covers — a local risk score, a local pathway state — is legitimate and must be in the interface specification. **Data the standard does cover**, placed in a Z-segment because the supplier's internal model did not match the standard one, is the damaging case: the meaning is unavailable to any generic consumer, and it is usually discovered when a third system is connected, eighteen months late. **Version bridging** carries a field from a later version of v2 than the interface is nominally on.

The deeper point generalises well past Z-segments. HL7 v2's optionality is so broad that **two conformant implementations can be mutually unintelligible**: most fields are optional, most code sets are suggested rather than required, and repetition and component structure vary. The standard's own answer is the *conformance profile* — a formal, machine-processable constraint on a message structure, with cardinality, length, value sets and usage fixed — whose methodology comes largely from the United States National Institute of Standards and Technology [6], and which integration engines can check [5].

What "we are HL7 compliant" does and does not promise

It promises, at best, that the product can emit and ingest messages whose segment and field structure matches some version of the standard. It does not promise that the product uses standard segments for standard content, populates the fields you need, uses the code systems you use, implements merge, unmerge, cancel and correction events, acknowledges at application level, will accept an unexpected segment, or that two instances of the same product, configured by different teams, behave the same way. That last is the most expensive, because **the configuration is the interface**, and no compliance claim covers it.

> **WHAT TO REQUIRE INSTEAD**
>
> A named conformance profile or implementation guide, with its version. A message-by-message, field-by-field specification including every Z-segment, with usage and cardinality. An explicit list of supported trigger events, naming merge, unmerge, cancel-admit, cancel-discharge, result correction and result cancellation. Acknowledgement semantics, and the behaviour on an unknown segment, code or identifier. And conformance evidence from a test facility where one exists — in South Africa, the attestation the framework contemplates [2].

The South African framework names HL7 version 2.3.1, 2.5 and 2.6, CDA Release 2.0, the version 3 reference information model as the 2014 basis, and the HL7 context management standard at version 1.4 [2]. Naming three versions of v2 is a realistic acknowledgement of the installed base rather than a failure of nerve, and it has a consequence to design for: **the integration layer must version-bridge, so the national specification cannot assume one wire format.** The context management reference synchronises patient context across several applications on one clinician's screen — a 1990s answer to a fragmented estate, superseded by the launch context described below, and a true statement about the installed base being described.

CDA, and the document model

Clinical Document Architecture is an XML document standard derived from the HL7 version 3 reference information model. A document has a header — who, about whom, when, what type, by what authority — and a body, and the body may be narrative text, structured entries, or both: the "levels" distinction, in which a level 1 document is an attested blob with a structured header and a level 3 document carries coded entries.

Three reasons it still matters. **The attested document is a legal artefact.** A discharge summary signed by a named clinician at a point in time is what a court, an inquest or a complaint process asks for; a set of mutable resources is not. CDA's design — a header asserting authorship and legal authentication, with the narrative block as the *authoritative* human-readable content and the structured entries derived from it — encodes that directly. Chapter 13 owns the record as a legal object; this is the standard built for it. **It is what document-sharing infrastructures carry.** The cross-enterprise sharing family treats a document as an opaque byte stream, so any format can be shared, but the content profiles that give the infrastructure meaning are CDA-based: medical summaries, laboratory reports, scanned documents, all three named in the South African framework [2][7]. **And it survives**: a CDA document with a stylesheet is readable in twenty years without the originating system, which a FHIR resource graph is not unless the terminology service and the profile are archived too. That is a retention argument.

CDA is bad at incremental update, querying, anything granular, and developer experience, and its poor reputation comes mostly from being used as a general-purpose exchange format rather than an attested document format. **Documents and resources are complementary, not competing.**

FHIR, and the version question that is usually asked backwards

The published release history [8]:

Version	Number	Published	Status
DSTU1, DSTU2	0.0.82, 1.0.2	30 September 2014, 24 October 2015	Historical
STU3	3.0.2	21 February 2017	Standard for trial use
R4	4.0.1	27 December 2018	First normative content
R4B	4.3.0	The milestones table gives 27 December 2018; the R4B change log gives 28 May 2022	Staging release
R5	5.0.0	26 March 2023	Trial use
R6	—	Not published	See below

The R4B date inconsistency is in the source, not introduced here; if you cite that page, say so rather than silently picking one. And **R6 is not published.** The management group's release plan runs draft-for-comment ballots in 2024 and 2025, a first normative ballot in January 2026, a second in July 2026 out of cycle, and tentative further ballots in March and June 2027 — with no stated publication date, and any issue requiring a substantive change forcing another ballot [9]. An architecture document assuming R6 availability is speculating.

So the comparison drawn in slide decks — R4 against R5 — is the wrong one. **The real comparison is R4 against R4 plus implementation guides**, and the reason is regulatory gravity rather than technical merit. United States federal regulation adopts, by name and version: FHIR Release 4.0.1; the US Core implementation guide at STU 3.1.1, whose adoption expires 1 January 2026, and at STU 6.1.0; the SMART application launch framework at Release 1.0.0, also expiring 1 January 2026, and at Release 2.0.0; and FHIR Bulk Data Access at version 1.0.0 STU 1 [10]. Because the certified-product market is built to that regulation, R4 is where the implementation guides, the test tooling, the servers and the supplier interfaces are. South Africa's framework cites Release 4 and Release 3 for specific profiles and omits a general FHIR version [2].

> **THE VERSION RULE**
>
> Choose the version your implementation guides are written against, not the newest one. A FHIR version is not a platform upgrade; it is a change to the contract with every connected party, and the ecosystem moves at the speed of its implementation guides rather than the speed of the specification.

FHIR is four things at once and the confusion comes from not separating them: a **content model** of roughly 150 coarse-grained resources with extension points; a set of **exchange styles** — REST, messaging, documents, operations and services are all in the specification, and the common error is to think FHIR means REST; a **conformance and tooling framework** in which structure definitions, value sets, capability statements and implementation guides are themselves resources, which is what makes FHIR different in kind from v2 because the constraints are machine-readable and testable; and a **terminology binding mechanism** in which coded elements bind to value sets with a strength, which is where chapter 12's problem lands.

A bare resource is nearly useless for interoperability because so much is optional. A *profile* constrains it; an *implementation guide* packages profiles, value sets, operations, examples and narrative into a versioned, publishable, testable artefact. The hierarchy runs from the base specification through cross-jurisdiction guides — the international patient summary being the significant one — to national guides, which are the real contract in each market. For South Africa the equivalent layer is the framework's National Interoperability Specifications, described as prescriptive and conformance-assessed [2]. So the first question on any national integration is: *is there a published national specification for this use case, and has it been through conformance testing?* The honest answer for most use cases in 2026 is that the framework exists and the specification inventory is not publicly enumerated — the same finding

practitioners made of the earlier framework in a 2023 study, one of them calling it "just a shopping list of standards", lacking guidance on architecture [11].

A façade over a legacy database, against a FHIR-native record

The most consequential distinction in this chapter for a design decision, and suppliers blur it deliberately.

A **façade** maps queries into a legacy schema and renders the results as FHIR resources. You get standard query syntax, standard serialisation, a recognisable developer experience, and conformance with an interface mandate. What you cannot get by adding more façade: write capability beyond what the legacy system's own rules allow, so a façade cannot create a condition if the source has no problem list; granularity the source does not have, so a free-text blood pressure emits an observation with a text value and nothing more; referential integrity, versioning and provenance; and **terminology**, because a façade over locally-coded data emits local codes in FHIR wrappers. That last is the field's most common disappointment: the interface is conformant and the content is not interoperable. A **FHIR-native record** gives real granularity, versioning, provenance, subscription and a write path, and costs a semantic ceiling set by the resource definitions and a migration problem when the version changes. Most mature estates land on the honest middle: a façade for external interface obligations, a *separate* clinical repository for depth, and an analytic model for secondary use — three models, with explicitly governed mappings between them [12].

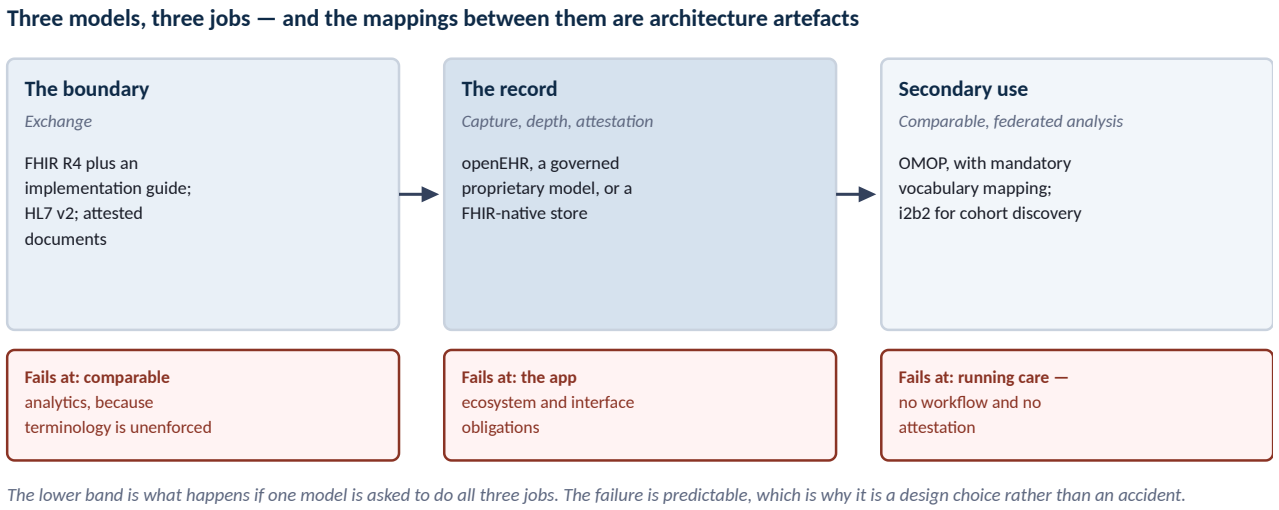


Figure 16.2 Three models, three jobs. The lower band is the predictable failure when one model is asked to do all three.

SMART, bulk data, subscriptions

SMART is the authorisation and launch layer that lets a third-party application run inside a clinical system against that system's FHIR interface, with specifications for application launch, backend services, bulk data and the decision-support hook pattern [13]. The clinical system's authorisation server issues a scoped token using OAuth 2.0; the token carries *launch context* — which patient, which encounter, which user — so the application does not have to ask; and scopes are expressed in a FHIR-specific grammar covering resource type, read or write, and patient-level against user-level against system-level access [14][15]. SMART solves, with a token and a launch URL, the problem desktop context synchronisation addressed in the 1990s — so an estate that still needs the latter has not adopted SMART, a useful maturity diagnostic. The regulation names launch 1.0.0, expiring 1 January 2026, and 2.0.0 [10]; build against 2.0.0.

Bulk data, also called flat FHIR, exists because the ordinary interface handles population-scale extraction badly. It enables access to patient-level data across a population, targeting population health management, quality measurement, surveillance, machine-learning dataset construction and research [16], adopted in regulation at version 1.0.0 STU 1 [10]. Its shape is asynchronous, and three consequences follow: it is a *batched* capability, not a substitute for a data pipeline and not low-latency; export scope is patient, group or system level, and group matters because it maps to a cohort, so defining the group is a governance act rather than a query; and the files are large and identifiable, which makes their transfer, storage and deletion a security design problem rather than an interface one. **Subscriptions** let a client be notified when data matching a criterion changes; the mechanism was substantially redesigned between R4 and R5, with the R5 topic-based model made available to R4 implementers through a backport guide, which is why "we support FHIR subscriptions" needs qualifying by version *and* by model.

What the specification deliberately does not standardise

Every item here becomes a local design decision the specification will not help with.

- **Workflow and process.** There is a task resource and a service request, and no standard order state machine, escalation rule or statement of who is accountable.
- **Identity resolution.** There is a patient resource with identifiers; how to match, what threshold to use and how to merge are not addressed. The IHE profiles standardise the transactions and still do not address the matching. Chapter 11 owns this.
- **Authorisation policy.** SMART standardises the token and the scope grammar; *what a given user may see* is entirely local, and consent is modelled as a resource with enforcement unspecified.
- **Which code to use, and narrative against structure.** Many bindings are extensible or preferred, so local codes are conformant; and unlike CDA, FHIR does not make the narrative authoritative, which leaves the legal-record question open.

- **Deduplication, record lifecycle and operational semantics.** Two sources asserting different medication lists are both carried; retention, archival, legal hold, correction against amendment, rate limits, pagination guarantees and consistency under concurrent write are all out of scope.

The commonest failure of the FHIR era is to treat the specification as if it were an architecture. It is a very good content and exchange standard with a conformance toolchain. It is not a design.

DICOM, which is not comparable to HL7

DICOM is maintained by the National Electrical Manufacturers Association and is unusual in being simultaneously a data format, a network protocol, an information model, a conformance framework and a device specification, published in numbered parts [17]. The South African framework cites three of those parts and DICOM 2011 for radiology results [2].

The comparison that helps: HL7 v2 standardises *messages between information systems*; DICOM standardises *objects produced by devices*, plus the services for moving them, plus what a conformant device must document about itself. That is a much harder guarantee, and it is why DICOM interoperability works at a level HL7 v2 interoperability does not.

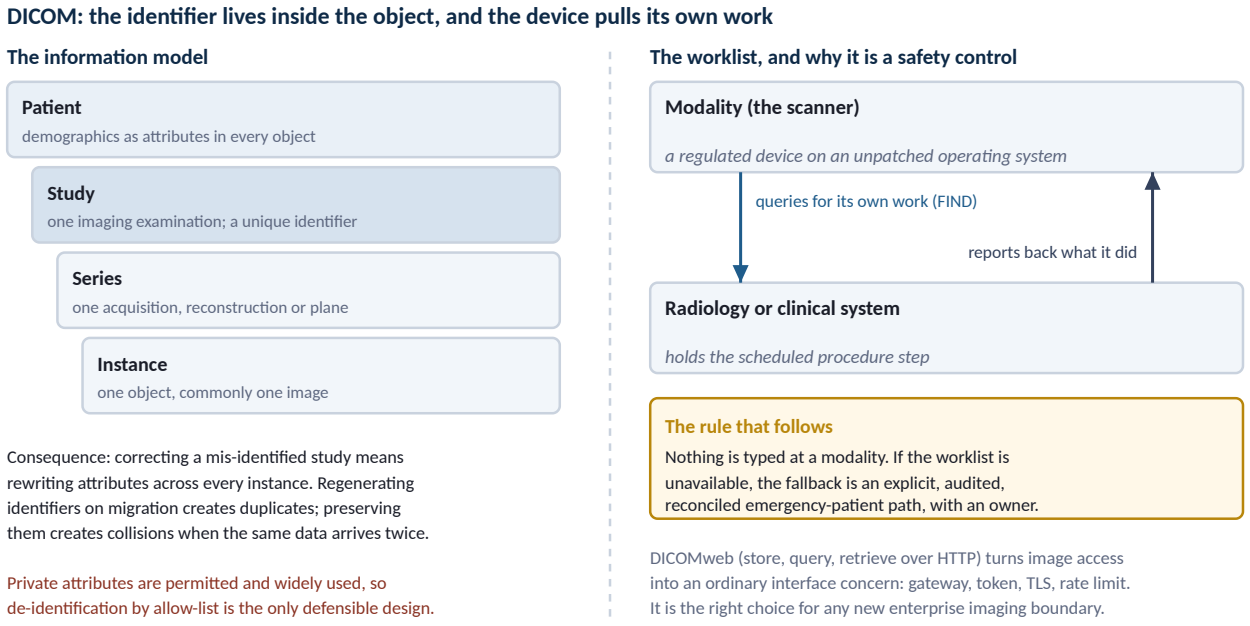


Figure 16.3 DICOM puts the patient identifier inside every object, which is why the worklist pull is a safety control rather than a convenience.

Described rather than reproduced, the information model is a hierarchy from patient to study to series to instance, each level carrying an identifier that is globally unique by construction. Three things follow. **The patient and study identifiers are inside every object**, so correcting a mis-identified study means rewriting attributes across every instance — which is why the worklist is a safety control rather than a convenience. **Identifiers must never be regenerated casually**: migration tools that regenerate them create duplicates, and tools that preserve them create collisions when the same data arrives twice by two routes, which is the most common cause of pain in archive consolidation. And **private attributes are permitted and widely used**, which is why de-identification by allow-list is the only defensible approach.

A device's **conformance statement** declares which service-object pairs it supports, in which role, with which transfer syntaxes and which optional attributes [18]. For an architect that statement is the contract: demand it before signing, read it, keep it as an architecture artefact. It tells you exactly what will and will not work, which is more than any other healthcare standard offers.

The **modality worklist** exists because a radiographer once had to type the patient's name and identifier into the scanner, and every transcription error produced a study attached to the wrong patient. The standard's answer is a query against a worklist information model, with demographic and procedure attributes pushed into the images automatically [19], plus a mechanism by which the modality reports back what it did and with what parameters [20]. The point generalises past imaging: **DICOM solved the "stop people typing identifiers into devices" problem with a pull pattern** — the device queries for its own work — which is the right pattern whenever an endpoint is unreliable, intermittently connected or operated under time pressure. The rule that follows is absolute: nothing is typed at a modality, and if the worklist is unavailable the fallback is an explicit, audited, reconciled emergency path with a named owner, not free typing.

DICOMweb is the web-service family — store, query and retrieve over HTTP — exposing the same information model. Its significance is that image access stops being a specialist network concern and becomes an ordinary interface concern with a gateway, a token, TLS and rate limiting, which makes enterprise image viewing in a clinical portal or on a mobile device feasible without a DICOM stack on the client. It is the right choice for any *new* enterprise imaging boundary.

Structured reporting carries report content — measurements, findings, coded observations — as structured objects rather than prose. It is widely implemented for machine-generated content such as dose reports and computer-aided detection output, and very narrowly used for the radiologist's narrative report, which in most estates remains free text produced by dictation and transmitted as an HL7 v2 result or document message. An architect who plans a clinical data strategy on the assumption that radiology findings arrive coded is planning on something untrue in 2026 — though the research behind this volume could not locate a current adoption survey, so treat that as practitioner consensus rather than a measured finding.

The de-identification problem

The standard includes de-identification profiles defining attribute-by-attribute handling and confidentiality options [21]. The hard problems are not in the standard. **Burned-in annotation** — identifiers rendered into the pixel data, common in ultrasound — is removed by no attribute-level process. **Private tags** may carry identifiers, so allow-listing known attributes and passing private ones through will leak. **Re-identification by reconstruction** is real: a head CT renders as a recognisable face, and defacing changes the clinical content. **Date shifting** needs a consistent offset per subject to preserve intervals, which makes de-identification a *stateful service*, not a filter. And **identifier management** is a trap: identifiers must be replaced consistently or the objects will not group, and keeping a mapping table recreates the linkage you were breaking.

The Western Cape provincial data centre's approach to the equivalent problem in non-imaging data is the pattern to borrow: named identifiers stored separately from clinical attributes and linked by an internal key, with **the key perturbed per released dataset** so two separately released datasets cannot be cross-linked [22]. And de-identification is a *regulated activity* when it is the basis for lawful secondary use, which puts the pipeline inside the accountability story under the privacy statute. Chapter 18 owns the legal frame.

IHE, and the profiles an architect actually meets

IHE is not a standards development organisation. It writes *integration profiles*: specifications that select existing standards, define **actors** — the roles a system plays — define **transactions** between actors, each bound to a specific standard with specific constraints, and specify the optional behaviours a system may declare. Systems declare which actors and options they implement, and the declarations are testable at connectathons [3].

This matters in South Africa whether an architect intends it or not, because **the gazetted framework is in structure an IHE framework.** Its core scope is the IT Infrastructure domain, its table of standards lists more than twenty profiles by acronym, and an annexure maps profile, actor and transaction [2]. Reading it without IHE literacy is not possible.

Cross-enterprise document sharing

The foundational profile, and the one whose vocabulary became the industry's default way of talking about exchange. Its actors [7]: a **Document Source** producing documents and supplying metadata; a **Document Repository** storing them persistently and registering the metadata; a **Document Registry** holding one metadata entry per document, including a pointer to where it is stored, and answering consumer queries; a **Document Consumer** that queries the registry then retrieves from repositories; and a **Patient Identity Source** for authoritative identifiers.

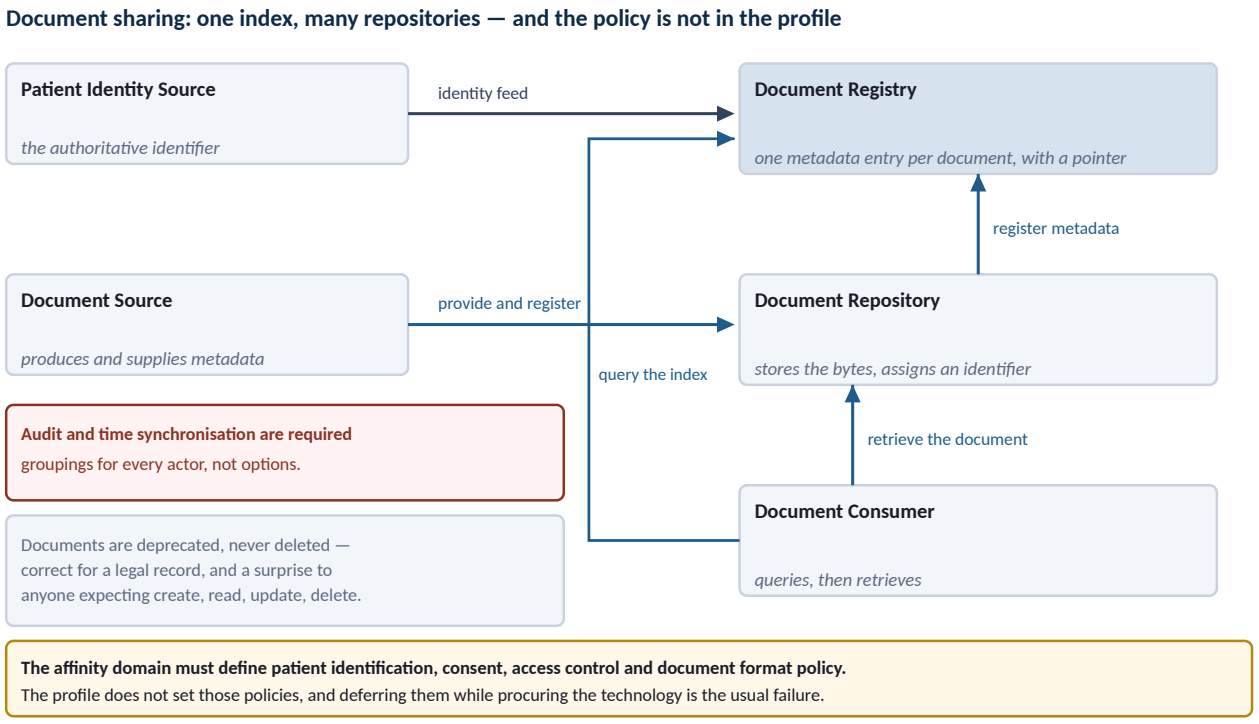


Figure 16.4 The document-sharing actors. The registry indexes many repositories; the affinity-domain policy, which the profile does not supply, is the hard part.

The separation of registry from repository is the profile's central idea, and it is why the pattern scales: one registry can index many repositories, so documents stay where they were produced and only the index is centralised. Three features are worth carrying into any federated design in any industry.

The **affinity domain** is a group of enterprises that agree common policies and share infrastructure, and each must define policies for patient identification, consent, access control and document content. **The profile does not set those policies.** The commonest failure in document-sharing programmes is to procure the technology and defer the affinity-domain policy, and the policy is the hard part. **Documents are never deleted** — availability status is approved or deprecated, deprecated documents stay queryable, and relationships are typed as replacement, addendum or transformation, which is correct for a legal record and a surprise to anyone expecting create, read, update and delete. And **audit and time synchronisation are mandatory groupings**, not options. That is the most transferable governance lesson in IHE, because a federated infrastructure without a trustworthy audit trail and without synchronised clocks is not merely incomplete, it is unaccountable.

The profile is also explicit about its limits: it does not track orders or their workflow, provide a security model, manage the affinity domain, federate domains, or give access to *parts* of a document [7].

The profile set, and one acronym collision

Profile	What it does, and where you meet it
Identifier cross-referencing, and demographics query	Links local identifiers to one used for exchange, and finds a patient from demographics. Asserts sameness; does not merge. The slot the national patient index occupies, named in three generations
Audit trail and node authentication; consistent time	Node authentication, encrypted transport, audit of activity, and clock synchronisation. Mandatory groupings, routinely absent
Cross-community access and patient discovery	How one exchange connects to another, and how a patient with records in two communities is found
Mobile access to documents; query for existing data; care services discovery	Document sharing for lightweight clients, FHIR access to granular clinical data, and a FHIR directory of facilities and practitioners. All three named in the framework; the last is the master facility list slot
Patient administration management; document subscription; mobile alert communication	The v2 registration and encounter feed, profiled; and notifying an interested party that a document exists or something happened — the basis of the notification pattern below
User assertion; patient privacy consent; sharing value sets	Who the human behind a transaction is; recording a consent policy, with chapter 18 owning the legal test; and publishing value sets for download

The acronym for sharing value sets collides exactly with the acronym for South Africa's Stock Visibility System. Both appear in national architecture documents, they are unrelated, and one is a logistics application that never touches the clinical record. Write them out in full.

The profile that resolves the document-against-interface argument

Document sharing serves the *producer* well and the *consumer* badly: consumers receive documents whose content is not decomposed usefully, or which lack the one item needed. IHE's data-element extraction profile addresses it, and its framing is the clearest statement of the trade available anywhere: document-level access preserves context and shows who attested to the content, while element-level access answers questions across documents — all allergies over a time span, for instance [23]. An extractor retrieves documents from the sharing infrastructure, decomposes them into candidate clinical resources, creates or updates those resources in a queryable clinical data source, and **creates a provenance record for each resource identifying the source document and the extraction time**; a consumer queries data elements, receives the provenance, and uses it to retrieve the source document.

Its stated limitations are the honest limits of *any* document-to-resource extraction design: the mapping from document content to data elements is out of scope, must be defined per deployment and may be imperfect; text not expressed as data elements will not appear in element-level queries; and trigger logic, access control and consent enforcement are all deployment-specific [23].

> **THE LESSON, WHICH APPLIES FAR BEYOND IHE**
>
> Any design that derives structured data from attested documents must carry provenance back to the document, and must accept that the derived data is a lossy, policy-bound projection rather than an equivalent. Many programmes described as FHIR enablement are performing exactly this extraction without the provenance discipline, creating clinical data whose basis cannot be re-examined.

openEHR, and the honest comparison

openEHR is a specification set for clinical data persistence based on *two-level modelling*: a small, stable reference model of generic structures underneath, and clinical content above it as **archetypes** — formal maximal-dataset definitions of a clinical concept, authored by clinicians — and **templates**, which constrain and compose archetypes for a specific form, document or message. Because content models are *data* rather than code, the software does not change when the clinical content does, and querying is by archetype path rather than physical schema, so queries survive schema evolution. Archetypes are governed collaboratively in a public repository with review workflow, versioning and publication status [24] — the most transferable idea in openEHR: **a governed, versioned, internationally reviewed library of clinical content models, separate from any product.**

Partisan comparisons with FHIR are useless. Here is the structured one.

Dimension	openEHR	FHIR
Designed for	A complete clinical record persisted over decades	Exchanging data between systems over the web
Content governance	Archetypes in a public, reviewed, versioned, clinician-authored repository	Profiles and implementation guides, per jurisdiction and project
Semantic ceiling, change without code	High; templates are configuration	Set by the resource definitions; resources are not configuration
Query	Archetype-path query language	Resource search, limited for complex questions
Ecosystem and regulation	Smaller; skills scarce; no binding national regulation anywhere	Very large; named in regulation at specific versions

The finding that decides it is not about data models at all. The first serious attempt to count openEHR deployment — a vendor and community survey running from October 2025 to March 2026 — reports eleven of sixteen contacted vendor organisations responding, headquartered in nine countries and reporting deployments in 22 countries, with a community survey of 29 responses across 17 countries bringing the total to 28 countries [25]. Regional coverage is real — Sweden 19 of 21 regions, Norway 3 of 4 health regions, Finland 16 of 21 wellbeing services counties — with national records built or procured on it in Ireland, Malta, Greece, Jamaica and Slovenia.

But no country was found to name openEHR in binding national regulation. Slovenia's law names its system, not the standard; Malta's commitment is contractual; Ireland's framework mentions openEHR only as an example. The authors describe the pattern, driven by practitioners and vendors rather than mandates, as "structurally fragile", and identify lack of openEHR-specific knowledge as the most consistent barrier, cited by 21 of 29 community respondents [25]. Carry their caveats: vendor data is self-reported, unverified and commercially interested, institution counts are not comparable — Australia appears as 170 institutions in one unverified aggregate, Germany's entries as roughly 165 hospitals, an upper bound of about 9% of its 1,841 — and eleven respondents may be half the market [25].

> **THE COMPARISON TURNS ON SKILLS, NOT SPECIFICATION MERIT**
>
> openEHR is a legitimate and in some respects superior choice for the clinical data repository, and it carries a skills risk and an ecosystem risk that FHIR does not. For a South African provider the skills risk is the binding constraint: the research behind this volume found no evidence of a South African openEHR implementer market, and the adoption survey's country list does not include South Africa. The practical architecture is FHIR at the boundary, with the repository decision made on skills availability rather than elegance.

OMOP and i2b2, for secondary use

The Observational Medical Outcomes Partnership common data model is an *analytic* model: a fixed relational schema into which heterogeneous source data is transformed, with all clinical content mapped onto a standardised vocabulary [26]. Its principles are deliberately the opposite of FHIR's — person-centric, event-based, fixed-schema, and with **vocabulary standardisation mandatory rather than optional**, source codes retained *and* mapped to standard concepts, which is why analyses are comparable across sites and FHIR extracts are not. Analytics travel to the data rather than the reverse: a study package written once runs unchanged at every site holding an instance, so a network can answer a question without pooling patient data — the strongest answer available to the tension between research value and data sovereignty. The older cohort-discovery platform built on a star schema and a concept ontology remains widely deployed in academic centres, with published work on querying its data through the newer tooling [27].

The Western Cape provincial data centre is a local version of this layer, harmonising to standard code sets rather than to the analytic model's own vocabulary [22]. Whether a South African analytic layer should adopt that model is live, and its value is realised in a *network*, which exists only if more than one South African site adopts it.

The patterns

Exchange topologies

Four, and the choice among them is political and legal far more than technical. **Centralised** makes query simple at the price of a single point of compromise holding a population's clinical data. **Federated with a central index** minimises what is centralised, but availability then depends on participants being up and the index still reveals that a person attended a particular facility — sensitive when that facility is a psychiatric hospital or a termination-of-pregnancy service. **Fully federated**, with no index, makes discovery expensive and identity across communities hard. **Consent-mediated** has the strongest legal basis, and needs a break-glass path.

The second is the document-sharing pattern and the dominant architecture internationally, and it is South Africa's stated position: the gazetted framework describes a federated health information architecture combining shared and local infrastructure [2], and the national digital health strategy proposes a national platform with shared services including integration, terminology, security, registries, repositories and analytics [28]. The Western Cape's daily patient master index feed to the national system since June 2020 is that pattern operating [29]. Regional scoping reviews consistently find policy ahead of implementation across Africa [30], and a 2026 assessment finds the privacy statute gives limited guidance on data ownership, stewardship and interoperability [31].

Document sharing against an interface

Document sharing gives attestation and degrades gracefully, and costs coarse granularity, no query across documents without extraction, and a dependency on metadata quality — which makes metadata governance the whole game. An interface gives granularity, query, write and real time, and costs an available, secured, versioned endpoint at every participant, no inherent attestation, local consent enforcement, and unmanageable version skew.

> **THE DECISION RULE**
>
> Where the exchange is *an act of clinical communication between accountable professionals* — referral, discharge, report, summary — use the document model. Where the exchange is *a query for a specific fact to support a decision now* — current medicines, latest creatinine, allergy list — use the interface model. Where you need both from one substrate, use extraction with provenance and accept that the extracted data is a lossy projection.

OpenHIE, and what each component is

OpenHIE is the reference architecture most relevant to South Africa and the rest of the continent, and it is genuinely well specified [32][33]. Read against the national digital health strategy's own architecture proposals, it is also the thing the strategy describes in policy language [28].

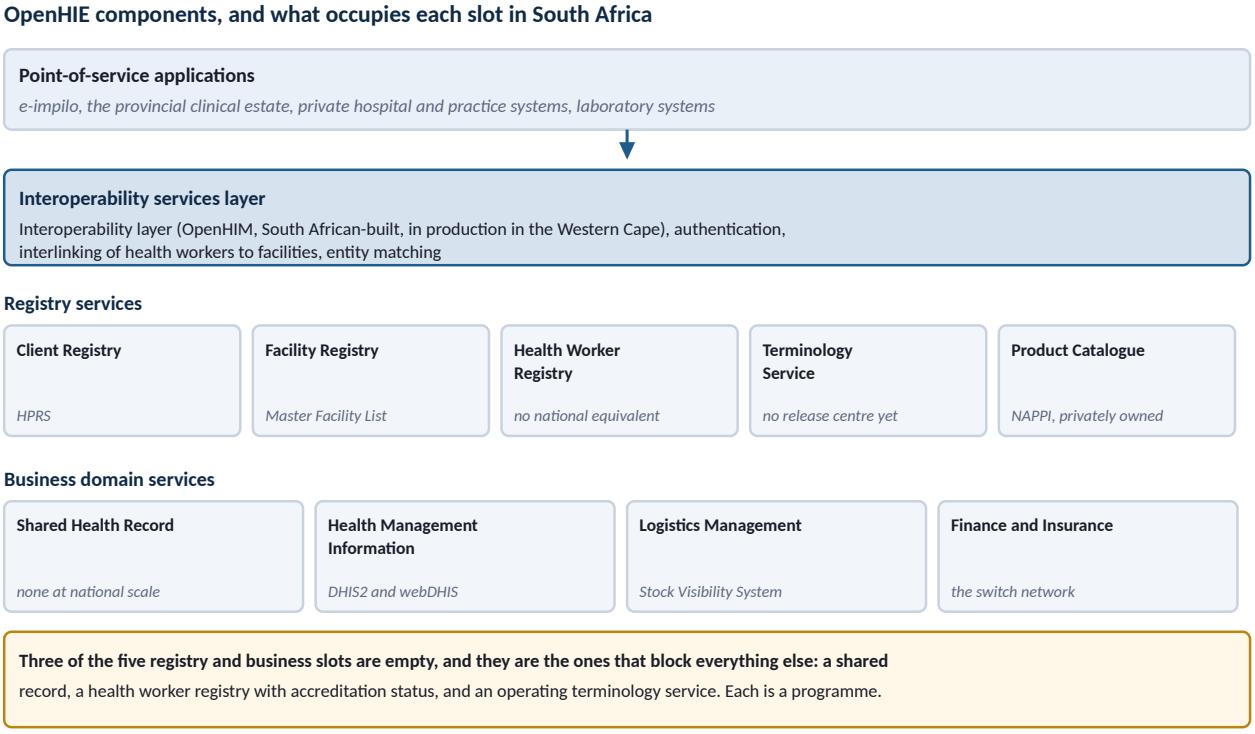


Figure 16.5 OpenHIE components with their South African occupants. The empty slots are the ones that block everything else.

The **interoperability services layer** receives all communications from external services in a health geography and orchestrates message processing, with authentication of sending systems, an interlinking service tying health workers to facilities, and entity matching to find duplicate patient or facility records. The **registry services** — a client registry, a facility registry, a health worker registry, a terminology service and a product catalogue — are the components an architect otherwise forgets, and each is a master data domain that programmes discover eighteen months in. The **business domain services** are a shared health record of normalised person-centric records validated against the registries, an aggregate health management information service, a logistics service for commodity visibility, and a finance and insurance service holding centralised claims data, which is the slot a single-payer fund's claims system would occupy [32].

Use it as the reference diagram rather than a commercial one for three reasons. It separates registries from repositories from the exchange layer, with a component per concern, which is the single most important structural idea in health information exchange. It names the components that get forgotten. And it is standards-referencing rather than standards-defining, so it composes with IHE profiles and FHIR [32]. Its honest limitations: it is written for public-sector exchange in low- and middle-income settings, says little about integration inside a hospital or a private provider's revenue cycle, and assumes more central coordination than a mixed public-private system has.

The shared care record

A shared care record is the applied form of exchange: a defined set of information about a person, assembled from several providers and presented to a clinician in another organisation at the point of care. It differs from an exchange in that it is *curated*, and four decisions decide whether it succeeds. **Scope** — a small, complete, current dataset beats a comprehensive patchy one. **Presentation** — it is read in minutes by someone who does not know the patient. **Currency indication** — a two-year-old medicines list shown indistinguishably from today's is dangerous, so each item needs its age and its asserting organisation. And the **consent model and break-glass**, which chapter 19 treats as a control. It belongs embedded in the local clinical system, because clinicians will not log into a second one.

Event-driven clinical notification

Distinct from exchange, and the higher-value capability: when something clinically significant happens to a person, tell the people who need to know.

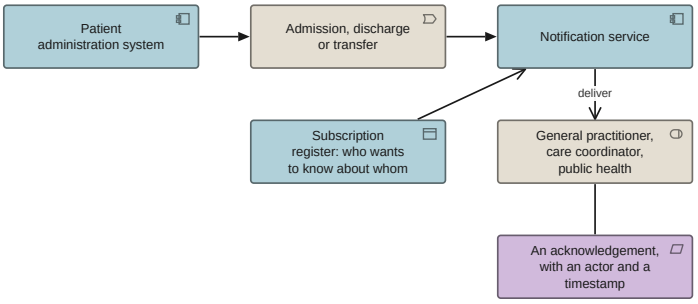


Figure 16.6 Event-driven clinical notification: four components, one of them a subscription register whose currency is a privacy control, and an acknowledgement almost nobody records.

The canonical cases are notifying a general practitioner that their patient has been admitted or discharged; a care coordinator that a patient has attended an emergency centre; a specialist clinic that a result is critically abnormal; and public health that a notifiable condition has been diagnosed. The components are an event source, usually the admission-discharge-transfer feed; a subscription registry recording who wants to know about whom and under what conditions; a notification service; and a delivery channel. The standards are FHIR subscriptions and the document-subscription and mobile-alert profiles, both named in the South African framework [2].

It needs only the feed that already exists plus a subscription list, and requires no shared record, document repository or terminology harmonisation. **For a health system at an early stage of integration maturity, event notification is the highest-return first capability**, against the instinct to build the record first. Three failure modes are specific to it: **subscription list currency**, because an out-of-date practice registration list sends notifications to the wrong practice, which is a privacy breach rather than a nuisance; **notification fatigue**, because an unfiltered feed is ignored within weeks; and **no acknowledgement path**, so nobody knows whether the notification was acted on — the closed-loop failure chapter 9 named in results reporting.

The South African position, and the contradiction in it

The gazetted framework is the most important single document for a South African health architect, and it is almost never read properly.

Its legal status: the Director-General published the 2021 framework under section 74(1) of the National Health Act, which covers building and maintaining health information systems, in *Government Gazette* No. 47337, Notice 2667, on 21 October 2022 [2], superseding the 2014 framework gazetted in *Government Gazette* No. 37583, General Notice 314 [34]. Its scope is bounded and stated: patient-centric interactions *between* health information systems, applying to systems that interact with or upload to shared national infrastructure, and explicitly not legal, regulatory, policy or care-process interoperability [2].

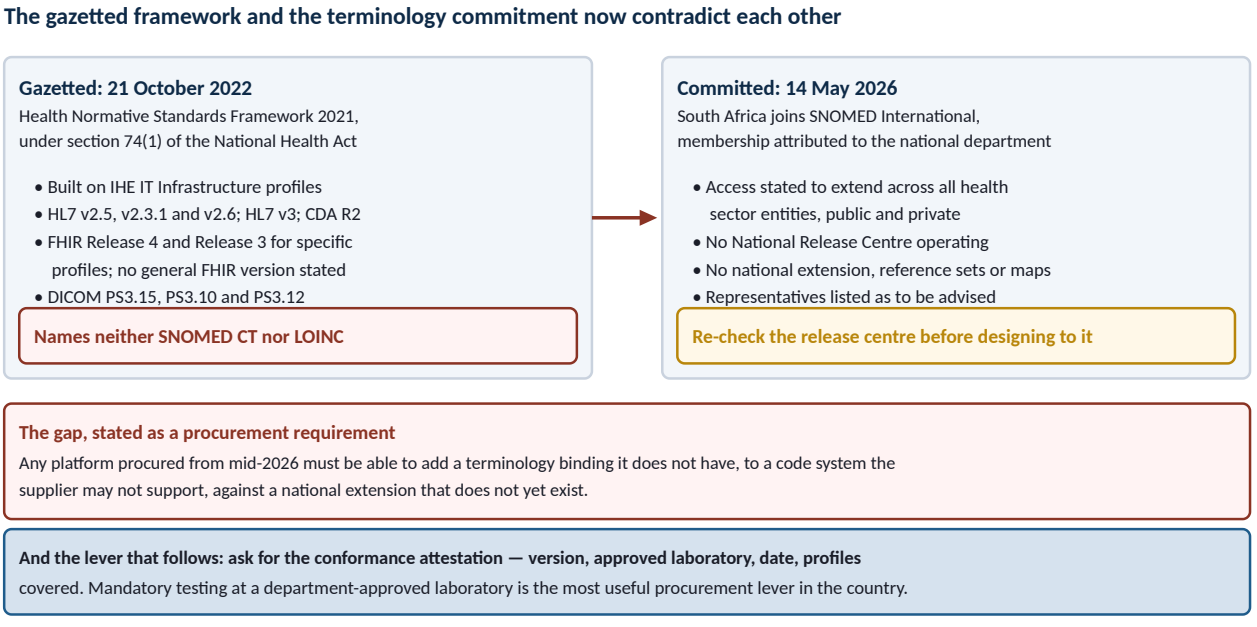


Figure 16.7 The gazetted framework names neither SNOMED CT nor LOINC; the terminology commitment of 14 May 2026 assumes both. The conformance attestation is the lever that survives the gap.

The framework is built mainly on IHE IT Infrastructure profiles. It references HL7 version 2.5, 2.3.1 and 2.6, HL7 version 3, CDA Release 2.0, FHIR Release 4 and Release 3 for specific profiles, and DICOM. **It names neither SNOMED CT nor LOINC.** The international disease classification appears only in the acronym list and is not tied to any mandated profile [2].

Set that against the terminology commitment. South Africa joined SNOMED International on or about 14 May 2026, with membership attributed to the National Department of Health and stated to extend access across all health sector entities, public and private [35][36]. But the National Release Centre

is not yet operating: no national extension, no national reference sets, no published national map, and representatives listed as to be advised [35]. Nothing is mandated.

So the terminology commitment and the gazetted standard now contradict each other, and the consequence is a procurement requirement rather than a drafting curiosity. **Any platform procured from mid-2026 must be able to add a terminology binding it does not have, to a code system the supplier may not support, against a national extension that does not exist.** Chapter 12 owns what the architect must therefore write down. The framework will have to be revised, though whether a revision is scheduled could not be established and remains unverified.

Four findings follow. **It is a standards reference, not an architecture**: the prescriptive content is deferred to implementation guides which, use case by use case, mostly do not exist publicly [2][11]. **There is no mandated clinical terminology in South Africa**, so until the framework is revised terminology is an organisational decision rather than a national one [2][35]. **It is an IHE-shaped framework**, so IHE literacy is a precondition for reading it — and it names three versions of HL7 v2 and no single FHIR version, so the integration layer must version-bridge by design. And **conformance testing and attestation are contemplated**, which is more than most national frameworks offer, and that is the lever.

There is also a good piece of method in the department's own material that answers the "shopping list" criticism. Its layered specification model runs from National Interoperability Use Cases — high-level information-sharing needs tied to national health priorities — to Business Instances, specific workflow and data-flow examples, to Implementation Outlines, subsets of workflow steps and transactions drawn from IHE profiles, to Technical Use Cases at transaction level [37]. That is the bridge from a list of standards to a specification an implementer can conform to, and an architect writing a national-facing integration should write to that shape whether or not a national specification exists.

> **THE MOST USEFUL PROCUREMENT LEVER IN THE COUNTRY**
>
> The framework requires that health information system developers, suppliers and vendors comply with the latest framework, undergo conformance testing at a laboratory approved by the National Department of Health, and receive an attestation, with integration into the national system depending on the department's terms and case-by-case approval [2]. So: **ask for the attestation document, by version, laboratory, date and profiles covered.** It converts "we are interoperable" into a document that either exists or does not, and chapter 17 gives the rest of the evidence tests. One caution belongs in the deliverable: whether an approved laboratory is operating, and whether any attestation has ever been issued, could not be established in the research behind this volume, and only a departmental enquiry will settle it.

One further South African fact reframes the opportunity: the private sector runs a mature, high-volume, real-time, nationally interoperating transaction network, and it carries *money*, not clinical meaning. There is no clinical interface layer between the private and public sectors at national scale [11], while the connectivity between every provider and every funder already exists.

Why every hospital has an integration engine

The intuitive answer is that healthcare lacks standards. It is wrong: healthcare has an unusual *abundance* of standards, and what it lacks is enforced constraint, which is what conformance profiles and implementation guides exist to supply. An engine exists for four reasons: **version heterogeneity**, the same standard at several versions with no coordinated upgrade in prospect; **dialect translation**, because two conformant v2 implementations can be mutually unintelligible; **fan-out with independent availability**, one admission feed reaching fifteen systems each of which may be down alone and each of which must receive every message eventually; and **operational visibility**, because somebody must answer "did this result reach the ward system, and if not, where is it", a *message-level* audit requirement the engine is the only place to satisfy.

Two failure modes matter most. **The engine becomes the business logic layer**, as clinical rules, routing decisions and data enrichment accumulate in transformation scripts invisible to clinical governance — the largest architectural risk in a typical provider estate, almost never on a risk register, and fixed by the standing rule that **transformations may change representation, not meaning**. And **the inventory does not exist**: the contingency-planning guidance in the United States safety-assurance guide series requires interface inventories to be reviewed every six to twelve months, and has a practice on restarting interfaces [38]. Hub by default, point-to-point by exception with a documented reason, and device telemetry never in the clinical engine.

The seam map, read as layers

Chapter 15 gave the seam map as systems and what joins them. Here it is as layers, and the contract.

Seam	Layers that govern it	What the contract actually is
Patient administration to everything	HL7 v2 admission messages; identifier cross-referencing or its FHIR form	The site message specification, plus whether merge and unmerge are implemented downstream
Clinical system to laboratory, and back	HL7 v2 order and result messages; the observation and unit code systems	A conformance profile if you insisted on one; otherwise the interface spreadsheet, plus the amended-result and unmatched-result rules
Clinical system to modality, modality to archive	DICOM modality worklist; DICOM store or its web form	The device conformance statement, plus the private-tag and identifier policy
Clinical system to pharmacy	HL7 v2 pharmacy messages, or the FHIR medication request	The local dictionary mapping, which has no national basis
Provider to funder	The national funder-claim message; the national disease classification table	The switch's own schemas and its accreditation step
Provider to province, nation or surveillance	Document sharing or cross-community access; FHIR; CDA; national notification channels	The affinity-domain policy, which usually does not exist yet, and the absence of an acknowledgement back

Read down the last column. Not one entry in it is a standard. That is the chapter.

What to take to the client

The deliverable from an interoperability workstream is not a list of standards. It is four things, all cheap.

A layered seam register. For each seam: the syntax layer, the semantic layer, the profile or implementation guide, and the local specification, each with its version and the date it was last confirmed. The last column is the contract, and naming it as such changes who feels responsible for it.

A specification ownership position. Every interface specification is a contractual deliverable owned by the organisation, in durable form, including code mappings and test cases. Chapter 17 prices the alternative.

An attestation requirement. Ask every supplier for conformance evidence against the framework, by version, laboratory, date and profiles covered — and record the answer, including "none", which is also evidence.

One capability, built first. In an early-stage estate, build event-driven notification before anything else. It needs the feed you already have and a subscription list, and it produces clinical value in weeks rather than years.

THE CHAPTER IN TEN LINES

- Most seams are governed by two or three standards at different layers at once, plus a profile, plus a local interface specification that is the actual contract.
- Ask which layers govern a seam and what the contract is, not which standard governs it. The contract is never a standard.
- HL7 v2 remains the backbone because it fits the dominant hospital events, is cheap at the endpoint and survives bad networks; its optionality means two conformant implementations can be mutually unintelligible.
- We are HL7 compliant promises only that messages parse. The configuration is the interface, and no compliance claim covers configuration.
- The real FHIR comparison is R4 against R4 plus implementation guides. R4 is 4.0.1 of 27 December 2018; R5 is 5.0.0 of 26 March 2023 and trial use; R6 is not published and has no fixed date.
- Choose the FHIR version your implementation guides are written against. United States regulation names 4.0.1, US Core STU 3.1.1 and 6.1.0, SMART launch 1.0.0 and 2.0.0, and Bulk Data 1.0.0 STU 1.
- FHIR deliberately does not standardise workflow, identity matching, authorisation policy, which code to use, narrative precedence or record lifecycle. It is not a design.
- DICOM puts the identifier inside every object, so nothing is typed at a modality and identifiers are never regenerated on migration. The conformance statement is the contract.
- The honest openEHR comparison turns on skills, not specification merit: no country names openEHR in binding national regulation, and the adoption pattern is described by its own surveyors as structurally fragile.
- The 2021 gazetted framework, published 21 October 2022, is built on IHE profiles with HL7 v2.5, 2.3.1 and 2.6, v3, CDA R2, FHIR R4 and R3 and DICOM, and names neither SNOMED CT nor LOINC.
- South Africa joined SNOMED International on or about 14 May 2026 and the National Release Centre is not yet operating, so any platform bought from mid-2026 must be able to add a terminology binding it does not have.
- The conformance-attestation regime, with mandatory testing at a department-approved laboratory, is the most useful procurement lever in the country. Ask for the document by version, laboratory, date and profiles.
- In a low-maturity estate, event-driven notification is the highest-return first capability: it needs only the existing admission feed and a subscription list.

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CHAPTER 17

Technology providers, and how to buy

Named as examples, never as recommendations. The deliverable is the evaluation method, not the answer.

Every product and company named in this chapter is named as an **example** — an illustration of a category, a commercial pattern or an architectural consequence. Nothing here is a recommendation and nothing is a ranking, and where one supplier appears more often than another that reflects the amount of verifiable public material about it rather than its merit. **Say the same sentence in your own deliverable**, because a reader about to run a procurement will otherwise read a name as an endorsement.

That matters more here than in most industries, for three reasons.

The decision is effectively irreversible on any horizon an architecture practice plans against. The published market data makes that visible rather than theoretical. For contract year 2025, research into United States acute-care share found that the number of hospitals making a net-new electronic health record decision fell by roughly 40% against 2024 and nearly 50% against 2023, and that only two health systems with more than ten hospitals made an enterprise-wide decision at all [1]. A market in which the largest buyers almost never choose again is not a market in which "we will revisit this in three years" is a real option.

A recommendation in a reference book is read as a licence to skip the evaluation. The questions at the end of this chapter are the deliverable; the names are the scaffolding. **And the clinical consequence of a wrong platform choice is not a slow report**: it is a missed allergy, a duplicated patient record, a result that lands in the wrong chart. No external party can carry that risk for the organisation that signs.

Chapter 15 opened on the four-way sourcing question — build, buy, adopt open source, or consume a state service. This chapter is mostly about the *buy* branch, with a section on open source, and it is organised around three questions the sales conversation is arranged to prevent you from answering. **What do I inherit?** The data model, the identifier scheme, the terminology, the integration surface, the hosting arrangement, the upgrade obligation. **What can I change, and what does changing it cost later?** Customisation the supplier supports, tolerates, or treats as voiding support. **How do I get out?** Extraction rights, format, cost, and who owns the specification of the interfaces that make the estate work. A supplier who answers all three in writing is rare; one who refuses the third has told you something.

There is no measurable global market, and that is a finding

An architect asked for "the global electronic health record market" is being asked for something that does not exist in measurable form. The only annually-refreshed, methodologically consistent share dataset the research could find is a **United States acute-care** series. Label it as such every time you use it.

For contract year 2025 that series reported, as summarised independently by two trade publications [1][2]:

Supplier	Share of US acute hospitals	Share of beds	Net hospital change in 2025
Epic	43,7%, up from 42,3%	56,9%	Plus 77 hospitals, 18,679 beds
Oracle Health	21,9%, down from 22,9%	20,4%	Minus 56 hospitals, 14,676 beds
MEDITECH	14,7%	12,5%	Minus 7 hospitals, 1,015 beds
TruBridge	7,6%	Not stated	Plus 2 net
Altera Digital Health	2,9%	Not stated	Plus 1, a 155-bed hospital
Other	9,2%	Not stated	Not stated

Three things in that table are easy to miss. **Share is measured in incompatible units.** Epic at 56,9% of beds against 43,7% of hospitals means its average customer is substantially larger than the market average, so a reference visit to a 900-bed academic centre tells you little about the configuration effort at a 120-bed district hospital. Ask what the *median* new implementation looks like, not the flagship. **Share movement is not satisfaction**, and the same research reports both: about 30% of sampled Oracle Health customers say the supplier is not part of their long-term plans and only about 35% are firmly committed [2]. **And retention can be rebuilt**: MEDITECH's reported legacy-retention rate — the proportion of legacy-platform customers making a go-forward decision who stay — was 84% for 2025, against 63% in 2024 and 30% in 2023 [1]. A supplier with a large legacy estate and a credible migration path is a different counterparty from one with no path, so ask for the legacy-to-current migration count by year.

Outside the United States the evidence thins to nothing usable. Europe yields analyst *ordinal rankings* and no comparable share series [11]. The United Kingdom's most recent *measured* general practice shares are a 2016 census published in 2018, giving one product about 55 to 56%, a second about 34% and a third about 9% across 7,526 English practices; a 2026 commercial analysis offers different figures without stating whether the basis is practices or patients, and the two are not reconcilable [3][4]. For Africa and the Gulf, nothing.

> **STATE THE SOUTH AFRICAN ABSENCE AS A FINDING**
>
> No South African share dataset for clinical platforms exists, in either sector. That is a property of the market rather than a gap in this book's research, and it has a direct consequence: **a South African supplier's share claim cannot be checked.** Treat it as unevidenced unless the supplier names the research house, the measurement unit, the geography, the date and the denominator. An architect who quotes United States share figures in a South African steering committee is quoting a different market.

Ownership and names have moved, repeatedly

A current-state catalogue that records supplier names without *acquisition dates and product-line provenance* will be wrong within eighteen months. The useful field is not "Supplier" but "Supplier as at <date>", plus "Product line origin". The ledger below was verified on or about 31 October 2026, and several entries post-date May 2026.

What you may have written	What it is now
Cerner	Oracle Health. Oracle agreed the acquisition on 20 December 2021 for about US$28,3 billion, closing June 2022; the flagship platform is still called Millennium, so a contract says one name, the invoice another and the interface specifications a third [6]
Allscripts; CPSI	Altera Digital Health, the Harris-owned successor to the Allscripts hospital business, selling Sunrise, Paragon and the web-enabled Paragon variant; and TruBridge [2]
EMIS	Enlivio Health for the corporate entity, with EMIS Web retained as the product name, rebranded 25 September 2026 after an acquisition reported as closing 13 March 2026, itself following a 2022 purchase [3][4][5][8]
Vision; Medicus	OneAdvanced-owned from August 2025; and Doctolib-owned from May 2026, Medicus having been the first new core general practice system approved in England in about 25 years, in June 2025 [3]
Agfa	Split. The hospital record, ORBIS, went to Dedalus in a transaction announced December 2019 and finalised by May 2020, while **Agfa retained its imaging IT business**, which is the product you meet in a radiology procurement [10][11]
IBM Micromedex	Merative Micromedex, since the sale completed 30 June 2022 — though South African sources, including one private group's own material, still use the older name, so record both [9][19]
Alfabet; MEGA; LeanIX	Bizzdesign products, the Alfabet addition announced 9 January 2025; and SAP LeanIX [12][13]
InterSystems	Two different things. See below

One caution on that table. The reported value of the Agfa transaction differs between sources — €975 million in one account and €945 million in another — and the research could not reconcile them, so both are recorded and neither is used as a figure [10][11]. The ownership record for one mid-market suite could not be verified at all; where that is true, say so rather than asserting.

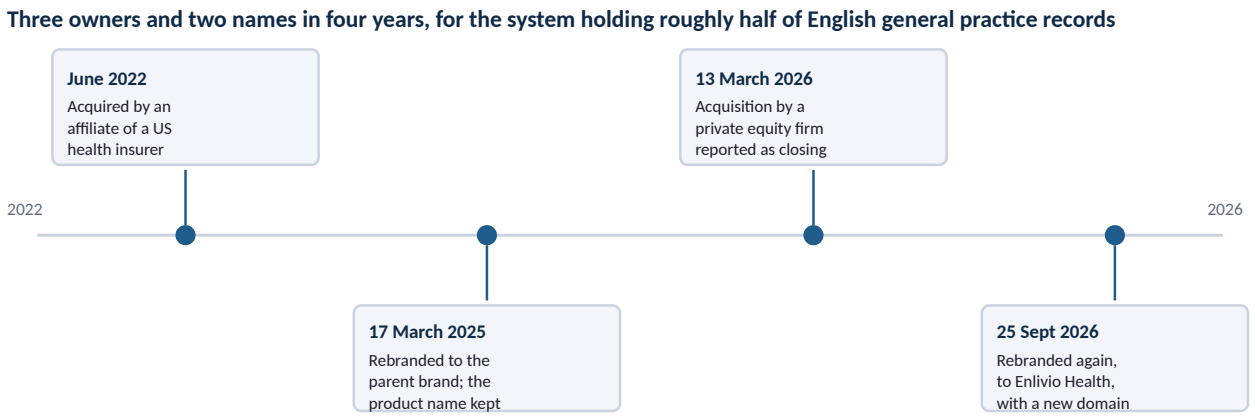


Figure 17.1 Concentration is extremely durable; ownership is not. Write the contract against the product and the data, not against the company.

Three consequences follow, none of them trivia. The catalogue field discipline above. **Support and roadmap commitments transfer with uncertain fidelity**: when a product line moves to a general private-equity owner, or to a platform company that wants it as a feed into an infrastructure business, the first thing to change is the investment profile of the parts the new owner does not consider strategic. And **the exit clause is written against a counterparty that may not exist in its current form when you need to invoke it** — an extraction right that depends on the goodwill of a named account team is not a right.

InterSystems must be separated into two. As a *platform* supplier it sells IRIS, IRIS for Health and HealthShare Health Connect, with the 2026.1 release announced as generally available [14]. As an *application* supplier it sells TrakCare, the system underpinning the National Health Laboratory Service's national laboratory operations [15] — which makes it, by test volume, arguably the most consequential single clinical software installation in the country. Holding both positions is legitimate and common, and it is a structural point to surface: **if the integration engine and a major application come from one supplier, the cost of an interface and the cost of a module become the same negotiation.** The mitigation is separately priced, separately terminable line items, and interface specifications as deliverables in a standard format rather than as engine configuration.

One naming trap will catch a South African architect reading a provincial inventory: a parliamentary reply tabled 1 August 2025 records "Meditech" in use in three Free State public hospitals [17], while the American supplier of that name states it has operated in South Africa since 1982 but names no South African customer and gives no installation count [16]. Establish which is meant before you draw it.

And the most-misattributed fact in South African private healthcare: the flagship private clinical platform in the country is a ten-year internal build on a licensed base platform, iMedOne from Deutsche Telekom Clinical Solutions, not either of the two American suites it is usually credited to [18][19][20][21].

What a suite costs the architecture

Organise this by what you inherit rather than by supplier: the inherited properties are what the architecture has to live with, and they are not on the feature matrix.

The five things you inherit that are not on the feature matrix

The identifier model	Can it hold a national identity number, a check-digited local folder number and a scheme membership number at once, with validity periods? *the question to ask*	If not, you will build a shadow index, and the shadow index becomes the real master
The terminology bindings	Which code systems are bound where, and can a code system the product does not support be added without a supplier development project? *the question to ask*	Bindings fixed at purchase go out of line with the national terminology position inside the term
The consent and access model	Is consent an attribute of the patient, the encounter, the document or the relationship? *the question to ask*	That choice decides whether break-glass, sensitive-record restriction and withdrawal are possible
The audit model	What is logged, at what granularity, retained how long, and can it be exported to our own security monitoring in a standard format? *the question to ask*	Every read of a clinical document, with user, role, patient, reason. Full audit is not an answer
The reporting latency chain	Which store does each dashboard read, and what is the lag between the operational store, the relational mirror and the warehouse? *the question to ask*	A view that mixes layers is internally inconsistent in a way that is very hard to diagnose

Figure 17.2 The five inherited properties, the question that exposes each, and what it costs when the answer is wrong.

The data model. One published example makes the shape concrete. A major suite's real-time clinical database, its relational reporting layer and its warehouse layer are three distinct stores [7]: a hierarchical operational store that ordinary analytics tooling cannot query, a relational mirror on a lag, and a dimensional warehouse on a further lag. **Latency between those tiers is an architecture decision with clinical consequences**, almost never discussed in the sales cycle. Every dashboard, quality indicator and capacity model reads one of the three, and a view that mixes them is internally inconsistent in a way that is very hard to diagnose later. Model it as named stores with stated lags, in the repository, on day one.

The identifier model is the one that most often produces a shadow index. The Western Cape public-sector arrangement is instructive because it is documented: facilities issue a nine-digit folder number whose last digit is a modulus-10 checksum, printed on barcoded labels, and the provincial patient master index is derived from the hospital patient administration system [30]. If the platform cannot carry a check-digited local folder number *alongside* a national identifier *alongside* a scheme membership number, with validity periods, you will build a shadow index and the shadow index will become the real master. Ask to see the identifier data structure, not the identifier screen.

The terminology bindings matter now for the reason chapter 16 gives: the gazetted framework names neither of the two code systems the national terminology commitment implies [24]. **A platform whose bindings are fixed at purchase will be out of line with the national terminology position inside the life of the contract.** Ask whether a code system the product does not support can be added without a supplier development project.

The integration surface. The commercially interesting fact about the largest suite is that the terms on which third parties may read and write to its record have become litigated in its home market [7]. Do not treat litigation as proof of anything alleged; the narrow, solid point is that those terms are *contractual rather than technical*, so they are negotiable and belong in your contract rather than your assumption set. From the other direction, the same research house reports that customers of a mid-market suite want broader platform capability **specifically to reduce their dependence on third-party tools** [2]. That is the central dynamic of the suite market stated by the buyers: every gap becomes a third-party system, every third-party system an interface, and every interface a cost and a failure mode. The decision is *which five gaps am I prepared to fill with interfaces, and what do those cost over ten years*.

The customisation you are allowed could not be established from a citable source for any suite in this research, and saying so is more useful than guessing. What *is* established is the international localisation record, and it is the part a South African architect should read closely. A 2016 Danish rollout across 18 hospitals drew criticism over training and translation; a Finnish deployment had cost about €625 million by November 2022, with clinicians filing complaints about usability and patient safety; a Norwegian deployment was the subject of a government audit reportedly finding it put patients at risk and cost more than expected; an English teaching-hospital installation cost £200 million in 2014 and suffered serious stability problems [7]; and a Swedish region postponed its rollout again on 21 January 2026, saying the system "is not usable" [6]. The honest reading is **not** that these implementations failed, but that every one of those difficulties is a localisation and clinical-practice-alignment difficulty rather than a software defect — and that the published cost figures are an order of magnitude above what a first-time buyer budgets. So: who pays for the localisation of order sets, formularies, tariff structures and coding; who owns the translated content; what happens to it at upgrade; and which published international programme is the closest analogue by size, clinical mix and funding model? **Ask for the analogue by name, and ask to speak to it without the supplier present.**

The upgrade cadence, and why "cloud or not" is the wrong question. Every suite now answers "cloud", and the answer is uninformative. Replace the deployment-model table with five narrower questions, whose answers determine the operating model, the security boundary and the exit cost.

1. **Who holds the instance** — dedicated single-tenant, multi-tenant, or a tenant inside another organisation's instance? This determines who decides when you upgrade.
2. **Who holds the cloud contract** — the software supplier reselling hyperscaler capacity, or you contracting the hyperscaler directly with the supplier as a managed service? This determines who is liable in an outage and who can be audited.
3. **Where does the data sit, physically, and under which law?**
4. **What is the upgrade obligation** — mandatory on a schedule, mandatory within a window, or elective with a supported-version floor? A mandatory-on-schedule arrangement is incompatible with clinical regression testing of content before each release, and that must be resolved in the contract rather than discovered in year two.
5. **What is the egress position** — who pays to get the data out, in what format, and how long does the supplier retain it after termination?

What an interface costs, and who owns its specification. This is the most reliably under-estimated line in any healthcare procurement. The pricing usually comprises a one-off development or enablement fee, an annual support fee per interface, a change fee when either end changes, and frequently a fee for the test environment and the test cycle. It is priced at all because a suite supplier's incentive is to supply the whole estate, so an interface to a third-party departmental system is a product it sells reluctantly.

There is a South African counter-example, and it is the benchmark to hold every other proposition against. The largest claims switch publishes a single XML specification with schemas suppliers use to validate their requests and responses, provides an integration pack, runs a test environment available at all hours that mirrors live funder rules and pricing with dummy beneficiary data, performs a live-environment accreditation, and maintains a dedicated integrations team [22]. **That is what good looks like**: one published specification, a conformance artefact, a test environment and an accreditation step.

> **WHO OWNS THE SPECIFICATION MATTERS MORE THAN WHAT IT COSTS**
>
> If the interface specification is a supplier artefact — engine configuration, a proprietary mapping document, a "we will handle it" arrangement — the organisation cannot rebuild the interface against a different platform, test it independently, or describe it to a replacement supplier. **The specification must be a contractual deliverable, owned by the organisation, in durable form: message structure, field mappings, trigger events, code mappings, acknowledgement and error handling, retry semantics, and the test cases that prove it.** It is the only portable asset an integration programme leaves behind, cheap to require at the outset and impossible to obtain afterwards.

Two further inherited properties complete the set. **The consent and access model** — whether consent is an attribute of the patient, the encounter, the document or the relationship — determines whether a break-glass path, a sensitive-record restriction or a withdrawal of sharing consent can be implemented at all. **The audit model** needs a specific answer rather than "we have full audit": every read of a clinical document logged with user, role, patient, document, timestamp and reason, retained for a stated period, exportable to the organisation's own security monitoring in a standard format.

Open source, honestly

The honest position is more interesting than either the enthusiast's or the sceptic's: some of these projects run at national scale with institutional backing and a real support market, others are archived, and the difference is almost entirely about who maintains them rather than the quality of the code. So the question is not "is it good software" but four questions. **Who maintains it, and is that body funded? Who runs it at scale, and can I speak to them? Who will support me, under contract, in my jurisdiction, for ten years? And what does the implementation cost, against a licence cost of nil?**

Two of the four due-diligence questions, answered for six options as at 31 October 2026

Question	OpenMRS	Bahmni	OpenEMR	HospitalRun	DHIS2	OpenHIM
Who maintains it, and is that body funded?	A registered non-profit; core teams include the SA Medical Research Council	Could not be established. Only a July 2017 coalition proposal was retrievable	The OpenEMR project, with a community wiki	Nobody. Archived by its owner 9 January 2023, read-only	The HISP Centre, University of Oslo, a WHO Collaborating Centre	A Cape Town non-profit, with UN development programme support
Evidence of life	Platform 2.7.0 on 9 Dec 2024; 3.2.1 on 23 Jan 2025	Not established	Project-published certification for version 7.0.0	Last status date 2 July 2020; README still claims a v2	75-plus countries, 69 at national scale at end-2022	In production in a provincial data centre

Check the repository, not the website. A project website is marketing; a commit history is evidence.
And check for the contradiction: a v2 announced inside an archived repository is a project that stopped mid-transition.

Figure 17.3 Two of the four questions, answered from public evidence. Red cells are the finding, not a judgement on the code.

OpenMRS is a Java web application under the Mozilla Public License 2.0 with an added healthcare disclaimer, owned and licensed by a registered non-profit, with core teams drawn from several institutions including the South African Medical Research Council — one of its founders came from there, so this is not a foreign project South Africa might adopt [25]. Its release cadence is the most reassuring fact available: a platform release on 9 December 2024 and a stable release on 23 January 2025 [25]. What ministries deploy, though, is a *country distribution* with its own forms, concepts, reports and release train, so an architect evaluating OpenMRS is evaluating a distribution whose maintainer — frequently a local implementing partner on donor funding — is the real counterparty. And the concept dictionary *is* the data model, configurable to the point of being a liability: two installations can be semantically incompatible while both valid, so concept-dictionary governance is not optional and does not come with the software.

Bahmni, a distribution built on OpenMRS, is the cautionary case for question one. The research could not establish from a fetchable source who maintains it today: the most recent retrievable governance statement is a July 2017 coalition proposal in which its original developer is referred to as a *past* contributor [26]. That is not a dismissal; it is the exact finding a procurement needs. **If a researcher with a day to spend cannot answer who maintains it, a procurement team with a deadline cannot assume it.** Ask in writing, require a named body, a named funding source and a named release manager, before the pilot rather than after.

OpenEMR is the one option here for which the research found a *project-published certification artefact against a regulator's criteria* rather than a self-assessment: the project publishes the requirements it met for United States ambulatory certification of version 7.0.0 [27]. That is not a South African compliance statement and does not substitute for conformance testing at a department-approved laboratory [24], but the existence of a conformance artefact at all distinguishes it from most products in this market. It is also an ambulatory product, so deploying it in a hospital means meeting the gaps at the inpatient boundary — bed management, theatre, medicines administration, discharge — and filling them with interfaces.

HospitalRun is archived, and belongs here as a due-diligence lesson rather than as an option. The repository was archived by its owner on 9 January 2023 and is read-only. It shows 705 commits, 991 stars and 378 forks, a project-status table whose most recent date is 2 July 2020, and a README stating that the 1.0.0-beta version is no longer supported and that version 2 is under development — a statement the archived status contradicts, with nothing saying whether that development continues elsewhere [28]. It is still cited in general-interest listings of open-source hospital systems, so an architect can be handed it in 2026 by someone who read a listicle and not the repository. Two lessons generalise. **Check the repository, not the website**: ninety seconds on the archive banner, the last commit date, the number of distinct recent committers and the age of open issues beats a day of project documentation, because a website is marketing and a commit history is evidence. And **check for the contradiction**: a version 2 announced inside an archived repository is a project that stopped mid-transition, which is the worst state to adopt into, because the documentation describes a system that does not exist.

Two options that belong here and are usually missing. DHIS2 is developed by a university centre that is a World Health Organization Collaborating Centre, licensed under a permissive three-clause licence, used by ministries of health in more than 75 low- and middle-income countries with 69 at national scale as at the end of 2022 — and it **originated in Cape Town**, as a research project between two universities whose first software was built for Cape Town districts [29]. Three dates for that origin circulate, 1994, 1995 and 1998, and they are not reconcilable from the available sources, so state the range and the substantive point: the global standard for routine health information reporting in low- and middle-income countries was born in a Cape Town district health office [29]. **And be explicit that DHIS2 is not an electronic health record.** It handles aggregate facility reporting and, through its tracker, individual-level longitudinal data; neither is a clinical record of care. Deploying it and believing you have a record is one of the most common and expensive architectural errors in African health informatics, and the error runs both ways — using the tracker for clinical workflow it was not designed for, and building a clinical system without planning the aggregate reporting obligation. The second missing option is OpenHIM, the integration mediator of chapter 16, South African-built and in production in a provincial health data centre [30][31].

> **THE HONEST TOTAL COST OF "FREE"**
>
> The licence line goes to nil. These do not. **Terminology and concept-dictionary governance**, a standing function rather than a project task. **Localisation** — forms, order sets, formulary, coding, report definitions, language. **Hosting and operations to clinical availability**, because an availability failure in a clinical system is a clinical event and the engineering to prevent it is identical whether the software is proprietary or free. **Integration** — pathology, radiology, pharmacy, funder claiming, identity, each a build. **Security and patching**: the national laboratory service's own account of the June 2024 ransomware incident was that it had not been able to update systems or apply security patches [32], a lesson about operational capacity rather than licence models. And **support continuity**: a commercial supplier's insolvency is an event with a legal process, while an implementing partner's loss of a donor grant is an event with no process at all. Open source converts the licence line into an internal capability obligation. If the organisation can fund and retain that capability the trade is often strongly favourable; if the saving is counted without the standing team, it has bought an unsupported clinical system.

It is the instance set, not the second instance

Almost every healthcare software contract is negotiated for one production instance. The organisation then discovers it needs more, each with a licence, hosting, support, refresh and administration cost that was not in the business case.

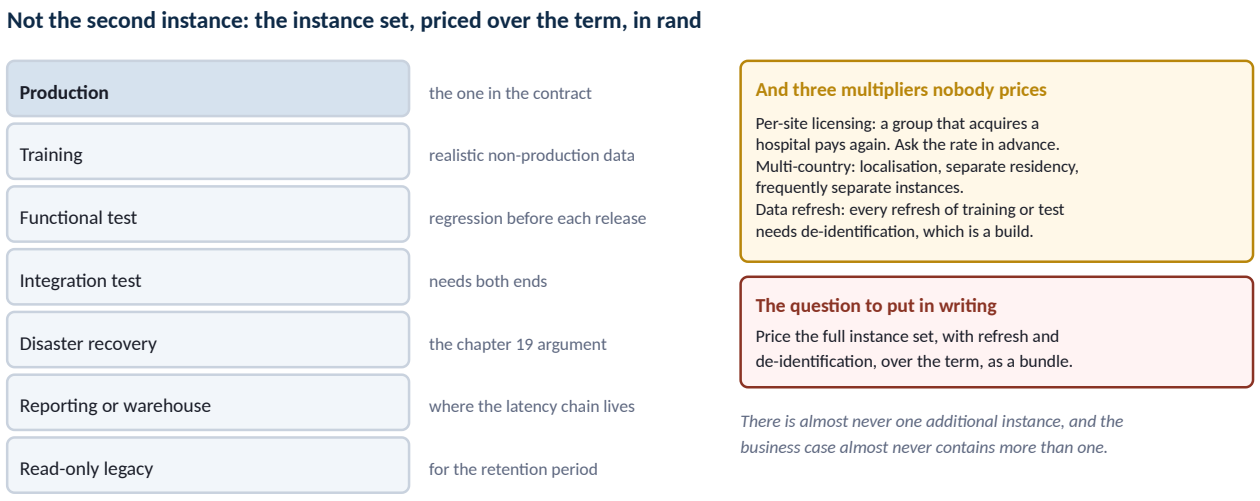


Figure 17.4 Seven instances, three multipliers, and one question. The business case almost never contains more than the first line.

There is almost no sourceable pricing, and saying so is the finding

Apply the rule that no price appears without a source and a date, and this is the result: **the research behind this volume found no published list price for any clinical platform and none for any enterprise architecture repository tool.** That is the market norm for enterprise software rather than a scandal, and it has two consequences. Benchmarking requires peers, not websites — and in a market the size of South Africa the peer set is small enough that an organisation should ask the supplier for South African references and then ask those references about commercial terms directly. And the architect's contribution is not a price; it is a model. The dated anchors the research does establish are orders of magnitude, worth carrying because they calibrate a business case.

Anchor	Figure and date	What it is
International programme totals	About €625 million by November 2022 for a Finnish region; £200 million in 2014 for one English teaching hospital; £77 million paid in 2018 for two English primary care products together	Programme and installation totals, in euro and sterling, against software covering most of a nation's general practice [7][8]
South African public ICT line items	About R15,000 per computer, about ten per facility, a R13 million national computer budget for the year, and R20 million for mobile routers plus two years of data from September 2025	2025 rand, from a parliamentary reply [17]
Provincial connectivity and donor funding	R29 million Eastern Cape connectivity budget, and about R435 million of PEPFAR funding, 2025	2025 rand, from committee evidence [35]
The national primary care record rollout, and laboratory incident remediation	Estimated to exceed R50 million, June 2026; and at least R300 million including R164 million for switches, firewalls and security, March 2025	Rollout *services* rather than software, and 2025 rand [32] [33]

No South African clinical platform licence cost could be established at all, and that is recorded here as an absence rather than filled with an estimate.

What replaces a price list is the **licence-metric model**. Healthcare software is licensed on more different bases than almost any other category, and the basis matters more than the rate because it determines how cost behaves when the organisation grows, shrinks, merges or changes its service mix.

Metric	Where it bites, and what to ask
Per bed	Beds are a poor proxy for load: a 200-bed hospital with large day-surgery and outpatient volume generates far more encounters than its bed count suggests, and bed numbers are politically managed. Ask what a licensed bed is, who counts them, and what happens when a ward closes
Per named, or per concurrent, user	Named users are predictable and catastrophic where the population includes rotating junior doctors, agency nurses, locums, students and visiting specialists; concurrent suits shift working and requires the organisation to know its own peak concurrency, which it usually does not. Ask how a named user is reassigned and how quickly, and ask for the concurrency measurement method and the true-up mechanism
Per encounter, episode, admission, facility or site	Activity-based cost rises exactly when activity rises, which in a publicly funded setting is when the budget does not; per-site makes the second site expensive. Ask what counts as an encounter — a telephone follow-up, a cancelled appointment, an outpatient attendance against an admission — and what counts as a site: a legal entity, a building, or a registered facility
Per transaction or funder claim	The switch model. One South African switch states more than 107 million transactions processed or switched annually, and another reported more than R30 billion in transactions processed in real time each year as at 2017 [44][45]. At that volume a fraction of a cent per transaction is a material revenue line, and the provider's cost is proportional to its claiming activity
Per module, or content subscription	A base platform plus priced capabilities; and reference data — interaction content, terminology, tariff and coding files — charged annually, independent of usage, the safety function stopping when the subscription does

The architect's job is not to find the cheapest metric. It is to model the organisation's own ten-year activity trajectory against each proposed metric and identify which produces an unaffordable number under the most likely growth scenario. That modelling takes a day, it is almost never done, and it is the single highest-return commercial contribution an architect makes to a procurement.

And then the written questions. The research behind this volume sets out 34, grouped so they can be lifted into a procurement document, every one answerable in writing: the data model and identity, including merge and unmerge behaviour and what happens to images, laboratory accessions and transfusion records attached to a losing identity; integration, including the itemised interface schedule and the specification-ownership requirement above; deployment, hosting and residency; availability and resilience; exit; people and delivery; and safety and governance, including the hazard log and the named clinical safety officer who signed it. Two of the 34 settle more than the rest. **Will you provide a test extract during the term, on request, so we can verify it is usable before we depend on it?** — because an untested extraction right is a promise. And **name a customer running this product and these interfaces, in production, on the same versions, in this country, and arrange for us to speak to their integration engineer without you present.** A supplier who cannot or will not arrange that has given you the answer.

WHY EXIT BELONGS IN THE ARCHITECTURE, NOT IN THE LEGAL ANNEX

An exit clause is normally treated as a legal matter handled after the architecture is agreed. That sequence is wrong, because **the exit clause determines what the architecture can do**: if the organisation cannot extract its clinical data in usable form, every later option — a warehouse, a population health platform, a research dataset, a migration, a federated record, a model — is contingent on the incumbent's cooperation. Healthcare makes it harder than other sectors for four reasons. Retention periods are long and statutory, so the plan must answer what happens to records the organisation must keep longer than it keeps the system. The data is not in one place, so "give us our data" has three answers with different content, and a warehouse extract is a reporting projection rather than the record. **Referential meaning lives in configuration, so the configuration is part of the data** — an extract of coded values with no code-set, order-set, template, user or role definitions is uninterpretable. And the audit trail may be the thing you need most and the thing least likely to be extractable, because a medico-legal enquiry five years after migration asks who saw what, when. The evidence that this is live rather than theoretical: about 35% of sampled customers of one major platform "might leave or want to leave but can't" [2] — the definition of a failed exit clause, reported at scale, in the largest healthcare software market in the world.

Two South African commercial facts that change a design

Currency. The most practically useful commercial fact in this research is a sentence from a mid-sized South African hospital group. It consolidated several on-premises datacentres into a hosted private cloud, and chose private cloud **partly because many suppliers billed in United States dollars, which made hosting costs hard to forecast against rand volatility** [18]. A dollar-denominated contract transfers foreign-exchange risk onto a rand revenue line for the life of the contract. Over a seven-year platform term that is not a rounding error and is not hedgeable at reasonable cost by a mid-sized group. It is a legitimate, documented reason a South African organisation chooses an architecture a European or American reference model would call retrograde. **Price every option in rand over the full term, state the foreign-exchange assumption, and treat dollar-denominated recurring cost as a risk with a named owner.**

Funding discontinuity, stated as a principle. Grant-funded components have operating lifetimes shorter than the systems they fund: about R435 million of PEPFAR funding to the Eastern Cape was issued in January 2025 and expected to be withdrawn in September 2025 [35], and a donor digital health accelerator was tied to a 2023 to 2025 grant cycle [34]. Running costs on two- and three-year grant cycles; systems with ten- to twenty-year lifetimes.

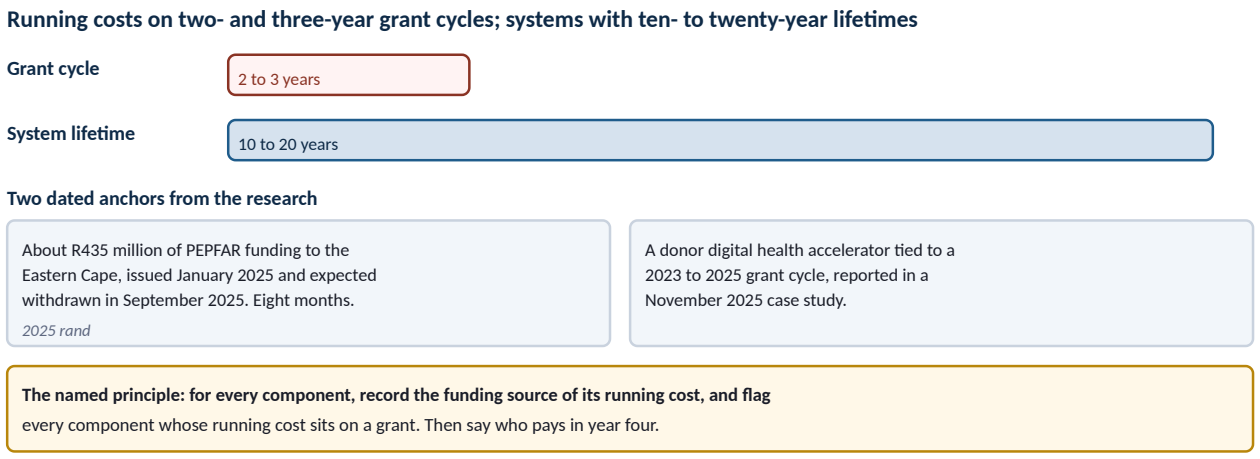


Figure 17.5 Two dated anchors for the mismatch between a grant cycle and a system lifetime, and the principle that follows from it.

THE NAMED PRINCIPLE

For every component in the estate, record the funding source of its running cost, and flag every component whose running cost sits on a grant. Then answer, in the architecture document, who pays in year four. It is a one-column addition to an application catalogue, and the difference between an estate that degrades predictably and one that stops.

Enterprise architecture tooling, briefly and squarely

The healthcare job differs from the generic one in five ways that determine which tool characteristics matter. The application portfolio is unusually large relative to the organisation's size and unusually fragmented, so **bulk import from spreadsheets, repeatedly, with reconciliation, is the primary data-acquisition mechanism rather than a nice-to-have**. The *interface* inventory is the valuable artefact, so the metamodel must represent interfaces as first-class objects with direction, payload, standard, trigger and criticality. The data objects are clinically specific and the standard metamodels do not contain them, so the tool must let you add attributes to extension types rather than merely add the types. The safety and regulatory overlay has to be traceable, so attaching and governing evidence against objects, with dates and owners, matters more than an attractive diagram. And the audience includes clinicians, who will not log in, so read-only browsable publication is the delivery mechanism and the viewer licensing model is a commercial question with a practical consequence.

On the same evidence standard as everything else in this chapter: **the research found no verifiable evidence that any enterprise architecture tool ships maintained healthcare reference content** — a capability model, a clinical process library, an application taxonomy, a standards metamodel extension or a profile catalogue. Several suppliers reference reference content generically. Assume you are building the healthcare content yourself, and treat any claim of it as a claim to inspect: ask to see it, ask when it was last updated, ask who maintains it.

On pricing, the same finding as for clinical platforms. One supplier publishes a pricing *page* describing a modular structure — a base platform including reports, diagramming, an interface and viewer licences, plus optional add-ons — and three licence roles, with viewers listed as unlimited, authors able to administer the repository, and contributors able to edit within a permitted scope. **It gives no figures and no currency** [36]. For every other tool named in the research, no supplier-published pricing was located at all. The **licence role model is therefore the lever, not the headline price**: unlimited viewers answer the clinician-audience requirement directly, and a scoped contributor role maps onto twenty departmental system owners each maintaining their own entry. So model your actual user population by role before asking for a quote, and ask every supplier to price against that same population so the quotes are comparable. The currency point applies here too: a tool is a small line item next to a clinical platform, and a multi-year recurring one.

Two ownership facts belong with the clinical ledger. An architecture repository owned by an application supplier is a repository with a commercial interest in the conclusions [13] — not an accusation, a governance observation, and the mitigation is ordinary: the organisation owns the data, the extraction right is contractual, and the conclusions are reviewed by people without a stake in the application roadmap. Raise it explicitly where the organisation's own resource-planning spine comes from the same supplier. And one group now holds three overlapping repository products while developing a fourth, new platform, by its own stated plan [12] — the configuration that most often produces a future migration for customers of the products not chosen as the strategic line, while a long-standing process-analysis product was launched as a standalone business in January 2025 as its former group was dismantled [38]. **Ask, in writing: which product is the strategic line, what is the committed support horizon for the others, and what are our terms if you migrate us?**

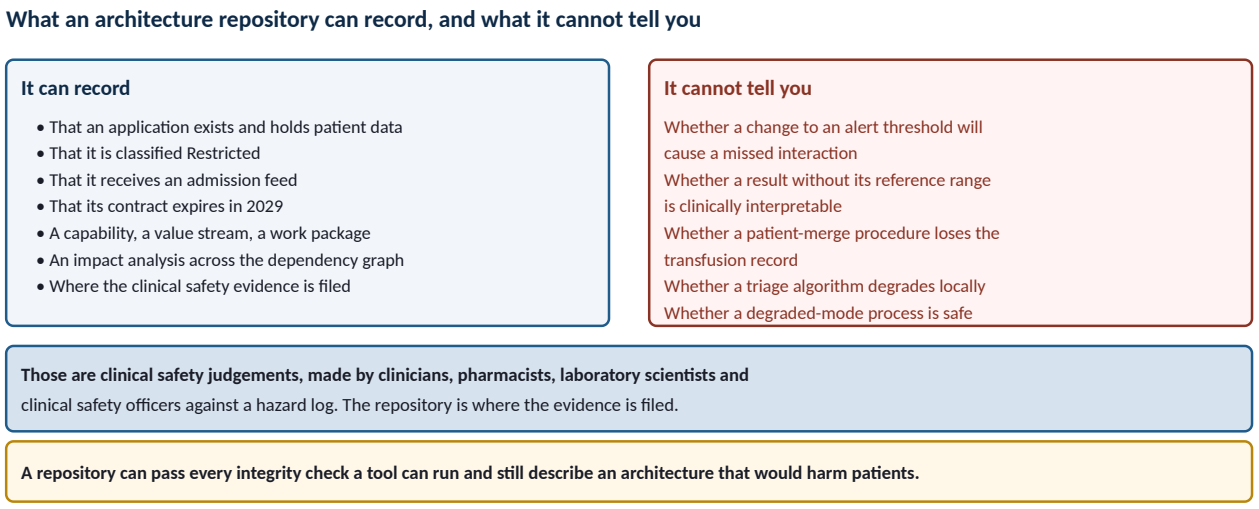


Figure 17.6 The honest limit of architecture tooling. It applies equally to every product named in this chapter.

The practical implication is the useful part, and almost nobody does it: model the safety artefacts as objects — hazard log, clinical safety case and its approval, impact assessment, conformance attestation, device classification — attach them to the applications and interfaces they cover, give them owners and review dates, and report on the ones missing or expired. That is a thing the tooling is genuinely good at.

Eight claims not to accept at face value

The research behind this volume describes these as seven claims and then sets out eight. Each is usually made in good faith and none is self-evidently false. The evidence test is the point.

The claim	What would settle it
"We are fully interoperable"	Interoperability is a property of a *pair* of products at *specific versions* exchanging *specific content* for a *specific purpose*. Require a named use case, a named counterparty at a named version and a named profile; a conformance attestation from a department-approved laboratory, with version, date and profiles covered [24]; a field-level interface specification you can read; and a production reference at the same versions in the same country. The sharpest single question: *name the field in your message that carries the national patient index identifier, and show me the message*
"We support FHIR"	A version number; the implementation guide the product conforms to and the profiles it publishes; the capability statement from the actual instance rather than the documentation; a public conformance test result naming the test kit and date [23]; a sandbox your engineers can call; and read *and* write demonstrated for the resources the use case needs, because write support is where most implementations stop. The South African framework is profile-based, so a FHIR interface not wrapped in the profiles it specifies is not aligned to it [24]
"Our AI achieves X% accuracy"	The verbatim intended-use statement — screening, triage, detection, diagnosis and decision support are different claims; the regulatory clearance by jurisdiction and indication, with the document; prospective performance in a population like yours, with positive and negative predictive value at *local* prevalence; the training population composition including equipment; the failure mode when the model is uncertain; the monitoring plan and what triggers withdrawal; and a clinical safety case with a named signatory [42]

The claim	What would settle it
"We guarantee 99,9% uptime"	The measurement definition and the component it is measured at; the exclusion list read in full, with power and third-party network checked; actual monthly availability for the last 24 months for customers of similar size, with the incident reports; the tested recovery objectives and the date of the last end-to-end restore; and the designed degraded-mode operation in writing
"This will be your single source of truth"	Do not try to settle it. Replace it with a **source-of-truth register**: per significant data element, which system is authoritative, which hold copies, the direction and latency of each copy, and who owns the reconciliation. Even inside one platform there are several stores with different latencies [7]; departmental systems stay authoritative for the specimen, the image, the cumulative dose and the dispensed item; a retrospective adjudication can change the financial truth after the account closes [40]; and identity is distributed whether you like it or not [30]
"Go-live in twelve weeks"	A reference implementation of comparable scope, scale and country, with actual and planned dates and the reason for the difference, from the customer; the definition of go-live — first facility, last department, or decommissioning of the predecessor, the last being the only one that matters and never the one quoted; the staffing assumption including clinical backfill; the adoption measure, because "live" is not "used"; and what was descoped. One South African patient-portal study found 42% of 150 participants registered and 26% logged in, against 96% in the comparison group, concluding implementation in the current public environment is not feasible without adaptations [39]
"We have N% market share", and "we are certified"	For share: the measurement unit, the geography, the source, the date and the denominator — share in the one consistent dataset differs by thirteen percentage points between hospitals and beds for a single supplier [1][2], and no South African dataset exists. For certification: the certificate itself, the body, the scope statement, the validity dates and the last audit's findings, because a certificate scoped to the corporate head office does not cover the clinical platform [43]

> **THE STAFFING ARITHMETIC BELONGS IN THE BUSINESS CASE**
>
> The national primary care record rollout specifies, per province, a technical lead, a project lead and a clinical trainer, and per district two teams of two, for about 3,083 facilities, with at least two days per facility for training and go-live and a follow-up about a month later, over 18 months [33]. Do the arithmetic that implies: of the order of ten thousand facility-days of field work in eighteen months, across eight provinces, delivered by several separately awarded providers. **That is a logistics and workforce programme wearing the costume of an information technology project.** The private comparator is the same shape — 40 workstreams across three countries over seven years [19][21]. Three realities follow. The scarce skill is clinical informatics rather than technology, and in South Africa that pool is very small and internationally mobile. **Clinical backfill is the largest unbudgeted cost**, and without a backfill line the programme is delivered by exhausted volunteers. And the partner's continuity is a risk you own, which for an open-source or locally built option may mean a small non-profit on project funding.

What to take to the client

Four deliverables, none of which requires naming a preferred supplier. **A sourcing position per capability**, applying chapter 15's four-way question, with the permanent internal capability each option requires stated alongside it. **A priced instance set and a licence-metric model**, over the contract term, in rand, with the foreign-exchange assumption stated and a named owner for the currency risk. **A specification and exit position**: interface specifications as owned contractual deliverables, and a *tested* extract obtained during the term rather than promised for the end of it. And **a funding-source column** on the application catalogue, flagging every component whose running cost sits on a grant, with the year-four answer written down.

And one sentence to keep in the document itself: **the deliverable is the evaluation method, not the answer.**

THE CHAPTER IN TEN LINES

- Vendors appear here as examples, never as recommendations. Say the same sentence in your own deliverable, or a name will be read as an endorsement.
- The platform decision is effectively irreversible: net-new decisions in the one measured market fell about 40% against 2024 and nearly 50% against 2023 in contract year 2025.
- There is no measurable global market. The only consistent annually-refreshed share series is United States acute care; Europe yields ordinal rankings, the UK a 2016 census, Africa and the Gulf nothing.
- The South African absence of share data is a finding, because it means a South African supplier's share claim cannot be checked.
- Record supplier as at a date, plus product line origin: Cerner is Oracle Health, Allscripts is Altera, CPSI is TruBridge, EMIS is Enlivio Health, Alfabet and MEGA are Bizzdesign products, LeanIX is SAP LeanIX, and Agfa split.
- Drop the deployment-model table. Every suite answers cloud. Ask instead who holds the instance, who holds the cloud contract, where the data sits, what the upgrade obligation is, and what the egress position is. And make the interface specification a contractual deliverable the organisation owns.
- HospitalRun was archived on 9 January 2023 and is read-only while its README still claims a version 2 in development. Check the repository, not the website, and check for the contradiction.
- DHIS2 and OpenHIM belong in any honest open-source list for this market, and DHIS2 is explicitly not an electronic health record. Separate InterSystems as platform from InterSystems as application.
- Price the instance set, not the second instance: production, training, functional test, integration test, disaster recovery, reporting and read-only legacy, each with a recurring de-identified refresh.
- There is almost no sourceable pricing, and saying so is the finding. Give the dated anchors, state the absence, and replace the price list with the licence-metric model and the written questions.
- A dollar-denominated contract puts foreign-exchange risk on a rand revenue line for the whole term. One documented South African group chose private cloud for exactly that reason.
- Donor funding discontinuity is a named principle: record the funding source of every component's running cost, flag the ones on a grant, and answer who pays in year four.
- No enterprise architecture tool was found to ship maintained healthcare reference content, and none publishes a list price. The repository is where safety evidence is filed, not where safety judgements are made.

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CHAPTER 18

Regulation, privacy and the records duty

The default is prohibition. Everything a South African health architecture does with a record is an exception that needs a name.

In most industries privacy is a constraint applied to an architecture after the architecture exists. In South African healthcare it *is* the architecture, for four sourced reasons. Health data is **special personal information** under POPIA, so the default is prohibition with named exceptions rather than permission with limits [1]. Unauthorised access to a health record is a **criminal offence** under National Health Act s17(2), punishable by a fine or up to one year's imprisonment, independent of POPIA [3]. Since **6 March 2026** a dedicated regulation has governed health-information processing — binding funders, administrators, managed-care organisations, employers and pension funds, and not providers [4]. And retention periods reach **twenty-five years** while the instruments setting them disagree, so the retention schedule drives the platform choice rather than the other way round [8][9].

Chapter 5 named the regulators and said who inspects what. This chapter is what each can do, and what that requires of a design. It runs as the powers run: POPIA, then the National Health Act because where the two differ it usually wins, then the Information Regulator's enforcement machinery, the Office of Health Standards Compliance, the Health Ombud and SAHPRA — and last the records duty, where chapter 13 owns the record as an object and this chapter owns the duty to keep it and the authority to destroy it.

POPIA, properly

What "special personal information" actually changes

Section 26 opens with a prohibition. A responsible party may, subject to section 27, **not** process personal information concerning religious or philosophical beliefs; race or ethnic origin; trade union membership; political persuasion; **health or sex life**; **biometric information**; or criminal behaviour, meaning an alleged offence, proceedings relating to one, or the disposal of such proceedings [1].

Read items five and six together, because they bite in a way architects do not expect. A hospital using fingerprint or facial identification at registration is processing **two** categories of special personal information about one person, under two authorisations. The public sector's biometric identification capability, at 82 primary health care facilities as at 31 March 2025, is in that position [5].

POPIA's eight conditions for lawful processing apply to health data as to everything else: accountability at s8, which is what requires an Information Officer; processing limitation at ss9–14; purpose specification at ss13–14; further processing at s15; information quality at s16; openness at ss17–18; security safeguards at ss19–22; and data subject participation at ss23–25 [6]. What Chapter 3 Part B adds is the prohibition and the authorisations that lift it: s27 generally, and s28 to s33 category by category, with **s32** carrying health and sex life [7]. Section 27's grounds, on the best secondary account available, are consent; the establishment, exercise or defence of a right in law; obligations under international public law; research or statistics in the public interest with safeguards; information deliberately made public; and ss28–33 [10].

> **THE TEXT OF SECTION 27 COULD NOT BE RETRIEVED**
>
> The research behind this volume could not retrieve the text of POPIA s27 itself; the account above is assembled from secondary sources describing it. Everything resting on it — and the 2026 regulations rest on it directly — should be checked against the Act's own words before it reaches a deliverable [4][7].

Section 32, set out

This is the provision that makes healthcare lawful, and the paraphrases in circulation are wrong. The s26 prohibition does not apply to processing by these parties [2]:

- **s32(1)(a)** — medical professionals, health care institutions or facilities, or social services, where necessary for **the data subject's proper treatment and care**, or for **administering the institution or practice**;
- **s32(1)(b)** — **insurers, medical schemes, scheme administrators and managed health care organisations**, where necessary for assessing the risk to be insured or covered provided the data subject has not objected, for performing an insurance or medical scheme agreement, or for enforcing contractual rights and obligations;
- **s32(1)(c), (d) and (e)** — schools making special arrangements for a pupil; bodies managing a child's care; and public bodies implementing prison sentences or detention measures;
- **s32(1)(f)** — **administrative bodies, pension funds, employers, or institutions working for them**, where necessary for implementing laws, pension regulations or collective agreements that create health- or sex-life-dependent rights, or for reintegrating or supporting workers affected by sickness or work incapacity.

Then five subsections. **s32(2)** permits processing under s32(1) only by responsible parties **bound by confidentiality** — by office, employment, profession, legal provision or **a written agreement with the data subject**. **s32(3)** requires a party permitted to process but **not** so bound to treat the information as confidential. **s32(4)** extends the authorisation to processing necessary to **supplement** health-information processing under s32(1)(a). **s32(5)** prohibits processing **information on inherited characteristics** about the data subject it came from, unless a serious medical interest prevails or the processing is for historical, statistical or research activity. And **s32(6)** permits more detailed rules **for subsections (1)(b) and (f)** — the regulation-making power, whose scope is the next section.

THREE MISREADINGS A CAREFUL AUTHOR WOULD MAKE

On the retrieved text of s32 [2]: **s32(3) is a confidentiality duty, not a restriction on further processing** — it catches parties who process under s32(1) without already owing a duty of confidence, while further processing is governed by s15. **s32(4) does not concern children**; children are dealt with in **ss34–35**, and what s32(4) does is extend the authorisation to processing that *supplements* s32(1)(a). And **s32(5) concerns inherited characteristics** — genetic information about relatives, in substance — not consultation or codes of conduct. None of the three is pedantic: each changes what a design must implement, and the third is the only place in POPIA that constrains genetic data.

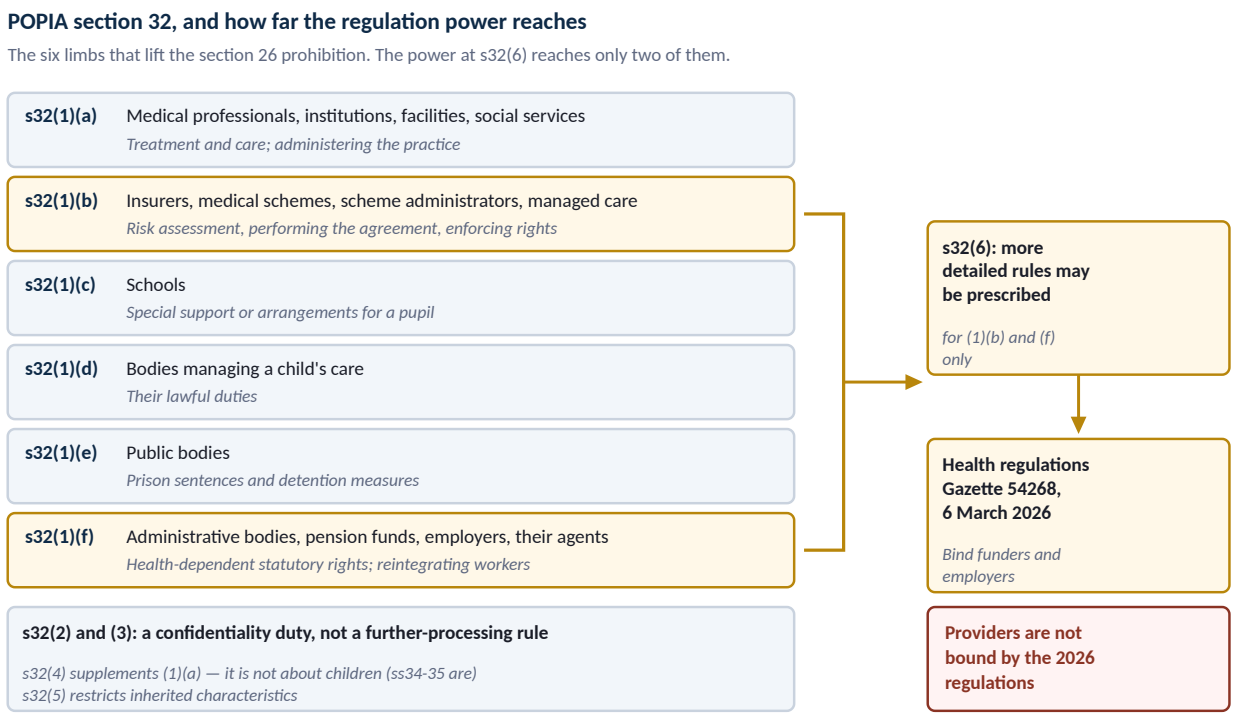


Figure 18.1 Section 32's six limbs, and why the regulation power at s32(6) reaches only funders and employers. The health regulations in force since 6 March 2026 therefore bind schemes, administrators, managed care organisations and employers, and not hospitals, clinics, laboratories or practitioners.

Why the March 2026 regulations do not bind a hospital

The instrument is the *Regulations Relating to the Processing of Data Subjects' Health Information by Certain Responsible Parties, 2026*, published in **Government Gazette No. 54268 on 6 March 2026, effective the same day, with no grace period** [4]. It was issued under **s112(2)(c)** of POPIA to help interpret **s32(6)**, after a draft put out for comment on 26 September 2025 [10].

The bound parties are responsible parties **and operators** in these categories: insurers, medical schemes, scheme administrators, managed health care organisations, administrative bodies, pension funds, employers, and institutions acting for employers, administrative bodies or pension funds [4].

Note who is not in that list: hospitals, clinics, laboratories and practitioners. The regulations give detailed rules for the s32(1)(b) and (f) categories — funders and employers — because that is all s32(6) permits. Providers remain governed by s32(1)(a), the confidentiality duty in s32(2)–(3), the National Health Act and professional rules. An architect at a provider told "the new health regulations require this of us" should check the scope; one at a scheme, administrator, managed care organisation or large employer should assume they apply in full, with nothing to transition.

Two changes between draft and enactment matter. **References to sex-life information were removed**, so the regulations cover health information only and sex-life information remains governed by s32 without regulations. And **employer** was broadened: no longer limited to employers working for administrative bodies or pension funds, no longer tied to the Occupational Health and Safety Act, and now any person or organisation paying others for work under a wage or salary contract [4].

The requirements are short and each one is a design instruction:

The regulation requires	What that means in a design
Processing only where s27 permits it	A named authorisation per processing activity, not a general policy statement
Safeguards against loss, damage, unauthorised destruction and unlawful access	The s19 obligation restated, with destruction as a named failure mode
Records kept secure and confidential, and **disposed of so they cannot be accessed or disclosed afterwards**	A destruction requirement with an evidentiary standard; deletion from an application is not disposal
Processing subject to a legal, employment, professional, office or written duty of confidentiality	The duty must be *created* where it does not already exist
Safeguards aligned to generally accepted security practices for the sector	An external standard imported without being named
No transfer abroad unless POPIA s72(1) is satisfied	A transfer register with a recorded ground per flow

Two rows need a note. On **security practices**, two South African sources name the same standard: the 2022 revision of the record-keeping guidance points at **ISO 27799**, the health-specific information security management standard [9], and the Medical Protection Society's guidance at **ISO/IEC 27002, as SANS 27002:2008, and ISO 27799** [24]. ISO 27799 is the most defensible answer available; this volume names it and describes its purpose rather than reproducing its content. On the **confidentiality duty**, a practitioner has one by virtue of profession, but a claims clerk, a developer with production access, a switch-house operator and an offshore support engineer do not — so contracts and operator agreements are compliance artefacts rather than boilerplate [2][4].

> **ONE THING TO CHECK AGAINST THE GAZETTE**
>
> The secondary source reporting the final text refers to "sections 36(1)(b) and (f)" where the context plainly indicates section 32. That is almost certainly a transcription error in the reporting rather than in the gazette, but it is the kind of error that propagates into deliverables. Read the gazette before quoting a subsection number [4].

Which law wins, and the rule to implement

The National Health Act's confidentiality regime runs alongside POPIA and does not defer to it [3]: **s14** makes health information about a patient confidential, with disclosure only on **written** consent, under a court order or law, or where non-disclosure poses a serious threat to public health; **s15** permits a health worker or provider with access to disclose personal information to others where **necessary for a legitimate purpose within the ordinary course and scope of their duties and where disclosure serves the patient's interests**; **s16** allows examination for treatment with the patient's authorisation, and study, teaching or research with authorisation from the patient, the establishment head **and** the research ethics committee, unless no identifying information is accessed; and **s17** requires controls against unauthorised access, with the specified failures made offences.

The interaction rule is what matters, and the practitioner guidance states it plainly: **whichever law gives the patient more protection or rights prevails**, which "often means the NHA overrides POPIA", with the recommendation to build a matrix of applicable laws per issue [7].

The difference that matters most is this. **POPIA s32(1)(a) permits processing for treatment and care without consent. National Health Act s14 requires written consent for disclosure to a third party.** These are not in conflict — s14 governs *disclosure outward*, s32 governs *processing* — but a design reading only POPIA will under-protect, and one reading only the National Health Act will block the legitimate internal care coordination s15 expressly permits. Note the form requirement too: s14(2)(a) requires consent **in writing**, so a verbal consent recorded in a note does not satisfy it.

The rule to implement: **internal, care-necessary, duty-bound sharing under s15 and s32(1)(a); external disclosure under s14 with written consent, a court order, a law, or a serious public health threat.** Everything else is an exception needing a named authority. One analysis draws a further distinction between "health", meaning the present, and "medical history", meaning the past [7]; it is contestable and architecturally unhelpful, because a record does not know when it became history. Design to the stricter standard throughout.

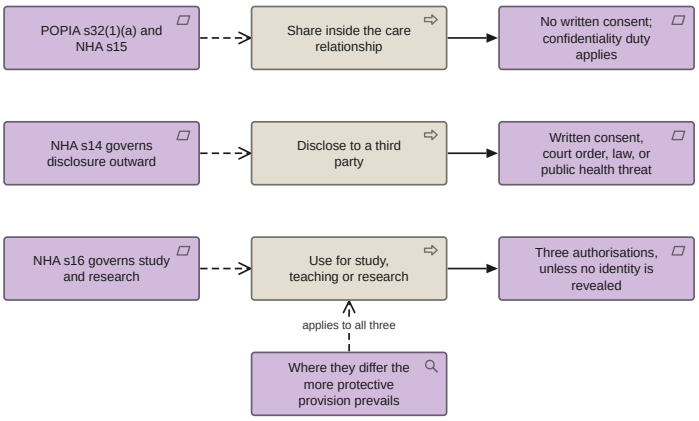


Figure 18.2 Three uses of one patient's record, and the authority each needs. Internal care-necessary sharing runs on POPIA s32(1)(a) with National Health Act s15; disclosure outward needs written consent under s14; research needs three authorisations under s16 unless no identity is revealed.

Two consequences. First, **consent is not the access-control mechanism**: a system that gates clinical access on a consent flag has mis-modelled the law and will block the emergency care s5 requires, while a system with no consent object cannot govern disclosure or research. Both failures are common and

opposite, so hold the consent object and support withdrawal states while relying on the statutory authorisation for clinical processing — the more so because whether a patient may withdraw consent for special personal information is itself unresolved, s27 being silent on withdrawal [6]. Second, s16(2) is the most load-bearing sentence in South African health research data architecture: where research does not reveal the patient's identity, the three authorisations are not required [3], so **a de-identified extract is a materially different legal object from a pseudonymised one**. The national ethics guidelines pull the other way [26], and on the prudent reading **an internal clinical data warehouse used for research is in scope for ethics review**, needing a register of approved uses with ethics references, dataset specifications and expiry dates.

Cloud: two mechanisms, not one

This is the most common gap in South African healthcare cloud designs, and it is a gap of exactly one application form.

Section 72(1) sets the conditions for transferring personal information outside South Africa. In substance: the recipient is subject to a law, binding corporate rules or binding agreement providing adequate protection on principles substantially similar to POPIA's, with similar onward-transfer restrictions; or the data subject consents; or the transfer is necessary to perform a contract with the data subject, or one concluded in their interest; or it is for their benefit and consent is not reasonably practicable [6]. The 2026 regulations make compliance with s72(1) mandatory for health information going abroad [4].

Section 57(1)(d) is separate, and is the one that gets missed. It requires **prior authorisation from the Information Regulator** for transferring special personal information to a third party in a foreign country that does not provide an adequate level of protection [2] — not a contractual condition but an **application to the Regulator before processing begins**, with a lead time. Both must be dealt with, and the position is not entirely settled: the commentary on the draft regulations observed that it appeared to overlook s57(1)(d) altogether, and the research behind this volume found no contrary authority [10][6]. Give a client both readings and take the conservative one.

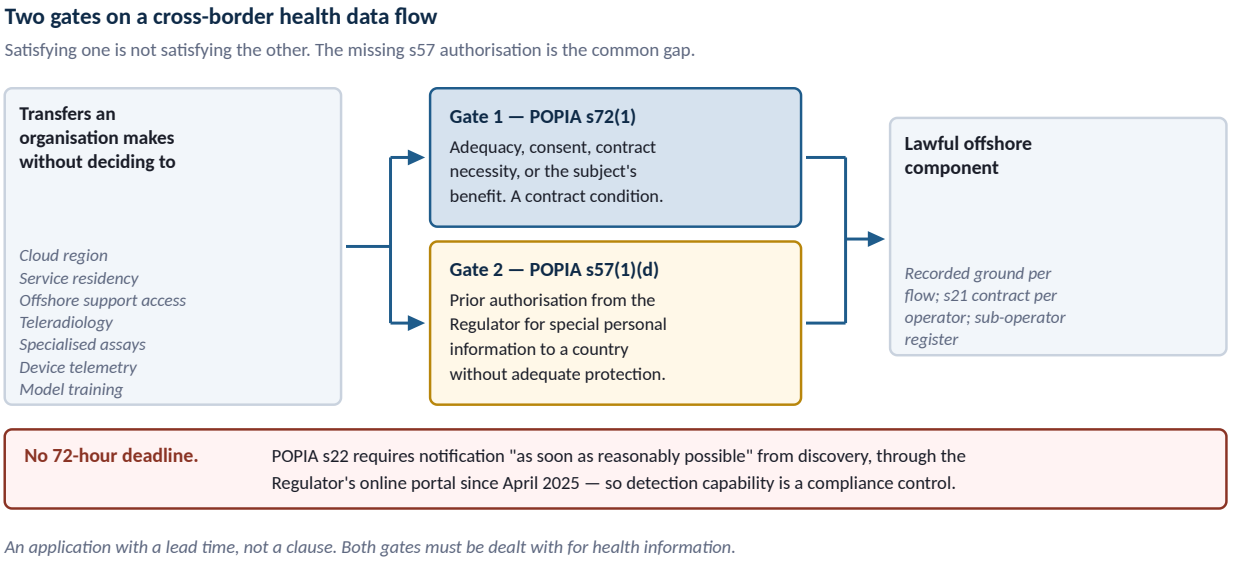


Figure 18.3 The two cross-border mechanisms for health data, the transfers an organisation makes without deciding to, and the absence of any 72-hour breach deadline in POPIA.

The transfers an organisation actually makes are mostly invisible to it [6]: cloud hosting in a foreign region; a clinical application delivered as a service whose data residency is elsewhere; teleradiology reporting; central laboratory testing for specialised assays; a group's consolidated reporting; a device manufacturer's telemetry; a model-training pipeline. And the one nobody counts: **vendor support access from an offshore team is a transfer even though nothing is copied**, because the processing occurs outside the Republic.

Each needs a recorded ground, and the register of processing activities with operators named is what makes them visible — and what makes findable the **s21 written contracts** POPIA requires with every operator [6]. Three clauses are routinely absent: a **named list of the data elements processed**, so a breach assessment can be scoped; **sub-operator rights and a register of them**, because sub-operators are where a transfer happens without anyone deciding to transfer; and **breach notification to the responsible party expressed in hours**, because its own clock runs from its own discovery and an operator that notifies in a week has consumed the window.

Breach notification: there is no 72-hour deadline

Section 22 requires the responsible party to notify the Regulator **and** the affected data subjects of a security compromise. **There is no fixed 72-hour deadline in POPIA.** The standard is "as soon as reasonably possible" after discovering the compromise, subject to the needs of a criminal investigation, and since **April 2025** notifications to the Regulator go through its **online portal** rather than by email [6].

Three consequences. The clock starts at **discovery**, which makes detection capability a compliance control; an organisation cannot start it when it finishes investigating. The notification must identify affected data subjects **individually**, which is impossible without the access logging National Health Act s17 already requires — the logging obligation and the notification obligation are the same obligation seen from two ends. And the **channel must be chosen per data subject with knowledge of what was exposed**, because a notification posted to a household address that reveals an HIV diagnosis or a

termination of pregnancy is a second disclosure. That makes the breach assessment data-element-aware, which is a data-architecture capability rather than a security one [6].

De-identification has no South African standard

There is no South African equivalent of a safe-harbour list of identifiers to remove. POPIA's definitions of "de-identify" and "re-identify" both turn on information that identifies a data subject, that can be identified **through a reasonably foreseeable method**, or that can be linked to identifying information by such a method; and the reasonableness standard is **objective and tied to the responsible party's position** [16]. The exclusion clause removes POPIA's application to information de-identified to the extent that it cannot be re-identified — and does not say whether the test is applied to the whole world or to the specific responsible party. The best available South African analysis answers that contextually [16]: pseudonymisation achieves non-identifiability **only where the holder lacks the linking dataset**; data still identifiable after coding is only partially pseudonymised and legally unchanged, because **partial pseudonymisation is not pseudonymisation**; **for the party holding the key the dataset is still personal information**, including for internal research, while **for a recipient with no key it is non-personal in their hands**; and a provider should review the dataset's history for any past collaboration that could supply a re-identification key, document that review, and record it in the transfer agreement.

So **the same dataset has two different legal characters in two different hands at the same time.** De-identification is therefore not a transformation. It is a **pairing** of a transformation with a statement of who holds the key, and the artefacts are the transformation specification, the key custodian, the key's protection, the transfer agreement recording that the recipient has no key, and the documented history review.

And the transformation is worth less than it looks. A generative modelling study of re-identification likelihood, tested on **210 populations**, predicted individual uniqueness with **area under the curve from 0.84 to 0.97** and estimated that **99,98% of Americans would be correctly re-identified in any dataset using 15 demographic attributes** [17]. That is United States data, and the paper's conclusion is that even heavily sampled anonymised datasets are unlikely to meet a strict anonymisation standard. For a longitudinal record it is worse: **a record is a sequence, and sequences are near-unique**, so the dates on which a person attended a facility over five years are a high-entropy fingerprint, and shifting them by a per-patient offset preserves intervals while destroying seasonality, while generalising to month or year destroys the intervals that make the data useful. **Rare attributes are self-identifying**, and suppressing them removes precisely the patients the research is about. **Free text cannot be de-identified reliably.** And **genomic data is not de-identifiable in any meaningful sense**, because the sequence is the identifier — which is what POPIA s32(5) exists for.

What the Information Regulator can actually do

Chapter 5 gave the enforcement record. The mechanism behind it is what matters for a design.

The ladder runs: an **enforcement notice** requiring specified remedial action within a stated period; then, on failure to comply, an **infringement notice** carrying an **administrative fine under s109**, with a statutory maximum of **R10 million**; and non-compliance with a notice is additionally an offence [11] [27]. The point most clients miss is at the top: the Regulator has stated that **an organisation that complies within the period stated in an enforcement notice cannot be fined for that processing** [27]. That makes the notice a genuine remediation opportunity, and gives the architecture of *being able to remediate quickly* a direct commercial value.

The two dominant findings are s22 and s15(1) [11][12][27]. Section 22 is notification failure — not the breach, the failure to report it — which makes detection and notification capability the single control with the greatest return available to an organisation. Section 15(1) is further processing incompatible with the purpose of collection, which is precisely what an unmanaged analytics platform does when a clinical record collected for care is loaded into a warehouse for operational reporting. That compatibility argument has to be made and documented, not assumed.

Two markers for the health sector specifically. The Regulator required the **National Health Laboratory Service** to produce details of the protection measures it had in place when the June 2024 attack occurred, saying it was **not investigating the hacking itself but checking compliance with POPIA** [14]; the outcome was not established by the research behind this volume. And in May 2026 it signalled a shift toward its own **compliance assessments**, with **insurance and health among the named priority sectors** [27], against more than **8,000 security compromise notifications** since POPIA's enforcement provisions commenced [11]. One further fine is worth carrying because its facts are the shape of a patient-portal risk: a **misconfigured setting at a bank let 696 customers view the home loan applications of 88 other people**, users reported it in March and May 2022, access was removed for the reporters **but the flaw was neither fixed nor reported**, and the R1 million fine followed on 27 October 2026 [15][6].

The R5 million precedent, and why it is about licences

The most directly usable POPIA precedent for a health architect concerns neither health nor data modelling. The **Department of Justice and Constitutional Development was fined R5 million** after a 2021 ransomware attack that exposed personal information from over a thousand files. The department had **let its security software licences lapse, including the tool that could have flagged the unauthorised access**. The Regulator issued an enforcement notice requiring **proof within 31 days that its antivirus, security monitoring and intrusion-detection licences had been renewed**, and the fine followed when that proof was not produced [13][14]. The rand figure is a 2025 figure and the matter is disputed before the courts [11].

A South African public body was fined R5 million for, in substance, letting its security tooling licences lapse. That sentence converts licence and patch management from operational hygiene into a quantified compliance exposure, and it is the most concrete number available to an architect arguing for a security renewal budget.

The remediation order as a design specification

The most useful artefact in the enforcement record is the list of remedial steps a notice demands, with its deadlines. The Central Johannesburg technical and vocational education college notice is the worked example [12][28]:

Required action	Deadline
Register the Information Officer, with proof	31 days
Designate and register deputy information officers, with proof	31 days
Notify affected data subjects of the security compromise, with proof	31 days
Send a written apology by email and every other channel used, with proof	31 days
Submit a POPIA compliance framework, or develop and submit one	31 days, or 120 if developing one
Take action against the employee whose conduct led to the unlawful sharing	60 days
Run POPIA awareness and training for all employees, with proof	90 days

The compliance framework must comprise a **privacy policy, a retention policy and schedule, an incident response policy, and an information privacy and security policy** [12]. Note the second: the retention schedule is a named deliverable in a regulator's remediation order, which is why the last part of this chapter treats it as an architecture artefact. A client that cannot produce those items inside a month has no defence.

The rest of the regime, as powers

The Office of Health Standards Compliance

What the Office can do, in sequence: publish an annual inspection strategy and notify selected establishments in advance; inspect; certify a compliant establishment; issue a **compliance notice stating the nature and severity of the non-compliance with timeframes for correction**; schedule a **re-inspection** when those lapse; and, on continued non-compliance, initiate enforcement [23]. In 2023/24 it issued **618 certificates of compliance**, 563 of them (91,0%) within 15 days of the final report against a 100% target, and **45 written warnings** with enforcement initiated within 10 days in all 45 cases; **79 establishments** went to re-inspection [22].

Three readings follow. An **inspection is an evidence request against the facility's own records** — the domains name health records, responses to adverse events, medicines and supplies, medical equipment, infection prevention and control, waste, facilities, human resources and occupational health and safety [23] — so an inspection-ready facility is one whose operational data can be produced, dated and attributed on demand. A **compliance notice is a dated remediation workflow with a re-inspection trigger**, the same shape as an enforcement notice. And **certificates have validity and must be tracked**.

One honest gap. Section 47 — all establishments must meet standards prescribed by the Minister after consulting the Office — is recorded in the consolidated Act as **awaiting proclamation**, while the Office inspects against norms and standards regulations **promulgated in 2018 and effective in 2019** and a 2021 academic source cites s47(1) and s47(3) as operative [3][31][33]. The commencement position could not be resolved; do not assert either way, and design to the domains, which are stable.

The Health Ombud

The Health Ombud sits inside the Office's structure and takes serious complaints about poor-quality care from public **and** private facilities. Its throughput is the binding constraint: **34 of 63 complaints, 54%, resolved through assessment within 30 working days against a 70% target**, the shortfall attributed to having only four assessors [22]. The design consequence is routing. A complaint may arrive from the facility, the Office's complaints centre, the Ombud or a professional council, and s18(4) requires referral of misdirected complaints [3], so a complaints register needs inbound and outbound routing, a working-day clock and a resolution outcome. It also needs **linkage but not merger with the patient safety incident register**: the two are different records with different owners, legal consequences and access rules, so auto-creating one from the other is attractive and wrong.

SAHPRA on devices and software

SAHPRA regulates products, not providers, and the shape of its power over software is unusual enough that most architects get it wrong in both directions. **The regime is an establishment licensing regime, not a product registration regime.** Authorisation runs through a **medical device establishment licence under s22C** of the Medicines and Related Substances Act 101 of 1965, in three types — manufacturer, distributor and wholesaler — and no device may be manufactured, distributed, imported, exported or sold without one, a valid licence also being a prerequisite for bidding on national and provincial tenders [20][21]. **Product registration has not begun**: the process is still in development and the Registration Call-Up Plan **has not been published** [18][20].

Three consequences fall out. There is **no national device registry to reference**, so a hospital's device master is its own. **Supplier licence status is a procurement control**, with an expiry date on the supplier master, because licences are held by suppliers rather than users. And because the pharmacovigilance and field safety route is SAHPRA's, recalls arrive from a regulator to the market and must be matched against the hospital's own device and lot holdings, which a hospital without lot-level traceability cannot do [20].

There is no dedicated software classification rule. The guideline operates four risk classes, A to D, with the manufacturer classifying and giving a rationale and SAHPRA making the final determination, and it reaches software by principle rather than by rule: software meeting the device definition "is also an active medical device" because it relies on an energy source; software intended to operate, control or influence a device's functions generally takes that device's classification; accessory software is classified separately. For in-vitro diagnostic software the guideline's worked examples put stand-alone software interpreting first-trimester screening results, and stand-alone software staging disease severity through an algorithm, both at **Class C** [19]. The absence of a rule is a real difference from the European Union, which has one specifically for software, and it means classifying standalone clinical software here requires an argument from principles an architect must construct and defend.

Reliance on reference jurisdictions. For Class C and D devices SAHPRA expects premarket approval or registration from one of **six reference jurisdictions — Australia, Brazil, Canada, the European Union, Japan and the United States** — or WHO prequalification, and may still impose further requirements [18]. The consequence is structural: **a locally developed higher-risk clinical software product has no straightforward South African route to market**, because it has no reference-jurisdiction approval to rely on, which pushes local innovation toward lower-risk classifications and non-device framings.

The 26 September 2025 position on models that update

SAHPRA's communication to industry on AI and machine-learning enabled medical devices, issue MD08-2025/2026, dated 26 September 2025, is more developed than most architects know exists, and it answers the question architects ask [18].

- **Classification.** AI software that directly drives or influences clinical decisions, where an error could cause significant harm, will likely be **Class C or D**. Generative AI devices are classified by the potential harm of their intended use.
- **Evidence required.** ISO 13485:2016 quality management including IEC 62304 software lifecycle processes and, where applicable, model-training governance; risk management aligned to ISO 14971 **including AI-specific risks such as algorithm error and data drift**; a technical file with the device description, algorithm logic, risk analysis, verification and validation results and cybersecurity controls; and clinical evidence of real-world performance.
- **The algorithm version is locked at licensing**, with changes made only through a controlled process.
- **Minor updates** — bug fixes and cybersecurity patches that do not alter AI logic — can go through the manufacturer's certified quality management system change control.
- **Major updates likely require regulatory review.** Changes that expand the intended use or alter core algorithm logic should not be made without notifying SAHPRA.
- **Anticipated changes** need a change control plan that **SAHPRA approves**, covering their scope; **data triggers for updates, such as drift beyond a threshold**; the implementation protocol and validation testing on independent datasets; and boundaries, such as no new indications without a new submission. **There is no formal predetermined-change mechanism yet**, SAHPRA being in step with international harmonisation and watching the United States model.
- **Validation and subgroup requirements.** Clinical validation on **independent test datasets** and representative populations and settings; performance measured in real-world conditions including how clinicians interact with the output; **subgroup analysis across sex, age and ethnicity**; **where training data came from elsewhere, evidence that performance holds for local populations**; dataset gaps justified; known biases disclosed.
- **Post-market.** Continuous performance monitoring against premarket expectations, monitoring for **drift and emerging bias**, **recording cases where clinicians overrode the device**, and incident reporting including near-harm.
- And the sentence to quote back: **"patient safety takes precedence over rapid deployment of untested updates."**

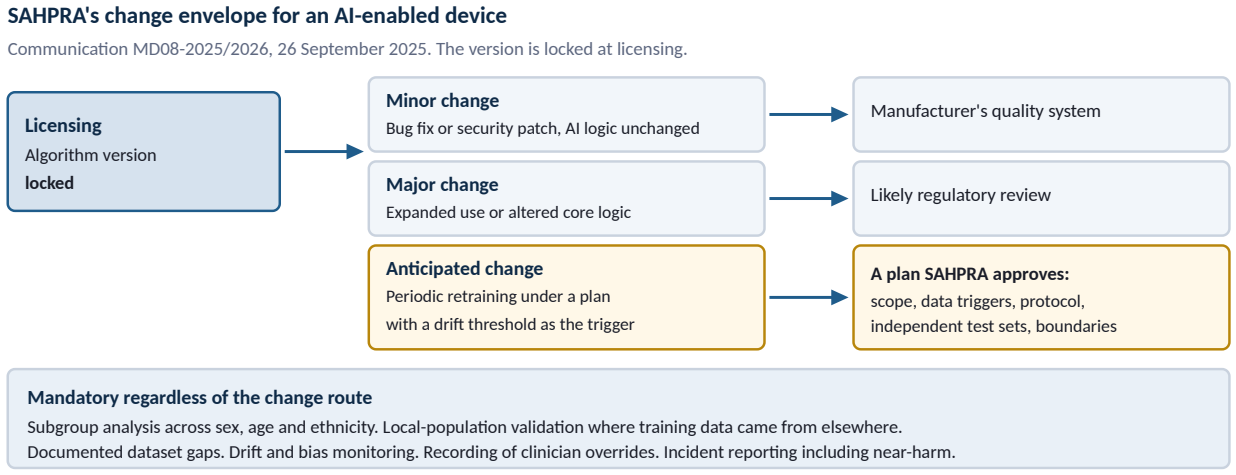


Figure 18.4 Three change routes for a licensed AI-enabled device under SAHPRA's September 2025 position, and the post-market obligations that apply whichever route a change takes.

> **FIVE QUESTIONS TO ASK OF ANY IMPORTED AI PRODUCT**
>
> What is your SAHPRA risk class and the rationale? Which reference jurisdiction has approved it, if it is Class C or D? **What version is locked at licensing, and is there an approved change control plan with its drift triggers?** What is the subgroup performance by sex, age and ethnicity, and **what South African validation exists if the training data came from elsewhere?** And what does the product record when a clinician overrides it? The last two are the ones vendors most often cannot answer, and the fourth is the one SAHPRA has made mandatory.

Three further points. **The substance is the right internal standard whether or not the regulator reaches the activity.** A prediction model built in-house is not obviously a device placed on the market, and an establishment-licensing regime addresses manufacture and distribution rather than internal use — but locked versions, approved change control, independent test sets, subgroup analysis, local validation, drift monitoring and override recording are the right internal discipline regardless [18]. Put the other way round: **a hospital that writes its own clinical decision support may be manufacturing an unregistered medical device.**

The patch boundary. The manufacturer owns the software state of a regulated device, because its conformity and risk analysis are predicated on a defined configuration; the organisation owns the risk of operating it on its network under IEC 80001-1; and the gap is closed by the responsibility agreement that standard requires [29]. Because changes that do not alter intended function or AI logic go through the manufacturer's quality system, a security patch is generally **not** a re-submission event [18]. So "we cannot patch because it is a medical device" is usually wrong; the correct statement is "we cannot patch unless the manufacturer has validated and released the patch", and procurement is the only lever, used before purchase, to get a patching cadence, a vulnerability disclosure route and an end-of-support date into the contract.

And one gate that no longer exists. The certificate-of-need scheme would have required authorisation to acquire prescribed technology; its sections were declared invalid and severed in *Solidarity Trade Union and Others v Minister of Health and Others* (CCT 240/24), ZACC 19 of 2026 [32]. There is therefore **no national gate on acquiring high-cost technology** other than provincial licensing for private facilities and ordinary budget control in the public sector. Chapter 7 has the judgment; chapter 9 has the Schedule 5 to 8 handwriting rule, which is carried second-hand inside the Council's record-keeping guidance and leaves the electronic-signature question open, so the paper path must be designed [8].

The records duty, and the authority to destroy

Chapter 13 owns the clinical record as an object: what must be in it, who may write in it, how it is corrected rather than overwritten, and who may read it. This section owns two narrower things — the duty to keep it, and the authority to get rid of it.

Where the duty sits, and what it is subject to

National Health Act s13 places the duty on the **person in charge of the health establishment**, not on the clinician: that person must ensure a prescribed health record is created and maintained for every patient. And the duty is expressly **subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act 2 of 2000** [3]. That subordination is where the destruction decision actually sits, and it is why a provincial hospital cannot set its own retention schedule.

Section 17(1) requires control measures against unauthorised access to records and to the systems holding them. **Section 17(2)** then makes a list of things offences, punishable by a fine, up to one year's imprisonment, or both — and it reads like a requirements specification because that is what it is: failing to perform the s17(1) duty; falsifying a record; creating, changing or destroying one without authority; **failing to create or change a record when properly required**; providing false information for inclusion; copying part of a record without authority; **linking a patient's identifying details to their condition, treatment or history without authority**; unauthorised access to a record or system, including intercepting transmitted information; **connecting a record-keeping system to other systems or terminals without authority**; and **modifying or impairing the programme used to keep records** [3].

Four of those change a design. **Under-documentation is capable of being an offence**, an unusually strong statutory footing for a completeness requirement. **Linkage requires authority**, so a master patient index, a clinical data warehouse or a research linkage must point at the authority it operates under. **Integration requires authority** too, so an integration project's authorisation record is a legal artefact. And **configuration change is in scope**, which brings unauthorised change and unmanaged patching inside a criminal provision.

"The law says six years" is wrong twice over

There is no South African health-record retention instrument. Every period in circulation traces to **HPCSA Booklet 9**, professional guidance made under the Health Professions Act 56 of 1974 — not a statute and not a regulation [8][9]. And the guideline disagrees with itself across versions:

Category	2016 text	September 2022 revision
General adult records	**At least 6 years** from dormancy (cl. 9.2)	**Ideally indefinite, particularly if electronic**; otherwise 6 years from dormancy
Minors, including obstetric records	Until the **21st birthday** (cl. 9.3.1)	At least until the 21st birthday
Mentally incapacitated patients	**Lifetime** of the patient (cl. 9.3.2)	For the patient's lifetime
Occupational health records	**20 years** after treatment (cl. 9.4)	20 years after treatment
Long-latency exposure conditions	Minimum **25 years** recommended (cl. 9.6)	Minimum of 25 years recommended
Closed or deceased private practices	At least **12 months** (cl. 10.4, 10.5)	At least 12 months in safe keeping

Three provisions travel with the periods: statutory periods take precedence (cl. 9.8); records in provincial hospitals and clinics may be destroyed **only with authorisation from the relevant Deputy Director-General** (cl. 9.5); and retention decisions should weigh storage costs against the value of records for defence, research and teaching (cl. 9.7) [8].

"Ideally indefinite, particularly if electronic" is a change in professional posture and it is architecturally consequential, converting retention from a schedule into an archive strategy on the reasoning that electronic storage is cheap — and addressing neither POPIA's requirement that information not be retained longer than necessary nor the 2026 regulations' disposal requirement.

So "the law says six years" is wrong twice over: it is not the law, and the guideline has changed its own position. It is wrong a third time on prescription. The Prescription Act 68 of 1969's three-years-after-majority rule is where the age-21 period comes from, but the general three-year period for adults runs from when the claimant knew or ought reasonably to have known of the claim, which in a delayed-diagnosis case can be many years after the consultation [25]. **A six-year retention period does not guarantee the record will exist when the medico-legal claim arrives**, and the Law Reform Commission's finding that **15 of 20 reviewed cases took more than five years to finalise** compounds it [33].

THREE THINGS TO ESTABLISH BEFORE YOU QUOTE A PERIOD

Which version of Booklet 9 is in force could not be determined. A September 2016 version and a version marked "Revised: September 2022" are both in circulation, and a 2025 practitioner webinar describes its material as current to October 2025 without a version number [8][9][25]. Obtain the current booklet from the Council. A retention schedule built on a cited secondary source is a compliance artefact built on sand.

A National Health Act regulation prescribing retention periods could not be located, despite s13's reference to "a prescribed health record" implying one exists — the single most important unresolved item in the research behind this volume. What would settle it: the regulations made under s90, obtained from the Department of Health.

The Institution of Legal Proceedings against Certain Organs of State Act's notice periods could not be sourced, and they bear on how long a provincial hospital's records stay litigation-relevant.

One quiet constraint sits in that table. **Twenty-five years is longer than any clinical system an architect will design will survive**, so occupational health records are an archival-migration problem from the first day of the design — and they must be kept apart from the clinical record of the same person, under two legal bases: the employer processing under s32(1)(f), the provider under s32(1)(a) [2][4].

Reconciling POPIA with the professional guideline

POPIA s14 requires that records not be kept longer than necessary for the purpose, with exceptions where law requires or authorises retention, a contract requires it, the data subject consented, or retention is for historical, statistical or research purposes with appropriate safeguards; thereafter they must be **destroyed, deleted or de-identified** so that reconstruction is impossible [6].

Take a provincial hospital record of an adult treated in 2010 and dormant since 2011. It is deletable under POPIA, because the purpose is served; retainable under POPIA, because law requires or authorises retention and the professional guidance and the Archives Act are the law pointed at; required to be kept under the guideline until 2017 and arguably beyond; and not destroyable at all without a Deputy Director-General authorisation and a disposal authority. All four are true at once.

Reading one — retention wins. POPIA s14's exception for retention required or authorised by law is engaged, so the record is retained for the statutory or professional period and POPIA adds nothing beyond requiring that the period be stated and applied. The architecture then needs a **retention schedule by record class with the authority cited for each period**, and automated deletion only after the longest applicable one.

Reading two — minimisation wins for the copies. The professional periods attach to the *patient record* as the profession understands it, and not obviously to every derived copy: the warehouse extract, the analytics mart, the research dataset, the integration engine's message store, the ten-year backup, the access log. On this reading POPIA s14 governs those, their purpose expires, and they must be deleted or de-identified. That yields the single most useful retention rule available: **the retention period of a copy is a function of the copy's purpose, not of the source record's class.** A message queue holding clinical messages for replay has a purpose measured in days. It is routinely retained for years, because nobody ever assigned it one.

The authority to destroy

Destruction is where the chain of authority becomes visible, and it runs differently for the three kinds of organisation this volume covers. Driehoek Health Group and Rietvlei District Health Service are fictional teaching examples used throughout.

Organisation	What authorises destruction	What the evidence must be
Provincial facility — Rietvlei District Health Service	The Archives Act: public records may not be destroyed without the National Archivist's authority, and disposal authorities are issued against **approved retention schedules**; plus the Deputy Director-General authorisation under the guideline cl. 9.5	An approved schedule, a disposal authority referencing it, the Deputy Director-General's authorisation, and a register of destroyed records
Private hospital group — Driehoek Health Group	The guideline's periods as applied in the organisation's own schedule, with POPIA s14 governing derived copies	A documented destruction procedure, a signed disposal authority, a register of destroyed records, and a confidentiality agreement with any disposal contractor
Scheme, administrator or employer	POPIA s14, plus the 2026 regulations' requirement to dispose of records so they cannot be accessed or disclosed afterwards	Evidence of secure, timely disposal across every copy — backups, replicas, warehouses, test environments and operator-held copies

The published practice that produces that evidence is unglamorous: a records management policy with a named responsible person; documented destruction procedures with **a signed disposal authority**; shredding or incineration for paper and overwriting or physical destruction for media; **a register of destroyed records**; and a confidentiality agreement with any outside disposal contractor [24]. **That register, with its signed disposal authority, is the artefact that lets a facility say "this record no longer exists and here is the lawful reason".** Without it a missing record is indistinguishable from a lost one — and chapter 1 has the arithmetic of what that costs a provincial department in conceded medico-legal claims.

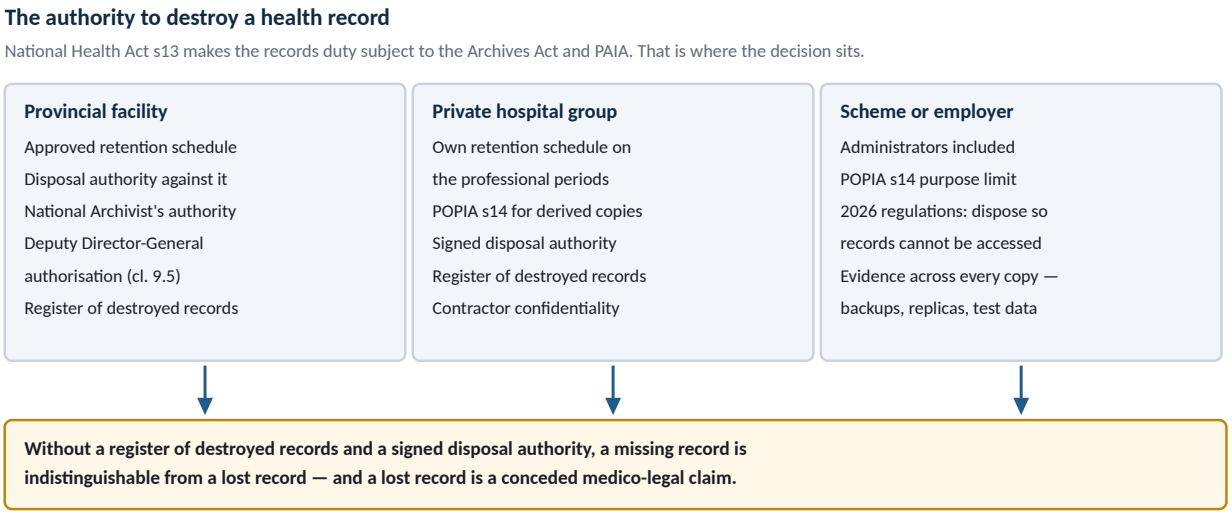


Figure 18.5 Three destruction-authority chains, and the single artefact all three need. Figures and clause references are from the professional record-keeping guidance and the 2026 health regulations.

All of that implies a **retention schedule held as data, not as a document**: record class; the authority for the period; the period; the trigger event — last contact, dormancy, a 21st birthday, a date of treatment, a date of death; the disposal method; the approval required; and the evidence of disposal. The last column is the one almost nobody has, and the one the Regulator's remediation order names.

One case needs stating because systems get it wrong. POPIA protects living natural persons, so on its own terms a deceased patient's record is not personal information [6], while the National Health Act's confidentiality duty is not expressed as lapsing on death, professional confidentiality survives it, and the record stays relevant to a medico-legal claim brought by an estate. Both readings are defensible and the answer is the same under both: **do not delete on death, do not relax access controls on death, and record death as a state of the person rather than as a trigger for anything.**

What follows for a design

Eleven requirements follow from this chapter, each justified today by an instrument in force today.

Requirement	Authority
Default-deny access on a care relationship, with **break-glass and audit** for emergencies	POPIA s32(1)(a); NHA s15 and s5 [2][3]
Complete access logging, **including reads**	NHA s17(2) as an offence; POPIA s22 [3][6]
Append-only records with attributed, reasoned amendment trails	The record-keeping guidance, 2016 and 2022 [8][9]
Item-level and age-dependent access rules	Own access from 12; parental access under 16 on the patient's written authorisation; termination-of-pregnancy information to the patient at any age [8][30]
Occupational health records separated from clinical records for one person	POPIA s32(1)(f) against s32(1)(a) [2][4]
A retention schedule with five classes, a disposal workflow, the Deputy Director-General gate and a destruction register	The guidance cl. 9; the Archives Act; the indemnifier's guidance [8][9][24]
Evidenced destruction across every copy	The 2026 health regulations [4]
Written confidentiality duties for everyone without a professional one	POPIA s32(2)–(3) [2][4]
An operator map with a legal basis per hop, switches included	POPIA s21 [6]
s72(1) conditions plus s57(1)(d) prior authorisation, before any offshore component	POPIA s72 and s57 [2][6]
A four-document compliance framework, a registered Information Officer and registered deputies	What an enforcement notice demands in 31 days [12]

And one framing point for a client conversation. **None of those eleven requires National Health Insurance, a strategy or a transformation programme to justify.** Every one is mandated now, by a named instrument, with an enforcement record, dated fines and a regulator that has named health as a priority sector for its own 2026–2027 assessments [11][27]. The compliance case for good records architecture in South African healthcare is already made, dated, gazetted and enforced. The work is to write the schedule down and hold the evidence.

THE CHAPTER IN TEN LINES

- Privacy is not a constraint on a South African healthcare architecture. It is the architecture, because the default under POPIA is prohibition with named exceptions.
- Section 32(3) is a confidentiality duty, section 32(4) is about supplementary processing and not children, and section 32(5) is about inherited characteristics.
- The regulation power is section 32(6) and it reaches only subsections (1)(b) and (f), which is why the 6 March 2026 health regulations bind funders and employers and not providers.
- POPIA and the National Health Act both govern health data and the more protective provision prevails, which often means the Act. Section 32(1)(a) permits processing for care; section 14 requires written consent to disclose to a third party.
- Cloud has two gates, not one: the section 72(1) transfer conditions and prior authorisation from the Regulator under section 57(1)(d). The missing section 57 authorisation is the commonest gap in South African healthcare cloud designs.
- There is no 72-hour breach deadline in POPIA. The standard is as soon as reasonably possible from discovery, through the Regulator's online portal since April 2025.
- De-identification has no South African standard and no safe-harbour list. Identifiability is contextual, so the same dataset is personal information in one pair of hands and not in another.
- The dominant enforcement findings are section 22 notification failure and section 15(1) incompatible further processing. A public body was fined R5 million for, in substance, letting its security licences lapse.
- SAHPRA has no dedicated software classification rule, product registration has not begun, and for higher-risk classes it relies on six reference jurisdictions.
- For an AI-enabled device the algorithm version is locked at licensing, anticipated changes need an approved plan with a drift trigger, and local-population validation is required where the training data came from elsewhere.
- There is no South African health-record retention instrument. Every period traces to a professional guideline that changed its own headline position between 2016 and 2022.
- The records duty sits on the person in charge of the establishment and is subject to the Archives Act and PAIA, which is where the destruction authority lives.
- The retention period of a copy is a function of the copy's purpose, not of the source record's class.

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CHAPTER 19

Safety, downtime and supplier concentration

Three of the six sourced attacks were on a supplier. Plan for weeks, and name the dependencies you do not own.

Chapter 1 made the claim: downtime in healthcare is clinical. This chapter is the evidence, and what follows from it.

Seven documented events do the work. Six are ransomware or cyber incidents; the seventh is a power failure, and including it is deliberate, because it shows that security spending and availability spending are different spending. **Three of the six cyber incidents were attacks on a supplier rather than on a hospital** — the claims clearing house, a pathology joint venture and a national laboratory service — and in each case the clinical harm was distributed across providers who had no control over the compromised entity. That is the dominant failure mode, and it is **supplier concentration outside the organisational boundary**. No amount of internal hardening addresses it.

Five architectural consequences run through the rest of the chapter. Recovery objectives must be derived from **clinical** tolerance rather than financial loss. The degraded mode is a **designed state**, not a failure. The paper process is **part of the architecture** and must be resourced, trained and tested. Segmentation is justified on **clinical safety** grounds. And audit is a **primary** control rather than a compensating one, because the access model cannot be tight enough to prevent.

The seven incidents, in detail

Health Service Executive, Ireland — Conti, 2021

Initial access on **18 March 2021** on the post-incident timeline; systems crypto-locked on **14 May 2021**; the attackers supplied a decryptor on **20 May** [1]. That is **nearly two months of dwell time** between the foothold and the detonation — the window in which detection should have happened and did not [2].

Patient care was disrupted for months, appointments were cancelled, and blood tests could not be performed or their results received [1]. Infected systems were wiped and restored, and the military was called in to help. The review found that the decryption key "allowed for an accelerated recovery process", and that **the organisation's backups were only *periodically* copied to offline tape**, so some data might have remained encrypted without the key [1].

That is the single most important sentence in this chapter. **A national health service was saved by the attacker's decryptor rather than by its own backups.**

About **$48 million** had been spent at the time of reporting — $14m on infrastructure, $6m on cybersecurity and strategic support, $17m on vendor support for applications, $9m on productivity software — with a forecast total of about **$110 million**, and the following year's budget adding $48m for cybersecurity on top of $69m of operating expenditure [1]. Those are United States dollars, as reported. The review assessed the organisation's cybersecurity maturity against an international framework as "a very low level", with **no chief information security officer** and weak patching, monitoring and anti-malware defences [1].

Change Healthcare, United States — ALPHV/BlackCat, 2024

The most instructive incident for an architect, because the failure was not at a hospital at all. It was at a shared service nobody had modelled as single-point-of-failure infrastructure.

Entry was through **compromised credentials for a low-level customer support employee, used to log into a Citrix remote access portal that lacked multi-factor authentication**, despite company policy requiring it on externally facing systems. The attackers created privileged accounts, moved laterally and exfiltrated data — and **the backups were reachable, so they could be disabled** [3][4].

Initial access in February 2024, with sources giving the 11th, the 12th or the 17th; ransomware deployed around **21 February**, detected when files were encrypted; exfiltration confirmed **7 March**; clearing house services **fully restored only on 19 November 2024**; and a final affected-individual estimate of **192,7 million** as at 31 July 2025, published 6 August 2025 [3][5].

Claims and payment processing halted across the industry, along with prior authorisations for care and for prescriptions. Providers could not submit funder claims for roughly two months; an American Hospital Association survey found **86% still experiencing funder-claim submission disruption two months on**; some providers reportedly could not file for up to eight months, and some funder claims were then rejected for missing filing deadlines [3]. A **$22 million ransom was paid**; the group allegedly performed an exit scam and the data was not deleted; an affiliate later passed it to a second group, which demanded a further payment that was not made. The parent company's 2024 cost estimate was revised to **$2,87 billion** [3].

Synnovis, London — Qilin, June 2024

A ransomware attack on a pathology services provider to several National Health Service trusts in London, in June 2024, disrupting pathology across those trusts [6][7].

Thousands of appointments and procedures were cancelled. **One patient died unexpectedly during the attack, and a patient safety investigation found several contributing factors, including "a long wait for a blood test result" caused by the disruption to pathology services.** An integrated care board figure showed **170 patients harmed**, mostly classed as low harm [6][7]. Commentary on the entry route has focused on missing multi-factor authentication [6]. The provider issued its breach notification only later [8].

This is the citation to use when a board asks whether cyber risk is a clinical risk. It is also, again, a shared-service concentration failure — the same structural lesson as the claims clearing house, with an exact South African analogue.

Ascension, United States — Black Basta, May 2024

The best-documented account of what downtime actually looks like in a large multi-hospital system.

An employee unknowingly downloaded a malicious file, giving the attackers a foothold they used to move laterally and encrypt files. Unusual activity was detected **8 May 2024**; a suspected cyberattack was announced **9 May** with disconnection from business associates; ransomware affecting hospitals was confirmed **13 May**. **Hospitals were still on downtime procedures on 24 May**, with record access restored in the first market that day, five markets by 5 June and the rest expected by 14 June. Exfiltration from a small number of servers — 7 of 25,000 — was confirmed 13 June, with no evidence that clinical systems were compromised; the breach portal was updated on 19 December 2024 to **5,599,699** affected records [9][10].

What was down: record access, telephone systems, patient portals, and the systems for ordering tests, procedures and medications. Staff used pen and paper. Some hospitals **diverted ambulances**, and **pharmacies closed** so that many hospitals could not fill prescriptions. **Laboratory results that normally took 30 to 60 minutes sometimes took several hours.** Nurses reported heavier patient loads of up to five or six patients each and concerns about medication errors; elective procedures were postponed; facility volumes fell 8% to 12% in May and June against the prior year. **Record access took roughly six weeks to restore** [9].

One honesty note: the principal source is internally inconsistent on the organisation's hospital count, giving 136, 140 and 142 in different places, and gives two dates for the interim regulatory report [9]. Neither affects the duration or the clinical findings.

National Health Laboratory Service, South Africa — BlackSuit, June 2024

The South African incident every architect working here should be able to describe.

The **BlackSuit** group gained access on **21 June 2024** after an employee clicked a phishing link, and ransomware encrypted applications on computers and servers **at about 03:00 on Saturday 22 June**, causing a full outage of internet, intranet and the **TrakCare** laboratory information system — with the central data warehouse holding **decades of archived results** also inaccessible [11][12]. The organisation says the ransom was not paid; patient information was on a separate server and was not compromised; but **the data warehouse was rendered useless and TrakCare was unusable, so doctors could not see test results** [12]. Contemporaneous reporting recorded that **backup data had been deleted** [13]. The acting information technology manager told Parliament: **"We were not able to update our systems or put security patches in place"** [12]. At least **R300 million** was being invested in systems, including **R164 million for switches, firewalls and security** — March 2025 rand figures [12].

The fallback was **manual collation and communication of results**, by telephone, messaging application, email, fax, courier and paper — and the clinical academics who wrote it up note that some of those routes may conflict with POPIA [11][14]. The organisation had **no business continuity plan**, and its downtime procedure covered only laboratory tests, with no contingency for a cyber breach [11]. All of it ran during an **mpox outbreak** [15].

> **TWO DURATIONS, AND THEY DISAGREE**
>
> The research behind this volume gives two figures for how long the June 2024 outage lasted, and they do not reconcile.
>
> The clinical academic account states the systems were **"wholly inaccessible for 40 days"** from 22 June, and were still only partially functional when the paper was written — received 19 August 2024, accepted 4 September 2024 [11].
>
> The operator announced **full restoration on Wednesday 21 August 2024**, after a rebuild of systems and infrastructure, with no dates given for individual systems [16]. That is **about 60 days from 22 June**. An earlier projection of 15 July 2024 was missed, and some critical services were back "in a phased and controlled manner" from around 1 August [14].
>
> Both are reasonable readings of different things — full inaccessibility against announced full restoration — but neither source reconciles them and no technical post-incident review was published. **Print both.** Forty days to partial function, about sixty to announced restoration, in the organisation that provides pathology for most of South Africa's public health sector.

No public post-incident review exists, and the entry vector beyond the phishing click, whether Active Directory was destroyed, and whether a ransom was demanded are not established in the public record [11][12]. The absence of a published review is itself worth remarking on. One academic assessment of the untold costs exists at a stated reference but could not be retrieved for this volume, and is likely the best South African source on the clinical and financial cost of a health cyber incident [17].

The same organisation, 16–18 March 2026 — a power failure

Twenty months later, the same national laboratory service failed again, and not through an attack.

On **Monday 16 March 2026 at around 21:00** its information systems failed. Public health facilities nationwide could not access laboratory results. **TrakCare** — named as the system underpinning national laboratory operations — was back up in the **early hours of Wednesday 18 March** [18]. The operator attributed the disruption to **"erratic power supply challenges"** around Sandringham, where its head office is located, called it an infrastructure issue, and said it was **not a cybersecurity incident**. The chief executive of the South African Medical Research Council linked it to a power outage at the head office where the servers holding patient results are housed, and said the organisation **appeared to have no uninterruptible power supply or other backup system** [18].

Specimen registration was mainly affected, which the operator said could delay testing and turnaround times; new investigations had to be deferred; emergency services were reportedly hardest hit and staff relied on printouts from laboratories; **the effect varied by facility depending on local contingency measures**; and a Cape Town doctor said staff had received no formal communication about the problem [18].

Four findings an architect should take from the pairing. **Security investment and availability investment are different investments**, and R164 million of switches, firewalls and security does not buy an uninterruptible power supply — a board that has approved one frequently believes it has approved both. **Specimen registration was the failure point, not result reporting**: the front of the process is the dependency that stops work, and architects design result-delivery resilience and forget registration resilience because registration looks administrative. **"The effect varied by facility depending on local**

contingency measures" is the whole argument for local fallback design in one clause. And **nobody told the clinicians**, which makes incident communication to clinical users part of the architecture of a clinical system, and it is almost never designed.

One more, which matters because it is the wrong conclusion to draw. **The vendor is not the problem and the vendor is not the solution.** The laboratory platform caused neither incident; a phishing click and a power failure did. An architect who responds to this case study by proposing a platform change has misdiagnosed it [18].

Life Healthcare, South Africa — June 2020

Included because the South African private hospital sector has been hit too, and because the systems affected were precisely the patient administration and revenue cycle core.

The group announced a cyberattack affecting its **admissions systems, business processing systems and email servers** at its Southern African operation, reported on 9 June 2020, during the COVID-19 pandemic. The extent of data compromise was undetermined at the time. The company said patient care was unaffected, with hospitals and administrative offices continuing to operate with some delays after switching to backup systems [19].

"Patient care was unaffected" is a statement about clinical continuity under a degraded administrative estate, and it is the outcome a good downtime design produces.

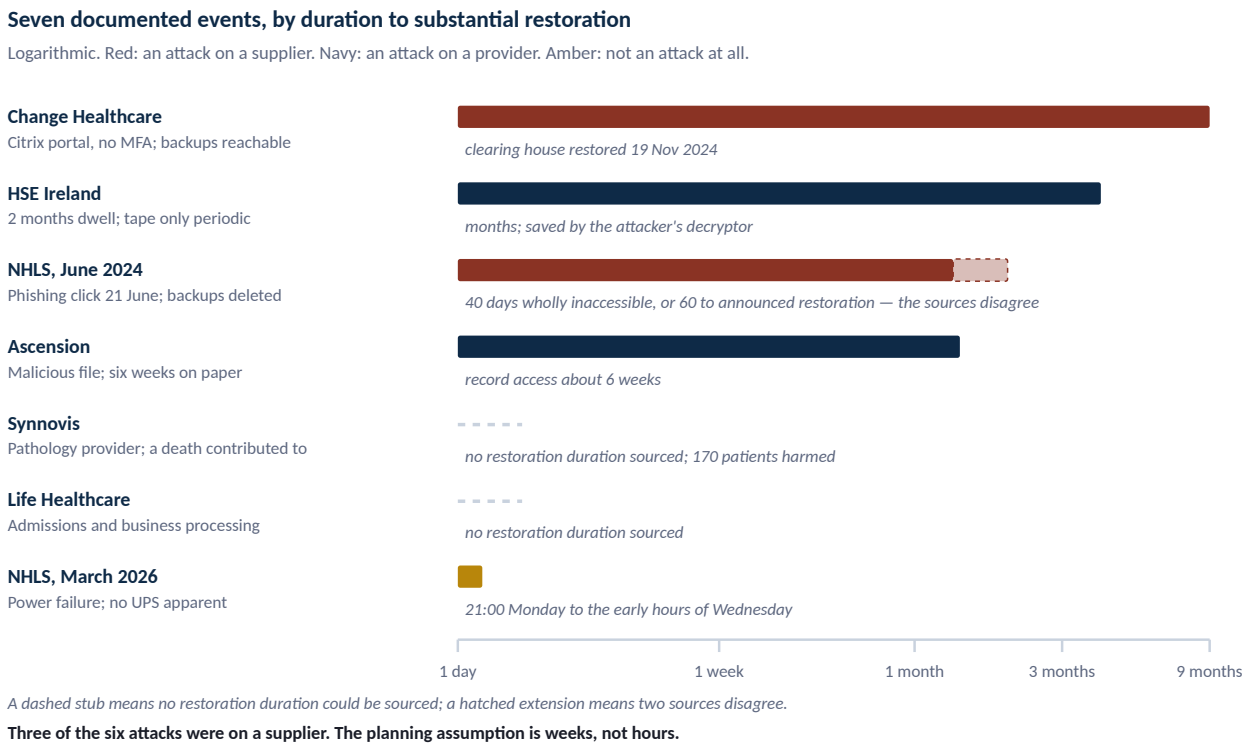


Figure 19.1 Seven events on a logarithmic duration scale, with entry vector and documented clinical consequence. Where a source gives no restoration duration the bar is omitted from the argument and the gap is stated.

What the seven have in common

	HSE Ireland	Change Healthcare	Synnovis
Year	2021	2024	2024
Dwell before detonation	about 2 months	days to about 10 days	not sourced
Entry vector	not isolated in sources used	credentials on a portal without multi-factor authentication	commentary focuses on the same gap
Backups	only periodically copied to offline tape	reachable by attackers and disabled	not sourced
Time to substantial restoration	months	clearing house fully restored about 9 months	not sourced
Documented clinical harm	care disrupted for months	care and prescription authorisation halted	**patient death contributed to; 170 harmed**

	Ascension	NHLS June 2024	NHLS March 2026	Life Healthcare
Year	2024	2024	2026	2020
Cause	malicious file downloaded by an employee	phishing click, ransomware	erratic power supply, not an attack	not sourced
Backups	not sourced	backup data reported deleted; infrastructure rebuilt	no uninterruptible power supply apparent	switched to backup systems
Time to substantial restoration	record access about 6 weeks	**40 days wholly inaccessible, or about 60 days to announced restoration**	about 29 hours	not sourced
Documented clinical harm	ambulance diversions, pharmacy closures, result delays	result delays and backlog during an outbreak	deferred investigations; emergency services hardest hit	"patient care unaffected"

Five generalisations are defensible from that pair of tables.

Multi-factor authentication on externally facing access is the control whose absence is named most often. The clearing house's own chief executive testified it was missing on the compromised portal despite policy requiring it; commentary on the pathology attack focuses on the same gap [3][6].

Backups fail in the ways attackers intend them to. Reachable backups get disabled; periodically copied offline tape leaves a gap; and in one South African case the backup data was reported deleted. The requirement is offline or immutable, frequent, and **restore-tested** [1][3][13].

Restoration is measured in weeks to months, not hours. Every incident with a sourced restoration timeline ran to weeks or months. Only the power failure was measured in hours — and it was a power failure.

Shared services are the concentration risk. Three of the six cyber incidents were attacks on a supplier, and in each case the clinical impact was distributed across many providers with no control over the compromised entity [3][6][11].

The clinical harm is in the diagnostic loop. Pathology and result delivery feature in three of the seven. The architecture's most protected path should be *order placed → specimen taken → result returned → result acknowledged*, because that is empirically where the harm occurs. Chapter 9 has that stream.

> THE PLANNING ASSUMPTION
>
> **Weeks, not hours.** Every resilience decision in a clinical estate should be tested against a multi-week outage of a system the organisation may not own. An architecture whose degraded mode is designed for an afternoon has been designed against the wrong event.

The read-only record

The highest-value single resilience investment in a clinical estate, and the one most often omitted because it looks like duplication.

The specification is published and it is specific: **a read-only backup record, updated at least hourly, tested weekly, and able to print** [20].

Why it matters so much: in a clinical outage the *write* path can be substituted by paper at a cost in effort, but the *read* path cannot be substituted at all. A clinician without the record does not know the allergies, the current medications, the last potassium, the resuscitation decision or the diagnosis. Paper can capture what happens next; it cannot reconstruct what came before. **So the read path is the one that must survive, and it can survive far more cheaply than the write path**, because a read-only copy needs no transactional integrity, no integration and no high availability on write, and can be hosted and secured differently.

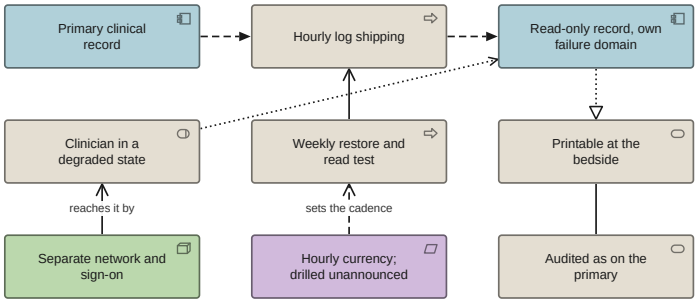


Figure 19.2 The read-only record as a separate failure domain. Hourly currency, weekly testing, printable output, separate authentication and network path, and audited like the primary.

Six design notes that decide whether it works.

Hourly currency is good enough for almost all clinical read needs and is achievable with log shipping or snapshot replication. It must be **independently reachable** — separate authentication, separate network path, separate hosting — or it fails with the primary; a read-only copy inside the same failure domain as the primary is theatre. It must be **printable**, because the next step after reading is often carrying information to a bedside. It must be **drilled**, because clinicians will not find it under pressure if they have never used it. An immutable, append-only read replica is hostile to encryption, which makes it the natural substrate for the ransomware scenario specifically. And **access to it must be audited like the primary**, because a downtime copy with relaxed access control is a standing privacy risk — and under National Health Act s17(2) unauthorised access to it is as criminal as unauthorised access to the original. Chapter 18 has that provision.

Downtime procedures, and the paper that is part of the architecture

The best freely available control set covers planned and unplanned record unavailability through **13 recommended practices in three domains**, each scored on a five-level implementation scale from 0% to 91–100%, with an explicit box for gaps the current product cannot close [20]. It is United States guidance and should be labelled as such, but nothing in it is jurisdiction-specific. The numbers worth carrying:

Practice	The figure it names
Disaster recovery plan	Reviewed at least annually; large providers should have a remote warm site able to run the full record **within 8 hours**, tested **quarterly**
Power	An uninterruptible supply sustaining the record for at least **10 minutes**, tested monthly; a generator with **2 days** of on-site fuel, tested monthly, sited where it is unlikely to flood
Paper forms	Enough on **each unit** to care for patients for at least **8 hours**, including orders, medication, laboratory and imaging documentation — **plus a process to reconcile paper back into the record**
Backup	Daily encrypted off-site backups with restore tests, monthly being optimal; backups kept **separate from normal storage to resist ransomware**; a **read-only record updated at least hourly, tested weekly, able to print**
Identity	Temporary record numbers for patients registered during downtime, with a reconciliation process, and standard identity checks for specimens, medicines, imaging and result delivery
Drills and training	Paper-based ordering and charting training, a job aid, and **unannounced downtime drills at least yearly**; interface inventories reviewed every six to twelve months
Declaring downtime	A written policy reviewed every two years; the **warm site ideally activated before 2 hours** of unplanned outage; paper copies of the policy on units and off-site
Response time	Monitored, optimally under **2 seconds**
Functional downtime	A **daily synthetic test order**; an hourly mean above **5 seconds**, or three standard deviations above the mean, signals functional downtime
Extended outages	For unplanned outages over **24 hours**, a multidisciplinary root-cause or failure-mode review assessing the effect on patient care

Four things to add to that list.

"Functional downtime" is the concept most architects are missing. A system responding in eight seconds is clinically down even though every monitor is green. The daily synthetic transaction and the five-second threshold make it measurable, and the idea transfers to any industry.

The paper reconciliation process is the hard part, not the paper. After six weeks on paper somebody must enter or scan weeks of clinical documentation, and the record's integrity depends on how that is done. The clinical academics writing up the South African laboratory outage warn specifically that data entered during downtime may be lost or affected by manual entry errors — a data-integrity consequence of an availability incident that outlives the incident [11]. Design the reconciliation path with the downtime path.

The 8-hour paper stock figure must be read against the six-week reality. It is a replenishment unit, not a reserve. An architecture needs a *supply* plan for extended paper operation, including printing capacity that does not depend on the failed estate.

Downtime drills must be unannounced and must include clinicians. A technical failover test is not a downtime drill.

Recovery objectives derived clinically

The standard practice of deriving recovery time and recovery point objectives from financial loss per hour produces the wrong answers here, because the loss function is neither monetary nor linear. The method matters more than the principle, so here is the method, in six steps.

One: decompose by clinical function, not by system. Not "the record", but identity resolution; allergy and medication list retrieval; order placement; result delivery; result acknowledgement and escalation; imaging retrieval; medication administration recording; the theatre list; blood product cross-match validity.

Two: for each, ask the clinical question — how long before a patient is harmed? The answers differ by orders of magnitude.

Three: ask what the manual substitute is and how long it holds. A manual process that works for four hours does not work for four weeks, and the South African laboratory experience is the evidence: manual collation of results worked, and produced a backlog [14].

Four: set objectives per function and design per objective. Most functions do not need the most expensive resilience; two or three do, and a read-only replica covers a surprising proportion of the short-tolerance read functions cheaply.

Five: express recovery point objectives in clinical terms too. "We may lose one hour of data" means a clinician may not see the last hour of observations, orders and administrations — which in critical care is a safety event. For some functions the acceptable recovery point is zero, and that is a design constraint rather than a wish.

Six: state the degraded-mode design for each function explicitly, and get it clinically approved. The approval is the artefact that makes the design defensible.



Figure 19.3 Clinical recovery tolerance by function rather than by system. The ladder is illustrative of the method; the bands are the orders of magnitude the research describes, not a schedule to adopt unchanged.

Segmentation, identity and break-glass

The argument for segmenting a clinical network is stronger than elsewhere, and it should be made on clinical safety grounds rather than information security grounds. The standard that makes that argument available is IEC 80001-1, which addresses risk management for networks incorporating medical devices and assigns the duty to the **health care delivery organisation** across safety, effectiveness and security together [21]. Chapter 18 has the responsibility agreement it requires.

Eight segments are a reasonable starting architecture: **medical devices that cannot be patched**, the most restrictive, with tightly controlled egress, no general internet access and monitoring rather than agents, because agents cannot be installed on regulated devices; **medical devices that can be patched**, on a managed cadence; **clinical workstations**, which need broad access and are the primary phishing target; **clinical servers and integration**; **administrative and corporate**, which is where email lives and therefore where compromise begins; **guest and patient wireless**, fully isolated; **operational technology** — building management, pneumatic tube, nurse call, access control, theatre integration, audiovisual — frequently forgotten, frequently flat and frequently reachable; and **backup and recovery infrastructure, which must not be reachable from the segments it protects**. That last one is the clearing house lesson stated as a design rule [3].

Three healthcare-specific constraints. **Devices depend on network services in undocumented ways** — time, name resolution, address assignment, specific multicast behaviours, and in older devices hard-coded addresses — so segmentation must be preceded by dependency discovery and, under IEC 80001-1, the change is risk-assessed. **Clinical workflow crosses segments constantly**, so segmentation drawn without a workflow map produces a design that is bypassed within weeks. And **emergency override must exist and must be auditable**, because in a mass casualty incident a network control that blocks clinical access becomes a clinical hazard.

Why audit is the primary control

In a hospital, thousands of staff may *legitimately* need to read a given record, and which thousands is not predictable in advance. An unplanned admission at 3am may legitimately be read by the emergency physician, the medical registrar, the intensive care team, the radiographer, the laboratory scientist, the pharmacist, the ward nurse, the physiotherapist, the discharge coordinator and the clinical coder. Role-based access control cannot express that set, because the determining variable is **clinical relationship**, which is dynamic.

The consequence is uncomfortable and should be stated rather than finessed: healthcare access control is nearly always broad, and the integrity of the record depends on **audit** rather than on prevention. Four mitigations genuinely help. **Relationship-based filtering** from rostering and assignment data, so the record is open to those with a current care relationship and broader access requires a reason — which makes the roster feed a security dependency. **Sensitive-record flags** for records with specific risks, which need an access model of their own. **Just-in-time elevation** rather than standing privilege. And **strong authentication at the clinician level**, including on clinical workstations, because the multi-factor lesson applies internally too.

Break-glass is the mechanism by which a clinician overrides an access restriction when clinical need requires it, with the override recorded. Seven design requirements decide whether it works [22]. It must be reachable **in seconds**, by a clinician who has never used it, without a help desk call — a break-glass path that requires a ticket is not one. It must **require a reason**, from a short list plus free text, captured at the moment of use. It must be **time-bounded**: this record, this shift, not permanently. It must **notify** the owning service, the care team and the privacy office, because notification at the time is a far stronger deterrent than retrospective audit. **Every use must be reviewed**, on a defined cadence and a defined proportion, ideally all of them. **Its**

use rate is a design metric: if break-glass is used often, the access model is wrong and the mechanism has become the normal path, at which point it provides neither access control nor deterrence. And it must never be the route for ordinary cross-team access, which is the failure the review exists to catch.

Auditing a record thousands may legitimately read

Given all of that, the audit log is a control rather than a log, and it has to be built like one.

What it must contain: who, meaning an individual, never a shared or service account acting on a human's behalf without the human's identity carried through; what record and what part of it; when, from a synchronised clock; from where, meaning device and location; by what route — application, interface, database, report, bulk export; and why, where a reason was required.

What must be audited and usually is not: reads, not just writes, because reads are where privacy breaches happen; **bulk and report access**, because a single query returning forty thousand records is a larger exposure than forty thousand individual views and is logged less well, if at all; **direct database access**, because administrators and reporting accounts bypass application audit entirely; **integration traffic**, because who configured a route and what it carries is an access-control question; **break-glass**; and **administrative actions** — access grants, role changes, audit configuration changes. Audit of the audit.

What makes an audit programme work rather than merely exist: proactive, risk-targeted review rather than complaint-driven review, where the highest-yield patterns are staff accessing their own record, a family member's record on a shared surname or address, a colleague's record, the record of someone in the news, access by someone with no care relationship on the roster, access clusters following a publicised incident, and access immediately before or after employment events — a pattern set that is an established practice area in its own right [30]. A named owner and a defined cadence. **Patient-facing transparency where feasible**, because letting a patient see who accessed their record is the strongest deterrent available and converts an internal control into an accountable one. Retention long enough to be useful, because breaches are discovered late. And tamper resistance, because an audit log a privileged user can edit is not evidence.

Two notes on clock and format, which are security properties and are usually treated as operational ones. **Time synchronisation is a security control**: without it, audit records from different systems cannot be ordered and the trail cannot establish sequence, which is why the relevant profile family makes consistent time a required grouping, and why the gazetted South African interoperability framework names the audit profile at all [23][24]. And **the audit record must be in a common format** or it cannot be correlated; an estate where each system logs in its own format has logs, not an audit trail. Central collection must also be assumed hostile-resistant, because a log shipped to a collector inside the same failure domain as the audited system does not survive the incident it exists to document.

The generalisable idea is worth stating plainly, because it transfers out of healthcare: **in a federation where no single party can enforce authorisation policy on the others, mutual node authentication plus a shared, standardised, time-synchronised, centrally collectable audit record is the enforceable control.**

Name the single points of failure you do not own

This is the chapter's instruction, and it should appear in every architecture document.

Every architecture must name its shared-service single points of failure, including the ones it does not own. Three of the six cyber incidents above were attacks on a supplier, and in each case the affected providers had no control over the compromised entity and no ability to shorten the outage [3][6][11].

In South Africa four are immediate:

- the **National Health Laboratory Service**, for public-sector pathology — reported at 265 laboratories and about 8,000 staff serving every public hospital and clinic in the country, on one named platform [25], and down twice in twenty months;
- the **claims switch network**, for private-sector revenue;
- **HPRS**, for national patient identity;
- **SyNCH**, for chronic medicines.

For each, the architecture document should record four things: the dependency, stated as a named service rather than a vendor; the **clinical functions** that stop when it stops; the designed degraded mode and how long it holds; and the owner of the relationship through which an outage is communicated. No individual hospital architecture can mitigate a national dependency, which is exactly why the dependency has to be **declared** in every provincial and facility architecture rather than assumed away.

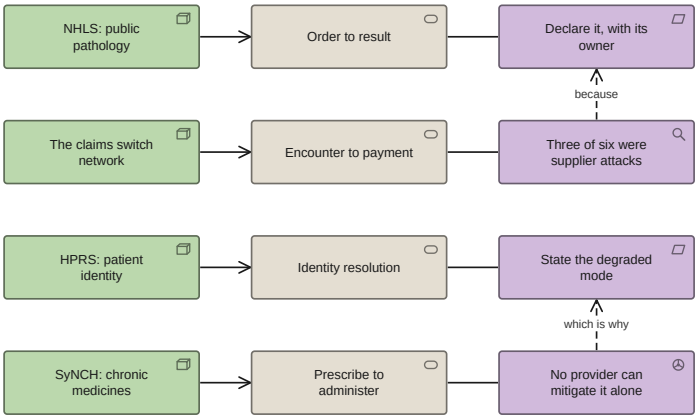


Figure 19.4 Four South African shared services that sit outside a provider's boundary, the clinical function each one stops, and the declaration every architecture owes them.

A note on the contracting consequence, because it is where the claim gets made. A committed availability figure of 99,9% or four nines does not settle any of this, and the South African evidence shows why. It is measured on the component rather than the service, so a platform can meet its number while a clinician cannot see a result because the failure is in a feed, a network segment or an identity service. **The exclusions do the work** — planned maintenance, force majeure, third-party network failure, customer-side infrastructure and power are routinely excluded, and **power is exactly what failed in March 2026**. The remedy is a service credit, and a credit does not treat a patient. And availability is not recoverability: both laboratory incidents were recovery problems, and in 2024 the data warehouse did not recover at all. What settles it instead is the measurement definition, the exclusion list read in full, the actual monthly availability record for the last 24 months with the incident reports, the tested recovery objectives with the date of the last end-to-end restore test, and the power and failover design at the hosting site with its test records [18][12].

Maturity: the model everybody quotes, and the one that fits

What EMRAM measures, and what it does not

The Electronic Medical Record Adoption Model is the most widely cited maturity model in health information technology, classifying a dimension of digital health from stage 0 to stage 7 [26]. What can be sourced about its mechanics is limited and matters: **stages 1 to 5 are self-assessed** through an online tool; **stage 6 is validated through a conference call** with an inspector; **stage 7 requires a prior stage 6 validation and an onsite visit by three inspectors**. Acute care stage 7 requires pervasiveness of use, including **more than 90% computerised provider order entry, more than 95% closed-loop medication administration**, and more than 95% for blood products, human milk and specimen collection; physician documentation using structured templates above 90%; a documented governance structure with examples of governance in action; and a filmless imaging department [27].

Seven limits, and the book should carry them because EMRAM is used uncritically in business cases.

1. **It measures adoption and pervasiveness of specific technologies, not outcomes.** The sourceable stage 7 criteria are percentage thresholds. An organisation can satisfy them while delivering poor care, and a stage 3 organisation can deliver excellent care.
2. **It is normative about *how*, not just *what*.** The criteria embed the integrated acute-hospital electronic record, which was the right model for North American academic medical centres and is a poor fit for a district hospital whose binding constraints are connectivity, staffing and medicines supply. A provincial hospital that invested in an excellent laboratory results pathway, a reliable patient index and a working paper-backed downtime capability would score poorly and be better run.
3. **It is partly self-assessed**, which makes cross-organisation comparison at the lower stages weak. A self-assessed stage 5 is not a validated stage 5, and the distinction matters commercially because vendors quote customer stages.
4. **The criteria are proprietary**, so an organisation cannot fully plan against them without engaging the publisher, and an independent reviewer cannot verify a claim. The research behind this volume could not source the requirements for stages 0 to 4 from the publisher's own material at all, and unofficial summaries in circulation are not authoritative [27].
5. **It measures the acute hospital.** Primary care, community services, mental health and the longitudinal record across providers are addressed by other models or not at all. For a health system whose architecture problem is *between* organisations, it measures the wrong unit.
6. **Pervasiveness thresholds can drive the wrong behaviour.** A 95% closed-loop administration target is a strong incentive to scan, and chapter 9 has the published evidence that barcode-assisted administration moves error between categories and costs time. A target expressed as a scanning percentage, with no corresponding measure of the error categories that worsen, is an incentive to a partially understood intervention.
7. **It says nothing about resilience. No stage requirement for a read-only record, tested downtime procedures or clinically derived recovery objectives could be sourced for this volume** [27]. After the seven incidents above, that is a serious omission in a model used to judge digital maturity.

Two further gaps the research could not close: whether any South African organisation holds a validated stage, and at what level — no validation announcement naming one was found, and if none exists that is itself a finding — and the content of the model's announced revision, whose pages could not be retrieved [26][27].

Use it, then, as a **vocabulary** and a **gap-finder**, not as a target. Its stage structure is a genuinely useful way to have a structured conversation with a clinical executive about what is missing. Pursuing a validated stage is a legitimate goal only when the organisation independently wants the capabilities the stage requires; pursuing it for the badge is a well-documented way to buy capability that is not used.

The better fit for this readership

There is a genuine alternative, and it fits a South African or broader African context better because it measures a **health system** rather than a hospital, and because it is free and open: the **Health Information Systems Interoperability Maturity Toolkit**, developed by MEASURE Evaluation with the Health Data Collaborative's digital health and interoperability working group. Version 1.0 was developed in 2017 and the toolkit was updated in January 2019 with lessons from early adoption in Ghana and Uganda [28][29].

It has **three domains divided into 18 subdomains**. Leadership and governance covers the governance structure, interoperability guiding documents, compliance, data ethics, monitoring and evaluation, **business continuity**, financial management and financial resource mobilisation. Human resources covers policy, capacity in skills and numbers, and capacity development. Technology covers national enterprise architecture, technical standards, data management, subsystems, operations and maintenance, communication networks and hardware. Maturity runs over **five levels: Nascent, Emerging, Established, Institutionalized and Optimized** — the toolkit's own spellings [28][29].

An assessment is self-administered, though an external party can lead it: decide whether an assessment is needed; define its scope; form an oversight team; do a desk review of existing documents; assemble a broad assessment team including stakeholders beyond the health ministry; collect data in a workshop where members answer individually or in small groups and then reach consensus; analyse, scoring subdomain levels and mapping them onto the model; validate; and plan actions with responsibilities, timelines and resources against priority gaps [28].

Three things make it a better fit than a stage model.

The lowest-subdomain rule is architecturally honest. Domain levels are set by the **lowest subdomain** and the overall level by the **lowest domain** [28]. Interoperability maturity genuinely is limited by its weakest component — an excellent enterprise architecture with no governance structure or no skilled staff does not produce interoperability — and taking the minimum rather than the average resists the averaging that makes most maturity scores meaningless.

"Business continuity" is a named subdomain under leadership and governance. After the seven incidents above, a maturity model that treats continuity as a governance responsibility rather than an infrastructure footnote is measuring the right thing.

Human resources is a first-class domain. The evidence elsewhere in this volume is that skills are the binding constraint in South African health informatics, and a model that scores them is more useful than one that does not.

The research behind this volume found **no published South African assessment using the toolkit**, and offers a provisional reading — Emerging to Established on technology and leadership, with the lowest-subdomain rule probably pulling the overall level to **Emerging** — which it flags clearly as opinion rather than a sourced assessment [28]. Treat it that way. Running the assessment properly would be a genuinely valuable piece of work, and it is a better use of a reader's week than chasing a stage.

Twelve things to benchmark instead

This is the most immediately useful page in this chapter, and possibly in the book, for a practising architect. Every measure below is cheap to collect, hard to game, and directly connected to a failure mode documented somewhere in this volume. Six of the twelve are drawn directly from the published contingency-planning practices; the rest are this volume's own synthesis of the failures above [20].

#	Measure
1	Percentage of laboratory orderables mapped to LOINC, and percentage of incoming results carrying a LOINC code and a unit of measure code
2	Median and 95th-percentile time from critical result authorisation to clinician acknowledgement, and the proportion never acknowledged
3	Count of items in integration exception queues, and median age of the oldest item — the single best proxy for integration health
4	Master patient index duplicate rate and false-merge rate, plus review-queue depth and age
5	Interface inventory completeness, and the date of the last review
6	Date of the last unannounced downtime drill, and whether clinicians participated
7	Date of the last successful restore test from offline backup
8	Whether a read-only record exists, its currency, and the date it was last used in anger
9	Break-glass use rate per thousand record accesses, and the percentage reviewed
10	Percentage of medical devices with a known software version, a known support status and a named patching owner
11	Synthetic transaction response time, hourly mean
12	Proportion of clinical decision support alerts suppressed, with a documented clinical owner for each suppression

None of those twelve is a stage. None requires a licence, a consultant or a survey. Each one is a number an organisation can produce in a fortnight and defend in a board paper, and each one moves when something real changes. Measures 6, 7 and 8 together are the resilience position in three lines — and they are the three a stage model does not ask for.

THE CHAPTER IN TEN LINES

- Downtime in healthcare is clinical, and a ransomware attack on a pathology supplier has been found to have contributed to a patient death, with 170 patients harmed.
- Three of the six sourced cyber incidents were attacks on a supplier. The dominant failure mode is supplier concentration outside the organisational boundary.
- The planning assumption is weeks, not hours. Every incident with a sourced restoration timeline ran to weeks or months.
- A national health service was saved by the attacker's decryptor rather than by its own backups, because its offline tape copies were only periodic.
- The June 2024 national laboratory outage is reported both as forty days wholly inaccessible and as about sixty days to announced full restoration. The two figures disagree and both should be printed.
- The March 2026 outage of the same service was a power failure, not an attack. Security investment and availability investment are different investments.
- A read-only record with hourly currency, weekly testing, printable output and independent reachability is the highest-value single resilience investment in a clinical estate.
- Recovery objectives are derived by clinical function, by asking how long before a patient is harmed and how long the manual substitute holds.
- Thousands of staff may legitimately read one record, so access control is broad and audit is the primary control rather than a compensating one.
- Every architecture must name its shared-service single points of failure including the ones it does not own: here the national laboratory service, the claims switch network, HPRS and SyNCH.
- EMRAM measures adoption and pervasiveness, is partly self-assessed, has proprietary criteria, measures the acute hospital, and contains no sourced resilience requirement at all.
- The interoperability maturity toolkit is a better fit: free, health-system level, with business continuity as a named subdomain and a lowest-subdomain scoring rule.
- Twelve cheap, hard-to-game measures beat a stage number, and three of them are the organisation's whole resilience position.

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CHAPTER 20

Modelling healthcare in ArchiMate

Forty-two elements, eleven relationships, and no element for a retention period, a code system or a consent that expires.

The notation is small. Forty-two element types, eleven relationship types, two kinds of junction. An architect who has spent a decade in it stops noticing how small, and the healthcare engagement is where that stops being harmless, because a provider's architecture turns on facts the notation has no place for: a classification, an effective date, a version, a retention period, a device risk class, a code system. All of those are properties in a tool, not elements in a diagram, and a model that does not say so produces views that look complete and answer nothing.

This chapter does two things. It fixes the element names, because the names in circulation are wrong and the wrong ones are in the material you will inherit. Then it gives the patterns — the dozen drawings a healthcare architect actually needs, and what each one has to carry in properties to be worth drawing.

The names, exactly

Start with the correction, because it travels into every deliverable.

Element names in ArchiMate 4 are unprefixed. There is one `Service` element, one `Process`, one `Function`, one `Event`, one `Role`, one `Collaboration`. Each is *placed* in a layer, and the layer is expressed by position and fill colour, not by the element's name. So a results-distribution capability of an integration engine is **a `Service` in the application layer**. A hospital's outpatient offering is a `Service` in the business layer. There is no layer-prefixed service element in either layer, and writing one in a deliverable tells a reviewer that the author has worked from pre-version-4 training material.

Two families *are* layer-specific, and the asymmetry is the thing to memorise:

- **The interface family.** `Business Interface`, `Application Interface` and `Technology Interface` are three distinct elements. A clinician-facing enquiry counter, a FHIR endpoint and a network port are not the same kind of thing and the notation agrees.
- **The object family.** `Business Object`, `Data Object` and `Artifact` are three distinct elements, in that order of descent: the thing the business talks about, its representation in an application, the file or table on a disk.

Beyond those, `Business Actor` and `Product` belong to the business layer; `Application Component` to the application layer; and `Node`, `Device`, `System Software`, `Equipment`, `Facility`, `Communication Network`, `Distribution Network`, `Artifact` and `Material` to the technology layer. The motivation group has `Stakeholder`, `Driver`, `Assessment`, `Goal`, `Outcome`, `Principle`, `Requirement`, `Value` and `Meaning`. Strategy has `Resource`, `Capability`, `Value Stream` and `Course of Action`. Implementation has `Work Package`, `Deliverable` and `Plateau`.

Element names: what carries a layer in the name, and what does not

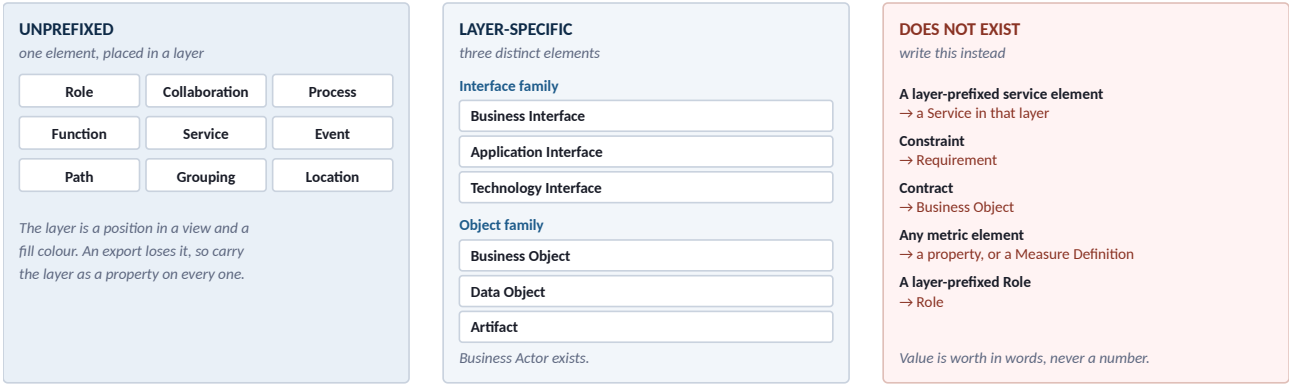


Figure 20.1 The naming correction, in one table. The nine unprefixed elements take their layer from where they are placed, so the layer has to be carried as a property or it is lost on export. Two families are layer-specific. Business Actor exists and the role element is plain Role, which is the asymmetry people get wrong.

THE FIVE NAMING ERRORS, AND WHAT TO WRITE INSTEAD

There is no layer-prefixed service element. Write "a `Service` in the application layer" or "a `Service` in the business layer". The layer is a placement, not part of the name.

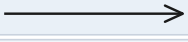
There is no `Constraint` element. Use `Requirement`. A constraint in the ordinary English sense — records must be retained, Schedule 5 to 8 prescriptions must be handwritten, special personal information may not leave the country without prior authorisation — is a `Requirement` with a restrictive statement. Earlier versions of the notation had a separate element; version 4 does not, and a model built on one will not import.

There is no `Contract` element. Use `Business Object`. This matters more in healthcare than elsewhere, because the objects an architect most wants to call contracts are a funder agreement, an interface specification, a service-level undertaking and a standard-plus-version — and all four are business objects that happen to be contractual. Model the counterparty as a `Business Actor`, the agreement as a `Business Object`, and the obligation it creates as a `Requirement`.

There is no metric element, and `Value` is not a number. `Value` in the motivation group means worth to a stakeholder, expressed in words. There is nowhere in the notation to put 74,6% or R50,2bn or a 24-hour recovery objective. Every quantity in a healthcare model is a property on an element, or an attribute of a `Business Object` that *defines* the measure. This is the single most common cause of a repository that cannot answer a board question.

There is no layer-prefixed variant of `Role`. `Business Actor` exists; the role element is plain `Role`, used in every layer. This is worth stating because `Business Actor` does exist, so the pattern looks like it should extend, and it does not. A clinical coder, an admitting officer, a data steward and a system account are all `Role`.

The eleven relationships, and what each is for in a provider model

Composition		A node composed of a device and its system software
Aggregation		A collaboration aggregating the components in a seam
Assignment		A role to the behaviour it performs; an actor to a role
Realization		A data object realising the clinical record; a component realising a service
Serving		An external service into the process that depends on it
Access		A process reading or writing an object. Direction carries read and write
Influence		An assessment on a requirement. The only soft relationship
Association		Everything the other ten cannot say, which is why it is overused
Triggering		One behaviour causing the next, within a stage
Flow		What moves between them: a referral, a result, a claim
Specialization		A kind of. Not an instance of, which the notation cannot express

Plus two junctions, and or. There is no cardinality on any of them, so a conceptual data model drawn only in this notation loses every multiplicity in it.

Figure 20.2 The eleven relationships, with what each is for in a provider model. Association is the one to watch: it carries everything the other ten cannot say, so a model in which most relationships are associations has not decided anything yet.

Two spellings. The relationship is `Realization` and the generalisation relationship is `Specialization`, both with a `z`, because those are the specification's own strings and a repository import matches on them. This volume is written in British English and these two words are the exceptions; write them the specification's way inside backticks and the British way in prose.

The inherited-material problem, which is yours

The risk is not that you will get this wrong. It is that the material you are handed already has. Pre-version-4 training content, vendor reference models and populated spreadsheets use the layer-prefixed service names freely, and the supplied worked example is no exception: two of its sheets carry object-type names for elements that do not exist in the notation — the sheet numbered `O15`, which holds ten externally-consumable services realising the ten level-one capabilities, and the sheet numbered `O19`, which holds ten vendor agreements. Both are legitimate content. Both are mislabelled as element types.

Handle them by number, not by name. In a deliverable, write: "`O15`'s ten rows become `Service` elements in the business layer; `O19`'s ten rows become `Business Object` elements with a `Business Actor` for each counterparty." A reader who knows the workbook will follow it, and nobody inherits the wrong element name from your document. If you rename the sheets, say so in a change note, because somebody's formula is pointing at the old tab — and in the supplied file one of those tab names ends in a trailing space, so the formulas are fragile already.

What the notation will not do

Six limits. Each one is a place where a healthcare architect reaches for something and finds nothing, and in every case the remedy is the same shape: record it as a property, and say in the view that you have.

There are no attributes. An element has a name, a documentation field and whatever properties the tool lets you define. The notation itself carries none. So the facts that decide a healthcare design — a data entity's classification, its retention period, the code system a diagnosis was coded against, a device's risk class, the version of a protocol, the effective date of a tariff — are all tool properties. Two consequences follow, and both are practical. A property is

invisible in a view unless the tool is configured to print it, so a diagram showing a `Data Object` called Diagnosis tells a reviewer nothing about whether it is classified correctly. And a property set is the first thing lost in an export to a spreadsheet, which is how a repository ends up with fourteen data entities and no classification anywhere.

There is no cardinality. `Association`, `Aggregation` and `Composition` carry no multiplicity. So "one patient has many encounters, and an encounter belongs to exactly one patient" is not expressible. A conceptual data model drawn only in this notation loses every cardinality in it, which is why the data chapter's model has to be maintained as a model with cardinalities and *then* drawn, not drawn and then interrogated. The supplied repository demonstrates the cost: its entity relationship sheet records a patient-to-encounter cardinality of many-to-many, and nothing in the drawing would have caught it.

There is no time on an element. `Plateau` is the only temporal construct and it describes a state of the whole estate at a point in the roadmap, not the validity period of one object. There is no way to say that this tariff applies from 1 April, that this protocol version was in force on the date of care, that this registration lapses in March, or that this consent expires. Effective dating is the single most pervasive requirement in South African health data and the notation has no expression for it at all. Put a valid-from and a valid-to property on every element whose identity includes a period, and treat the absence of those two properties as a modelling defect rather than an omission.

There is no distinction between a type and an instance. A `Device` element is a kind of device, not a particular one. The distinction between a device model — manufacturer, classification, intended use — and a device instance — serial number, location, last calibration — cannot be drawn, and nor can the lot or batch that is the unit of a recall. The same applies to a protocol and its versions, and to a practitioner and their registrations. Model the type in ArchiMate and hold the instance in the system of record; what the architecture model owns is the *statement that both exist and which system holds each*.

There is no state machine. The notation has `Event` and `Triggering` and `Flow`, which let you draw a sequence. It has nothing to express a state an object can be in. "Sent but not acknowledged" — the state that makes a referral and a diagnostic result auditable, and the one most estates do not record — cannot be drawn. The workaround that holds up in review is a `Business Object` for the tracked request with its permitted states listed in the documentation field, and a `Requirement` naming the state that must be recorded. The requirement is what gets argued about, which is the point.

There are no swimlanes. A lane boundary is where handoffs fail, and in a provider the handoff is almost always the finding. That is why the level-two process models in this volume are drawn in BPMN. Use ArchiMate for the value stream, the stages, the roles and the systems, and change notation when the boundary is the point.

Three places a healthcare fact can live, and what has no expression at all



The remedy for the middle column is a property set agreed on day one. The remedy for the right-hand column is to say so, in writing, at the front of the document.

Figure 20.3 Where a healthcare fact can live. The middle column is a property set to agree in the first week. The right-hand column is what the notation cannot express at all, and the only honest response is to say so at the front of the document and name where each fact is maintained instead.

> **DERIVED RELATIONSHIPS, AND WHY A CLEAN REPOSITORY CAN BE WORTHLESS**
>
> The notation permits relationships to be *derived*: where a chain of structural relationships runs from A to B, a single relationship from A to B is valid, and a tool will tell you so. This is useful and it is also the mechanism by which a repository passes every integrity check and says nothing. A relationship can resolve perfectly at both ends, satisfy every derivation rule, and pair two objects that have nothing to do with each other. No tool will catch it, because the error is semantic and the tool only tests structure. The next chapter is a repository in exactly that condition, and the three questions that find it are: does the subject of one end appear anywhere in the other end's name; would the person who owns one end recognise the other; and if the relationship were deleted, would anybody notice.

Seven patterns worth having in your head

What follows is not a reference model. It is seven drawings that come up on every provider engagement, with the properties each needs. Two invented organisations carry the examples, both fictional: **Driehoek Health Group**, a mid-size private hospital group of nine acute hospitals and two day clinics with about 1,750 registered beds; and **Rietvlei District Health Service**, one public health district with a district hospital, three community health centres, 22 fixed clinics and a mobile service.

Consent: three objects, three owners, one withdrawal that does not work backwards

Consent to treatment, consent to disclosure and consent to research are different objects with different legal bases and different consequences of withdrawal. Modelling them as one is the classic error and the supplied repository makes it, with a single entity whose definition contains the word "and".

Draw three `Business Object` elements, not one. Give each its own `Role` for the owner — clinical governance for treatment, the information officer for disclosure, the research office for research — and its own `Requirement` for the constraint that actually binds it. The constraints are not the same, and that is the argument the drawing is for:

Object	The binding constraint, drawn as a requirement element	Why the model needs it separate
Consent to treatment	Capacity is a function of age and an assessed maturity: under section 129 of the Children's Act a child over 12 with sufficient maturity and capacity may consent to their own treatment [2]	The assessment is itself a clinical act with an author and a date, so the object has a sub-object; a boolean cannot carry it
Consent to disclosure	Section 14(2)(a) of the National Health Act permits disclosure on the user's consent **in writing** [1]	A verbal consent does not satisfy it, so form is a required property and not a nicety
Consent to research	Section 16(1)(b) requires three authorisations — the user, the head of the establishment and a research ethics committee [1]	Three relationships on one object; a model with one approver cannot represent it

Then draw the object the repository never has: a **disclosure event**. Information given to a named recipient, on a date, for a purpose. It is a `Business Object` accessed by the process that made the disclosure, and it is what makes a withdrawal actionable — because withdrawal of a disclosure consent stops future disclosure and cannot unsend what was sent, so the only way to act on it is to know who received what. It is also what makes the data subject's participation right operable.

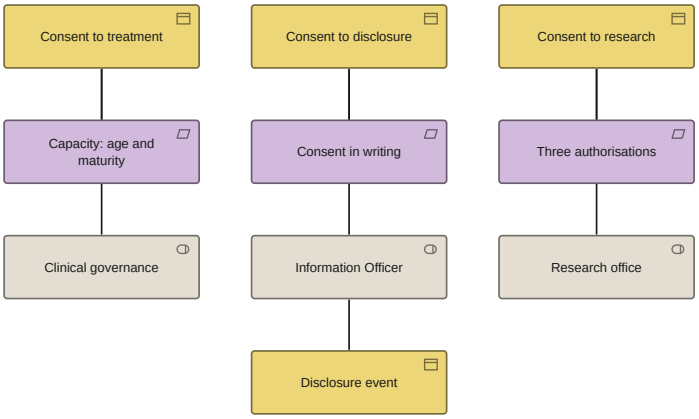


Figure 20.4 Consent as three objects. Each has its own owner and its own binding constraint, and the constraints differ — which is the argument the drawing exists to have. The disclosure event is the object no supplied repository has, and it is what makes a withdrawal actionable, because withdrawal stops future disclosure and cannot unsend what was sent.

Properties on each consent object: scope, version, granted date, expiry, withdrawal date, the person withdrawing, and the form. The two most often missing are withdrawal and expiry, and their absence means the organisation cannot prove it stopped processing when it was told to.

The record as one object across three layers

The clinical record is simultaneously a clinical instrument, a legal instrument and a billing instrument, and the notation handles this better than most people expect — provided the three layers are drawn as three elements and not one.

- A `Business Object` for **the clinical record as a legal construct**: what the provider must retain, can be compelled to produce, and must be able to reconstruct as it stood at a point in time. This is the element the retention `Requirement` attaches to, and the element a medico-legal claim is answered from. It is not identical to any one system's contents.
- One or more `Data Object` elements for the record *in* each application, each with a `Realization` to the business object. A record held across an electronic patient record, a laboratory system, an imaging archive and a paper folder is four data objects realising one business object, and drawing it that way is how the retrieval problem becomes visible.
- `Artifact` elements for what is actually on storage, where the residency and encryption properties live.

The `Requirement` elements that attach to the business object are the useful part. Append-only amendment with an attributed, timestamped, reasoned correction, because the professional guidance requires correction by strike-through with the correction, date, full signature and reason — which means a system implementing an in-place update of a clinical note is non-conformant [3][4]. Separation of billing records from patient care records, which the same guidance requires at paragraph 13.7 and which almost no integrated hospital information system honours [3]. Retrievability, because in an Auditor-General review of paid medico-legal claims, departments conceded in three of nine cases because the record could not be found [5].

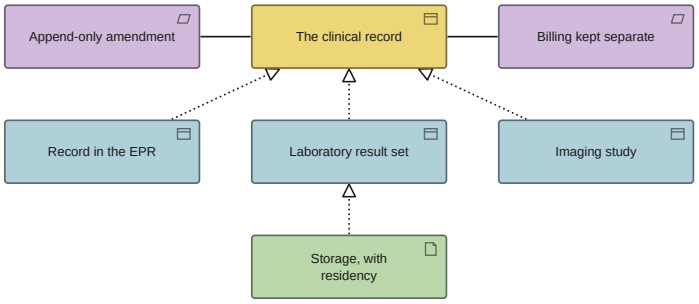


Figure 20.5 The record across three layers. One business object carries the legal construct and the retention requirement; the data objects are the record as each system holds it; the artifact is where residency and encryption live. Drawn this way, the retrieval problem and the billing separation requirement are both visible.

> **WHAT THIS DRAWING SETTLES**
>
> Three arguments, in one picture. First, what "we need an electronic health record" means, because the business object and the data objects are visibly different animals. Second, where retention applies — to the business object, which means every data object realising it inherits a period, and a warehouse extract that nobody assigned a purpose to is suddenly in scope. Third, the separation requirement, because once the billing objects are drawn as separate business objects with a separate access population, "the revenue cycle and the clinical record are different stores" stops being an opinion and becomes a modelled constraint with a professional-guidance citation behind it.

A pathway against a protocol version

A protocol is the organisation's adopted standard. A pathway is one patient's instantiation of it. Collapsing them is why nobody can answer "was this patient managed according to protocol?", because the answer requires knowing which version was in force on the date of care.

Draw the protocol as a `Business Object` and the pathway as a `Process`, joined by an `Association` and not a `Realization`: the process does not realise the protocol, it *follows* it, and the difference is the deviation. Put the version in the protocol object's name or in a property, and accept that the notation cannot hold the version set — so what the model owns is the statement that a protocol register exists, that version is part of protocol identity, and which system holds the register. In practice, for most South African providers, the honest answer to that last question is a folder on an intranet with the version in the filename and the approval in an email.

Two properties on the protocol object earn their place. **The national source and its version**, because the citable national artefacts exist and are dated — the Primary Healthcare Standard Treatment Guidelines and Essential Medicines List, eighth edition, 2024 [6], and the National Integrated Maternal and Perinatal Care Guideline signed on 4 October 2024 [7]. And **the local modification**, because a provider that has modified a national guideline has taken on the clinical risk of the modification, and the modification must be traceable.

The element that completes the pattern is a `Requirement` for **deviation with a documented reason**. Deviation is the clinically and legally important fact, not an exception to suppress. Draw it, and the conversation about whether the pathway system should have a free-text reason field happens in Phase B rather than in build.

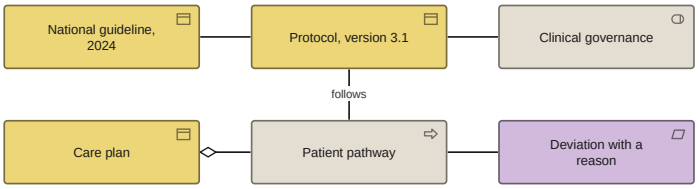


Figure 20.6 A pathway against a protocol version. The process follows a standard rather than realising it, the version is part of the protocol's identity, and deviation with a documented reason is a requirement rather than an exception to suppress. The notation cannot hold the version set, so the model records which system does.

Identity: the components, and the one that is a role

Identity is where an architect is most likely to draw a single box labelled "master patient index" and lose the argument. The components are four, and only three of them are software.

- An `Application Component` for the index itself.
- A `Service` in the application layer for **identity resolution**, consumed by registration, ordering, results and claims. Drawing the service separately from the component is what lets the model say that the service may be provided by a national system the organisation does not own.
- A `Data Object` for the **identifier cross-reference**: tuples of identifier type, value, assigning authority and validity period. Assigning authority is not optional — a folder number without a facility is meaningless and a membership number without a scheme is ambiguous. The identifier types a South African provider must carry include the national identity number, the ten-digit health patient registration number, the facility folder number per facility, programme register numbers, a passport plus country, and a provisional identifier for the unidentified patient.
- A `Role` for the **review queue**. This is the component that is a person, and it is the one that gets dropped. A probabilistic engine produces a score and two thresholds: an auto-link threshold and a review threshold, with a human adjudicating between them [8]. An unworked review queue converts

the whole design into "treat as different people", which is the setting that maximises false duplicates. Queue depth and queue age are the two operational measures that say whether the index is working.

Then three `Requirement` elements, because each is a decision somebody must take and sign:

1. **The thresholds are a clinical safety decision.** A false merge puts two people's clinical data in one record; a false duplicate splits one person's across two. Neither is acceptable and they trade off. The governance record should show that a clinical risk owner approved the numbers, and in practice they are set by an implementation consultant during configuration and never revisited [8].
2. **High-risk data gates the merge.** A merge that would combine two records with different ABO blood groups, or one carrying a recorded fatal allergy, does not auto-merge regardless of score.
3. **Merge and unmerge are first-class operations.** The subsumed identifier is retained as a permanently resolvable alias, because external systems hold it, and a merge that invalidates it breaks every downstream reference.

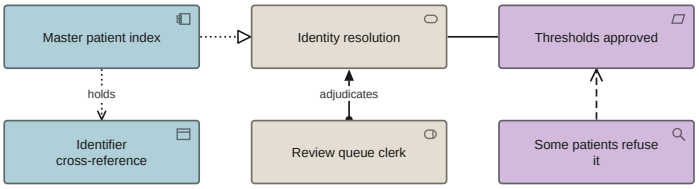


Figure 20.7 Identity resolution as four components, one of which is a person. The review queue is the element that gets dropped, and an unworked queue converts a probabilistic design into the setting that maximises false duplicates. The assessment records that for some patients reliable linkage is a risk they will refuse.

One more element, and it is the one that distinguishes a healthcare identity design from a customer master data design. Draw an `Assessment` recording that identity resolution is not an unqualified good. South African research records stakeholders saying that key populations fear law-enforcement access to health data and are unlikely to permit health workers to identify them on government systems [9]. For some patients reliable linkage is a risk they will refuse, and an index without an access-control and consent counterpart is half a design.

A seam, with the standard that does not actually govern it

The instinct an architect arrives with is one standard per seam, and it produces a wrong model of this industry. Most seams are governed by two or three standards at different layers at once, plus a profile that constrains them, plus a local interface specification that is the actual contract. The order-to-laboratory seam is governed by a messaging standard for syntax, an observation code system and a unit code system for semantics, a conformance profile for constraint, and a site-specific interface specification including local segment extensions for reality. Ask "which standard governs this seam" and you get a one-word answer that is useless. Ask "which layers govern this seam, and what is the actual contract" and you get the four objects you have to draw.

So the seam pattern is four elements and one deliberate omission:

- A `Collaboration` for the two or more components that jointly do the work. Note the name: it is plain `Collaboration`, not a layer-prefixed one, and it is placed in the application layer by position. Its behaviour is a `Process` or a `Service` assigned to it.
- An `Application Interface` at each end — these *are* layer-specific, so the name is right as written — with the direction and the trigger as properties.
- A `Business Object` for **the interface specification**, which is the contract. Not a `Contract` element, because there is none. Its properties are the standard, the version, the profile, the local extensions, the owner, and the date it was last agreed.
- A `Requirement` for what the seam must do that the standard does not require: acknowledge receipt, carry the assigning authority with every identifier, survive the other end being unavailable for a week.

The omission is a line labelled with a standard's name. A standard is not a relationship. Writing "HL7 v2" on the arrow between two components is the commonest way a seam map becomes decorative, because the question that gets asked in every incident and every procurement — what talks to what, in which direction, carrying what, and what breaks if it stops — is answered by the properties of the interface specification object, not by a label on a line. In an estate held together by point-to-point feeds the interface inventory is the valuable artefact, not the application inventory, and the metamodel has to represent interfaces as first-class objects with direction, payload, standard, trigger and criticality.

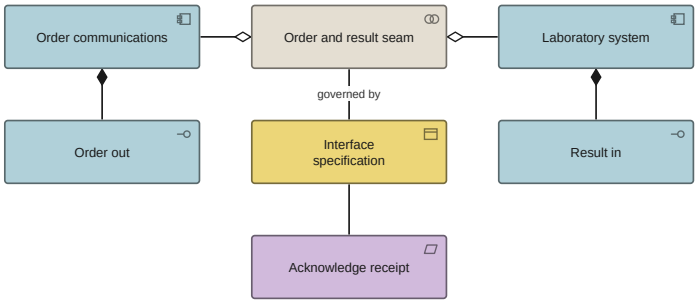


Figure 20.8 A seam. The contract is a business object with the standard, version, profile and local extensions as properties — not a label on a line, because a standard is not a relationship. The requirement carries what the seam must do that no standard requires of it.

The supplied worked example shows what happens when they are not. Its interface sheet holds twenty-three rows with columns for managed, logged, type and frequency, and sixteen of the twenty-three are mutually inconsistent across those four columns — two direct database reads labelled "managed", five managed links that are not logged, eight unmanaged links that nonetheless produce correlated logs, and two rows whose type is batch or file and whose frequency is real time. The columns are right. Nobody checked them against each other, because no relationship in the model depends on them.

A Measure Definition, in a notation with no metric element

This is the pattern the notation most obviously lacks and most needs. There is no metric element and `Value` is not a number, so the question is where a measure lives.

The answer is that the *definition* is an object and the *number* is a property. Draw a `Business Object` called a Measure Definition, carrying the numerator, the denominator, the exclusions, the boundary rule, the version and the period over which it applies. Attach it with an `Association` to the `Driver` or `Goal` it informs, and with an `Access` relationship from the process or application that computes it. The baseline and target are properties on the definition, not elements.

The evidence that this must be an entity rather than a convention is published and specific. On theatre utilisation there is no simple mathematical relationship between the four calculation methods, with raw utilisation of 80% against touch-time utilisation of 60% on the same list — on one specialty, in one Trust, over fifteen weeks [10]. There is no conversion. The same holds for occupancy, length of stay, readmission and mortality. A repository that records a baseline without the rule that produced it has recorded a number nobody can reproduce, and the supplied worked example does exactly that: a lineage-coverage measure with a baseline of 23% that reconciles to no figure anywhere else in the file.

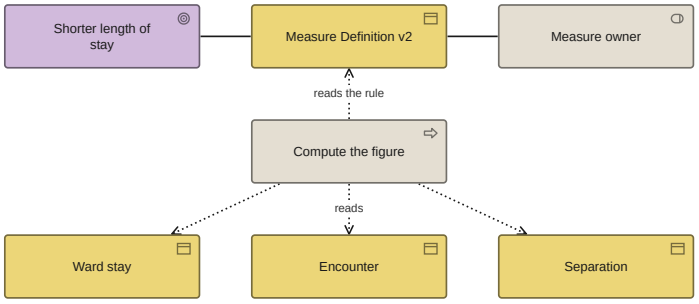


Figure 20.9 A Measure Definition in a notation with no metric element. The definition is an object; the baseline and target are properties on it. The computing process reads the rule and the lowest-level timestamped facts, which is what makes a definitional change a re-computation rather than a migration.

Two South African measures make the case on their own. **Readmission within 30 days**, absent a cross-facility identifier, measures patient loyalty to one facility rather than discharge quality, because a readmission to a different hospital is invisible. And the national aggregate reporting system's completeness has two published values from one assessment — facility reporting 99,5% against data element reporting 49,7% [11] — so a dashboard tile labelled "completeness" is meaningless without its definition attached. Draw the definition, and the tile has a provenance.

The regulated device: `Node`, `Device` and `System Software`

This is the pattern most often got wrong, and the error is a single box.

A regulated medical device that contains software is **a `Node` composed of a `Device` and a `System Software`**, tied to `Requirement` and `Assessment` elements. Modelling it only as a `Device` loses the regulatory constraint, which is the whole reason for recording it in an architecture repository at all. The three technology elements carry the physical unit, the embedded software and the addressable whole; the motivation elements carry why it is in the repository.

The `Requirement` elements come from the South African regulatory position, and it is more developed than most architects expect:

- **A risk class, determined by rule.** Devices are regulated under the Medicines and Related Substances Act through the general regulations for medical devices and in-vitro diagnostics, with four risk classes for non-IVD devices and a parallel IVD scale; classification is by rule, the manufacturer or distributor determines the class, the higher class applies where more than one rule is engaged, and accessories are classified separately from the device they accompany [12].
- **No dedicated software classification rule.** Software as a medical device is classified by principle as an active device. There is no separate rule set for it [12].
- **A locked algorithm version.** For a device containing a machine-learning model, the regulator's September 2025 position is that the algorithm version is locked at licensing; minor updates run through a certified quality system, major updates are likely to need review, and an anticipated change plan requires approval covering scope, implementation protocol, validation on independent datasets, boundaries, and data triggers such as drift beyond a threshold [13].
- **Local-population validation and subgroup analysis.** The same position requires subgroup analysis across sex, age and ethnicity, local-population validation where the training data came from elsewhere, documented dataset gaps, and recording of clinician overrides [13]. That last set is the first question to ask of any imported product with a model in it.

The `Assessment` elements carry two facts an architect must not leave implicit. **Product registration is not yet operating** — the regulator has not begun product registration and the relevant regulation has not been called up, so what is in force is an establishment licence regime covering manufacturers, distributors and wholesalers [13]. Which means the second assessment: a provider's device master **cannot be populated from a national product**

register, because there is not one, so the asset register plus the supplier's declarations are the only available source of truth, and the register is a claim rather than a fact.

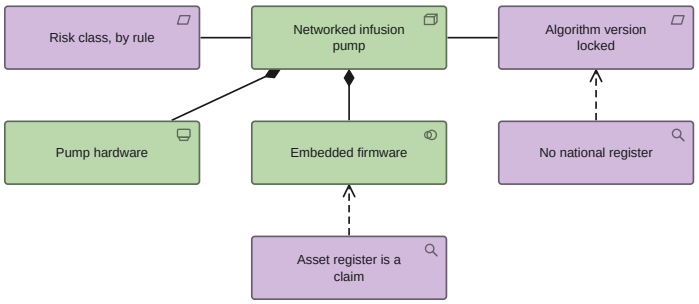


Figure 20.10 A regulated device as a Node composed of a Device and a System Software, tied to requirements and assessments. Modelling it only as a Device loses the regulatory constraint, which is the whole reason it is in the repository. The two assessments are the facts an architect must not leave implicit.

Properties on the device elements: risk class and the rule engaged, intended use with its provenance — the regulator's guideline takes intended use from the instructions for use, labelling, advertising and technical documentation, so it is a *claimed* attribute [12] — calibration due and last calibration, maintenance schedule, software version, patch level, and for an implant the patient link. Without the last of those a recall cannot be executed: the organisation can find the stock and not the patients.

A shared service the organisation does not own

The last pattern is the one that changes a resilience conversation, and it is three elements.

Every provider depends on services outside its own boundary, and in South Africa the four that matter are named: the national laboratory service, the claims switch network, the national patient registration system and the chronic medicine dispensing and distribution service. None is owned by the provider. All four are single points of failure, and the planning assumption the evidence supports is weeks rather than hours.

Draw the external organisation as a `Business Actor` outside a `Grouping` that bounds the provider. Draw what it provides as a `Service` — in the business layer if the dependency is an organisational one, in the application layer if it is a system interface — with a `Serving` relationship into the provider's `Process`. Then draw the `Assessment` that says the service is a single point of failure the organisation does not own, and a `Requirement` naming the manual substitute and how long it holds.

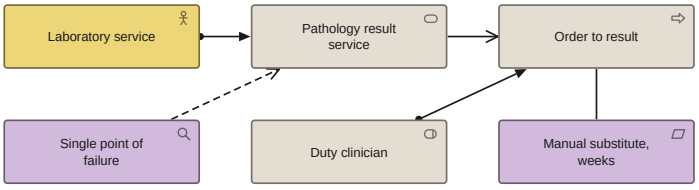


Figure 20.11 A shared service the organisation does not own, with the provider boundary left out for legibility. A dependency inside the boundary gets a recovery objective; a dependency outside it gets a substitute, and the substitute is a clinical design decision with a named person who invokes it.

The pattern earns its place because of what it makes drawable. A dependency inside the boundary gets a recovery objective. A dependency outside it gets a *substitute*, and the substitute is a clinical design decision rather than a technical one: during the 2024 national laboratory outage, results were delivered by telephone, message, email, fax, courier and paper [14][15], every route producing an observation with a correct effective time and an unrecorded available time. Model the substitute and that data-quality consequence is visible before the outage rather than after it.

> **THE FIVE QUESTIONS A VIEW OF AN EXTERNAL DEPENDENCY HAS TO ANSWER**
>
> Who holds the service. What the provider's own process does when it stops. How long the substitute holds — hours, days or weeks. What data the substitute loses, and whether the loss is recoverable. And who, by name and role, decides to invoke it. A view that answers those five is worth more than a view with fifty boxes in it, and it fits on one page.

Which views to draw, and in what order

The specification's appendix offers twenty-five viewpoints. The supplied worked example's deck draws one of each, which is the wrong instinct — twenty-five views is a catalogue, not an argument, and a catalogue is read once. Draw six.

1. **A layered view of one value stream, end to end.** Not the whole estate. One stream, with its stages, the roles that perform them, the applications each stage touches and the data objects each reads and writes. This is the view that finds the stage with no system behind it, and in a provider there is always one.

2. **An information-structure view of the entities that cross organisational boundaries.** Patient, encounter, diagnostic order and result, referral, transfer summary, consent. Each with its classification and its master system as visible properties. This is the view a privacy officer can read, and the only one that makes a misclassification obvious.
3. **A service-realisation view for the identity service.** Which capability it realises, which applications consume it, which component provides it, and — the point — whether the provider owns the component.
4. **A motivation view for the obligations that are actually in force.** Drivers for the statutes and professional rules, requirements for what each demands, assessments for the current position, and nothing aspirational. The discipline is that every requirement traces to a driver with a citable instrument behind it, which excludes most of what a motivation layer usually contains.
5. **A capability map, at level one only, with a maturity property and an owner.** Level two and three decompositions are where generated content accumulates, and a map with thirty generated children is worse than a map with ten real parents.
6. **A deployment view of the external dependencies.** The previous pattern, drawn once for all four shared services, with the boundary explicit.

Everything else waits until somebody asks a question it would answer. A view with no question behind it is maintained by nobody and is wrong within a quarter.

What the repository cannot tell you

One paragraph, and it is the most important one in the chapter, because it bounds everything above.

A repository can record that an application exists, that it holds patient data, that it is classified Restricted, that it receives an admission feed and that its contract expires in 2029. It can model a capability, a value stream, a work package and a principle. It can run an impact analysis across a dependency graph and tell you what breaks if one interface stops. It cannot tell you whether a configuration change to a drug-interaction alert threshold will cause a missed interaction. It cannot tell you whether a result that arrives without its reference range is clinically interpretable. It cannot tell you whether a patient-merge procedure loses the transfusion record. It cannot tell you whether an imported triage model degrades in the local population. It cannot tell you whether a degraded-mode process is clinically safe.

Those are clinical safety judgements. They are made by clinicians, pharmacists, laboratory scientists and clinical safety officers, against a hazard log, with a documented safety case. The repository is where the evidence of those judgements is filed and traced. It is not where the judgements are made, and no amount of modelling rigour substitutes for them. An architect who presents a clean repository as evidence of a safe architecture has made a category error, and the next chapter is the demonstration: perfect referential integrity over content that would harm patients.

What the repository *is* genuinely good at, and what nobody does, is the inverse. Model the safety artefacts as objects — a hazard log, a clinical safety case and its approval, a data protection impact assessment, a conformance attestation, a device classification — attach each to the applications and interfaces it covers, give it an owner and a review date, and report on the ones that are missing or expired. That is a query, it takes a fortnight to set up, and it is the highest-value thing an architecture repository does in a provider.

One tooling note, because it will come up in the first month and the honest answer is short. This volume's research found no verifiable evidence that any enterprise architecture tool ships healthcare reference content as a shipped, maintained, documented artefact — no healthcare capability model, no clinical process library, no messaging or terminology metamodel extension, no profile catalogue. Several vendors reference reference content generically. Assume you are building the healthcare content yourself, and treat any vendor claim of healthcare reference content as a claim to be inspected: ask to see it, ask when it was last updated, and ask who maintains it. On pricing the position is the same and worth saying plainly: no published list price was located for any enterprise architecture repository tool, and the one vendor that publishes a pricing page publishes a modular structure and three licence role types with no figures and no currency [16]. What determines cost in a provider estate with twenty departmental system owners is the licence role model, not a headline price, so model the actual user population by role before asking anyone for a quote.

Keeping it honest

Four conventions, each of which takes an hour to agree and saves a rebuild.

Record the layer as a property. Because `Service`, `Process`, `Function`, `Event`, `Role` and `Collaboration` are unprefixed, the layer exists only as a position in a view and a fill colour. An export loses it. A spreadsheet populated from an export loses it. So carry a layer property on every unprefixed element, set it on creation, and validate it. This is also, incidentally, how the inherited layer-prefixed names got into circulation in the first place: somebody needed the layer in a list and put it in the name.

Name an element for what it is, not for the sheet it came from. A row in a generated workbook whose description reads "externally consumable service realising clinical care delivery" is a sentence with a capability name dropped into it. Promoting it to an element preserves the emptiness and gives it a reference number. Rename, or do not import.

Review every relationship for meaning, not resolution. Take a sample of twenty, and for each ask whether the owner of one end would recognise the other. A pass rate below about three-quarters means the relationships were generated and the model is decorative. The next chapter is what that looks like from the inside.

Say what the model does not carry. One page at the front of every architecture document, listing what is held as a property rather than drawn, what has no expression in the notation at all — time, cardinality, state, quantity, instances — and where each of those facts is actually maintained. It is the cheapest credibility a modelling deliverable can buy, and it stops the second most common failure of a healthcare repository: a reader assuming that because the picture is complete, the decisions behind it have been taken.

The most common failure is the one the next chapter is about.

THE CHAPTER IN TEN LINES

- Element names in ArchiMate 4 are unprefixed. There is one Service element, placed in a layer, so write a Service in the application layer.
- The interface and object families are layer-specific. Business Actor exists, and there is no layer-prefixed variant of Role.
- There is no Constraint element, use Requirement. There is no Contract element, use Business Object.
- There is no metric element and Value is not a number, so every quantity is a property or an attribute of a Measure Definition object.
- Inherited material carries the wrong names: the supplied workbook's sheets O15 and O19 name element types that do not exist. Handle them by sheet number.
- The notation has no attributes, no cardinality, no validity period on an element, no type-instance distinction, no states and no swimlanes. Say so at the front of the document.
- Relationships can be derived, which is how a repository passes every integrity check and still says nothing.
- Consent is three objects with three owners and three different binding constraints, plus a disclosure event that makes withdrawal actionable.
- The clinical record is one business object realised by a data object in every system that holds part of it, and that is where retention attaches.
- A regulated device is a Node composed of a Device and a System Software, tied to Requirement and Assessment elements. A Device alone loses the regulatory constraint.
- A shared service outside the boundary gets a substitute rather than a recovery objective, and the substitute names who invokes it.
- Draw six views, each with the question it settles. Twenty-five is a catalogue and a catalogue is read once.
- No repository can tell you whether a configuration is clinically safe. It is where the evidence of that judgement is filed, not where it is made.

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CHAPTER 21

Ravensmere Health Network, end to end

Four hundred and thirty relationships, every one of which resolves, and eight sheets built by pairing two lists in row order.

You have been handed a repository. It is a spreadsheet of 105 sheets covering Ravensmere Health Network — a fictional provider group, so stated on its own index sheet — and it has 820 records, 430 relationships and thirteen report sheets. Every relationship in it resolves. Both ends of all 430 rows name an object that exists, by reference and by name, with no dangling pointer and no name that disagrees with the reference it carries.

It is also, in substance, close to worthless, and the hour in which you work out why is the most valuable hour of your first week.

This chapter is that hour, written out. It is not an exercise in finding mistakes. Every defect below is a caricature of a pattern documented in the research behind this volume — exaggerated for teaching, but exaggerated in a direction a reader will recognise within a month of starting work. The point is not that this repository is bad. The point is that the mechanisms that produced it are the normal mechanisms, and that your own first deliverable, written in good faith, will contain two or three of them unless you know what to look for.

The twenty minutes before you read anything

Count. Do it before you form a view, because the counts are the only thing in the file that cannot argue back.

The shape is regular. Sixty-four object sheets numbered `O01` to `O66`, with `O34` and `O51` missing and nothing anywhere accounting for the two gaps; one industry-extension sheet; twenty-five relationship sheets `R01` to `R25`; thirteen report sheets `RPT01` to `RPT13`; an index; and a checks sheet. Row 1 is a sheet title, row 2 an organisation-and-domain banner with a record count, row 4 the headers, and data from row 5. The first data column on every object sheet is `Ref`.

Three mechanical traps, in the file itself, before any content question arises.

Twenty-two tab names are cut at exactly 31 characters, which is the spreadsheet format's limit, and the cut lands mid-word. `O26` is tabbed "Logical Technology Componen". That truncation has propagated *into the sheets*: row 1 of `O26` also reads "Logical Technology Componen", and row 1 of `O27` reads "Physical Technology Compone", because the title cell was written from the truncated tab name rather than from the object type's real name. Index rows 40 and 41 reproduce the same truncation. So the file has lost the names of two of its own object types, in three places each.

One tab name ends in a trailing space. `R04` was cut after a word boundary, so the tab carries an invisible final character. Any formula, lookup or script written against the name without it silently fails to find the sheet.

The index disagrees with the tabs on case. Index row 79 names the extension sheet with a lower-case initial where the tab has an upper-case one, and the index is the only place in the file that spells it that way. An index-driven traversal of the workbook therefore fails on exactly one sheet, and it is the sheet that declares the industry content.

None of this is interesting in itself. It matters because it tells you how the file was made: by a generator, writing titles from tab names, with nobody reading the result. Every content finding below is downstream of that.

What is in the file, and what the first twenty minutes should establish

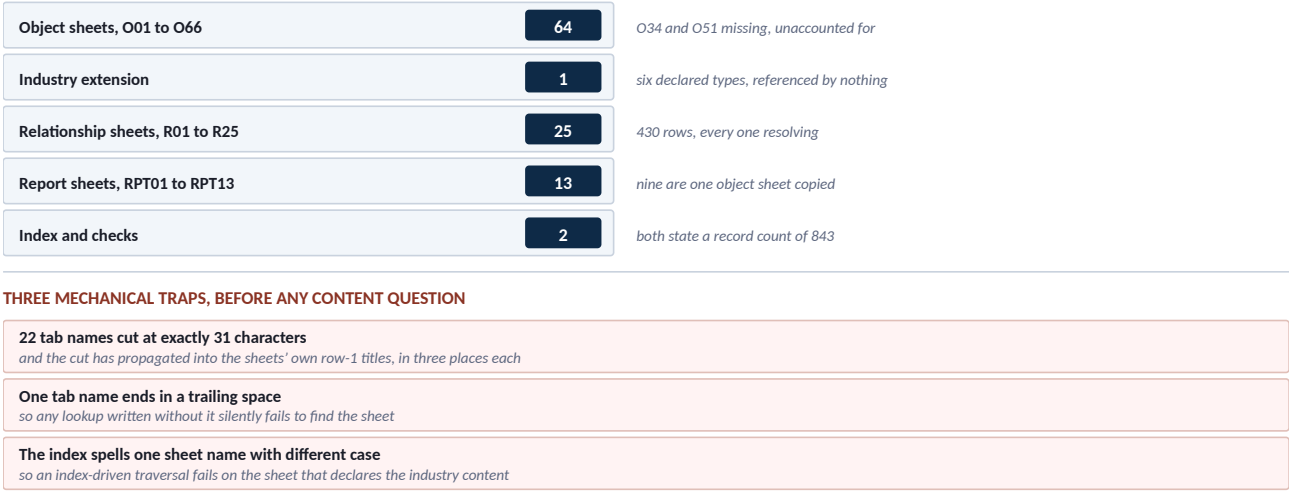


Figure 21.1 The shape of the supplied repository, and the three mechanical defects visible before any content question arises. All three point the same way: the file was written by a generator, with titles taken from tab names, and nobody read the result.

What it claims about itself

The file is unusually self-aware, which makes it a better teaching instrument than a merely empty one. It has a checks sheet with sixteen measures and thresholds. It has a dashboard report with eighteen measures. It has an orphan-analysis report whose stated job is to find what nothing references. And its index makes two explicit quality claims.

The first claim, at index row 8, is true: *every relationship row names both ends by reference and by name, so a broken link is visible without a lookup.* Verified across all 430 rows, and across every cross-reference column carried inside the object sheets as well. Nothing dangles.

The second claim, at index row 10, is that *the figures are internally consistent and describe no real organisation.* The second half holds. The first half does not, and the rest of this chapter is largely the demonstration.

> **THE ONE SENTENCE TO TAKE FROM THIS CHAPTER**
>
> A link can resolve perfectly at both ends and still be wrong, and nothing in a repository's own checks tests whether the two ends of a resolved link belong together. Referential integrity is a property of the pointers. Meaning is a property of the pairing. Every tool tests the first. No tool tests the second.

The arithmetic, which you should redo

Four of the file's own numbers are wrong and three more are defensible only under a definition the check does not state. Redoing them takes half an hour and buys you the right to be believed on everything else.

The record count is wrong in three places. The index banner, the dashboard at `RPT01` row 6 and the checks sheet at row 6 all say 843 records. The sheets hold 820. And the index contradicts itself: rows 15 to 79 give a per-sheet record count for each of the 65 object sheets, every one of those 65 figures matches its sheet exactly, and they sum to 820. The banner sits directly above a list that adds to a different number. The difference, 23, is exactly the row count of `R06`, the interface sheet — which suggests the interfaces were counted into the object total. That last step is inference; the fact is that 843 is 23 too high.

Three work packages are reported past their end date. The correct figure is zero. `O30` flags `WP-02`, `WP-06` and `WP-08` as past end date, and the dashboard and the roadmap report repeat it. The earliest end date anywhere on the sheet is February 2029. Not one of the twelve work packages has reached its *start* date either; the earliest start is January 2027.

Two platform components are reported past vendor support. The correct figure today is one. `O27` flags `PTC-04`, support ending May 2026, which is genuinely past support; and `PTC-06`, support ending December 2026, which is not. The technology-currency report says explicitly that two components are running past vendor support *today*, so this is an error rather than a matter of reading. `PTC-06` becomes correct in January 2027.

Then three figures that are defensible but mislabelled, and the right answer is to give every reading rather than pick one:

The check	What its label says	What the file supports
Control objectives without evidence, 6	Taken literally, the count of objectives with no mechanism is 2	On the regulatory report's own stated test, no mechanism or a mechanism in breach, it is 6. On the test an auditor would apply, not one of the 7 is evidenced
Data entities without lineage, 5	Correct for lineage recorded as absent	8 of 14 lack full lineage, including the patient master. And the metadata sheet separately records lineage coverage as full, which cannot be true alongside either figure
Vacant architecture roles, 0	True of the 18 architecture roles, all filled	Silent on 8 data-ownership titles that exist nowhere else in the file, which leaves 10 of 14 data entities owned by a title the repository does not define

And one report that is not wrong so much as absent from its own job. `RPT13`, whose title and gloss are *objects that nothing else references*, holds two rows. Both are correct — two applications have no business-layer service mapped to them — but the test it ran was applications only. On its own stated test, twenty of the sixty-five object sheets appear in **no relationship sheet whatsoever**, by reference or by name: 206 records, including every principle, every constraint, every business process, every contract, every deliverable, every governance body, gate, decision and change request, and every delivery risk. The whole governance layer and the whole motivation-constraint layer hang off nothing. A reader can check that without arguing about leaf nodes, which is why it is the reading worth stating.

The file's own figures, against a count of the sheets

What the file checks	It says	The sheets say	Where it is also stated
Records	843	**820**	*Index banner, RPT01 row 6, ZZ row 6*
Work packages past end date	3	**0**	*RPT01 row 21, RPT08 footnote*
Platform components past support	2	**1**	*RPT01 row 15, RPT10 footnote*
Control objectives without evidence	6	**2 / 6 / 7**	*ZZ row 15, RPT01 row 17*
Data entities without lineage	5	**5 / 8**	*ZZ row 16, RPT05 footnote*
Vacant architecture roles	0	**0, of 18**	*ZZ row 17, silent on 8 titles*
Orphans found	2	**206 records**	*RPT13, on its own stated test*
Relationship instances	430	**430**	*Index, RPT01 row 8, ZZ row 8*
Object types populated	65	**65**	*Index, RPT01 row 5, ZZ row 5*
Level 1 capabilities	10	**10, and 10+30+60**	*RPT01 row 22, R01 holds 90*

Red: wrong on any reading. Amber: defensible only under a definition the check does not state — give every reading. Blue: verified correct.

Figure 21.2 The file's own self-checks against a count of the sheets. Four figures are wrong on any reading, three are defensible only under a definition the check does not state, and three are correct and worth keeping. Redoing this arithmetic takes half an hour and buys standing for everything else.

Eight sheets built by pairing two lists in row order

This is the finding, and it is worth more than the other eleven put together.

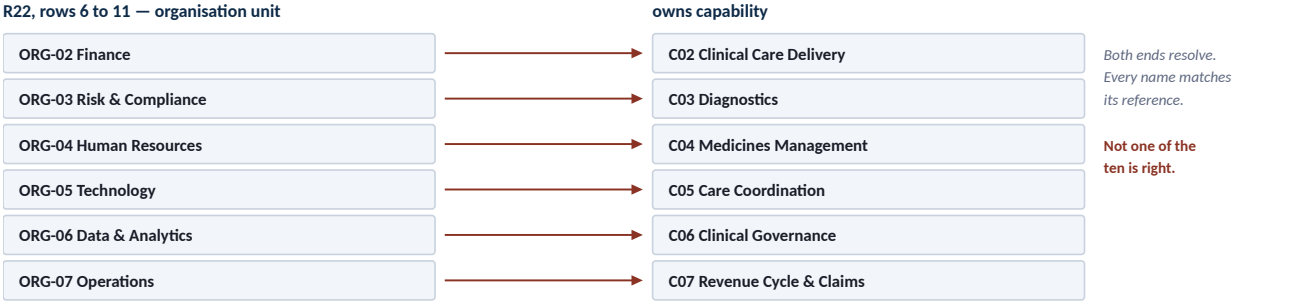
Eight relationship sheets — `R04`, `R07`, `R09`, `R13`, `R20`, `R22`, `R23` and `R25` — were built by taking two lists and joining them by row position. Every reference resolves. Every name matches its reference. The integrity check passes. And the model says things no provider would recognise.

- `R22`, ten rows, organisational units against capabilities in index order. Row 8: Human Resources owns Medicines Management. Row 14: Supply and Logistics owns Population Health and Analytics. Row 12: Commercial owns Workforce and Rostering. Not one of the ten is right, and Internal Audit and Legal own no capability at all.
- `R25`, five rows, striding the application list. Row 7: Emergency Care is supported by the billing system. Row 8: Diagnostic Services is supported by the patient portal — while the laboratory system and the imaging system support no product at all, and one of the products is literally "laboratory and imaging services".
- `R07`, twelve rows, applications joined to physical components by number. The patient administration system runs on the image store. The billing system runs on the medical device gateways. The patient portal runs on the backup and archive tier. The component actually named "integration engine servers" hosts the imaging system, while the integration engine sits on the cloud landing zone. Ten of twelve are wrong; only two survive.
- `R09`, thirty-two rows, requirements against building blocks, offset by seven. Row 12's requirement is "Patient mastered once" and it reaches a workforce-and-rostering capability uplift. Row 36's requirement is to adopt medical device integration and it reaches terminology services. A building block named "Patient mastering" exists; the requirement that names that exact subject reaches something else. Zero of thirty-two pair correctly by subject.
- `R13`, thirty-seven rows, work packages cycled through the building-block list. The consequence is systematic: `WP-02`, "Data ownership and mastering", delivers one of the ten data blocks. `WP-04`, "Technology currency", delivers no technology block at all. `WP-03`, "Integration platform", delivers no integration block.
- `R23`, six rows, increments against the odd-numbered projects. All six increment names differ from the project said to deliver them: the increment called "Analytics and evidence" is delivered by the resilience project while the project called "Analytics and reporting" delivers no increment at all.
- `R20`, twelve rows, applications sited at locations by number, then wrapping. It also contradicts `R07` plus `O27` for eleven of twelve applications: one sheet says where each application runs and another says where each physical component sits, and the two agree for exactly one application.
- `R04`, ten rows, services joined to applications by number. Six of ten are plainly wrong — the clinical-care service reaches the registration and scheduling system rather than the clinical record; the clinical governance and safety service reaches theatre management rather than the incident system; the population-health service reaches the patient portal rather than the data warehouse.

The pattern behind it. Nobody sat down and wrote these. A list of ten was laid against another list of ten, in order, by a generator or by a hand filling a template to a deadline. That is not a hypothetical mechanism; it is how a repository gets populated when the deliverable is a populated repository rather than an answered question. And it is invisible to every automated check, because the pointers are immaculate. The notation's own derivation rules make it worse: a chain of valid structural relationships derives a valid relationship, so the tool will confirm these pairings are *permitted*. Permitted is not the same as true.

> **THE THREE QUESTIONS, AND THE SAMPLING RULE**
>
> Take twenty relationship rows at random. For each: does the subject of one end appear anywhere in the other end's name; would the person who owns one end recognise the other; and if the row were deleted, would anybody notice. Score it. Below about three-quarters passing, the relationships were generated and the model is decorative, and you should say so in writing in week one rather than in month four when somebody has built a plan on it.

How eight relationship sheets were built: two lists, joined by row position



THE SAME MECHANISM, ON SEVEN MORE SHEETS

R04	10 rows	services to applications by number; 6 of 10 wrong
R07	12 rows	applications to physical components; 10 of 12 wrong
R09	32 rows	requirements to building blocks, offset by seven; 0 of 32 correct
R13	37 rows	work packages cycled through the block list; none delivers its own subject
R20	12 rows	applications to locations, then wrapping; contradicts R07 for 11 of 12
R23	6 rows	increments to the odd-numbered projects; all six names differ
R25	5 rows	striding the application list; the billing system supports emergency care

Figure 21.3 The mechanism behind the largest finding, drawn once. Two lists of ten, joined by row position. Every reference resolves, every name matches its reference, the integrity check passes, and Human Resources owns medicines management. Seven more sheets were built the same way.

Not one of the fourteen data entities is mastered where it could be

`O20` holds fourteen data entities, each with a class, a definition, a classification, an owner, a master system and a lineage value. Column 7 is the master system. Reading each application's own description from `O23`:

O20 column 7: the master system recorded, against what each system is

Row	Data entity	Master system recorded	What that system is, in the file's own words
5	**Patient**	ePrescribing & Meds Admin	*Prescribing, verification and administration*
7	**Clinical Observation**	Revenue Cycle / Claims	*Coding, billing and claim adjudication*
8	**Diagnosis**	PAS	*Registration, scheduling and patient administration*
10	**Medication Order**	PAS	*Registration, scheduling and patient administration*
11	**Diagnostic Order / Result**	Rostering	*Staff scheduling to demand and skill*
16	**Claim**	EHR / EPR	*The clinical record of truth*
17	**Consent**	Integration Engine	*HL7 and FHIR message routing*
18	**Medical Device**	Theatre Management	*Scheduling and utilisation of theatres*
6, 12, 14	**Encounter, Care Plan, Practitioner**	Clinical Data Warehouse	*Governed analytics across the care record*

Rows 7 and 16 are exactly swapped: the clinical record of truth masters the billed episode, and the billing system masters the clinical measurement. R05 rows 5 to 18 reproduce all fourteen assignments verbatim.

Figure 21.4 The mastering map as the file records it. The pattern it caricatures is real: entities end up mastered where the first system was integrated rather than where they belong. The remedy is not to fill in the right answer, because several of these entities have no single plausible owner.

Rows 7 and 16 are the pair to hold up in a workshop: the clinical record of truth masters the billed episode, and the billing system masters the clinical measurement. **They are exactly swapped.** And `R05` rows 5 to 18 reproduce all fourteen assignments verbatim, so the error is consistent across two sheets and would survive any cross-check the file performs on itself.

Three further mastering problems compound it.

`O43`, the master data rule sheet, **names a different golden source from `O20` for eight of the ten entities it covers.** Only Diagnosis and Medication Order agree. The rule for Patient is typed "golden source" and names the rostering system.

`O39`, data migration, has six rows in which **neither the from-system nor the to-system is the entity's master** per `O20`. One migrates the patient master into the imaging system. Another migrates prescriptions into the incident and risk system.

And nine of the fourteen entities are mastered in an application whose own disposition is retire, replace or consolidate. The registration system is being retired while it masters two entities and is the golden source for four; nothing in the file records where those move to, and one of the source-to-target

mappings maps Claim *into* it.

The pattern behind it. This is the documented warehouse and master-data failure mode: entities end up mastered where the first system was integrated rather than where they belong. The first system integrated in a provider is almost always patient administration, so its grain and its identifiers become the enterprise's, and everything loaded afterwards is forced onto them. The caricature here is that the assignment is random rather than first-come, but the lived version produces the same deliverable problem: a mastering map that reflects integration history and nobody's decision.

The teaching answer is not to fill in the right system. For several of these entities **there is no single plausible owner, and asserting one teaches a falsehood.** Practitioner has four candidates — human resources for the person, credentialling for registration, the clinical directorate for privileges, billing for practice codes. Care Setting, Ward, Bed and Location have four between estates, nursing operations, the clinical directorate and finance. Care Plan is genuinely contested. The deliverable is the governance decision with the candidates listed, not the answer.

The billing-and-clinical inversion, which a professional rule prohibits

The swap in rows 7 and 16 is not merely untidy. HPCSA Booklet 9 requires at paragraph 13.7 that billing records be kept **separate** from patient care records [1]. That is an explicit segregation requirement from the professional regulator, and it is the strongest available authority for a design principle that is otherwise hard to sell: the revenue cycle and the clinical record are different stores with different access populations, and the clinical record should not be reachable from a billing role.

A repository that masters the clinical observation in the billing system has modelled the prohibited arrangement. A repository that masters the funder claim in the clinical record has modelled it in the other direction. Both are in this file, on two rows of one sheet, and it carries a control objective for funder and coding rules whose evidence value is "in breach".

Two things follow for your own work. First, cite the paragraph. An architect arguing for separation of stores from first principles loses to a vendor arguing for a single integrated product; an architect citing a professional rule that the practitioners in the room are personally bound by does not. Second, note what the file's own vocabulary does to the argument. Its entity is called `Claim`, bare, defined as "a billed episode submitted to a funder" — so it means a funder claim. In a healthcare repository the bare word is ambiguous, because the other claim in this industry is the medico-legal claim against the provider, and the records duty answers to that one. Rename it. A repository in which `Claim` could mean either is a repository in which the retention requirement cannot be scoped.

Classification: the identifier is more open than the thing it identifies

`O20` column 5 holds the classification. Six entities that are patient-identifying or clinical by their own definition in the adjacent column are classified open:

- Row 6, Encounter: `Public`, defined as "a contact between patient and service, with its setting".
- Row 9, Procedure: `Public`, "a coded intervention performed".
- Row 16, Claim: `Public`, "a billed episode submitted to a funder".
- Row 18, Medical Device: `Public`.
- Row 8, Diagnosis: `Internal`, "a coded condition attributed to a patient".
- Row 11, Diagnostic Order / Result: `Internal`.

Against which row 13, Clinical Protocol, is `Restricted` — the strictest value anywhere in the file — for "an evidence-based pathway the organisation follows", which identifies nobody. **The document the organisation publishes is locked down harder than the diagnosis attributed to a named patient.** Only two of the fourteen entities are `Restricted`, and one of them is the protocol.

It gets worse one level down. `O37` holds sixty data attributes, and twenty-six of them are classified more openly than their parent entity. Nineteen are `Public` outright. The one to put on a slide is the first row of the sheet: **`ATR-001`, the patient Identifier, classified `Public` beneath an entity classified `Restricted`.** The Diagnosis entity's six attributes are `Public` except two, so every identifying and dating attribute of a coded condition attributed to a patient is open. The Practitioner entity is `Confidential` and its name, status and effective date are `Public`.

Why it matters beyond tidiness: information concerning a person's health is special personal information under POPIA, whose processing is prohibited unless an authorisation applies [4]. A classification column that calls a coded diagnosis `Internal` and its identifier `Public` has not made a filing error. It has recorded, in the repository the organisation will hand to an auditor, a position its own regulator prohibits.

The reason this is a teaching defect rather than a typing error is that **the file states the rule it breaks.** Principle `PRN-14`, at `O05` row 18, reads: classification travels with data; classification is a property of the data, not its location. An attribute cannot loosen what its entity set, on the repository's own principle. Nothing would catch it: the data classification standard at `O32` row 7 is recorded as current with conformance not assessed, and the data-platform enforcement point at `O65` row 8 records that classification is not applied at read.

The pattern behind it. Classification is set per row by whoever filled the row, at the moment they filled it, with no inheritance rule and no validation. That is the ordinary condition of a classification exercise run as a spreadsheet column, and the lived version is less dramatic only because the entity and its attributes are usually filled by the same person on the same afternoon. The architectural remedy is the one the file names and does not implement: classification inherits downward, an attribute may tighten and may not loosen, and the read path enforces it.

Consent and medical device: one row each, and nothing else

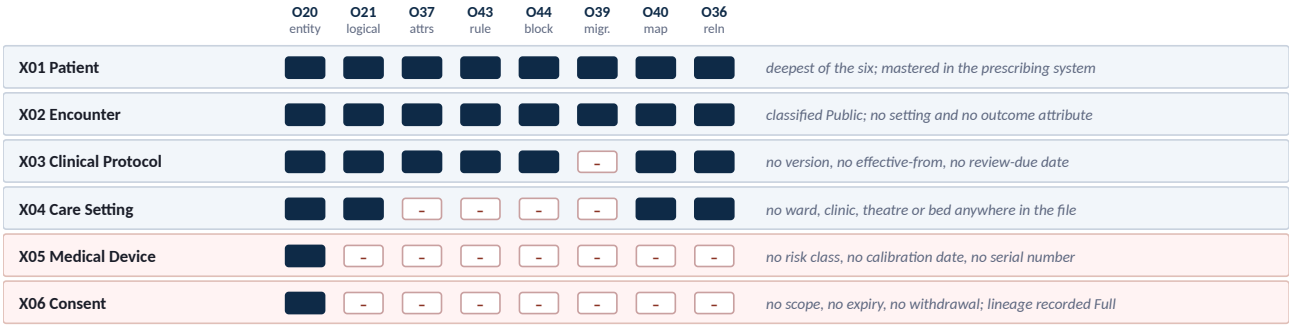
The file declares six industry-extension types — patient, encounter, clinical protocol, care setting, medical device and consent. Each appears as one row of `O20` and nowhere else, and the extension sheet itself is referenced by nothing. But the depth behind the six is wildly uneven, and the two with no depth at all are the two a provider most needs.

Consent, `E13`, appears on exactly three sheets: `O20`, `R05` twice, and the data-ownership report. No logical data component. No attributes — the attribute sheet stops at Practitioner. No master data rule, no data building block, no migration, no mapping, no entity relationship, no requirement. Its lineage is nevertheless recorded as *full*, the strongest value on the sheet, for an entity with no lineage path anywhere. There is no consent scope, no type, no grant date, no expiry, no withdrawal and no version. Its definition reads "recorded permission for treatment **and** for data sharing" — and that conjunction is the error, because consent to treatment and consent to disclosure have different legal bases, different scopes and different consequences of withdrawal [2]. Meanwhile the file asserts the obligation: a driver names consent and lawful basis, and the matching control objective's evidence value is "in breach". The obligation is stated and no structure exists to meet it.

Medical Device, `E14`, is the same shape. Three sheets. No attributes, no mastering rule, no building block, no relationship. Classified `Public`. The word "calibration" appears twice in the entire file — in this entity's definition and in the extension type's — and "risk class" appears once. There is no device register, no calibration date, no maintenance schedule, no risk class, no serial number and no manufacturer, although the extension sheet says the type holds calibration, maintenance and risk class. The nearest real content is infrastructure rather than devices: a physical component called "medical device gateways", whose sheet's class values are compute, storage, network, database and end user — so a medical device cannot be recorded there without being called a compute component.

The pattern behind it. An entity with no attributes is a placeholder for an argument nobody has had. Both of these are arguments with known participants and known difficulty. Consent requires the information officer, clinical governance and the research office to agree that there are three objects rather than one, and then requires somebody to own expiry and withdrawal. Medical device requires clinical engineering, procurement and regulatory affairs to agree who holds the register — and in South Africa it requires accepting that **the register cannot be populated from a national product register, because the regulator has not begun product registration**, so the asset register plus supplier declarations are the only source of truth [3]. The risk class itself is determinable — classification is by rule, the manufacturer or distributor determines the class, and the higher class applies where more than one rule is engaged [6] — so the absence here is a decision nobody took, not information nobody could get. Those are hard conversations. A row with a reference number and a definition looks like progress and defers both.

Six declared extension types, and how much structure sits behind each



The extension sheet itself is referenced by nothing. An entity with no attributes is a placeholder for an argument nobody has had.

Figure 21.5 How much structure sits behind each of the six declared industry types. Two are carried at the depth the rest of the data model uses, two are partial, and the two a provider most needs a structure for exist as one row and nothing else.

Three defects that read as a single failure of review

Production holds no personal data. `O48` has four rows. Production: personal data present, no. Pre-production: yes. Test: no. Development: yes. A provider's live patient-record estate is recorded as holding no personal data while two non-production environments do. The file's own security position says the opposite is the problem: non-production masking is recorded absent, the non-production provisioning enforcement point records that most refreshes are unmasked, and the policy "personal information is protected wherever it is / applies across production and non-production" is recorded as breached. Two readings are available. Either the column means "unmasked copies present", in which case production's "no" is the only honest cell on the sheet; or it means what its header says, in which case production is wrong. The header text says "personal data present", which supports the second.

Resilience tiering is inverted. `O49` gives each application a tier, a recovery objective and a tested flag. Tier 1 carries a 24-hour objective and Tier 3 a 15-minute one: the tier label and the recovery objective move in opposite directions across the whole sheet. Set against application criticality, it gets worse. The electronic patient record — "the clinical record of truth" — is rated **medium** criticality with a 24-hour objective, while the billing system and the incident system are both rated critical. Both critical applications sit below Tier 1. And nine of twelve applications have **never been tested**, including both critical ones and the clinical record. The corresponding objective, recovery objectives tested on cadence, is recorded as not started, and its measure carries a baseline of 43 with the unit "count" for a measure with twelve possible subjects.

The plan costs more than the alternative it was chosen for. The work package sheet, the project sheet and the project-delivers-work-package relationship all agree: $61.3m, and each project's cost, start and end equal its work package's in all twelve rows. This is the cleanest arithmetic in the file after the capability decomposition. The architecture alternative marked *recommended* is costed at $46m. The plan that implements it is 33% more expensive, and more expensive than three of the five alternatives including two rejected as cheaper. No sheet reconciles the $15.3m.

Those three have one thing in common. Each is a single cell, or a single column pair, that contradicts something stated elsewhere in the same file. None requires domain knowledge to find. All three survived because nothing read the file as a whole — which is the condition a repository is in whenever its reports are copies of its object sheets, and nine of the thirteen reports here are exactly that, a single sheet with one column dropped or one footnote added.

The delivery layer, read as a plan, against a reference date of 25 October 2026

O30, twelve work packages

Earliest start on the sheet: January 2027
Earliest end on the sheet: February 2029
Not one has reached its start date
Seven read Delivered
Three read Past end date; the true figure is 0
WP-05 reads Not started for approval and Delivered for delivery, on the same row

Cost, against the option it implements

O30 work packages	**$61.3m**
O56 projects	**$61.3m**
R14 project delivers work package	**$61.3m**
O54 ALT-B, marked Recommended	**$46m**
Unreconciled gap	**+$15.3m, 33%**

Above two options rejected as cheaper. No sheet reconciles it.

THREE TRANSITIONS IN THE WRONG ORDER

TA-1 Governed and supportable	starts 2027-03	ends 2031-07	*finishes last*
TA-2 Integrable	starts 2027-11	ends 2029-09	*starts after TA-3*
TA-3 Business visible	starts 2027-06	ends 2031-02	*holds the 2nd earliest package*

O66 dates its six increments as Month 10, 16, 22, 30, 42 and 56. There is no programme baseline date anywhere in the file, so the two timebases cannot be reconciled. Do not invent a start date: the increment landing at month 10 is delivered by the latest-starting project.

Figure 21.6 The delivery layer against the reference date. Seven packages delivered before they start, a programme costing a third more than the option it implements, three transitions in the wrong order, and two timebases one of which has no origin.

The plan, read as a plan

Four more findings sit in the delivery layer, and they are the ones a programme director will care about first.

Seven work packages are delivered before they start. `O30` carries delivery status for twelve packages whose earliest start is January 2027. Seven read "Delivered". `WP-05` reads "Not started" for approval and "Delivered" for delivery, two columns apart on the same row, for work beginning in March 2027. Three more read "In flight" for approval and "Delivered" for delivery. The project sheet carries the same dates and the roadmap report reproduces the table unchanged.

The transition architectures run out of order. Three transitions hold seven of the twelve packages. The last transition contains the second-earliest-starting package and begins before the middle one. The first transition finishes in July 2031, after both the transitions that are supposed to follow it. A transition architecture is a state the estate passes through, and on these dates the estate reaches state three before it has finished reaching state one. Five packages sit in no transition at all, among them the most expensive on the sheet.

There are two incompatible time axes. The increment sheet dates its six increments as "Month 10", "Month 16" and so on, and there is **no programme baseline date anywhere in the file**, so those cannot be resolved to a calendar. The only check available is internal ordering, and it fails: the increment that lands at month 10 is delivered by the latest-starting of the six projects, and the one that lands at month 22 by the earliest-starting. Do not invent a start date to reconcile them. Report that the file holds two timebases and that one of them has no origin.

Three of the five value streams contradict their own decomposition inside the same file. `O14` declares a stage count per value stream. `O16` holds the processes. Two streams declare five stages and one declares six, while `O16` gives **every** stream exactly four processes, named "stage 1" to "stage 4", regardless of what `O14` says. Twenty-four stages are declared; twenty exist. And no process has a name beyond the value stream's name plus a number, a performer, or a system.

That last one is the cleanest small illustration in the file of a declared count maintained by one hand against a decomposition maintained by another. It is also the defect most likely to appear in your own work, because a count in a summary column is the easiest thing in a repository to leave behind when the layer beneath it changes.

> **EXIT CRITERIA THAT ARE COUNTED AND NOT WRITTEN**
>
> `O53` records "5 exit criteria", "6 exit criteria" and "7 exit criteria" against the three transitions. Eighteen exit criteria are counted. Not one of them appears anywhere in the file. The governance gate sheet defines four gates and the compliance assessment sheet exercises one of them, three times, all three verdicts non-compliant, all three with evidence recorded as "no evidence produced". Thirty-four of the thirty-seven building blocks are never assessed.

The reason several of these defects could not have been avoided

One finding explains a cluster of the others, and it is the one to raise first with a client, because it is a question about them rather than about their spreadsheet.

`O09` holds twelve organisational units: executive, finance, risk and compliance, human resources, technology, data and analytics, operations, commercial, customer service, supply and logistics, internal audit, legal. **There is no medical directorate, no nursing, no pharmacy, no allied health, no clinical governance unit and no diagnostics unit.** The actor sheet's eighteen actors are executives, architects and delivery heads: no clinician, no clinical safety officer, no pharmacist, no medical records manager. The role sheet mirrors the actor sheet one for one.

That single absence produces several of the defects above mechanically. `R22` cannot assign a clinical capability to a plausible owner because there is no clinical function to assign it to. `O23` makes human resources the owner of the imaging system, internal audit the owner of theatre scheduling and customer service the owner of prescribing, because those are the units that exist. The eight "head of" data-ownership titles in `O20` exist nowhere else in the file because the posts they name have no unit to sit in. And the delivery risk sheet holds eight programme-generic risks and not one clinical risk, although the file records a control objective for clinical safety standards whose evidence value is "in breach".

The same absence shows in the motivation layer, where the generic base was never rewritten for the industry. The file is a provider healthcare repository whose drivers, goals, objectives and measures talk about customers: customer expectation, growing revenue from existing customers, a single view of customer across channels, a cross-sell rate improved from baseline, a cross-sell ratio with a baseline of 48. A hospital does not cross-sell to patients, and "single view of customer across channels" is a retailer's problem; a provider's problem is patient identity matching across settings and providers, which is a different design with a different failure mode — and in South Africa a measured one: in 13,499 infant tests across Tshwane, only 4% of infants matched exactly on all four demographic details and of those tested twice, 18% were tested at the same facility both times and only 5% had the same folder number on both occasions [5]. The data-set sheet is worse: four of its six sets are not healthcare at all, and three of those four are a customer analytics set, a finance consolidation set and a risk measurement set described as "exposure and position data", which is a trading-book phrase. There is no clinical data set, no patient data set and no safety data set. And the twenty-four principles are the generic framework set — primacy of principles, maximise benefit, common use applications, technology independence, ease of use. **Not one is about clinical safety, consent, the clinical record or the patient**, and all twenty-four are owned by a body that is neither an actor nor a role in the file.

The pattern behind it. A repository is populated from a template, and a template carries the organisation it was written for. The residue is not sloppiness; it is the visible edge of a reusable asset. Which is useful, because it tells you exactly how much of the file was written for this client: the ten level-one capability descriptions and the ten level-one capability names. Everything else was either generated from them or inherited.

The views deck

A deck of twenty-seven slides accompanies the file: a title, a contents page, and twenty-five views, one per viewpoint of the specification's appendix, each with a title, a one-line subtitle, the drawing and a footer naming the viewpoint it was created with.

Five of its captions were written for a different industry and never rewritten. Two name an organisation that does not appear anywhere in this repository — it is the invented organisation of this series' insurance volume — and describe what that organisation sells and what resources it holds. A third gives the value-stream view a value stream called "settle a claim", which is not one of the five the file holds; all five of those are clinical. A fourth describes the process-cooperation view as "the claim process end to end", while the processes on the sheet are clinical and the nearest value stream is encounter to payment. A fifth says the capability map decomposes *twelve* level-one capabilities, where the sheet holds ten.

Say this plainly to the client, and say it early, because it is the single most damaging thing in the package and the easiest to fix. A deck that names another organisation has been in front of somebody. It tells a reader that the artefact is a reused template, which retrospectively discounts every other claim in it — including the ones that are true, and the referential integrity claim is true. Do not reproduce the other organisation's name in anything you write: describe the defect by slide number and by what the caption asserts.

Two cautions on method. The deck is the supplied worked example and ships unchanged, so flag it rather than editing somebody else's file silently. And note that the file itself is clean of cross-industry naming: the organisation name, the insurance vocabulary, none of it appears in the 105 sheets. The leak is in the deck only, which is consistent with a workbook regenerated for this industry and a deck whose captions were hand-edited and incompletely.

What to do with it in week one

Five days. The deliverable is not a corrected repository, because correcting it would take a quarter and would produce a file nobody had agreed to.

Day one: count, and write the counts down. The arithmetic above. Produce a one-page note of every figure the file states about itself, with your own count beside it, and the reading you took where the label is loose. Give every reading rather than picking one. This note is the thing that buys you standing for the rest of the engagement, and it is the only deliverable in the first week that nobody can argue with.

Day two: sample the relationships. Twenty rows, the three questions, a pass rate. Then state the conclusion in one sentence: the relationships in this repository resolve and do not mean anything, and therefore no impact analysis, no roadmap dependency and no traceability claim built on them is usable. Expect this to be unwelcome, and expect it to be accepted, because it is checkable in ten minutes by anybody who doubts it.

Day three: the mastering map, as a question. Take the fourteen entities and, for each, list the candidate owners and the candidate master systems rather than asserting one. Mark the ones where there is genuinely no single plausible owner. This converts the file's worst sheet into the agenda for a governance workshop, which is the deliverable the research supports.

Day four: the three cells that contradict the file. Production holding no personal data. The inverted resilience tiering with nine untested applications. The plan costing more than the alternative it implements. Each is a single sentence with a sheet and a row behind it, each is checkable, and each belongs to a different executive. They are the three findings most likely to get the repository work funded.

Day five: the absence. The organisation has no clinical structure in its own repository, and several of the defects follow from it. Put it as a question: which clinical functions should appear as organisational units, and who in each owns data. That question, asked in week one, is worth more than any model you could build in the same five days.

Then the structural work, which is not week one and which the file tells you the shape of. The six declared extension types need somewhere to live — consent with scope and expiry, medical device with risk class and calibration, a clinical coding system, and a retention clock on the clinical record. That is new sheets, not new columns, and the numbering decision for them should be taken before anyone populates anything, because reconciling it afterwards costs more than taking it now.

THE HONEST SUMMARY TO PUT IN FRONT OF A STEERING COMMITTEE

This repository has perfect referential integrity, a correct relationship count, a correct object type count, a correct and fully reconciled capability decomposition, and correct interface counts. Those are real and they are worth keeping. It also has a headline record count 23 too high, eight relationship sheets whose rows resolve and do not mean anything, no data entity mastered where it could be, patient identifiers classified more openly than the patient, no structure for consent or for a medical device, production recorded as holding no personal data, inverted resilience tiering with the clinical record at a 24-hour objective and untested, a programme costing 33% more than the option it implements, and an accompanying deck carrying captions written for another organisation in another industry. A repository can pass every integrity check a tool can run and still describe an architecture that would harm patients. That is the whole of the lesson, and it is why the next chapter starts with questions rather than with models.

THE CHAPTER IN TEN LINES

- Count before you read. The file states 843 records; the sheets hold 820, and the index's own per-sheet figures sum to 820.
- Three work packages are reported past their end date and the true figure is zero; two components past support and the true figure today is one.
- RPT13 reports two orphans. Twenty of the sixty-five object sheets, 206 records, appear in no relationship sheet at all.
- Eight relationship sheets were built by pairing two lists in row order. Every reference resolves and almost nothing means anything.
- Not one of the fourteen data entities is mastered in a system that could own it, and the clinical observation and the funder claim are exactly swapped.
- The swap models an arrangement HPCSA Booklet 9 prohibits at paragraph 13.7, which is the strongest authority available for separating the two stores.
- Twenty-six of sixty attributes are classified more openly than their entity, including the patient identifier, against the file's own principle PRN-14.
- Consent and medical device exist as one row each. An entity with no attributes is a placeholder for an argument nobody has had.
- Production is recorded as holding no personal data while two non-production environments do, and resilience tiering is inverted, with the clinical record at a 24-hour objective and never tested.
- The programme costs 33% more than the alternative marked recommended, and no sheet reconciles the gap.
- The organisation has no clinical structure in its own repository, which is why several of the other defects could not have been avoided.
- The views deck carries five captions written for another industry, two of them naming another organisation. Flag it; do not reproduce the name.
- A repository can pass every integrity check a tool can run and still describe an architecture that would harm patients.

SOURCES FOR THIS CHAPTER

- **1.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition (paragraph 13.7, billing records to be kept separate from patient care records) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **2.** National Health Act 61 of 2003, consolidated text (s14 confidentiality and written consent to disclosure; s16 authorisations for access for research) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **3.** SAHPRA communication to industry on artificial intelligence and machine learning medical devices, MD08-2025/2026, 26 September 2025 (product registration of medical devices and IVDs not yet begun; regulation 8 not called up; the establishment licence regime applies instead) https://www.sahpra.org.za/wp-content/uploads/2025/09/MD08-20252026_-SAHPRA-Communication-to-Industry-AI-Medical-devices_Acknowledgements.pdf
- **4.** Protection of Personal Information Act 4 of 2013, section 26, prohibition on processing of special personal information (health information among the prohibited categories absent an authorisation) https://popia.co.za/section-26-prohibition-on-processing-of-special-personal-information/
- **5.** Haeri Mazanderani et al., South African Medical Journal 2018;108(9):729-733 (Tshwane, May 2016 to May 2017, 13 499 infant HIV PCR tests; 4% exact match on all four demographic details; 5% of repeat tests at the same facility sharing a folder number) https://scielo.org.za/scielo.php?script=sci_arttext&pid=S0256-95742018001000017
- **6.** SAHPRA, Guideline for the Classification of Medical Devices and IVDs, SAHPGL-MD-04 version 5, 27 February 2025 (classification by rule; the manufacturer or distributor determines the class; the higher class applies where more than one rule is engaged) https://www.sahpra.org.za/wp-content/uploads/2025/03/SAHPGL-MD-04_v5-Guideline-for-Classification-of-MD-and-IVDs.pdf

CHAPTER 22

Your first ninety days

A worked reference design, a decision order, and the eight things to leave behind.

Ninety days is not long enough to fix anything in a health organisation, and it is exactly long enough to write down what nobody has written down. That is the whole of the job, and the volume's central claim tells you why: almost nothing that matters in South African healthcare is specified nationally, so the decisions are the organisation's own to make, and the only question is whether they are made explicitly and recorded, or made implicitly in a configuration screen by somebody who has left.

Chapters 10 to 21 did the teaching. This chapter is the order of work. It assumes the reader has been handed a populated repository, a sponsor with a date, and a quarter.

Read one worked design before you start

There is a published South African patient-level health data architecture, built in a provincial department with provincial money, documented by the people who built it, including its compromises. It is the **Western Cape Provincial Health Data Centre**, and the right use of it is not as an aspiration but as a reference design: a set of decisions somebody already made under the constraints the reader is about to meet.

Its shape, from the published data centre profile [1]. It queries the systems that already exist rather than replacing them: the provincial patient administration system as the core, polled daily for new patients, admissions, outpatient visits and discharge summaries; the national laboratory service for digitised pathology; pharmacy dispensing where it is digitised; bespoke electronic programme registers for HIV and tuberculosis; a community database of household and community health worker records; radiology and laboratory report metadata summarising repositories held on the source systems; and an antenatal text-message programme supplying self-enrolled pregnancy data. Scale, at the profile's date: about 30 TB on commodity database infrastructure across eight virtual servers, about eight million unique people having sought public-sector care over the preceding decade, about 15 million attendances or admissions a year, a facility base of 52 hospitals, 272 provincial primary care clinics and 82 city clinics, and around 50 approved data releases a year split about evenly between operational and research requests [1]. A separate published account of its clinical-facing portal records an extended pilot with over 200 clinicians, over 17 000 patients viewed in six months and an average of 150 patient searches a day [2].

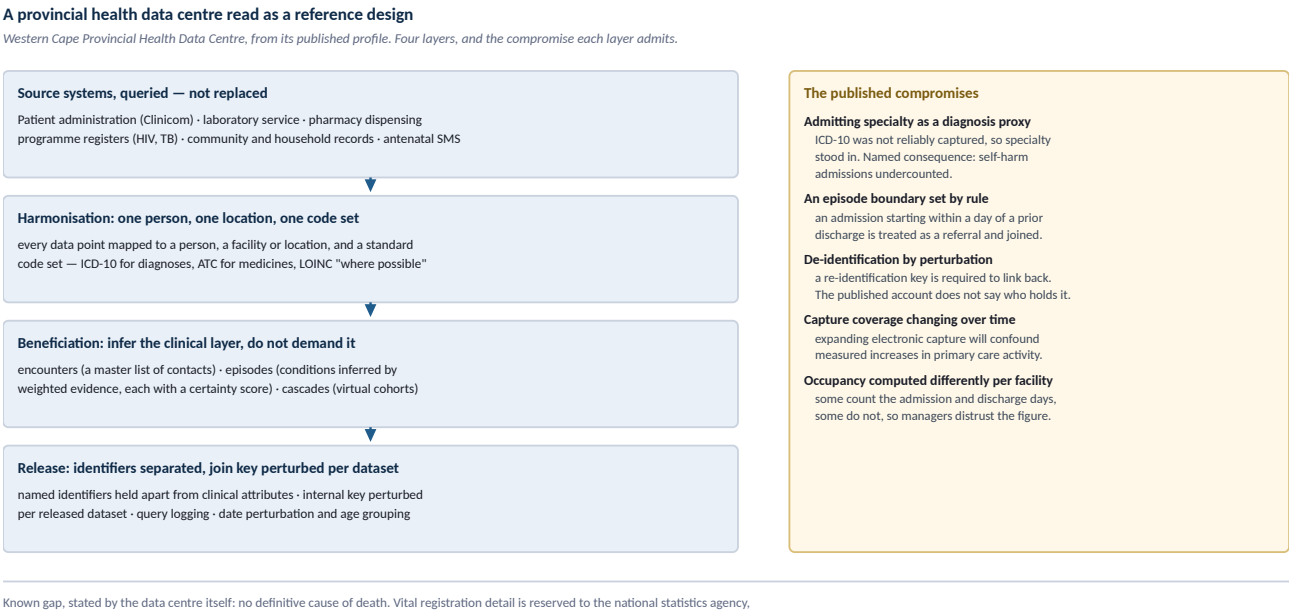


Figure 22.1 The Western Cape Provincial Health Data Centre as a reference design, with the five compromises its own publications state.

The decisions worth copying

Eight, and the profile states them as decisions rather than as features [1].

1. **Build on the patient administration system that is already there** rather than replacing it.
2. **Make the shared identifier a real identifier** — a namespace, a check digit, a barcode. The provincial folder number is nine digits with a modulus-10 check digit, printed as a barcode.
3. **Link rather than merge, store certainty, and keep the link table.** Linkage is mainly deterministic with explicit certainty levels, canonical identifiers with corroboration ranked highest, edit-distance text comparison as a supplement, and a probabilistic score built but held out of production pending validation. Links live in a lookup table, chains are revised by transitive closure, and historical links are reused.
4. **Blacklist the known bad identifier values.** Blacklists and whitelists for recurring errors are part of the design, not a data-cleansing afterthought.

5. **Map to standard code sets at ingestion, accepting "where possible".** ICD-10 for diagnoses, ATC for medicines, LOINC for laboratory tests where possible. The qualifier is in the published text and should stay in the reader's design.
6. **Infer the clinical layer rather than demanding that source systems code better.** Three inferred layers: encounters, as a master list of contacts from a household visit to an admission; episodes, as health conditions inferred by weighting and combining evidence from several sources, each carrying a certainty score, *because diagnosis coding is limited*; and cascades, as virtual continuously-updated cohorts tracking characterisation, comorbidity, retention in care, control and outcome.
7. **Separate identifiers from attributes and perturb the join key per release**, so that two separately produced datasets cannot be cross-linked.
8. **Put the governance in the design**: named oversight committees for information technology, data governance and health research; a standard release procedure; automatic query logging; identifiers released apart from attribute data; and a reasoned, published position on consent — named data shared only for direct care, research releases requiring research-committee approval and almost always patient consent, patients not asked to consent to inclusion because the centre is treated as care infrastructure, with a stated plan to inform patients about anonymised use and eventually offer an opt-out.

Decision 3 is the one most architects get wrong, and it is worth stating as a rule. A link is reversible and carries a certainty value; a merge is a structural transformation that is only reversible if it was designed to be. Decision 6 is the one most architects do not believe is allowed. It is: inferring a condition from several weak signals, with the inference's certainty recorded, is a legitimate architecture, and it is the only one available where coding is unreliable.

The compromises, which are the teaching material

A reference design that publishes only its decisions teaches technique. This one publishes its compromises, which teaches judgement. Five, from the published account of its mental health admissions dashboard [3].

A proxy was used where coding was unreliable, and the bias was named. The dashboard defines a psychiatric admission as an admission to a specialised psychiatric hospital *or* an admission with admitting specialty recorded as psychiatry at any other public hospital. The stated reason is that ICD-10 coding was not always reliably captured, so specialty was used instead. The authors then state the consequence: this excludes psychiatric patients admitted to emergency centres or to other specialties, **undercounting patients, especially those admitted for self-harm** [3]. That is the shape of the deliverable. Not "data quality is poor", which is unactionable, but: here is the proxy, here is the direction of the error, here is the population it loses.

The episode boundary was resolved by an explicit rule. The cohort is defined per admission rather than per facility record, because one admission can span several facilities; and an admission starting within a day of a prior discharge is treated as a referral and joined for length-of-stay purposes [3]. Chapter 4 set out why spell, episode and admission cut the same care at different places. Here is a provincial department choosing a cut, writing it down, and living with it. Note what the rule does to a readmission measure: a transfer dressed as a readmission is removed, and a genuine readmission on day two of discharge is removed with it.

De-identification was done by perturbation, with a key held somewhere. Patient master index identifiers are perturbed to de-identify the dashboard dataset, with a re-identification key required to link back [3]. The published account does not say who holds that key, and this research could not establish it — an unverified point, and what would settle it is the data centre's own release procedure or a data transfer agreement. It matters because a dataset that retains a re-identification key is a materially different legal object from one that does not: section 16(2) of the National Health Act dispenses with the three research authorisations only where the study does not reveal the user's identity [4], and POPIA's identifiability test is contextual rather than a list of fields to strip. The practical artefact is therefore a *pairing* — the transformation specification together with a statement of who holds the key — and the reader's version of this design must name the custodian that the published one does not.

Capture coverage changes over time, and that confounds every trend. The account states plainly that primary healthcare data capture has limited coverage and that **gradual expansion of electronic capture will confound measured increases in primary care activity** [3]. This is the single most important caveat in longitudinal routine-data analysis and it is almost never stated. Any dashboard showing primary care activity rising over a rollout period is showing, in unknown proportion, more care and more recording of care. There is no adjustment that fixes it; there is only a denominator of reporting facilities, stated on the chart.

The same provincial data produces different occupancy depending on the facility's convention. The dashboard's own authors record it: facilities calculate bed occupancy differently — some include the admission and discharge days, some do not — so facility managers may distrust dashboard figures that differ from their own [3]. Both are right under their own rule. Nobody can tell, because the rule is not attached to the number. That is the published South African case for Measure Definition as an entity, and chapter 13 introduced it.

> **THE DELIVERABLE SHAPE**
>
> Each of those five is a sentence of the form: *here is the compromise, here is its named consequence, here is what would have to change to remove it.* An architecture document in a South African health organisation that contains twenty such sentences is worth more than one that contains a target state diagram. The reader's first ninety days produce the twenty sentences.

The data centre also names its own gap, which is the last thing to copy. It cannot access definitive cause of death, because cause-of-death detail is reserved to the national statistics agency and may not be released in identifiable form [1]. A reference design that states the thing it cannot do is the artefact to imitate.

The order of the work

Decision order is not teaching order, and the evidence is specific about the difference. Identity, terminology and integration strategy must be settled **before** the application strategy, because those three determine whether a multi-vendor estate is survivable at all; and in a low-maturity estate the highest-return first capability is event-driven clinical notification rather than a shared record. Chapter 8 stated that as a sequencing fact. Here it is as a work plan.

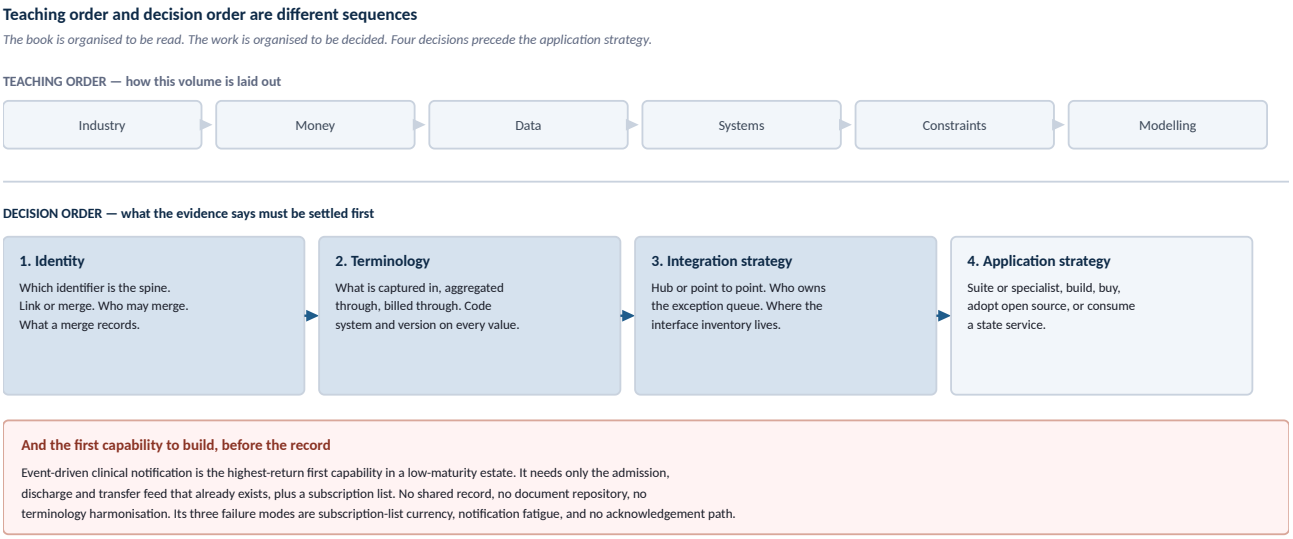


Figure 22.2 Teaching order against decision order, and the capability that comes before the record.

Notification is worth one paragraph of justification, because every instinct points the other way. It needs an event source, which is the admission, discharge and transfer feed that already exists; a subscription registry recording who wants to know about whom and under what condition; a notification service; and a delivery channel. It needs no shared record, no document repository and no terminology harmonisation. The South African gazetted framework already names profiles for document subscription and for mobile alert communication, so the standards selection is not an open question [5]. Its three failure modes are known and all three are governance rather than engineering: subscription-list currency, because an out-of-date practice list sends a notification to the wrong practice and that is a privacy breach rather than a nuisance; notification fatigue, because an unfiltered feed is ignored within weeks; and no acknowledgement path, so nobody knows whether the notification was acted on. Chapter 9 showed the same closed-loop failure in three streams. Building the acknowledgement into the first notification flow is the cheapest demonstration available that the architecture function changes outcomes.

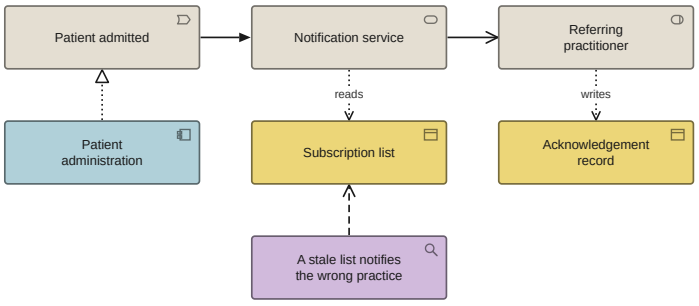


Figure 22.3 Event-driven notification modelled with the elements that exist: an Event, a Service in the application layer, and the subscription list and acknowledgement as Business Objects.

Days 1 to 15 — count, and collect

Nothing is decided in the first fortnight. Four registers are started, and they are started by asking for documents rather than by running workshops.

What to ask for, on the first day

Ask for	From	What it actually tells you
The interface inventory, and the date last reviewed	Whoever runs integration	Whether there is an integration function or only an integration server
The exception queues, with counts and the age of the oldest item	The same person	The best proxy for integration health, available in an afternoon
The identifier types in use, with their assigning authorities	Registration and health records	Whether anyone has ever written the identifier layers down
Every statutory and programmatic report sent out, with recipient and clock	The information officer and clinical managers	The outward-flow obligations: the only immovable requirements in the estate
The three figures the executive argues about, with the query behind each	Finance and the information office	Where the measure definitions are missing
The last unannounced downtime drill, and whether clinicians took part	The chief information officer	Whether the planning assumption is hours or weeks
The list of shared services the organisation depends on and does not own	Operations	The single points of failure outside the perimeter
The retention schedule, and the authority cited per period	Health records	Almost always: there is none, or it cites a period with no instrument behind it

That list is deliberately all documents. Nothing on it requires a decision, a budget or a steering committee, and a request for a document that does not exist is itself a finding with a date on it.

What to read

The organisation's own instruments first: its licence or certification correspondence, its last Auditor-General or internal audit management letter on records, its operator agreements with administrators and laboratories, and the clinical protocols it has adopted with the national versions they derive from and the local modifications made to them. Then four external instruments, read once and properly: the gazetted normative standards framework, because it is the most important single document for a South African health architect and is worth the afternoon; the morbidity coding standards and the funder regulation that makes a diagnostic code a condition of payment [6]; the National Health Act's record sections, because section 17(2) reads as a data architecture requirement specification with a criminal penalty attached [4]; and the professional record-keeping guidance, obtained in its current edition rather than quoted from a secondary account [7].

What to count

Four counts, all cheap, none requiring the organisation's cooperation beyond read access.

- **Interfaces, and interfaces with a named owner.** The ratio is the finding.
- **Identifier types actually present in the patient index**, including the defaulted and placeholder values. A placeholder identity number appearing thousands of times is a blacklist entry waiting to be written, exactly as the reference design does it [1].
- **Facilities or wards reporting zero of something that cannot be zero.** A primary care facility reporting no patient safety incidents in a year is a reporting failure, not a safe facility.
- **Days between a result being authorised and a clinician acknowledging it**, for the subset where an acknowledgement is recorded at all. Where none is recorded, that is the count.

What to write down

Start the four registers, in the dullest possible form, and circulate them weekly with the date on them: the system inventory, the interface inventory, the outward-flow register, and the identifier catalogue. Appendix 23 gives the column sets for the last two. The discipline that matters is versioning them from the first week, because a register's value is its change history and nobody starts that retrospectively.

What not to decide

Not the target application estate. Not the integration product. Not cloud. Not the terminology. Not whether the organisation needs a master patient index, which it certainly does, because the question that matters is what its merge policy will be and that cannot be answered in a fortnight.

Days 16 to 45 — name the boundaries

The second block converts the registers into named positions. The output is three short documents, each two to four pages, each dated and attributed.

A vocabulary note. Not a glossary — chapter 4's definitions are already written and should be adopted verbatim rather than renegotiated. The note records which boundary definitions *this* organisation uses, where two legitimate definitions exist, and what the organisation's choice costs. Three entries earn their place immediately: where an inpatient stay begins, which has at least four defensible answers; whether home leave interrupts an episode; and whether the admission and discharge days both count in a bed-day. The last one is the occupancy problem the reference design published about its own province [3].

The measure definitions for the figures already in dispute. Not all of them. The three the executive argues about, written as numerator, denominator, exclusions, boundary rule, period and version. The published evidence that this must be an entity rather than a convention is theatre utilisation: on one specialty in one hospital trust over fifteen weeks, there is no simple mathematical relationship between four calculation methods, with raw utilisation of

80% and touch-time utilisation of 60% on the same list [8]. Carry that denominator — one specialty, one trust, fifteen weeks — because an architect who quotes the finding without it has committed the error chapter 2 teaches against.

An identity position paper. The South African evidence is published, quantified and specific, and it is the only argument an architect needs. In a Tshwane cohort of 13 499 infant HIV polymerase chain reaction tests between May 2016 and May 2017, only 4% of infants matched exactly on all four demographic details; the first name matched in 5%; 74% were registered as "baby of"; of those tested twice at the same facility only 5% had the same folder number; at least 21% could not be linked at all; and the laboratory service's own linking algorithm scored sensitivity of 4,7% with a positive predictive value of 40,5% [9]. Four identifier layers exist rather than two — the national identity number, the ten-digit health patient registration number, the facility folder number, and programme register numbers [10]. The national index is real and large, reaching 3 266 public establishments by 31 March 2025, which breaks down as 3 150 primary health care facilities, 113 hospitals and three general practices, with biometrics at 82 primary health care facilities [11] — and it allocates offline during power outages and produces duplicates when power returns, remediated by periodic central clean-ups [12]. The paper's conclusion writes itself: a nationally-issued identifier allocated by a system that must function offline is a locally-issued identifier with a national format, so the organisation's own resolution layer is not optional and no national crosswalk exists to replace it.

One further paragraph belongs in that paper and is routinely omitted. Identity resolution is not an unqualified good. Published South African stakeholder research records key populations fearing law-enforcement access and being unlikely to permit health workers to identify them on government systems [13]. For some patients reliable linkage is a risk they will refuse, so a master patient index design without an access-control and consent counterpart is half a design.

What will already be urged on you by day 30

What is urged	Why it is urged	Why it waits
Choose the suite, or confirm the one already chosen	A procurement window, usually real	Identity, terminology and integration strategy determine whether the choice survives
A data lake, because reporting is slow	The reports are genuinely slow	A lake over undefined measures produces faster disagreement
A patient app or portal	It is visible and demonstrable	The authorisation defect is the documented failure mode, and it is a privacy breach at scale
An artificial intelligence pilot	Somebody saw a demonstration	No model is maintainable without the identity and measure layers under it
Migrate to cloud	A hosting contract is expiring	Two statutory mechanisms apply, one of which is an application to the Regulator with a lead time
Rationalise the programme registers	They are obviously duplicative	They exist because the national routine system is aggregate; supply the missing capability instead
A terminology licence and a mapping project	The country joined the international terminology body in May 2026	The national release centre is not yet operating, nothing is mandated, and the gazetted framework names neither of the two terminologies most architects expect [5][14]

The last row deserves its own note, because it is the most consequential piece of timing in the volume. South Africa joined SNOMED International on or about 14 May 2026, which changes the licensing economics [14]. But no national extension, no national reference sets and no published national map exist, and representatives were listed as to be advised; this is unverified and what would settle it is a designation announcement from the Department of Health, so **re-check the status of the National Release Centre before designing to it.** Meanwhile the gazetted framework, published in October 2022, is built on integration profiles with several versions of the messaging standard, the document standard and the resource standard, and names neither SNOMED CT nor LOINC; the diagnostic classification appears only in its acronym list [5]. The correct instruction is therefore *be ready, not early*: store the code system and the version on every coded value, keep the clinical assertion separate from its classification, and build terminology as a service rather than as tables inside an application database.

Days 46 to 75 — decide, in order

Three decisions, in the order the evidence gives, each recorded as a dated decision with the authority cited and the alternatives that were rejected.

Identity. Link or merge, and under what certainty. The asymmetry drives the configuration: a false merge puts two people's clinical data in one record and the consequences are clinical, while a false duplicate splits one person's data and the consequences are also clinical but differently. A false merge is harder to detect, because a merged record looks complete. So the thresholds are a clinical safety decision requiring a named clinical risk owner's signature, the review queue needs a service-level target and a measured depth, and a merge that would combine two records with contradictory high-risk data should not auto-merge whatever the score. Two rules go in the decision and nowhere else is as cheap to put them: **a merge must be reversible by construction**, which means source identifiers and source demographic values are never overwritten, only superseded; and a merge is a recorded act with an authority, because creating, changing or destroying a record without authority and linking a user's identifying details to their condition without authority are both offences under section 17(2) of the National Health Act, carrying a fine or up to one year's imprisonment [4]. That is the strongest available argument for access and merge auditing, and it is not a privacy argument.

Terminology. Four statements, no more. What is captured in; what is aggregated through, by which governed mapping; what is billed through; and that the code system plus version travels with every coded value. The South African constraint is that a diagnostic code is required on a funder claim by a funder regulation, and the obligation is conditional on payment — an uncoded claim is unpayable, not unlawful [6] — so the organisation is choosing between a clinical capture terminology it is not required to have and a billing classification it cannot avoid. Decide, and record that the decision anticipates a national terminology position that does not yet exist.

Integration strategy. Hub or point to point; who owns the exception queue by name; where the interface inventory lives and who reviews it on what cycle; and what the conformance requirement in the next contract will be. The last of those is the most useful procurement lever available in the country and is routinely unused: the gazetted framework contemplates conformance testing at a Department-approved laboratory with an attestation issued [5].

Whether such a laboratory is operating could not be established in this research, and what would settle it is a departmental list of approved facilities — so the clause should require the attestation and name an alternative test regime if none is available.

Then, and only then, the application strategy. Chapter 15 gave the four-way sourcing question; chapter 17 gave the buying discipline. The decision this chapter adds is about *sequence*, and it is that the application strategy is a consequence of the three decisions above rather than an input to them.

Days 76 to 90 — prove one thing, and baseline the rest

The last fortnight is not a presentation. It is one working capability and one baseline.

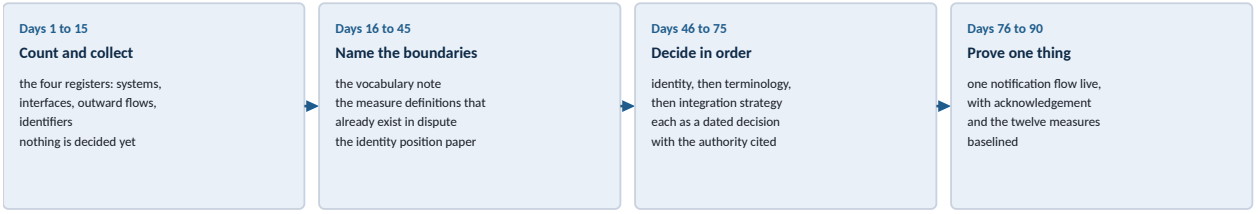
One notification flow, with its acknowledgement. Choose the narrowest clinically valuable case the estate can already support: in a private group, telling a referring practitioner that their patient has been admitted and discharged; in a district service, telling a clinic that a patient it referred has attended the hospital. Build the subscription list with an owner and a review cycle. Record the acknowledgement. Measure the proportion never acknowledged, which is the number that changes the conversation.

The baseline. Chapter 19 closes on twelve cheap, hard-to-game benchmark measures, each tied to a documented failure mode. They are the baseline, and the instruction here is narrow: take the first measurement inside the ninety days, record the method with each figure, and do not improve anything yet. A baseline taken after remediation has begun is not a baseline. Several of the twelve will come back as "not measurable", and that is the finding to report rather than a gap to apologise for.

One honest maturity statement. The acute-hospital adoption stage models measure adoption and pervasiveness rather than outcomes, are partly self-assessed, and contain no sourced resilience requirement. The freely published health-system interoperability maturity toolkit is a better fit for this readership: health-system rather than hospital level, five levels, three domains and eighteen subdomains, with business continuity as a named subdomain and a scoring rule that sets each domain at its **lowest** subdomain rather than its average [15]. The lowest-subdomain rule is what makes it honest, and it is why the statement an architect can defend after ninety days is usually a single level with one named binding constraint behind it.

Ninety days, four blocks, eight written deliverables

Nothing here needs a budget. Everything here is an artefact that survives the architect leaving.



THE EIGHT ARTEFACTS, AND WHO SIGNS EACH	
System and interface inventory, with a date of last review	*The integration owner*
Outward-flow register: recipient, trigger, clock, instrument	*The information officer*
Identifier catalogue: type, assigning authority, validity	*The identity data owner*
Identity decision: link or merge, thresholds, who may merge	*A named clinical risk owner*
Terminology decision: capture, aggregate, bill, and versions	*The clinical coding lead*
Measure definitions for the figures already in dispute	*The measure owner, per measure*
Retention schedule as data, with the authority per period	*The accounting officer*
Entity ownership decisions, including the contested ones	*The data governance body*

Figure 22.4 The four blocks of the first ninety days and the eight artefacts they produce, each with the role that signs it.

What to refuse to decide, and how to say so

Refusal needs a form of words, because "we need more information" reads as a stall. The form that works is: *this decision has a named prerequisite, here it is, here is the date we expect it, and here is what we will do in the meantime.*

Refuse to decide	The named prerequisite	What to do in the meantime
The target application estate	The identity, terminology and integration decisions	Publish the four registers and the three position papers
The retention periods	The current edition of the professional guidance, which disagrees with itself across versions [7]	Build the retention schedule as data, with the authority column empty and visible
Whether the record may be hosted outside the country	The transfer conditions and, separately, prior authorisation for special personal information going to a country without adequate protection	Record both mechanisms in the design and start the application if it will be needed
The case-mix or grouping approach	A national procedure code set and a grouping method, neither of which exists in routine use	Capture the procedure data that any future grouping would need
Whether hospital-acquired conditions can be measured	A present-on-admission flag, which South African routine data does not carry	Say so on every dashboard that implies otherwise
The national terminology binding	A National Release Centre that is not yet operating [14]	Store code system and version on every coded value

Two of those refusals are worth more than the rest. The present-on-admission one, because a dashboard that reports hospital-acquired condition rates computed from South African routine data is reporting a number that cannot be computed, and somebody will act on it. And the retention one, because the honest position — that every period in circulation traces to a professional guideline rather than to a statute, and that the guideline's own successive editions disagree, while the destruction authority for a public record sits with the National Archives instrument that the Act's record-keeping duty is made subject to [4][7] — is more useful to an accounting officer than a confident six-year rule that is wrong twice over.

The one argument that buys the work

A records programme in a South African public health organisation is justified medico-legally before it is justified clinically, and the arithmetic is published. In an Auditor-General audit of 27 facilities across nine provinces, 15 reported missing or lost records, 17 did not track records after issue and 25 had inadequate storage; and in three of nine paid medico-legal claims reviewed, departments conceded because they could not locate the medical record [16]. Provincial medico-legal contingent liability at 31 March 2026 was published as R50,2bn by the Auditor-General and R50,6bn by National Treasury — two official values, neither wrong, different measurement bases on the same date [16][17]. Set either against the **R6,5 million** budgeted for "Digital Health System" inside the 2026/27 direct National Health Insurance allocation of R1,427bn [18] — and note that the 2026 Budget Review gives the direct allocation for 2026/27 as R475m with R1,485bn as the three-year total [19], so the two published presentations disagree and R6,5m is most plausibly a medium-term figure too. Cite both. The argument is unaffected on either reading. Chapter 13 has the full arithmetic. The sentence that survives a steering committee is that a retrieval capability which prevented one concession of the order of R20 million would fund more digital health than the national line carries.

In a private group the equivalent argument is different in form and identical in structure. The immovable requirements are the funder regulation that makes a diagnostic code a condition of payment [6], the notifiable-conditions clocks, the breach-notification duty and the records duty — all in force today, all with an enforcement record, and none of them contingent on a statute that has not commenced. Chapter 7 made that case about National Health Insurance and this chapter only inherits it: build the dependencies regardless of the legal outcome, because each has independent value.

What ninety days cannot do

It cannot fix the identifier. It cannot produce a clinical data warehouse worth querying. It cannot deliver a terminology service, because the national position it would bind to does not exist. It cannot change a vendor contract signed before the architect arrived. It should not attempt a target state diagram for an estate whose current state has no interface inventory.

What it can do is leave behind eight artefacts that outlast the architect, each with an owner's name and a date, and one working capability with an acknowledgement in it. That is a defensible quarter in this industry, and it is more than most engagements leave behind.

THE CHAPTER IN TEN LINES

- Ninety days cannot fix anything in a health organisation and is exactly long enough to write down what nobody has written down.
- Read the Western Cape Provincial Health Data Centre as a reference design: it queries the systems that exist, makes the shared identifier a real identifier, links rather than merges, and stores certainty.
- Blacklist the known bad identifier values, map to standard code sets at ingestion accepting where possible, and infer the clinical layer rather than demanding better source coding.
- Separate identifiers from attributes and perturb the join key per released dataset, and name the custodian of the re-identification key that the published account does not name.
- Its five published compromises are the teaching material: a specialty proxy that undercounts self-harm, an episode rule joining admissions within a day of a discharge, perturbation with a key held somewhere, expanding capture confounding primary care trends, and occupancy computed differently per facility.
- The deliverable shape is a sentence naming the compromise, its consequence, and what would have to change to remove it.
- Decision order is not teaching order: identity, terminology and integration strategy are settled before the application strategy.
- Event-driven notification is the highest-return first capability in a low-maturity estate, and its three failure modes are governance rather than engineering.
- The first fortnight asks for documents and starts four registers: systems, interfaces, outward flows and identifiers. Nothing is decided.
- The second block produces three short dated documents: a vocabulary note, measure definitions for the figures already in dispute, and an identity position paper.
- Refusal needs a form of words: the decision has a named prerequisite, here it is, here is when we expect it, here is what we do meanwhile.
- Take the first measurement of chapter 19's twelve benchmark measures inside the ninety days, record the method with each figure, and improve nothing yet.
- A records programme in a South African public health organisation is justified medico-legally before it is justified clinically, and the arithmetic is published.
- Leave eight artefacts with an owner's name and a date, and one working capability with an acknowledgement in it.

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APPENDIX A

The cross-maps

The mapping tables to keep beside you, and nothing argued that was not argued earlier.

These are the tables an architect keeps beside them. Nothing here is argued; every position was argued in an earlier chapter, named where the map needs it. Six maps, each split where one table would run too long or too wide, with a header row on every part.

Which map to reach for

Six maps, the question each answers, and the chapter that argued the position behind it.

MAP	WHAT IT HOLDS	THE QUESTION IT ANSWERS	ARGUED IN
One	The sheets against ArchiMate 4	*"What element is this sheet, and does that element exist?"*	Chapter 20
Two	Systems against journeys	*"Which journeys does this system carry, and what must it hold?"*	Chapter 9
Three	Entities against candidate owners	*"Who could own this, and what must the decision record?"*	Chapters 5 and 10
Four	The outward flows	*"What must we send, to whom, on what clock, under which instrument?"*	Chapter 18
Five	Seams against the layers that govern them	*"Which layers govern this seam, and what is the actual contract?"*	Chapter 16
Six	South African instruments	*"What does this instrument actually require, and of whom?"*	Chapters 7 and 18

Rejected: a framework content metamodel against the sheets; a glossary cross-map; a defect register; a vendor map.

Figure 23.1 The six maps, the question each answers, and the chapter that carries the argument behind it.

Map one — the repository's sheets against ArchiMate 4

The supplied repository is a workbook of sheets, each a flat list of one object type. ArchiMate 4 carries 42 elements. The two do not line up, and several sheets are named after elements that do not exist. Chapter 20 explains why; this is the lookup.

The rule that resolves most of them: ArchiMate 4 element names are **unprefixed**. There is one `Service` element, one `Process`, one `Function`, one `Event`, one `Role` and one `Collaboration`, each placed in a layer rather than renamed by it. So an interface engine's published capability is *a `Service` in the application layer*. The interface and object families are the exception and genuinely are layer-specific: `Business Interface`, `Application Interface` and `Technology Interface`; `Business Object`, `Data Object` and `Artifact`.

Motivation and business sheets

Ten map straight across and need no note: O01 Driver, O02 Goal, O05 Principle and O06 Requirement to the elements of those names; O09 Organization Unit and O10 Actor to `Business Actor`; O12 Location to `Location`, which is a common-layer element rather than a business one; O13 Business Capability to `Capability` in the strategy layer; O14 Value Stream to `Value Stream`; and O18 Product to `Product`. Three more need only the unprefixed name: O16 Business Process is a `Process`, O17 Business Function is a `Function`, and O35 Business Building Block is a `Grouping`, a framework concept rather than an element. The rest need a decision.

Sheet	What the sheet holds	Element to use	Caution
O03 Objective	Time-boxed ends with a target	`Goal`, or `Outcome` where the row records a result achieved	There is no `Objective` element
O04 Measure	The metric attached to an objective	None	**There is no metric element**, and `Value` is not a number. Carry the metric as a property and the rule behind it as a Measure Definition, held as a `Business Object`
O07 Constraint	What a solution may not do	`Requirement`	**There is no `Constraint` element.** Negative phrasing is the only difference
O08 Assumption	What is taken as true	`Assessment`	No assumption element; an assessment with a stated confidence is the nearest honest fit
O11 Role	Roles performed	`Role`	Unprefixed. There is no `Business Role` element
O15	Ten service rows in the business layer	A `Service` in the business layer	**The sheet is named after an element that does not exist.** Quote the sheet number and describe the element
O19	Ten agreement rows	`Business Object`	**There is no contract element.** The agreement is an object; its parties are modelled by association

Data, application and technology sheets

Seven map straight across: O23 Application Component to `Application Component`; O22 Physical Data Component to `Artifact`; O38 Data Set to `Data Object`, or `Artifact` where it is a file; O39 Data Migration to `Work Package`; O44, O45 and O46 Building Blocks and O48 Environment to `Grouping`; and O50 Network Link to `Communication Network`, or `Path` where it is logical. Five more resolve in a line: O21 Logical Data Component is a `Data Object`, or a `Grouping` where it is a container; O37 Data Attribute is a property of the `Data Object` rather than an element; O40 Source To Target

Mapping is a `Flow` relationship with the rule held as a `Business Object`; O42 Metadata Element is a property, or a `Business Object` where it is governed; and O43 Master Data Rule is a `Requirement`, or a `Business Object` where the rule itself is the artefact.

Sheet	What the sheet holds	Element to use	Caution
O20 Data Entity	Fourteen entities	`Business Object` conceptually, `Data Object` once logical	The same name at both levels is the commonest modelling error here
O24 Information System Service	Twelve service rows	A `Service` in the application layer	There is no layer-prefixed service element
O25 Technology Service	Twelve service rows	A `Service` in the technology layer	As above
O28 Platform Service	Eight rows	A `Service` in the technology layer	As above
O26 Logical Technology Componen	Seven rows; the title is truncated at 31 characters	`Node`	The truncation has propagated into the sheet's own first row
O27 Physical Technology Compone	Twelve rows; also truncated	`Node` with `Device` and `System Software`	A regulated clinical device needs all three plus `Requirement` and `Assessment` (chapter 20)
O36 Data Entity Relationship	Eleven rows	None; this is an `Association` relationship	—
O49 Resilience Tier	Twelve rows	None; `Requirement` plus `Grouping`	Tier is a classification, so it belongs on the requirement

Realization, governance and security sheets

Eight of these map directly and need no note: O30 Work Package to `Work Package`; O31 Deliverable to `Deliverable`; O53 Transition Architecture and O66 Architecture Increment to `Plateau`; O54 Architecture Alternative to `Course of Action`; O59 Compliance Assessment to `Assessment`; and O47 Reference Model Entry and O55 Building Block to `Grouping`. Six more resolve in a line: O52 Implementation Factor is a `Driver`, or an `Assessment` where it records a judgement; O58 Governance Gate is an `Event`, with a `Process` for the review that runs at it; O61 Governance Change Request is a `Business Object`, or a `Work Package` once accepted; O63 Control Objective is a `Goal`, or a `Requirement` where it is testable; O64 Security Policy is a `Principle` with the `Requirement` elements that implement it; and O65 Enforcement Point is a `Node`, or a `Device` where it is physical. The rest need a decision.

Sheet	What the sheet holds	Element to use	Caution
O29 Gap	Two rows	An `Assessment` held between two `Plateau` elements	There is no gap element in the 42
O32 Standard	Sixteen rows with a conformance column	`Requirement` where conformance is mandated, `Principle` where it is preference	No standard element. Map five is the companion: a standard governs a seam
O33 Architecture Contract	Eight rows	`Business Object`, or `Deliverable` where a work package produces it	As O19: no contract element
O41 Security Control	Eight rows	`Requirement`, realised by a `Node` or a `Service`	There is no control element
O56 Project	Twelve rows	`Work Package`	No project element; the distinction from O30 is organisational
O57 Governance Body	Five rows	`Business Actor`, or `Collaboration` for several parties	Unprefixed: collaboration is not layer-named
O60 Governance Decision	Seven rows	`Business Object`	A decision record is an object with a date and an author
O62 Delivery Risk	Eight rows	`Assessment`, associated with the `Driver` it threatens	There is no risk element
EXT, six industry types	One row each, held in O20 and nowhere else	Mostly `Business Object`; the device type needs `Node`, `Device` and `System Software`	The care setting type classifies service, not place (chapter 4)

> **THE ELEMENTS THAT DO NOT EXIST, IN ONE PLACE**
>
> There is no layer-prefixed service, process, function, event or collaboration element; no `Business Role`; no `Constraint`; no contract element; no `Objective`; no metric element; no gap, assumption, standard, risk, control or project element. Where a sheet is named after one of these it is not wrong about the *content* — it names a framework concept the notation expresses differently.

Map two — systems against the journeys that touch them

Chapter 9 names the stages of eight journeys. This is the same information read from the system inward, the direction an application portfolio review runs in. Stage names are chapter 9's exactly.

System class	Journeys and stages it carries	What it must hold	Where it is commonly absent
Patient administration	Referral to treatment (accept, redirect or decline; book); encounter to payment (establish liability); emergency arrival and triage (register); admission to discharge (admit; occupy)	Identifier cross-reference; the encounter; the ward stay with start and end; waiting list and clinic diary	Nowhere. The gap is the ward stay
Clinical record	Referral to treatment (attend and treat; back-refer and close); admission to discharge (treat; discharge); transfer between facilities (compose the summary)	The clinical assertion separate from its code; authorship; correction by amendment	Much of public primary care, where the record is a paper folder
Order communications	Order to result (request)	Orderable item, clinical question, priority, placer identifier	Much of the public estate, where the requisition is paper
Laboratory information system	Order to result (accession and check; analyse and authorise; report and route)	Accession number cross-referenced to the placer order; value, unit, range, authoriser	Never absent, but owned by another organisation in both sectors
Radiology and imaging	Order to result, where the orderable is an image	The scheduled procedure step; the study identifier; the report as an object distinct from the images	Report coding; structured reporting is rare
ePrescribing and medicines administration	Prescribe to administer (prescribe; administer, or record not given)	Dose, route, frequency, duration, indication; actual dose and time; the reason not given	The administration record, the entity most often missing entirely
Pharmacy dispensing and stock	Prescribe to administer (verify; dispense or supply; reconcile)	Product, quantity, batch, expiry, days supply, pharmacist	Patient-level dispensing in much of the public sector, where only stock movement exists
Theatre management	Admission to discharge (treat)	Five timestamps, not two; implants and consumables; cancellation reason	Implant capture, usually manual
Revenue cycle and claims	Encounter to payment (code; assemble and price; submit; adjudicate; reconcile and resubmit)	The funder claim distinct from the invoice; tariff by funder, provider, code and effective date	Remittance: the national claim standard defines no such message
Incident and risk	Incident to improvement, all eight stages	The minimum field set; a severity code; an action tracker with owner and date	The action tracker, and any denominator at all
Patient portal	Referral to treatment (book)	Authenticated identity bound to the clinical identity; an authorisation model tested against another patient's data	Authorisation testing, the documented failure mode
Integration engine	Every journey crossing a system boundary	The interface inventory; the exception queue with an owner; message-level operational visibility	The exception queue owner, and the inventory's review date
Clinical data warehouse	All journeys, for measurement	The measure definition, versioned; lowest-level timestamped facts, not pre-cut aggregates	The measure definition
Triage module or tracking board	Emergency arrival and triage (triage; re-triage; stream and treat)	A series of timestamped categories, not one value; discriminator checks	Most emergency centres, where triage is paper and a sticker
Pre-hospital record	Emergency arrival and triage (arrive); transfer between facilities (move)	A pre-hospital identifier, and the step that links it to the encounter	The link to the encounter, almost everywhere
Bed bureau or placement	Transfer between facilities (find and secure a bed; obtain acceptance)	The acceptance, with actor and timestamp	Most of the country: a telephone call, with no measure

Map three — entities against their candidate owners

Chapters 5 and 10 establish that for several entities there is no single plausible owner, and that asserting one teaches a falsehood. The deliverable is the governance decision. This map lists the candidates so the decision can be put to a governance body as a choice rather than a question.

Entity	Candidate owners	What each claims	What the decision must record
Practitioner	Human resources; credentialling; the clinical directorate; billing	HR owns employment; credentialling owns registration verification; the directorate owns privileges; billing owns practice codes	Which function is accountable for the *person* record, and who must be told when a registration lapses
Care Setting, Ward, Bed and Location	Estates; nursing operations; the clinical directorate; finance	Estates owns the place; nursing the staffed complement; the directorate the specialty; finance the funded count	Which of the four bed counts the occupancy figure uses, and who may change a complement
Care Plan	Genuinely contested between services	Each service that contributes a pathway claims the plan that contains it	Whether the patient has one plan assembled from pathways, and who reconciles conflicting goals
Medicines master	Pharmacy, for the dispensing catalogue; finance, for the price; **nobody, for the clinical classification**	Two sourced owners and one absence. A third through a procurement catalogue is a defensible inference, not a finding	Who owns the therapeutic classification, and whether the organisation builds its own dictionary
Measure Definition	No current owner anywhere	The entity does not exist in the supplied repository, so no function has claimed it	A named owner per measure, with a version and effective date
Disclosure event	No current owner anywhere	Information officers claim the duty; no system claims the record	Where the record that information went to a named recipient on a date for a purpose is held

Map four — the outward flows

At minimum twenty-one statutory and programmatic flows leave a South African provider. Each is a data interface with a schema, a clock and a legal consequence, and filling this table for a real organisation is the best use of a Phase C workshop here. Chapter 18 carries the duties.

Surveillance and registries

Flow	Recipient	Trigger	Clock	Instrument
Notifiable medical conditions, Category 1	Sub-district focal person, to the national institute	Diagnosis or clinical suspicion	Immediate by telephone; written or electronic within **24 hours**	Notifiable conditions regulations, Gazette 41330, 15 December 2017, regulation 13 [1]
Notifiable medical conditions, Category 2	As above	Clinical or laboratory diagnosis	**7 days**	As above [1]
Notifiable medical conditions, Category 3	Focal person, from the laboratory	Laboratory diagnosis	**7 days**	As above [1]
Notifiable medical conditions, Category 4	Focal person, from the laboratory	Laboratory diagnosis	**1 month**	As above [1]
Outbreak or unusual cluster	Focal person	Detection	Immediate, written within **24 hours**	As above, regulation 13(5) [1]
Medical scheme notifiable-conditions return	National department	Funder claim received	**Monthly**	As above, regulation 13(4) [1]
Cancer notification, establishment	National Cancer Registry	Confirmed diagnosis	**3 months**	Regulation 380 of 2011 under the National Health Act [2]
Cancer notification, clinician, patient under 15	National Cancer Registry	Diagnosis	**3 months**	As above [2]
Cancer notification, laboratory	National Cancer Registry	Laboratory confirmation	**Quarterly**	As above [2]
HIV programme indicators	Programme register, to the national routine system	Monthly register close	**Monthly**	District health management information system policy, 2011 [3]
Tuberculosis programme indicators	Register, to the national routine system	Monthly register close	**Monthly**	As above [3]
National Indicator Data Set	The national routine system	Monthly facility close	**Monthly**	As above [3]

Deaths, products, funders and the Regulator

Flow	Recipient	Trigger	Clock	Instrument
Maternal death	The confidential enquiry, and as a notifiable condition	Death	Per enquiry guidance; notifiable-conditions clock **7 days**	National Health Act; notifiable conditions regulations [1] [4]
Perinatal death	The perinatal problem identification programme	Death	Per programme	Programme instrument
Child death	The child problem identification programme	Death	Per programme	Programme instrument
Death certification	Home Affairs, onward to the statistics agency	Death	At death, before burial	Births and Deaths Registration Act 51 of 1992, regulations 11 and 14 [5]
Adverse drug reaction	The certificate holder, onward to the medicines regulator	Suspected serious reaction	Holder: **15 calendar days** from receipt of the four minimum data elements	Medicines Act 101 of 1965, regulation 40; pharmacovigilance guideline version 10, 22 September 2025 [6]
Device incident	The medicines regulator	Adverse event, incident or near-harm	Per the device vigilance guideline	Medicines Act; Regulation 1515, Gazette 40480, 9 December 2016 [7]
Medical scheme claim	Switch, onward to the scheme	Service rendered	Per contract; submission window and payment terms are statutory	Medical Schemes Act 131 of 1998, regulation 5(f) [8]
Security compromise notification	The Information Regulator's online portal, and the data subjects	Discovery of a compromise	"As soon as reasonably possible" — **there is no 72-hour deadline**	POPIA s22; Regulator portal from April 2025 [9]
Access-to-information annual report	The Information Regulator	Annual	Per the Regulator's call	PAIA 2 of 2000 [10]

Map five — seams against the layers that govern them

The question "which standard governs this seam" has a misleading answer: most seams are governed by two or three standards at different layers at once, plus a profile, plus a local interface specification that is the actual contract. Chapter 16 makes the argument. These tables are the reframing — which layers govern the seam, and what is signed. Chapter 16 carries what breaks at each of them.

Identity, encounter and the clinical seams

Seam	Syntax layer	Semantic layer	Profile or guide	The actual contract
Patient index to everything	Messaging standard, or a resource API	Identifier type and assigning authority	Identity cross-reference and demographic query profiles	The local interface specification, with its extension segments
Patient administration to departmental systems	Messaging standard, admission and transfer events	Encounter and location vocabularies, usually local	Patient administration management profile	The local specification, plus the event subset actually sent
Scheduling to departmental systems and portal	Messaging standard, scheduling events	Service and slot vocabularies, local	—	The local specification
Clinical record to laboratory, orders	Messaging standard, or a resource API	Orderable catalogue; observation naming where adopted	Laboratory testing workflow profiles	The site interface specification and the catalogue version
Laboratory to clinical record, results	Messaging standard, or a resource API	Observation naming and units	As above	As above
Clinical record or imaging system to modality	Imaging standard, worklist service	Procedure and body-site vocabularies	Scheduled workflow profiles	Each device's conformance statement
Modality to image repository	Imaging standard, storage services	Study and series identifiers	As above	The conformance statement

Medicines, money and the outward seams

Seam	Syntax layer	Semantic layer	Profile or guide	The actual contract
Clinical record to pharmacy	Messaging standard, or a resource API	Product identification; **no national medicines dictionary exists**	—	The local specification and product file version
Devices to critical care systems	A device communication standard in principle, vendor protocols in practice	Observation naming, where used	—	The vendor's specification
Theatre to patient administration and coding	Local, or a messaging standard	Procedure coding, for which **no national set is in routine use**	—	The local specification
Rostering to access control	Rarely standardised	—	—	Nothing
Provider to funder	The national claim message, through a switch; **the operative wire format is frequently the switch's own XML, validated against its schemas**	Diagnostic classification; product and tariff code sets	The switch's integration pack and accreditation step	The switch's schemas and the accreditation test
Provider to region or nation	Document exchange, or a resource API	Document type and section vocabularies	Cross-enterprise document sharing profiles	The national framework's implementation guides, where published
Provider to surveillance	National channels	Condition lists by category	—	The national system's own form

Map six — the South African instruments against what each requires

Chapter 18 carries the duties and chapter 7 the status of those that have not commenced. This map answers one question per instrument: what does it *actually* require, and of whom.

Statutes and regulations: records, privacy and access

Instrument	Status	What it actually requires	Who it binds	Architectural consequence
National Health Act 61 of 2003, ss13 to 17	In force	A record for every user; **written** consent for third-party disclosure; access only for a legitimate purpose within the care relationship; control measures against unauthorised access; ten listed offences, including linking a user's details to their condition without authority [4]	Every health establishment	Role-based access with break-glass, reason and audit. A recorded authority for every merge and every new integration
National Health Act, ss36 to 40	**Invalid and severed, 18 May 2026** (CCT 240/24, ZACC 19 of 2026) [13]	Nothing. Never commenced; no regulations made	Nobody	Past tense. Private facility supply is now regulated only by provincial licensing
National Health Insurance Act 20 of 2023	Assented May 2024; **no section in force** [19]	Nothing yet	Nobody	A contingency. Re-check the Constitutional Court judgment, heard 5 to 7 May 2026, reserved at 31 October 2026
POPIA 4 of 2013	In force; enforcement against private bodies has begun	Health data is special personal information, prohibited unless authorised; breach notification through the Regulator's portal; transfer conditions; and, separately, prior authorisation for special personal information going to a country without adequate protection [10][11]	Responsible parties and operators	The scheme is responsible party, the administrator operator. Cloud has two mechanisms, and the second has a lead time
POPIA health regulations, Gazette 54268, **6 March 2026**	In force same day, no grace period	Processing rules for health information; sex-life references removed from the final text [12]	Insurers, schemes, administrators, managed care organisations, administrative bodies, pension funds, employers	**They do not bind providers**
PAIA 2 of 2000	In force	Access on request; an annual report to the Regulator [9]	Public and private bodies	The record must be retrievable on someone else's clock
National Archives and Record Service Act 43 of 1996	In force	No destruction of a public record without the National Archivist's authority, against an approved disposal schedule; National Health Act s13 makes the record-keeping duty subject to it [4]	Public bodies	The destruction-authority constraint, not a retention period

Statutes and regulations: licensing, notification and products

Instrument	Status	What it actually requires	Who it binds	Architectural consequence
Medical Schemes Act 131 of 1998, reg 5(f)	In force	A diagnostic code on every funder claim. The obligation is **conditional on payment**: an uncoded claim is unpayable, not unlawful [8]	Schemes, and anyone claiming from one	Effective-dated code validity is an architecture requirement
Norms and standards regulations, effective 2019	In force	Seven chapters and 22 regulations, which the health standards office inspects against [20]	Health establishments	Three compliance instruments run at once with incompatible results. Establish which one a compliance claim means
Regulation R158 of 1980 under the Health Act 63 of 1977	Operative in practice; **s44 of the enabling Act was wholly repealed in 2012**, authority delegated by Proclamation R152 of 1994	Provincial licensing of private facilities, with bed and theatre counts per site [18]	Private establishments	Licensed beds are a fifth bed count, distinct from physical, commissioned, staffed and occupied
Notifiable conditions regulations, Gazette 41330, 15 December 2017	In force	Notification by category, with clocks, under criminal sanction [1]	Practitioners, laboratories, establishments, schemes	Four clocks in one interface; category drives the clock, so it is versioned reference data
Medicines Act 101 of 1965, with Regulation 1515	In force	Device and in-vitro diagnostic classification; an establishment licence under s22C. **Product registration is not yet operating** [7]	Manufacturers, importers, distributors	Software as a medical device is classified by principles as an active device
Children's Act 38 of 2005, s129	In force	A child over 12 with sufficient maturity and capacity may consent to their own treatment [17]	Providers	Consent capacity is age *and* an assessed maturity, and the assessment is a recorded clinical act
Electronic Communications and Transactions Act 25 of 2002	In force	Electronic signature and data message rules [4]	Everyone	The signature question for a scheduled prescription is open, so a paper path must be designed

Professional, funder and framework instruments

Instrument	Status	What it actually requires	Who it binds	Architectural consequence
Professional record-keeping guidance (Booklet 9)	**Guidance, not law.** The 2016 text and a September 2022 review draft disagree on retention	Retention periods by record class; correction by strike-through with date, signature and reason, never deletion; separation of billing from clinical content [15]	Registered practitioners, and through them their employers	**"The law says six years" is wrong twice over.** Obtain the current booklet, and build the retention schedule as data with an authority column
Health normative standards framework, Gazette 47337, 21 October 2022	Gazetted under National Health Act s74(1)	A reference set of standards; prescriptive content deferred to implementation guides, with conformance testing at a Department-approved laboratory and an attestation issued [14]	Systems interacting with or uploading to national infrastructure	The attestation is the most useful procurement lever in the country. It names neither of the two terminologies most architects expect
National terminology position	Joined the international terminology body on or about **14 May 2026**; the national release centre is **not yet operating** [16]	Nothing is mandated: no national extension, no reference sets, no published map	Nobody yet	Be ready, not early. Store code system and version on every coded value. **Unverified: re-check the release centre's status**
Routine health information policy, 2011	Policy	A monthly facility close and submission of the national indicator set [3]	Public facilities	The system is aggregate. Patient-level questions it was not designed for are categorically unanswerable
National claim message	Published national standard, last revised November 2016	A claim-side message only. **It defines no remittance or payment-advice message** and no link from a response to a payment [8]	Providers and funders transacting through a switch	Remittance arrives in whatever format each funder chooses, which is why denial analysis is hard here
Prescribed minimum benefit list	Regulatory, and not settled	Cover in full for a defined set of diagnosis and treatment pairs, and for a chronic disease list	Schemes	The pair count is published as 271; the chronic list as both 26 and 27. Hold it as effective-dated reference data

What is not in this appendix, and why

Four maps were considered and left out.

A framework content metamodel against the repository's sheets. The repository *is* that metamodel rendered as a workbook, so the map would be close to one to one and carry no information. It would also mean printing a framework's own decomposition, which this volume does nowhere. Chapter 20 describes the relationship in prose.

A glossary cross-map. Chapter 4 defines the terms and the glossary appendix collects them. A third listing is a fourth place for them to diverge.

A repository defect register. Chapter 21 names the documented pattern behind each defect, which is the part that transfers. A bare list does not.

A vendor or product map. Product names and ownership move, several within the research period. A dated ledger belongs where the date and the caveat can travel with it, which is chapter 17.

One map here is weaker than the rest. Map two's "where it is commonly absent" column aggregates what the research observed rather than a surveyed figure; no South African system-presence census exists, and what would settle it is a provincial or group-level application inventory, which is one of the artefacts chapter 22 asks for.

THE CHAPTER IN TEN LINES

- Six maps: the repository's sheets against ArchiMate 4; systems against journeys; entities against candidate owners; the outward flows; seams against the layers that govern them; and the South African instruments.
- ArchiMate 4 element names are unprefixed. There is one Service element, one Process, one Function, one Event, one Role and one Collaboration, each placed in a layer rather than renamed by it.
- Interfaces and objects are the exception and genuinely are layer-specific.
- Several repository sheets are named after elements the notation does not contain. Quote the sheet number and describe the element.
- There is no metric element, and Value is not a number. Carry the rule behind a figure as a Measure Definition held as a business object.
- Read the journey map from the system inward; it is the direction an application portfolio review runs in.
- For Practitioner, for care setting and bed, for the care plan and for the medicines master there is no single plausible owner. The deliverable is the governance decision.
- At minimum twenty-one statutory and programmatic flows leave a South African provider, each with a recipient, a trigger, a clock and an instrument.
- Most seams are governed by two or three standards at different layers at once, plus a profile, plus a local interface specification that is the actual contract.
- The POPIA health regulations bind funders, administrators and employers. They do not bind providers.
- The diagnostic coding obligation is conditional on payment: an uncoded funder claim is unpayable, not unlawful.
- Retention periods come from professional guidance whose own editions disagree, while destruction authority for a public record sits with the archives instrument.
- A framework content metamodel against the sheets, a glossary cross-map, a defect register and a vendor map were considered and left out.

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- **18.** KwaZulu-Natal Department of Health, licensing of private health care facilities (Regulation R158 of 1980) https://www.kznhealth.gov.za/licensing-of-private-health-care-facilities/
- **19.** The Presidency, statement on National Health Insurance Act litigation https://www.thepresidency.gov.za/presidency-statement-national-health-insurance-act-nhi-litigation
- **20.** Office of Health Standards Compliance, Quarterly Bulletin, 2 May 2024 (the norms and standards regulations: seven chapters, 22 regulations) https://ohsc.org.za/quarterly-bulletin/

APPENDIX B

Glossary

One hundred and eighty-two entries: the sixty-two chapter 4 settled, and everything it did not, with the gaps named rather than filled.

This glossary is the volume's single vocabulary. Chapter 4 defines sixty-two terms, and those definitions are carried here unchanged in substance; where this appendix compresses one, it compresses and does not vary it. Everything chapter 4 did not define has been added: the abbreviations an architect meets on a seam diagram, the standards and profiles, the code systems, the South African instruments and the bodies that administer them, the public-sector systems, and the clinical, operational and funder vocabulary.

Three conventions. Where a term has two legitimate meanings, both are given and the chapter that settles the volume's usage is named. Where the research could not establish a current edition, an operative version or a status, the entry says so rather than asserting one — there are 217 such gaps on record across the four research files, and a dozen land here. Every version, date, count and licensing position carries a citation.

It reproduces no standard's content, and appendix 25 says where each one lives and what access costs.

The terms with two meanings

Term	The two meanings	What this volume does
Claim	A request for payment directed at a funder; a demand for damages against a provider	Chapter 4: always qualified, as *funder claim* or *medico-legal claim*, never bare
Electronic health record	Strictly, a longitudinal record of a person across providers; loosely, any clinical system	Chapter 4: the strict sense only. The single-organisation record is an EMR or EPR
Provider	The National Health Act's *health care provider*, which is a person; also used of a practice and of an organisation	Chapter 4: never unqualified. Say person, practice or health establishment [1]
User	The Act's word for the patient; also a person logged in to a system	Chapter 4: say *patient* or *system user*, never the Act's word for the other [1]
Admission	The event that opens an inpatient spell; loosely, the stay itself	Chapter 4: an event, never a stay
Bed	Physical, commissioned, staffed or available, occupied, and in the private sector licensed	Chapter 4: a funded, staffed capacity unit, held as a dated time series
Tariff	A national price list; contract data per funder, per agreement, per period	Chapter 6: contract data. There has been no national tariff since 28 July 2010 [51]
SVS	The IHE Sharing Value Sets profile; the national Stock Visibility System	Chapter 16: context only. The collision is real and nothing resolves it [37]

A

- **Administration (of a medicine).** The event in which a dose was actually given, or deliberately not given. The entity most often omitted and the only one that answers the clinical question; its most valuable attribute is the reason not given.
- **Administrator.** A company accredited by the Council for Medical Schemes to run membership, collection, adjudication, payment and reporting for a scheme, and an *operator* under POPIA [14].
- **Admission.** An event, not a duration: the event that opens an inpatient spell, with a method, a source and a timestamp.
- **ADT (admit, discharge, transfer).** The HL7 version 2 message family for patient administration. Its merge and unmerge events are the architecturally important ones and the most often unimplemented downstream.
- **Adverse drug reaction reporting.** Under regulation 40 made under the Medicines and Related Substances Act 101 of 1965 the submission duty sits with the holder of a certificate of registration, not the clinician, on a 15-calendar-day clock for serious South African cases [57].
- **Adverse event.** In the South African safety vocabulary, the *harmful subset* of a patient safety incident, which is the general term [79].
- **Appointment.** A booked future encounter. It is *not* an encounter: it may become one, be cancelled, or be not attended, and deleting cancelled appointments destroys the denominator for non-attendance analysis.
- **ATC (Anatomical Therapeutic Chemical classification).** The World Health Organization's five-level classification of active substances by the system they act on [33]. It measures consumption and cannot be prescribed from: it identifies no product, strength, form or pack.
- **ATNA (Audit Trail and Node Authentication).** The IHE audit-and-node-authentication profile, and a required grouping for every document-sharing actor — which makes audit non-optional by specification [13].
- **Authorisation.** The funder's prior approval of an admission, an expected length of stay and often a procedure. It is the funder's unit of counting, and matches neither spells nor episodes.

B

- **BHF (Board of Healthcare Funders).** The private funder industry body, whose practice numbering is a practical precondition of claiming — so a private facility needs a provincial licence *and* a funder-industry practice number before it can trade [16].
- **Bed.** The unit of capacity, not a physical object: a funded, staffed capacity unit, which is why a hospital can have more physical beds than beds. Commissioning and closure are dated events, so bed complement is a time series.

- **Bed-day.** One patient-day of occupancy. Bed-days are additive and insensitive to where a stay boundary falls; averages are neither.
- **Benefit option.** The priced benefit design within a medical scheme, whose rules and contribution table need the Council's approval before they change [14].
- **Block exemption.** The draft interim block exemption for tariff determination in the healthcare sector, which would permit collective determination of tariffs and of standardised diagnosis, procedure, device and treatment codes. Published for comment 14 February 2025; **not in force** at 31 October 2026 [68].
- **Booklet 9.** The HPCSA's *Guidelines on the Keeping of Patient Records*, and the only South African instrument from which retention periods can be read. The 2016 text gives at least six years from dormancy; a September 2022 revision says ideally indefinitely. **Which edition is operative could not be established** [2][3].

C

- **Capitation.** Payment of a fixed amount per enrolled person per period, with the provider carrying utilisation risk. South Africa does not operate at scale the five it needs: enrolment, attribution, risk adjustment, payment-decoupled encounter capture and leakage detection.
- **Care plan.** A patient-specific, forward-looking statement of intended care: goals with target dates, planned activities with statuses, responsible parties and a review date. The planned activity and the actual event are different objects, linked.
- **Care setting.** The *kind* of care delivered — emergency, acute inpatient, day case, outpatient, theatre, intensive care, maternity, rehabilitation, community, virtual. A classification of service and not a place.
- **Case mix.** The classification of activity into resource-homogeneous groups so volume can be adjusted for complexity. South Africa has none in routine use, and the private standards body's group subcommittee is dormant [27].
- **CCSA (Complete CPT for South Africa).** The locally adapted procedure code set used in private claiming. The South African Medical Association holds CCSA copyright and the American Medical Association holds CPT copyright; its maintainer's standards say they apply "for Data and not for Billing", and the private standards committee declines to recommend it nationally [28].
- **CDA (Clinical Document Architecture).** HL7's XML standard for an attested clinical document, with a header asserting authorship and legal authentication; the gazetted framework names Release 2.0 [37].
- **CDL (Chronic Disease List).** One of the three components of prescribed minimum benefits. The Council gives 26 conditions [50]; a 2026 scheme guide gives 27 [66], and **the discrepancy was not resolvable from public sources.**
- **CDS (clinical decision support).** Any mechanism that brings knowledge to a decision. The question is never what it does but where the knowledge lives and who is accountable for changing it.
- **CDS Hooks.** The specification by which a clinical system calls external decision-support services at defined workflow points, over HTTPS with JSON payloads; a 2.0 release exists [4].
- **CDW (clinical data warehouse).** An institutional store assembled from source systems for reporting, quality improvement, costing and research [42]. It must not be served from the transactional clinical systems.
- **Certificate of need.** The never-commenced licensing scheme in sections 36 to 40 of the National Health Act, declared invalid and severed on 18 May 2026 in *Solidarity Trade Union and Others v Minister of Health* (CCT 240/24), ZACC 19 of 2026 [59].
- **Claim (funder sense).** A submission to a third-party funder requesting payment of part of an invoice under a benefit arrangement. It is not the invoice, and in the private sector it travels through a switch [26].
- **Claim (medico-legal sense).** A demand for damages against a provider or a provincial department arising from alleged negligent care: an unrelated object with an unrelated lifecycle, counterparty and evidential standard.
- **Claim line.** The granular request: a code, a quantity, an amount and modifiers. The South African line-item qualifier carries more than ten kinds of code, so one code column with a type discriminator carries more than ten meanings [26].
- **Classification.** A code system that sorts every case into exactly one mutually exclusive, exhaustive category so cases can be counted; mono-hierarchical, with residual categories. ICD is the canonical example.
- **Clinical coder.** The role that reads the record, applies the classification's sequencing and tie-break rules, *queries the clinician where the record is ambiguous*, and assigns codes. A registered qualification exists [76]; no statute requires employing one.
- **CMS (Council for Medical Schemes).** The regulator under the Medical Schemes Act 131 of 1998, supervising schemes, administrators, managed care organisations and brokers [14]. It has no power over insurance products such as gap cover.
- **Code list (identifier scheme).** A scheme that names a specific thing unambiguously — a medicine product, a billing provider, an observation.
- **Coded against recorded.** The recorded text is the clinician's assertion; the code is a classification of it, made later, by someone else, usually for a non-clinical purpose. Hold both, linked, with the coder, the date and the coding purpose.
- **COHSASA.** The Council for Health Service Accreditation of Southern Africa, a voluntary body whose published list shows 49 accredited facilities across ten countries, 29 South African, against a public estate of roughly 3,700 establishments [73].
- **COIDA.** The Compensation for Occupational Injuries and Diseases Act 130 of 1993, funding treatment of injured employees on an annually gazetted per-diem-plus-carve-out scale. Minor and major theatre are billed **per minute**, which makes theatre time a revenue field [75].
- **Consent.** At least three entities. *Consent to treatment* concerns an intervention, and withdrawal stops it. *Consent to disclosure* concerns information, recipients and purposes, and withdrawal stops future disclosure but cannot unsend what was sent. *Consent to research* is a third object.
- **Consultant episode.** A subdivision of the spell by clinical responsibility: the period a patient spends in one consultant's continuous care, with no concurrent episodes [6]. The definition is English.
- **Contribution.** What a member pays a medical scheme: a function of benefit option, dependant composition and in some schemes an income band. Not a risk-rated price, and no individual risk score stands behind it.
- **CPT (Current Procedural Terminology).** The American Medical Association's billing nomenclature for professional services, and licensed content — which is why CPT-derived material cannot simply be loaded into a system or shipped in an extract [28].

D

- **DDD (Defined Daily Dose).** The assumed average maintenance dose per day for a medicine used for its main indication in an adult [34]: a statistical unit, not a recommended dose.
- **De-identification.** Transforming data so a subject is no longer identifiable. South Africa has no de-identification standard and no safe-harbour identifier list, and POPIA's test — identification by a reasonably foreseeable method — makes identifiability **contextual** [47].
- **Designated service provider.** The provider a scheme nominates for a benefit, whose use gives full prescribed-minimum-benefit cover without a co-payment. Voluntary use of another attracts a shortfall; emergencies are exempt [66].
- **Deterministic matching.** Declaring an identity match when specified fields agree exactly, in a tiered rule set. Explainable and defensible to a clinician; brittle against transposed digits, married names and approximate dates of birth.
- **DHIS2.** The open-source aggregate health management information platform from the Health Information Systems Programme centre at the University of Oslo, licensed BSD 3-Clause, used by ministries in more than 75 countries and at national scale in 69 as at end-2022 [69]. **It is not an electronic health record.**
- **DICOM.** Maintained by the National Electrical Manufacturers Association, and unusual in being simultaneously a data format, a network protocol, an information model, a conformance framework and a device specification [15]. The gazetted framework cites it for radiology results [37].
- **Diagnostic order.** A clinician's request for an investigation, identified by an order identifier at the placer *and* an accession number at the laboratory: two identifiers owned by two organisations.
- **Diagnostic result.** One or more observations plus an interpretation plus a report. Observations are corrected individually and reports reissued as versions; collapsing the two makes turnaround unmeasurable.
- **Disclosure event.** The record that defined information about a named person was disclosed to a named recipient, for a stated purpose, on a stated date, under a stated authority. Chapter 13 introduces it; it makes POPIA's openness and participation rights operable.
- **Dispense.** A pharmacy event issuing a quantity of a specific product against an order. A dispense is against a *product*, with a lot number and an expiry date — the two attributes every recall depends on.
- **DTP (diagnosis-treatment pair).** The second component of prescribed minimum benefits. The count is **271** [50]; the commonly repeated 270 appears in no primary source retrieved for this volume.

E

- **e-impilo.** The National Department of Health's own bespoke electronic medical record, phase one formally live 28 March 2025 [39], being rolled out in eight provinces with mandatory integration to patient registration, the laboratory system, SyNCH and the Master Health Facility List. **None of its architecture is public** [40].
- **EDIFACT.** The United Nations syntax in which the South African medical scheme claim standard is published [26]. The published standard is EDIFACT; the operative wire format at a switch is frequently the switch's own XML, validated against the switch's schemas [36].
- **Effective dating.** Holding a validity period on a reference-data row so a historical record can be read against the rules in force when it was made. In South African coding this is a requirement: code validity carries start and end dates [23].
- **EHR (electronic health record), strictly.** A longitudinal, cross-provider record of a person assembled from many organisations — no single owner, a consent model, and an identity-resolution problem an organisational record does not have.
- **EML (Essential Medicines List).** The national list of medicines that may be used, published with the Standard Treatment Guidelines; the primary-care volume is at its eighth edition, 2024 [29]. A formulary is not a product catalogue and not a terminology.
- **EMR / EPR.** The clinical record held by *one* provider organisation: notes, observations, orders, results, prescriptions, assessments, care plans.
- **EMRAM.** The HIMSS Electronic Medical Record Adoption Model, eight stages from 0 to 7 [67]. It measures adoption and pervasiveness rather than outcomes, is partly self-assessed, and contains no sourced resilience requirement.
- **Encounter.** The atomic interaction between a patient and the health system in which care is delivered or assessed, with a start, an end, a setting, a responsible practitioner and a reason. Encounters nest and overlap, because real care does.
- **Episode of care.** The association of a patient with a provider organisation over a period, for a problem or a programme, spanning many encounters [7]. No South African provider has a construct spanning providers.
- **External cause code.** The ICD-10 code recording how an injury happened. It can never occupy the primary position, must accompany every injury and poisoning code, and its activity character links to occupational-injury funder eligibility [22].

F

- **FHIR (Fast Healthcare Interoperability Resources).** HL7's current standard: a resource content model, four exchange paradigms, a machine-readable conformance framework and a terminology binding mechanism. Release 4 is 4.0.1 of 27 December 2018, the first with normative content; R5 is 5.0.0 of 26 March 2023; **R6 is not published** [8][9].
- **Folder number.** The facility-scoped patient number every South African facility keeps anyway, and not stable even within a facility: of infants tested twice at one facility in a published study, only 5% carried the same folder number both times [72].
- **Further processing.** POPIA section 15: processing beyond the purpose of collection, which must be shown compatible with it [47]. The second commonest finding in the Information Regulator's enforcement record, and precisely what an unmanaged analytics platform does.

G

H

- **Health care provider.** The National Health Act's term for a *person* who provides health care [1]. Not a place and not an organisation.

- **Health establishment.** The Act's term for the *place*: the regulated unit that holds a licence, owes the records duty under section 13 and may be inspected [1].
- **HIE (health information exchange).** The infrastructure by which records are discovered and retrieved across organisations, in four topologies — centralised, federated with a central index, fully federated, consent-mediated. South Africa's stated architecture is the second [37].
- **HL7 version 2.** The message standard first published in the late 1980s, with versions from 2.1 to 2.9 still in production, and still the backbone of intra-hospital integration. The gazetted framework names 2.3.1, 2.5 and 2.6 [37].
- **HMS² (Health Management System 2).** The Eastern Cape provincial record system [78]. **Its own public material names no developer or vendor**, which is a finding rather than an omission.
- **HNSF (Health Normative Standards Framework).** The gazetted South African interoperability framework: the 2021 HNSF, published 21 October 2022 in *Government Gazette* 47337 under section 74(1) of the National Health Act, built mainly on IHE profiles and naming HL7 version 2, CDA R2, FHIR R4 and R3, and DICOM. **It names neither SNOMED CT nor LOINC** and sets no deadline [37].
- **HPCSA (Health Professions Council of South Africa).** The council under the Health Professions Act 56 of 1974, with twelve professional boards. It distinguishes *scope of profession*, under section 33, from *scope of practice*, the boundary of an individual's competence under ethical rule 21.
- **HPRN.** The ten-digit Health Patient Registration Number allocated by the national registration system. **Its check-digit or validation structure is not published** so far as this research could establish.
- **HPRS (Health Patient Registration System).** The national patient index: demographics only, keyed primarily on the South African identity number, installed at 3,266 establishments at 31 March 2025 — 3,150 primary care facilities, 113 hospitals and three general practitioner practices [39]. It sits beneath the clinical systems.

I

- **ICD-10.** The World Health Organization's statistical classification of mortality and morbidity, and the operative South African classification. The coding standards record adoption in 1996 and implementation in July 2005 [21]; other accounts give 1 July 2005. The decisive date is **1 July 2014**, when age and gender validation began rejecting claims.
- **ICD-10 Master Industry Table.** The South African national implementation artefact: the codes used here with age and gender indicators, morphology codes, and business-rule flags for clinical use, primary position, asterisk, dagger and sequelae validity, each with South African effective dates. **The January 2014 edition is the latest located** [23].
- **ICD-11.** The successor classification, adopted 2019 and effective from 1 January 2022 for member-state reporting [30], and far more terminology-like: a foundation with derived linearisations, post-coordination with extension codes, API-first distribution. South Africa's collaborating centre states ICD-10 remains current and **no transition plan was found** [25].
- **ICD-O.** The oncology classification, coding tumour topography *and* morphology; in South Africa third-edition morphology codes travel inside the Master Industry Table [23].
- **Ideal Clinic.** The Department's own primary-care self-assessment programme, awarding Silver, Gold and Platinum status in parallel with the regulator's inspections. It certified 33,3% of facilities in 2019 while the regulator found 14,7% meeting the minimum standard in 2019/20 [63].
- **IHE (Integrating the Healthcare Enterprise).** Not a standards development organisation: it writes *integration profiles* selecting existing standards, defining actors and transactions, and specifying declarable options [12]. The same problem is re-profiled as technology generations change, which is why there appear to be four standards for one thing.
- **Information Regulator.** The regulator for POPIA and PAIA. Administrative fines run to a statutory maximum of R10 million, and health is a named priority sector in its 2026–2027 plan [81].
- **Integration engine.** The message broker, transformer and router every hospital has. It exists not because healthcare lacks standards but because of version heterogeneity, dialect translation and message-level operational visibility.
- **Invoice (account).** The provider's assertion of amounts due for services rendered, directed at whoever is liable. One stay can produce several invoices; one invoice can span several encounters.
- **Insurer.** A licensed insurance undertaking regulated under insurance law, selling products such as gap cover and hospital cash plans. Not a medical scheme — different statutes, different regulators.

L

- **LIS (laboratory information system).** The system that owns the **specimen**, not the test: accession and identity, the worklist, instrument interfacing, autoverification, reference ranges, quality control, authorisation and sign-out, the report and the archive.
- **LOINC.** The Regenstrief Institute's code system for naming observations — what was measured, not the value. Each term is defined by a fixed set of axes, so two laboratories' "sodium" may be different terms. More than 109,000 concepts, version 2.82 announced 23 February 2026, **royalty-free** [31] [32].
- **Location.** The physical place, hierarchically from site to building to department to room to bed, with capacities, physical characteristics such as negative pressure or piped oxygen, and an operational status. A ward occupies a location and is not one.

M

- **Main condition.** The condition diagnosed *at the end of the episode of care* that is primarily responsible for the need for treatment, with a tie-break on severity, then resource consumption, then order of recording [21]. Primary is a property of a diagnosis within an episode, not of the diagnosis.
- **Managed care organisation.** A separately accredited entity performing protocols, pre-authorisation, case management, provider networks, formularies and utilisation review. Such arrangements covered 99,19% of beneficiaries on 2024 reporting [77].
- **Master Household Index.** A FHIR-based index for linking patient data across systems, documented in the Department's own 2025 publication [78]. **Its scope and operational status are unverified.**

- **Measure Definition.** The versioned, effective-dated, owned rule by which a derived figure was computed: numerator, denominator, exclusions, boundary rule and period. An entity, not an audit field: the four theatre utilisation methods have no mathematical relationship to each other [71].
- **MEDCLM.** The South African medical scheme claim message: version 0, release 912, agency ZA, revision 13.4, November 2016 [26]. It defines **no remittance message and no link from a claim response to a payment**, and still carries SNOMED II and III fields for pathology.
- **Medical scheme.** A not-for-profit mutual entity registered under the Medical Schemes Act 131 of 1998, owned by its members, which holds the money and carries the risk [14]. **Not an insurer**: community rating within an option, no health-status underwriting, open enrolment, statutory reserves.
- **Medication order (prescription).** A prescriber's instruction that a patient receive a medicine, with a dose, route, frequency, duration and indication. The transition that breaks models is *stopped*.
- **Medication reconciliation.** A point-in-time assertion of what the patient is actually taking, compiled from several sources. Not a view over the orders: an independent clinical assertion whose valuable output is the set of **discrepancies**.
- **MediKredit.** A South African claims switch which **owns and maintains NAPPI** [35] and publishes the schemas vendors validate against [36]. The national private medicine code is therefore a licensed dependency under every medicines service in the sector.
- **MediSwitch.** A South African claims switch, described by a reseller as operating under the Altron HealthTech group. **The acquisition date and terms could not be established.**
- **Member and dependant.** The funder's subject, in four levels: scheme, benefit option, member, dependant. Not the patient — a model using the membership number as a patient identifier loses the patient at every policy change [26].
- **MHD (Mobile access to Health Documents).** The FHIR-based IHE equivalent of cross-enterprise document sharing, for lightweight clients, named in the gazetted framework [37].
- **mCSD (Mobile Care Services Discovery).** The FHIR-based IHE directory profile for facilities, organisations, practitioners and services, named in the gazetted framework [37], and the slot a facility registry fills.
- **MPI (master patient index).** The service that resolves which records refer to the same person. A duplicate means a clinician sees half a record; a **false merge** means a clinician sees another person's clinical data. Keep a link table with a certainty level, not a merged record [42].

N

- **NAPPI (National Pharmaceutical Product Index).** The de facto South African product identifier for medicines and surgical consumables, identifying a **pack** rather than an ingredient or a dose [35]. It serves claiming and dispensing, not prescribing, interaction checking or exposure analysis.
- **National Core Standards.** The February 2011 quality instrument with seven domains, used until 2019 [80]. It did not cleanly succeed into one instrument; three now run at once and they disagree.
- **NHI Act.** The National Health Insurance Act 20 of 2023, signed 15 May 2024. **No section is in force**, by presidential undertaking of 20 February 2026 made an order of court on 24 February 2026, and there is no published data dictionary [47].
- **NHLS (National Health Laboratory Service).** The dominant public-sector pathology provider and host of the communicable diseases institute — in effect a *national* laboratory information service, which is an asset most countries lack and a concentration risk most do not carry [70].
- **NHRPL (National Health Reference Price List).** The reference schedule funders and providers negotiated around until its enabling regulations were set aside on 28 July 2010 [51].
- **NICD.** The National Institute for Communicable Diseases, hosted by the laboratory service, which publishes case definitions and receives notifiable condition notifications [56].
- **NMC (notifiable medical condition).** A condition reportable under the 2017 regulations made under the National Health Act, in four categories with clocks from 24 hours to a month [55]. The 2017 gazette lists 22, 21, 7 and 1 entries; the 2025 brochure gives 27 and 18, and **no amending notice accounting for the difference could be found** [56].
- **Norms and standards regulations.** The instrument the health standards regulator actually inspects against: promulgated 2018, effective 2019, seven chapters and 22 regulations [58].
- **Nursing register categories.** The Nursing Council maintains registers for registered nurses and midwives, enrolled nurses and auxiliary nurses, and is explicit that its statistics count **people registered**, including job seekers and non-practisers [60] — so every ratio describes a register, not a workforce.

O

- **OCS (order communications system).** The capability carrying a clinician's request to a diagnostic or treatment service and the result back. The state that is never closed is *result acknowledged*.
- **OHSC (Office of Health Standards Compliance).** The establishment quality regulator, inspecting against the norms and standards regulations and issuing certificates, compliance notices and enforcement [58]. The regime is incomplete rather than new: as recently as 2022 there were no finalised inspection tools for private acute hospitals or general practitioner practices.
- **openEHR.** A specification set for clinical data persistence on two-level modelling: a stable reference model, with clinical content as archetypes and templates governed in a public knowledge manager [44]. A 2026 survey found **no country naming it in binding national regulation** [45].
- **OpenHIE.** The reference architecture most relevant to South Africa, separating registries from repositories from the exchange layer [43]. Its reference interoperability layer, OpenHIM, is South African-built.
- **Operator.** Under POPIA sections 20 and 21, anyone processing personal information on a responsible party's behalf under a **written contract** obliging equivalent safeguards [47]. The healthcare list is long and usually unexamined, from the laboratory and the switch to the cloud host.
- **OMOP CDM.** A fixed, person-centric, event-based analytic schema with mandatory mapping onto a standardised vocabulary [46]. Its value is realised in a *network*.
- **Organisation.** The legal entity: holds licences, is party to contracts, holds the practice code, is the responsible party under POPIA. Not a site, a hospital or a department.

P

- **PACS (picture archiving and communication system).** The system owning the images and the reading workflow: storage, retrieval, display, hanging protocols and comparison with priors.
- **PAIA.** The Promotion of Access to Information Act 2 of 2000. The records duty in section 13 of the National Health Act is expressly subject to it and to the National Archives Act, which is where destruction authority sits [1].
- **PAS (patient administration system).** The system of record for the administrative identity of the patient and for the encounter. It owns the institutional patient index, the bed state, the waiting list and the clinic diary, and **no clinical content**.
- **Pathway.** The instantiation for one patient of a standardised route through care. Deviation with a documented reason is the clinically important fact, not an exception to suppress.
- **Patient.** A *role* a person occupies in relation to a provider organisation, as distinct from the **Person**, who is the human being. "Deceased" is a state of the Person.
- **Payable-unit nomenclature.** A code set naming a payable unit of work, which is not the same object as a clinical act: one procedure generates several payable lines and one line can cover several procedures.
- **PDQ and PDQm.** The IHE patient demographics query profiles, in SOAP and FHIR generations, which locate a patient and their identity from demographics [37].
- **Person.** The human being, as distinct from the **Patient**, which is a role occupied in relation to a provider organisation. A deceased person is no longer a data subject under POPIA while their record remains legally regulated — unresolved in the South African literature rather than settled [47].
- **PHDC (Provincial Health Data Centre).** The Western Cape's clinical data platform and the best worked South African reference design: link rather than merge, store a certainty level, blacklist placeholder identifiers, perturb the join key per release, harmonise to ICD-10, ATC and LOINC [42].
- **PHISC (Private Healthcare Information Standards Committee).** The private-sector standards body, with subcommittees for clinical coding, information governance, message standards, pharmaceutical coding, the electronic health record, and devices and diagnostics, and a dormant diagnosis-related group subcommittee [27].
- **PIX and PIXm.** The IHE patient identifier cross-referencing profiles, which assert that two local identifiers refer to the same person. They **standardise the transactions, not the matching** [12].
- **PMB (prescribed minimum benefits).** The benefits every registered scheme must fund in full on every option: any emergency medical condition, 271 diagnosis-treatment pairs, and the Chronic Disease List [50]. A determination is a rules evaluation over codes, networks, protocols, option and accumulator state; a code alone does not confirm cover.
- **POPIA.** The Protection of Personal Information Act 4 of 2013. Health information is **special personal information**, so the default is prohibition under section 26 unless an authorisation applies [49]; section 32(1)(a) authorises processing for the proper treatment and care of the data subject [47].
- **Post-coordination.** Assembling a meaning from several terminology concepts using a defined grammar. It creates equivalence testing, querying by hierarchy rather than value list, and governance of what expressions are permitted — and a field sized for one code cannot hold an expression.
- **Practice code number.** The identifier that routes the money. It names a **billing entity** — an individual, a group practice, a hospital or a laboratory — and not a clinician. **The governing standardisation manual could not be retrieved.**
- **Practitioner.** A person authorised to deliver health care in a defined scope. There is no single plausible owner of the entity — human resources, credentialling, the clinical directorate and billing each own part — so the deliverable is the governance decision.
- **Practitioner role.** The practitioner at one organisation, in one specialty, at one location, for a period, with a set of privileges. One clinical event can carry up to six roles that may all be the same person.
- **Present on admission.** The flag stating whether a condition existed at admission or arose during the stay. **Neither the South African public morbidity coding standards nor the private claim message mandates it** [21][26], so hospital-acquired condition rates are not computable here.
- **Probabilistic matching.** Scoring field-level agreement by how informative each field is, with an auto-link band, a no-match band and, between them, a **human review queue**.
- **Protocol.** An organisation-adopted, evidence-based specification of how a defined clinical situation should be managed, identified by an identifier **plus a version**, because "was this patient managed according to protocol?" is meaningless without the version in force.

R

- **Record (the legal record).** What the provider must retain, can be compelled to produce, and must be able to reconstruct as it stood at a point in time. Not identical to any one system's contents and not identical to a screen [1][2].
- **Reference set (SNOMED CT).** SNOMED CT's mechanism for everything subset-like — a value set, a language preference, an ordered list, a map [17]. Nobody deploys all of SNOMED CT.
- **Referral.** A request from one practitioner or service to another to assume or share care. Its auditable states are **sent** and **acknowledged**: a referral sent and never acknowledged is a patient lost to follow-up.
- **Registration (of a practitioner).** The authorisation itself, with its own lifecycle: register, category, number, first registration date, status, conditions, endorsements. Status is live — a lapsed practitioner's orders should not be actionable [61].
- **Regulation 5(f).** The provision made under the Medical Schemes Act 131 of 1998 requiring a diagnostic code on a claim [24]. A **funder** regulation and conditional on payment: an uncoded claim is unpayable, not unlawful. The coding standards cite separate regulations in *Government Gazette* 20556 of October 1999 [21].
- **Remittance (payment advice).** The funder's statement of what it has paid, declined or adjusted, line by line, with reason codes. The entity organisations most often fail to model, and the national claim standard is the reason [26].
- **Responsible party.** Under POPIA, the party determining the purpose and means of processing. In a scheme arrangement the **scheme** is the responsible party and the administrator an operator, so an administrator change is a data migration with a regulatory deadline [47].
- **RIS (radiology information system).** The system owning the radiology workflow: request, vetting and protocolling, scheduling, the examination record, the dose record and the report.

- **Road Accident Fund.** A statutory funder of care for road accident injury; the public fee schedule classifies its patients as externally funded full-paying patients [52]. **Its tariff instrument and claim volumes could not be retrieved.**

S

- **SaMD (software as a medical device).** Software intended for a medical purpose, and a regulated product in its own right. **SAHPRA has no dedicated software classification rule**, classifying it by principles as an active device, and product registration is not yet operating [62].
- **SAHPRA.** The South African Health Products Regulatory Authority, regulating medicines, devices and diagnostics, clinical trials, radiation control and pharmacovigilance [53]. Its 26 September 2025 position on artificial-intelligence-enabled devices locks the algorithm version at licensing and requires local-population validation where training data came from elsewhere [54].
- **SANC (South African Nursing Council).** The council under the Nursing Act 33 of 2005. Registration is a legal precondition of employment in nursing, and its online register is the most usable practitioner verification source in the country [61].
- **Schedules (of medicines).** The classification determining who may supply a medicine. **The scheduling detail could not be sourced.** One consequence is decisive: the professional rule in the Council's record-keeping guidance requires handwriting and a personal original signature for Schedules 5 to 8, while Schedules 1 to 4 may be typed [2].
- **Security compromise.** POPIA's term for a breach, notifiable to the Regulator and to affected data subjects "as soon as reasonably possible" after **discovery**. There is **no 72-hour deadline**, and since April 2025 notifications go through the Regulator's online portal [48].
- **SEP (single exit price).** The gazetted maximum private-sector medicine price: an item-level, effective-dated list that changes for thousands of items on one date, with a published registry [74]. Any system holding medicine prices must be built for a bulk, dated price event.
- **Separation.** The event that ends a stay — a discharge, a death or a transfer out — and the denominator of average length of stay.
- **Severity Assessment Code.** The four-level severity scale in the South African patient safety instrument, which drives how deeply an incident is investigated [79]. **"Sentinel event" does not appear in the national guideline at all.**
- **SINJANI.** A KwaZulu-Natal provincial health reporting system. **Its data model, its relationship to the national aggregate platform, its vendor and its vintage are not publicly documented**, and neither the national strategy nor the gazetted framework mentions it [38].
- **SMART on FHIR.** The authorisation and launch layer letting a third-party application run inside a clinical system against its FHIR API, with a token carrying launch context [10].
- **SNOMED CT.** The clinical terminology: concepts with permanent identifiers, a poly-hierarchy that makes hierarchy queries possible, logical definitions permitting post-coordination, and national content published as namespaced extensions. International editions are monthly, the October 2026 edition carrying release date 1 October 2026, and concepts are inactivated rather than deleted, so store the identifier **and the release version** [17]. Use is free in member countries, and elsewhere an annual affiliate licence applies [18]. **South Africa joined on or about 14 May 2026 and the National Release Centre is not yet operating** [19][20].
- **Special personal information.** POPIA's category covering, among other things, health or sex life. Processing is prohibited by section 26 unless authorised; the section 32 authorisations are the health-specific ones [49].
- **Specimen.** The sampled material, with a type, a collection date-time, a collector, a site, a condition on receipt and a chain of custody. An entity, because one order may require several specimens and one specimen may serve several orders.
- **Spell.** The continuous stay: one patient's unbroken period as an inpatient of one provider, starting when a responsible clinician assumes responsibility after a decision to admit — which can be before formal admission completes — and ending at the provider discharge [5].
- **SVS (Stock Visibility System).** The national public-sector stock reporting system: parallel logistics that never touches the clinical record. The acronym collides with IHE's Sharing Value Sets profile, named in the gazetted framework [37].
- **Switch.** The intermediary that validates and routes claims between providers and funders. An operator under POPIA, a single point of dependency for revenue, and **where format conformance is actually enforced** — so the switch's rules are the operative schema [36].
- **SyNCH.** The national chronic medicine dispensing system, and one of the four mandatory integrations named for the national primary-care record [40].

T

- **Tariff.** Contract data: the combination of funder, provider, service code and effective date. Not an attribute of a service — a single price column on a service master is a defect.
- **Terminology.** A code system that lets a clinician record what was observed or done as precisely as they can express it, with a defined and computable meaning: poly-hierarchical, formally defined, built for capture and inference.
- **Terminology service.** The service holding code systems, versions, value sets and maps, and answering expansion, validation and hierarchy questions. Infrastructure, not tables in an application database.
- **TIER.Net.** The national electronic HIV and tuberculosis management register, reporting aggregates upward [41]. Its design idea is a pattern worth naming: one set of data definitions, with paper, electronic and networked implementation tiers against them.
- **TrakCare.** The InterSystems application that is the National Health Laboratory Service's system of record, and the system rendered unusable in the June 2024 ransomware incident [70].
- **Triage.** Assigning a clinical priority at unscheduled presentation. The South African Triage Scale gives five categories from a composite physiological score, with discriminator lists that escalate a patient regardless of score, and **mandatory continuous re-triage**, so a triage record is a *series* [64]. Its thresholds are the only sourced numeric service-level targets for any South African clinical process [65].

U

- **UCUM.** The Unified Code for Units of Measure. A system that stores a number and free-texts or discards the unit is the commonest route to a clinically significant data error in a consolidated record.

- **UID (DICOM unique identifier).** The globally unique identifier carried at study, series and instance level. Regenerating them creates duplicates and preserving them creates collisions if the same data arrives by two routes [15].
- **Unmerge.** Reversing an identity link, and the test of whether a master patient index design is serious: if the product cannot unmerge, the architecture must not merge.
- **UPFS (Uniform Patient Fee Schedule).** The public-sector fee schedule, with patient classification by means test. A subsidised outpatient attending several clinics in one day is assessed **once, at the most expensive clinic**, while full-paying patients are assessed per visit per clinic [52].
- **User.** The National Health Act's word for the patient: the subject of its consent, records, confidentiality and complaints provisions [1]. Never use it for a system user.

V

- **Visit.** An administrative construct: one attendance at one place on one day. The unit of registration, which in outpatient settings frequently coincides with an encounter, which is why it is mistaken for one.
- **VNA (vendor-neutral archive).** Long-term image storage held independently of the viewing system, in standard DICOM, so the archive outlives the picture archiving system. It typically performs *tag morphing*, a data-altering function on a regulated clinical artefact.

W

- **Ward.** An organisational unit with a nursing establishment, a bed complement and a specialty, which occupies a location but is not one, and is reconfigured without moving.
- **Ward stay (bed occupancy period).** The location-time fact: this patient, in this bed, in this ward, from this time to that time. The only construct from which occupancy can be computed.

X

- **XCA (Cross-Community Access).** The IHE federation primitive: how one exchange connects to others and forwards requests onward, named in the gazetted framework [37].
- **XDS.b (Cross-Enterprise Document Sharing).** The foundational IHE document-sharing profile, whose central idea is the **separation of registry from repository**: one index can point at many repositories [13]. Documents are deprecated, never deleted.
- **XUA (Cross-Enterprise User Assertion).** The IHE profile conveying information about the user behind a SOAP transaction, using SAML, to support access control, consent and audit [12].

Z

- **Z-segment.** A locally defined HL7 version 2 segment, outside the standard [11]. Distinguish genuinely local data, which belongs in the interface specification, from data the standard *does* cover placed locally because the vendor's model did not match.

THE CHAPTER IN TEN LINES

- Chapter 4's sixty-two definitions are carried here unchanged in substance; this appendix adds the rest.
- Where a term has two legitimate meanings, both appear and the chapter that settles the volume's usage is named.
- Claim is never bare: write funder claim or medico-legal claim.
- Admission is an event, never a stay; spell and consultant episode are English definitions adopted for want of South African ones.
- Bed has at least five senses, and bed complement is a dated time series rather than a number.
- Contribution belongs to a medical scheme; premium belongs to a named insurance product.
- A dozen entries say the research could not establish an edition, a version or a status, and name what would settle it.
- The Master Industry Table, Booklet 9 and the morbidity coding standards all have unresolved current editions.
- The gazetted interoperability framework names neither SNOMED CT nor LOINC, and sets no compliance deadline.
- South Africa joined SNOMED International on or about 14 May 2026 and the National Release Centre is not yet operating.
- No glossary entry reproduces a code, a value set, a message layout or a resource definition.
- Appendix 25 carries the access, licensing and retrieval position for every standard and body named here.

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APPENDIX C

Standards, bodies and where to find them

Sixty-nine primary sources, the access position for each, and eleven things nobody could retrieve.

This appendix exists so a reader can go and check. The prose is therefore short and the retrieval position is long: for each standard, profile, code system, regulator, council, industry body and public entity, what it is, what it governs, what access costs, whether South Africa is a member or has adopted it, and where the primary source actually lives.

Three categories, and conflating them is the commonest error in an architecture document. **Syntax and transport standards** say how a message or resource is structured and moved: HL7 version 2, HL7 CDA, FHIR, DICOM. **Semantic standards** say what the content means: SNOMED CT, LOINC, ICD, ATC. **Profiles and implementation guides** are constrained, use-case-specific selections from the first two, with named actors and transactions: the IHE profiles, FHIR implementation guides, national specifications [1]. The third category is what actually makes systems interoperate and the one procurement documents omit. "We are HL7 compliant" is a claim about the first category and carries almost no information.

The layer, and what it governs	The South African position
1 · Syntax and transport How a message or resource is structured and moved. HL7 version 2, HL7 CDA, FHIR, DICOM.	**Gazetted** The framework names HL7 version 2.3.1, 2.5 and 2.6, CDA Release 2.0, FHIR R4 and R3, and DICOM.
2 · Semantics What the content means. SNOMED CT, LOINC, ICD-10, ICD-O, ATC.	**Not gazetted** The framework names neither SNOMED CT nor LOINC. ICD-10 reaches practice only through the national Master Industry Table.
3 · Profile or implementation guide A constrained selection, with named actors and transactions. IHE profiles, FHIR implementation guides.	**Gazetted in outline** The framework is IHE-based, and its National Interoperability Specifications are prescriptive but not publicly enumerated.
4 · The local interface specification Field by field, trigger by trigger, with local segments, acknowledgement semantics and error handling.	**Nobody else writes this** It is the actual contract. Every seam has one, whether or not anyone has written it down.

Figure 25.1 Four layers govern every seam, and the local interface specification is the only one South Africa has not partly specified. Read it with the gazetted framework open: layer 1 is named, layer 2 is absent, layer 3 is prescriptive but not enumerated, and layer 4 is the architect's own deliverable.

South Africa has institutionalised that distinction better than most countries and then not supplied the content. The gazetted framework states that implementation guides, which it calls National Interoperability Specifications, are prescriptive, are assessed for conformance to the framework, and are tested at a laboratory approved by the National Department of Health, with an attestation issued [2]. **No list of approved laboratories and no inventory of published specifications could be retrieved**, and whether any attestation has ever been issued could not be established. The regime is still the most useful procurement lever in the country, because it converts an interoperability claim into a testable artefact — but ask for the attestation, the version tested against, the date, the laboratory and the profiles covered, and expect the answer to be revealing.

One convention runs through what follows. Where a source is paywalled, licensed, member-only, blocked to automated retrieval or simply not published, this appendix says so. An honest "not retrievable" is worth more than a confident summary, and the research behind this volume recorded 217 such items across four files. Several of them are standards an architect would assume were open.

The syntax and transport standards

HL7 version 2. The message standard that carries admission, discharge, transfer, order, result, scheduling, charge and document events inside a hospital, in versions from 2.1 to 2.9, all still in production somewhere. *Access:* **the research behind this volume did not establish HL7 International's current licensing terms** for the version 2 specifications; obtain them from hl7.org before assuming the text may be redistributed. *South Africa:* the gazetted framework names versions 2.3.1, 2.5 and 2.6, which is a realistic acknowledgement of the installed base and a requirement that the integration layer version-bridge [2]. *Where it lives:* hl7.org. Practitioner explanations of the locally defined Z-segment mechanism are open [3].

HL7 CDA. The XML standard for an attested clinical document, derived from the version 3 reference information model, with a header asserting authorship and legal authentication. *Access:* **as for version 2, the licensing terms were not established here.** *South Africa:* the framework names CDA Release 2.0, and lists the medical summary, laboratory report and scanned-document content profiles among its backward-compatibility items [2]. *Where it lives:* hl7.org.

FHIR. The current HL7 standard — a resource content model, four exchange paradigms, a machine-readable conformance framework and a terminology binding mechanism. *Access:* **free and openly published**, including every resource definition, profile and implementation guide, which makes it the one

major clinical standard with no commercial barrier. *Versions:* Release 4 is 4.0.1 of 27 December 2018, the first with normative content; R4B is 4.3.0, and **HL7's own history page is internally inconsistent about its date**; R5 is 5.0.0 of 26 March 2023; **R6 is not published** and its publication date is not fixed, with normative ballots run and further ones tentative [4][5]. *South Africa:* the framework cites Release 4 and Release 3 for specific profiles and intentionally omits a general version reference [2]. *Where it lives:* hl7.org/fhir, and the version directory at hl7.org/fhir/history.html [4].

SMART App Launch, Bulk Data and CDS Hooks. The authorisation and launch layer, the population-scale export API, and the pattern by which a clinical system calls external decision support. *Access:* all free and openly published. *Versions that matter:* United States regulation names FHIR 4.0.1, US Core STU 3.1.1 and 6.1.0, SMART App Launch 1.0.0 and 2.0.0, and Bulk Data v1.0.0 STU 1 — which is why R4 is where the tooling is [6]. *Where it lives:* docs.smarthealthit.org and cds-hooks.hl7.org [7][8].

DICOM. Simultaneously a data format, a network protocol, an information model, a conformance framework and a device specification, published in numbered parts by the National Electrical Manufacturers Association. *Access:* the current edition is **published in full, as HTML and PDF**, and was retrieved directly for this volume. *South Africa:* the framework cites the security and media-storage parts and DICOM 2011 for radiology results [2]. *Where it lives:* dicom.nema.org, with Part 4 at dicom.nema.org/medical/dicom/current/output/html/part04.html and Part 15 at the same path [9][10]. The artefact to demand from a supplier is the **conformance statement**, which no other healthcare standard provides an equivalent of.

The profiles

IHE. Not a standards development organisation. It writes integration profiles that select existing standards, define actors and transactions, and specify declarable options, then tests implementations against each other at connectathons [1]. *Access:* **free**. The IT Infrastructure Technical Framework is published in full. *Where it lives:* profiles.ihe.net, with the document-sharing chapter at profiles.ihe.net/ITI/TF/Volume1/ch-10.html, the revision 17 volume 1 PDF at ihe.net, and the document-to-element bridge profile at profiles.ihe.net/ITI/mXDE/volume-1.html [11][12][13].

The profiles South Africa has gazetted. The framework is, structurally, an IHE framework. Its table of standards lists the audit and node authentication, cross-community access, cross-community patient discovery, document exchange by media and by reliable messaging, cross-enterprise document sharing, the medical summary, laboratory report and scanned-document content profiles, cross-enterprise user assertion, mobile care services discovery, mobile access to health documents, the patient identifier cross-referencing and demographics query families in all three generations, patient administration management, sharing value sets, document subscription and a patient master identity registry profile, among others — and notes that some are still in trial implementation and may change [2]. *Consequence:* a supplier arriving with an excellent FHIR API and no IHE profile support is not aligned to the gazetted framework.

The code systems

SNOMED CT. The clinical terminology, maintained by SNOMED International, a member-owned body. *Release cadence:* monthly international editions; the October 2026 edition carries release date 1 October 2026, and the release notes are the artefact to read rather than marketing material, because they record inactivations and hierarchy remodelling [14]. *Access and licensing:* **no charge for use in member countries or territories**, with deployments registered through the national release centre or the member licensing and distribution service; in non-member territories a licence is applied for annually with a statement of usage, charges based on amount of use and World Bank classification, fee exemptions by application, and **separate permission and fees for web applications and browsers** [15]. *South Africa:* **a member since on or about 14 May 2026**, attributed to the National Department of Health and stated to extend access across public and private providers and academic, research and civil-society institutions [16][17]. **The National Release Centre is not operating** — no national extension, no reference sets, no published map, representatives listed as to be advised [16]. **The cost of access under South African membership is not stated anywhere retrievable**; the licensing page is generic [15]. *Where it lives:* snomed.org, with the member page at snomed.org/members/south-africa and the release-centre guide on the SNOMED International documentation site [16][18].

ICD-10 and the Master Industry Table. The classification is the World Health Organization's; the operative South African artefact is the national Master Industry Table, maintained by the Ministerial ICD-10 Task Team and published as spreadsheet and comma-delimited files, carrying the codes used here with age and gender indicators, morphology codes, business-rule flags and South African effective dates [19]. *Access:* the Master Industry Table and its user guide are **published free** on the Department's site; the three WHO ICD-10 volumes are a separate matter, and the guide is explicit that the table replaces neither them nor ICD-10 training [19]. *The edition problem:* the guide describes a biennial 1 July cycle and references a table published on 1 January 2014 for implementation from 1 July 2014. **No later edition could be located on the Department's site**, and the position should be confirmed with the Department before any edition is quoted as current. *Where it lives:* health.gov.za — the user guide at health.gov.za/wp-content/uploads/2021/02/cd-10mit_userguide_january2014.pdf [19].

The South African coding rule set. Five documents, three maintainers, no single machine-readable source: the morbidity coding standards, **version 6 of April 2014, which states that it is revised annually and for which no later version was found** [20]; the Technical User Guide, version 2.00 of June 2014, which carries the code limits, serialisation rules and the external-cause constraint [21]; the Master Industry Table with its effective dates [19]; circular No 1 of 2012, which cites Regulation 5(f) [22]; and the private sector's addendum, **at version 7 of October 2024, which is current while the national standards may not be** [23]. *Access:* all free. *Where they live:* health.gov.za for the first four, phisc.net for the addendum.

ICD-11. Adopted by the World Health Assembly in 2019 and in effect from 1 January 2022 for member-state reporting [24]. *Access:* **distributed through an API rather than primarily as a file**, which is itself a change an architecture must absorb. *South Africa:* the WHO Family of International Classifications collaborating centre states that ICD-10 is the current South African standard and that ICD-11 is in its work plan, without saying what the country has done [25]. **No dated South African transition plan exists in any published sense.** *Where it lives:* who.int/standards/classifications and whofic.org.za/icd [24][25].

LOINC. Names observations — the question an observation answers, across a fixed set of defining axes. Maintained by the Regenstrief Institute, updated twice yearly in February and August, more than 109,000 concepts, version 2.82 announced 23 February 2026 [26]. *Access:* **royalty-free**, which makes a laboratory catalogue mapped to LOINC the lowest-friction semantic win available to any provider. *South Africa:* **no national mandate, no national subset and no national mapping could be found**, and the gazetted framework does not mention LOINC at all [2]. The Western Cape data centre harmonises laboratory tests to LOINC "where possible" [27]. *Where it lives:* loinc.org, with the starting point at loinc.org/start [28].

ATC and the Defined Daily Dose. A five-level classification of active substances by the system they act on, maintained by the World Health Organization collaborating centre in Oslo, with the defined daily dose as its statistical companion unit [29][30]. *Access:* **published openly** on who.int; terms for the printed index and associated tooling were not established here. *What it is not:* a prescribing dictionary. It identifies no product, strength, form or pack. *Where it lives:* who.int/tools/atc-ddd-toolkit.

NAPPI. The de facto South African product identifier for medicines and surgical consumables, identifying a pack. *Access and ownership:* **owned and maintained by MediKredit, a claims switch**, and distributed under commercial terms — a licensed dependency sitting under every private-sector medicines service, formulary and interaction check [31]. *The architectural gap:* South Africa has a pack-level product identifier, a regulated price register, a public-sector procurement catalogue, a national formulary and a regulator's registration register, and **no national ingredient-level medicines dictionary with virtual-product and dose-form structure**. *Where it lives:* medikredit.co.za/products-and-services/nappi.

CCSA and CPT. The procedure code set used in South African private claiming is CCSA, derived from the American Medical Association's Current Procedural Terminology. *Access and ownership:* **CCSA copyright is held by the South African Medical Association and CPT copyright by the American Medical Association**; any reprinting of CPT content in a product or publication requires an AMA licence, and training that uses CCSA is subject to SAMA terms [32]. *Two statements to read twice:* the CCSA standards say they apply "for Data and not for Billing", and **the private standards committee expressly does not recommend CCSA as a national procedural coding standard** [32]. Two bodies, one of them a copyright holder. *Public sector:* the Uniform Patient Fee Schedule, published provincially and free [33]. *Where the standards live:* phisc.net; the code set itself is licensed content and is not published there.

The smaller code systems. ICD-O codes tumour topography and morphology and is the cancer registry standard; in South Africa its third-edition morphology codes travel inside the Master Industry Table [19]. **ICF** classifies functioning rather than disease. **ICHI** is the World Health Organization's free, natively post-coordinated intervention classification, explored in South African surgical research with **no indication of national adoption** [34]. **UCUM** is the units code system and the quiet essential. **ICPC-2** appears as a field in the South African claim message [35]. *Access:* all free except the ICD volumes. *Where they live:* who.int for the WHO family; unitsofmeasure.org for units.

The record and analytic specifications

openEHR. A specification set for clinical data persistence on two-level modelling, with clinical content governed as archetypes and templates in a public, reviewed, versioned knowledge manager — the most transferable institutional idea in it. *Access:* **specifications and archetypes are free and open**; the scarce resource is modelling skill, not licence. *Adoption, honestly:* the first serious count, a 2026 vendor and community survey, found deployments reported in 28 countries across both surveys, with heavily caveated self-reported figures — and **no country naming openEHR in binding national regulation**, a pattern the authors call structurally fragile, with knowledge scarcity the most consistent barrier [36]. *South Africa:* **not in the survey's country list, and no South African implementer market could be evidenced.** *Where it lives:* openehr.org and ckm.openehr.org [37].

OMOP and i2b2. OMOP is an analytic common data model with mandatory mapping onto a standardised vocabulary, which is why OMOP analyses are comparable across sites and FHIR extracts are not; i2b2 is the older cohort-discovery platform. *Access:* **both free and open source.** *South Africa:* **no South African OMOP instance could be established**, and OMOP's value is realised in a network. *Where it lives:* ohdsi.github.io/CommonDataModel and github.com/OHDSI [38][39].

OpenHIE. The reference architecture closest to the South African and wider African problem, with a component per concern — client, facility, health worker, terminology and product registries, a shared health record, an aggregate reporting service, logistics, and finance and insurance — composed over IHE profiles and FHIR rather than competing with them. *Access:* **free.** *South Africa:* its reference interoperability layer, OpenHIM, is South African-built and in Western Cape production. *Where it lives:* guides.ohie.org/arch-spec [40].

The device and software standards

IEC 80001-1:2021. Risk management for IT networks incorporating medical devices, second edition, September 2021, and addressed to the **healthcare delivery organisation** rather than the manufacturer. It is the standard that lets an architect argue for network segmentation on clinical safety grounds. *Access:* **purchased**, per copy, from the standards bodies; this volume therefore describes the mechanism and reproduces nothing. *Where it lives:* iso.org, catalogue entry for standard 72026 [41].

IEC 62304, ISO 14971 and ISO 13485. The medical device software lifecycle standard, operative as the 2015 edition with amendment 1 and with a draft 2026 edition in progress; the risk management standard it derives its safety classes from; and the quality management system standard that wraps both. *Access:* **all purchased and licensed.** *Why a hospital architect should care:* a hospital that writes its own clinical decision support may be manufacturing an unregistered medical device, and SAHPRA requires evidence of all three in combination for artificial-intelligence-enabled devices [42]. *Where they live:* iso.org and iec.ch catalogues; **the clause-level detail was deliberately not retrieved for this volume because the documents are licensed.**

SAHPRA on software and artificial intelligence. The 26 September 2025 communication to industry is more developed than most architects expect: no formal predetermined-change mechanism yet, the **algorithm version locked at licensing**, minor updates through a certified quality system, major updates likely needing review, anticipated-change plans requiring approval, data triggers such as drift beyond a threshold, validation on independent datasets, subgroup analysis across sex, age and ethnicity, **local-population validation where training data came from elsewhere**, documented dataset gaps, and recording of clinician overrides [42]. *Access:* **free.** *Where it lives:* sahpra.org.za, with the communication published under wp-content/uploads/2025/09 [42].

The South African instruments

The Health Normative Standards Framework. The 2021 framework, published 21 October 2022 in *Government Gazette* 47337, notice 2667, by the Director-General of Health under section 74(1) of the National Health Act, superseding the 2014 framework [2][43]. *What it does not do:* it **names neither SNOMED CT nor LOINC**, ICD-10 appears only in its acronym list, it sets **no compliance deadline and no transition period**, and a footnote says the

version implementers must follow is the 2020 framework, which is inconsistent with its own title [2]. *Access:* **free.** *Where it lives:* health.gov.za/wp-content/uploads/2022/10/HNSF_Gazette_21_October_2022.pdf [2].

The National Digital Health Strategy 2019–2024. The policy frame, which requires conformance testing of all health information systems and proposes extending the normative framework into a Health Enterprise Architecture. **No Health Enterprise Architecture document could be located**, and **no gazetted successor strategy was found.** *Access:* free. *Where it lives:* health.gov.za [43].

The medical scheme claim. The published national standard is the EDIFACT message MEDCLM, version 0, release 912, controlling agency ZA, revision 13.4, November 2016, which **defines no remittance or payment-advice message** [35]. *The practical correction:* the switch publishes schemas that software vendors validate their XML requests and responses against, with an integration pack and a live-environment accreditation step, so **the operative wire format is frequently the switch's own XML** [31]. Build to the switch. *Access:* the standard is free from phisc.net; the switch's integration pack is commercial. *Where it lives:* phisc.net.

The statutory and regulatory instruments. The **National Health Act 61 of 2003** is the frame — sections 13, 14, 15, 16 and 17 produce the records, confidentiality and access requirements, and section 17(2) makes unauthorised access a criminal offence [44]. **POPIA** and its **health regulations**, in force from *Government Gazette* 54268 of 6 March 2026 with no grace period, which bind funders, administrators, managed care organisations, administrative bodies, pension funds and employers and **not health care providers** [45]. **HPCSA Booklet 9** on records, and **Booklet 10** on telehealth, revised December 2021 [46][47]. The **norms and standards regulations**, promulgated 2018 and effective 2019, seven chapters and 22 regulations [48]. The **notifiable medical conditions regulations**, *Government Gazette* 41330 of 15 December 2017 [49]. **Regulation R158 of 1980**, under which provincial facility licensing still rests, made under a section of the Health Act 63 of 1977 that was wholly repealed in 2012 — **the regulation itself was not retrieved, and two provinces cite different amendment years** [50]. The **NHI Act 20 of 2023**, signed 15 May 2024, of which **no section is in force** [51]; chapter 7 carries the undertaking and the order of court that settle that. *Access:* all free except Booklet editions, which are published but whose currency is contested. *Where they live:* saflii.org and gov.za for statutes, health.gov.za and hpcsa.co.za for the rest.

The regulators, councils and public entities

Body	What it governs	Where to look, and what is retrievable
Health Professions Council of South Africa	Individual practitioners across twelve professional boards, under the Health Professions Act 56 of 1974; scope of profession and scope of practice; records and telehealth guidance	hpcsa.co.za. **Its statistics and scope pages are blocked to automated retrieval and an annual report PDF failed**, so registration counts in circulation are secondary [46]
South African Nursing Council	Nurses and midwives under the Nursing Act 33 of 2005; registers that are a precondition of employment	sanc.co.za. The online register is usable; **the statistics pages publish no figures**, and the Council states its numbers count people registered, not people nursing [52][53]
South African Pharmacy Council	Pharmacists, support personnel and pharmacy premises	pharmcouncil.co.za
Council for Medical Schemes	Schemes, administrators, managed care organisations and brokers under the Medical Schemes Act 131 of 1998; prescribed minimum benefits	medicalschemes.co.za. Annual and industry reports are published; **the practice-code standardisation manual is blocked to automated retrieval** [54] [55]
SAHPRA	Medicines, biologicals, devices and diagnostics, trials, radiation control, pharmacovigilance, under the Medicines and Related Substances Act 101 of 1965	sahpra.org.za. **The full device classification guideline could not be retrieved**, and **product registration is not yet operating** [42][56]
Office of Health Standards Compliance, and the Health Ombud	Establishment quality under the National Health Act as amended by Act 12 of 2013	ohsc.org.za. **The site publishes no inspection or complaint statistics and only one quarterly bulletin was listed**; current performance comes from parliamentary presentations [48]
Information Regulator	POPIA and PAIA; administrative fines to a statutory maximum of R10 million	inforegulator.org.za. Media statements are published; **the enforcement-notices section displayed no notices**, and section 72 guidance could not be surfaced [57]
Competition Commission and Tribunal	Private market structure, the Health Market Inquiry, the draft tariff block exemption	compcom.co.za. **The 2004 switching-market Tribunal judgment could not be retrieved from the law report site** [58]
National Health Laboratory Service, and the NICD	Public diagnostic pathology; communicable disease surveillance and case definitions	nhls.ac.za and nicd.ac.za. **Laboratory and test-volume figures needed to size the concentration risk were not stated in the sources retrieved** [59]
Statistics South Africa and Home Affairs	Death certification under the Births and Deaths Registration Act 51 of 1992, and cause-of-death coding	statssa.gov.za; the certification guide at samrc.ac.za [60]

The industry bodies

PHISC, the Private Healthcare Information Standards Committee. The maintainer of the private-sector message, coding and pharmaceutical standards, with subcommittees for clinical coding, information governance, message standards, pharmaceutical coding, the electronic health record, and devices and diagnostics — and a **dormant** diagnosis-related group subcommittee, which is why South Africa has no case-mix grouping [61]. *Access:* standards published free. *Where it lives:* phisc.net.

The Board of Healthcare Funders. The funder industry body. Its practice numbering is in practice a second key alongside a provincial licence before a private facility can trade [62]. *Where it lives:* bhfglobal.com.

The South African Medical Association. Holder of CCSA copyright and publisher of the doctors' coding manual, which is the main procedural billing reference for most practitioners [32]. *Where it lives:* samedical.org.

COHSASA. Voluntary third-party accreditation across Africa, with 49 accredited facilities in ten countries, 29 of them South African, on accreditation cycles of two, three and four years — a minority practice against a public estate of roughly 3,700 establishments [63]. **Whether the international accreditation body has accredited the COHSASA standards could not be confirmed.** *Where it lives:* cohsasa.co.za.

SAMED and HASA. The device industry association, whose establishment-licence guide is the clearest free account of the SAHPRA licensing regime [56]; and the private hospital association, which was a party to the 2010 tariff litigation [64]. *Where they live:* samed.org.za and hasa.co.za.

The maturity instruments, and the tooling

HIMSS EMRAM. Eight stages, 0 to 7. *Access:* **proprietary criteria, and validation is run by HIMSS**; the stage requirements for the lower stages and the revised stage themes **could not be retrieved**, and the announcement pages for the model's revision returned errors [65]. *South Africa:* **no validated South African stage achievement could be found**, which is itself a finding. *Where it lives:* himss.org/maturity-models/emram.

The Health Information Systems Interoperability Maturity Toolkit. Developed by MEASURE Evaluation with the Health Data Collaborative, version 1.0 in 2017 and updated January 2019: **free and open**, health-system rather than hospital level, five levels from Nascent to Optimized, three domains and 18 subdomains, **business continuity as a named subdomain**, and a scoring rule that sets each domain at its lowest subdomain [66]. **No published South African assessment using it could be found.** *Where it lives:* measureevaluation.org and healthdatacollaborative.org [66][67].

The enterprise architecture tooling. One finding governs this whole category: **this research found no published list price for any enterprise architecture repository tool.** One vendor publishes a pricing page describing a modular structure and licence role types — base platform with unlimited viewers, authors and scoped contributors, plus add-ons — **with no figures and no currency**; for every other tool named in the research no vendor-published pricing was located at all [68]. Ownership has also moved: one repository vendor acquired two others and states it will keep three products in parallel while developing a fourth platform [69]. And a second finding matters more than price: **no verifiable evidence was found that any enterprise architecture tool ships maintained healthcare reference content** — no healthcare capability model, clinical process library, healthcare application taxonomy, HL7 or FHIR metamodel extension, or IHE profile catalogue; the one vendor pricing page that could be retrieved names no standards or frameworks in its text at all [68]. Treat any such claim as a claim to be inspected: ask to see it, ask when it was last updated, ask who maintains it, and verify metamodel extensibility, interface modelling and evidence governance in a proof of concept against the organisation's own data. No product is recommended here, and the absence of detail about any of them reflects what could be sourced rather than a judgement.

> **TWO THINGS TO RE-CHECK BEFORE YOU RELY ON THIS APPENDIX**
>
> **The Constitutional Court's National Health Insurance judgment.** Cases CCT 251/25 and CCT 269/25 were heard on 5 to 7 May 2026 and judgment was **still reserved at 31 October 2026**, the research date. No section of the NHI Act was in force at that date, and chapter 7 carries the undertaking and the order of court that put it beyond argument [51]. Check the Constitutional Court's judgments index before quoting the legal position, and check it again before any procurement decision that turns on it.
>
> **The status of the SNOMED CT National Release Centre for South Africa.** SNOMED International's member page describes the centre as being established, names no launch date and lists General Assembly and Member Forum representatives as to be advised [16]. Whether a South African extension namespace has been allocated, whether national reference sets or an ICD-10 map are planned, and what access costs under the member licence all remain unestablished [15][16]. This determines whether an organisation adopting SNOMED CT now should expect national content inside its planning horizon. Re-check the member page and ask the Department's clinical coding advisory working group directly.

Not retrievable, and why

The research behind this volume recorded 217 items it could not establish. These are the ones that bear directly on a standards or governance decision, and each is a question to put to a named body rather than a gap to fill with an assumption.

- **Whether any laboratory is approved to perform conformance testing under the gazetted framework, and whether any attestation has ever been issued.** The framework describes the process; nothing confirms an operating facility [2]. The whole prescriptive force of the framework rests on this.
- **The inventory of published National Interoperability Specifications.** No departmental register or publication list was found [2].
- **Whether the gazetted framework is scheduled for revision to add SNOMED CT and LOINC.** Not found.
- **The current edition of the ICD-10 Master Industry Table**, and whether the morbidity coding standards have been revised beyond version 6 of April 2014 [19][20].
- **The cost of SNOMED CT access under South African membership.** The licensing page is generic [15].
- **The full SAHPRA device classification rules for classes A to D**, and what a hospital that holds and uses devices must do, as distinct from a manufacturer, distributor or wholesaler [56].
- **The current legal basis, if any, for electronic signature of Schedule 5 to 8 prescriptions.** The handwriting rule is available only second-hand, quoted inside the Council's records guidance, and the position under the Electronic Communications and Transactions Act 25 of 2002 is open [46].
- **The HPCSA's current telemedicine guidance**, and the operative edition of Booklet 9 [46][47].
- **The Council for Medical Schemes standardisation manual** on practice code numbering, discipline codes and place-of-service codes — blocked to automated retrieval [55].
- **The architecture of the KwaZulu-Natal provincial reporting system**, and the current version levels of the provincial patient administration systems. Neither is publicly documented [43].
- **The clause-level content of the IEC and ISO device standards**, which are licensed documents. A consultancy making clause-level claims should hold licensed copies [41].

Everything else in this appendix is sourced and dated. Where two sources disagree — and in this industry they constantly do — both are given above, and neither is averaged away.

THE CHAPTER IN TEN LINES

- Three categories of standard: syntax and transport, semantics, and profiles. Only the third makes systems interoperate.
- FHIR, DICOM, LOINC, IHE, openEHR, OMOP and OpenHIE are free; the IEC and ISO device standards are purchased; HL7 version 2 and CDA licensing terms were not established here.
- SNOMED CT is free to use in member countries, but the cost of access under South African membership is stated nowhere retrievable.
- CCSA copyright sits with the South African Medical Association and CPT copyright with the American Medical Association; CCSA is positioned for data and not for billing.
- NAPPI is owned and maintained by a claims switch, so the national private medicine code is a licensed commercial dependency.
- The gazetted interoperability framework is IHE-based, names neither SNOMED CT nor LOINC, and sets no compliance deadline.
- Its conformance-testing and attestation regime is the best procurement lever in the country, and no approved laboratory or issued attestation could be confirmed.
- The ICD-10 Master Industry Table is free and effective-dated; the latest edition anyone could locate is January 2014.
- No published list price exists for any enterprise architecture tool, and no tool was shown to ship maintained healthcare reference content.
- Two items must be re-checked: the Constitutional Court's NHI judgment, reserved at 31 October 2026, and the status of the SNOMED CT National Release Centre.
- Eleven standards or governance facts could not be retrieved at all, and each is a question for a named body rather than a gap to assume away.
- Several regulators publish less about themselves than Parliament does about them; for current performance, read the committee papers.

SOURCES FOR THIS CHAPTER

- **1.** IHE Profiles for Health Information Exchange https://www.iheusa.org/sites/iheusa/files/wysiwyg/inline-documents/IHE_Profiles_for_Health_Information_Exchange.pdf
- **2.** Health Normative Standards Framework 2021, Government Gazette 47337, Notice 2667, 21 October 2022 https://www.health.gov.za/wp-content/uploads/2022/10/HNSF_Gazette_21_October_2022.pdf
- **3.** HL7 Z-segments, practitioner explanation (Rhapsody) https://rhapsody.health/resources/hl7-z-segments/
- **4.** HL7 FHIR version history and directory https://www.hl7.org/fhir/history.html
- **5.** Release Plan for R6, HL7 FHIR Management Group https://confluence.hl7.org/spaces/FMG/pages/256509111/Release+Plan+for+R6
- **6.** 45 CFR 170.215, standards adopted for application programming interfaces (United States) https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-D/part-170/subpart-B/section-170.215
- **7.** SMART Health IT documentation index (App Launch, Backend Services, Bulk Data) https://docs.smarthealthit.org/
- **8.** CDS Hooks 1.0 specification, HL7 https://cds-hooks.hl7.org/1.0/
- **9.** DICOM standard, current output, Part 4 service class specifications https://dicom.nema.org/medical/dicom/current/output/html/part04.html
- **10.** DICOM standard, PS3.15 security and system management profiles https://dicom.nema.org/medical/dicom/current/output/html/part15.html
- **11.** IHE IT Infrastructure Technical Framework, Volume 1, chapter 10 (XDS.b) https://profiles.ihe.net/ITI/TF/Volume1/ch-10.html
- **12.** IHE IT Infrastructure Technical Framework Revision 17, Volume 1, integration profiles https://www.ihe.net/uploadedFiles/Documents/ITI/IHE_ITI_TF_Rev17-0_Vol1_FT_2020-07-20.pdf
- **13.** IHE mXDE, Volume 1 (mobile cross-enterprise document data element extraction) https://profiles.ihe.net/ITI/mXDE/volume-1.html
- **14.** SNOMED CT October 2026 International Edition release notes, SNOMED International https://conf.spaces.snomed.org/wiki/spaces/RMT/pages/1242693633/SNOMED+CT+October+2026+International+Edition+-+SNOMED+International+Release+notes
- **15.** Get SNOMED CT, licensing (SNOMED International) https://www.snomed.org/get-snomed
- **16.** SNOMED International, South Africa member page https://www.snomed.org/members/south-africa
- **17.** SNOMED International announcement, South Africa adopts SNOMED CT, 14 May 2026 https://forums.snomed.org/t/major-milestone-for-africa-south-africa-adopts-snomed-ct-to-drive-digital-health-transformation/1332
- **18.** SNOMED CT National Release Center Guide, SNOMED International https://confluence.ihtsdotools.org/display/DOCNRCG/SNOMED+CT+National+Release+Center+Guide
- **19.** User guide to the ICD-10 Master Industry Table, January 2014, National Department of Health https://www.health.gov.za/wp-content/uploads/2021/02/cd-10mit_userguide_january2014.pdf
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